



User Manual

Maxsurf Modeler 2025
MOSES Modeler 2025
SACS Hull Modeler 2025
19 June 2025

License & Copyright

Modeler Program & User Manual

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Licensing

For licensing information look at the latest release note or Installation guide (InstallationGuide.pdf stored in same location as the executable).

Contents

License & Copyright	iii
Licensing	iii
Contents	v
Differences between versions	11
Coordinate System	11
Maxsurf Modeler	11
MOSES Hull Modeler	11
SACS Hull Modeler	12
About This Manual	12
Chapter 1 Introduction	13
Modeler	13
Structure	13
Definitions	14
Concepts	14
Workflow	14
Chapter 2 Basic Principles	15
The Spline and Spring Analogy	15
B-Spline Properties	18
Bibliography	18
Chapter 3 Getting Started	19
Installing Modeler	19
Learning Modeler	19
Chapter 4 Modeler Windows	20
View Windows - General	20
Modeler Coordinate System	20
Setting the Frame of Reference	21
Setting the Zero Point	22
Setting the Vessel Type	23
Setting the Windage Surfaces	25
Modeling windage profile shape on CL	28
Setting Units	29
Taking Measurements	30
Setting up the Grid	30
Drawing Settings	35
Displaying Contour Lines	36
Zoom, Shrink, Pan and Home View	39
Saved Views	41
View Windows	43
Body Plan Window	43
Plan and Profile Windows	47
Perspective Window	48
Part Window	52
Tool Panes	52
Assembly Pane	53
Properties Pane	55
Table Windows	58
Table Windows – General	58
Control Points Window	64
Markers Window	65
Curves Window	66
Surfaces Window	66
Offsets Window	66
Part Window	69
Graph Windows	70
Graph Windows - General	70

Curve of Areas Window	71
Calculations Window	72
Expressions	73
Statement Syntax	73
Built-in Variables	73
Built-in Functions	74
Calculation Units	75
Solving Calculations	75
Saving Calculations	76
Modeler Settings	76
Modeler Preferences	76
Color and Font	78
Auto-saved backup recovery files	79
Chapter 5 Using Modeler	79
Working with Markers	80
Working with Markers – Concepts	80
Working with Markers – Procedures	81
Generating Markers for Developable Surfaces	92
Marker to Surface Measurements	93
Working with Curves	95
Curve Types in Modeler	95
Curve Stiffness	97
Curve Properties	98
Curve Operations	100
Working with Surfaces	106
Surface Types in Modeler	107
Surface Stiffness	108
Surface Appearance	110
Surface Properties	111
Outside Arrows	113
Surface Materials and Skin Thickness	114
Surface Precision	115
Displaying Curvature on Contours	117
Rendering a Surface	118
Surface Operations	123
Surface Creation	134
Trimming Surfaces	139
Bonding Surfaces	149
Generating NURBS Surfaces from Parts	159
Working with Mesh Surfaces	159
Add Mesh from 3 or 4 markers	160
Mesh from simple geometry	160
Add Mesh from planar curve	165
Add Mesh from extruded curve	166
Extrude Mesh	168
Mesh visibility	169
Locking/Unlocking a Mesh	169
Deleting a Mesh	169
Importing a Mesh	169
Exporting a Mesh	169
Adding a node to a Mesh	169
Deleting a node from a Mesh	169
Snap Node(s) to Surface	169
Selecting a triangle from a mesh	170
Deleting a triangle from a mesh	171
Flip adjacent tri pair	171
Mesh node properties	171
Outside Arrows	171

Inverting a Mesh	171
Compact Mesh nodes	171
Moving a Mesh Surface	171
Move Mesh Freeform	172
Size a Mesh Surface	172
Flip a Mesh Surface	173
Rotate a Mesh Surface	173
Align Mesh Surfaces	173
Offset a Mesh Surface	174
Split a Mesh	175
Join Mesh Surfaces	176
Boolean operations for closed Meshes	177
Mesh Repair	179
Mesh bonding	180
Mesh meshed from a NURB surface	185
Mesh from Parametric-based NURB Surface mesh	188
Tips for creating a NURB surface that will mesh well	190
Mesh from Structural Parts	193
Detailed Mesh Statistics	194
Mesh rendering options	195
Quadmshes	197
Quadrilateral and Triangle mesh elements	198
Generating quadrilateral meshes	198
Quadmshes can be added on structural parts	199
Import and export	199
Mesh operations	199
User interface changes	200
Web Meshes	200
Working with Control Points	200
Adding Control Points	201
Deleting Control Points	202
Control Point copy and paste	202
Moving Control Points	203
Align Control Points To First Selected	204
Aligning Control Points	204
Smooth Controls or Straighten Controls	207
Manipulating Groups of Control Points	209
Compacting Control Points	213
Grouping Control Points	215
Displaying Control Points	215
Control Point Weight	217
Advanced - Transpose	221
Surface Fitting	221
Surface Fitting – Concepts	222
Fitting a NURB Surface to Markers (Manual fit)	226
Fitting a NURB Surface to Markers (Automated fit)	230
Fitting a NURB Surface to Marker stations (Automated fit)	231
Fitting a NURB Surface to Markers (Least squares fit)	234
Measuring Surface Error	234
Mesh Surface from Markers	235
Calculations	247
Hydrostatics	247
Calculate Girth	249
Calculate Areas	250
Using Parametric Transformation	251
Parametric Transformation - Concepts	251
Parametric Transformation – Procedures	252
Search Parameters	252

Scaling factors.....	254
Hull Shape Comparison.....	255
Parametric Transformation Restrictions.....	255
Americas Cup Yachts	256
Working with Structure	257
Preparing the design for generating structure	257
Working with Frames.....	258
Adding a Frame.....	258
Adding a Frame Opening	261
Calculating Frames.....	266
Frame Visibility	267
Deleting Frames	268
Modifying Frames	268
Moving Frames.....	268
Copying Frames	268
Working With Decks	269
Adding a Deck	269
Deleting a Deck.....	270
Adding a Deck Opening	270
Drawing Decks.....	271
Deck Visibility.....	271
Modifying Decks.....	272
Moving Decks	272
Copying Decks.....	272
Deck Limitations	272
Working with Stringers.....	272
Stringer Tips and Tricks	273
Adding Stringers.....	273
Generating Stringers.....	274
Generating a Family of Stringers	275
Modifying Stringers.....	276
Longitudinal Girders.....	277
Adding Points To Stringers	282
Moving Stringer Points	283
Deleting Stringer Points	285
Modifying Stringer Points.....	285
Duplicating Stringers	286
Breaking Stringers.....	288
Joining Stringers	288
Breaking Stringers at Bulkheads	288
Mirroring Stringers	290
Girth Centreline.....	291
Calculating Stringers	291
Calculating Stringer Inverse Bending Lines	292
Working with Stringer Shapes	294
Sections Library.....	294
Setting Shapes & Cutouts	295
Stringer Orientation	295
Working with Upstands	296
Calculating Upstands.....	297
Upstand Cutouts.....	297
Upstand Orientation	297
Working with Inclined Sections.....	297
Adding Inclined Sections	297
Displaying Inclined Sections.....	298
Working with Plates	299
Working with Plates - Concepts.....	299
Working with Plates – Procedures	299

Plating a Whole Surface	299
Adding a Plate	302
Deleting a Plate.....	304
Defining Triangular Plates	304
Plate clash detection.....	305
Plate Development Methods and Options.....	306
Calculating Plates	310
Girth Differences.....	311
Plate Strain.....	312
Plate Development and Rolling Jigs.....	313
Plate User Coordinate Systems (UCS).....	315
Plate Inset Lines	316
Plate Templates.....	316
Plate Pin Jigs.....	317
Shell Expansion.....	318
Plate Displays	319
Working with Parts	322
Calculating Part Weights.....	322
Nesting Parts	322
Exporting Parts	325
Materials and Section Libraries	327
Materials Library	327
Sections Library.....	329
Data exchange.....	330
Input of Data.....	331
Pasting (General)	331
Importing Background Images.....	331
Importing DXF splines and polylines as curves	335
Importing DXF background	335
Importing DXF Markers	336
Importing IGES Curves and Surfaces.....	337
Importing Rhino .3dm files.....	338
Output of Data	348
Printing	348
Copying	351
Export Lines Plan	352
Offsets Data Output.....	352
Animation Files.....	353
Exporting a Modeler Design.....	354
Exporting Parts to Rhino	359
Exporting Parts to Multiframe.....	360
Chapter 6 Modeler Reference	364
Toolbars	365
Customising Toolbars.....	365
Modeler Toolbars	368
Menus	369
File Menu	370
Bentley Cloud Services Submenu	372
Edit Menu	372
View Menu	373
Marker Menu.....	374
Mesh Menu.....	376
Controls Menu.....	379
Curves Menu	381
Surfaces Menu	382
Frame Menu	385
Deck Menu.....	385
Stringer Menu.....	386

Plate Menu	388
Display Menu	388
Data Menu.....	391
Window Menu	393
Help Menu.....	394
Right Click Menu	395
Surface Selection Right Click Menu	395
Ribbon	395
Appendix A – Data Export.....	396
Supported File Formats	396
Importing.....	396
Exporting.....	397
Pictures and Text.....	397
Copying the drawing	397
Copying Text	398
2D Drafting Data	398
3D Drafting Data	399
3D Surface Definition	400
Exporting to ShipConstructor Hull.....	401
Appendix B – Surface Algorithms	401
Appendix C – Command Keys	402
Modifier Keys.....	402
Special Keys.....	402
Control Point Weights.....	403
Menu Shortcuts.....	403
Appendix D – Moving Files Across Platforms	404
Moving Files From Macintosh to Windows.....	404
Moving Files from Windows to Macintosh	405
Appendix E – MOSES Objects.....	406
User Interface.....	406
Introduction.....	407
MOSES Model.....	407
Hydrodynamic model.....	408
Structural model	408
Definition of MOSES Objects	408
Automatic PART, COMPARTMENT and PANEL objects	409
Geometry symmetry of PLATE and PANEL objects	409
Node Reuse	412
Verifying the Model and Object data	413
Saving Model and Object data	413
Table UI and Column editing	414
Common columns.....	416
BODY table.....	416
PART table	417
COMPARTMENT table.....	417
Piece: PLATE table	417
Piece: PANEL table	417
Piece: TUBTANK table.....	418
BEAM table.....	418
CLASS table	418
POINT table.....	419
Index.....	420

Differences between versions

There are several different versions of Modeler; here the differences between them are explained.

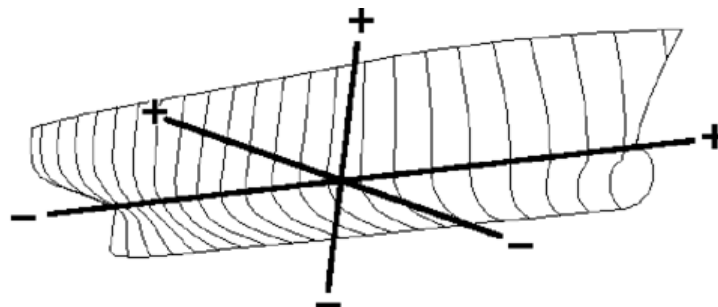
Coordinate System

All Modules: Body plan, view from behind (stbd side of vessel on right-hand side of screen)

All Modules with the exception of Stability: Plan view from bottom (stbd side of vessel on upper half of screen); Stability has the Plan view from top (stbd side of vessel on lower half of screen)

All Modules: Profile view from stbd (bow on right-hand side of screen)

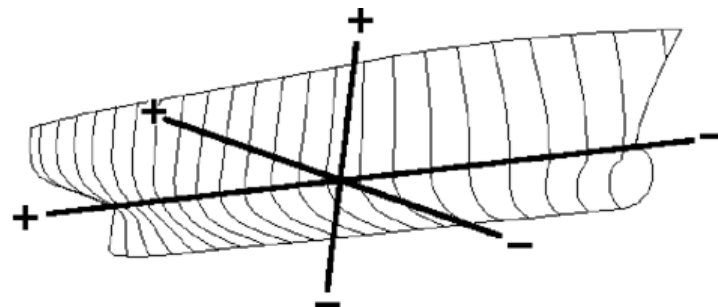
Maxsurf Modeler



Longitudinal:	+ve forward	-ve aft
Transverse:	+ve stbd	-ve port
Vertical:	+ve up	-ve down

The zero-point can be defined at any location in the model

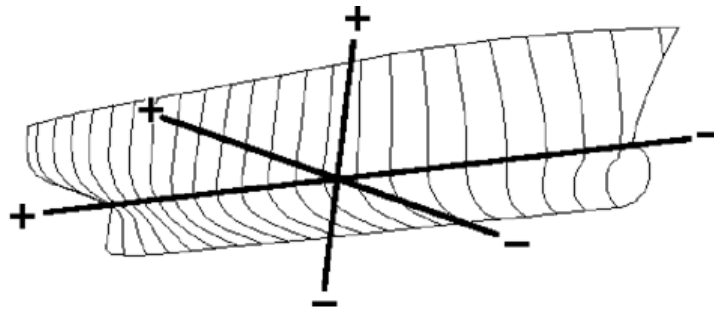
MOSES Hull Modeler



x-Axis (longitudinal):	+ve aft	-ve forward
y-Axis (horizontal):	+ve stbd	-ve port
z-Axis (vertical):	+ve up	-ve down

The zero-point should be defined to coincide with the furthest forward and the lowest part of the model.

SACS Hull Modeler



x-Axis (longitudinal):	+ve aft	-ve forward
y-Axis (horizontal):	+ve stbd	-ve port
z-Axis (vertical):	+ve up	-ve down

About This Manual

This manual describes the Modeler program and is divided into five chapters. It introduces you to the features of Modeler in a step-by-step manner. Many of the examples given use designs supplied with your Modeler application and contained in a folder titled 'Sample Designs'.

If you are unfamiliar with your computer you should read your owner's manual. This will introduce you to commonly used terms such as click and drag, and the basic methods for using any Windows application.

The chapters in this manual are as follows:

[Chapter 1 Introduction](#): What is Modeler?

[Chapter 2 Basic Principles](#): An introduction to the concept of the NURB surface and how its shape is controlled by a net of control points.

[Chapter 3 Getting Started](#): Directs you to sources for installing and learning to use Modeler.

[Chapter 4 Modeler Windows](#): Outline of different features available in each of the windows in Modeler.

[Chapter 5 Using](#): Further explanation on how to manipulate surfaces, do calculations and import and export data.

[Chapter 6 Modeler Reference](#): A list of each menu item and command, which may be used as a quick reference whenever you are using the program.

Many of the examples given in the manual have intentionally been kept as simple as possible so that you can clearly see how each of Modeler's functions behaves. By reading the manual and experimenting with the sample designs, you will very quickly become an expert user.

Note

In case you are new to Modeler it is highly recommended to read the first two chapters of this manual. This will give you a good understanding of fundamental concepts and functions in Modeler.

Chapter 1 Introduction

Modeler

Modeler is a powerful three-dimensional surface modelling system for use in the field of marine design. It provides you with a clear and familiar environment in which to work, allowing for systematic experimentation and rapid optimisation of any new design.

Modeler's multiple surface capabilities, allowing any number of surfaces to be modelled in any given design, offer scope for the creation of a wide range of hull forms. Combined with the built-in hydrostatic calculations, you have the tools to experiment with shapes and explore design parameters.

Highly accurate output is produced in the form of hull lines, transfer files for other programs, and comprehensive offsets tables. Data transfer to other programs in the Modeler suite is from a Modeler design file, alleviating the need for the re-entry of data once a design has been finalized, and removing the possibility of loss of accuracy through the use of incomplete hull offset files.

The above features offer you a truly integrated system. Each of the modules in the Modeler range work from the same data file and each share a common user interface.

Modeler, as well as its downstream analysis modules, offers an Automation interface. There are also several automation samples available in the Modeler program files directory.

Structure

The integration of the structural modelling capability (previously in the Structure module) provides the flexibility of creating the vessel primary structure while you are designing and fairing the hull shape. In this manual, when referring to this aspect of Modeler we shall refer to Modeler-Structure.

Modeler-Structure is a structure modelling utility that allows the designer to model the primary structure within the 3D NURB surface model. All parts created in Modeler-Structure are parametric, meaning that the parts will automatically update to changes in the surface model. This allows the designer to model the structure concurrently with the hull shape. For example, the designer can start defining structural elements while the hull form still requires final fairing.

It is important to realise that Modeler-Structure is a naval architecture tool which compliments a traditional CAD package such as Microstation. It allows the primary structure to be modelled in earlier stages of design and thus contributes to the optimisation/iteration routine; not only by optimising the structural arrangement but also indirectly by providing accurate and up-to-date weight calculations for stability and performance analysis. Finally it can then be used to produce highly accurate production output, such as developed plates, stringers and frame contours, when the design is finalised. Modeler-Structure has several export formats available that allow the designer to work in a flexible CAD/CAM environment.

Modeler-Structure's features include:

- Automatic or manual generation and fitting of stringers.
- Stringer shape and cutout definition.
- Stringer development.
- Frame generation.
- Deck generation.
- Plate development of developable plates.
- Plate expansion of non-developable plates.
- Plate forming templates.
- Plate pin jigs.

- Export of all parts to ShipConstructor, Autocad and other CAD or CAM systems.

Definitions

Stringers – stiffeners of hull shell plate. Wherever stringers pass through a frame a cutout will be placed in the frame to allow the stringer to pass through. The stringer paths may also be used as plate boundaries or as reference lines on plate developments.

Cutouts - The Section Library contains a database of stringer shapes and cutouts. When a stringer passes through a frame, one of the cutouts from the library will be used to place the cutout in the frame. The library also contains a list of materials.(see below)

Upstands - An upstand is a short line that runs from the intersection of a stringer and a frame in the orientation-direction of the stringer. It allows you to override the orientation and cutout shape of a stringer at a particular frame.

Frames – Frames are transverse or inclined structural components located at a particular section or inclined section. They can contain any number of openings including an interior opening at a particular web depth.

Decks – Decks are horizontal components located at a particular waterline. They can contain any number of openings.

Plates - Plates are 4 sided regions on a NURBS surface defined by four contours which form its boundaries. Plates have a thickness which is defined by their material types.

Materials - Each part is made from a particular material. A database of materials is stored in the Section Library which is opened automatically when the program starts up. The library also contains a list of shapes and cutouts. (see above)

Concepts

Modeler-Structure allows you to define where parts will be located by referring to construction lines on the surface of the hull. The construction lines you can use are sections, waterlines, buttocks, diagonals, inclined sections, feature lines, parametrics, intersections, surface curves and edges. Using Modeler-Structure involves selecting one or more construction lines to define the location on the design where each part is to be created.

Modeler-Structure allows you to create a complete and accurate set of parts including stringers, frames, decks and plates. You can then export these parts via DXF file to a CAD system for further detailing or directly to an NC CAM system for cutting.

Modeler-Structure does not include every detailing feature you may require. It is designed to be used in conjunction with a CAD system such as ShipConstructor or Autocad so that any additional special details can be added after the basic part geometry has been created in Modeler-Structure. You can also use your CAD system to prepare stringer shapes and cutout shapes. These shapes can be imported into Modeler-Structure via DXF files.

Workflow

The procedure for using Modeler-Structure to build up a complete structural model is as follows;

- Define where frames will be located and what shape they will be by selecting sections and using the Add Frame command.
- Define where stringers will be located by drawing or generating stringers.
- Assign cutout shapes to stringers by selecting stringers and using the Shape & Cutout... command.
- Calculate the stringer shapes by using the Calculate Stringers command.

- Calculate the upstands at the intersection between the stringers and the frames using the Calculate Upstands command.
- Use the Calculate Frames command to generate the frame shapes including cutouts for stringers which pass through them and cutouts for internal spaces.
- Define where decks will be located and what shape they will be by selecting waterlines and using the Add Deck command.
- Define plates by selecting four contours defining the plate boundary and using the Add Plate command.
- Develop plates by using the Calculate Plates command.
- Copy frames, stringers, decks and plates to DXF files using the Export command.

The suggested procedure above is just one way of building up your parts. In fact you can use any command in any order and add and delete any type of part at any time. The suggested sequence above simply ensures that you have created frames before generating stringers between frames, created stringers before finding upstands and so on. As you become more proficient with Structure, you will probably want to define all of your parts and then use the Calc All Parts command to calculate everything in the design at highest precision.

Chapter 2 Basic Principles

This chapter outlines the concepts on which the program is based.

Hulls, appendages and superstructures are defined in Modeler using one or more surfaces. Typically a surface is used between discontinuities in the design. For example, a three surface design of a sailing yacht might be made up of one surface for the hull, one for the keel and one for the rudder. A five surface design for a workboat might use one surface for the upper hull from sheer line to chine, a second surface for the chine, a third surface for the lower hull from chine to keel, a fourth surface for the deck and a fifth surface for the transom. Designs may contain up to 1200 surfaces in Modeler Advanced and up to 20 surfaces in Basic Modeler.

Surfaces are defined in Modeler by the position of a set of control points that collectively form a control point net. Movement of these control points allows you to manipulate a surface into a desired shape.

Central to the process of modelling designs using Modeler is an understanding of how control points may be used to attain the surface shapes that you wish to achieve, and this is best explained by the following analogy.

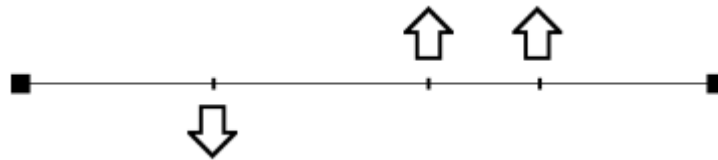
In this chapter:

- [The Spline and Spring Analogy](#)
- [B-Spline Properties](#)
- [Bibliography](#)

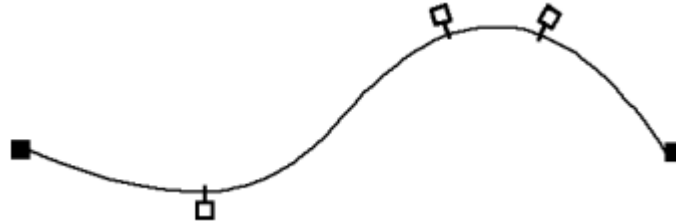
The Spline and Spring Analogy

Historically, naval architects have used flexible battens, or splines to draw smooth two-dimensional curves. By fixing the end points of the spline and applying a load at one or more points along the spline, a curve is created that can then be traced onto a drawing. The curve's smoothness will depend on the stiffness of the spline and the exact placement of the loads, but as long as a few simple rules are followed, the resulting curvature will be smooth.

The spline will initially lie straight on the drawing board.



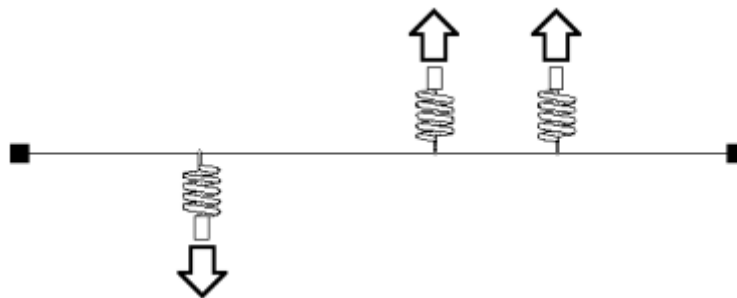
After dragging the spline at a number of points, the natural stiffness of the spline will result in the smooth curve used for drawing.



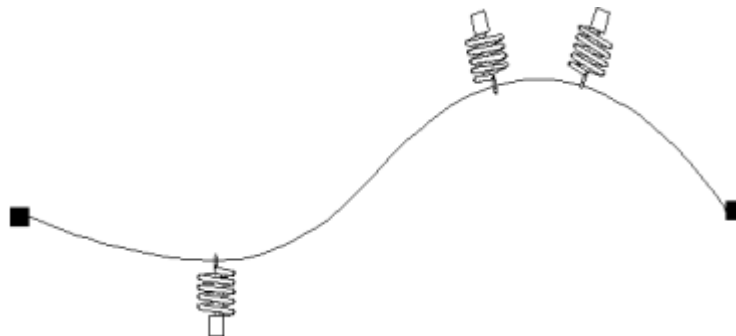
To generate its curves, Modeler uses a mathematical equation, the B-spline, which is analogous to this procedure. As with drafting splines, curves are defined by the position of their end points, the location and number of controlling points along the curve, and the stiffness of the spline.

Rather than a row of weights that sit along the spline, Modeler's curves are shaped by control points, which can be thought of as being attached to the spline by springs. When the control points are moved the inherent stiffness of the spline and the springs combine to keep the curve smooth. An obvious effect of this is that the control points do not lie on the curve created but rather the curve is attracted toward the position of the control points.

Thus, a spline, which is initially straight, is dragged to a new shape at a number of points.



The resulting curve is smooth with only the two end control points lying on the curve.

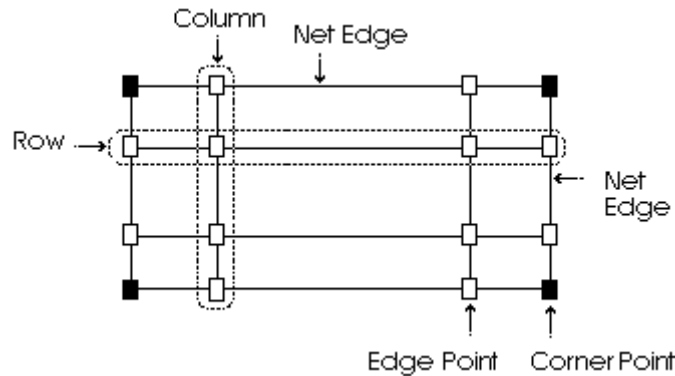


By moving the controlling points about, you can bend the spline to a given shape. The curvature of the spline would be free from irregularities, due to the elasticity of the springs and the stiffness of the spline itself. If the spline were made more flexible or stiffer, the curvature would correspondingly increase or decrease.

Although this is a two-dimensional example, Modeler uses an analogous procedure to generate its splines in three dimensions to form a surface.

Just as a row of two-dimensional control points can define a two-dimensional curve, a network of three dimensional control points can define an entire three-dimensional surface.

If you consider a network of three-dimensional controlling points, you can imagine that splines could be held along and across the net, hence defining a surface. Modeler does just this, and uses a network of three dimensional control points to generate a surface.

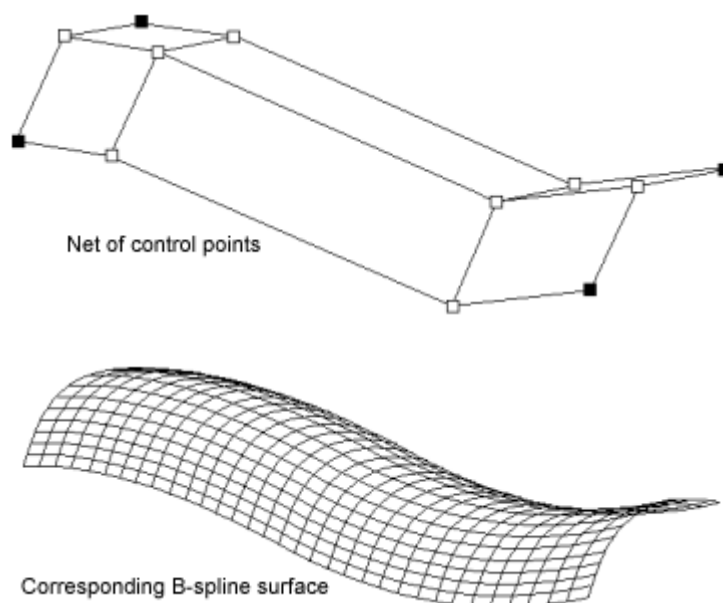


The net is formed by rows and columns of control points and has four edges and four corners. Any number of rows and columns of control points may be used, depending on the complexity of the desired surface. However it is usually desirable to use as few rows and columns as possible. The surface may have different stiffness in the row and column directions.

A surface is created by the generation of splines in three dimensions from the control points that make up the net.

The effect a control point has on a surface depends firstly on whether it is a corner, edge or internal control point.

- Corners of a surface are defined exactly by the position of the corresponding corner of the net.
- Edges are defined only by the control points on the corresponding edge of the net.
- Internal points of the surface may be influenced by many or all of the control points in the net.



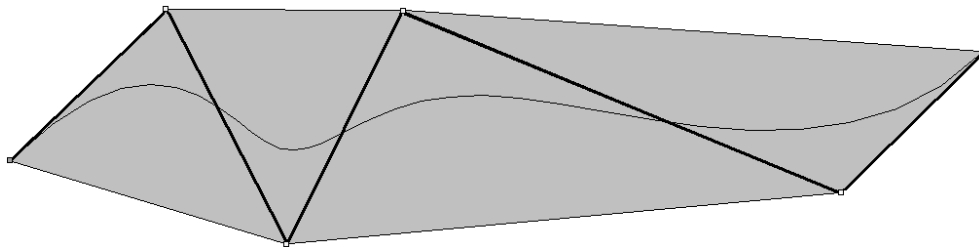
Any number of independent surfaces may be used in the definition of a design in Modeler, each with its own control point net. A control point only influences the surface to which it belongs. The only exception to this is the case where two surfaces are bonded together along a common edge. If a control point is moved on a bonded edge it will affect both of the surfaces sharing the bonded edge.

When using Modeler, remember that you are changing the position of control points relative to one another in the net to achieve a desired change in the surface. Modeler will then recalculate and display the new surface shape. Just as in the spring analogy you may only change the shape of a surface by moving control points in the net rather than directly moving the surface itself.

Continue reading [B-Spline Properties](#) on page 18.

B-Spline Properties

B-spline curves are "Variation Diminishing", meaning that it is guaranteed not to have more inflections than its net of control points. This means that if you reduce the number of inflections in a surface net to zero, there will be no inflections in the underlying surface.



Variation Diminishing, End Slope and Convex Hull (area highlighted in grey) properties of a B-spline

B-spline curves and surfaces always start and finish with the same slope as the start and finish of the accompanying net. The slope of the start and end points of the edges of a surface can always be exactly controlled using this property.

B-splines are also guaranteed not to extend outside of the convex hull (area highlighted in grey in the image above) of its control point net. This means that the surface cannot have bulges or hollows greater than the deflections of the control points.

Note

These properties combined make the Net an excellent fairness indicator. In general: if the Net is fair, then the underlying surface will also be fair.

More information on NURB surfaces can be found in the [Working with Curves](#) section on page 95.

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Chapter 3 Getting Started

This chapter briefly describes where to find the relevant resources to

- [Installing](#)
- [Learning](#)

Installing Modeler

Install Modeler by inserting the CD and running the Setup program, then follow the instructions on screen.

Learning Modeler

The best way to learn to use Modeler is to start the tutorials online at

[Bentley student center](#) (youtube)
[Formsys webinars](#) (youtube)
www.bentley.com .

Both new users as well as experienced Modeler users will benefit from working through the videos.

Continue reading:

- [Chapter 4 Modeler Windows](#)
Outline of different features available in each of the windows in Modeler.
- [Chapter 5 Using](#)
Further explanation on how to manipulate surfaces, do calculations and import and export data.
- [Chapter 6 Modeler Reference](#)
A list of each menu item and command, which may be used as a quick reference whenever you are using the program.

Chapter 4 Modeler Windows

This chapter explains the many functions available in the Modeler windows.

[View Windows - General](#): Generic features in Modeler that apply to all view windows, ie Body Plan, Profile, Plan and Perspective window.

[View Windows](#): A detailed explanation of the different windows that Modeler uses to display hull shapes, together with explanations of the various controls within the each of them.

[Tool Pane](#): Describes how to use the Assembly Pane to organise and manipulate curves and surfaces and the Properties Pane to view the properties of selected items..

[Table Windows](#): Explains both generic features available when working in any of the Table windows in the Modeler suite, together with explanations of Table window – specific features.

[Graph Windows](#): Explains both generic features available when working in any of the Graph windows in the Modeler suite, together with explanations of the Curve of Areas graph in Modeler.

[Calculations Window](#): Explains how to use the Calculations window to create custom calculations inside Modeler.

[Modeler Settings](#): Explains the Modeler preferences and display settings.

View Windows - General

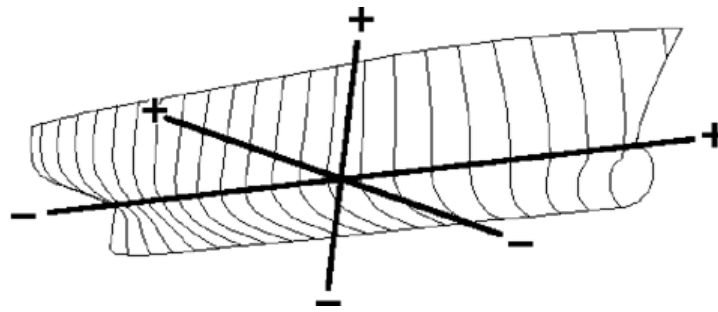
The purpose of this section is to cover the various controls and special features that each of Modeler's graphical windows contain.

This section describes:

- [Modeler Coordinate System](#)
- [Setting the Frame of Reference](#)
- [Setting the Zero Point](#)
- [Setting the Vessel Type](#)
- [Setting the Windage Surfaces](#)
- [Taking Measurements](#)
- [Setting up the Grid](#)
- [Displaying Contour Lines](#)
- [Zoom, Shrink, Pan and Home View](#)

Modeler Coordinate System

Modeler has defined its views to be consistent with Naval Architecture standards throughout all view windows.



+ve forward -ve aft
+ve starboard -ve port
+ve up -ve down

Window	View direction
Body plan	Looking forward from the stern
Profile view	Looking from starboard, bow to the right
Plan view	Looking from below, starboard above the centreline

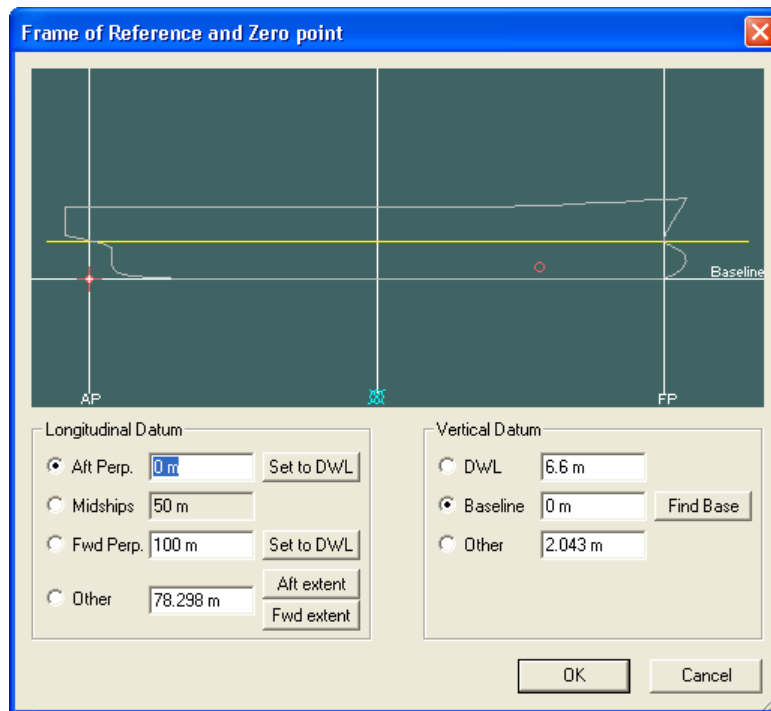
Setting the Frame of Reference

Frame of Reference - Concepts

The Frame of Reference dialog is used to specify the positions of the key locations used in ship design. These positions are:

- Forward Perpendicular
- Aft Perpendicular
- Amidships
- Datum Waterline (DWL)
- Baseline

The DWL is used for all calculations in the Calculations window, calculation of the Sectional Area Curve in the Curve of Areas window and calculation of surface areas.



The Frame of Reference dialog has functions to automatically locate the Baseline at the lowest point of the hull (the Find Base button) and to set the Fore and Aft Perpendiculars to the extreme ends of the DWL (the Set to DWL button).

In all cases where the Forward or Aft Perpendicular is updated, the Amidships position is calculated to be at the midway position between the perpendiculars.

Note:

All values are entered relative to the Zero Point. However, it may be that the zero point is set to the position of one of the elements in the Frame of Reference. In this case, the Zero Point is not updated until you click OK in the Frame of Reference dialog. See [Setting the Zero Point](#) below for more information.

Frame of Reference – Procedures

To set the Frame of Reference

- Click the Find Base button

This will find the lowest point of your surface model.

- Enter the DWL height relative to the current zero point:
For example: when your baseline is at -0.85 m and your draft is 0.5 m then you should enter -0.35 m.
- Press the Set to DWL button to set the Fwd perpendicular to the DWL.
- Press the Set to DWL button to set the Aft perpendicular to the DWL.
Alternatively you can manually update the Aft perpendicular to eg the rudder stock location if needed.
- Click OK

Please note that your Frame of Reference position values are dependent on the location of the zero point and changes when Setting the Zero Point will affect the Frame of Reference.

Setting the Zero Point

The zero point is the reference point that all measurements are taken from. You may choose the position of the longitudinal zero coordinate by selecting from the available options:

Forward Extremity.

The forward most point of all existing surfaces.

Forward Perpendicular.

This is specified in the Frame of Reference dialog where it may be automatically positioned at the intersection of the design waterline with the bow.

Amidships.

Midway between the fore and aft perpendiculars.

Aft Perpendicular.

This is specified in the Frame of Reference dialog where it may be automatically positioned at the intersection of the design waterline with the stern.

Aft Extremity.

The aft most point of all existing surfaces.

The vertical position of the Zero Point may be set to the Datum Waterline or to the Baseline. The positions of the forward and aft perpendiculars, Datum Waterline and Baseline are all specified in the Frame of Reference dialog.

The forward and aft extremities are recalculated every time you change the design. If you have used either of these points to define the longitudinal position of the zero point, then the zero point will move as the design is changed unless it is locked. Even if locked, the zero point will be recalculated if you set the frame of reference again.

The Zero point is always displayed in the Plan, Profile, Body Plan and Perspective Windows as a small vertical cross.

When the model zero datum is moved by the user in the Frame of Reference dialog, only the single value of the internal model zero point (x,y,z) is changed (as opposed to moving all objects in the design). Coordinate data displayed to the user are adjusted by this internal model zero:

user_x (longitudinal direction, positive forward)	= internal_x - zero_x
user_y (transverse direction, positive starboard)	= internal_y - zero_y
user_z (vertical direction, positive up)	= zero_z - internal_z

Very occasionally, it is useful to be able to view the internal model zero point. This is probably only useful if you are directly accessing the Stability results database where some of the data are stored in internal coordinates. It is possible to view the internal zero point in the About | Sys.Info dialog and in the Frame of Reference dialog (if the Shift key is held down when launching the dialog). The COM automation interface includes InternalZero property (FrameOfReference) which can be used to obtain the internal zero point.

Also see:

[Modeler Coordinate System](#) on page 20.

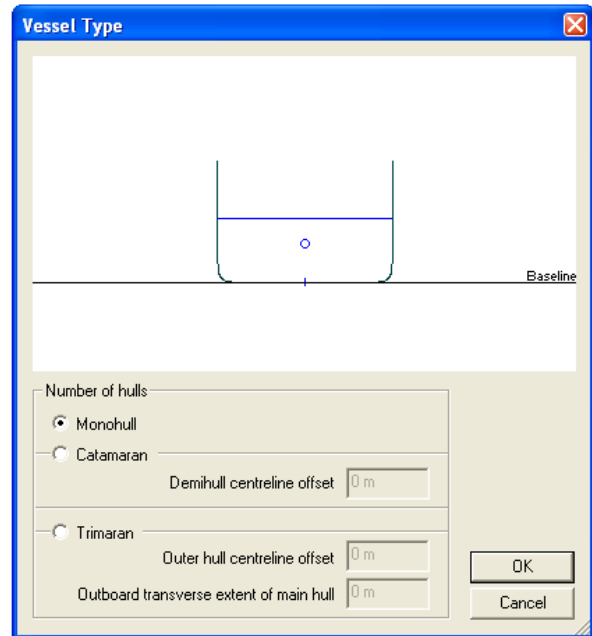
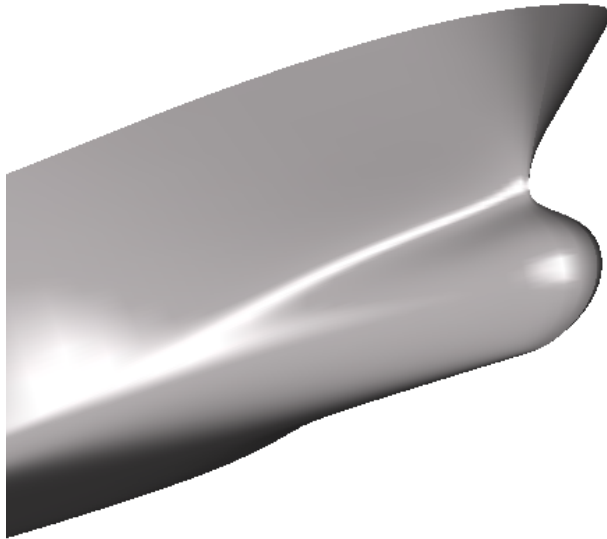
[Setting the Frame of Reference](#) on page 21.

Setting the Vessel Type

The vessel type: monohull, catamaran or trimaran, is defined in the Vessel Type dialog. This additional model information is used in other programs in the Modeler suite. Currently Resistance uses this information to define the panel arrangement for the slender body analysis. The other programs in the Modeler suite do not use the vessel type information yet.

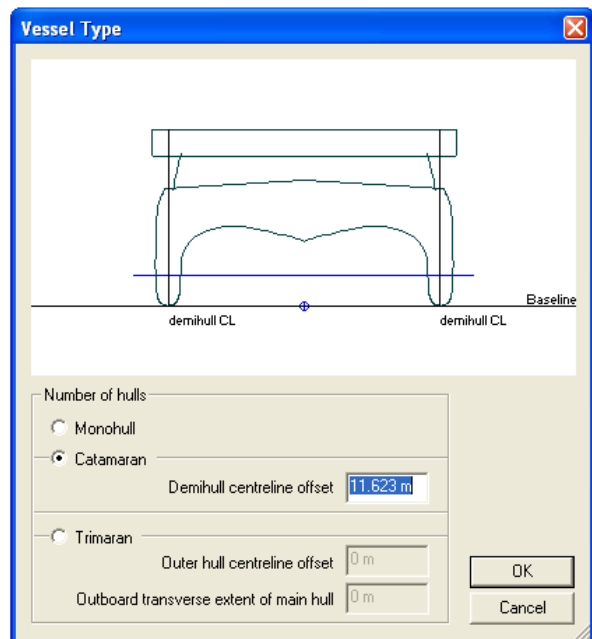
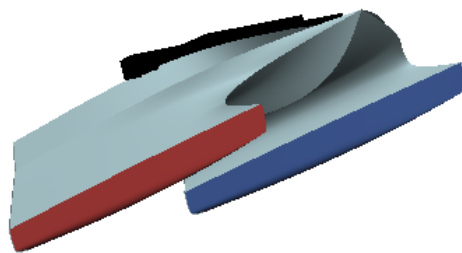
You should set up the Frame of Reference and Zero Point first. Then, when the design is finished, the Vessel Type can be setup. None of this data is changed automatically, so if you were to increase the demihull separation of a catamaran, it would be necessary to return to this dialog to update the data manually.

Monohull



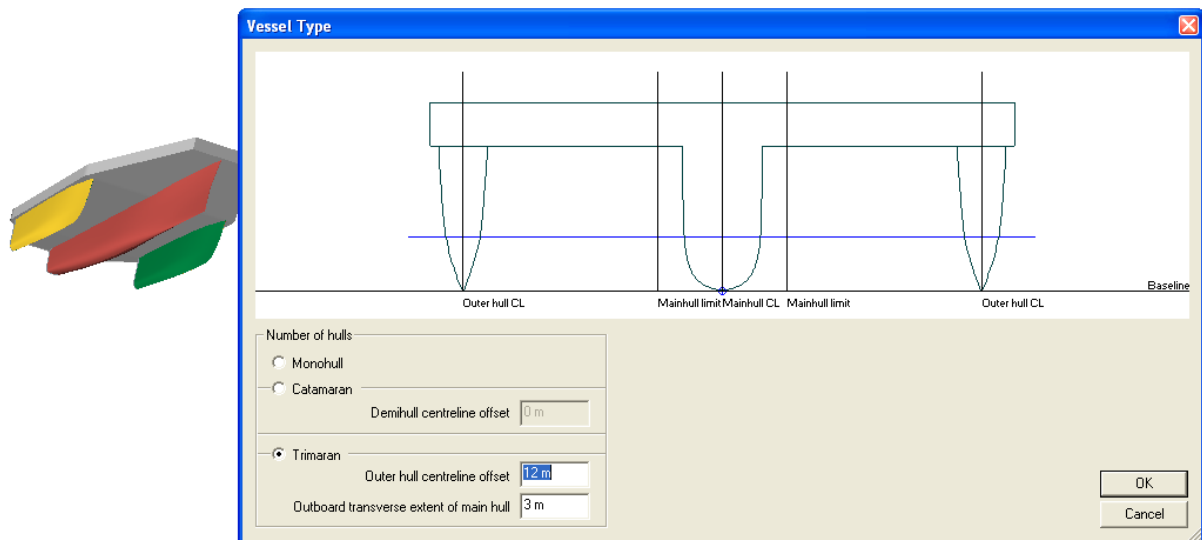
For the monohull vessel type there is nothing more to specify other than that the vessel is a monohull from the 'Number of hulls' radio items.

Catamaran



In the case of a catamaran, you should select the appropriate radio button and specify the offset of the local demihull centreline. Analysis programs, such as Resistance, will use the local demihull centreline when computing the analysis mesh for the model.

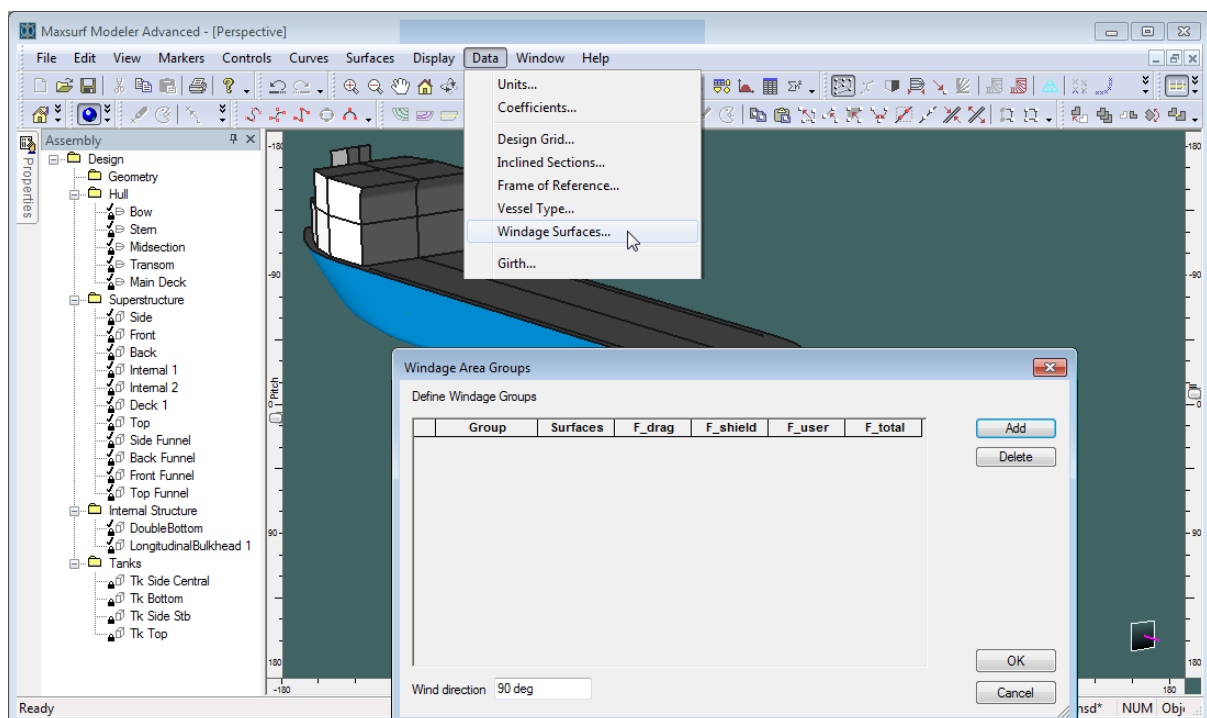
Trimaran



For trimarans, you must specify the centreline offset location of the outer hull and also the transverse extents of the main hull. Again these data are used in analysis programs such as Resistance. The transverse extents of the main hulls are used to clip the sections used in the analysis programs so it is generally sufficient to only include the underwater part of the main hull.

Setting the Windage Surfaces

Windage areas and underwater projected areas definitions have been added to the vessel model. These data may be defined and edited in both Modeler and Stability via the Windage Surfaces dialog in the Data menu.



Windage Surfaces dialog (Data menu)

If no Windage groups are defined, then the older system for the calculation of windage and lateral projected underwater area is used. That is the hydrostatic sections are projected into the transverse plane. The outer perimeter formed by joining the upper and lower limits of these projected sections is then used to calculate both the windage area of the hull and the underwater projected area. The zero-trim waterline at the current

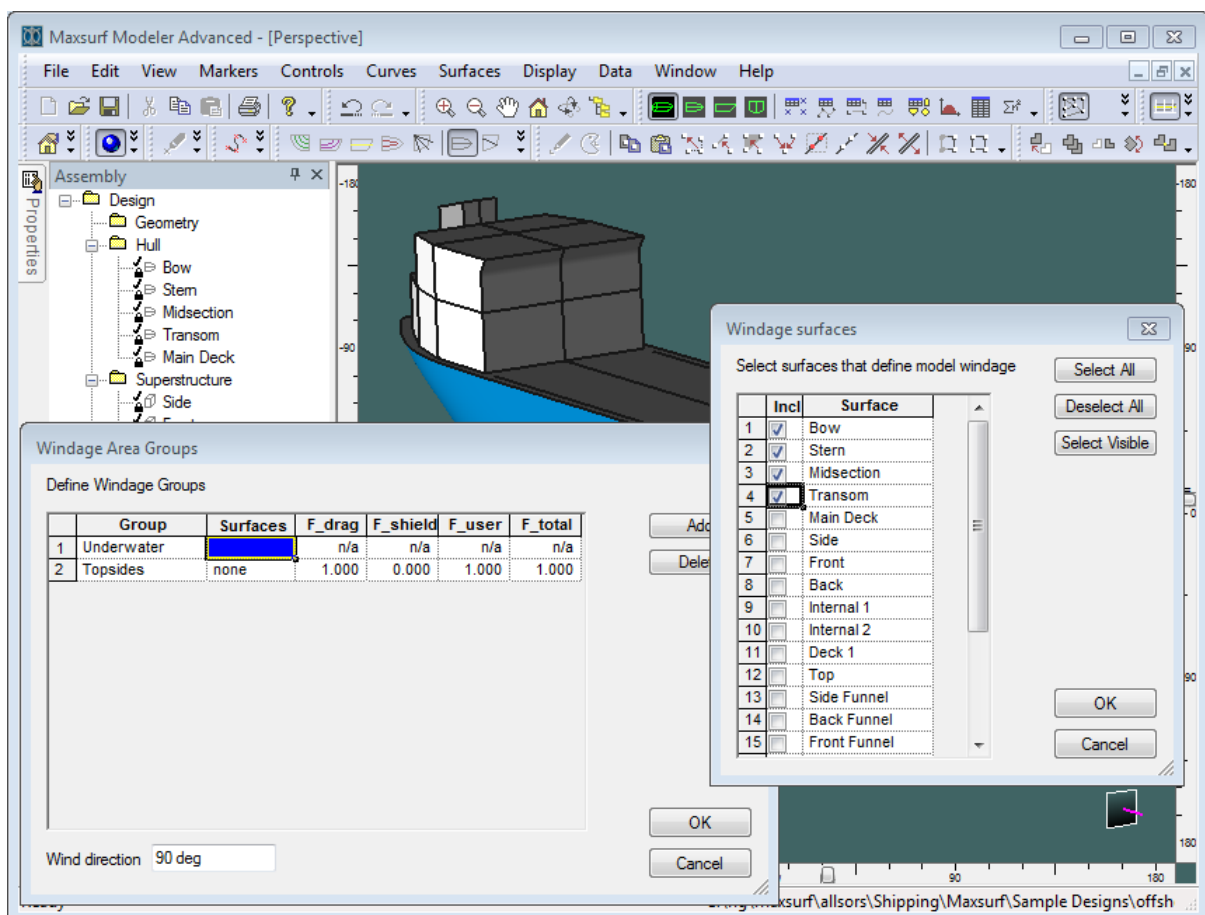
midship draft is used to determine which part of the projection is underwater and which part is windage area. Because of these limitations, the effects of vessel trim and "holes" in the model are not accounted for by this older method. The new method overcomes these limitations as well as adding new features.

Windage Groups

The concept of a Windage Group has been added. This groups together model surfaces which should be treated as a single object. There are always at least two Windage Groups and the first one defines the surfaces that should be used to calculate the underwater lateral projected area. Individual surfaces may be included in multiple Windage Groups. Apart from the underwater group, Windage Groups have various factors associated with them:

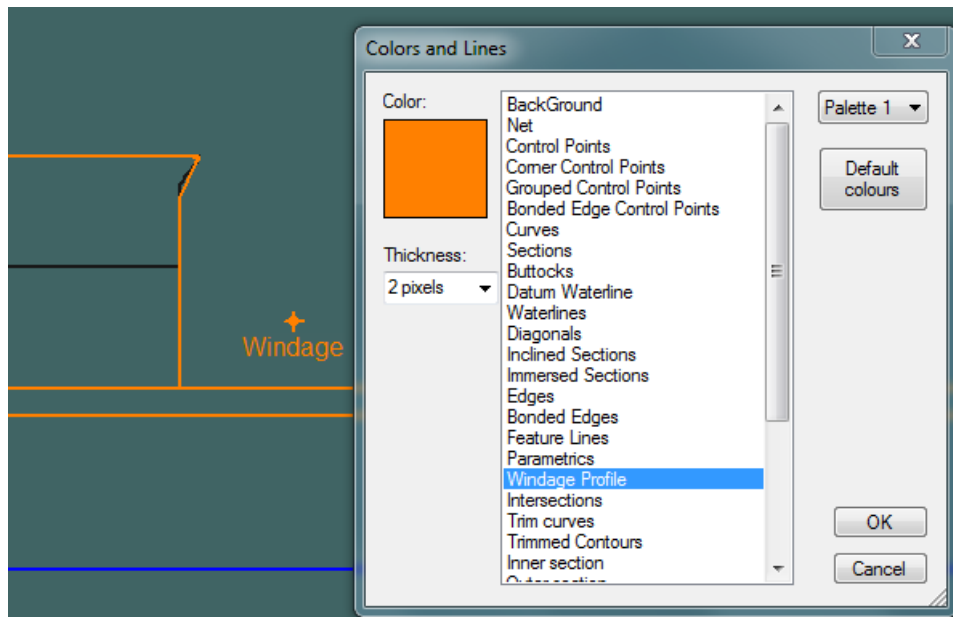
- F_{drag} : windage drag factor; default value 1.0
- F_{shield} : shielding factor; default value 0.0
- F_{user} : a user-defined factor; default value 1.0
- $F_{total} = F_{drag} \cdot (1.0 - F_{shield}) \cdot F_{user}$

Windage Groups may be added and deleted with the respective buttons in the dialog. The surfaces to be included in each group are defined in selected by double clicking in the "Surfaces" cell in the table, in a similar manner to the selection of boundary surfaces for Tanks and Compartments.



Windage Group definition and Surface selection

The color of the Windage Profile outline can be changed in the Colors dialog; the underwater profile is shown using the "Immersed Sections" color.

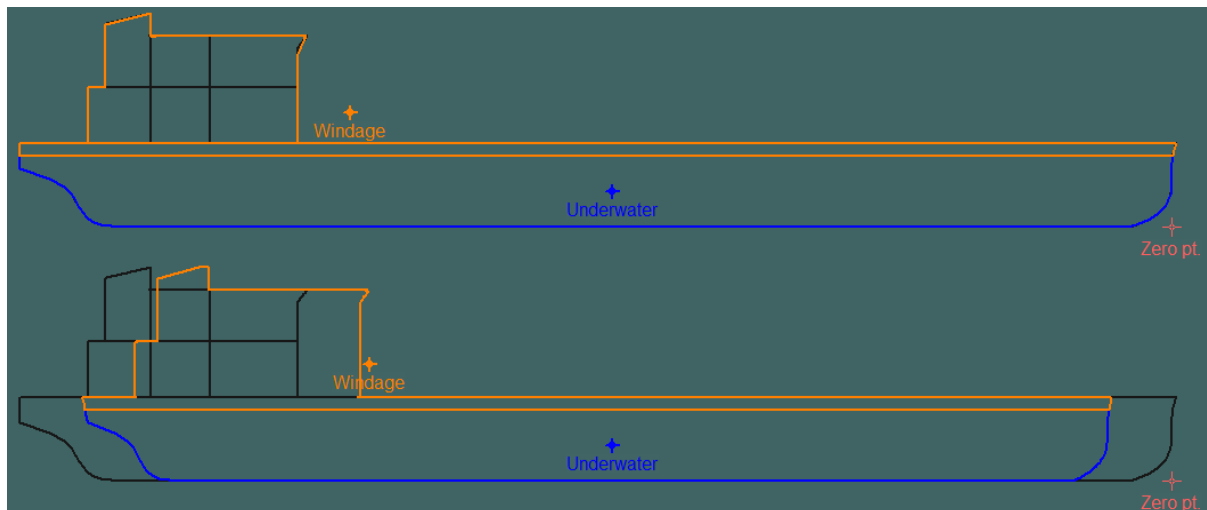


Color selection

Wind direction

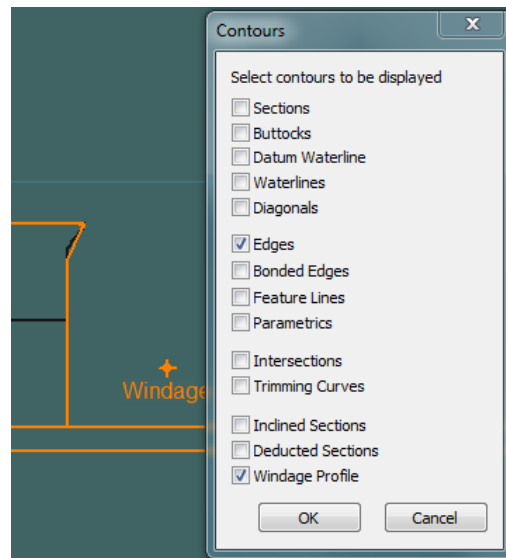
The Windage direction specifies the projection direction used for the surfaces: 90deg. gives a projection in the lateral plane; 0deg. gives a direction in the longitudinal plane. Angles between 0 and 180deg are allowed since the sign of the projection vector does not matter.

Note that to improve performance, the projected windage contour uses a fairly coarse surface mesh. This may result in the projected windage contour not exactly corresponding with the surface edges, but the effect on projected area and center of area is negligible. Due to the calculation method used for the projected contours, it is possible that some visual artifacts may be present but again these have negligible effect on projected area and center of area.



Windage projections viewed in profile at 90deg (upper) and 70deg (lower)

The windage profile may be displayed in Modeler by turning on "Windage Profile" in the Display | Contours dialog:

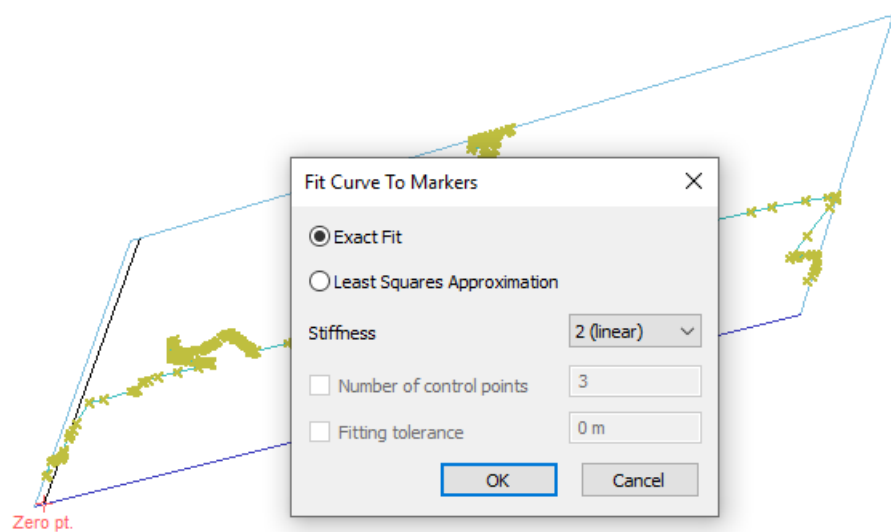
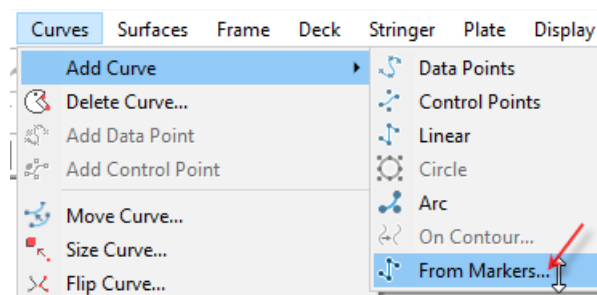


Modeling windage profile shape on CL

An effective way of modeling the vessel's windage profile is as an asymmetrical mesh on the CL buttock plane. If you have a DXF or set of points defining the profile, this can be achieved quite simply:

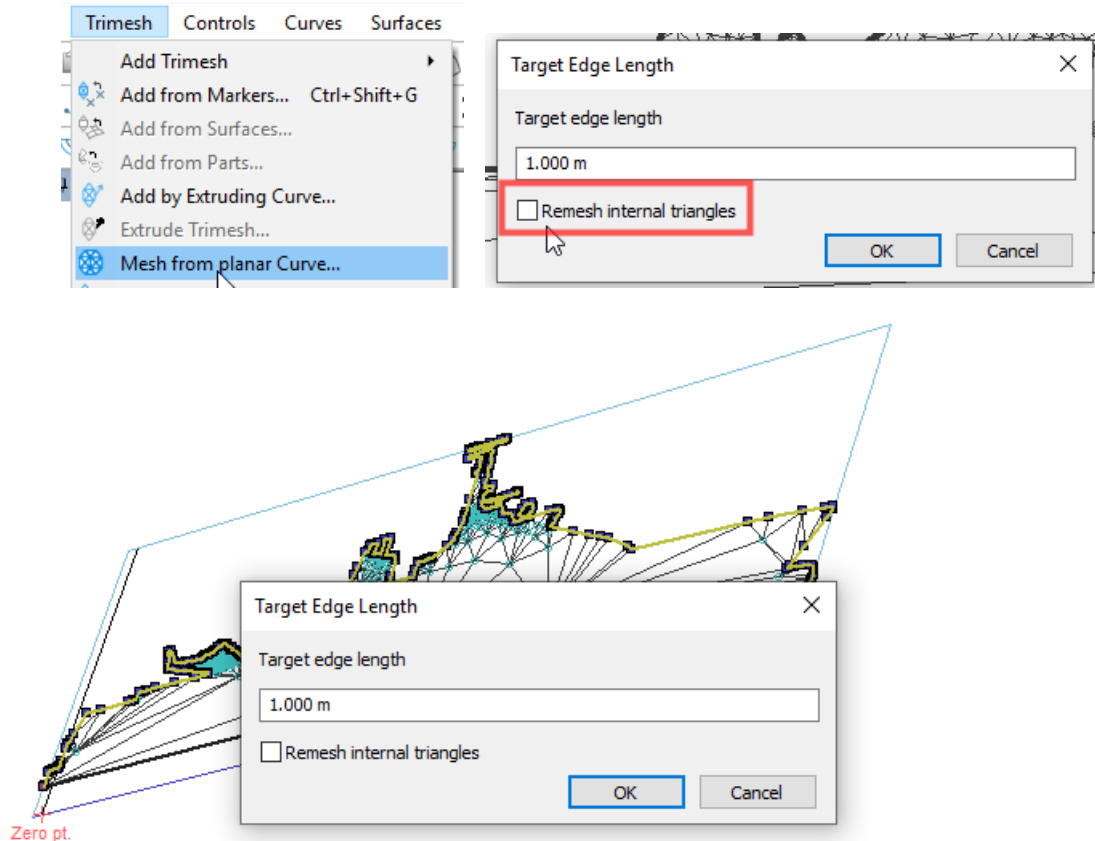
Import the DXF or point data to Marers

Create an exact linear curve from the imported Markers

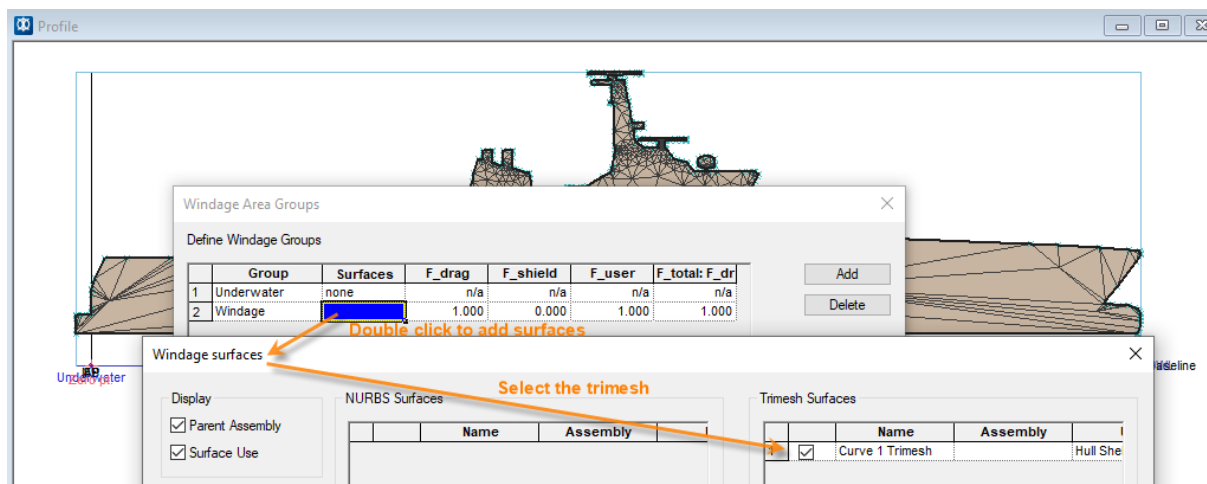


Create a planar, linear Curve from the Markers

Finally, create a mesh from the planar curve, ensuring that the remesh option is *not* selected, this ensures an exact fit to the curve points with the minimum number of triangles, thus making the calculation of projected area as fast as possible:



Generate a Mesh from the planar Curve without remeshing



Add a Windage group and add the new Mesh to it

Setting Units

The Units that Modeler will use to display measurement data and calculation results can be set in the Units dialog from the Data menu.

Also see the [Units](#) section on page 391 on how unit conversion is done in Modeler.

Taking Measurements

In the design views, other than the perspective view, Modeler displays a number of co-ordinates and measurements in the bottom-left corner of the design views:

↔ 6.913 m ↑ -3.85 m ↻ 330.9° ↗ 7.913 m

These give you the horizontal and vertical co-ordinates of the cursor and the angle and distance from the last clicked position. These are displayed in the current units. The angle is measured counter-clockwise from the positive horizontal axis.

The horizontal and vertical measurements depend on the current view: If you are in the plan view, they will be the longitudinal and transverse co-ordinates; in the profile view, the longitudinal and vertical co-ordinates; and in the body-plan view the transverse and vertical co-ordinates.

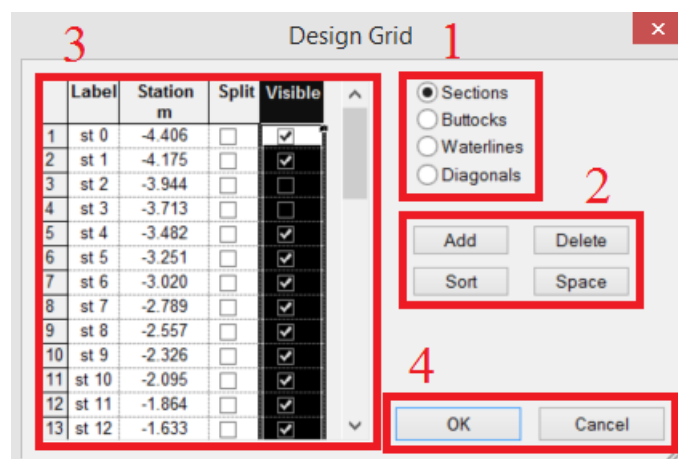
Both the Profile and the Body Plan windows display a horizontal line representing the Datum Waterline (DWL). The DWL is set using the Frame of Reference dialog in the Data menu. It is used as the waterline for computation of all data listed in the Calculations window, as well as the sectional area curve in the Curve of Areas window.

Setting up the Grid

Setting up the Grid - Concepts

The positions of the Sections, Waterlines, Buttocks, and Diagonals form what is called the Grid. The grid can be displayed on any window by selecting the Grid command from the Display menu. Elements can be added to the Grid by using the Design Grid function from the Data menu.

The positions for each of the grid elements are stored in individual lists, which may be selected by clicking on the labelled buttons in the top right hand corner of the Design Grid dialog box (region 1 of the dialog in image below).



The Design Grid dialog can be divided up into the following regions:

- Select the type of grid you want to work on, i.e. Sections, Buttocks, Waterlines or Diagonals
- Commands to Add, Delete, Sort or Space the grid type selected in 1
- Displays the current grid locations for the grid type selected in 1
- OK and Cancel to confirm or undo any changes made in the dialog

The positions of Grid elements are as follows:

Sections

Position of each section relative to the longitudinal zero point. Positive numbers are to the right (forward), negative numbers to the left (aft) of the zero point.

Buttocks

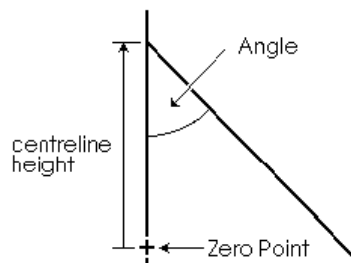
Position of each Buttock relative to the centreline of the hull.

Waterlines

Position of each waterline relative to the vertical zero point. Positive numbers are above, negative numbers below the zero point.

Diagonals

Two columns of figures are provided, centreline height and angle. The centreline height is the position of the origin of each diagonal relative to the vertical zero point. Positive numbers are above, negative numbers below the zero point. The angle is the angle between the diagonal and the centreline as shown below. Angles above 90 degrees will give diagonals above the centreline height.



Adding Grid Lines

To add one or more grid lines

- Select the row after which the new grid lines are to be added.
- Click the Add button.
- Enter the number of grid lines to be added to the list.



All new position fields will be initialized to zero.

The number of Grid elements is limited to the following maxima:

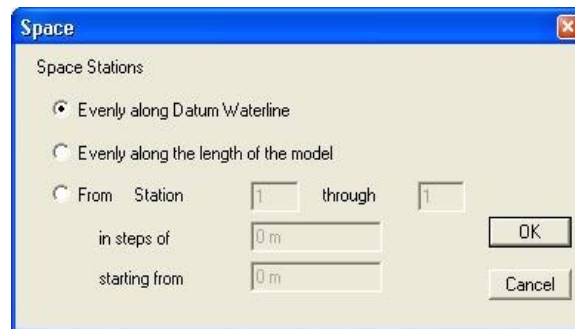
- 1000 Stations
- 100 Waterlines
- 100 Buttocks
- 50 Diagonals

Spacing Grid Lines

To space a group of stations along the Datum Waterline or overall length of the design,

- Click the Space button.

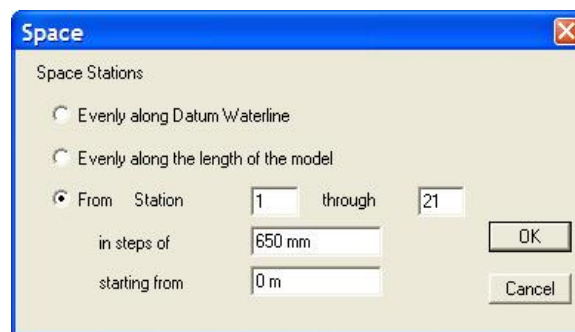
A dialog will appear with options for spacing the stations.



- Select Evenly Along Datum Waterline or Evenly along the length of the model, as desired.
- Click OK.

To space grid lines at a specific intervals

- Select From and enter the numbers of the grid lines to space



- Enter the required interval between grid lines.
- Enter a starting position to space from.
- Click OK.

You may also specify the range of grid positions by simply selecting those rows in the Design Grid dialog. In this case, the start and finish grid lines, a default spacing and a starting position are set up automatically.

Similar functions are available for spacing the waterlines and buttocks.

Editing Grid Lines

To edit a grid line position

- Select the cell corresponding to the position you wish to edit.
- Type in a new value.

Dimensions entered in units other than the current units are converted automatically.

Grid Labels can be specified by entering text into the first column of the table. Grid labels are used for labelling the grid when printing or plotting, and for inclusion in the Table of Offsets.

Sorting Grid Lines

To sort grid lines

- Click the Sort button.

Sorting occurs as follows:

Stations:	Longitudinally.
Buttocks:	Transversely.
Waterlines:	Vertically.

Diagonals: Vertically - centreline height only.

Copying and Pasting Grid Lines

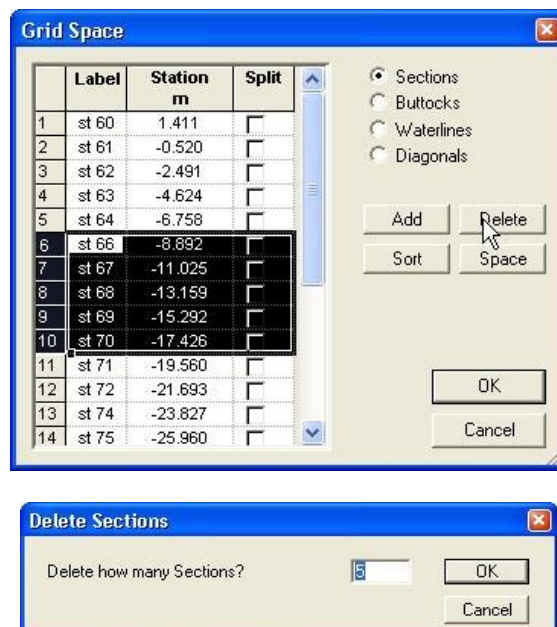
The Design Grid dialog allows copy and paste operations to take place on the tables of grid names and positions. As the main menu bar is not accessible from within the dialog, copy and paste are achieved using the standard command key equivalents, Ctrl C for copy and Ctrl V for paste.

Deleting Grid Lines

To delete one or more grid lines

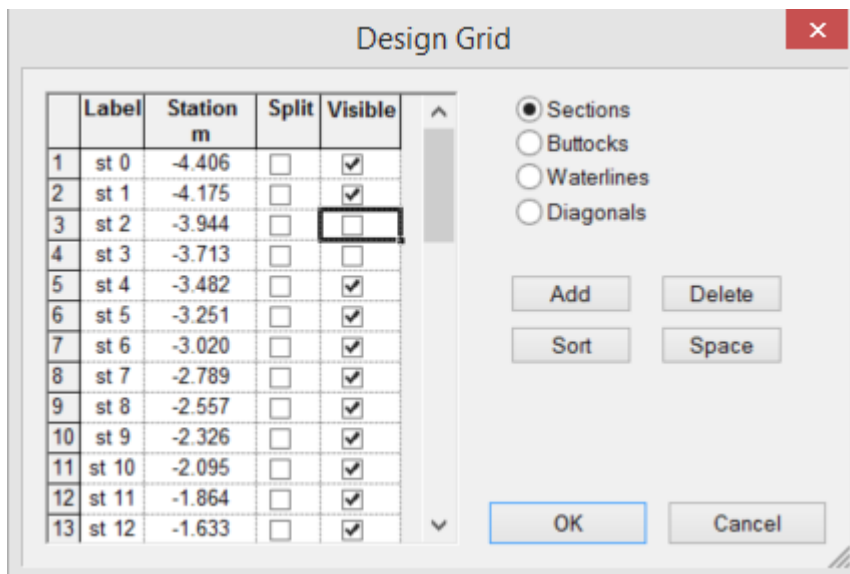
- Select the row corresponding to the last of the set of grid lines to be deleted.
- Click the delete button.
- Enter the number of grid lines to be deleted.
- Click OK.

If a range of rows is selected in the Design Grid dialog, the corresponding number of rows is automatically set in the deletion dialog. These rows will be deleted when you click the OK button.



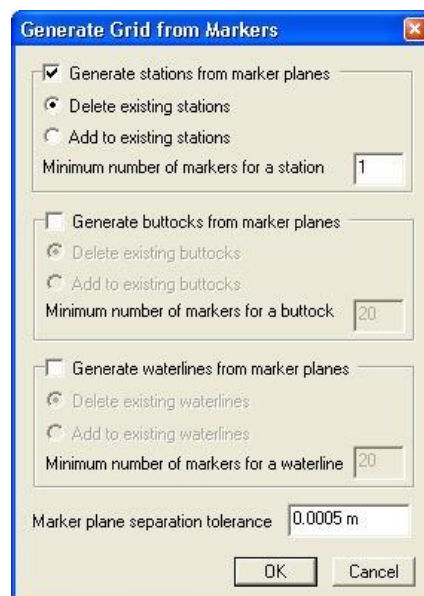
Setting visibility of individual contours

Individual Sections, Waterlines, Buttocks and Diagonals can be switched on or off via a visibility flag in the Design Grid dialog:



Generating a Grid from Markers

If you have imported or defined markers with common longitudinal, transverse or vertical positions, you can automatically generate a grid from these markers. The Generate Grid from Markers command will automatically generate the grid (sections, buttocks and waterlines) based on the marker data and also associate markers with the correct marker station index. (See [Working with Markers](#) on page 80). Sections are generated at the markers' longitudinal position; buttocks at their transverse offset; and waterlines at their vertical location:



Using the checkboxes you can select which items of the grid to generate (i.e. stations, buttocks and waterlines). Then, using the radio buttons, you have the option of deleting and replacing the existing grid data or adding to the existing grid data. You can also specify how many markers must be at the same longitudinal position, transverse offset or vertical location for a station, buttock or waterline (respectively) to be added. Finally, you can specify a tolerance for markers to be considered at the same plane: for example if the Marker plane separation tolerance were set to 0.001mm then markers up to 1mm apart (in longitudinal position, transverse offset or vertical location) would be considered to be in the same plane and the station, buttock or waterline location would be averaged.

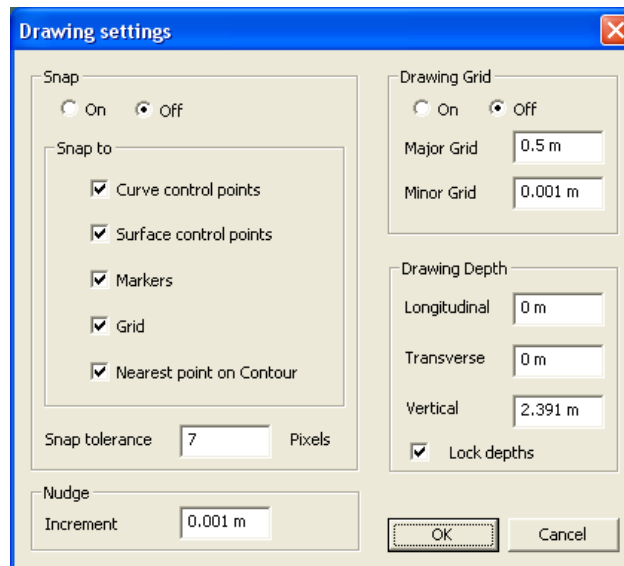
When [Importing DXF Markers](#) or GHS markers, Modeler can automatically generate the grid. The section grid is generated automatically only if no existing sections are defined. If sections are already defined then

imported markers that lie within 1cm of a section will be allocated to that section, otherwise they will be left unallocated.

Also see [Displaying Contour Lines](#) on page 36.

Drawing Settings

The Drawing Settings dialog box contains settings that aid in the drawing process. The dialog may be accessed from the menu (Display | Drawing Settings...) or from the tool bar button in the Data EX toolbar. Settings are grouped into the object snap settings and the drawing grid settings:



Object snap

When drawing a curve or moving a surface via dragging the control point it is sometimes useful to be able to snap the point exactly to an existing location, such as a marker. To do this you can use the object snap settings. To turn object snap on you can do it via the radio button in the drawing settings dialog or toggle it by pressing the f3 key. The f3 key can be toggled during drag and curve creation commands. The current status of the object snap is displayed in the bottom right hand corner of the application pane. You can select which objects you wish to snap to by checking the boxes in the "Sanp to" section of the drawing setting dialog box. The screen settings for how close the mouse needs to be to an object to snap is set in the snap tolerance edit box. The tolerance is measured in pixels and is set to 7 by default.

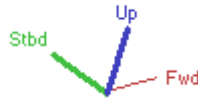
Drawing grid

Drawing a 3 dimensional object, such as a boat hull, in a 2 dimensional environment, such as the computer screen, can present visualisation and manipulation problems. In Modeler there is the concept of a drawing gird. The drawing grid comprises of three orthogonal planes, the section plane, the waterline plane and the buttock plane. These planes are the "depth" value on which you are currently working. For example, any control points added in the perspective view when the drawing grid is on will be added at the "depth" location of the grid. The grid can be switched on or off via the radio buttons (the drawing grid may also be displayed from the Display | Grid | Show Current Drawing Plane command from the menu). The grid depth is then set by entering values in the x, y and z edit boxes. The x value denotes the depth of the section drawing plane, the y value denotes the depth of the buttock drawing plane and the z value denotes the depth of the waterplane.

The grid increment distance is set in the "Major Grid" edit box. The "Minor Grid" is the value that a point position will be rounded to when dragged and dropped. For example if the minor grid is set to a value of

1mm and a control point is dropped at (3.2456 m, 1.9467 m, 2.9542 m) the point will be moved to the nearest 1mm Minor Grid point i.e. (3.246 m, 1.947 m, 2.954 m).

When the drawing plane is switched on and the perspective window is active, the most perpendicular plane to the current view angle is displayed. To check which plane is currently displayed you can refer to the UCS icon displayed in the bottom right hand corner of the screen:



The two bold lines indicate the currently active plane, so for example in the above orientation the section plane would be the current working plane. The position of the cursor will be displayed in the bottom left of the perspective window only when the drawing grid is displayed.

By default the drawing grid depth values are automatically snapped to the last selected control point or marker. To turn this feature off check the "Lock Depths" option in the dialog.

Nudge Increment

This is the value used when nudging a control point using the arrow keys. This number should be fairly low as nudging is normally used for the final fairing of the surface. If the ctrl key is held down when nudging there will be a 10 fold increase in the nudge increment.

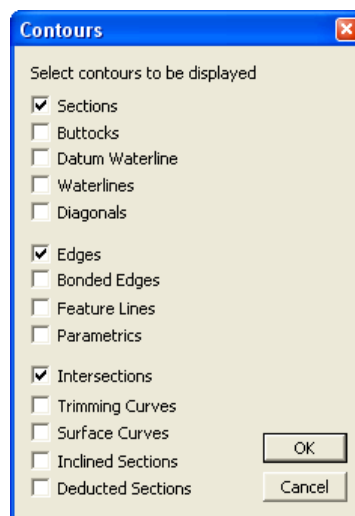
Displaying Contour Lines

Contour Lines – Concepts

Contours may be passed through your design to give greater definition of the surface. For example, you might wish to see your design with the sections displayed at the frame spacing you have chosen. To do this you would set the Grid to the required frame spacing and then show the section contours for the design. The selected contours are displayed on all visible surfaces.

Displaying Contours - Procedures

The Display | Contours option allows you to select which contours are drawn on the screen at any given time. Any combination of contours may be chosen from the contours dialog.



You can set the color and thickness of of contour lines by selecting Colors and Lines from the View menu. When drawing the contours on the surface, Modeler uses the Precision as set in Display | Precision. See [Surface Precision](#) on page 115 for important information on precision.

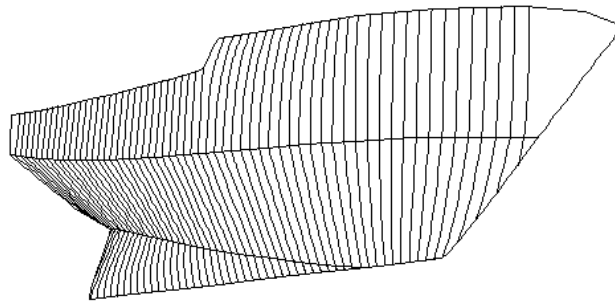
For information about inserting sections, buttocks, waterlines and more, see the [Setting up the Grid](#) section on page 30.

You may choose to display:

Sections

When the Sections are activated, all sections will be shown in the Body Plan and Perspective windows, otherwise only the current section will be displayed.

Shown below is a design with 100 stations auto-spaced along the Datum Waterline.



Sections

Buttocks

Buttocks can be displayed in the Profile and Perspective windows.

Datum Waterline

The shape of the Datum Waterline will be displayed in the Perspective window.

Waterlines

Waterlines can be displayed in the Plan and Perspective windows.

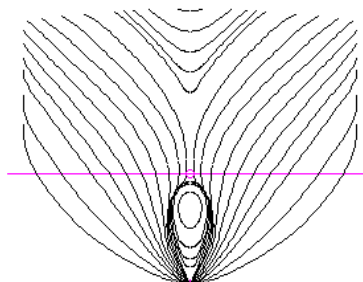
Diagonals

When the Diagonals are activated, Diagonals will be displayed in the Plan, Profile and Perspective windows. Diagonals are shown as true diagonals in all windows, except for the Plan view where they are displayed as perpendicular projections so that the diagonal's transverse offset in the Plan view is in fact the distance along the diagonal from the centreline to the hull surface.

When both waterlines and diagonals are selected, the Plan will separate them with the waterlines above the centreline and the diagonals below the centreline, in the traditional manner.

Edges

Normally all surface edges will be displayed. However in some cases you may wish to hide the surface edges in order to show only the sections. In this case, turning off the Edges display will give the following result.

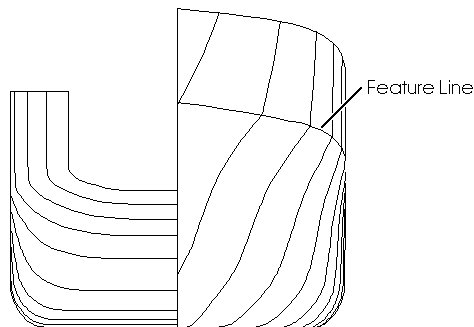


Bonded Edges

Show or hide all bonded edges.

Feature lines

Feature lines are created whenever sufficient control points are compacted together to form a hard edge within the surface. To display these features, select the feature lines option. The illustration below shows the use of a feature line to create a knuckle in a set of bow sections.

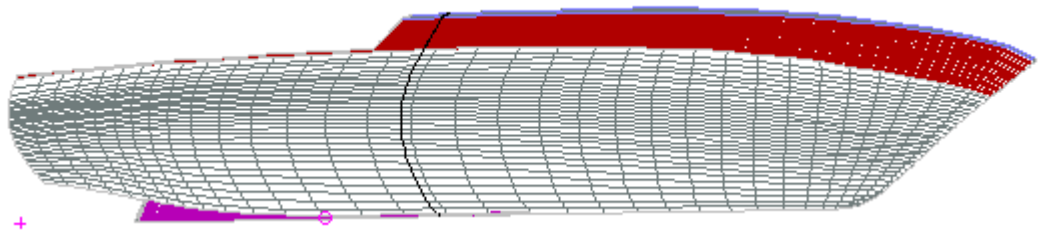


Parametrics

Selecting Parametrics displays the parametric surface, or the surface data calculated by Modeler from which other surface contours are derived. This view is useful for the examination of the surface as a whole.

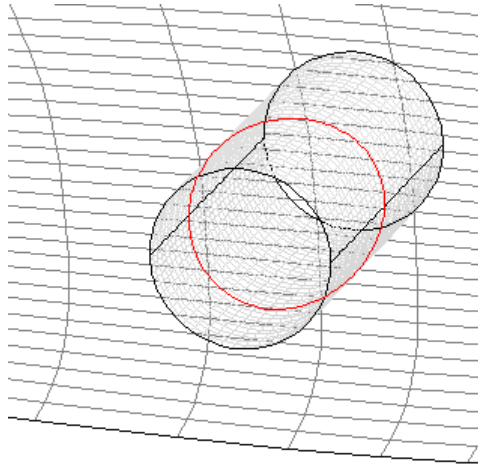
Parametric lines do not lie in any particular plane and give a good indication of any inconsistencies in surfaces that might arise due to misplaced control points.

The color of the parametric curves may be either the color of the surface or the color defined for parametric curves. Use the option in the Preferences dialog to select which you prefer.



Intersections

Selecting Intersections calculates and highlights all intersection lines between surfaces in the Plan, Profile, Body Plan, and Perspective windows.



The intersection line (red) of a bowthruster with the hull.

Trimming curves

Trimming curves are curves on the surface that can be used to define the boundaries of regions used to trim the surface. Trimming curves are created in 2 ways. Either they can be projected onto the surface from a Curve, else they can be imported from Rhino using the File | Import | .3DM Rhino file format. See [Importing Rhino .3dm files](#) on page 338 for more information.

Inclined Sections

Structure also allows the intersection of arbitrarily inclined planes with the hull. The resulting inclined sections can be displayed in Modeler.

Deducted Sections

These are sections, which have had the skin thickness of intersecting surfaces deducted. Skin thicknesses are calculated perpendicular to the surface. For more information, see [Outside Arrows](#) on page 113 and [Surface Materials and Skin Thickness](#) on page 114.

Selecting Contour Lines

You may select contours in the same way as you select control points (by clicking on them or by dragging a selection box around them). Once selected, it is possible to look at the curvature of the contour, see [Displaying Curvature on Contours](#) on page 117.

When the pointer is over a contour or surface edge, the name of the contour or edge will be displayed in the status bar at the bottom left of the main window.

Zoom, Shrink, Pan and Home View

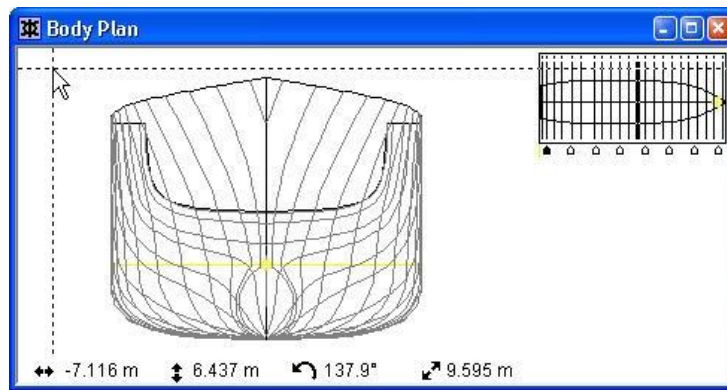
The Zoom, Shrink, Pan and Home View commands allow you to change the size of the drawing in the front window.

Zooming

The Zoom function allows you to work on any part of your design by enlarging a particular area to fill the screen.

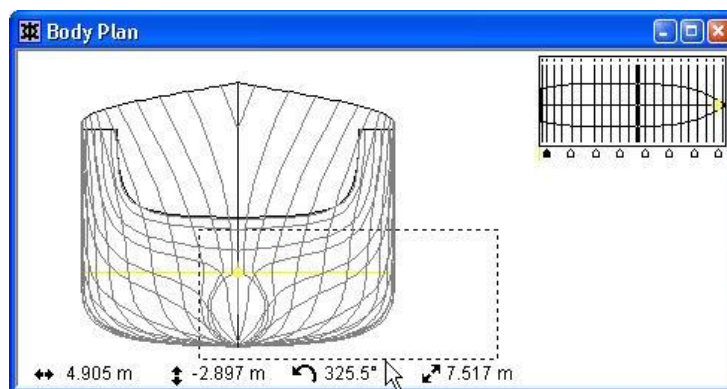
The zoom function works in any of the Modeler drawing windows. To zoom in on a design:

- Select Zoom from the View menu (or the keyboard equivalent Ctrl E).



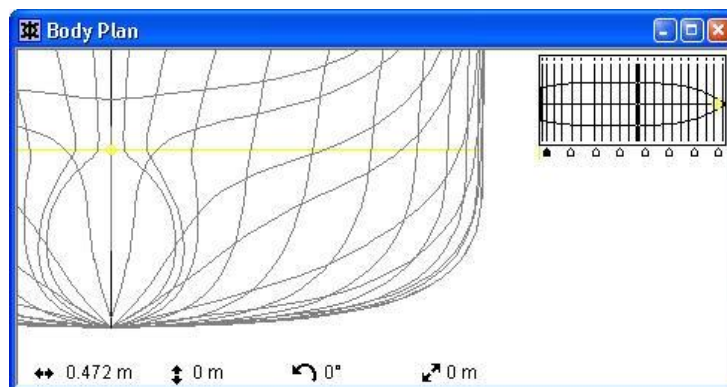
A set of cross hairs will appear that follow the movements of the mouse. These are used to set the starting position of the zoom rectangle.

- Click and drag the mouse in any direction.



The zoom rectangle will appear and grow as the mouse moves.

- Release the mouse button.



The view contained in the zoom rectangle will be enlarged to fit the screen.

There is a limit on how far you can zoom, and you will find that after continual zooming the image will not enlarge any further.

Note:

If, before releasing the mouse button, you decide that you do not wish to use Zoom, or that you wish to change the starting position of the zoom rectangle, simply return the cursor to within a few pixels of the starting position, release the mouse button, and the cross hairs will reappear.

Shrinking

Choosing Shrink will reduce the size of the displayed image in an active drawing window by a factor of two.

To shrink the displayed image:

- Select Shrink from the View menu
- or
- Use the keyboard equivalent, Ctrl R.

Multiple shrinks may be achieved by striking key R the amount of times that you wish to shrink while holding down the Ctrl key.

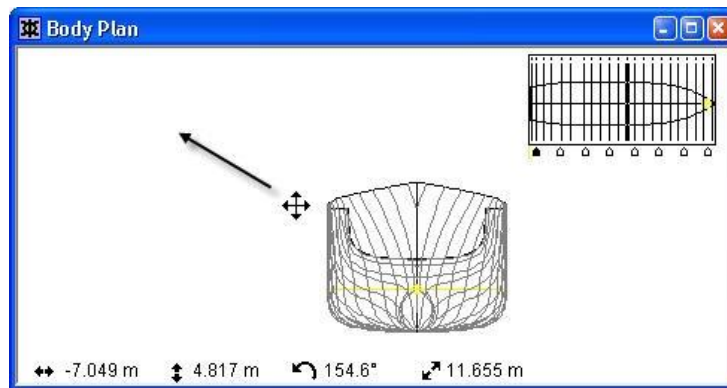
Panning

Choosing Pan allows you to move the image around within a drawing window.

To Pan an image:

- Select Pan from the View menu.

Move the arrow cursor to anywhere within the window.



- Click and drag to a new position.

The image will move with the cursor until the mouse button is released.

Mouse Wheel Support

You can use the mouse wheel to easily zoom into and out of a design. You can use this feature if you have a mouse with a 'wheel' in the centre of the mouse, i.e. Microsoft Intellimouse or Logitech Wheelmouse.

Holding down the mouse wheel to pan is also supported.

Zooming will be directed towards the current mouse position, for example if the mouse is above the bow zooming will be directed towards the bow.

Home View

When you select Home View a zoomed or panned image is returned to its original state. This state may be set at any time for any combination of zooming and panning by selecting Set Home View from the View menu.

Modeler starts up with default Home View settings, which are based on the Home Views in use the last time the program was operated.

Saved Views

Saved Views extends the Home View existing functionality.

Home View Rotations

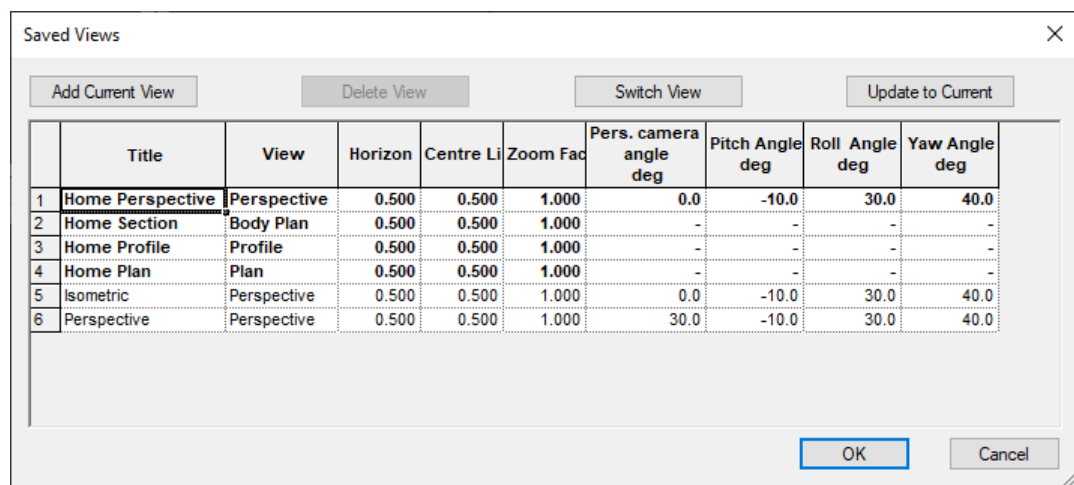
When setting the Home View in the Perspective window the view rotations are now stored as well as the zoom settings. All of these settings, and those of the Saved Views are saved to the model file.

Saved Views Dialog

In addition to the Home View for each of the view windows (Perspective, Section, Profile, Plan and Expansion) it is possible to save any number of user defined views.

The perspective angle is included in the model file Saved Views and these views are available in all Maxsurf applications when a model file is loaded. You can select a custom view by using the Saved Views dialog. The Saved Views for a model will be updated when the model file is saved. Because the model file is only saved from Modeler and Stability, changes made to the Saved Views in other applications are only temporary.

To add, edit or delete these views open the Saved Views dialog by clicking on "Saved Views" in the View menu. The following dialog will be displayed:



Default (bold) and Customized (rows 5, 6) Saved Views

Add Current View

To add the current view to the Saved Views list click on the Add Current View button and it will be added to the bottom of the list with the name "View n".

Edit a Saved View

The Title, Horizon, Centre Line, Zoom Factor and all three Angle fields are manually editable. Changes will be saved when you press "OK". Pressing "Cancel" will discard your most recent changes.

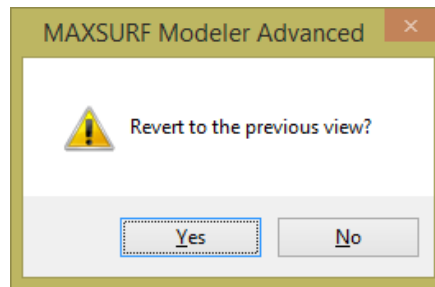
Delete View

To delete a Saved View click anywhere in its row and click the Delete View button. The Home Views for each window, shown in bold, cannot be deleted. If you have accidentally deleted a view, click "Cancel" to exit the dialog and your most recent changes will be discarded.

Switch View

To switch the current view to a Saved View click anywhere in the row of the view you would like to switch to and click the Switch View button. You can only switch to a Saved View that is in the same window that you are currently working in.

If you switch a view and then leave this dialog by clicking "Cancel" you will be prompted if you would like to revert to the view last shown before you opened the dialog.



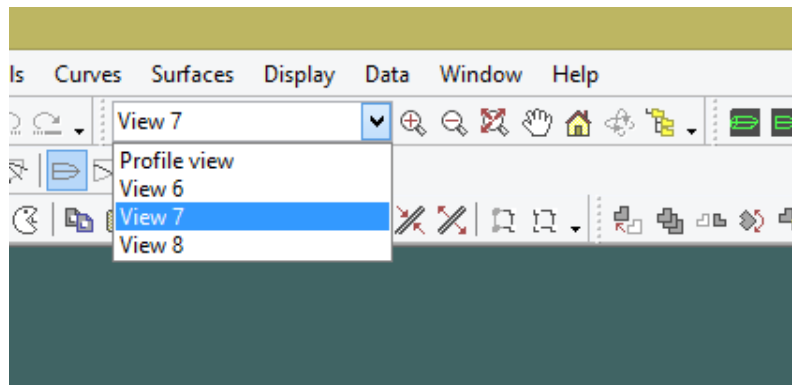
Revert view dialog

Update to Current

It is possible to update the view settings (zoom and rotation) of a row/view in the table to those of the window that you are using by clicking the Update to Current window.

View Toolbar Saved Views combo box

Rather than having to open the Saved Views dialog when you would like to switch Saved Views you can do so using the combo box in the View Toolbar.



View Toolbar Saved Views combo

Only the Saved Views that relate to the top window are shown.

View Windows

The Modeler application consists of a number of windows. Some windows contain tables with design data or calculations while others allow you to view the shape of your design. Modeler uses four of these "view windows" and each is slightly different to the other; not only in the direction from which the window allows you to view the model, but also in functionality. For example: in Perspective view, you can rotate and view the model in 3D whilst in the Profile view window you only get a 2D view of the side of the vessel.

In this section the different view directions and view window specific functionalities are described:

- [Body Plan Window](#)
- [Plan and Profile Windows](#)
- [Perspective Window](#)

Body Plan Window

The body plan window displays the transverse sections as seen from aft of the model. Starboard is on the right of the centreline and Portside on the left.

The number of sections and their longitudinal positions is specified in the Design Grid dialog in the Data menu.

In body plan view there are a number of options to visualise the model shape

- Display all sections
- Display one section at a time and use the control box to navigate through the sections
- Display half of the model, with or without split sections

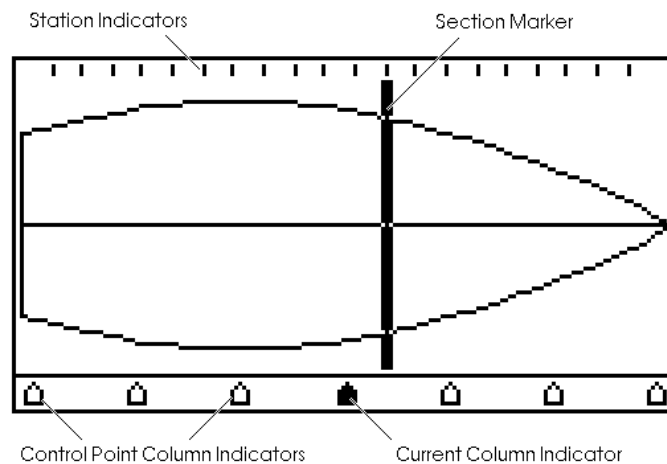
Display All Sections

To display all the sections, simply tick the sections box in the Contours dialog in the Display menu.

If the stations are turned off, only the current station will be displayed in the Body Plan and Perspective windows. In the Body Plan window the control box in the top right hand corner is used to navigate through the sections.

Control Box

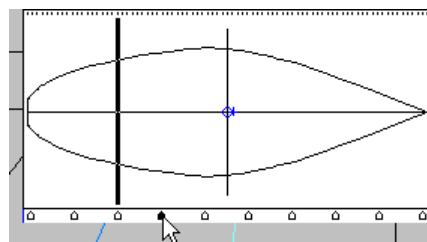
In the top right hand corner of the Body Plan window is the control box containing a miniature view of the plan shape of all currently visible surfaces. The purpose of the control box is to allow you to specify precisely what part of a surface you wish to view or modify, by allowing you to select different cross sections and columns of control points in that surface.



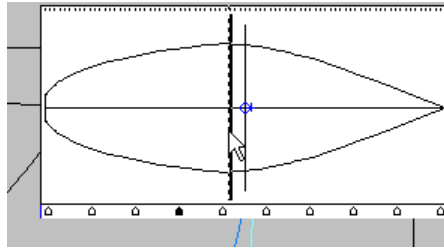
At the top of the control box several short vertical lines are drawn. These are the station indicators; there is one indicator for each of the stations specified in the Design Grid dialog.

The section marker line shows which section is currently displayed.

The "triangles" on the bottom display the control point columns for the surface you are currently working on. The triangles are spaced evenly along the bottom of the control box.



By selecting the triangles you select a column of control points and display the section that is closest to that column. Alternatively, a column is made the current column of its respective surface whenever one of its control points is selected in any of the drawing windows.



By clicking at the desired longitudinal position well inside the control box, the section closest to that longitudinal position is displayed. By using the arrow keys on your keyboard you can then scroll through the sections. The right arrow displays the next station forward and the left arrow displays the next station aft.

Note

When the cursor is within the control box the horizontal cursor coordinate at the bottom of the window refers to the cursor's longitudinal position on the plan shape.

Markers Display Options In Body Plan Window

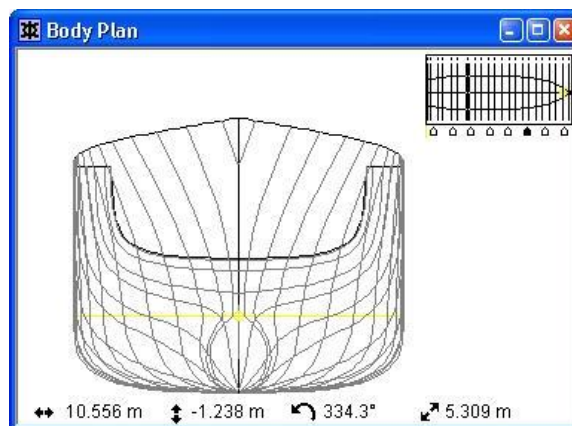
If markers are defined, there are three display modes, chosen from the Markers sub-menu in the Display menu:

- Hide all markers — all defined markers are invisible
- Markers for current station — only markers with the same longitudinal position as the current station are displayed
- Show all markers — all defined markers are displayed

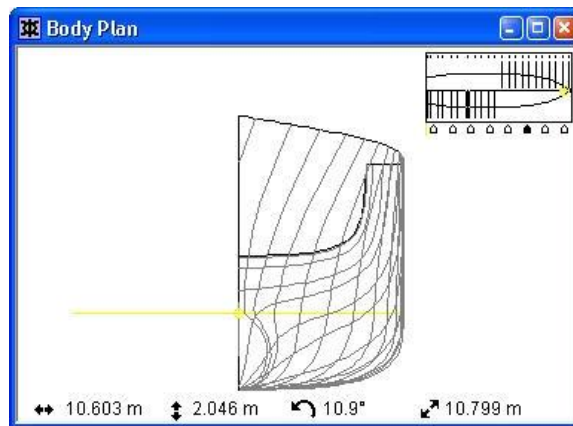
These display modes are particularly useful when trying to fit a surface to a table of offsets.

Half Hull Display

The half hull display selects whether both halves of a symmetrical surface are displayed. If half is turned on, only the starboard side of symmetrical surfaces will be displayed.



When the Half function is not activated, both sides of each symmetrical surface will be displayed. However, when Half is activated, the Body Plan window will show either a split section view or all sections on one side depending on whether Split Section Display is setup correctly. The Split Section Display option is described in greater detail in the next section.



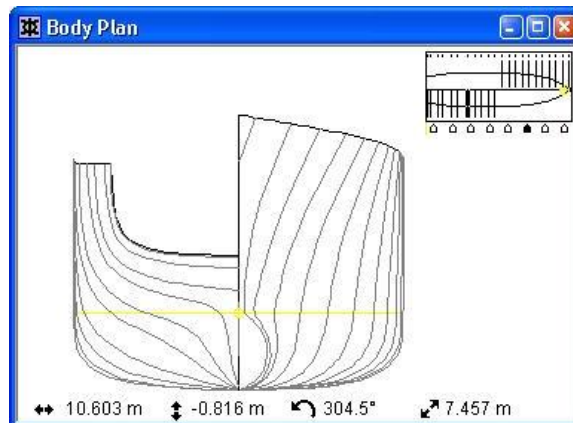
Note

Activating Half function changes the hull display in all view windows.

Split Section Display

The Split Section Display affects the way in which sections are displayed in the Body Plan window; if enabled, aft sections are displayed on the left and forward sections on the right.

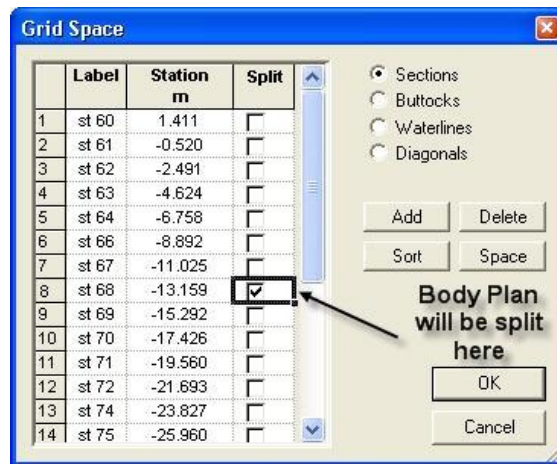
The Split Section Display option is only enabled when the half hull display is turned on. If Split Section Display is chosen for any surface, it will be displayed with the forward side of the hull on the right of the centreline and the aft half of the hull on the left. The split section option is specified independently for each surface in the surface's Properties dialog and the Surfaces window.



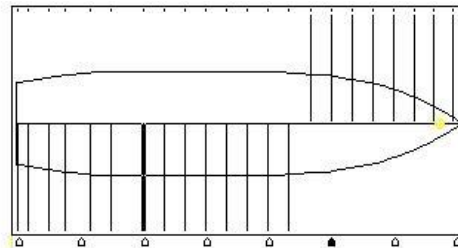
If Split Section Display is not chosen, all stations will be drawn over half of the hull on the right hand side of the centreline.

Defining Section Splitting

It is possible to specify at which section the body plan view will be split using a column in the Design Grid titled Split. Clicking in this column selects the station where the displayed station will change side in the Body Plan window.



The Control box in the body plan window will be drawn with the sections staggered to show where the split has been placed:



Plan and Profile Windows

Plan Window

The plan window displays the model as seen from below, with Starboard above the centreline and Portside below.

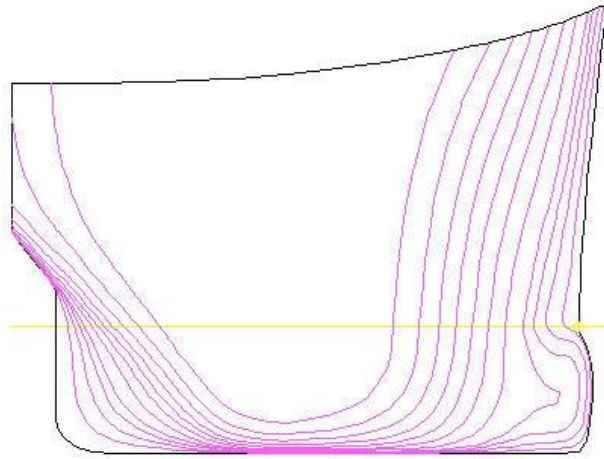
Profile Window

Displays the hull as seen as seen from the Starboard side, with the bow to the right of the screen.

Compress Function

By choosing Compress it is possible to foreshorten the display of the design in the Plan and Profile windows. This foreshortened display is valuable for fairing long slender hulls on screen.





The vertical axis is expanded relative to the longitudinal axis by a factor of four.

Perspective Window

Perspective Window – Concepts

The Perspective window is bordered by three sliders containing pointers that control the rotation of the displayed design. The model is rotated by dragging the sliders to the desired angle. The zero position for each ruler is at its centre and markers are given at 10°. The sliders also have elongated markers at 30° intervals. You may rotate the design in the Perspective window within the following limits:

Yaw	$\pm 180^\circ$
Pitch	$\pm 180^\circ$
Roll	$\pm 180^\circ$

The view direction sliders have several special features listed below:

- Left-clicking in an empty part of the slider will move the slider thumb one degree towards the mouse; you can hold the left mouse button down for the model to slowly rotate as the slider thumb moves towards the mouse position.



Mouse left-click or left-button-held-down in slider: thumb moves towards mouse

- Left-double-clicking on the slider thumb will move the thumb to the nearest 15 degree position



Mouse left-double-click on slider thumb: snap to nearest 15 deg. position

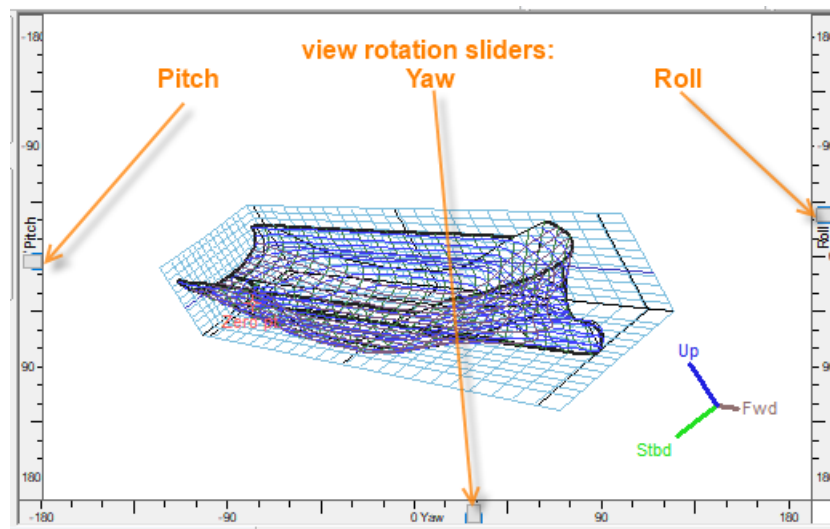
- Shift-left-double-clicking on the slider thumb will move the thumb to the zero degree position



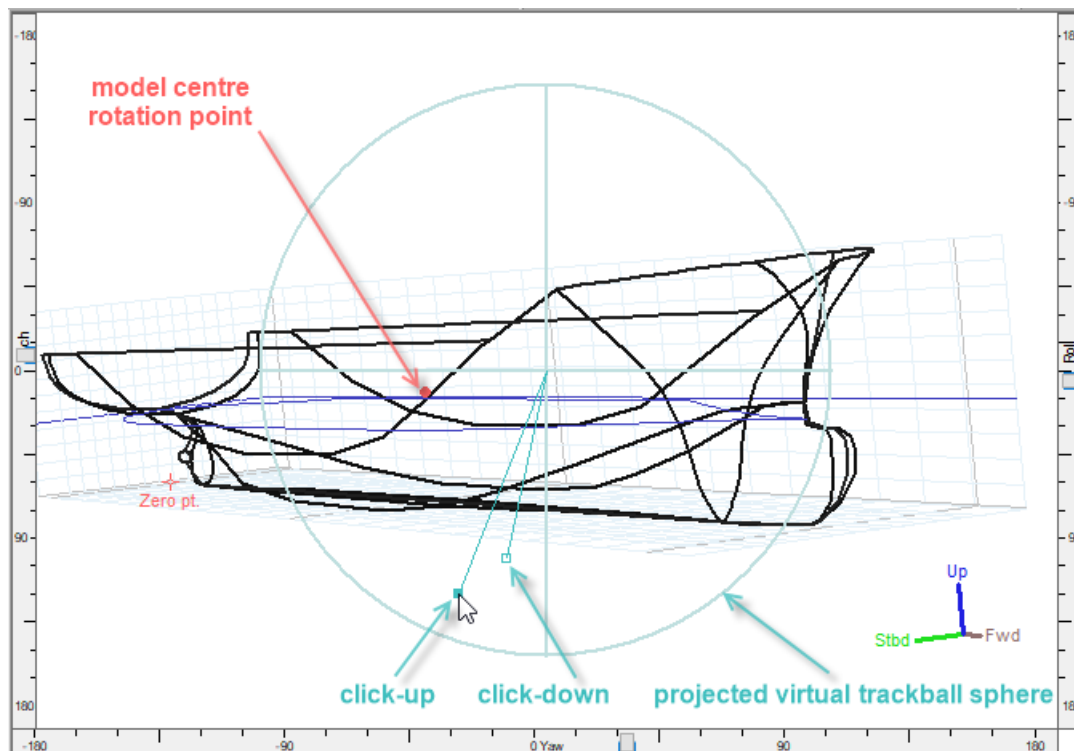
Mouse shift-left-double-click on slider thumb: snap to zero deg. Position

The view may be rotated about the three major axes using the slider controls located on the sides and bottom edge of the window. The model is first rotated around the yaw axis, then pitch and finally roll. This

can lead to 'gimbal lock' when the pitch angle is $\pm 90^\circ$. In this situation a degree of freedom is lost because the yaw rotation axis is aligned with the roll rotation axis



Slider for rotations around main axes



Virtual trackball view orientation control (for explanation only; not shown on screen)

Rotation about an arbitrary axis can be effected using the Virtual Trackball (either via the View | Rotate menu; Ctrl+Q shortcut or shift+middle mouse button). Conceptually you click on a point projected onto a virtual trackball sphere located at the centre of the screen and drag point on the sphere to another location. The rotation is about an axis is normal to the two vectors connecting the click-down and click-up points on the sphere to the centre of the sphere; the angle of rotation is the angle. It should be remembered that the virtual sphere is used only to determine the rotation axis and angle, the rotation always occurs about the same point located roughly at the centre of the model.

The Virtual Trackball can exhibit some slightly unexpected behavior if the path between the click-down and click-up points crosses the sphere boundary. To effect rotations about the axis perpendicular to the screen, it

is best to click and drag *outside* the sphere (ie near the edges of the window); for rotations about axes in other directions, click and drag fairly close to the window centre.

Rotate

Whilst the view direction can be changed using the sliders on the left, right and bottom of the window, some users may find it more convenient to use the Rotate command.

The Rotate function in the view menu activates the Rotate command, which is a virtual trackball, which lets you freely rotate a design in the perspective view.

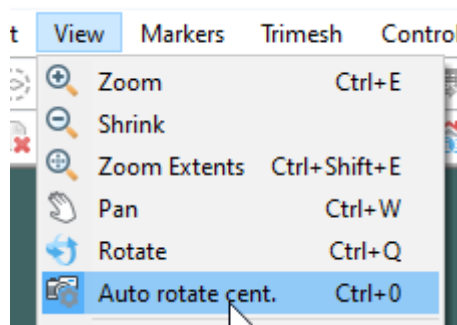
After selecting the Rotate tool, move the mouse to a location in the Perspective window and press the left mouse button. With the left button depressed, you can rotate the image by dragging the mouse around and the rotation is performed by projecting the movements onto a virtual sphere on the screen – essentially like a virtual trackball. Generally speaking, moving the mouse left and right rotates about a vertical axis while moving the mouse up and down rotates about a horizontal axis. The Rotate mode is exited when the left button is released.

The rotate function may also be initialised via the mouse. Hold the shift key down while pressing the middle mouse button to rotate the model.

View rotation centre auto update

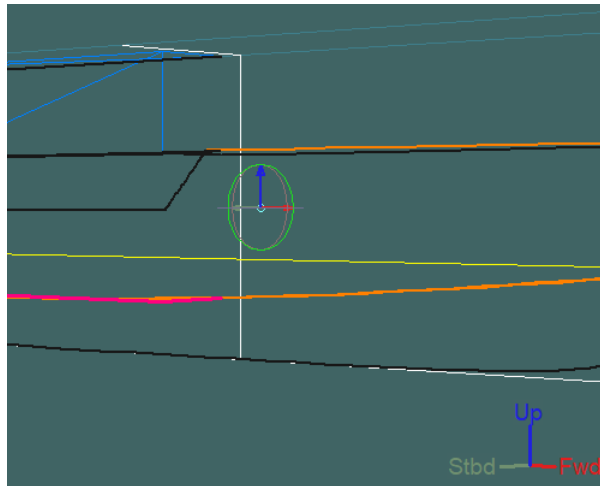
The centre about which the view rotates in the perspective view can be made to update automatically so that it is always approximately about the middle of the window. This means that when you are focused in on a part of the vessel near the bow, then when rotating, the vessel will rotate about a point which maintains the bow on the window. The rotation point is updated automatically by projecting a point in the middle of the window onto the vessel centerline plane. The projection direction used depends on the view orientation and is in the longitudinal, transverse or vertical direction, which ever is closest aligned with the current view direction.

To enable this automatic movement of the rotation centre use the command in the View menu (with the Perspective window active) or the Ctrl+0 (zero) shortcut.



View | Auto (update) Rotation Center command

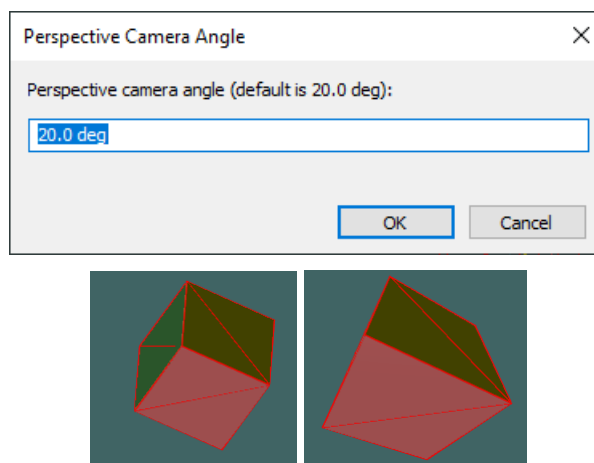
When rotating the model you will see the rotation centre and orientation icon. (The rotation center may be displayed permanently; the Ctrl+9 shortcut is used to toggle the permanent display)



Rotation centre icon is visible during rotations with sliders or virtual trackball

Perspective camera angle (field of view angle)

The perspective camera angle (field of view angle) can be changed; the default value is 20 degrees. An angle of zero will produce an isometric view without any perspective distortion whilst an angle of 130 degrees will produce a fish-eye view with a high degree of perspective distortion. You can change the value in the View | Perspective dialog:



View | Perspective (options) dialog

The default perspective angle is stored as a common preference for all Maxsurf applications and the value may only be changed in Modeler (as described above). This is the default value that will be used when creating new models or when loading older model files. From version 23.03 onwards, the model file includes the perspective angle along with the other view settings, and the value in the model file will override the previous default value.

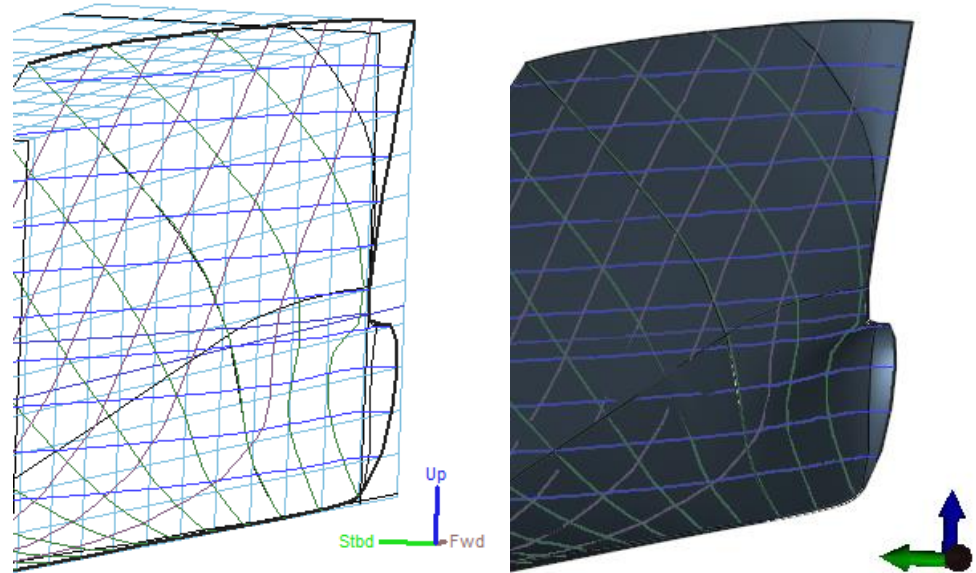
Moving and manipulating mode objects in Perspective view

As it is not possible to uniquely determine a three-dimensional coordinate from a two-dimensional screen position, Modeler limits movement of control points to a particular plane. It does this by determining which plane is most perpendicular to the direction of view, and only allows movement in that plane. You will see an axis indicator in the lower right corner of the window showing the current plane of movement.

For example, if the bow of the hull was pointing almost towards you, the sections plane would be most perpendicular, and points could be moved in the same plane as they could be in the Body Plan window (i.e. transversely and vertically, but not longitudinally).

In the perspective view, the control point constrain function (holding the Shift key whilst dragging the control point) constrains movement to longitudinal, transverse or vertical motion in the design's co-ordinate system (not the screen's horizontal and vertical directions).

An axis indicator is shown in the bottom right of the screen. This indicates the orientation of the main model axes in reference to the screen. The 'most normal' plane (the plane in which objects will be manipulated) is shown by the two axes which have the highest saturation (the pale axis is the one which is most perpendicular to the screen in the current model orientation).



Axis indicator (wireframe and rendered) 'Stbd' and 'Up' are saturated ('Fwd' is pale) so active working plane is equivalent to working in Body-Plan view

Also see:

[Rendering a Surface](#) on page 118.

Part Window

The Part window displays a 2D view of each part. You can select different surface contour lines to be displayed in the Part window. This may be helpful when exporting plate parts.

The Part window acts differently depending on the part that is displayed.

- When frames are displayed, the grid may also be displayed behind the frame.
- When plates are displayed, the Part toolbar will enable you to select different plate views. Also see: Plate Displays on page 319.

Up/Left and Down/Right arrow keys can be used to move back and forth between parts in the Part window.

When exporting from the Part window, there is an option to Copy All Parts. Each part will be placed on a separate layer in the DXF file named with the part's name.

Also see: Exporting Parts on page 322

Tool Panes

Modeler has two Tool Panes to help you manipulate and organise your model more efficiently.

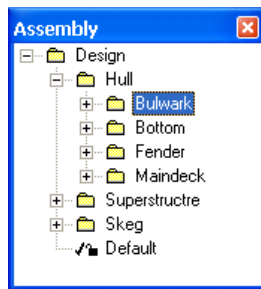
- [Assembly Pane](#)
- [Properties Pane](#)

Assembly Pane

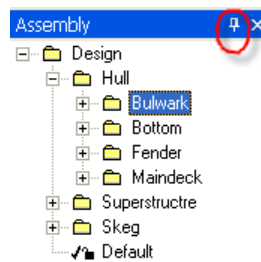
The Assembly Pane adds the ability to organise surfaces into groups (or: “assemblies”) and a quick access to surface related commands and settings such as visibility, locking and trimming through a right click menu. Grouping surfaces enables you to change the settings for several surfaces with one right click action.

The Assembly Pane in Modeler can be in different states:

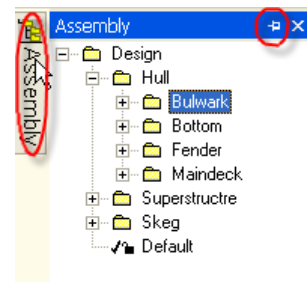
- Floating
- Docked
- Pinned – Auto Hide
- Hidden



Floating



Docked



Pinned
Auto Hide

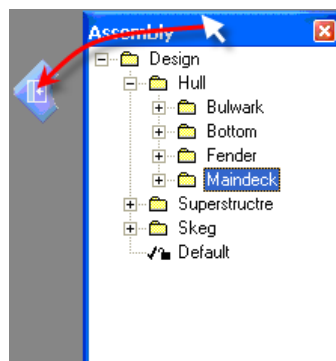
Opening the Assembly Pane

The Assembly Pane may be brought to the front by selecting View | Assembly Pane, pressing F2, or using the  button on the view toolbar.

The Assembly Pane contains a tree view of the surface hierarchy. This pane can be docked into the application window, or floating above it.

Docking the Assembly Pane

To dock a floating Assembly Pane, simply left click on the top window border and drag it onto any of the stickers that appear.



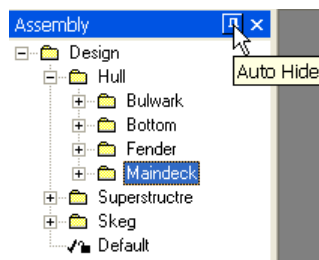
Drag the Assembly Pane onto the sticker to dock it.

When the Assembly Pane is docked, this process can easily be reversed to switch back to a Floating Assembly Pane. Simply select the top border of the docked window and drag it to any location on the screen.

Tip: when the Assembly Pane is docked, you can use the “Tile Horizontal” or “Tile Vertical” commands from the Windows menu to automatically fill the rest of the window with your favourite working windows.

Pinning or Auto Hiding the Assembly Pane

You can change the Assembly Pane from Docked to Pinned or Auto Hide and visa versa by clicking on the Pin icon on the top right of the window.



When the Assembly Pane is in the Auto Hide state, it will automatically hide itself to the outside of the screen if it is not being used. To show the dialog again, simply move your mouse cursor over the bar on the side of the screen (or click on it) and it will appear.

Editing Surfaces and Assemblies

The tree view has full support for inline editing, so names of assemblies and surfaces can be changed easily simply by slowly clicking twice on the name. Surfaces properties can also be edited by double clicking on the name, which then brings up the Surface Properties dialog box. You can also right-click on any item in the tree to modify it via context sensitive menus. Further, a surface's properties may be edited by double clicking on the surface name in the Assembly Pane.

Drag and Drop

You can move a surface into an assembly by dragging and dropping the surface icon in the tree onto the assembly. You can move an assembly into another assembly in the same way.

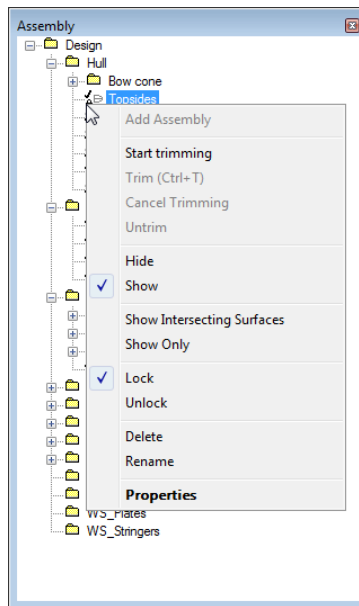
You can repeat this process as required to create any arrangement of assemblies you may require.

Context Sensitive Menus

Right clicking on an item in the tree view brings up a menu with commands for that item.

For surfaces this context sensitive menu contains:

Add Assembly	Adds a new assembly
Rename	Renames the currently selected surface or assembly
Show/Hide	Shows or Hides the currently selected surface
Lock/Unlock	Locks or unlocks the currently selected surface
Show Intersecting Surfaces	Shows only surfaces that intersect with the currently selected surface(s).
Trimming	Modify trimming on the currently selected surface, see Trimming Surfaces on page 139.
Properties	Shows Surface Properties Dialog for the current surface



Assembly Pane – Context Sensitive Menu for Surfaces

For assemblies this context sensitive menu contains:

Add	Adds a new assembly, with the same parent as the current assembly
Delete	Deletes the currently selected assembly
Rename	Renames the currently selected assembly
Show / Hide	Shows or Hides all surfaces that belong to the currently selected assembly, including any surfaces belonging to sub-assemblies of this assembly.
Show Only	Shows only the selected assembly items. All remaining assembly items are hidden.
Lock / Unlock	Locks or Unlocks all surfaces that belong to the currently selected assembly, including any surfaces belonging to sub-assemblies of this assembly.

Surface Icons in Assembly Pane

Icons are used next to each item to show its current state – i.e. unlocked/locked, visible/hidden, and also for assemblies, i.e. collapsed/expanded. Icons allow you to see at a glance what state a surface is in at any time. The tick icon indicates that the surface is visible; the padlock icon indicates that the surface is locked.

Also see: [Properties Pane](#).

Properties Pane

Properties Pane – Concepts

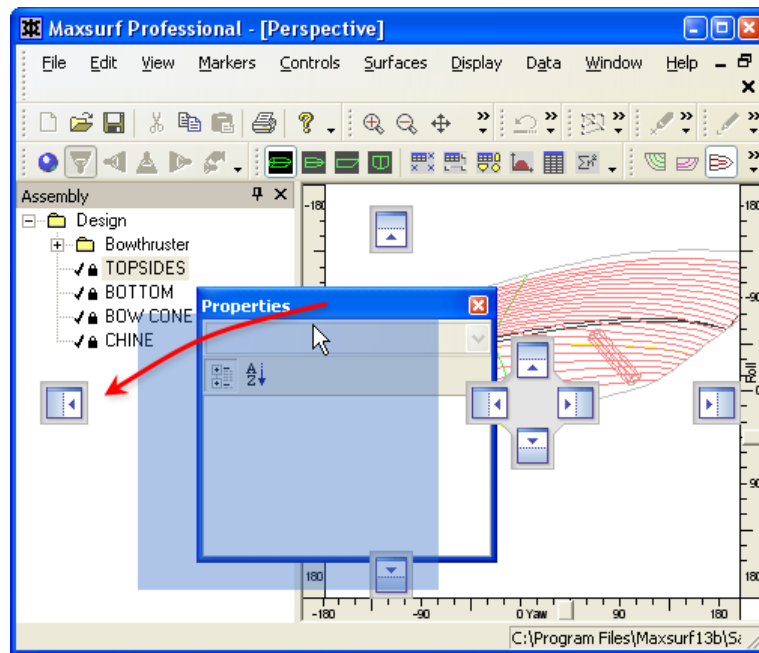
The Properties Pane allows you to manipulate properties of the currently selected item in the view windows.

Opening the Properties Pane

The Properties Pane can be shown by selecting Properties from the View menu, or pressing the F3 function key on your keyboard.

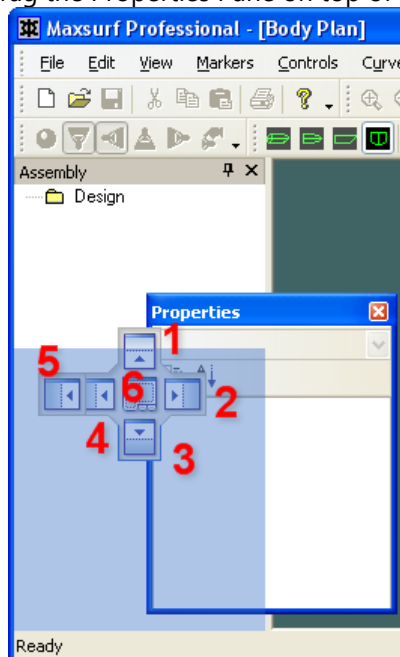
Properties Pane Options

The Properties Pane works very similar to the Assembly Pane in the way that it can be floating, docked or set to auto hide. The Assembly Pane and the Properties Pane can also be docked together into one pane (one above the other or side by side) or as tabbed panes behind each other.



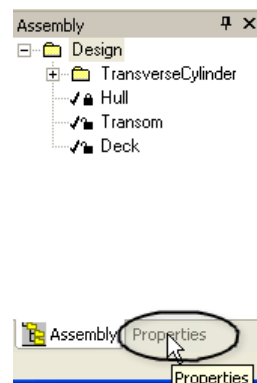
Regular docking and floating options for Properties Pane;
same as for the Assembly Pane.

When you drag the Properties Pane on top of the docked Assembly Pane, you will see the following:



Properties Pane docking position:

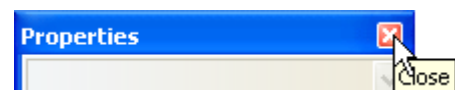
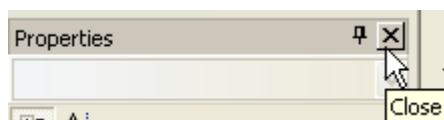
- 1: Above Assembly Pane
- 2: To the right of Assembly Pane
- 3: Below Assembly Pane
- 4: To the left of Assembly Pane
- 5: To the left of Assembly Pane as a separate pane
- 6: Underneath the Assembly Pane into a tabbed pane.



Properties Pane docking in the same area as the Assembly Pane has different options.

Closing the Properties Pane

The Properties Pane can be closed by clicking on the cross in the top right of the Pane.



Closing when docked

Closing when floating

Working with the Properties Pane

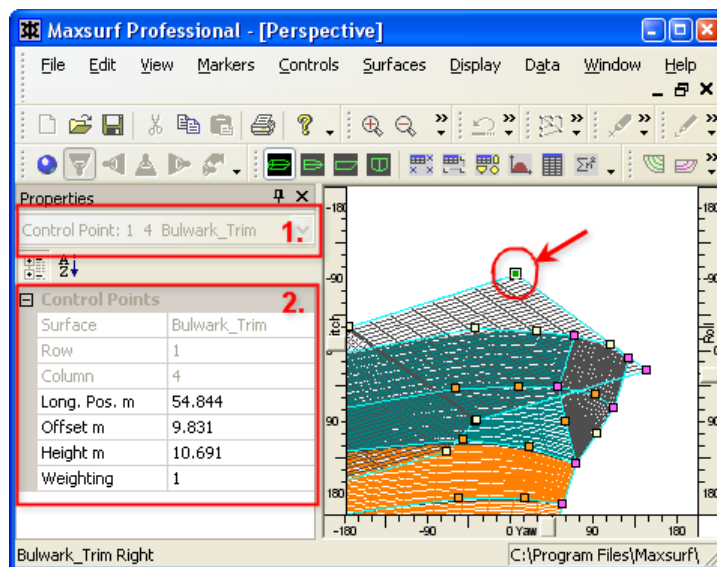
The Properties Pane can be used to view and modify the properties of the following objects:

Object	Selection method
Surfaces	Assembly Pane and rendered Perspective view One single surfaces can be selected at any time
Control Points	Multiple control points selection from any view window
Markers	Multiple control points selection from any view window

Single object selected

The image below shows a single control point selected in Perspective view. The Properties Pane can be divided up into two areas that show the following:

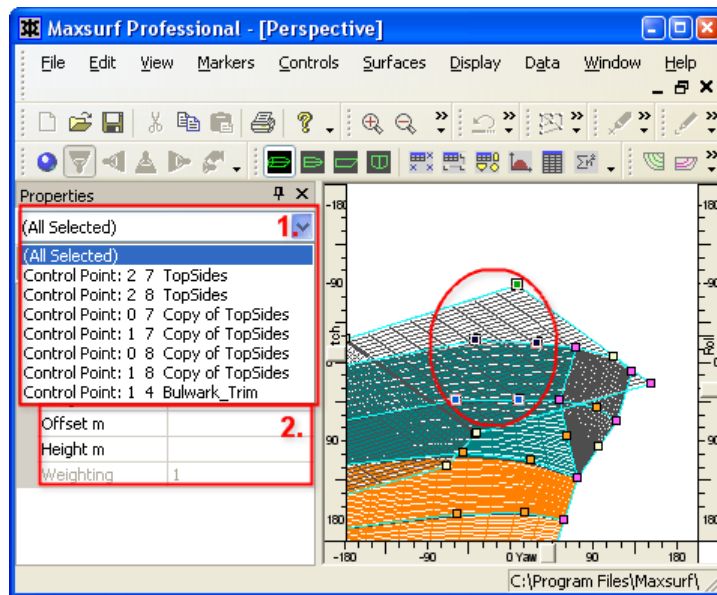
1. The object selected. In this case the row 1, column 4 control point on the Bullwark_Trim surface.
2. The properties of the object selected.
 - a. Some fields are greyed out which means they are read-only and cannot be edited. For example, in case of a control point object, these are the Surface that the control point belongs to and the row and column numbers.
 - b. The fields that can be edited are the position and the weight of the control point. For example, you can move the control point fwd to 55 m by simply selecting the Long Pos. value field and typing 55. The position of the control point will be updated in the view windows as soon as you hit enter or click outside of the Long Pos. value field.



The Properties of the selected control point are listed in the Properties Pane on the left.

Multiple Control Points Selected

The image below shows multiple control points selected. The Properties Pane now allows you to modify the properties of the entire selection, or select one particular object to be modified individually using the drop down in area 1.



When multiple control points have been selected, you can modify the properties of the selection or switch to individual control points using the drop down.

Only values that are the same for all selected objects are shown in area 2 in the Properties Pane. For example: all selected control points in this image share a control point weight 1.

You can modify the properties of a group of control points in a selection by typing in the value fields. For example, if you would like all control points to move to a Long Pos. of 60 m, you can simply type 60 in the Long Pos. field and all control points will move as soon as you hit enter or click outside the field.

Selecting, viewing and modifying the properties of surfaces and markers works in much the same way.

Note

You can not select more than one surface simultaneously.

Also see: [Assembly Pane](#).

Table Windows

There are several windows in Modeler and other programs in the Modeler suite that display tabulated data.

This section will first discuss the generic features available when working in Table windows in the Modeler suite and then discuss the following specific Table windows available in Modeler:

- [Control Points Window](#)
- [Markers Window](#)
- [Surfaces Window](#)
- [Offsets Window](#)
- [Structure Windows](#)

Table Windows – General

This section describes general features of tables in the Modeler suite.

Display Area

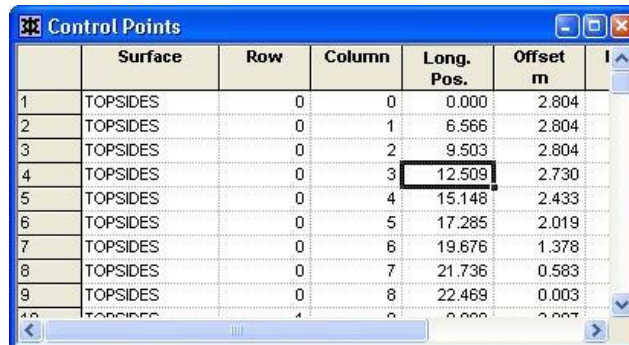
Using the scroll bars on the right and bottom-right of the window changes the displayed area. These will disappear if the whole table is visible. In addition, it is possible to use the zoom and shrink functions to increase or reduce the size of the table.

Selection

There are a number of ways of selecting data from the tables. You may wish to select data for copying or pasting to or from another application.

Single Cell

A single cell is selected simply by clicking once in the cell. The selected cell will be highlighted with a bold outline:

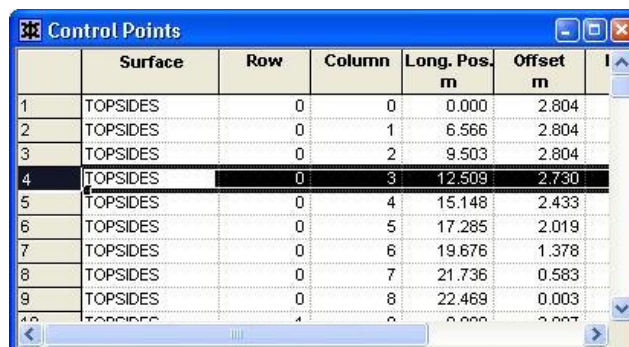


The screenshot shows the 'Control Points' window with a table. The cell at row 4, column 3 (Long. Pos. 12.509) is selected, indicated by a bold border.

	Surface	Row	Column	Long. Pos.	Offset m
1	TOPSIDES	0	0	0.000	2.804
2	TOPSIDES	0	1	6.566	2.804
3	TOPSIDES	0	2	9.503	2.804
4	TOPSIDES	0	3	12.509	2.730
5	TOPSIDES	0	4	15.148	2.433
6	TOPSIDES	0	5	17.285	2.019
7	TOPSIDES	0	6	19.676	1.378
8	TOPSIDES	0	7	21.736	0.583
9	TOPSIDES	0	8	22.469	0.003

Single Row

A single row is selected by clicking once in the grey, row number cell in the left-most column. The selected row will be highlighted by making the background black and the text white:

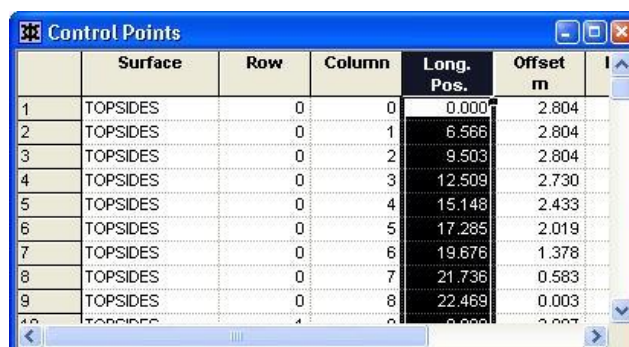


The screenshot shows the 'Control Points' window with row 4 selected. The entire row, including the row number '4' in the first column, has a black background and white text.

	Surface	Row	Column	Long. Pos.	Offset m
1	TOPSIDES	0	0	0.000	2.804
2	TOPSIDES	0	1	6.566	2.804
3	TOPSIDES	0	2	9.503	2.804
4	TOPSIDES	0	3	12.509	2.730
5	TOPSIDES	0	4	15.148	2.433
6	TOPSIDES	0	5	17.285	2.019
7	TOPSIDES	0	6	19.676	1.378
8	TOPSIDES	0	7	21.736	0.583
9	TOPSIDES	0	8	22.469	0.003

Single Column

A single column is selected by clicking once in the grey, column-heading cell in the top-most row. The selected column will be highlighted by making the background black and the text white:



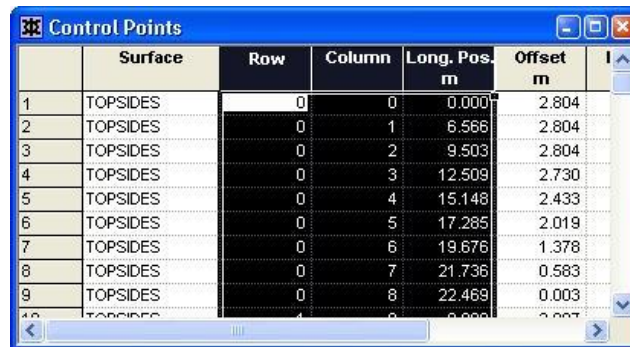
The screenshot shows the 'Control Points' window with the 'Long. Pos.' column selected. The entire column, including the header 'Long. Pos.', has a black background and white text.

	Surface	Row	Column	Long. Pos.	Offset m
1	TOPSIDES	0	0	0.000	2.804
2	TOPSIDES	0	1	6.566	2.804
3	TOPSIDES	0	2	9.503	2.804
4	TOPSIDES	0	3	12.509	2.730
5	TOPSIDES	0	4	15.148	2.433
6	TOPSIDES	0	5	17.285	2.019
7	TOPSIDES	0	6	19.676	1.378
8	TOPSIDES	0	7	21.736	0.583
9	TOPSIDES	0	8	22.469	0.003

Contiguous Rows and Columns

Contiguous rows or columns are selected by clicking in the grey, row number cell or the grey, column heading cell and dragging the mouse over the other rows or columns to be selected. Alternatively, you may click in the first row or column header and then click in the final row or column header with the Shift key

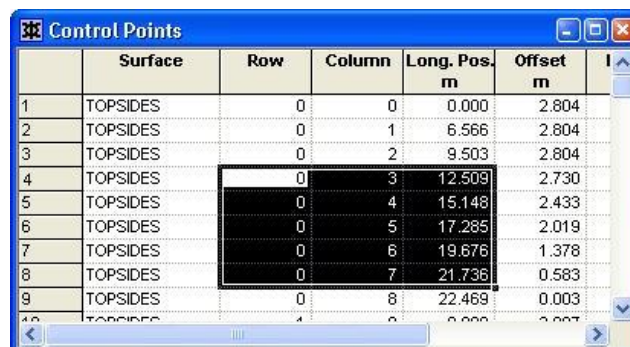
depressed. All the columns between, and including, the first and last will be selected. You may also use the scroll bars if you use the second method.



	Surface	Row	Column	Long. Pos. m	Offset m	I
1	TOPSIDES	0	0	0.000	2.804	
2	TOPSIDES	0	1	6.566	2.804	
3	TOPSIDES	0	2	9.503	2.804	
4	TOPSIDES	0	3	12.509	2.730	
5	TOPSIDES	0	4	15.148	2.433	
6	TOPSIDES	0	5	17.285	2.019	
7	TOPSIDES	0	6	19.676	1.378	
8	TOPSIDES	0	7	21.736	0.583	
9	TOPSIDES	0	8	22.469	0.003	
10	TOPSIDES	0	9	22.800	0.000	

Block of Cells

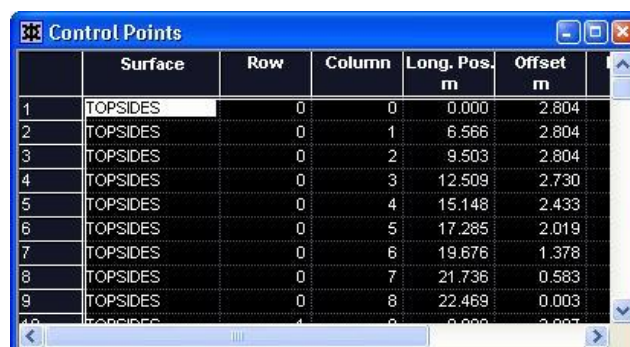
A rectangular block of cells may be selected by clicking and dragging over the cells to be selected. Alternatively you may click in the first cell and then click in the final cell with the Shift key depressed. You may also use the scroll bars if you use the second method.



	Surface	Row	Column	Long. Pos. m	Offset m	I
1	TOPSIDES	0	0	0.000	2.804	
2	TOPSIDES	0	1	6.566	2.804	
3	TOPSIDES	0	2	9.503	2.804	
4	TOPSIDES	0	3	12.509	2.730	
5	TOPSIDES	0	4	15.148	2.433	
6	TOPSIDES	0	5	17.285	2.019	
7	TOPSIDES	0	6	19.676	1.378	
8	TOPSIDES	0	7	21.736	0.583	
9	TOPSIDES	0	8	22.469	0.003	
10	TOPSIDES	0	9	22.800	0.000	

Complete Table

The complete table may be selected simply by clicking in the blank, grey, top-left cell:



	Surface	Row	Column	Long. Pos. m	Offset m	I
1	TOPSIDES	0	0	0.000	2.804	
2	TOPSIDES	0	1	6.566	2.804	
3	TOPSIDES	0	2	9.503	2.804	
4	TOPSIDES	0	3	12.509	2.730	
5	TOPSIDES	0	4	15.148	2.433	
6	TOPSIDES	0	5	17.285	2.019	
7	TOPSIDES	0	6	19.676	1.378	
8	TOPSIDES	0	7	21.736	0.583	
9	TOPSIDES	0	8	22.469	0.003	
10	TOPSIDES	0	9	22.800	0.000	

Sizing Columns

The width of the columns may be adjusted by two different methods. In either case, the column widths will be saved when you exit the program. When you re-start the program, the saved column widths will be re-loaded.

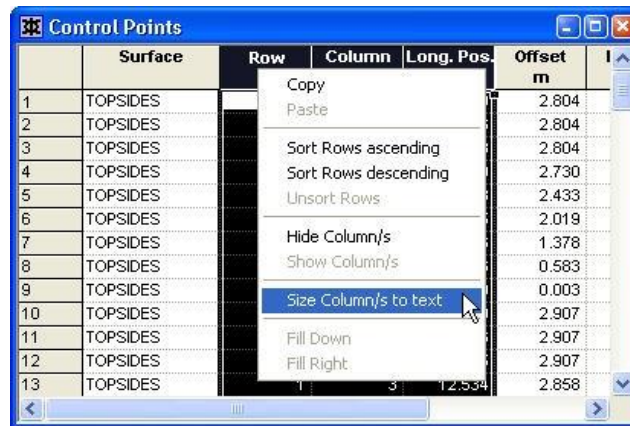
Dragging

The column widths may be adjusted by placing the cursor over the line between the two columns, in the greyed column heading area. The cursor will change to a re-sizing cursor and the column may now be re-sized by clicking and dragging the line to the new desired position. The column to the left will be adjusted.

You may re-size several columns together by first selecting a group of columns and then adjusting the width of any of them. When you release the mouse, all the selected columns will be re-sized.

Size to Text

Alternatively, the column widths may be sized according to the width of the text displayed in the column's header. To do this, select the columns to be re-sized and then right-click in one of the grey header cells of one of the selected columns. Select Size Column/s to text from the context menu:

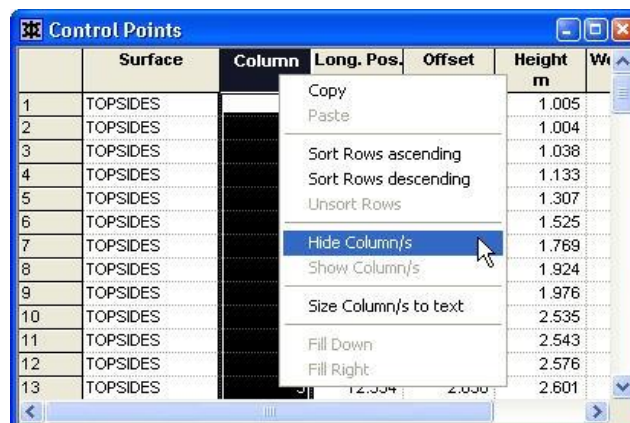


Customizing the Display

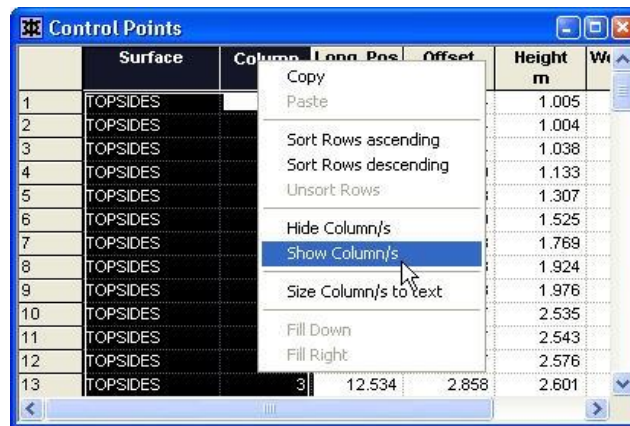
In some tables the format of the displayed data may be customised. In most tables it is possible to hide columns and sort rows. Some programs also have additional features, for further details of these additional functions please see the documentation for the relevant program.

Column Hiding

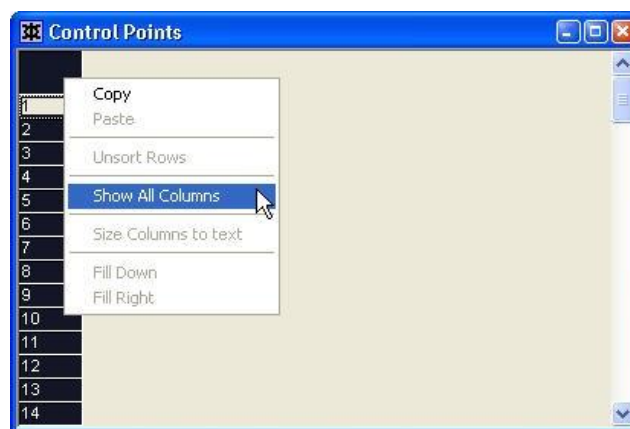
To hide a column or selection of columns, select the columns and right-click in one of the grey header cells of one of the selected columns. Choose Hide Column/s from the context menu:



To unhide the columns, select the columns on each side of the hidden columns, right click and choose Show Column/s:



If all the columns are hidden, right-click in the top-left grey cell and choose Show All Columns from the menu:



Row Sorting

The rows may be sorted according to the data in a particular column. Right-click in the grey header row of the column you wish to sort by, then select Sort Rows ascending or Sort Rows descending as required. To return the rows to their original order, right-click anywhere in the grey column header row and select Unsort Rows.

Font

The font used in the table may be changed using the Font command in the View menu.

Editing

There are several tools which aide editing data in a table. These are described in greater detail below.

Typing

Data may be typed in a cell by clicking in the cell and typing in the new data. This will overwrite the original entry in the cell. If you wish to edit the data in a cell, double-click to place the cursor at the position in the cell where you wish to start typing. Once editing, you can use the mouse to move the cursor by clicking once at the new insertion point or you can use the left and right cursor keys. If you wish to select certain characters in the cell, click and drag the mouse over the characters to be selected, these will be highlighted with a blue background and white text.

Copy

The entire contents of a cell or group of cells may be copied to the clipboard for pasting into another application or Modeler table. Select the cells to be copied and choose Copy from the Edit menu, or use the keyboard shortcut Ctrl+C or Ctrl+Ins.

If a group of cells have been selected, they will be copied in tab delimited format which enables direct pasting into applications such as MS Excel.

Note

Use Ctrl+Shft+C to include the table column-headings.

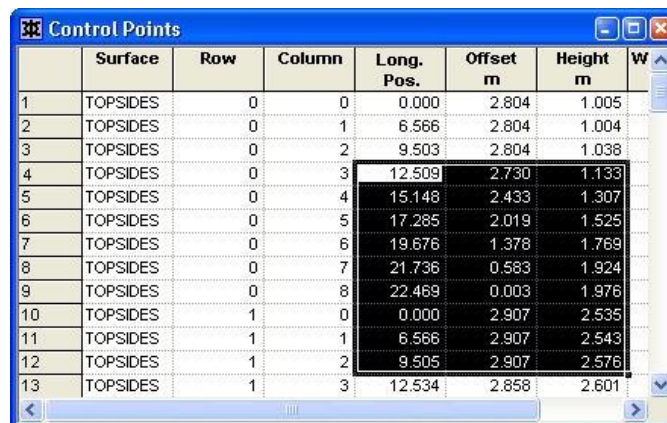
Paste

Data may be pasted from other tables or other applications such as MS Excel. Once the data has been copied, select the top-left cell of the area you wish to paste the data into. Choose Paste from the Edit menu, or use the keyboard shortcut Ctrl+V or Shift+Ins.

Make sure the table you are pasting into contains more or an equal number of rows and columns as the table you copied the data from.

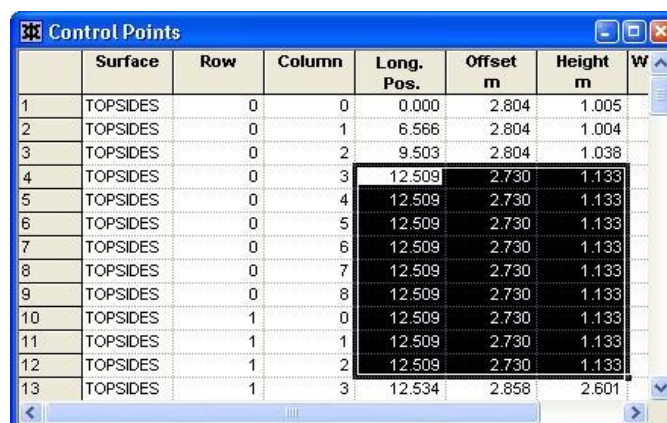
Fill Down

The data from a cell, or group of cells, may be copied into the cells below it. Select the group of cells you wish the new data to be copied into; include also the cells which contain the data you wish to fill down with. Right-click and select Fill Down or use the keyboard shortcut Ctrl+D.



	Surface	Row	Column	Long. Pos.	Offset m	Height m	W
1	TOPSIDES	0	0	0.000	2.804	1.005	
2	TOPSIDES	0	1	6.566	2.804	1.004	
3	TOPSIDES	0	2	9.503	2.804	1.038	
4	TOPSIDES	0	3	12.509	2.730	1.133	
5	TOPSIDES	0	4	15.148	2.433	1.307	
6	TOPSIDES	0	5	17.285	2.019	1.525	
7	TOPSIDES	0	6	19.676	1.378	1.769	
8	TOPSIDES	0	7	21.736	0.583	1.924	
9	TOPSIDES	0	8	22.469	0.003	1.976	
10	TOPSIDES	1	0	0.000	2.907	2.535	
11	TOPSIDES	1	1	6.566	2.907	2.543	
12	TOPSIDES	1	2	9.505	2.907	2.576	
13	TOPSIDES	1	3	12.534	2.858	2.601	

Selection and data before Fill Down

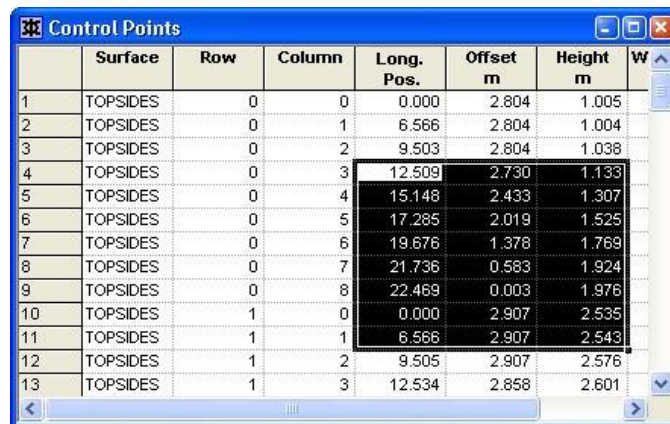


	Surface	Row	Column	Long. Pos.	Offset m	Height m	W
1	TOPSIDES	0	0	0.000	2.804	1.005	
2	TOPSIDES	0	1	6.566	2.804	1.004	
3	TOPSIDES	0	2	9.503	2.804	1.038	
4	TOPSIDES	0	3	12.509	2.730	1.133	
5	TOPSIDES	0	4	12.509	2.730	1.133	
6	TOPSIDES	0	5	12.509	2.730	1.133	
7	TOPSIDES	0	6	12.509	2.730	1.133	
8	TOPSIDES	0	7	12.509	2.730	1.133	
9	TOPSIDES	0	8	12.509	2.730	1.133	
10	TOPSIDES	1	0	12.509	2.730	1.133	
11	TOPSIDES	1	1	12.509	2.730	1.133	
12	TOPSIDES	1	2	12.509	2.730	1.133	
13	TOPSIDES	1	3	12.534	2.858	2.601	

Modified data after Fill Down

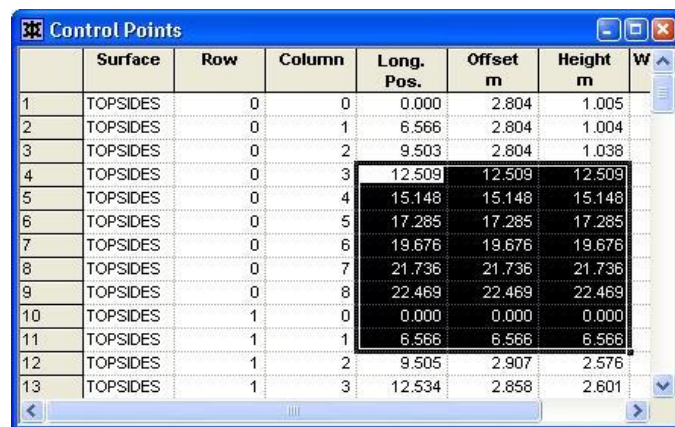
Fill Right

In a similar manner, data may be filled to the right using the Fill Right command:



	Surface	Row	Column	Long. Pos.	Offset m	Height m	W
1	TOPSIDES	0	0	0.000	2.804	1.005	
2	TOPSIDES	0	1	6.566	2.804	1.004	
3	TOPSIDES	0	2	9.503	2.804	1.038	
4	TOPSIDES	0	3	12.509	2.730	1.133	
5	TOPSIDES	0	4	15.148	2.433	1.307	
6	TOPSIDES	0	5	17.285	2.019	1.525	
7	TOPSIDES	0	6	19.676	1.378	1.769	
8	TOPSIDES	0	7	21.736	0.583	1.924	
9	TOPSIDES	0	8	22.469	0.003	1.976	
10	TOPSIDES	1	0	0.000	2.907	2.535	
11	TOPSIDES	1	1	6.566	2.907	2.543	
12	TOPSIDES	1	2	9.505	2.907	2.576	
13	TOPSIDES	1	3	12.534	2.858	2.601	

Selection and data before Fill Right

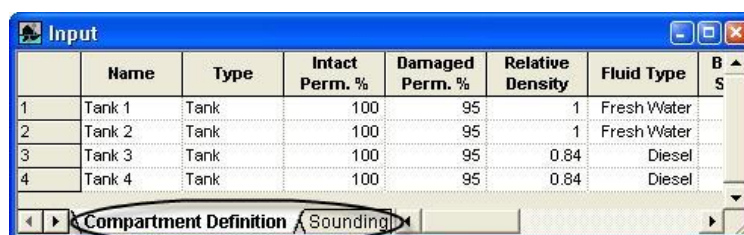


	Surface	Row	Column	Long. Pos.	Offset m	Height m	W
1	TOPSIDES	0	0	0.000	2.804	1.005	
2	TOPSIDES	0	1	6.566	2.804	1.004	
3	TOPSIDES	0	2	9.503	2.804	1.038	
4	TOPSIDES	0	3	12.509	12.509	12.509	
5	TOPSIDES	0	4	15.148	15.148	15.148	
6	TOPSIDES	0	5	17.285	17.285	17.285	
7	TOPSIDES	0	6	19.676	19.676	19.676	
8	TOPSIDES	0	7	21.736	21.736	21.736	
9	TOPSIDES	0	8	22.469	22.469	22.469	
10	TOPSIDES	1	0	0.000	0.000	0.000	
11	TOPSIDES	1	1	6.566	6.566	6.566	
12	TOPSIDES	1	2	9.505	2.907	2.576	
13	TOPSIDES	1	3	12.534	2.858	2.601	

Modified data after Fill Right

Multiple Tables

The tables used in the Modeler suite are similar to the Worksheets in MS Excel. Just like in Excel, you can switch between tables by clicking on the tab of the desired table at the bottom of the window (if there are multiple tables available in that window). Use the scroll arrows in the bottom left hand corner to view more tabs. The scroll bar on the bottom right is used to scroll through the currently displayed table.



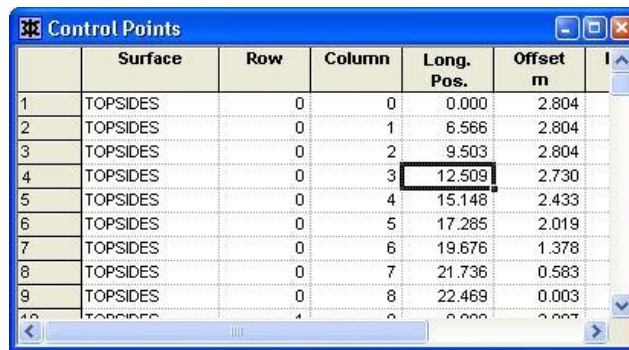
	Name	Type	Intact Perm. %	Damaged Perm. %	Relative Density	Fluid Type	B S
1	Tank 1	Tank	100	95	1	Fresh Water	
2	Tank 2	Tank	100	95	1	Fresh Water	
3	Tank 3	Tank	100	95	0.84	Diesel	
4	Tank 4	Tank	100	95	0.84	Diesel	

Example of multiple tables in one window (from Stability).

Control Points Window

The Control Points window allows the user to specify the control point positions directly using the keyboard rather than by using the mouse in one of the design windows. Whenever the Control Points window is selected the highlighted row is that of the current control point in the [Current Surface](#).

Only the control points of visible surfaces will be displayed. Those of locked, visible surfaces will be displayed in grey and cannot be edited. Only the control points of unlocked, visible surfaces can be changed.



	Surface	Row	Column	Long. Pos.	Offset m	I
1	TOPSIDES	0	0	0.000	2.804	
2	TOPSIDES	0	1	6.566	2.804	
3	TOPSIDES	0	2	9.503	2.804	
4	TOPSIDES	0	3	12.509	2.730	
5	TOPSIDES	0	4	15.148	2.433	
6	TOPSIDES	0	5	17.285	2.019	
7	TOPSIDES	0	6	19.676	1.378	
8	TOPSIDES	0	7	21.736	0.583	
9	TOPSIDES	0	8	22.469	0.003	
10	TOPSIDES	0	9	22.800	0.000	

The position, offset and height of a control point may be edited by selecting the relevant cell. It is also possible to copy and paste data from other applications such as spreadsheets. You may select rows, columns, the whole window, or an individual cell by using the mouse.

The Surface, Column, and Row columns are not editable. The Surface column displays the surface name as specified in the Properties dialog. The Column and Row columns specify the row and column numbers in the same order as their position in the net.

Editing Control Points

To edit a control point position numerically

- Select the control point you wish to edit.

Select the control point from any of the drawing windows.

- Select the Control Points window.
- Scroll through the list.

The selected control point will be highlighted.

- Edit the control point position.

Note:

Addition and deletion of control point rows and columns may not be performed in the Control Points window. To add and delete rows use the Add and Delete functions from the Controls menu in their respective drawing windows.

Setting Control Point Display format

In the Control Points window it is possible to view the points listed by row then column or by column then row. This is achieved by setting the Row then Column or the Column then Row items in the Preferences dialog (in the View menu).

This is an advantage when you wish to paste a single row of data into a surface such as in the case of an airfoil section being imported from a spreadsheet. In other cases you may wish to paste only a single column of data.

Alternatively, you may sort the data by any column by right clicking in that column's header and selecting from the pop-up context menu. Or you may sort in ascending order by double clicking the column's header.

Synchronization of Control Points and Design windows

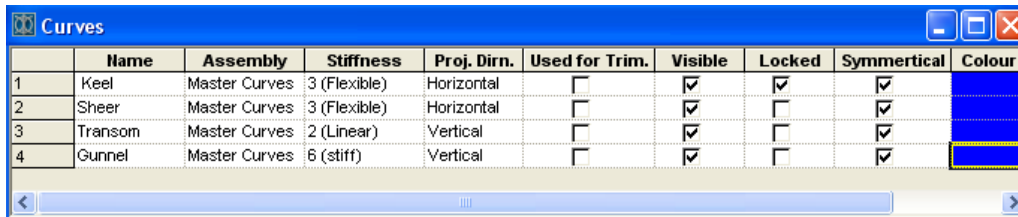
Single control points selected in the design view are selected in the Control Points table and control points selected in the Control Points table will be selected in the design views. This can make it easier to verify visually which control point is which.

Markers Window

See section on [Working with Markers](#).

Curves Window

The curves window provides quick and flexible access to all the curves' properties.



	Name	Assembly	Stiffness	Proj. Dirn.	Used for Trim.	Visible	Locked	Symmetrical	Colour
1	Keel	Master Curves	3 (Flexible)	Horizontal	☐	☒	☒	☒	
2	Sheer	Master Curves	3 (Flexible)	Horizontal	☐	☒	☐	☒	
3	Transom	Master Curves	2 (Linear)	Vertical	☐	☒	☐	☒	
4	Gunnel	Master Curves	6 (stiff)	Vertical	☐	☒	☐	☒	

Column hiding and sorting may be used in this window and this may be used to reduce the size of the window. This window may then be used as a very convenient way of turning on and off the visibility and locking of curves in a complicated design. Copy, Paste and Fill down may also be used in this table.

Surfaces Window

As with the curves window, the surfaces window provides quick and flexible access to all the surfaces' properties.



	Name	Assembly	Type	Use	Rows	Cols	Long. Stiff.	Trans
1	hull	Hull	NURB	Hull	5	7	6	
2	keel	Appendages	B-Spline	Hull	3	3	3	
3	bulb	Appendages	B-Spline	Hull	4	5	3	
4	transom	Hull	B-Spline	Hull	2	3	3	
5	cockpit	Hull	B-Spline	Hull	3	7	2	
6	deck	Hull	NURB	Hull	2	7	6	
7	rudder	Appendages	B-Spline	Hull	3	4	3	
8	coachroof	Hull	B-Spline	Hull	3	3	3	

Column hiding and sorting may be used in this window and this may be used to reduce the size of the window. This window may then be used as a very convenient way of turning on and off the visibility and locking of surfaces in a complicated design. Copy, Paste and Fill down may also be used in this table.

Offsets Window

The Offsets window allows you to view the offsets for a design. Offsets are found for the given station (or: section) as set in the Design Grid dialog from the Data menu.



	Contour	Long. Pos. m	Offset m	Height m
1	b 1	-6.610	0.200	0.038
2	b 2	-6.610	0.400	0.056
3	b 3	-6.610	0.600	0.093
4	b 4	-6.610	0.800	0.154
5	b 5	-6.610	1.000	0.242
6	b 6	-6.610	1.200	0.360
7	b 7	-6.610	1.400	0.520
8	b 8	-6.610	1.600	0.743
9	DWL	-6.610	----	----
10	wl 2	-6.610	1.238	1.000
11	wl 3	-6.610	1.702	0.900
12	wl 4	-6.610	1.640	0.800
13	wl 5	-6.610	1.567	0.700
14	wl 6	-6.610	1.481	0.600
15	wl 7	-6.610	1.378	0.500
16	wl 8	-6.610	1.256	0.400
17	wl 9	-6.610	1.106	0.300
18	wl 10	-6.610	0.913	0.200

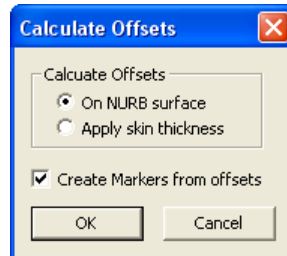
The offsets are displayed for one station at a time. See the [Displaying Offsets](#) section for instructions on how to change this display.

Calculating Offsets

To calculate offsets

- Select the Offsets window.
- Select Calculate Offsets from the Data menu.

A dialog will appear allowing you to choose whether or not to include skin thickness deduction and/or to automatically create a table of markers from the calculated offset points.



Offsets calculation may take some time depending on the number of surfaces and grid lines in use. Offsets are sorted by station position.

Note:

If your model contains bonded edges and you would like these to be included in the offsets calculation, you have to make sure the bonded edges are visible. Same for feature lines to calculate offsets at the position of a knuckle.

Skin Thickness Deduction for Offsets

Offsets can be calculated after deducting the skin thickness. The skin thickness is deducted on a surface-by-surface basis. In order to ensure this is done correctly please check the following:

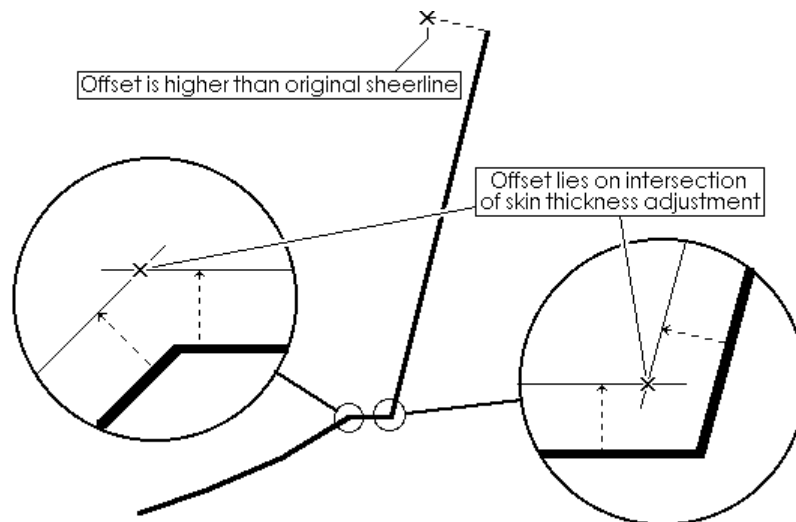
- Thickness specification of each surface
- Thickness direction of each surface, see [Surface Materials and Skin Thickness](#).
- [Outside Arrows](#) direction of each surface

If you have chosen to deduct skin thickness, Modeler will add or deduct that thickness perpendicular to the surface at each point on the hull, so that areas of the hull that are not parallel to the centreline will be correctly calculated.

Where an offset is for an edge or feature line, the following rules apply:

Edges

Skin thickness is taken along the surface perpendicular at the edge, rather than along a waterline, buttock or diagonal. An example of this would be for a sheer line, where the topside was not vertical, with the resulting offset higher than the original point. Builders should be made aware that the height specified does not correspond to the height of the sheer line.



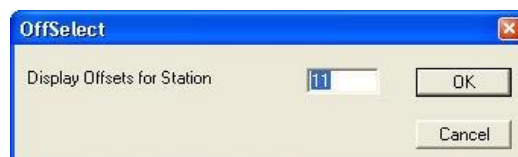
Feature lines

Offsets at feature lines such as knuckles and chines within a surface are taken at the intersection of the projection of the deducted surface, rather than on a radius around the feature point.

Displaying Offsets

To display different pages of the offset table:

- Choose Go To Offset... from the Data menu



- Enter the number of the station whose offsets you wish to view.
- Click OK

The Offsets window will change to display the offsets for the station requested.

Copying Offsets

Offsets may be copied for transfer to other programs. To do this

- Click in the top left hand corner of the Offsets table.

This selects and highlights the entire Offsets window for copying.

- Select Copy from the Edit menu.

Switch to your word processor or spreadsheet and choose Paste.

Customizing Offsets

Once a table of Offsets has been calculated in Modeler, it is possible to customize the names used for waterlines, buttocks, diagonals, feature lines and edges. Simply type the text required into the required cell in the Offsets window. These names will be used when the table of offsets is printed or written to a file.

Writing an Offsets File

It is also possible to write out a text file of the complete table of offsets. This can be read directly into other programs such as word processors or spreadsheets. To do this

- Make sure the Offsets window is in front.
- Select Save Offsets from the File menu.

- Specify a name for the offsets file and click on the save button.

The offsets will be saved in a TEXT file of that name and may be opened by any word processor or spreadsheet.

Part Window

The menu items for the tables in the Data window appear in the Window menu.

Stringers table

Stringer names and stringer properties can be edited in the Stringer table in the Data window.

Stringer points table

Stringer points names and associated properties can be altered in the Stringer Point table. Girths are displayed in the table of stringer points.

Upstands table

Upstands are sorted by frame and girth in the Upstand table.

Upstand girth locations and girth spacing's are displayed in the Upstand table.

Frames table

Frame names, positions and limits can be edited in the Frame table in the Data window.

Frames may be selected by highlighting the corresponding rows in the Frames table. You may then recalculate or delete these selected frames.

Decks table

Deck names, positions(heights) and limits can be edited in the Deck table in the Data window.

Plates table

Plate names and properties can be edited in the Plate table in the Data window.

Plates may be selected by highlighting the corresponding rows in the Plates table. You may then recalculate or delete these selected plates.

Plate Dimensions table

The diagonal dimensions of plates are displayed in the Plate Dimensions table.

Pin Jig table

Heights and edge distances for all the pins in the jig can be viewed in this table. The pin jig information is displayed for the plate currently being viewed in the Part window.

Parts table

The Parts provides weight, area, CG and perimeter information of the calculated parts as well as a summary total for the design. Perimeter and area values for Stringers refer to the perimeter and area of the stringer web.

The weight of a stringer is calculated

Note

The weight, area and perimeter information for parts on symmetrical surfaces will display the values for both sides combined. The CG however will be for one side of the vessel.

Graph Windows

There are several windows in Modeler and other programs in the Modeler suite that display Graph data. The [Graph Windows - General](#) section discusses generic features of graph windows in the Modeler suite. You can also find specific information on how to select different view options for the [Curve of Areas Window](#).

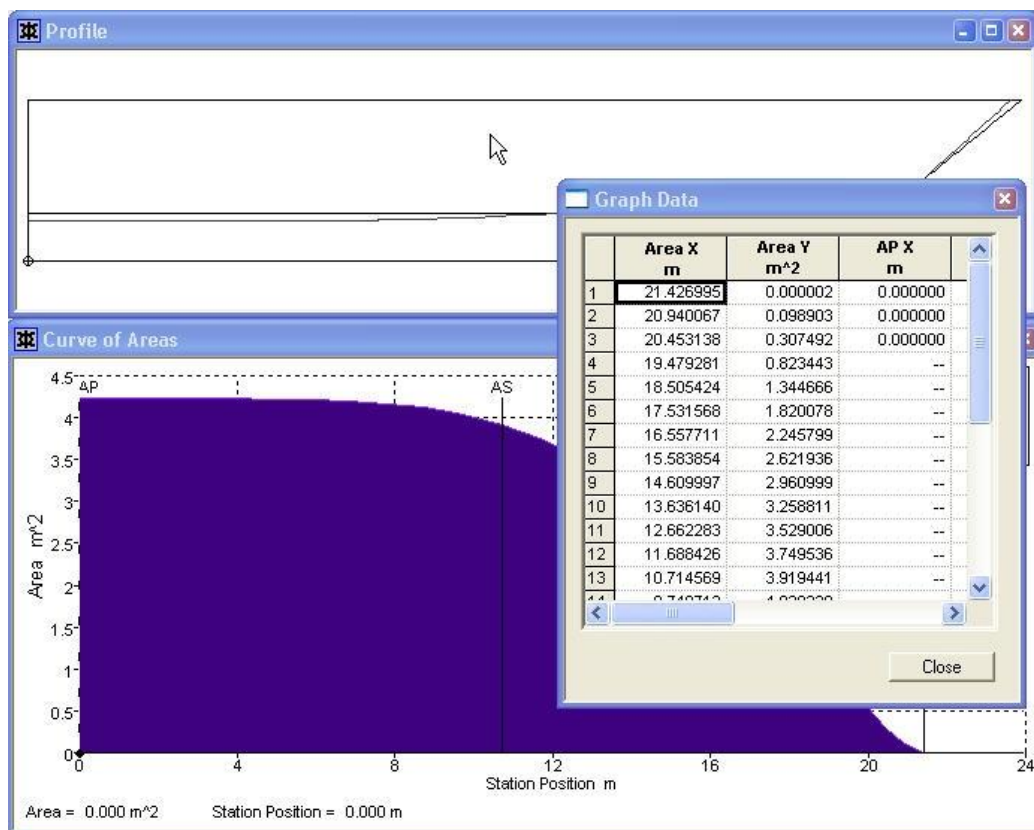
Graph Windows - General

Data Interpolation

Data may be read off the graph by clicking on the curve of interest. A slider is then displayed which may be dragged along the curve. The abscissa and ordinate are displayed in the bottom-left corner of the window.

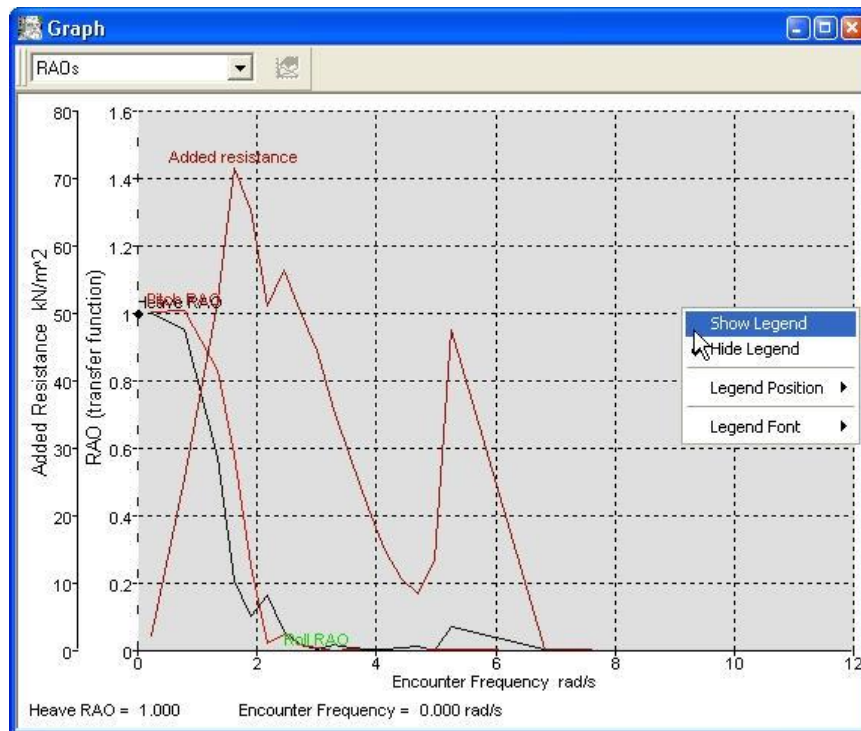
Graph data

The graphed data in any graph can be displayed in tabulated format by double clicking on the graph. This can be used to obtain the sectional area curve data for example. The complete set of data can be copied by right clicking on the top left grey cell to select the entire table, then selecting copy from the menu. If you wish the Copy operation to include the column headings, hold down the Shift key at the same time as you select the Copy command. The dialog may also be resized to best display the tabulated data.



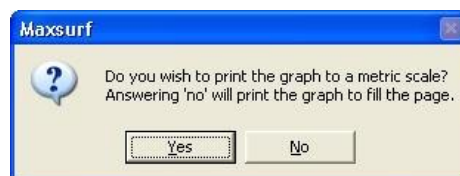
Graph legend

A graph legend may be displayed in the various graph windows. Right click in the graph view to obtain the context-menu that gives the various display options:



Scaled graph printing

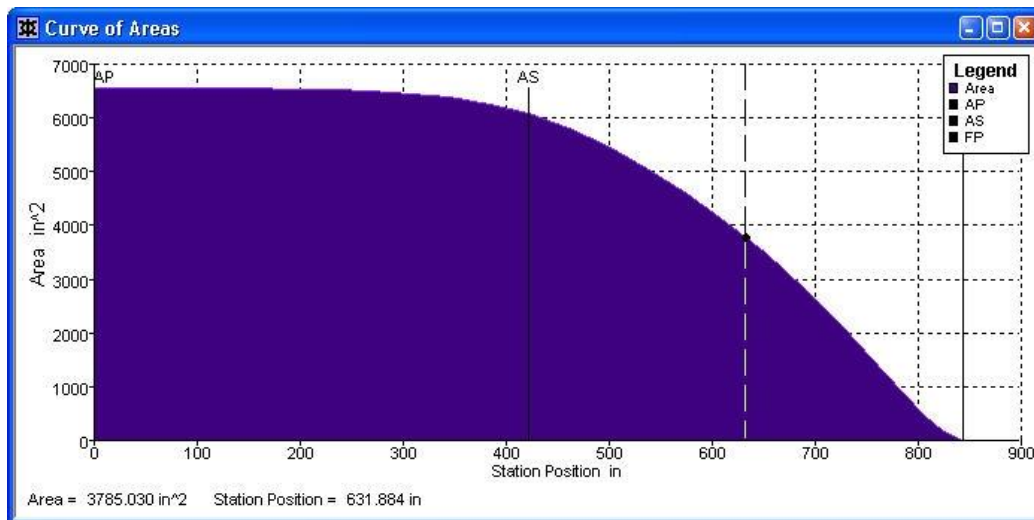
You can specify whether a graph should be printed to a scale (to facilitate taking measurements from the graph). Holding down the Shift key while selecting Print from the File menu, or using the toolbar button will display the following dialog. Answering "Yes", will mean that the graph is scaled so that one grid unit is a sensible number of centimetres or inches: either 1, 2, 2.5 or 5. The use of centimetres or inches will depend on whether the units used for length are metric or imperial. The setting will be remembered once the selection has been made; to change the setting, hold down the shift key again when selecting print to display the dialog. Choosing "No" will cause the graph to be printed filling the page and is the default behaviour.



Scaled print dialog if length units are metric

Curve of Areas Window

When a design is open the Curve of Areas window may be used to determine the Sectional Area Curve for the current datum waterline.



The form of Sectional Area Curve displayed may be chosen in the Preferences dialog in the View menu.

Three choices exist:

- Section Area.
Graphs sectional area against longitudinal position from the zero point.
- Section Area / Maximum Section Area.
Graphs sectional area normalised by maximum sectional area to allow the comparison of sectional area curves from different designs of similar displacement.
- Section Area / Volume.
Graphs sectional area normalised by volume^{2/3}. This allows the comparison of sectional area curves of designs with different displacements.

The Sectional Area Curve is made up from a number of sectional areas. Most are spaced equally along the datum waterline with additional half stations at each end. You can specify the number of sections used to display the curve by choosing the appropriate Sectional Area Curve Stations button in the Preferences dialog.

To display sectional area values along the curve, simply click on the movable crosshair and move it to the required position. The station position and current value for the curve will be displayed in the bottom left hand corner of the window.

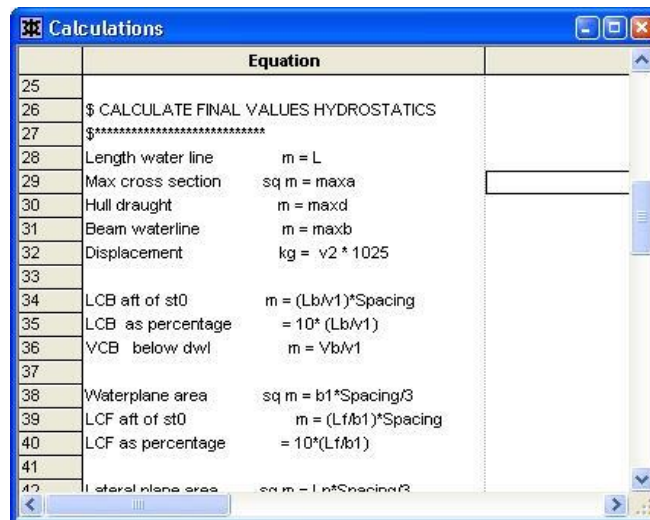
Calculations Window

Modeler has two ways of performing calculations on the model:

- Calculations from the Data menu
- Calculations in the Calculations Window

In this section the Calculations window will be discussed. For more information on the calculations available from the data menu, see the calculations section.

In the Calculations window, Modeler is capable of evaluating calculations entered by the user and solving them in a form much like a simple programming language. These calculations may be simple arithmetic expressions, or may include variables that are pre-calculated by Modeler. Formulae, together with relevant comments, are displayed on the left, with the result of each calculation displayed in the second column. You can also add comments (e.g. units of the result) in the right hand column.

**Note:**

- Although the calculations window allows user customisable calculations, more accurate results can be achieved by using the calculations from the data menu. This is recommended in case there is no need for customised calculations.
- The calculations sheets provided with Modeler are examples only. They should be checked thoroughly before use in your office, and your attention is drawn to the conditions of use outlined in the disclaimer at the front of this manual. Also ensure that you are using the Calculation sheet consistent with your current Units setting.

Expressions

Expressions use standard arithmetic notation and are evaluated based on the level of precedence of their operators, and their left to right ordering. Precedence levels determine the order of computation of expressions. For example, a multiplication (level three) would be computed before an addition (level four). The operators and their precedence levels are:

()	PARENTHESES	Level one
^	EXPONENTIATION	Level two
*, /	MULTIPLICATION, DIVISION	Level three
+, -	ADDITION, SUBTRACTION	Level four

Statement Syntax

Syntax for each statement should be similar to the following:



Built-in Variables

Built-in variables contain information about the volume enclosed by the surfaces being designed, and the plane specified by the Datum Waterline (DWL). Once the DWL has been positioned and the Frame of

Reference set up correctly, Modeler determines the waterline length and calculates data for 13 immersed stations (stations 0 to 10, plus half stations 0.5 and 9.5), in addition to several general variables. Station 0 is the forward-most station and Station 10 the aft-most. The variables available to be included in calculations are:

SPACING	Immersed Station Spacing
MAXA	Maximum Immersed Sectional Area
MAXB	Maximum Waterline Beam
MAXD	Maximum Draft
STAT0..STAT10	Immersed Station Position
SA0..SA10	Immersed Sectional Area
DR0..DR10	Draft
WLB0..WLB10	Waterline Beam
IGIR0..IGIR10	Immersed Girth
TGIR0..TGIR10	Total Girth
CAH0..CAH10	Transverse Centre of Immersed Area
CAV0..CAV10	Vertical Centre of Immersed Area (above DWL)

Half stations at 0.5 and 9.5 have the following variable names:

STAT0.5	Immersed Station Position at position 0.5
STAT9.5	Immersed Station Position at position 9.5
Likewise for the fields :	SA, DR, WLB, IGIR, TGIR, CAH, CAV.

Note:

- The station positions used for the volumetric analysis of the hull in the calculations are completely independent of the station positions set up by the Design Grid function from the Data Menu. Modeler determines the position of the STAT0 to STAT10 variables, and therefore the SPACING variable, purely on the position of the Datum Waterline you have specified.
- The calculations performed in the Modeler calculations window use 13 sections. The Hydrostatic and Area calculations available from the data menu use a number of sections that is dependent on the precision setting in Modeler, where for example medium uses 50 sections. The calculation results will be different each time a different number of sections are used. The hull shape is approximated most accurately at highest precision (200 sections).

Built-in Functions

Built-in functions provided are:

PI	Pi
SIN(x)	Sine of x
COS(x)	Cosine of x
TAN(x)	Tangent of x
LN(x)	Logarithm to base e of x
ARCTAN(x)	ArcTangent of x
SQRT(x)	Square root of x

Calculation Units

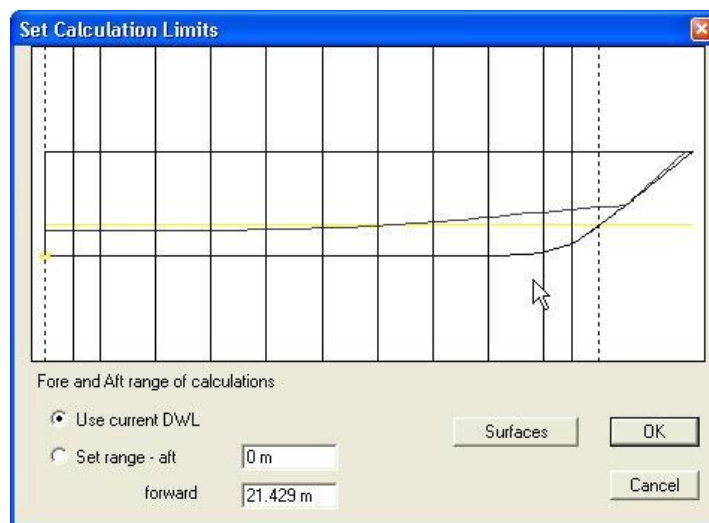
Calculations are solved in the current units as set by the Units function in the Data Menu. Please ensure that any expressions used are correct for the units in use.

Solving Calculations

To open and solve an existing calculations sheet

- Select the Calculations window.
- Select Open Calculations from the File Menu.
- Select a file of calculations to open from the disk. (Samples are provided in the Modeler\Calcs directory.)
- Click Open and the calculations sheet will be opened and displayed in the calculations window.
- Choose Solve Calculations from the Data menu.

A dialog appears that allows you to specify a longitudinal range over which to perform calculations. The dialog will have the Use current DWL option selected. If you click the OK button, calculations will proceed using the current waterline length of the hull.



If, however, you select the Set range option and specify new fore and aft limits for the analysis, the calculations performed when you click OK will be restricted to this zone of the hull. The Simpson's stations created by Modeler are displayed on the screen as a visual check. Note that calculations using Simpson's stations are approximate. While these are usually sufficient for initial design checks, for accurate calculations you should use Stability, which uses more accurate integration and allows a much larger number of stations to be used.

Note: When you are setting the range you may refer to stations on the grid directly by typing S followed by the number of the station you want. For example, if S3 is typed in, the longitudinal position for Station 3 of the grid will be automatically substituted.

When you click OK, Modeler will evaluate the arithmetic expressions in each statement of the calculation window and calculate the result, which is displayed in the second column.

Modeler also assigns the result to the variable name on the left, so that the result may be referred to in later statements.

If a division by zero has been found, NAN (not a number) will be displayed in the results column. If there is an error in your entered expressions, an error message will be displayed. Locate and correct the error and solve again to continue.

You may insert comments into your calculations by preceding a comment line with a \$ symbol. This line will then be ignored by the calculations routines.

There is no practical limit to the size of the calculations sheet that can be solved. To enable you to see all of a large sheet of calculations you may use the scroll bar on the right hand side of the Calculations window.

Saving Calculations

To save a calculations sheet

- Select the Calculations window.
- Select Save Calculations or Save Calculations As from the File menu.

If you choose Save As you will be prompted to enter a name for the new calculations sheet.

Modeler Settings

Modeler has a number of settings that give you control over how the program operates and displays results:

- [Modeler Preferences](#)
- [Color and Font](#)

Modeler Preferences

Modeler has a number of global preferences, which affect how the program works. These may be changed by selecting Preferences from the Edit menu.

Preferences

Control Point Size

☐ Small

☒ Normal

☐ Large

Control Point Window

☒ Row then Column

☐ Column then Row

Sectional Area Curve Type

☒ Area

☐ Area / Maximum Area

☐ Area / Volume^(2/3)

Graph Type:

☐ Line

☒ Area

☐ Point

Sectional Area Curve Stations

☒ Auto ☐ 25

☐ Grid ☐ 39

Trim Engine Version

☐ V13.xx

☒ V14.00 onwards

Refresh Options

☐ Update Current Window Only

☒ Update All Windows

☐ Curve of Areas Window Auto Update

Contour Tolerance

Curvature Porcupine Size %

Outside Arrow Size %

No. of Undo Levels

Sub-menu Length

☒ Use surface colour for drawing parametrics and markers

☐ Use surface colour for drawing other contours

☒ Show grid contours on all surfaces (not just hull surfaces)

☐ Show marker error with color gradient

☒ Clear cutting surfaces list for new surfaces

OK Cancel

The different settings are described below:

Control Point Size

The control point size may be set to either: huge (6 pixels), large (4 pixels), or small (2 pixels).

Graph Type

The Sectional Area Curve graph may be set to be a filled graph, a line graph, or a graph made up of discrete data points.

Sectional Area Curve Type

The sectional area curve may be set to graph:

- Section Area.
- Section Area / Maximum Section Area.
- Section Area / Volume^{2/3}.

Control Point Window Column Ordering.

In the Control Points window it is possible to view the points listed by row then column or by column then row. The reason for doing this is to simplify the pasting of single rows or columns into the Control Points window.

Sectional Area Curve Stations.

The sectional area curve may be set to use 25 or 39 stations or use the "auto" setting. The "auto" setting will increase the number of sections when you increase the [Surface Precision](#). Medium precision uses 50 sections and highest uses 200.

For complex hulls it is better to use more stations, while for simple hulls it will be faster to calculate fewer stations.

Refresh Options

Depending on the speed of your computer, select between: Update Frontmost Window Only; Update All Windows or Update All Windows Dynamically. On a slow computer, Update All Windows Dynamically may be too slow to use the program effectively.

Curve of Areas Window Auto Update specifies whether the window is updated every time the design is modified. If disabled the window is updated only when the curve of areas windows gets focus. This option should only be enabled if required as it will affect application performance.

Contour Tolerance

This value is used when computing surface contours using Highest Precision. A lower tolerance value will give more precise contours.

Curvature Porcupine Size

This value is used to scale the curvature porcupines displayed on contours.

Outside Arrow Size

This value is used to scale the outside arrows displayed on surfaces.

No. of Undo Levels

This sets the maximum number of 'undos', which are stored and may be set to a value between 2 and 100. Setting too high a value may cause your computer to slow down since this will require considerable memory, the amount of memory used depends on the complexity of the model.

Sub-menu Length

This sets the maximum number of items in the surfaces sub-menus before the More Surfaces... item is displayed.

Use Surface Color for Drawing Parametrics

The parametric surface contours may be displayed in the global parametric contour color or in the color of the surface. This option can be useful for better visualising the design and distinguishing the different surfaces.

Show Grid Contour on All Surfaces

A flag for turning grid contours on/off for internal structure surfaces. If this is OFF then grid contours (sections, buttocks etc) will only be drawn on hull surfaces (not internal structure surfaces).

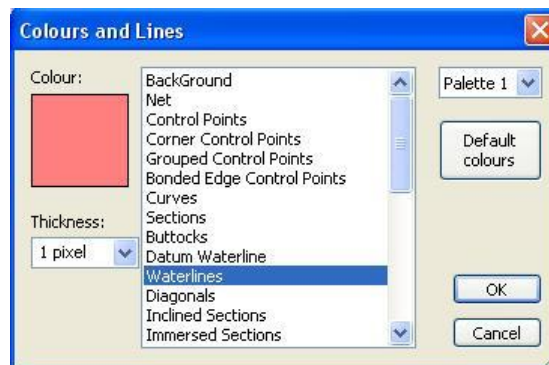
Color and Font

The color and font used in the windows may be changed with the Color and Font commands in the View menu.

The Color command allows you to set the color of lines, controls, and graphs.

Remember to always be careful when using color. It is very easy to get carried away with bright colors and end up with a garish display that is uncomfortable to work with. In general it is best to use a neutral background such as mid Gray or dull blue and use lighter or darker shades of a color rather than fully saturated hues.

To change the colors, select Color from the View menu.

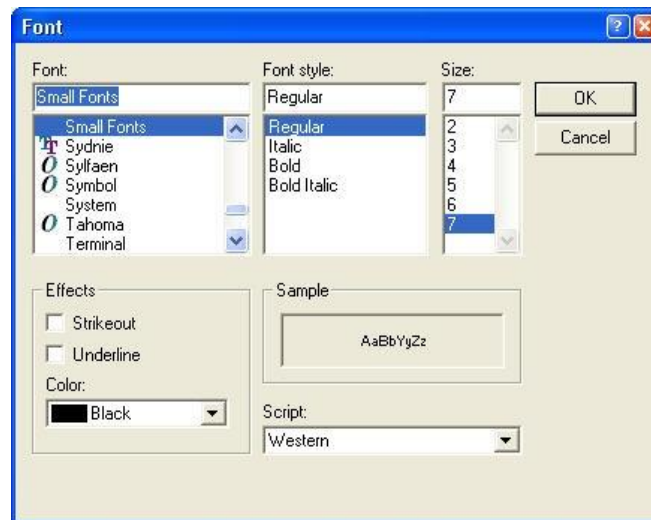


From the scrollable list, select the item whose color you wish to change. The item's current color will be displayed on the left of the dialog. To change the color, click in the box and select a new color from the palette. To change the thickness of a line select the contour type in the list and then choose the thickness of the line from the "Thickness" drop down control.

The colors of items in the four design views are all consistent. The colors used in the graph window may also be changed with the color command.

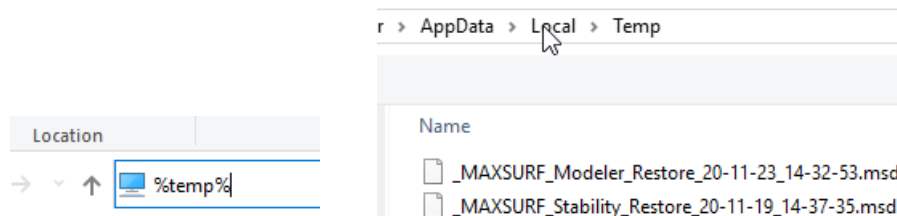
The Font command allows you to set the size and style of the text in the current window; the font may be independently set for all the windows.

Select Font from the View menu and select the desired font, style and text size.



Auto-saved backup recovery files

Recovery files are automatically saved. These files are typically:

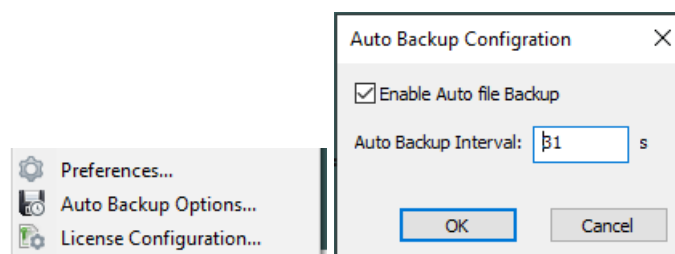


Typical location of auto-saved backup files

They are time-stamped in the format yy-mm-dd_hh-mm-ss and located in the first folder from the following list which to which the user has write permission:

1. user's temp folder (%TEMP%); typically C:\Users\USER.NAME\AppData\Local\Temp
2. the application folder; typically C:\Program Files\Bentley\Offshore\Maxsurf CONNECT Edition Vxx
3. the user's Documents folder; typically C:\Users\USER.NAME\Documents
4. a folder selected by the user

The auto-save of backup files is implemented in Multiframe, Modeler and Stability. If the model has been modified, this feature automatically saves the model after a specific time interval. You can adjust the auto-save interval (in seconds) or turn the feature off completely. The default auto-save interval is 120s. If necessary, these settings may be modified, through the Edit | Auto Backup Options... dialog as shown below:



Chapter 5 Using Modeler

The previous chapter covered a lot of the functions that are available to you when you are working in any one of the windows. This chapter will discuss specific functions in Modeler to manipulate the shape of the design, do calculations and work with external data.

[Working with Markers](#): Especially when you are planning to fit a NURB surface to a set of data points, this is a useful section to read.

[Working with Curves](#): The characteristics of the surfaces used in Modeler, commands that apply to surfaces such as trimming and bonding.

Working with Control Points: Tools available for manipulating a control point or group of control points to achieve the desired surface shapes.

Calculations: How to calculate hydrostatics, girths and surface areas in Modeler.

[Surface Fitting](#): Extensive guide covering the different tools that Modeler offers to fit NURB surfaces or a TriMesh surface to existing design data.

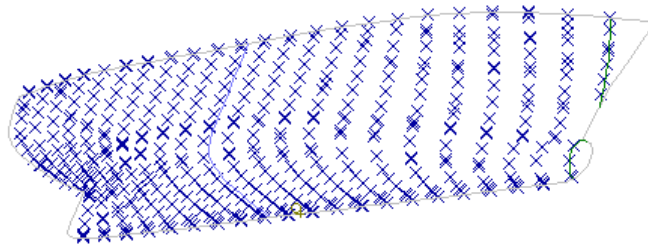
Using Parametric Transformation: Numerical modification of critical design parameters. This function is available in Modeler Advanced only.

[Input of Data](#): Different types and methods of input data.

[Output of Data](#): Different types and methods of output data.

Working with Markers

Markers are reference marks displayed on the screen. They may be offset data that has originated from an existing hull or simply limiting dimensions that need to be visible while developing a design.



See:

- [Working with Markers – Concepts](#)
- [Working with Markers – Procedures](#)
- [Generating Markers for Developable Surfaces](#)

Working with Markers – Concepts

Markers are purely graphical and have no effect on the calculation of surfaces. The maximum number of markers you may use is 30000. In practice you will usually use many fewer than this.

Markers may either be read in with a design from Fitting, read directly from a text file or by [Importing DXF Markers](#). Markers can also be entered manually using the Add Marker function in the Edit menu or copied and pasted into the markers table from another application such as a spreadsheet. They are stored as three-dimensional points in space and may be added, moved or deleted. All marker positions are stored in the design file whenever you save a design.

Each marker can be associated with a station in your design. Usually if you are importing markers that form the offsets of an existing design, you will set up the grid in Modeler to match the station spacing of the original design, and then set the station number of each marker to match its station in the offsets table. See

Move Selected Markers on page 375 for information on automatically generating a grid from a set of markers.

Markers may also be associated or linked to a specific surface and location in that surface. Once linked to a surface, the marker takes on the color of that surface. This function may be controlled with "Use Surface Color for Drawing Parametrics" control in the Edit | Preferences dialog. If this control is turned off, markers that have been linked to a surface will be displayed in the same color as the surface parametrics. This is not only useful for display purposes but can also be used to [Measuring Surface Error](#) and utilise for [Fitting a NURB Surface to Markers](#).

Next: see [Working with Markers – Procedures](#).

Working with Markers – Procedures

In this section:

- [Displaying Markers](#)
- [Marker Selection](#)
- [Adding Markers](#) one by one on screen, in the markers window
- [Pasting Markers from Excel](#), paste a large number of markers from an excel spreadsheet.
- [Deleting Markers](#)
- [Modifying Markers](#)
- [Sorting Markers](#)
- Linking Markers to a Surface
- [Linking Markers to a Section](#)
- [Saving Markers](#)
- [Reading Markers Files](#)

Also see

[Marker Menu](#) on page 374; summary of commands in Markers menu.

[Generating Markers for Developable Surfaces](#) on page 92

[Surface Fitting](#) on page 221

[Mesh Surface](#) on page 235

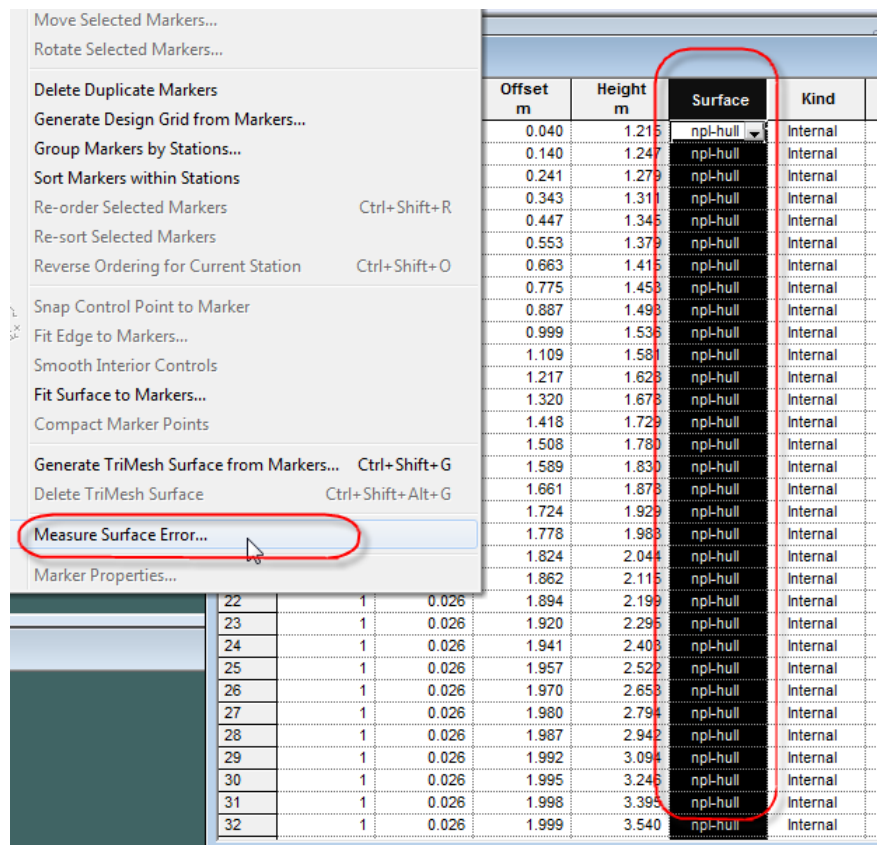
Displaying Markers

To turn all markers on or off

- Select the Markers function from the Display menu.

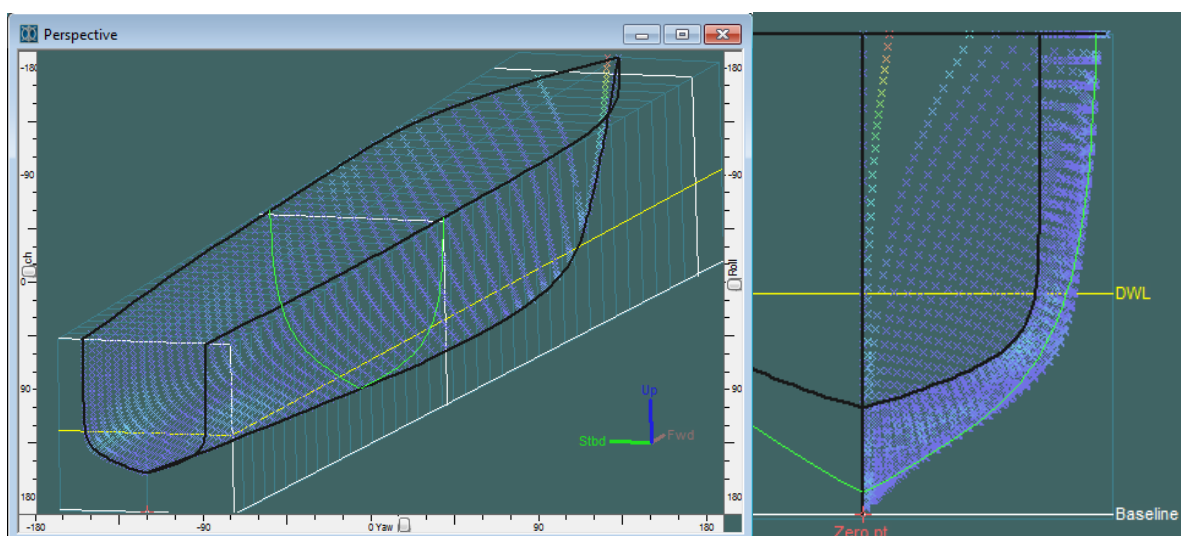
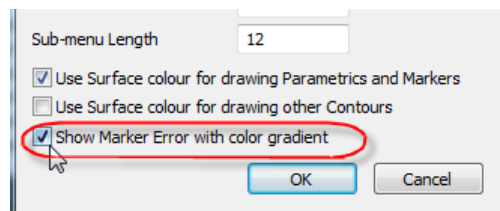
You can choose to display all markers or no markers, or you may choose to only show those markers whose station number is the same as the current station.

Other Markers display features, which are designed to assist when fitting surfaces to large Marker datasets (for example from laser scans), are available. The first is an option to color-code the Markers by their distance from their assigned surface. Ensure that all the markers of interest have the correct surface associated with them (Markers table, or Markers properties); then select Measure Surface Error from the Markers menu:



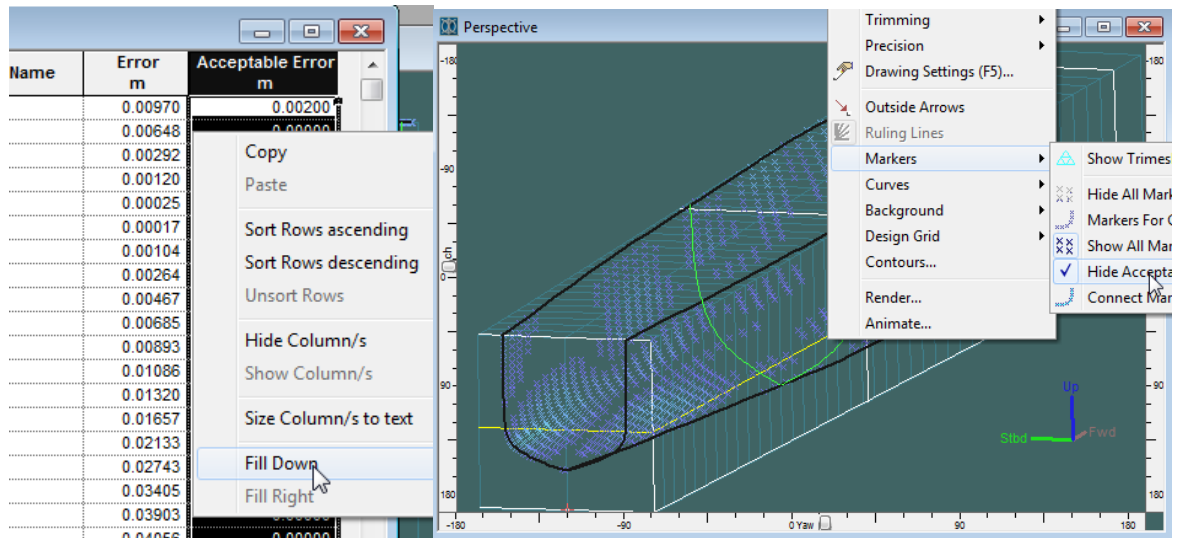
Measure the distance of the Markers from their associated Surfaces.

Then in the Edit Preferences dialog, select "Show Marker Error with color gradient"



Markers are now color-coded according to the distance from the surface (dark blue: closest; red: furthest away)

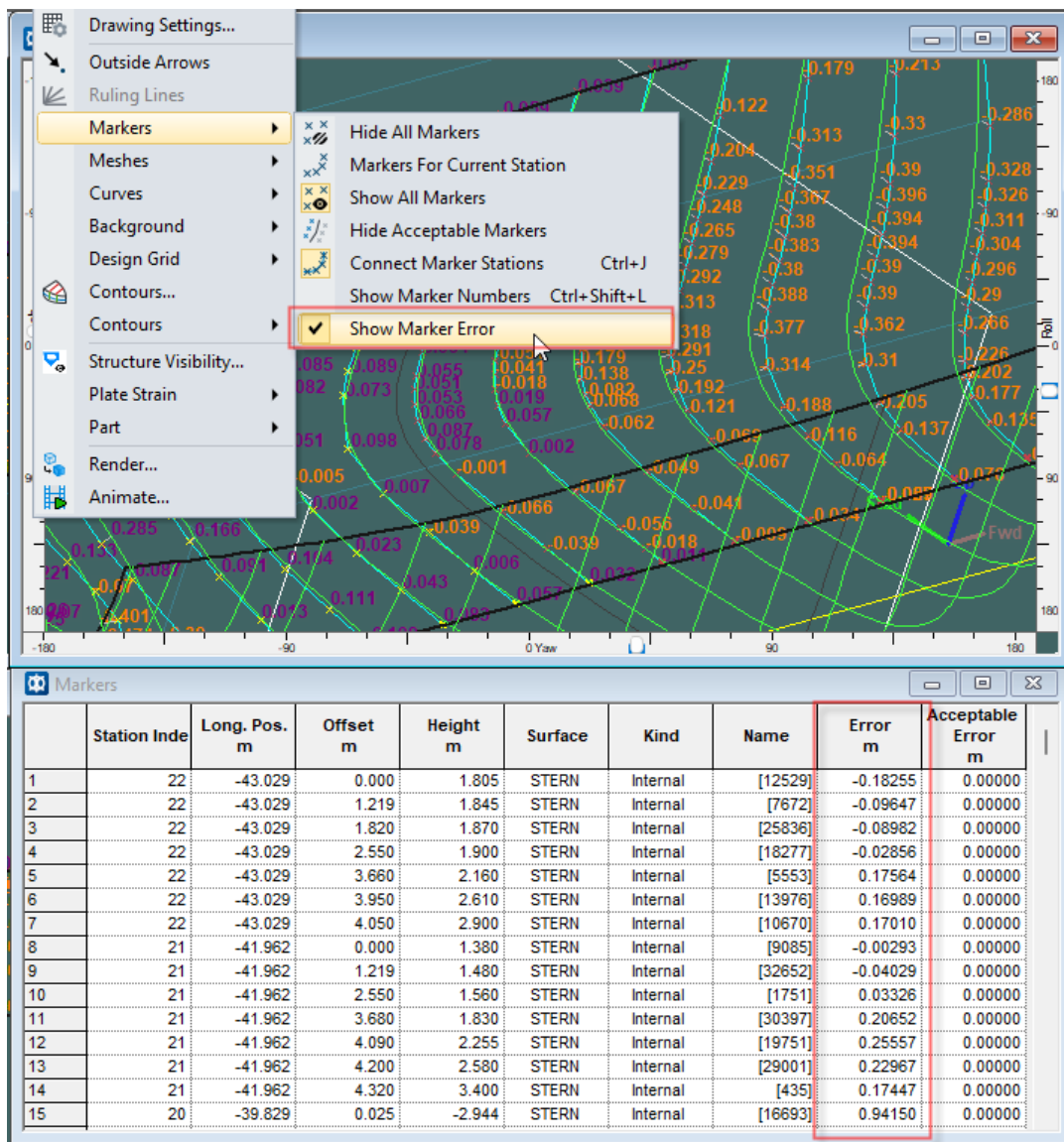
The second option is to hide markers that are below an acceptable distance from the surface. Enter the acceptable distance in the Markers table (the fill-down option can be convenient to do this). Then choose "Hide Acceptable Markers" from the Display | Markers menu:



Hide Markers that are within an acceptable distance of their associated surface.

The distance a Marker is from its associated Surface is signed. The value is positive if the vector from the Surface to the Marker is in the same direction as the local surface normal, and negative if it is in the opposite direction. If the convention of having the surface outside arrows pointing out is used, then a *positive* error means that the marker is *outside* the vessel and a *negative* error means that the marker is *inside* the vessel.

The value for each marker is tabulated, but may now be displayed on the model view. Markers with negative error are shown in the color used for "Inner sections" and Markers with positive error are shown using the color for "Outer sections".



Distance (error) from Marker to Surface is now signed and can be displayed

Marker Selection

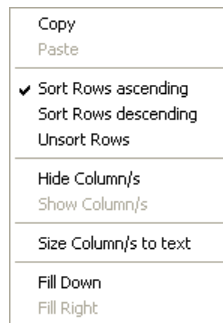
In the design views, markers may be selected in a similar manner to the control points: by simply clicking on them with the left mouse button or by dragging a box around them. Holding down the Shift or Control key while making the selection allows you to add more markers to the current selection or remove previously selected markers from the selection.

Selected markers are highlighted in the highlighting color.

Note:

You may have to lock the surfaces to hide the control points, as selection of control points takes precedence over selection of markers; i.e. if the drag box contains control points, these will be selected rather than the markers.

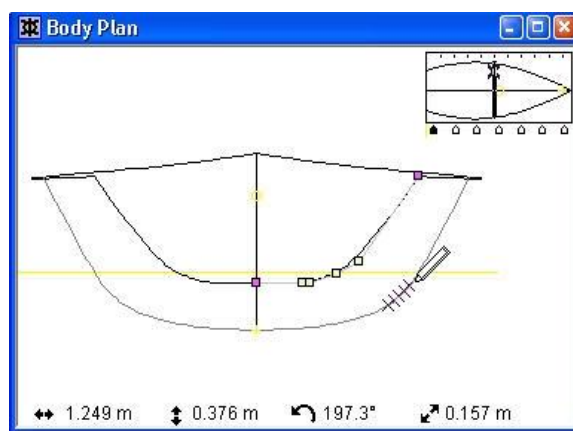
In the Markers table, you may only make a continuous selection. However, data may be sorted by right clicking in the column header and choosing from the pop-up menu to sort ascending or descending based on the contents of that column. The markers may be unsorted by right clicking in any of the column headers and selecting Unsort Rows from the pop-up menu:



Adding Markers

To add a marker to a drawing window

- Select Add Marker from the Marker menu or use the Ctrl+M shortcut key. A pencil cursor will be displayed.



- Click the mouse at the position of the new marker.

Use the position indicators to place the marker at the required position. The added marker will be associated with the current station as displayed in the control box.

Markers may also be placed graphically in the Plan or Profile views. For these 2D views, the third dimension takes the value of the last entered marker. For example, if you wanted to enter markers on the 2m waterline, first switch to the Plan view. Add a marker in the Plan view; double click the marker so that you can specify its height as 2m, then any subsequently added markers (in the Plan view) will have their height at 2m. Markers on Buttock lines may be entered in a similar fashion in the Profile view.

Alternatively, a marker can be added directly into the Markers window. To do this

- Select the Markers window
- If there is no Markers file yet, select File | New
- Highlight a row by clicking in the far left hand column.

	Station Inde	Long. Pos. m	Offset m	Height m	Surface	Kind	Name	Error m
1	6	-4.291	1.031	0.180	None			--
2	6	-4.291	1.072	0.212	None			--
3	6	-4.291	1.104	0.244	None			--
4	6	-4.291	1.155	0.281	None			--
5	6	-4.291	1.187	0.336	None			--

- Select Add Marker from the Edit menu.
- Enter the number of markers you wish to add and hit Return.

Rows will be added after the row selected, with cells initialised to zero.

Note:

You must have a design open before you can add markers.

Pasting Markers from Excel

If you have for example an excel spreadsheet with for example 1284 rows with offset points:

- Make sure the columns in Excel are sorted the same as the marker window i.e. Long pos. Offset, height.
- Copy all rows in Excel
- Edit | Add Marker and specify the number of markers you wish to add (1284)
- Click in row number 1 cell in the Long Pos. column in the Markers window
- Paste
- Check that all markers have been imported.

Deleting Markers

To delete a marker from a view window

- Select the marker(s) to be deleted
- Select Delete Marker from the Marker menu or use the Delete key on your keyboard
- Click yes to confirm you want to delete the selected markers.

The marker will be deleted.

To delete a marker from the Markers window

- Select the Markers window
- Select the row(s) of the marker(s) that you wish to delete
- Select Delete Marker from the Marker Menu or use the Delete key on your keyboard
- Click yes to confirm you want to delete the selected markers.

Rows will be deleted from the Markers window.

Modifying Markers

The co-ordinates in the Markers window may be edited to move a marker in space. To edit a marker

- Click on a cell in the Markers window.

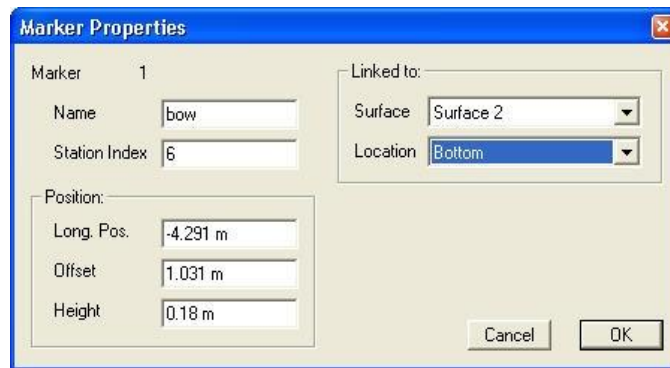


	Station Inde	Long. Pos. m	Offset m	Height m	Surface	Kind	Name	Error m
1	6	-4.291	1.031	0.180	None			--
2	6	-4.291	1.072	0.212	None			--
3	6	-4.291	1.104	0.244	None			--
4	6	-4.291	1.155	0.281	None			--
5	6	-4.291	1.187	0.336	None			--

- Enter the changed Station Number, Longitudinal Position, Offset, or Height.

Alternatively, to edit a marker directly from a view window

- Double click on a marker in one of the drawing windows.



A small dialog box will appear containing data about the marker. You may select and edit any of the marker's co-ordinates. When you click OK the marker will be updated to the new position.

You may also set the data for a number of markers together:

- Select the markers in one of the drawing windows.
- Select Properties from the Markers menu.

If you have selected several markers, those fields that contain exactly the same data for all the selected markers will be displayed with the common data. If the data differs, the field will be left blank. Editing the data in any field will update all the selected markers. This can be useful for making sure that all the markers on one station have identical longitudinal positions, or for specifying which section they should be linked to.

Sorting Markers

When to sort markers

The order in which markers are sorted along a section line is important when:

- the marker data is used in Fitting standalone to automatically generate a surface
- when you want to use the Display | Connect Marker Stations as a reference during manual surface fitting tasks
- When Mesh Surface

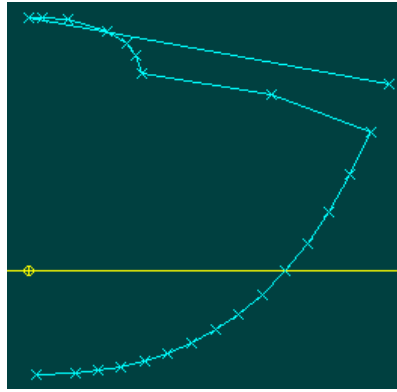
The order of the markers is irrelevant when you are using the Fitting a NURB Surface to Markers command in which case you can skip this section.

Sorting Markers - Concept

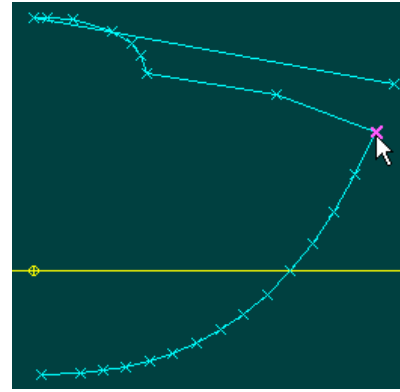
Markers that have been assigned to the same Station Index (see [Linking Markers to a Section](#)) can be sorted for each station using the Sort Marker Stations command from the Markers menu. This uses a nearest neighbour sort, which works in most instances, but can be confounded if the section shape is complex and/or the spacing between markers varies significantly. To view the connectivity (and ordering) of the markers turn on the Display | Connect Marker Stations option: Ctrl+J, . This is best done in the Body Plan view, with the Display | Show Markers for Current Station option selected ().

Sorting Markers - Procedures

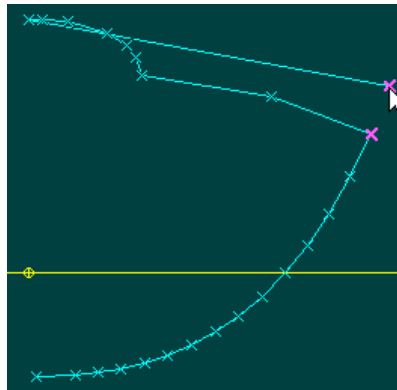
To re-order a range of markers, select them in the new desired order (from the keel outwards towards the deck edge (or deck centre line) and select Markers | Re-order Selected Markers (Ctrl+Shift+R). If the new order is the reverse of what is required, simply repeat the command without changing the selection, and the order will be reversed.



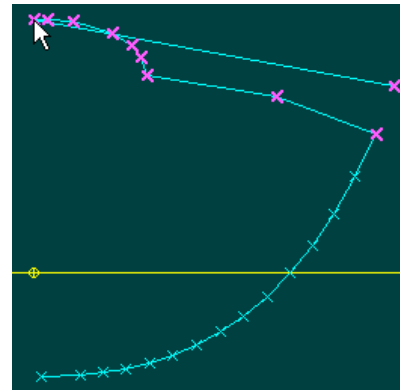
Markers incorrectly ordered



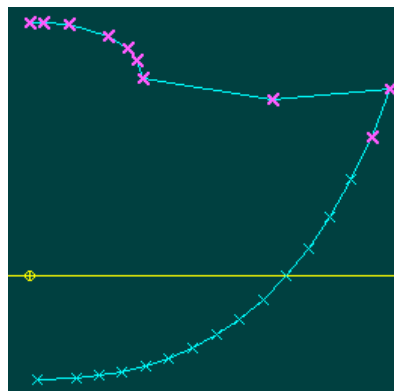
Start selecting Markers in correct order



Start with marker that should be closest to keel in the group

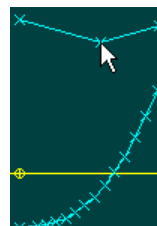
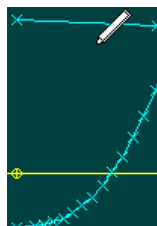


Markers selected in correct order. Use Re-Order Selected Markers command to re-order



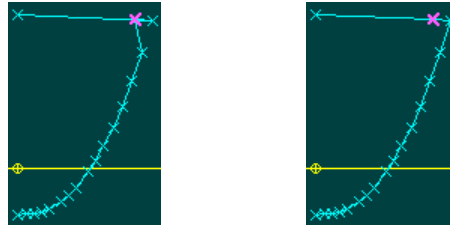
Points are now ordered in the selection order, if this is the inverse of what is required, simply repeat the Use Re-Order Selected Markers command without changing the selection.

When a marker is added, it will be inserted to so as to minimise the increase in section length. This normally ensures that it is inserted in the correct location.



Keyboard tools for Sorting Markers

A single marker can be moved along the ordering by using Ctrl+Cursor keys; left and down keys move the point one positions towards the start of the ordering whilst up and right keys move the point one position towards the end of the ordering; the Home key makes it the first point and the End key makes it the last point:



Before and after:

Moving the selected point one position towards the start of the curve corrects the marker ordering

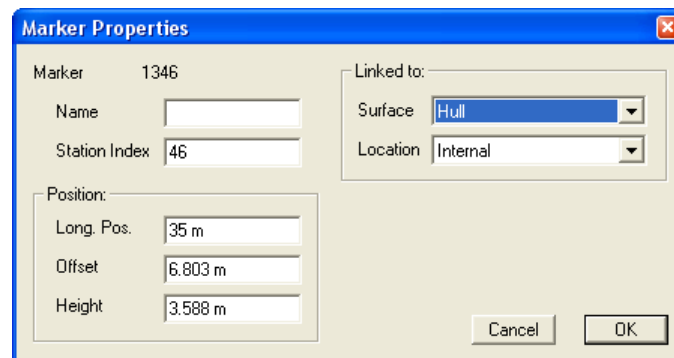
Re-Sort selected markers

If the marker data is evenly spaced and regular you may wish to use the Re-Order selected markers command. This command differs to re-order selected markers as it is an automatic process. This command will automatically reorder the markers in each station starting from the centreline lowest via a nearest neighbour approach. If you have a number of stations then you can select all the markers across all the stations and run this command. Modeler will automatically reorder the markers according to the station they are associated with.

Linking Markers to a Surface

Unless you have used the Generating Markers for Developable Surfaces to create your marker data, you will have to link the markers to the surface manually. To do this,

- Select the marker in any view window
- Select Marker | Marker Properties (or use the Properties Pane)
- And use the drop down selection list to select the surface you wish to link the marker to:



After a marker has been linked to a surface, its color changes to the surface color.

Location

The location of the marker on the surface can either be:

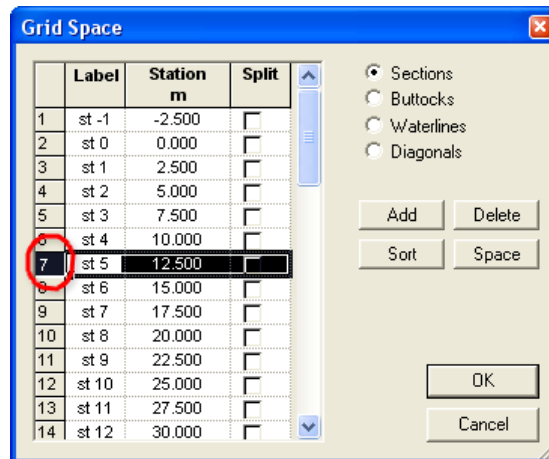
- Internal (in the middle of the surface)
- Edge (either Top, Bottom, Fwd or Aft edge)
- Corner point (either Top/Aft, Bottom/Aft, TopFwd or Top Aft)

For most surface fitting cases it is not necessary to set the Location for each marker and just linking it to the surface will suffice. Once markers have been associated with a particular surface edge, you can fit the surface edge to its markers by simply selecting a control point in the edge and selecting Markers | Fit edge to markers (see below). This is very useful as it allows you to easily change an edge's properties (number of

control points or stiffness) and refit the edge to obtain the best possible fit to the markers without having to always reselect the markers. It also makes it very easy to add or remove markers to those used for the edge fitting.

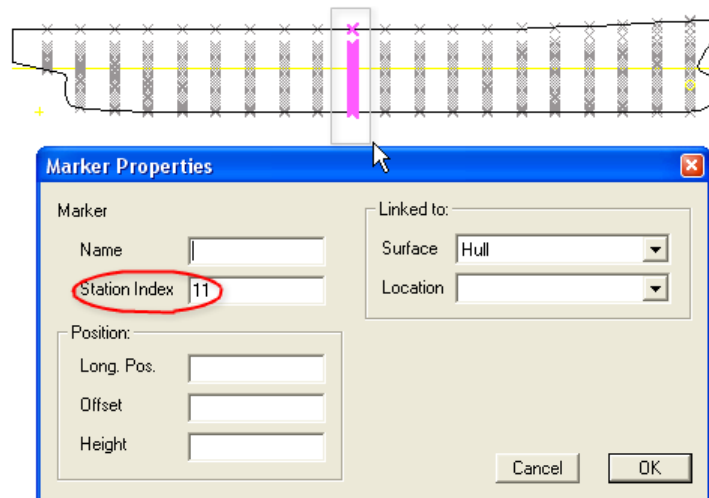
Linking Markers to a Section

In order to be able to compare the marker data with the surface sections, Modeler allows you to link markers to a section using the "station index". The station index is the number to the left of the Design Grid dialog:

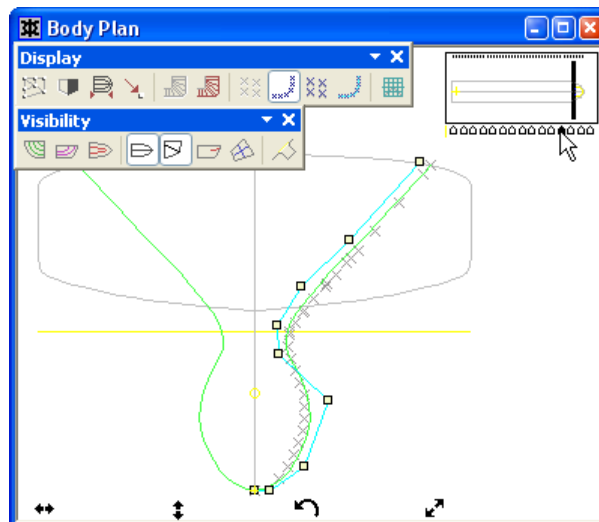


St 5 has station index number 7.

Linking the markers to each section is best done in Profile or Plan view using a selection window:



A group of markers that have been linked to one section is called a "Marker Station". Once a marker has been linked to a section, you can display each section and the associated markers (Marker Station) in the body plan view:



Displaying the markers for each section makes manual fitting of a surface to a set of markers a lot easier.

Automatically linking the Markers

Manually linking the markers to each section can be a tedious process. A very powerful tool to automatically link the markers to each section is the [Generating a Grid from Markers](#) command.

When working with point clouds of markers "Assign Markers to Stations" function can be used to automatically assign markers to stations if they lie within a user specified distance of a station.

Also see

Display | [Markers](#) on page 390
[Body Plan Window](#) on page 43

Saving Markers

When the Markers window is frontmost, you save the Marker data as a txt file:

- File | Save Markers

Reading Markers Files

It is possible to read a text file of markers into Modeler. This text file may be created from the Save Markers or Save Offsets commands in Modeler or created in a spreadsheet, word processor or other application.

To open a marker file:

- Switch to the Marker window
- Select File | Open and specify the text file you would like to open

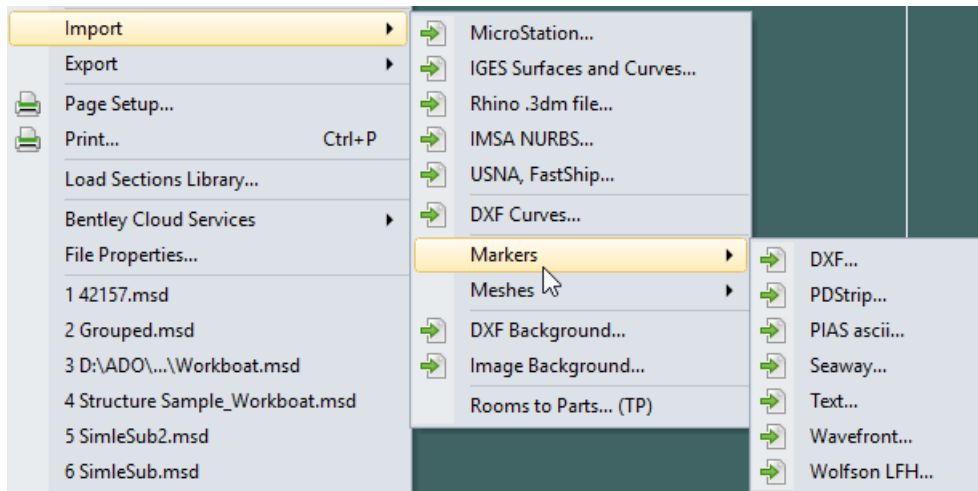
The format for the Markers file is tab delimited text. Each coordinate is separated by a Tab character, and each line ends with a Carriage Return character. An example would be:

1	2.33	-0.2	1.2
1	2.33	0.1	2.4
2	4.66	-0.7	1.0
3	7	0.3	1.4

Column 1 is the station number with which the marker is associated. This controls when the marker is displayed and when it is hidden when the Show Markers for Current Station command is active. Column 2 is the longitudinal position from the zero point, Column 3 the offset from the centreline and column 4 the height above the zero point.

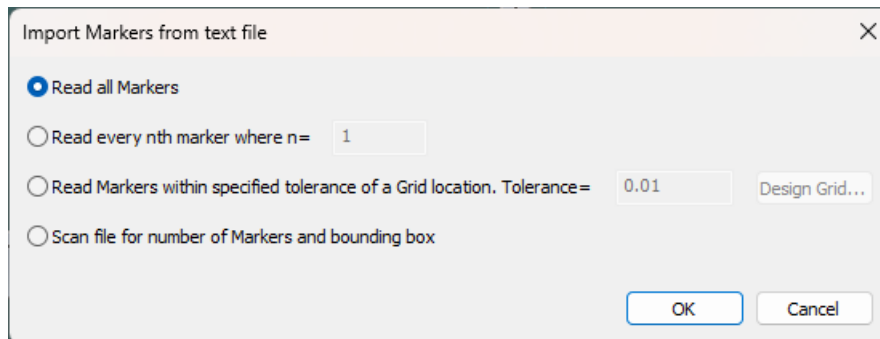
Importing Markers

Markers can be imported from a number of formats. See the File | Import | Marker menu:



Marker Import, supported file formats

Text and Wavefront options provide additional options which are useful when working with large point clouds. Modeler supports up to 64k Markers in a single model so it may be necessary to down-sample the point cloud.



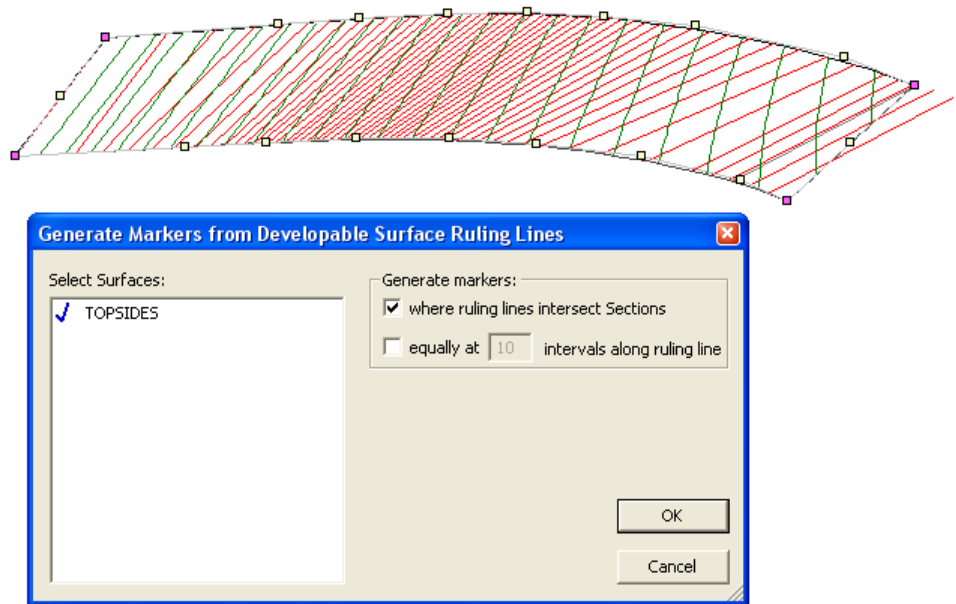
Options when loading Markers from a text file

Reading Markers from other file formats will read up to the maximum allowable number of Markers.

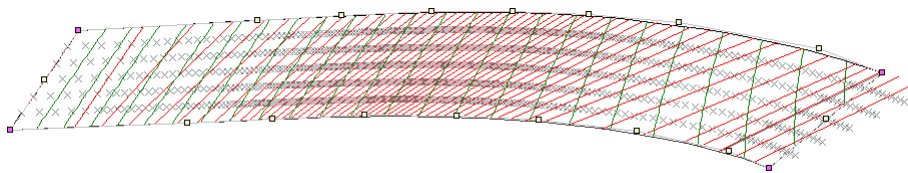
Please refer to the [Importing DXF Markers](#) section for more details on reading Markers from a DXF file. Also see: [Adding Markers](#).

Generating Markers for Developable Surfaces

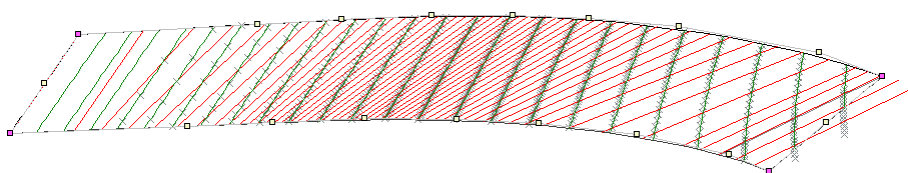
To generate markers that describe the shape of the developable surface you have to first set the surface type to Developable. This will create ruling lines from the surface edges from which the markers can be generated using the Markers | Generate Markers for Developable Surfaces command. This command is only available if there is at least one surface in the design with the surface type set to developable surface (see [Surface Types in](#) on page 107).



Markers are calculated for the surfaces selected in the list on the left of the dialog (only developable surfaces are listed). Markers can be generated where the ruling lines intersect stations previously defined in the Design Grid dialog, or at equal intervals along the ruling lines. The number of ruling lines calculated depends on the current surface precision setting, but during the generate markers command the precision is increased temporarily to highest.



Markers generated at equal intervals along the ruling lines – this is useful for the automated surface fitting (Markers | Fit surface to Markers) as it ensures that markers are generated even when the rulings are vertical and do not intersect stations.



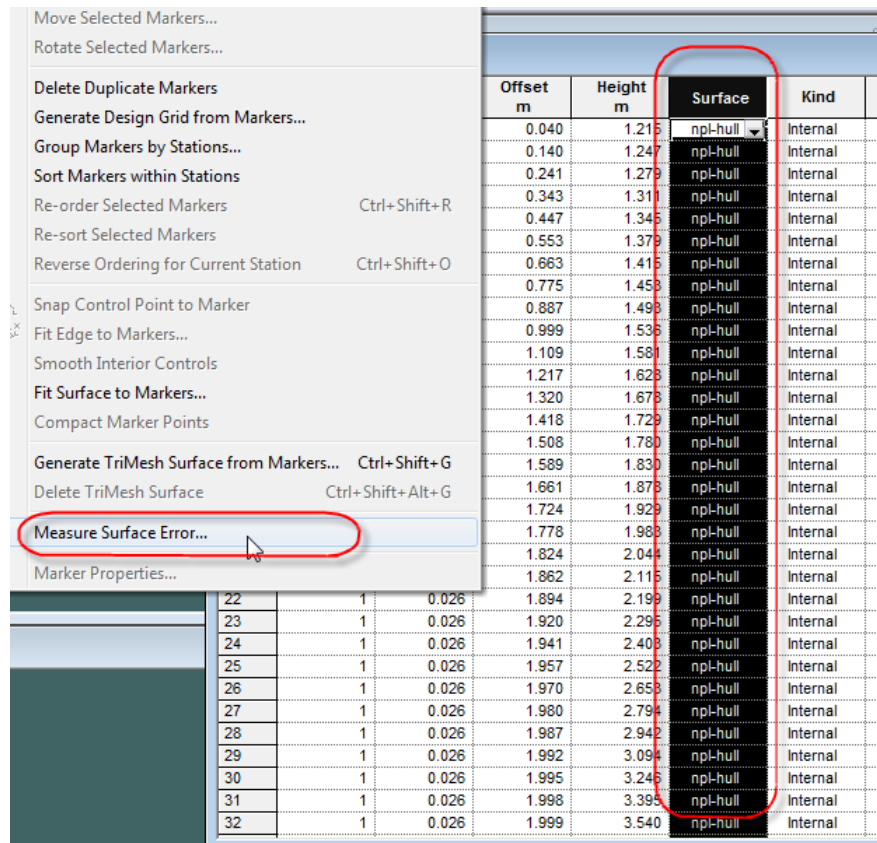
Markers generated where they intersect the sections. This is useful when manually fitting the surface to the markers in the body plan view.

The Markers generated by this function will be linked to the surface and sections automatically.

Marker to Surface Measurements

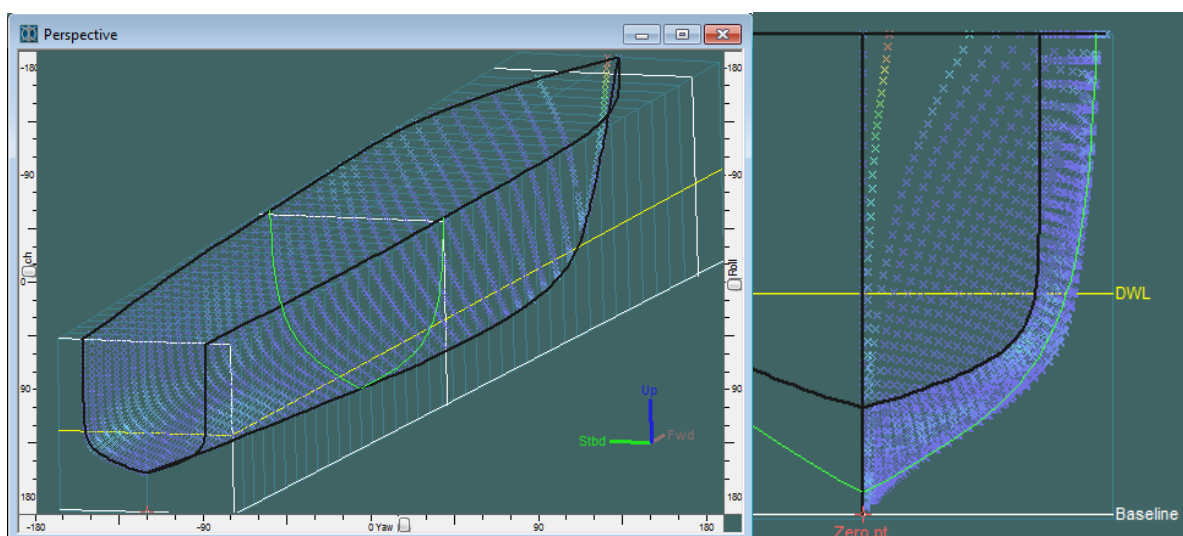
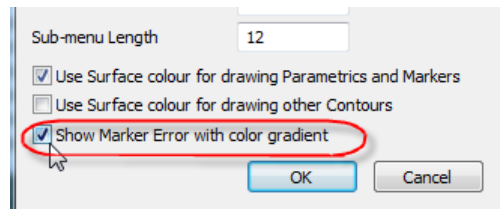
Two features are available which are designed to assist when fitting surfaces to large Marker datasets (for example from laser scans).

The first is an option to color-code the Markers by their distance from their assigned surface. Ensure that all the markers of interest have the correct surface associated with them (Markers table, or Markers properties); then select Measure Surface Error from the Markers menu:



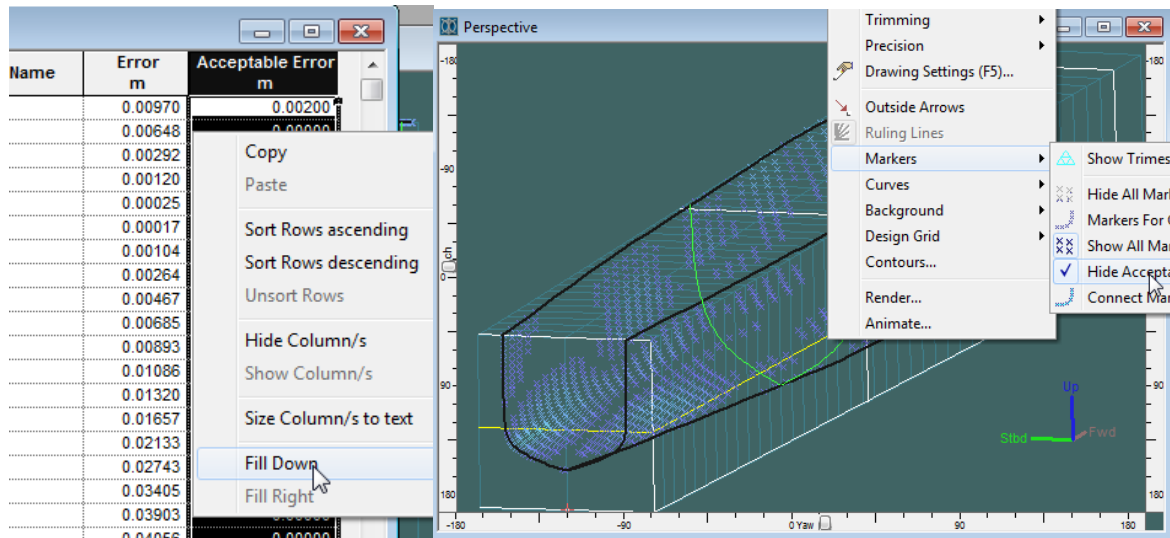
Measure the distance of the Markers from their associated Surfaces.

Then in the Edit Preferences dialog, select "Show Marker Error with color gradient"



Markers are now color-coded according to the distance from the surface (dark blue: closest; red: furthest away)

The second option is to hide markers that are below an acceptable distance from the surface. Enter the acceptable distance in the Markers table (the fill-down option can be convenient to do this). Then choose "Hide Acceptable Markers" from the Display | Markers menu:



Hide Markers that are within an acceptable distance of their associated surface.

Working with Curves

A Modeler design may contain any number of Curves. Curves are generally used in Modeler to aid in the creation of surfaces as well as trimming of surfaces. In this section the different curve types, operations and uses will be explained:

- [Curve Types in](#)
- [Curve Stiffness](#)
- [Curve Properties](#)
- [Curve Operations](#)

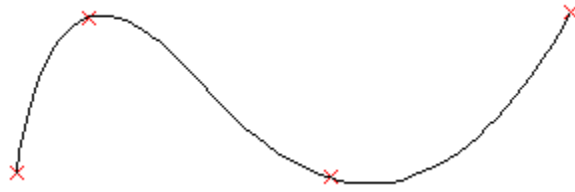
Curve Types in Modeler

All curves in Modeler are Non-Uniform Rational B-Spline curves (NURBS for short). For more information on NURBS see the section in working with surfaces ([NURBS Surface](#) on page 107).

The two main distinctions of curves within Modeler are curves generated from the user entering the position of the control points (termed "control point curves"):



and curves generated from the user entering the position of points through which the curve must pass (termed "data point curves"):

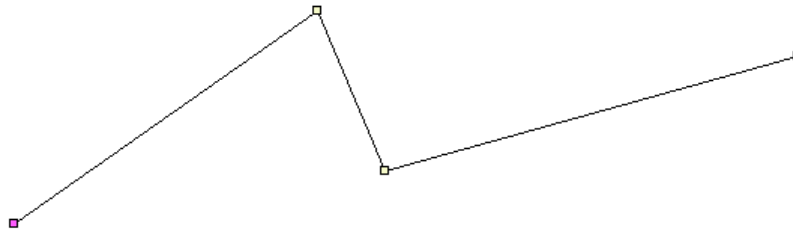


Data points added are actually a special type of marker that retain a link to the curve. So if a marker that is linked to the curve is moved the resulting curve shape will also adapt to the new position.

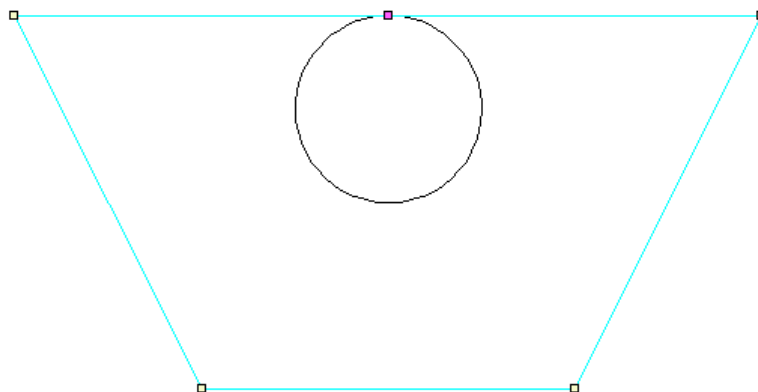
For more information on control points and how they are used to generate the curve shape see Section “Working with Control Points” on page 200.

Three other curve types that can be created with Modeler, that are a subset of the above two curve types, are linear curves (or polylines), circles and arcs.

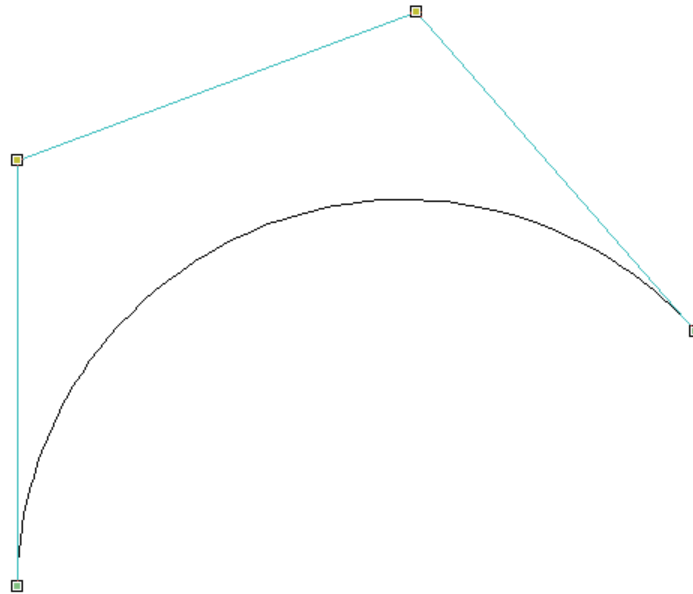
Linear curves are control point curves that have their stiffness (see [Curve Stiffness](#) on page 97) set to 2. In this case the curve is linear between control points and the curve passes directly through each control point:



A circle is defined by its centre point and radius. Modeler will then create a control point curve with the appropriate control point positions and weights to define a circle:



The final curve type is an arc. An arc curve is defined by a centre, start point and end point. The radius of the arc is determined from the distance between the centre point and start point. The angle of the arc is determined by the angle between the lines defined from the centre to the start point and the centre to the end point. Arcs between 0-90 degrees are defined by a 3 point NURB curve, arcs between 90-180 degrees are defined by a 4 point NURB curve. To create an arc choose Curve | Add curve | Arc from the menu or click on the add arc button from the curve tool bar. The three points can be defined either by entering the positions of the three points in the dialog or by clicking the “pick ->” button and graphically picking the points with the mouse.

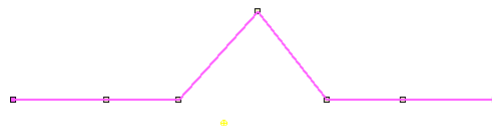


Curve Stiffness

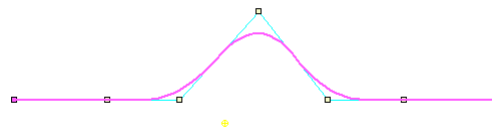
The stiffness of the curve is one of the parameters that defines the curve shape and is analogous to the selection of a different weight of spline when drawing a curve on the drawing board. For more information on stiffness and how it effects a NURBS curve or surface see [Surface Stiffness](#) on page 108.

Below is a series of pictures demonstrating the effect changing the curve stiffness has on the shape of the curve:

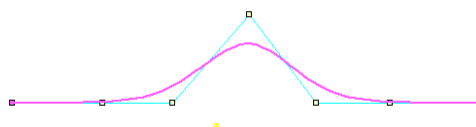
Stiffness 2 (Linear):



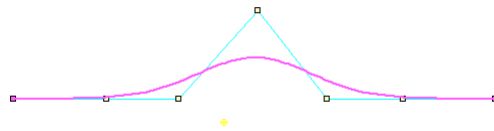
Stiffness 3 (Flexible):



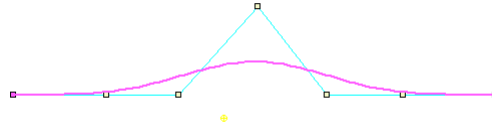
Stiffness 4:



Stiffness 5:



Stiffness 6 (stiff):



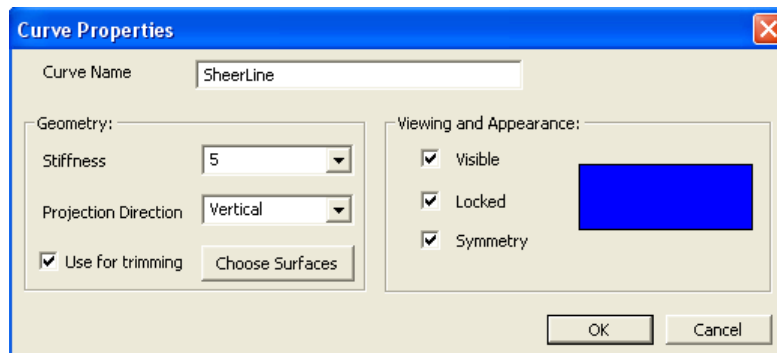
Curve Properties

Each curve that is created within a Modeler design has a unique set of properties associated with it. The Properties command allows you to view and modify these properties.

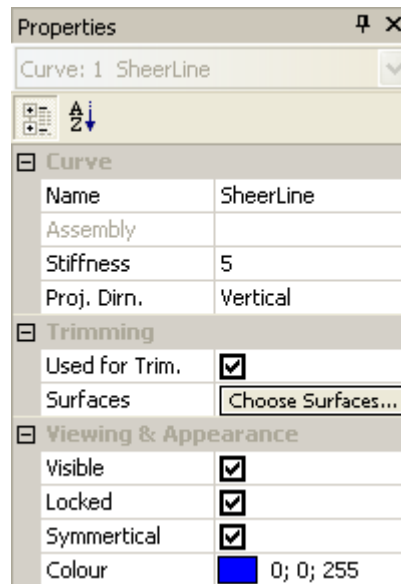
Note:

The properties of all the curves may be quickly viewed and modified in the Surfaces window or by right clicking on the surface from the Assembly Pane or in any view window. When a curve is selected its properties are also displayed in the property sheet.

Properties as viewed in the properties dialog:



And in the property sheet:



Curve Name

This may be any name typed in from the keyboard up to twenty characters long.

Stiffness

The curve stiffness. For more information see [Curve Stiffness](#) on page 97.

Projection Direction

When a curve is used to trim a surface the projection direction is used to determine how the curve cuts the surface. The default projection direction is dependent on the view in which the curve was created. If the curve was created in the body plan view the curve projection direction will be longitudinal, in the plan view, vertical and in the profile view horizontal. If a curve is created in the perspective view then the most perpendicular plane is taken as the projection direction. For more information on projection direction and trimming surfaces with curve see [Trimming Surfaces with Curves](#) on page 99.

Use for Trimming

Clicking on this button in the properties dialog will bring up a dialog containing a list of surfaces. From this list you can check the surfaces which you wish to trim by using this curve.

Visible

This determines whether the curve is visible or not. Visibility may also be set in the Visibility dialog from the Curves Menu.

Locked

This determines if the curve is locked or not. If a curve is locked none of its control points are displayed, and the curve may not be modified. Locking may also be set by right clicking on the curve in the assembly tree or from the property sheet.

Symmetry

This determines whether a curve is displayed with its image reflected about the longitudinal centreline or not. This property may also be set by right clicking on the object in the assembly view or from the property sheet.

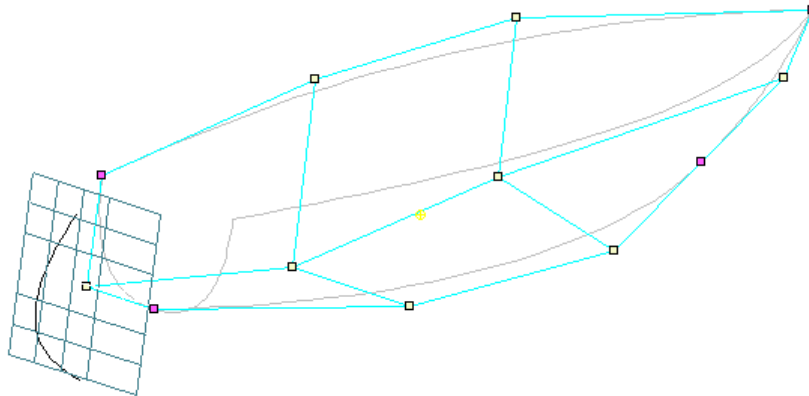
Curve Operations

Adding curve

One of four different curve types can be added from the Curve | Add Curve menu item (for more information on the curve types see [Curve Types in](#) on page 95). Each of the curve types may also be added from the Curve toolbar. In addition a Curve may be added from a Surface contour and a linear Curve may be added using a selection of Markers.

When manually adding Curves, they are constrained to be planar and in one of the principle orthogonal planes. Which plane they are added on depends on the current view. For example if the user is in the plan view then the curve will be added in the x-y plane. By default the z value of the plane will be 0.0m. However this z value may be set prior to positioning the curve ([see Setting the depth value of a curve drawing plane](#) on page 100).

A curve drawn in the perspective view will be added on the plane that is most perpendicular to the current point of view. To display the current drawing plane go to Display | Grid | Show Current Curve Drawing Plane. Then when you begin adding a curve the drawing plane will be indicated by the grid:



As with adding a curve in any of the orthogonal views the plane depth value may be specified by holding the shift key down when selecting an add curve command.

Curves can only be added on orthogonal planes (i.e. xy, yz, or xz planes), if a planar curve is required that is not on an orthogonal plane create the curve as normal and use rotation and translation operations to position the curve in the correct location.

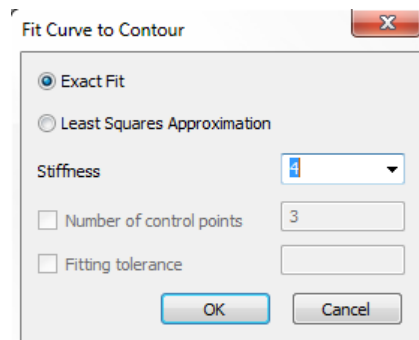
Setting the depth value of a curve drawing plane

To set the depth value (for example the z value of an x-y plane) hold down the shift key when selecting an add curve command from either the toolbar or the menu item. This will activate a dialog box asking you to enter the appropriate depth value (for example longitudinal position in the body plan view) for the drawing plane of the curve:



Add Curve on Contour

When one or more contours on a surface are selected then the Add Curve on Contour function becomes enabled from the Curve menu. When the command is executed the following dialog will appear:



The user has two options, "Exact Fit", which allows Modeler to choose the stiffness and number of control points of the resulting curve or "Least Squares Approximation" which allows the user to set these variables. If the Least Squares option is selected then Modeler will fit the closest curve according to the parameters set by the user. A fitting tolerance may also be set. If this option is set Modeler will determine if the resultant curve lies within this tolerance from the original contour. If it does not then Modeler automatically adjusts the number of control points until the curve fits within the specified tolerance.

Deleting curve

There are several ways to delete a curve. Graphically you can select a curve (or multiple curves) with the mouse and hit the delete key.

In the Assembly tree a curve can be deleted by right clicking on the curve and choosing delete.

In the Curve window a curve can be deleted by highlighting the Curve row and hitting the delete key.

A curve can also be deleted by selecting the curve and choosing the Curve | Delete Curve menu item.

Add data point

A data point can only be added to a data point curve. In one of the orthogonal views (plan, profile or body plan), select a curve and choose Curves | Add Data Point from the menu. The cursor will change to a pen and the user may now add the new data point.

Note:

When adding a data point to the curve the depth will be set at the same depth as the other data points in the curve. If there are a range of depths for the other data points in the curve then the depth value will be set to 0.0m

Add control point

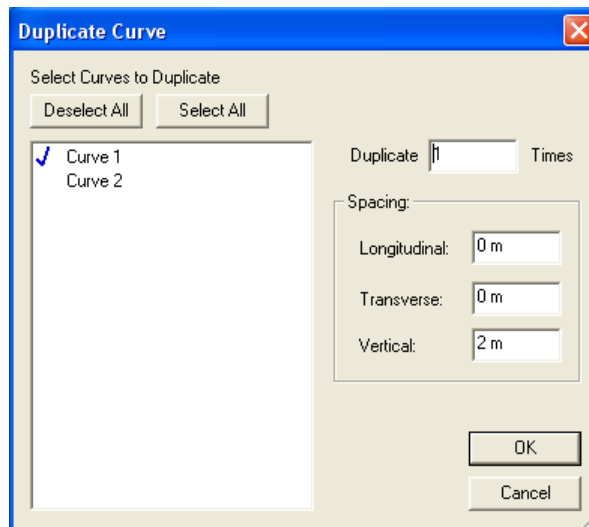
A control point can only be added to a control point curve. In one of the orthogonal views (plan, profile or body plan), select a curve and choose Curves | Add Control Point from the menu. The cursor will change to a pen and the user may now add the new control point.

Note:

When adding a control point to the curve the depth will be set at the same depth as the other control points in the curve. If there are a range of depths for the other control points in the curve then the depth value will be set to 0.0m

Move Curve

This command moves the selected curves.

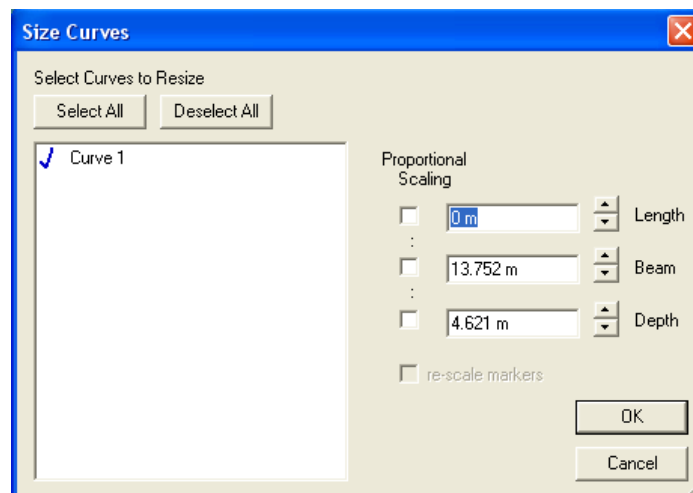


- Select move Curve from the Curves menu.
- Select the curves to be duplicated.
- Click OK

A number of duplicates may be made at one time and these can be offset from the original, and one another, by the specified longitudinal, transverse and vertical spacing's. In this way a linear array of the curves may be created. The curves that are selected in the dialog box by default are those which are currently selected in the drawing windows. To selected/deselect curves click to the left of the curve name.

The Size function allows a designer to scale and re-proportion a curve or group of curves by their principle dimensions. To resize the curves:

- Select Size from the Curves menu.



- Select which curves you wish to resize.

Do this by selecting the curve names from the check boxes on the left hand side of the dialog. When a single curve is selected the size displayed will be the overall dimensions of that curve. When multiple curves are selected the size displayed will be the overall dimensions of all selected curves.

- Select whether you wish to use proportional scaling.

For example, if you wish to maintain a length/beam ratio, tick the boxes of the dimension to be scaled proportionally (length and beam). When any of the dimension boxes are changed all boxes selected for proportional scaling will be changed in proportion.

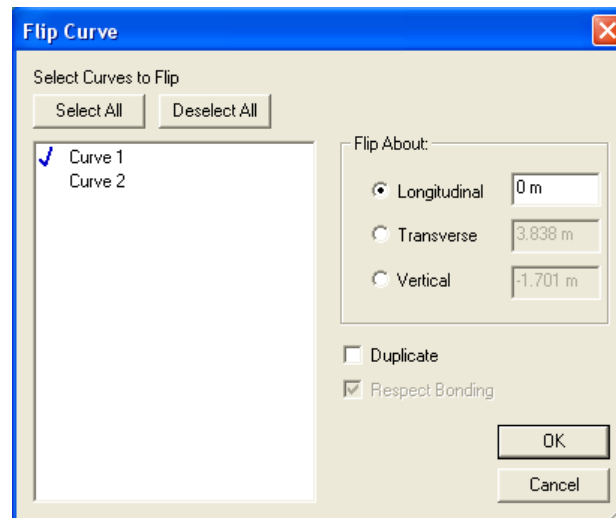
- Select and edit the appropriate size boxes to alter the size.

You may do this by either editing the text directly or by using the scale up / scale down arrows at the right hand side of the size boxes. Dimensions may be entered in any units by specifying the appropriate suffix.

Flip Curve

Curves may be flipped about specified planes: longitudinally, transversely or vertically. The actual mirror plane may also be specified, and need not be about the origin. At the same time you may also create a duplicate of the original curve.

- Select Flip Curve from the Curves menu.

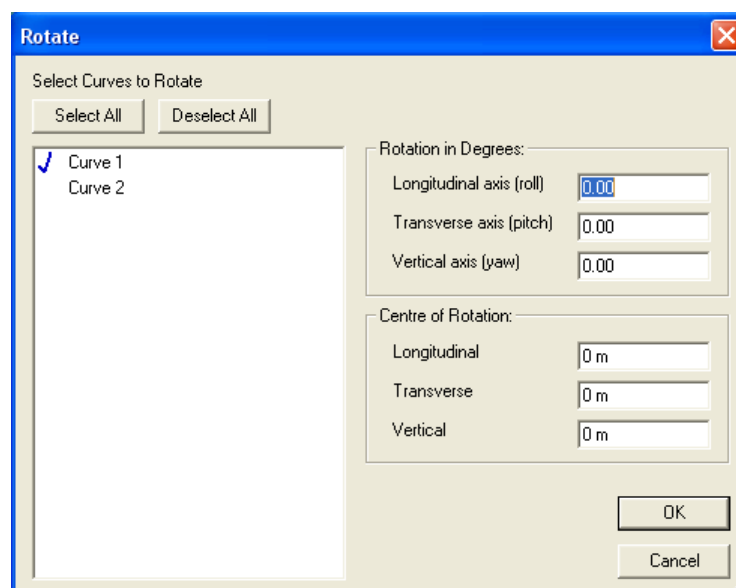


- Select the curves you wish to flip.

If you are duplicating the curves, they need not be unlocked since the original curves will remain unchanged. If you are not duplicating the curves then they must be unlocked.

Rotate Curve

The rotation dialog allows multiple curves to be rotated simultaneously around a specified centre of rotation. Curves to be rotated are selected by clicking in the list of surface names (default curves that are checked when the dialog is opened are those which are currently selected in the drawing windows).

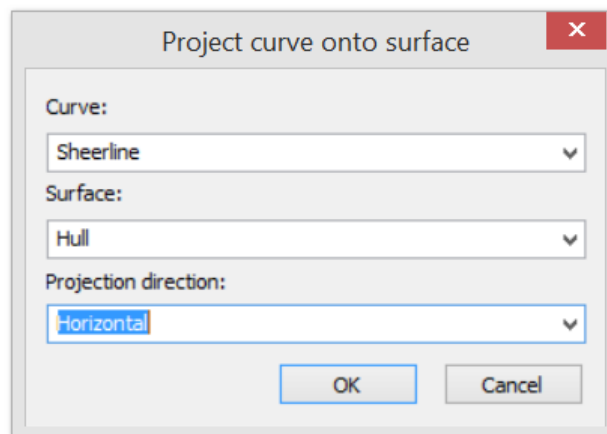


Reverse Curve

The reverse curve command will reverse the order of the control points in a curve, so that the first becomes the last and visa versa. This command will only become enabled when a curve is selected. The results of some functions that use curves (for example the surface skinning function) are dependent on the curve direction. In these cases the reverse curve function may be useful to orientate the curve in the correct direction, rather than creating a new curve. To determine current direction of the curve highlight an end curve control point, if in the properties window the Column is zero then that end is the start of the curve.

Project Curve onto Surface

A curve can be projected onto a surface. This function is available from the Curve menu and is enabled when there is at least one surface and at least one curve in the design. When the function is executed a dialog will appear:



The user needs to specify which curve is to be projected onto which surface and in which projection direction.

Break Curve

This command will break the curve into two curves at the currently selected control point. Both the newly created child curves will have the same properties (where possible) of the parent curve. Each of the child curves does not have end constraints imposed. The command will only be enabled in the menu if a curve control point is selected.

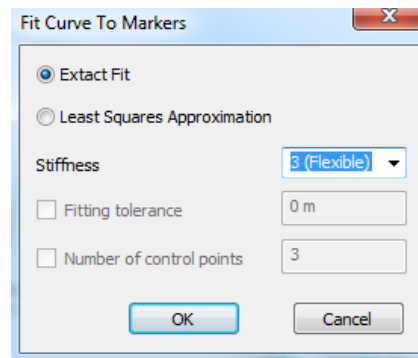
Join Curves

The join curves command will join the multiple selected curves into one curve. If two control points or two data points are selected from two separate curves the selection order is important as the second selected control point (data point) will be moved to the first selected control point (data point). When multiple curves are selected for joining, the properties of the first selected curve will be assigned to the joined curve. When multiple curves are joined they are joined end to closest end.

This command will only be enabled in the menu if two end control points from two separate curves are selected, two end data points from 2 different curves are selected, 2 or more curves are selected, 2 or more control points from different curves are selected.

Fit Curve to Markers

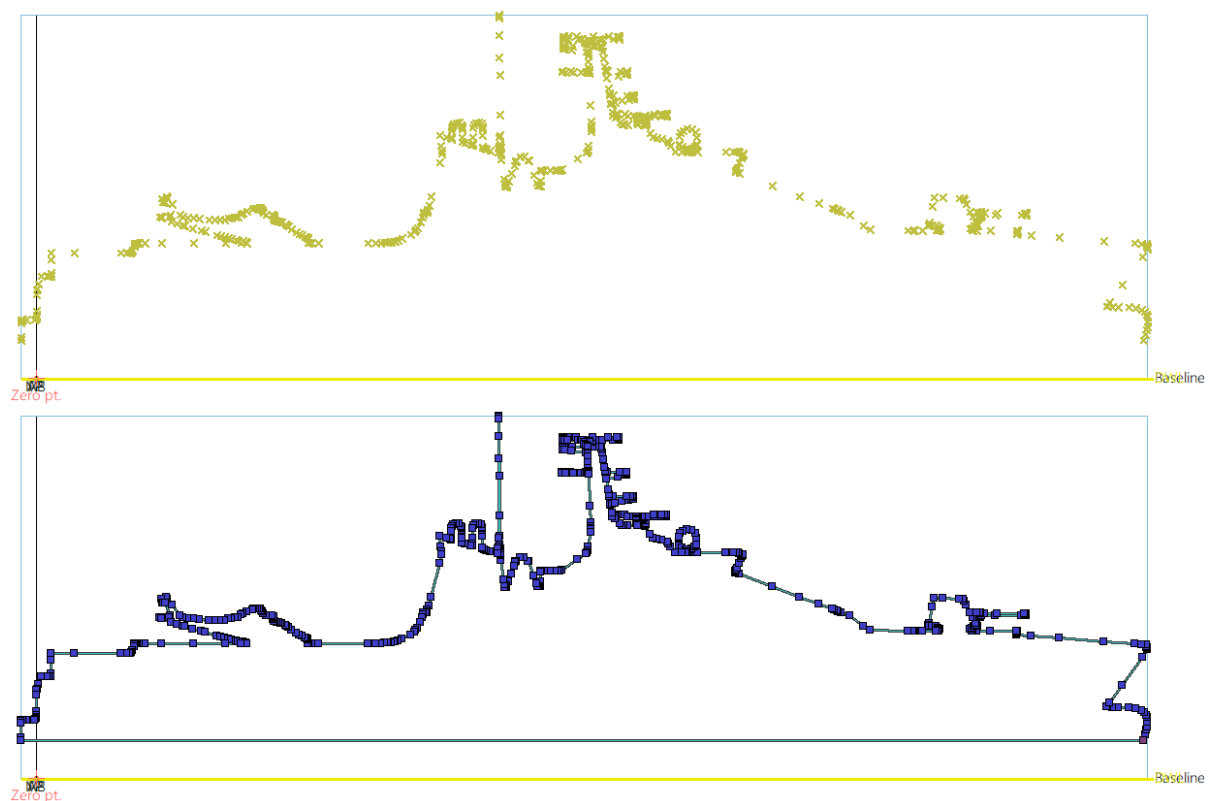
To enable the fit curve to markers menu option, 2 or more markers need to be selected. When the menu item is selected the following dialog will appear:



There are two options when fitting a curve to Markers, either an exact fit or a least squares approximation. The exact fit will create a control point curve of stiffness specified by the user that passes exactly through each of the markers selected in the order that they were selected.

If a least squares approximation fit is chosen then Modeler will fit a curve of best fit through the data points based on the information provided in the dialog box. When this type of fit is chosen the user may also specify the number of control points of the new curve as well as a fit tolerance. If the least squares curve deviates from the markers by more than the specified tolerance then Modeler will automatically raise the number of control points until the specified tolerance is met.

An 'Exact Fit' using a linear (Stiffness=2) curve can be used to create a curve that goes exactly through all the markers. For example, as the initial step for creating a windage profile mesh from an imported DXF profile drawing:



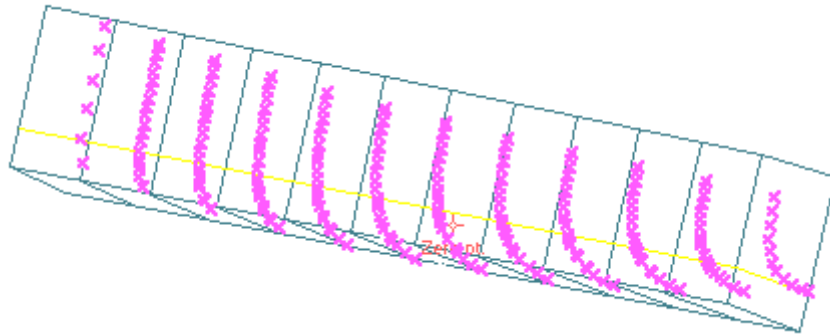
An exact, linear Curve fitted to selected Markers (imported from a DXF file)

Fit Curve to Markers Stations

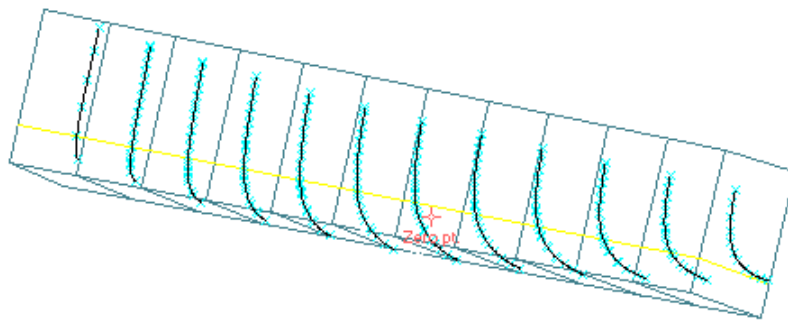
If Markers are assigned to stations and 2 or more markers are selected for a station then the fit curve to marker stations option will be enabled from the curve menu. The same dialog will appear as for Fit Curve to

Markers. Modeler will automatically arrange the markers for each station so that they are in order from centreline lowest and fit the curve through each set of markers for each station according to the parameters specified by the user.

To run this command, first ensure each marker you wish to fit a curve through is assigned to a station. Next select all the markers you wish to fit curves through:



From the Curves menu select Fit Curve to Marker Stations and the fit curve to Markers dialog will appear (see above). Specify the fitting requirements and click OK, Modeler will then fit a curve to each of the set of markers for each station:



These curves may then be used to skin a surface. See [Skinning Curves to Create a Surface](#) command in the Surfaces section of the Manual.

Working with Surfaces

A model in Modeler can be made up of any number of surfaces. This section will explain all about the different surface types and surface operations available in Modeler.

In this section:

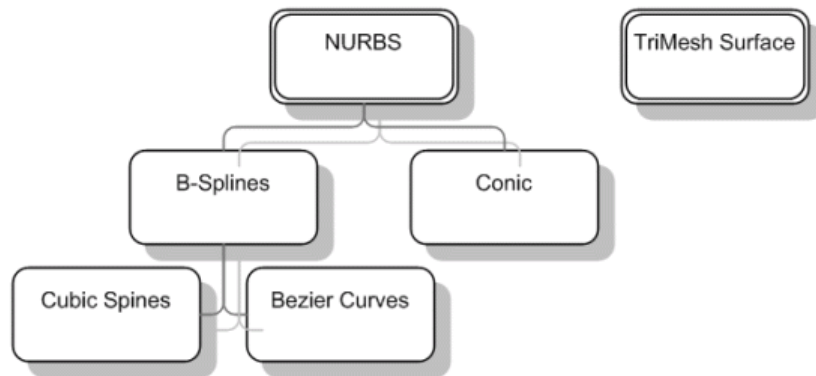
- [Surface Types in](#)
- [Surface Stiffness](#)
- [Surface Appearance](#)
- [Surface Properties](#)
- [Outside Arrows](#)
- [Surface Materials and Skin Thickness](#)
- [Surface Precision](#)
- [Displaying Curvature on Contours](#)
- [Rendering a Surface](#)
- [Surface Operations](#)

- [Trimming Surfaces](#)
- [Cutting Surfaces](#)

Surface Types in Modeler

Modeler allows you to create complex designs using any number of a range of surface types. The two main types that are supported in Modeler are:

- [NURBS Surface](#)
- [Mesh Surfaces](#)



NURBS Surface

NURB stands for Non Uniform Rational B-spline. The shape of a NURB surface is calculated mathematically from the following properties only:

- Control point positions
- Control point weights
- Surface stiffness in transverse and longitudinal direction

The principle behind NURB surfaces is explained in chapter 2 in the section [The Spline and Spring Analogy](#) on page 15.

Modeler is capable of creating complex models using any number of NURB surfaces with up to 25 rows of control points. Modeler allows the use of four surface types, which are all members of the NURB family:

B-Spline

This type of NURB surface has all control point weights set to unity. B-spline surfaces are ideal for many applications of ship design. However, they can not precisely represent simple conics such as circular, elliptical or parabolic curves. Modeler offers an alternative surface definition (Conic) that overcomes this disadvantage and allows you to combine conic curves with free form surfaces at will.

NURB

With this surface type, it is possible to precisely represent conic curves and surfaces. This is achieved by the addition of weights, which may be varied for each control point.

Conic

When this surface type is selected, a NURB surface is used and the control point weights are automatically calculated by Modeler to produce a precise conic surface. Use this surface only if you require precise conic surfaces, such as for a bow cone.

Note

The control point weights are only computed automatically for 3-point forms. If you are using 4-point forms you will need to make your surfaces NURB surfaces and type in the weights manually.

Developable

A developable surface is one that can be formed from a flat sheet without stretching the material. This material may be aluminium, steel, wood or even paper - the material chosen has no effect on whether or not the surface is developable. Examples of simple developable surfaces are cylinders and cones, as opposed to a sphere which is clearly not developable. A developable surface is created from the edges of a B-spline surface and a set of offset points can be generated at each station, See [Generate Markers for Developable Surfaces](#) on page 375. These offsets can then used to manually or automatically fit the B-spline surface to the developable shape. This allows you to create a surface that is developable within practical tolerances. Modeler can give you an indication of the level of developability by displaying a Gaussian curvature rendering of the surface. See [Curvature Display Using Rendering](#) on page 118.

The actual plate expansion calculation and export of the expanded plate shape to a CNC cutting machine or other CAD software, can be done by creating a plate ot plates from the surface.

Surface Stiffness

Surface Stiffness – Concepts

The choice of surface stiffness is analogous to the selection of a different weight of spline when drawing a curve on the drawing board. When drawing a smooth shallow curve you would select a stiff spline, and when drawing a curve featuring a high rate of change of curvature you would select a flexible spline.

Flexible splines are useful for modelling knuckles and discontinuities, otherwise try and work with the stiffest spline possible. Often a good compromise is to use flexible splines transversely and stiffer splines longitudinally.

Two factors influence the stiffness of a surface:

Setting of stiffness through the Properties dialog.

A number of stiffness settings allow for the specification of stiffness in the longitudinal (row) or transverse (column) direction.

Linear	Order 2
Flexible	Order 3
....	
Stiff	Order 10

There is no absolute value for the recommended surface stiffness since it is very much dependent on the surface shape you wish to achieve. In general, a longitudinal stiffness of 5 and a transverse stiffness of 4 offers a good starting point from where you can go up or down dependent on the surface shape you wish to achieve.

The stiffer the surface, the easier it is to get a fair surface. At the same time, a stiff surface is more difficult to achieve high curvature shapes with.

Number of control points.

You need at least the same number of control points in the direction of the surface stiffness. For example: for a *longitudinal* stiffness of 6 you need at least 6 *columns* of control points and for a *transverse* stiffness of 5 you need at least 5 *rows* of control points.

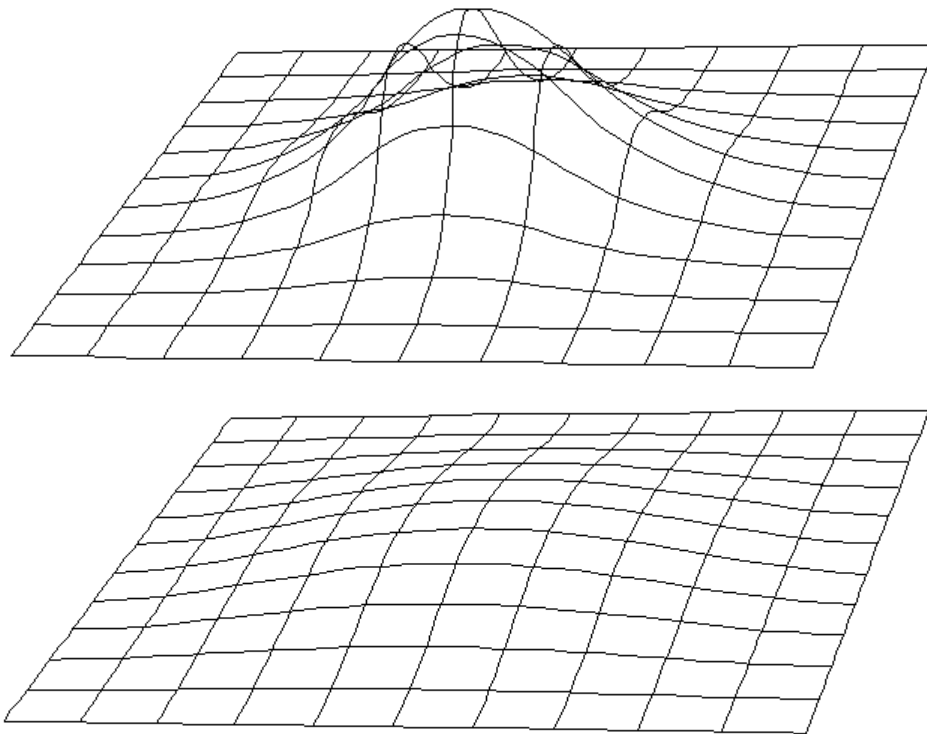
To get an idea why this is necessary, imagine a 2 column surface (only a forward and an aft edge column without any control points columns in between); the surface stiffness can only be linear.

Modeler will make sure the stiffness of the surface cannot be greater than the number of control points in that direction; i.e. if you have an order 6 surface and you reduce the number of columns from 6 to 5, Modeler automatically reduces the surface stiffness from 6 to 5.

Tip:

Use as little control points as possible on a surface that is as stiff as possible whilst achieving the required shape. This will result in a fair surface model.

The example below shows a surface with the same net but with two different stiffness's. One surface is set to flexible in both the longitudinal and transverse directions and the other is set to stiff.



Local Influence

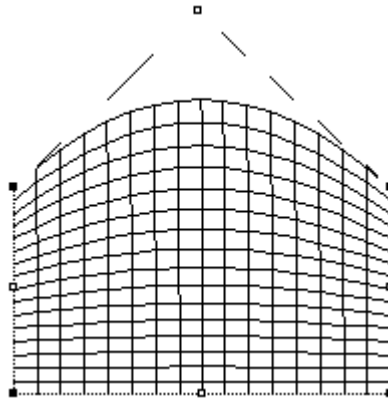
When using few control points, a control point movement may influence the whole surface, but when using many control points the control point's area of influence will be more local.

This is illustrated below, showing the extent to which the surface deforms due to a control point movement using a rectangular 3*3 control point net.

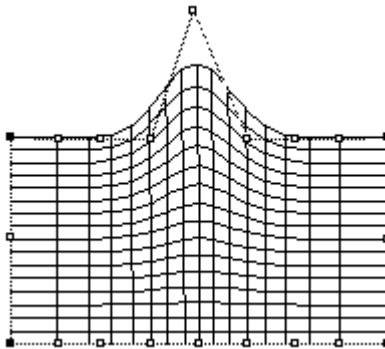
The movement of the middle control point on the top edge would result in, the following deformation.

Note:

The influence of the control point movement may be seen across the full width of the parametric surface.



If the net is increased to 9*3, keeping the stiffness (as set in the Properties dialog) the same, the result of a similar control point movement would be as follows.



Through the addition of extra control points:

- the surface has become more flexible
- the influence of the control point movement is much more localized.

Note:

The surface stiffness actually relates to the number of continuous derivatives of the surface. There are two special cases:

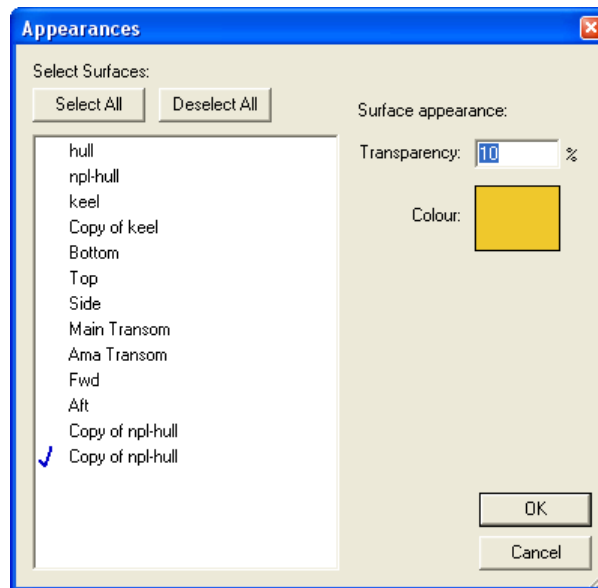
1. If the spline is order 4, the curvature is continuous and the spline is the same as a piece-wise cubic spline.
2. If the order is one less than the number of control points, it is a Bezier spline.

Surface Appearance

The surface appearance may be used to change the color of the smooth rendered view and the parametric curves of the surface.

To change the surface appearance:

- Select Appearance from the Surfaces menu.
- In the dialog, select the surfaces whose appearance you wish to modify.
- To modify the color, click in the colored square and select the new color from the pallet.

**Note:**

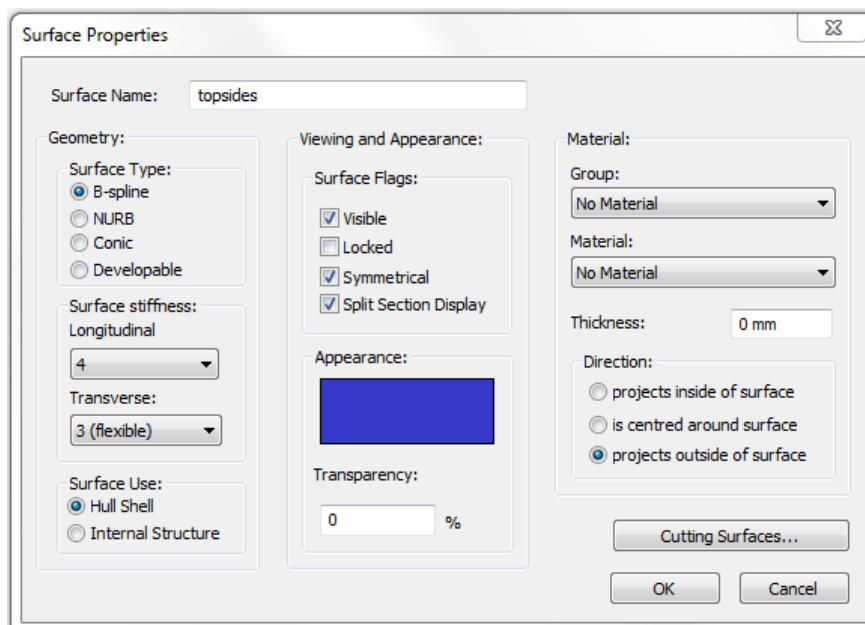
Some graphics cards may ignore the transparency option.

Surface Properties

Each surface that is created within a Modeler design has a unique set of properties associated with it. The Properties command allows you to view and modify these properties.

Note:

The properties of all the surfaces may be quickly viewed and modified in the Surfaces window or by right clicking on the surface from the Assembly Pane or in any view window.



It is possible for you to modify:

Surface Name

This may be any name typed in from the keyboard up to twenty characters long.

Surface Type

May be set to conventional B-spline, NURB, Conic or Developable surface.

Surface Use

This tells Modeler and Stability whether the surface is an internal structure or not. Surfaces that are to be used for internal tank or compartment walls in Stability must be defined as Internal Structure.

Note:

It is very important to specify "surface use" in order for Stability and Structure to interpret the Modeler model correctly.

Visibility

This determines whether the surface is visible or not. Visibility may also be set in the Visibility dialog from the Surfaces Menu.

Locking

This determines if the surface is locked or not. If a surface is locked none of its control points are displayed, and the surface may not be modified. Locking may also be set in the Locking dialog from the Surfaces menu.

Symmetrical

This determines whether a surface is displayed with its image reflected about the longitudinal centreline or not.

Split Section Display

Split Section Display is only activated in the Body Plan window, when the Half option is in effect. In this situation, if Split Section Display is activated, the surface will be shown with the forward sections on the right hand side and the aft sections on the left, as one would expect in a traditional hull body plan.

The position chosen for the break in the display of the sections depends on the station you have selected in the Design Grid dialog.

Stiffness

Sets the surface stiffness in the longitudinal (row) or transverse (column) direction. See also [Surface Stiffness](#) on page 108.

Material

This tells Modeler, Stability and Structure what material is used for construction, and how the thickness is calculated in regards to the hull shape.

Note

A sections library file (*.slb) must have been loaded when Modeler was started for there to be a list of materials to select. If no materials are available, you may simply type in the thickness. A new library can be loaded from the File | Load Sections Library... function. Further details are in the next section, Surface Materials.

Surface use

The table below shows how different surface classes (NURBS and TriMesh) and surface use (Hull and Structure) are used in the different Maxsurf modules.

	Surface types in design		
Surface types in design	NURBS	Mesh	Both

Modeler Hydrostatics at DWL	Visible "hull" surfaces	Visible "hull" Meshes	Visible "hull" NURBS and Meshes
Modeler structural elements	Selected surfaces	n/a	Selected NURBS surfaces
Stability watertight hull	Selected "hull" surfaces	Selected "hull" Meshes	Selected "hull" NURBS and Meshes
Stability Rooms	Selected "structure" surfaces	Selected "structure" Meshes	Selected "structure" NURBS and Meshes
Motions strip theory	Selected "hull" surfaces	Selected "hull" Meshes	Selected "hull" NURBS and Meshes
Motions panel method	n/a	Selected	Selected Mesh surfaces
Resistance regression	Selected "hull" surfaces	Selected "hull" Meshes	Selected "hull" NURBS and Meshes
Resistance slender body method	Selected "hull" surfaces	Selected "hull" Meshes	Selected "hull" NURBS and Meshes
VPP	All "hull" surfaces; longest surface used for IOR girth station measurements	n/a	As for NURBS only model

Outside Arrows

Another surface property is the Outside Direction. This direction is displayed in Modeler with Outside Arrows, which can be displayed from the Display menu or the Outside Arrows toolbar button.

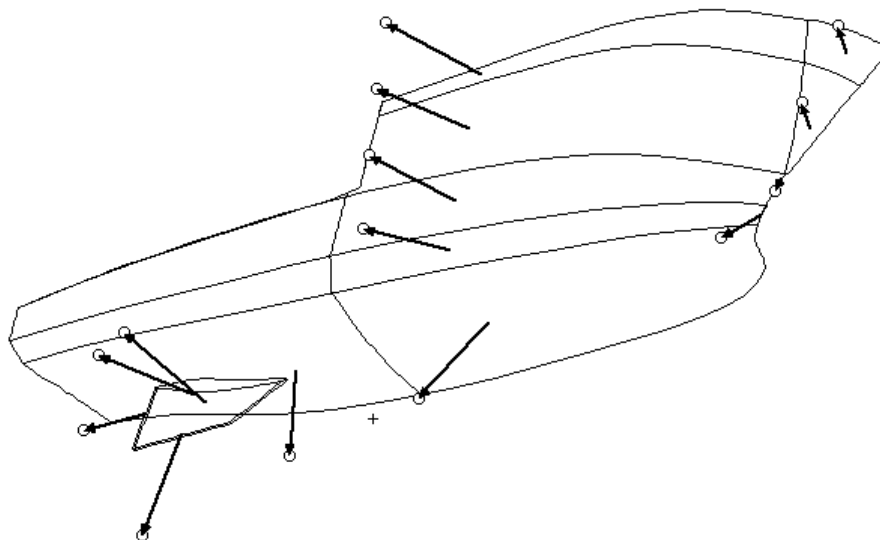
The Outside Arrows tell Modeler which is the inside and which is the outside of a surface. This is important for the following Modeler functions:

- Skin Thickness direction
- Curvature display
- Correct interpretation of model in Structure and Stability

Setting Outside Arrows

- Select Outside Arrows from the Display menu.

Selecting this menu item displays surface outside arrows in all of the display windows.



By clicking on the circles at the heads of these arrows it is possible to reverse their direction. Do this until all arrows point to the outside of the hull. It may be necessary to look at different views or to rotate the hull in the perspective window in order to see the directions clearly. Once you have specified the outside directions, save the design; these directions will be saved in the Modeler design file and are used automatically by Structure and Stability.

Note:

The length of the outside arrow may be changed in the Preferences dialog.

Surface Materials and Skin Thickness

Modeler provides a method for specifying skin thickness and material properties that are consistent and compatible with both Stability and Structure. This also allows sections with skin thickness deductions to be displayed in Modeler, printed from Modeler or exported to a DXF file.

Thickness can be either specified directly or selected from the Material pop-up menu. The Material pop-up menu will contain materials that have been loaded from the sections library. Structure may be used to modify the library; users who do not have Structure will have the default library.

The direction that the thickness projects from the line of the designed surface needs to be specified. Three choices are available:

- inside the designed surface
- centred on the designed surface
- skin outside the designed surface.

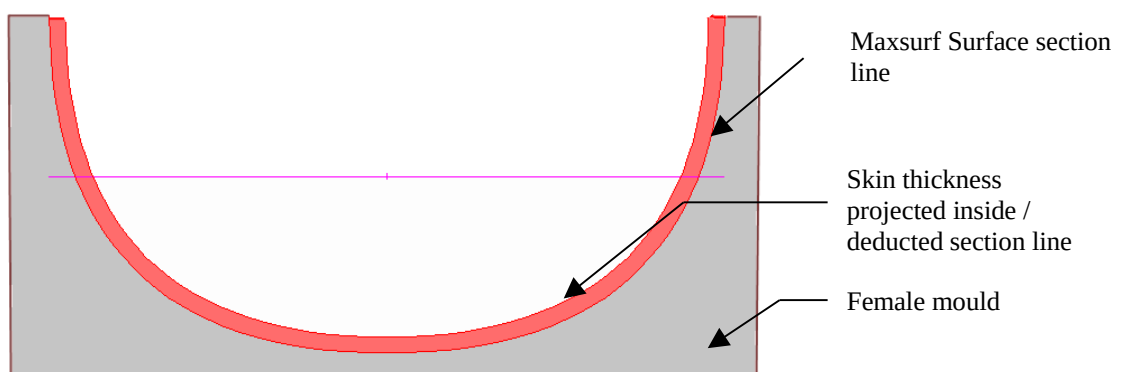
To do this Modeler needs to know what the inside and what the outside of each surface is. Make sure the Outside Arrows are set up correctly (pointing outwards, generally towards the "wet side" of the vessel). For more information on outside arrows see [Outside Arrows](#) on page 113.

Note: Skin thickness and hydrostatic calculations

In Modeler skin thickness is used for creating the table of offsets, however for hydrostatic calculation purposes, skin thickness is ignored. Stability can be used to calculate the effect of skin thickness on the vessel's hydrostatics by selecting "include skin thickness" in the opening dialog.

Skin Thickness Projected to the Inside

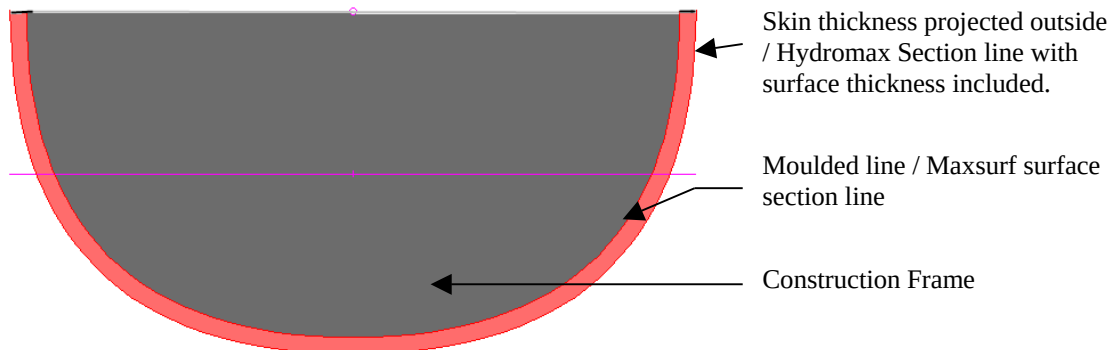
An example of a thickness deducted inside the hull would be a hull made of foam sandwich, where the Modeler surface modelled the outside of the hull. The skin thickness deduction would then give the line of the inside of the foam core and laminations.



Female mould GRP example

Skin Thickness Projected to the Outside

A typical steel or aluminium boat is designed to a moulded line, to the outside of the framing, but the inside of the hull plating. Another example is a GRP boat build from a male mould. The Skin Thickness would then be added on to give the line of the outside of the plating.



Frame construction method (Same for male GRP mould method)

Centred Skin Thickness

Plating centred around the designed surface would be unusual for the hull shell, but has been included for possible use in the specification of internal structure.

Display Skin Thickness and Direction

You can see the effect of the skin thickness and direction you have specified by displaying deducted sections. To display deducted sections

- Choose Contours from the Display Menu and select Deducted Sections.

If this option is selected, deducted sections will be displayed on screen and included in DXF and IGES file output.

Modeling Skin Thickness for MOSES

When generating models for MOSES, you should create the NURB surfaces so that they are on the outer part of the hull shell; so that the skin thickness (if applied) is projected to the inside of the vessel. The reason for this is that the TriMesh used in MOSES will lie on the NURB surface (the TriMesh cannot be automatically offset from the NURB surface); however compartments (which are defined by sections in MOSES) can be offset to account for the skin thickness of the hull-shell material.

Surface Precision

Surface Precision – Concepts

Each curve drawn by Modeler is made up of a number of short straight lines.

The precision selected corresponds directly to the number of line segments shown in a surface edge or contour, and also equates directly to the number of parametric lines shown when displaying parametric contours.

Lowest Precision

Lowest Precision may be used early in the design process for general design layout. It allows for fast modification of the shape, but lacks the necessary detail for optimising a design. It uses an 8 by 8 parametric mesh.

Low Precision

Low Precision is the suggested working precision for your design development and modification if you have a slow computer. It uses a 16 by 16 parametric mesh.

Medium Precision

Medium Precision is the suggested working precision for designs with more curvature, or with discontinuities, which need more precision for display. It uses a 32 by 32 parametric mesh.

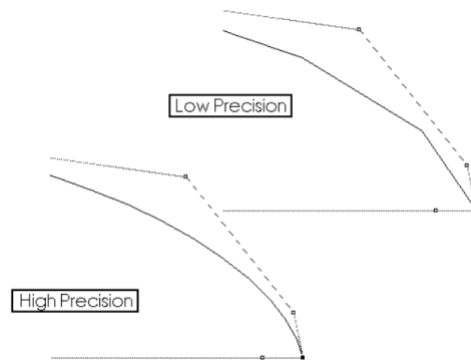
High Precision

High Precision may be used for complex shapes, or during printing or zooming when more curve definition is needed. It uses a 64 by 64 parametric mesh.

Highest Precision

Highest Precision may be used when extremely high definition is required, typically for plotting final lines plans. Highest Precision will take much longer to calculate than Low precision.

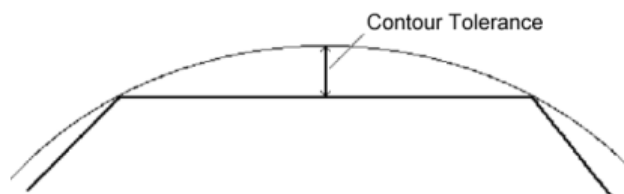
High and Highest precision are not fixed precisions, but adapt the number of segments to the amount of curvature in a surface.



Precision setting is saved with the model. If a model is saved in highest precision in Modeler, it will be by default opened in highest precision. This is also true when porting the design across applications (e.g. Stability, Structure etc).

Data Insertion

High and highest precisions start with a 64 by 64 parametric mesh and then insert points into the curve to reach the curve tolerance you have specified using the Contour Tolerance field in the Preferences dialog.



Data Reduction

At Highest Precision redundant points are also removed from the curve. For example, if a surface is flat, all of the co-linear points along the middle of a line will be removed, just leaving the required end points. This can considerably increase display speed in Modeler and reduce the size of the DXF or IGES file. This makes the curves much faster to manipulate in your CAD system.

Precision	Number of segments	Data reduction	Data insertion
-----------	--------------------	----------------	----------------

Lowest	8	✗	✗
Low	16	✗	✗
Medium	32	✗	✗
High	64	✓	✗
Highest	64	✓	✓

The number of curve segments apply both to lines drawn to the screen, Clipboard, and IGES files, as well as to output devices such as printers.

Surface Precision – Procedures

To change the precision that contours are calculated and displayed at:

- Select Display | Precision

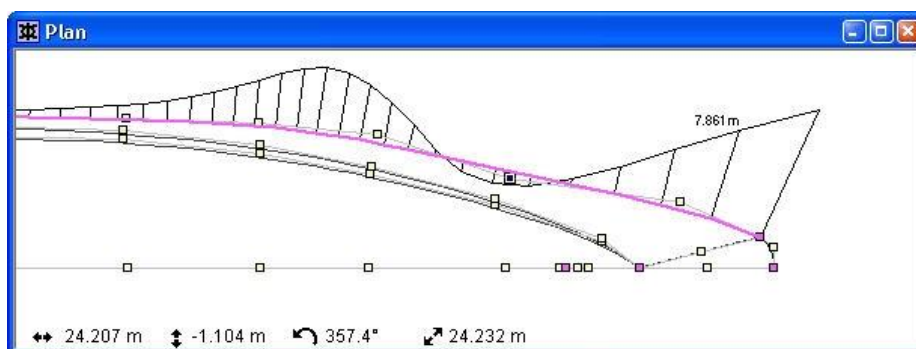
All contours on all surfaces will be displayed at the same precision.

Displaying Curvature on Contours

A way to display curvature on a Modeler design is to select a contour or group of contours, and use the Show Curvature command from the Display menu. Curvature may be displayed on all surface contours except intersection lines.

This will display "Porcupines" perpendicular to the curve. The length of the porcupine is inversely proportional to the square root of the radius of curvature at that point on the curve. I.e. the longer the porcupine line, the smaller the radius and a smaller radius means more curvature. The value of the smallest radius on a given curve is displayed numerically at the end of the porcupine located at the corresponding position on the curve. The number of porcupines is determined by the current [Precision](#).

Curvature may be displayed on any plane or three-dimensional contours e.g. waterlines or surface edges. Displaying the curvature of three-dimensional curves in the perspective view will highlight any torsion in the curve.



Display Curvature – Procedure

- Select a contour by clicking on it.

It will be highlighted with a different line thickness and color. To select more than one contour, shift-click on other contours.

- Select Show Curvature from the Display menu

To hide the display of curvature porcupines,

- Hide Curvature from the Display menu.

The lengths of the curvature porcupines may be scaled in the Curvature Scaling Factor field in the Preferences dialog. See [Modeler Preferences](#) on page 76.

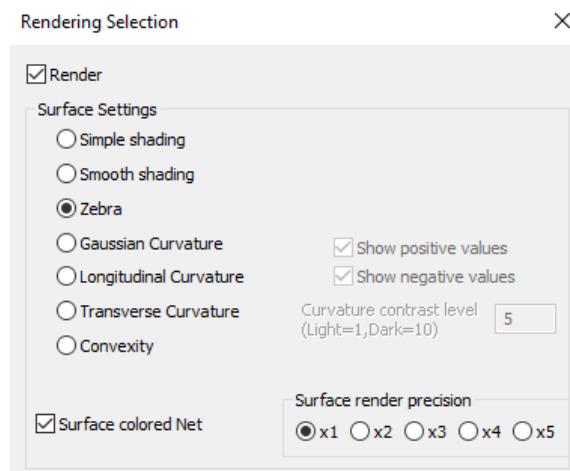
For other ways to display curvature on a surface, see [Rendering a Surface](#) on page 118.

Rendering a Surface

Rendering a surface can be used to display the shape and fairness of the surface. Modeler has different rendering types available. Each rendering type can be used for different purposes. See also: [Perspective Window](#) on page 48.

Curvature Display Using Rendering

Modeler has four different ways of calculating and rendering curvature on a surface. You control these settings using the Rendering command from the Display menu. Render may only be selected when the Perspective window is active.



To display the rendered view, turn on hidden surface elimination. It is then possible to display a shaded view of the design, or to use false coloring to highlight the curvature of the surfaces.

Hidden Surface Elimination

Selecting Hidden Surface Elimination performs surface shading over the surface, using a fixed light source. If Smooth Shading is selected, the base colors of the surfaces are those chosen in the Appearance option from the Surface menu.

Simple Shading

Shading using polygons is utilised. Surface contours may also be displayed with this option.

Smooth Shading

An alternative shading algorithm is used to provide smoother rendering. The colors of the surfaces may be changed in the Appearance dialog.

To examine the fairness of your design, Modeler can use false coloring to display four types of surface curvature. These have different uses and are described below:

Gaussian Curvature

Is the product of the maximum and minimum curvatures at a point on the surface. Gaussian curvature will indicate whether the surface is locally elliptical (positive Gaussian curvature, i.e. both curvatures in the same direction), whether it is hyperbolic (negative Gaussian curvature, curvatures with opposite signs i.e. saddle shaped), or whether it is developable (zero Gaussian curvature).

Gaussian curvature is a useful indicator of areas of twist in a surface, but is not directly linked to surface fairness. It is possible for a surface to be quite unfair and still be developable, yet the Gaussian curvature display will not reveal any problems.

Gaussian curvature can be used as an indication of whether a surface is developable. A developable surface is one that can be formed out of a flat sheet material by bending or rolling, without stretching or distorting the material. In this case, the surface will have a Gaussian curvature of zero at all points; i.e. the surface is only curved in one direction and straight in the direction orthogonal to that. See [Surface Types in](#) on page 107 for more information on what developable surfaces are and [Generate Markers for Developable Surfaces](#) on page 375 for information on how to model developable surfaces in Modeler.

You can adjust the brightness value to make the display more or less sensitive to the Gaussian curvature values. See [Error! Reference source not found.](#) on page **Error! Bookmark not defined.** for more information.

Note:

The Gaussian curvature gives an *indication* of developability, but should only be used to isolate areas where the plate will be less developable.

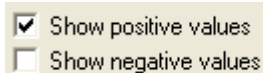
If you want to be certain of a plates' developability, load the surface into Structure and develop a surface plate. You can then examine the strain distribution on the plate; the plate is developable if there is no strain. A small amount of strain can be acceptable dependent on the builders' plate bending equipment and production techniques.

See the Structure manual for more information, especially regarding precision settings during plate development.

Longitudinal Curvature

This is a display of the curvature of each longitudinal parametric curve, taken perpendicular to the surface at each point along the curve.

This display is extremely useful for determining the longitudinal fairness of a design. You should look for an even graduation of color along the hull. Inflections can be detected by looking for changes from blue (positive curvature) to red (negative curvature). If you choose to show positive curvature values only, do so by using the following settings:



This will reveal areas of negative curvature by painting them black.

Transverse Curvature

Is the curvature of each transverse parametric curve, also taken perpendicular to the surface at each point along the curve.

This display is extremely useful for determining the transverse fairness of a design. You can mask off areas of positive or negative curvature in the same way as with the longitudinal curvature display.

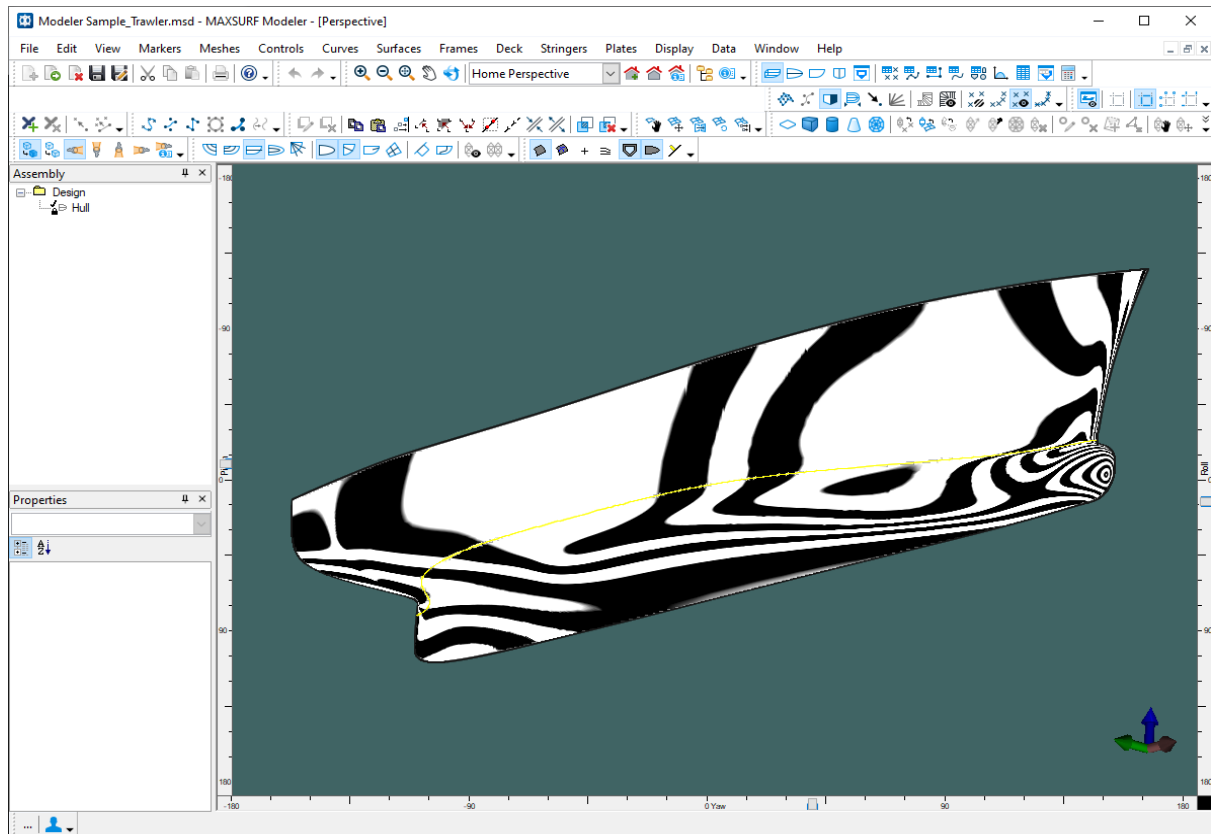
Convexity

The convexity check highlights any areas of negative curvature.

The convexity check calculates and displays the minimum curvature for each point on the surface. If this minimum curvature is less than zero, the surface is locally concave.

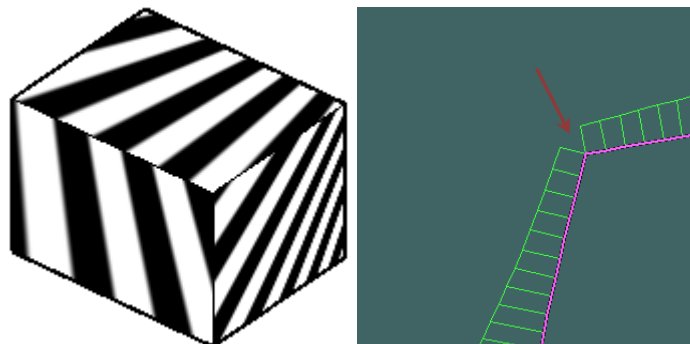
Zebra

Zebra mapping assists in generating extremely smooth NURBS surfaces helping to identify areas of discontinuity/poor continuity. At the bond between two surfaces, the way that the zebra stripes align and curve tells you a lot about the smoothness of the bond between the two surfaces.



Zebra stripes touch but do not line up. C0 Position.

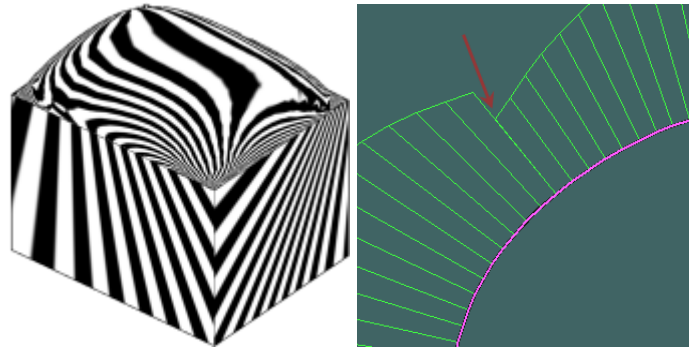
The position of the surface edges is collocated ie they touch. But the tangency and curvature do not match. If looking at the curvature porcupines on a contour across the bonded edges, there will be a gap or overlap between the porcupine ends on each side of the edge.



C0 Position continuity: zebra stripes and curvature porcupines

Zebra stripes touch and line up but are not smooth. C1 Tangent.

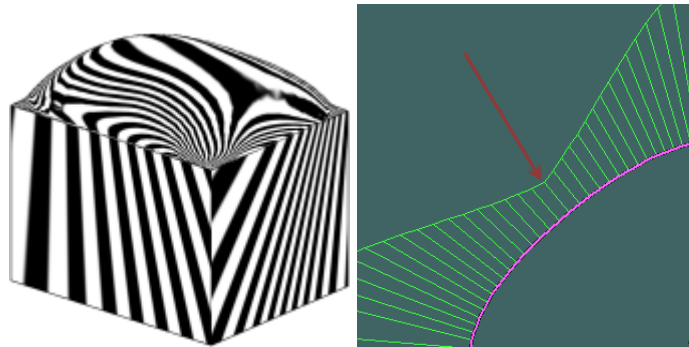
The position and tangent of surfaces are the same at the bond. This indicates C1 (or position and tangent continuity). The curvature porcupines on each side of the edge will be parallel but of different lengths.



C1 Tangent continuity: zebra stripes and curvature porcupines

Zebra stripes touch, line up and are smooth. C2 Curvature.

The position, tangent, and curvature of the surface edges are the same at the bond. This indicates C2 (or position, tangent and curvature continuity). It can be quite subtle to distinguish C1 and C2 from the zebra stripes. The curvature porcupines on each side of the edge will be parallel and the length.



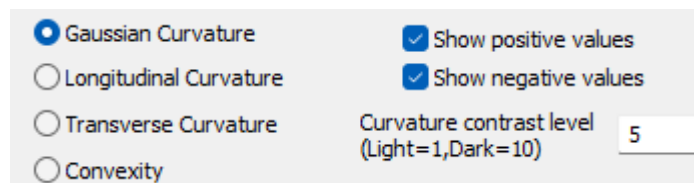
C2 Curvature continuity: zebra stripes and curvature porcupines

Note:

Before you use Rendering, make sure that you have set the direction of the [Outside Arrows](#) (see page 113). When the Outside arrow points outwards, positive curvature is displayed as blue and negative curvature as red.

Curvature Contrast Level

The curvature contrast level referred to in the dialog affects the way in which the colors in the display are mapped onto the curvature values.



Because the range of colors is small and the range of curvatures infinite you may need to change the coloring to suit the curves you are looking at. Changing the curvature contrast level re-maps the colors - if the image is very dark and it is difficult to distinguish different curvatures, try using a smaller number. On the other hand if the image tends to be all pale colors, try a darker tonal value.

For Gaussian curvature, a higher curvature contrast level will make the display more sensitive to non-zero Gaussian curvature, highlighting more areas. See [Curvature Display Using Rendering](#) on page 118 for more information.

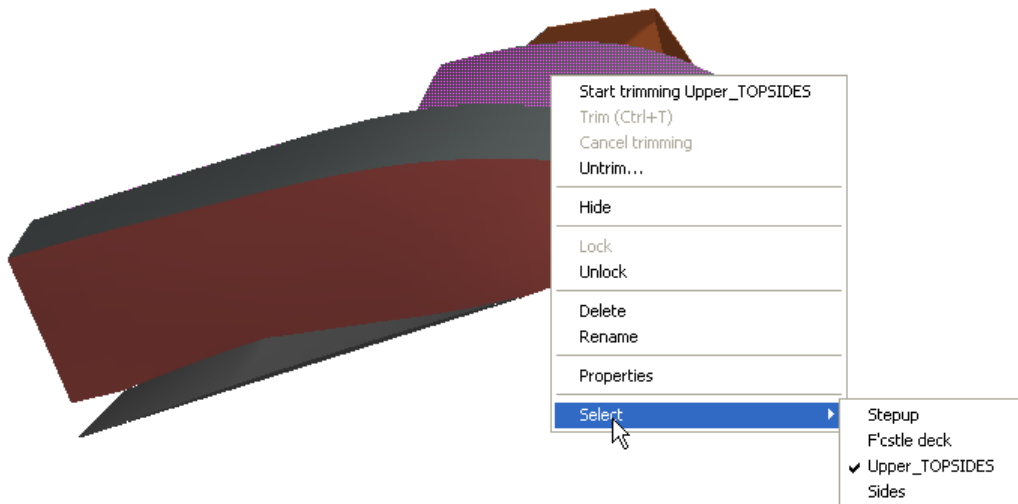
Rendering

In the perspective view you can view the model as smoothly shaded objects; options affecting the rendered drawing are set in the Rendering dialog.

Rendered Selection

When this option is selected, it is possible to select a rendered surface. This brings up the surface properties in the Properties Pane. You can also use the right-click menu to trim, lock or delete the surface etcetera.

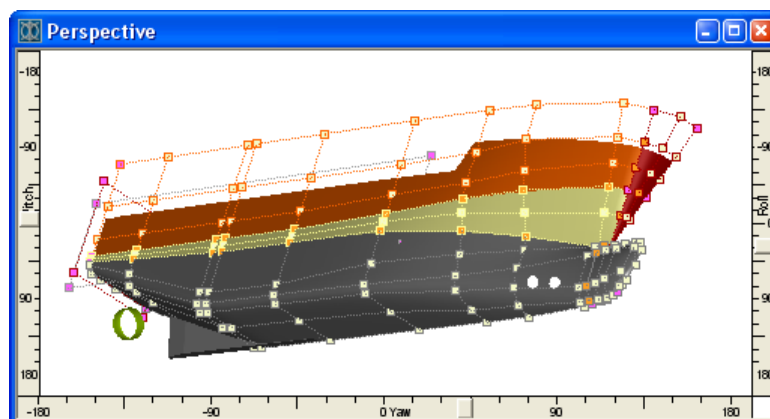
When there are several surfaces directly behind each other, a Select option will appear at the bottom of the right click menu which allows you to select any surface in the menu.



The color of the selection highlighting can be set in the View | Colors and Lines dialog.

Rendered Net Color Coded

When this option is selected, the control point net is displayed in the rendered surface color. It may be desirable to set the surfaces transparency to for example 30% to avoid a control point being hidden by the rendered surface.



Color coded control points.

Lighting Options

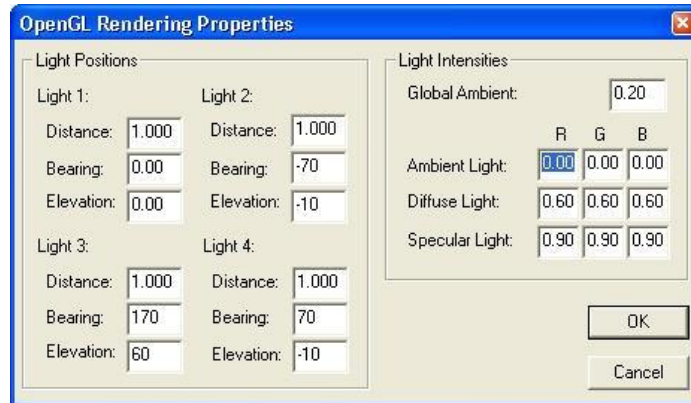
Modeler allows the user to edit the lighting settings used in the Perspective window when rendering is turned on.

Light Positions

Modeler has the option to use four different light sources. These can be turned on and off by clicking in the Render toolbar.



It is possible to vary the location of these lights by editing values in a lighting dialog. This dialog is displayed by clicking on the right most icon in the Render toolbar when the Perspective window is active and rendering is turned on.



Each light is specified by three values, distance, bearing and elevation. Distance is measured from the centre of the view in the perspective window. Bearing is taken relative to the direction from the eye to the object being viewed. For example, a light at 0 degrees is pointing in the same direction as the eye is looking at the object, 90 degrees is to the right of the object, -90 degrees is to the left of the object and 180 degrees is behind the object. Elevation is in degrees above the horizontal. A negative elevation value defines a light shining up from below the horizontal.

Light Intensities

Ambient, Diffuse and Specular values define what kind of reflection will be seen on the surface. If only ambient light values are specified, the object will be lit evenly from all directions and its reflected light will not appear to change in brightness as it is rotated. This is analogous to the sort of light seen on an overcast day. The result is no differentiation of color or brightness across the surface.

Diffuse light is directional but is made up of parallel rays. The effect on an object illuminated with a diffuse light is that it changes in brightness as it is rotated due to the change in the incident angle of the surface.

A specular light is one that comes from a specific point such as a light bulb. When an object is lit with specular lights, definite highlights will be seen moving across the surface as it is rotated.

In practice, the ideal combination seems to be a small amount of ambient light, and roughly equal contributions from the diffuse and specular lights.

Surface Operations

Modeler contains several commands that allow you to manipulate whole surfaces. These commands generally apply to the [Current Surface](#).

Current Surface

The current surface is the last surface selected or modified. If you are unsure which surface is the current surface, you can check the list of surfaces in the submenu that appears when you hold the cursor over Properties in the Surfaces menu. Alternatively, you can ensure that the surface you wish to work on is the current surface by selecting any control point in the surface.

Adding Shapes

Modeler has a number of standard shapes, which may be added to a design. They are as follows:

Box

This is a box shaped prism; the primary axis may be selected to be orientated in the longitudinal, transverse or vertical direction, this will determine where the open ends are. The length, width and height are the dimensions in longitudinal, transverse and vertical directions respectively.

Additional surfaces to close the open ends of the prism may be added if desired.

The box is placed on the centre line of the design. If the Symmetrical box is ticked, either a full or half model may be selected. For shapes that lie symmetrically on the centreline, it is best to select the Half Model on Centreline option. To add a complete box, away from the centreline, tick the Symmetrical box and select the Full Model button, then move the box away from the centre line to its desired location using the Move command from the Surfaces menu. If the box appears symmetrically on both sides of the vessel, you can then turn on the symmetry flag for the box's surfaces.

Cylinder

This is a circular or elliptical cylinder, the options work in the same way as the box.

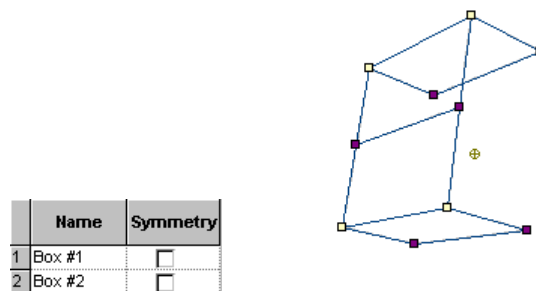
Sphere

This is a circular sphere of specified radius. The various symmetry options are the same as for the box.

Hemisphere

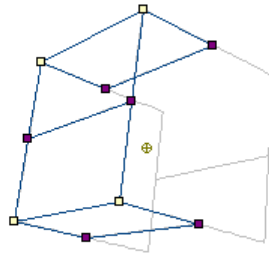
This is a circular hemisphere of specified radius, which may or may not be closed with a disk. The hemisphere may be orientated in the longitudinal, transverse or vertical direction.

The illustrations below show the different effects that may be achieved with the symmetry options. In addition, the front and back ends of the box could have been closed with two additional surfaces if the Close Ends option had been ticked. (Box#1 is the top part of the box and Box#2 is the bottom; Box#3 and Box#4 are the mirrored surfaces.)



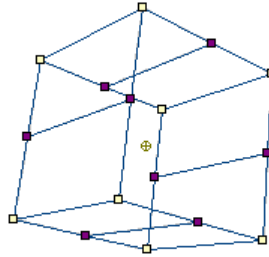
Non-symmetrical model

	Name	Symmetry
1	Box #1	☒
2	Box #2	☒



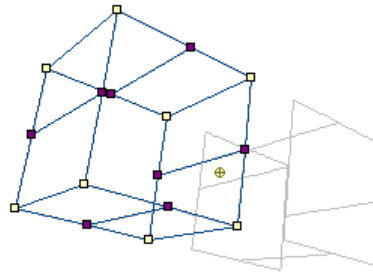
Symmetrical half model on centreline

	Name	Symmetry
1	Box #1	☐
2	Box #2	☐
3	Box #3	☐
4	Box #4	☐



Symmetrical full model on centreline made up of four non-symmetrical surfaces

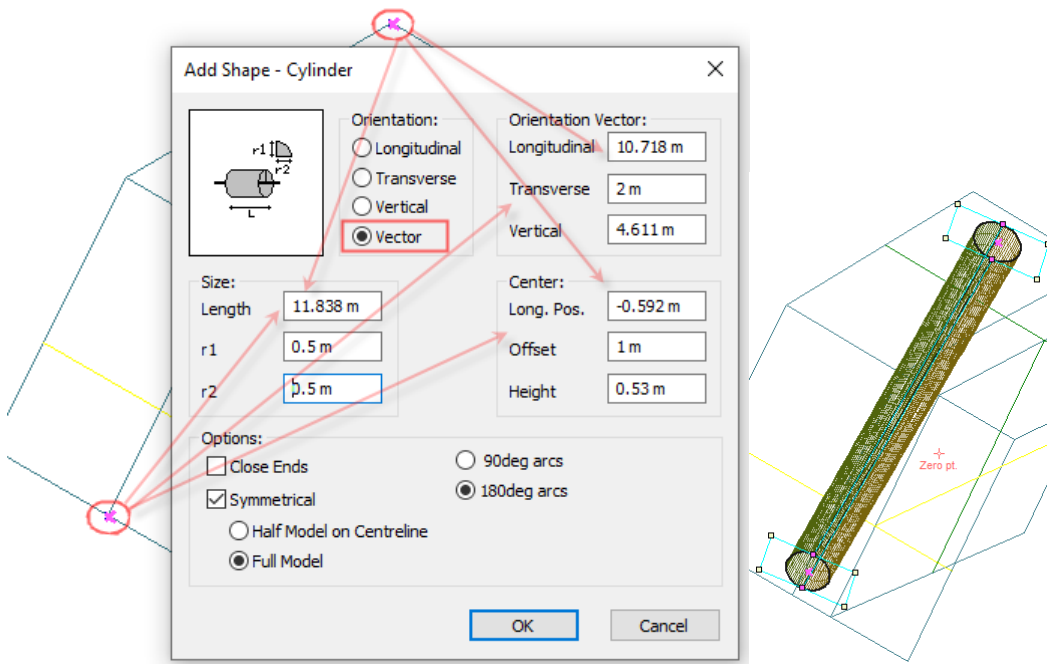
	Name	Symmetry
1	Box #1	☒
2	Box #2	☒
3	Box #3	☒
4	Box #4	☒



Symmetrical model offset from centreline after move command and changing symmetry flags

Select Add Surface to add a surface shape to your design. It may be renamed through the Surface Properties dialog. When adding one of the standard shapes such as a box or cylinder, it is possible to specify:

- Shape center (rather than defaulting to model center always)
- Shape orientation other than along one of the principle axis directions.
- Orientation, center and length will be set automatically if you first select two marker points which define the ends of the shape.

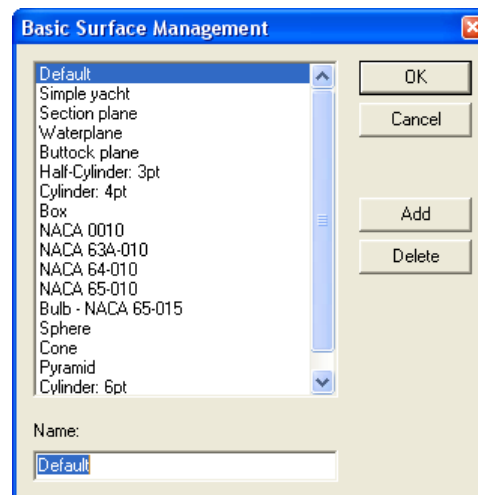


Adding Surfaces

When adding a surface, it is possible to select from a range of different surfaces in the Add Surface menu.

To add or delete surfaces from the list in the Add Surface menu, use the Edit Surface List command from the Surfaces | Add Surface menu. You can add surfaces to this list by holding down the shift key whilst saving the design.

To delete surfaces from the Surfaces list, select the surfaces to be deleted then click the Delete button.

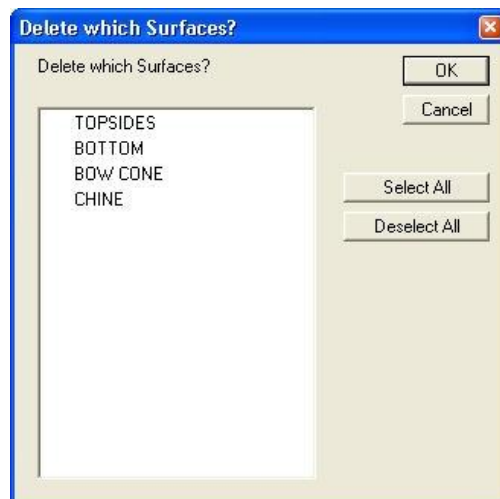


To add surfaces from your existing design to the surface list, click the Add button then choose the surfaces from your design that you wish to add to the list.



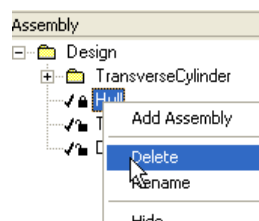
Deleting Surfaces

Choosing Delete Surface presents a dialog that allows you to select surfaces for deletion. Simply click on those surfaces you wish to delete and click OK.



Multiple surfaces may be deleted at the same time, and a surface that is bonded to another may be deleted without the need to unbond first.

You can also use right click command in the Assembly Pane to delete a surface:



Or you can delete any number of selected surfaces directly from the Surfaces window by selecting the surface(s) and pressing the delete key on your keyboard.

Moving Surfaces

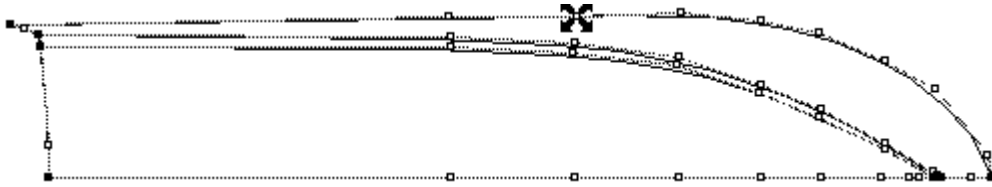
Surfaces may be selected and moved relative to one another. This can be achieved by freeform dragging of the surface or by typing in a numerical offset.

Freeform Movement:

Movement is made in the plane of the window that is being used, or in the case of the Perspective window, in the plane most perpendicular to the current view. To select a surface for movement

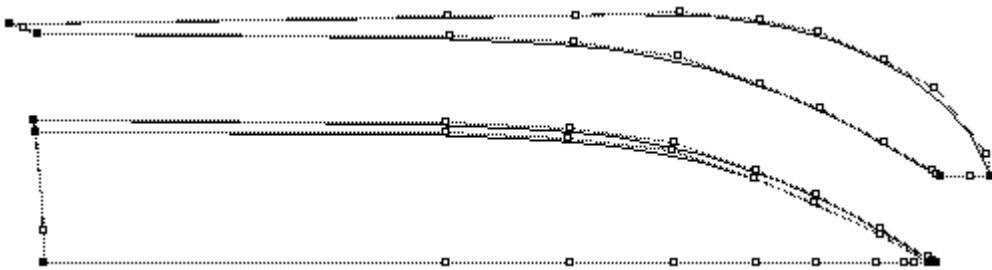
- Select Move Surface - Freeform from the Surfaces Menu.

The move cursor will be displayed.



- Position the cursor over any control point.
- Click and drag the surface to the new location.

An outline of the surface will display the position of the surface while it is being moved.



- Release the mouse button.

On releasing the mouse button the surface will be displayed again at the new location.

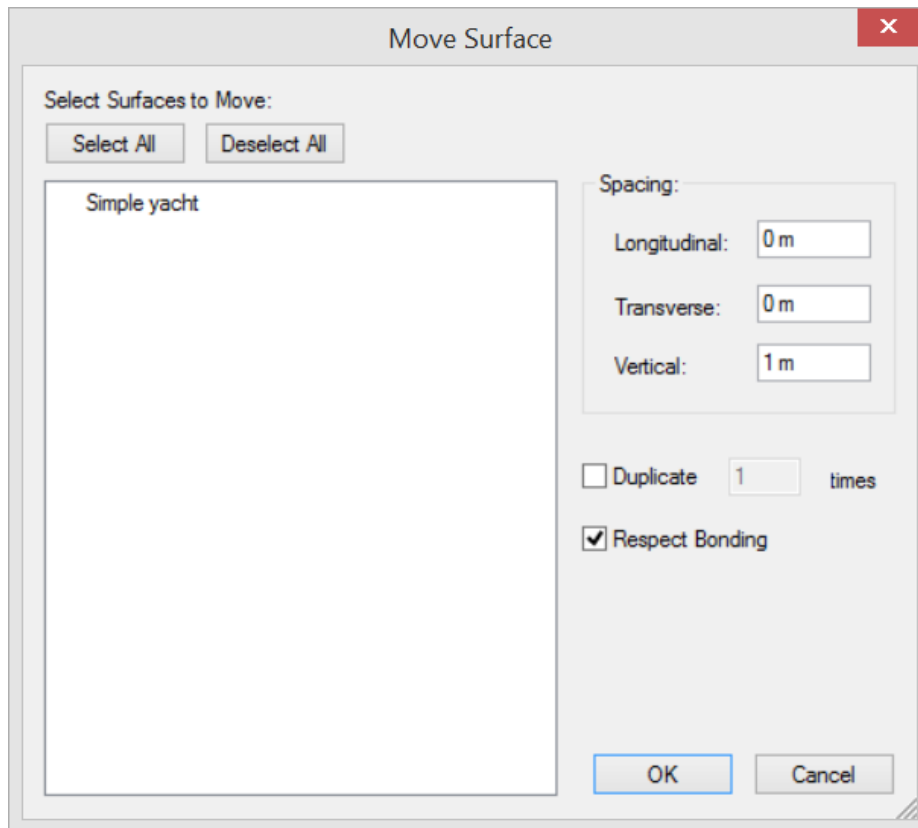
Move Surface will move all surfaces bonded to the surface that is being moved. Note that all these surfaces must be unlocked.

Numerical Movement:

To move surfaces by a specified amount:

- Select Move Surface - Numerical from the Surfaces Menu.
- Select the surfaces to be moved and specify the amount that they should be moved in the longitudinal, transverse and vertical directions.

Locked surfaces or those which are bonded to locked surfaces cannot be selected unless duplicate is selected.

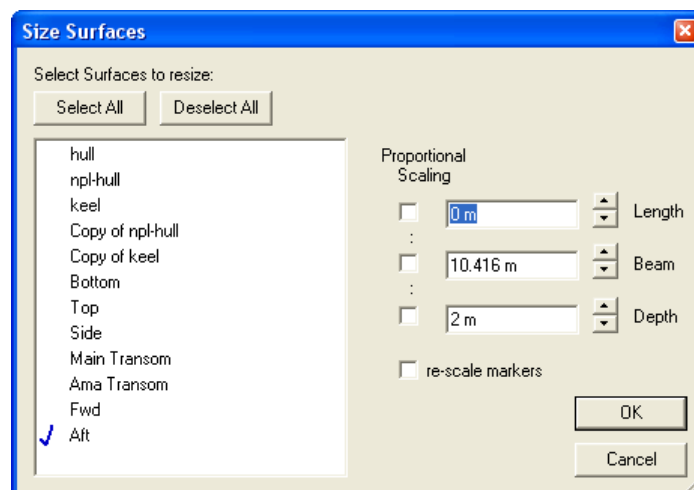


Size Surfaces

The Size function allows a designer to scale and re-proportion a surface or group of surfaces by their principle dimensions. This function is particularly useful when modelling geometrically similar variations from a parent design, and also allows precise specification of principle design parameters.

To resize a design

- Select Size from the Surfaces menu.



- Select which surfaces you wish to resize.

Do this by selecting the surface names from the check boxes on the left hand side of the dialog. When a single surface is selected the size displayed will be the overall dimensions of that surface. When multiple surfaces are selected the size displayed will be the overall dimensions of all selected surfaces.

- Select whether you wish to use proportional scaling.

For example, if you wish to maintain a length/beam ratio, tick the boxes of the dimension to be scaled proportionally (length and beam). When any of the dimension boxes are changed all boxes selected for proportional scaling will be changed in proportion.

Similarly, length, depth and beam may be scaled simultaneously, to produce a geometrically similar vessel, by ticking all three boxes. The desired length, depth or beam may be entered and all dimensions will be scaled proportionally. The dimensions will follow in proportion from the time that their proportional scaling boxes were ticked.

- Select and edit the appropriate size boxes to alter the size.

You may do this by either editing the text directly or by using the scale up / scale down arrows at the right hand side of the size boxes. Dimensions may be entered in any units by specifying the appropriate suffix.

Note:

Only the actual surfaces are changed by resizing. The design grid does not change as the hull is resized. This means that you may need to reset the grid to accurately reflect a new waterline length.

Markers may be scaled in the same proportion as the hull or may be left as they are. Tick the re-scale markers box to scale the markers with the surfaces.

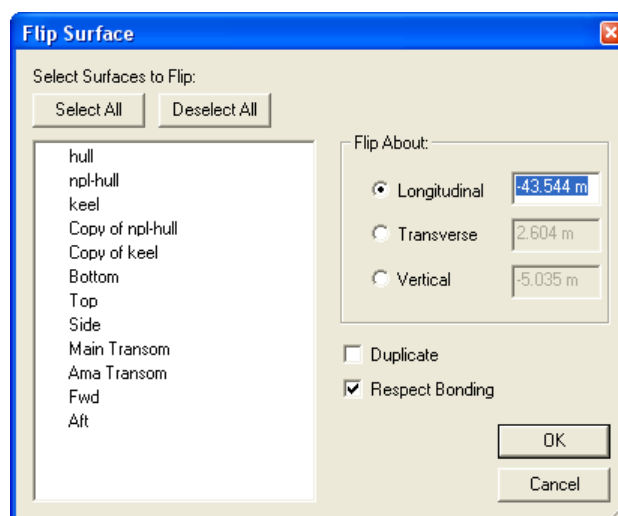
You can size different surfaces by different amounts. This can be done by:

- Selecting the first surface(s) you wish to resize
- Resize them (as described above)
- Deselect these surfaces and select the other surfaces you wish to resize
- Resize these surfaces
- Repeat for all the surfaces you wish to resize
- Click OK

Flipping Surfaces

Surfaces may be flipped about specified planes: longitudinally, transversely or vertically. The actual mirror plane may also be specified, and need not be about the origin. At the same time you may also create a duplicate of the original surfaces.

- Select Flip Surfaces from the Surfaces menu.



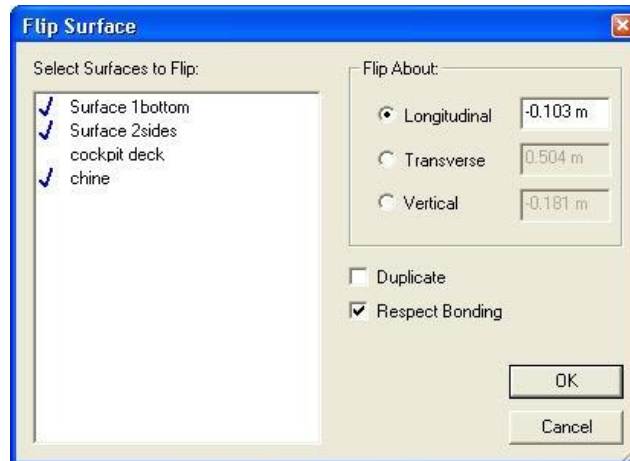
- Select the surfaces you wish to flip.

If you are duplicating the surfaces, they need not be unlocked since the original surfaces will remain unchanged. If you are not duplicating the surfaces then they must be unlocked. If you choose Respect

Bonding, bonded surfaces will be selected together. If you are not making duplicates, all the bonded surfaces must be unlocked.

- Specify the reflection plane.

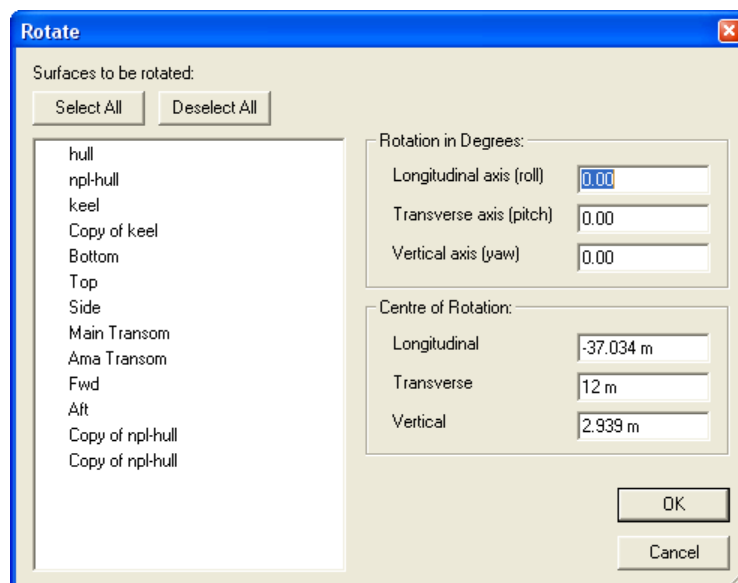
For example, if you wished to make a symmetrical catamaran hull, with a centreline spacing of 8m, you would design one side of the hull, move it transversely 8m then flip and duplicate it about the 8m transverse plane.



Flipping about the longitudinal direction could be used to produce a double-ended design.

Rotating Surfaces

The rotation dialog allows multiple surfaces to be rotated simultaneously around a specified centre of rotation. Surfaces to be rotated are selected by clicking in the list of surface names.



Aligning Surfaces

Two surfaces can be aligned by selecting one control point from each surface. The second control point will be moved, along with its entire surface, so that it aligns exactly with the first control point.

To align two surfaces

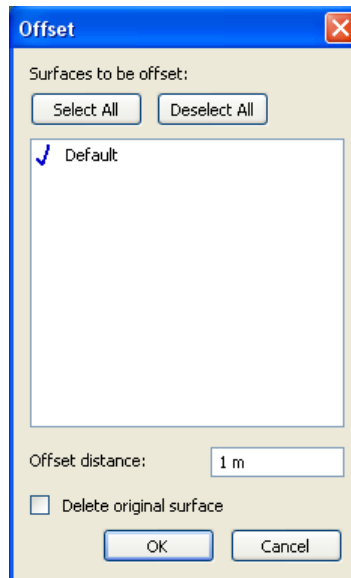
- Select a control point from the surface to remain fixed.
- Hold down the Shift key.
- Select a control point from the surface to be moved.
- Select Align Surfaces from the Surfaces menu.

The second surface you selected will be moved so that the two control points become coincident.

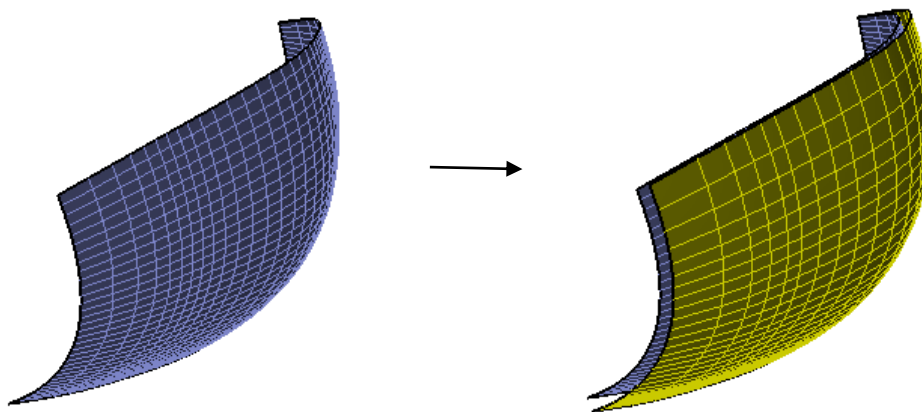
Offsetting Surfaces

The offset surface command from the surface menu is enabled when a design contains one or more surfaces.

To offset a surface choose Offset Surface from the Surface menu. The following dialog will appear:



Select the surface(s) you wish to offset. Only unlocked surface can be offset. If you check "delete original surface" the original surface will be deleted once the surface has been offset. Enter the distance you wish to offset the surface. The surface will be offset in the direction of the outside arrow of the surface. If you wish to offset the surface in the opposite direction to the outside arrow, set the offset distance to a negative number. Below is an example of a surface offset by 1m:

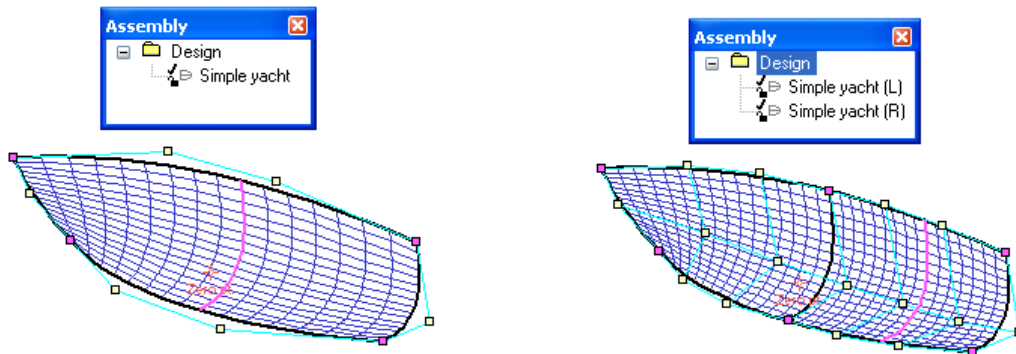


Splitting Surfaces

The Split Surface command from the surface menu is enabled when a parametric curve or a control point is selected on an unlocked surface.

To split a surface using parametrics, first turn on parametrics from the Display Contour dialog box (or by clicking on the parametrics button on the visibility toolbar). Make sure the surface you are trying to split is unlocked. From the Surfaces menu choose Split Surface. Two new surfaces will be created with a common edge at the position of the old parametric contour you selected. To join the surfaces back together you can use the Join Surfaces command as outline in the next section.

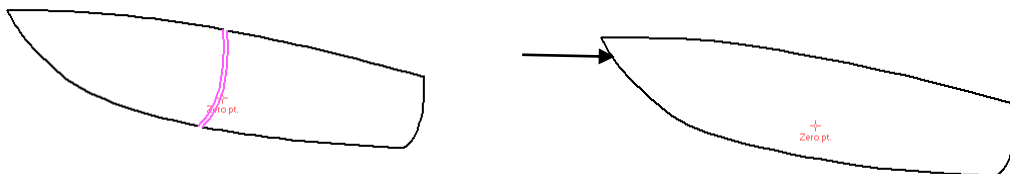
Surfaces may also be split by selecting a control point on the edge of the surface. The surface will then be split perpendicular to the edge at the parametric position of the selected control point. Alternatively you can select 2 control points that lie on a column or row of the surface. The surface will then be split along this row/column. The final way a surface may be split is by selecting a single control point from the body plan view with just one control point column displayed. The surface will then be split longitudinally at this parametric position.



If a bonded edge is split then the split will continue across to the bonded surface(s).

Joining Surfaces

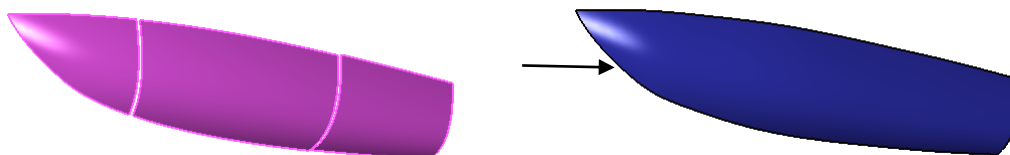
The join surface command will join 2 or more surfaces to create a single surface. Surfaces to be joined must be compatible along their joining edge. Compatible edges have the same stiffness and same number of control points. Two surfaces can be joined by selecting the edges of the two surfaces to be joined:



More than two surfaces can be joined by either:

- Multi selecting them in the rendered view and executing the join surfaces command.
- Multi selecting them by dragging a selection rectangle over the control points and executing the join surface command.

Surfaces will be joined in the order they are selected. Surfaces will be joined along their closest common edges.



Locking Surfaces

Surfaces can be locked or unlocked. When a surface is locked, changes to the surfaces can not be made and the control points are not displayed. You can lock a surface:

- Using right click command in the Assembly tree
- Using the Surfaces | Locking command
- Using the Surface Properties dialog or properties pane
- Using the Surfaces window and ticking the locking column flag

Surface Creation

Modeler surface menu contains several commands that allow the user to create a surface from parent curves or markers

Add NURB Surface from 3 or 4 markers

Use the “Approximate Surface to Markers” command with a selection of 3 or 4 Markers. In this case it will use the Markers to define directly the corner Control points of the Surface without performing a least-squares fit; if 3 Markers are selected then two of the Surface’s control points will be compacted to create a triangular Surface; if more than 4 Markers are selected then the normal least-squares fitting will take place.

Markers should be selected in the following order: bottom-right; bottom-left; top-left; top-right. This will result in the surface normal facing outwards from the screen. Markers need not be coplanar, and if not this will result in a warped surface being generated.

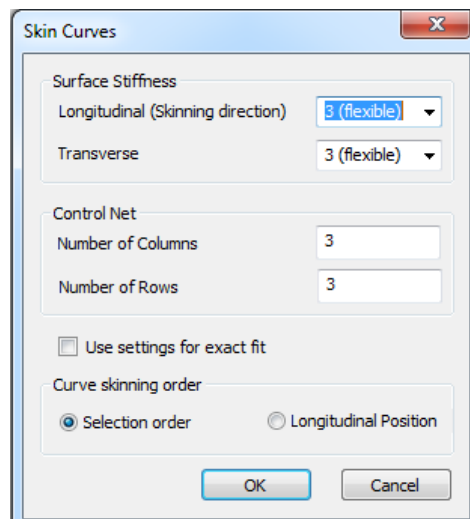


Marker selection and resulting NURB surface

Skinning Curves to Create a Surface

The skin surfaces command creates a surface by skinning around a set of selected curves. The order in which the curves are skinned is dependent on the selection order or the longitudinal position depending on the radio button selected in the dialog. The skin surfaces command is enabled when a design is open and 2 or more curves are selected.

When the command is executed the following dialog will appear:



The degree of the surface created in both directions needs to be specified in the Surface Stiffness edit boxes. The number of control point columns and rows in the control net needs to be specified by the user. The stiffness cannot be greater than the number of control points in longitudinal or transverse directions.

Modeler will automatically adjust corresponding values if this rule is broken. The parametrics of the surface will be created such that the u direction is from start curve to end curve.

The order in which the curves are skinned makes a difference to the result. You can either select curves individually in the order in which you wish them to be skinned (by holding the shift key down) or select all the curves at once by using a selection rectangle. If you use a selection rectangle the curves will be skinned in the order they were created, however there is also the option to skin them according to their longitudinal position. To do this select the "Longitudinal Position" radio button under the Curve skinning order options.

"Use settings for exact fit" automatically sets the stiffness in the longitudinal and transverse directions and the number of control point columns and rows so that the resulting surface passes through each of the curves exactly. When this option is checked Surface Stiffness and Control Net settings are not editable by the user.

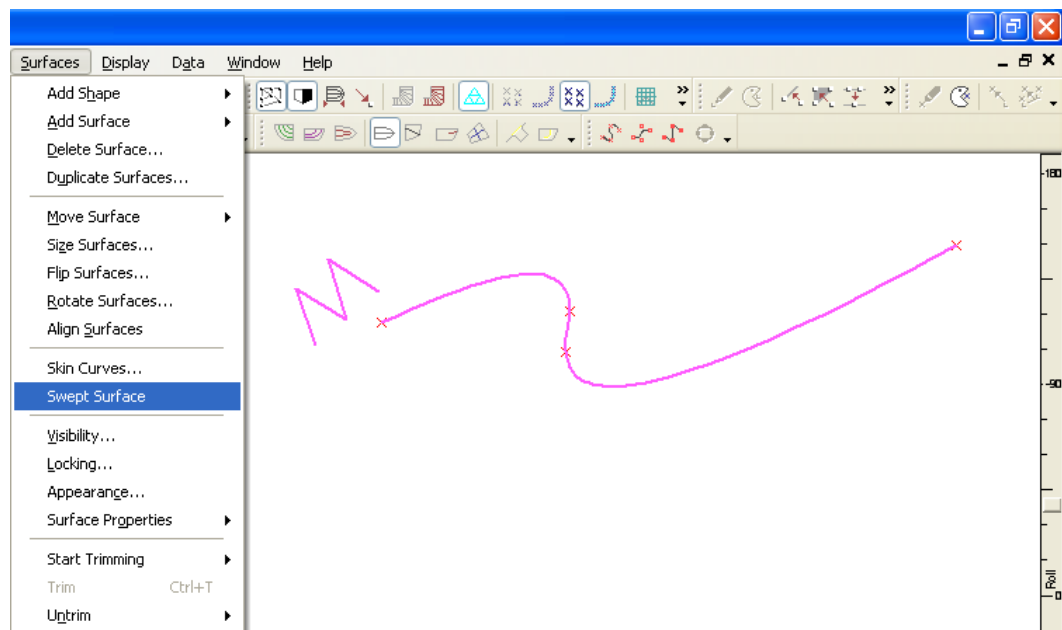
The skin surface command works best when the set of curves are all planar, for example a set of section curves or a set of waterline curves or a set of buttock curves. Mixing of planar curves (for example 5 section curves plus a bow profile curve) may cause unexpected results.

Note:

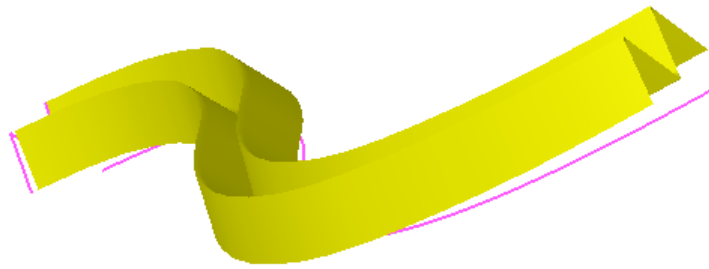
While the surface created using the "Use settings for exact fit" option will pass exactly through each of the curves it does not necessarily create a "regular" surface with a uniform net, a prerequisite for "smooth" surfaces.

Sweeping a Surface

The sweep surface command creates a surface from two curves. Selection order is important with this command. The first curve selected is termed the "section curve" and the second curve the "trajectory curve". The new surface is created by sweeping the section curve along the path of the trajectory curve. The swept surface command is enabled when a design is open and two curves are selected.

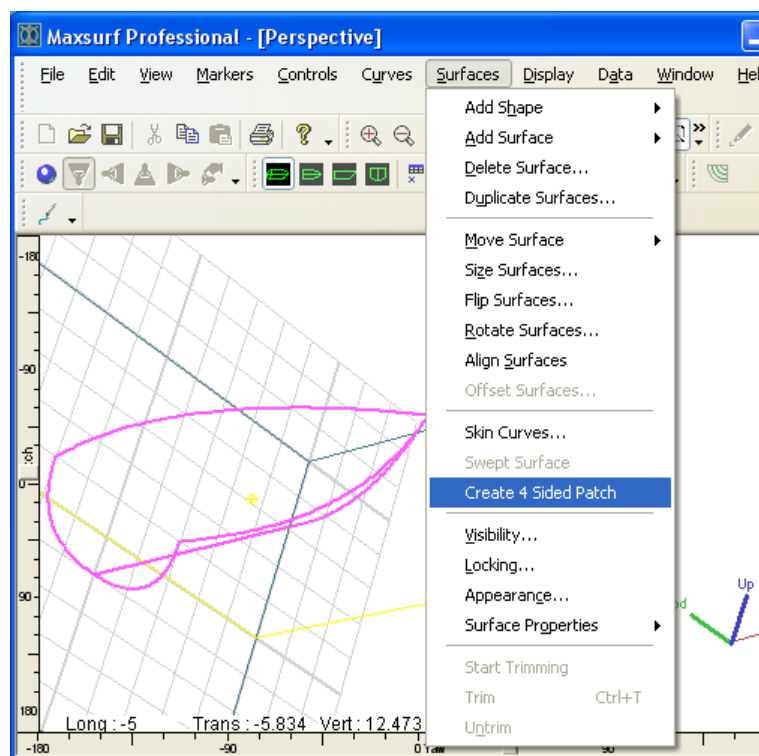


When the command is executed a new surface will be added to the design:

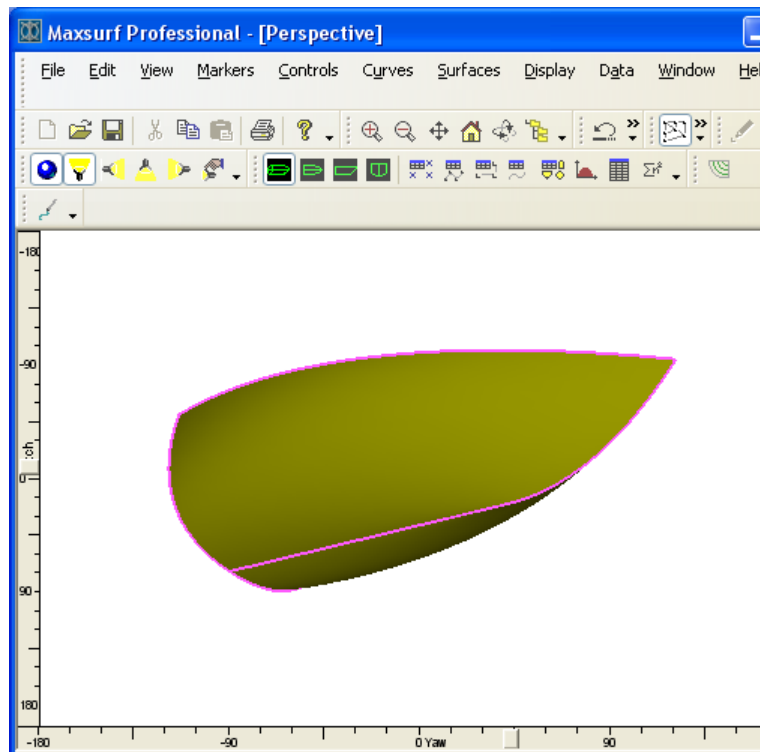


Creating a 4 Sided Patch

If four curves are selected the Create 4 sided patch function will be enabled from the Surfaces menu. The curves must create a closed loop (i.e. the last point of the first curve must be coincident with the first point of the second curve etc). Curves do not need to be compatible (i.e. do not need to share the same number of control points or stiffness). The curves do not need to be of the same type (i.e. control point curve or data point curve). The function is useful for a quick creation of a hull surface from four curves such as a transom curve, a deck edge curve a bow curve and a keel curve:

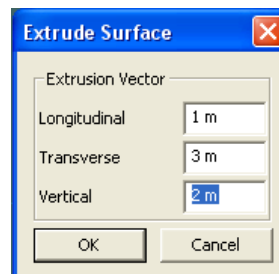


The smoothest fitting surface will be added which can then be refined by the user:



Creating an Extruded Surface

A curve or surface contour may be extruded to form a surface. To form an extruded surface create the section curve, highlight it (or select a surface contour) and choose Extrude Surface... from the surfaces menu. A dialog box will then appear, that asks for a direction vector:

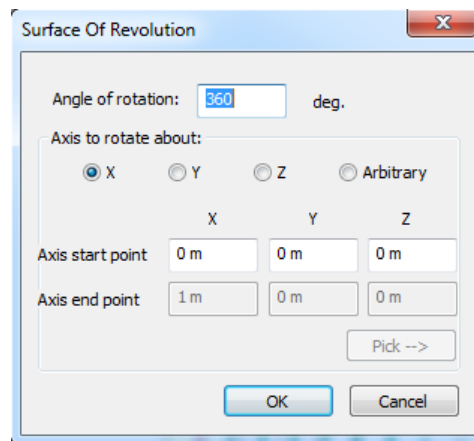


The direction of the extrusion is defined from the Direction Vector, for example a direction vector of Longitudinal=1.0, Transverse=0.0, Vertical=0.0 will extrude the surface along the longitudinal axis of the vessel. The extrusion distance is determined from the length of the vector. The extruded surface will be linear in the extrusion direction and will have a stiffness of 2.

Any number of curves or contours may be selected, an extruded surface will be created for each.

Creating a Surface of Revolution

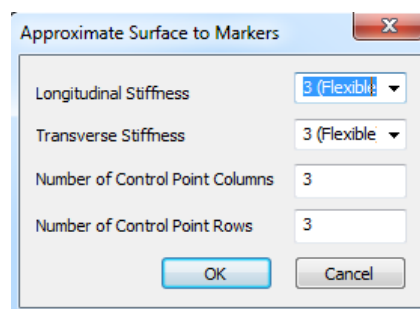
To enable the surface of revolution from the Surface menu at least one curve or one contour on a surface must be selected. When the command is executed the following dialog will appear:



The curve can be rotated about any of the 3 orthogonal axis or an arbitrary axis defined by the user. The arbitrary axis can be explicitly defined by entering the xyz coordinates of the start and end points or by clicking the pick button to graphically pick the start and end points. The angle to rotate the curve about the axis to create the surface also needs to be defined and is by default set to 360 degrees. Once the OK button is clicked the surface(s) will be created.

Approximating a Surface to Markers

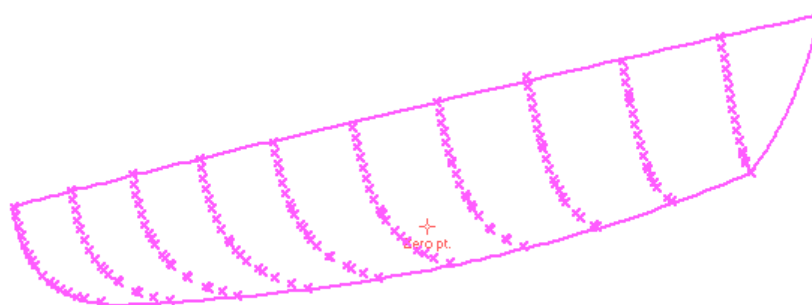
A surface can be approximated to a selected set of Markers. The surface will be a least squares approximation. The command is available from the Surfaces menu and is enabled when 4 or more markers are selected. When the command is executed the following dialog will appear:

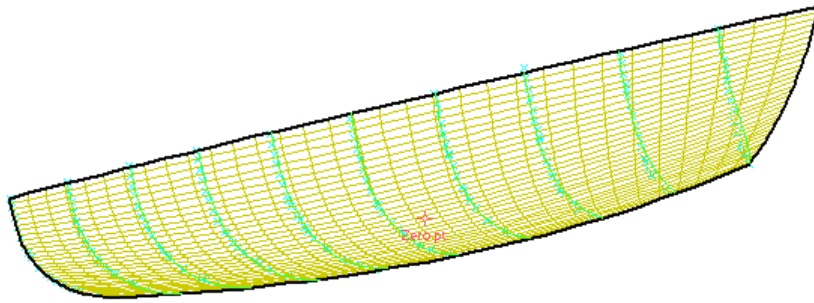


The user can then enter the stiffness and number of control points required for the surface in each of the parametric directions. For a NURBS surface the following rules must be obeyed and they are automatically implemented by Modeler:

- Number of control point columns must be greater than or equal to the longitudinal stiffness.
- Number of control point rows must be greater than or equal to the transverse stiffness.

The user may also wish to define the 4 surface edges by use of Modeler curves. To create a surface with surface edges defined simply select all four edge contours and the markers that you wish to fit the surface to and execute the command from the Surfaces menu.





The number of markers required to fit a surface is equal to the number of control point columns multiplied by the number of control point rows, which are defined by the user in the dialog when the command is executed.

Trimming Surfaces

A surface's boundary and any intersections with other surfaces create regions which may be made visible or invisible. A surface that has invisible regions is said to be trimmed. A surface that intersects the trimmed region may be referred to as the trimming surface.

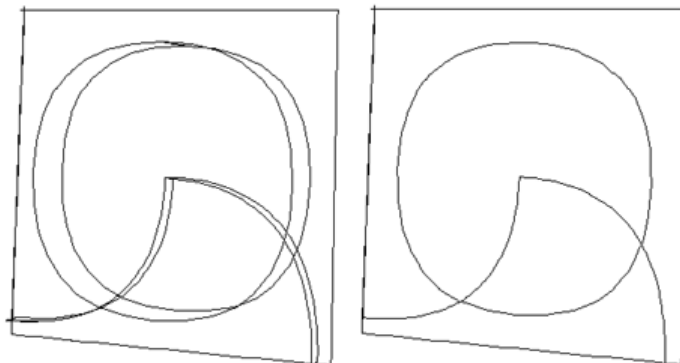
- [Trimming Concept](#)
- [When to Use Trimming](#)
- [Trimming Rules](#)
- [Trimming a Surface](#)
- [Displaying Trimmed Surfaces](#)
- [Untrim Surface](#)

Trimming Concept

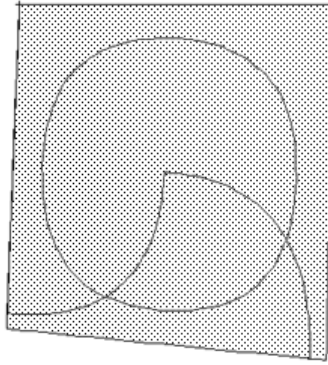
Modeler uses the concept of a 'region' to define an area of a surface that may be visible or invisible (trimmed off). These regions are defined by the boundaries of the surface and any intersections that the surface has with any other cutting surface. Cutting surfaces are user defined surfaces that are used to trim the surface.

Before trimming any surfaces, it is useful to check that the surfaces intersect properly. Since, if this is not the case, it will not be possible to correctly trim the surfaces. To display the surface intersections, select Contours from the Display menu and place a tick next to the Intersections item. Surface intersections are normally displayed in yellow. Check that these are as required.

Consider the sample shape below. The left picture below shows four surfaces: a planar surface which we wish to trim, a cylindrical surface and two arc-shaped surfaces. We wish to trim the planar surface to the boundaries defined by its intersections with the other three surfaces.

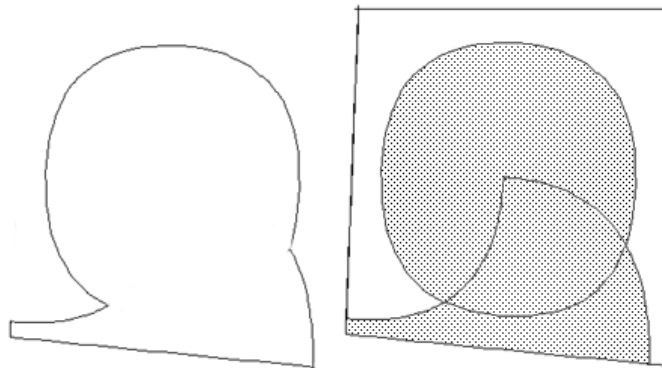


Initially, all of the regions on the surface are visible and selected. We need to de-select the regions that need to be trimmed off.



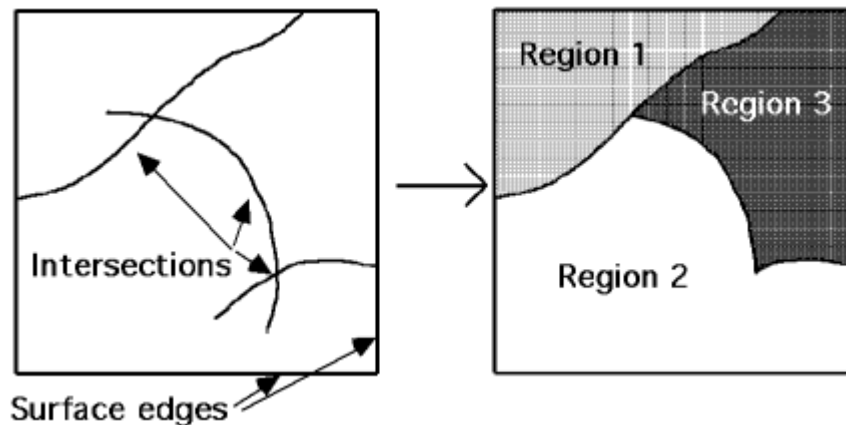
All regions are initially selected, regions to be trimmed off are removed by clicking in the unwanted region.

Once we trim off the unwanted regions we are left with the required shape.



Perhaps the most important thing to remember is that for a region to be used as a trimming region, it must be fully bounded. In other words, the perimeter of the region must be fully closed. If a perimeter is not closed, a region cannot be properly formed. The diagrams below explain the difference between closed and unclosed regions.

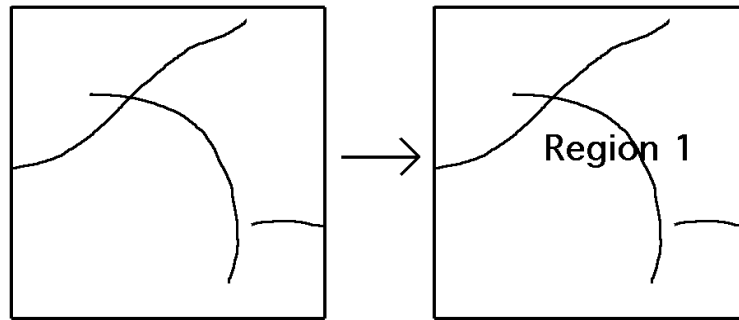
GOOD: (Closed regions)



POOR: (Unclosed regions)

Note:

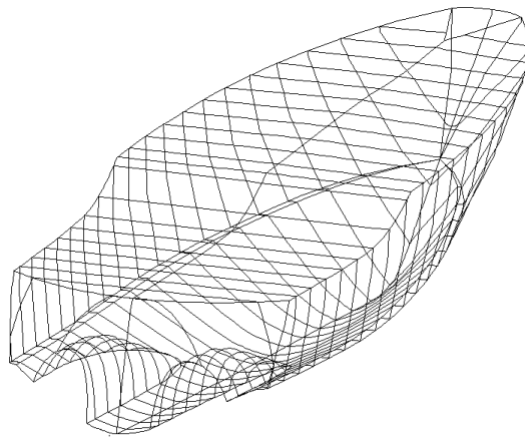
Modeler will find all the closed areas defined by the surface edges and any intersections. Even though two of the three intersections do not define fully bounded regions, when placed together, Modeler will find both regions 2 and 3.



In this example, the intersections do not define fully closed regions, so Modeler will find a single region, which covers the entire surface.

When to Use Trimming

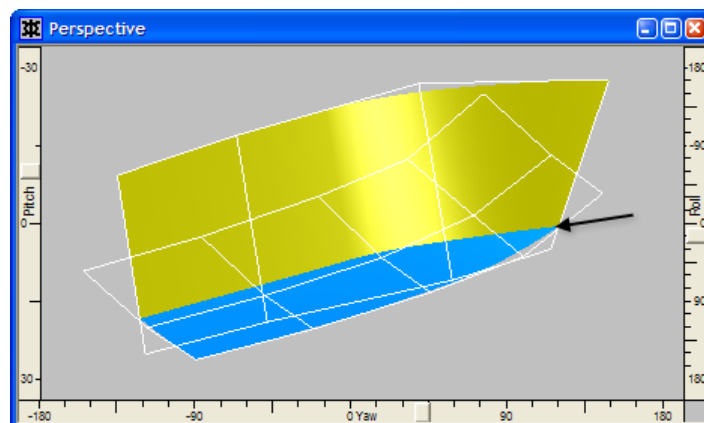
Trimmed surfaces allow complex shapes to be modelled in Modeler. Trimming is most likely to be useful when the surface is fair but there are discontinuities in the edges of the surface or a hole has to be cut in the interior.



Typical applications:

- Deck edges
- Transom
- Propeller tunnel
- Bowthruster
- Non-tangential Skeg / Keel - hull transitions
- Superstructure penetrations through decks

Another reason to use trimming could be when two surfaces are combined to form one shape, while they do not have the same properties such as stiffness or number of control point columns.



An example of a Chined hull using trimmed surfaces. It is difficult to control the shape of the intersection line using two trimmed surfaces. In this case it is often preferable to use bonding.

The disadvantages of using trimming for this purpose are:

- No or little control over the shape of the intersection line of the two surfaces, especially if the intersection line is quite shallow (eg a propeller tunnel with a smooth transition with the hull)
- It is more difficult to make changes to the model whilst maintaining trimming information. The trimming is easily lost.

It is often better to use bonded surfaces when two surfaces have to be combined into one shape.

Note:

In general it is recommended to use trimming to create discontinuities in the surfaces edge or holes in the surface interior.

Instead of trimming you can often also choose to use bonded surfaces. One of the main advantages of using bonded surfaces is the ability to join two surfaces and have control over the intersection shape. Models with bonded surfaces are easier to make changes to and export to other programs. See [When to Use Bonding](#) on page 150.

Trimming Rules

Trimming is an extremely powerful tool to model a wide variety of shapes using surfaces. However, trimming is also a common cause of difficulties with the surface model that can easily be prevented by following the guidelines below.

1: Closed regions

Make sure the region you wish to trim is completely enclosed by intersection lines or edge lines.

- Symptom: the region cannot be selected
- Check the intersection line carefully by zooming in.
- Check the centreline control points are on the centreline (do a multiple selection and a control properties, check offset = 0).

2: Overlap

Use sufficient overlap on the trimming surface. Plenty of overlap helps to identify remnants of untrimmed surfaces, makes trimming easier, makes it easier to change the trimmed surface without losing trimming information etc.

- Symptom: region cannot be selected, precision settings affect trimming. See images below for explanation why precision can affect trimming.
- Never put the trimming surface's edge exactly in the plane of the surface that you want to trim.

3: Over-compacted control points

Do not over-compact control points to create knuckles in the surface. See [Discontinuities and Feature Lines](#) on page 214.

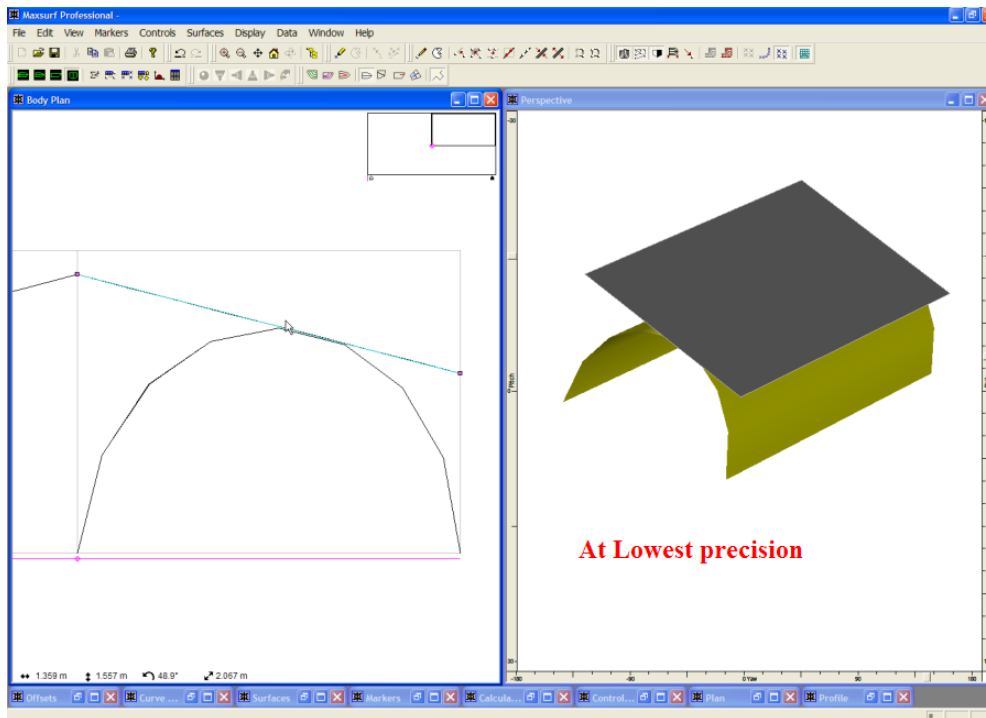
4: Surfaces exactly on top of each other

Do not place two surfaces exactly on top of each other.

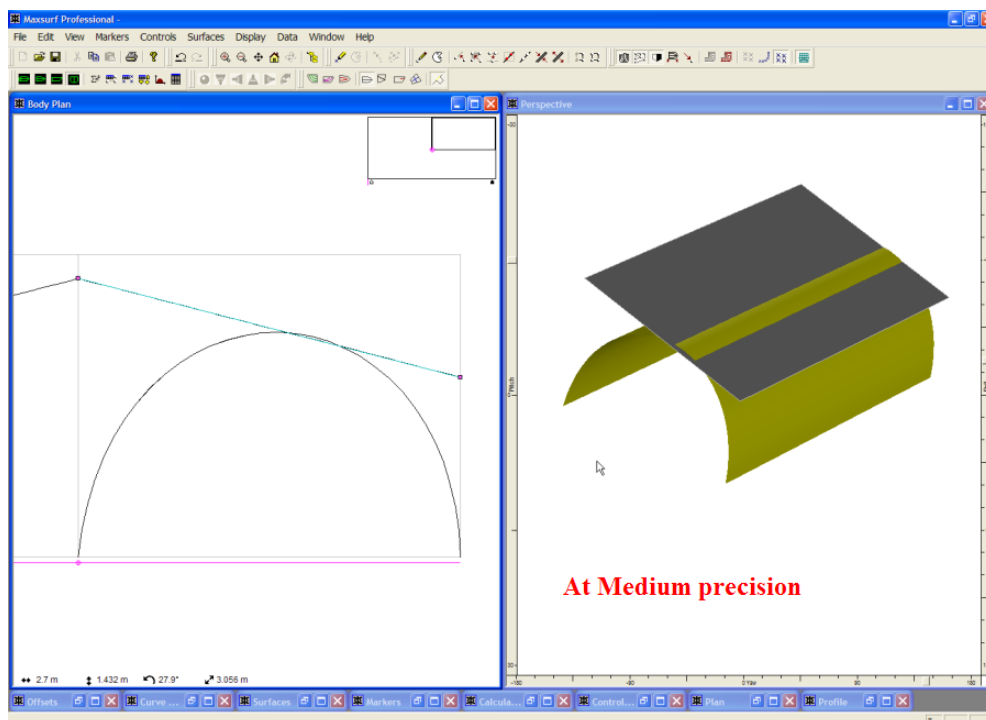
- Symptom: nothing happens after Start trimming is selected
- Make sure the surfaces are sufficiently spaced so that there is no maze of intersection lines visible.

5: Make sure trimming works at medium & highest precision

It is important to make sure the model trims at medium and highest [Surface Precision](#). This guarantees that the model can be transferred to other programs without any problems. Sometimes it may be better to use bonded surfaces as well, see [When to Use Bonding](#) on page 150.



No trimming region at lowest precision



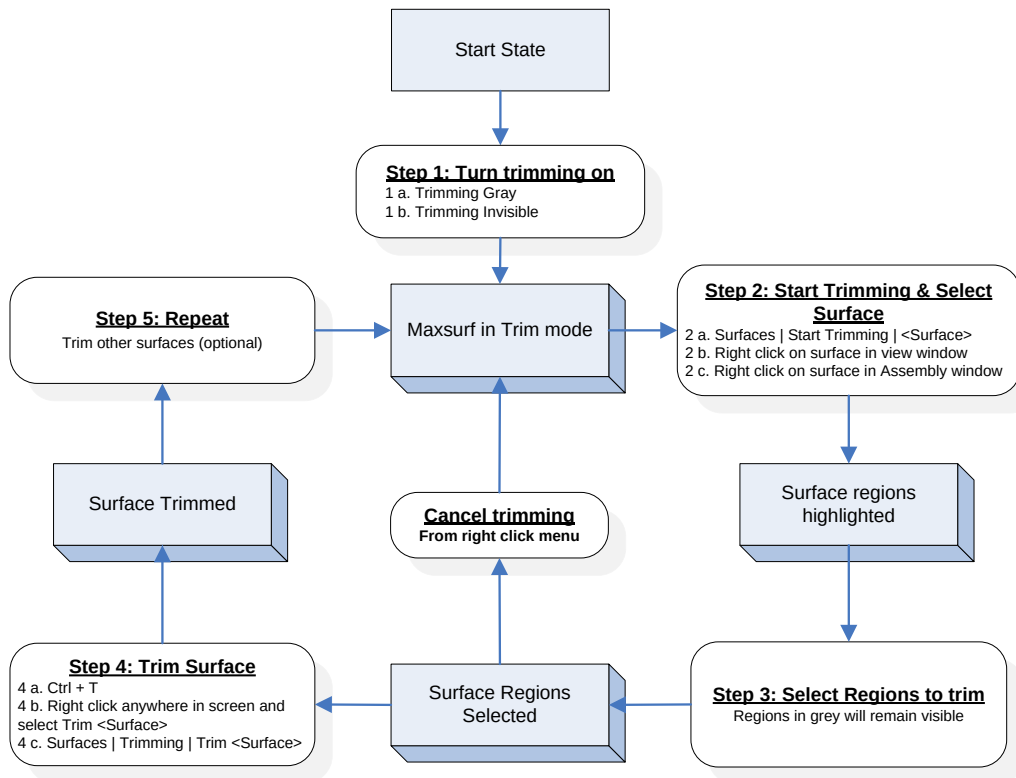
Trimming region at medium precision

Not only Modeler uses the trimmed surfaces for its calculations, also the other programs in the Modeler suite require the model to be trimmed properly.

All trimming rules originate from the trimming concepts, so make sure you have read the [Trimming Concept](#) section on page 139.

Trimming a Surface

First have a look at the following flowchart giving an overview of the trimming steps:



Note

It is important to note that after you have taken step 2, you must do steps 3 and 4 as well. Modeler will continue to ask you to select your regions and trim the surface until you have finished Step 4.

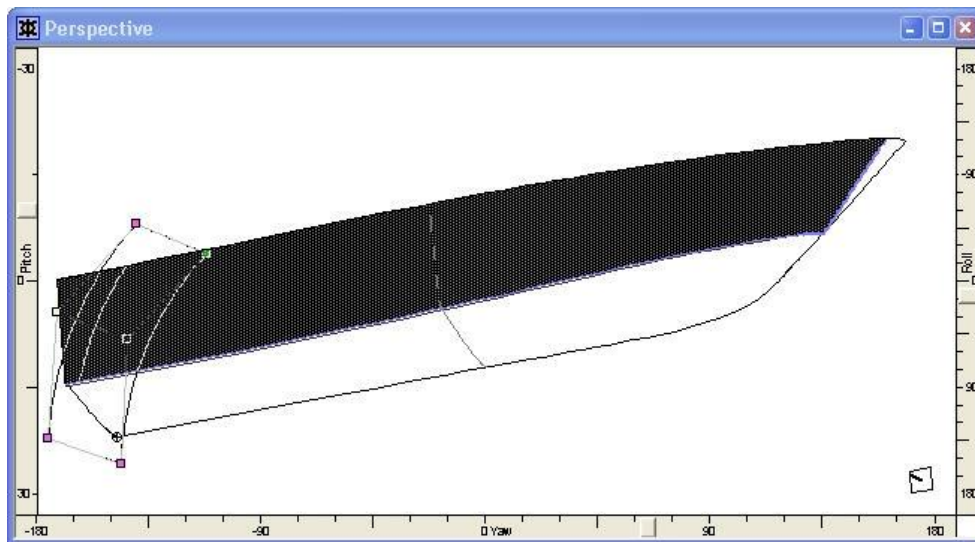
Example:

The following example shows a chined vessel. The boat has been modelled with a hull and a transom. The transom will be used to trim off the aft most sections of the hull.

To trim the hull surface

- Turn on trimming by selecting Trim Gray or Trim Invisible from the Display - Trimming menu.
- Select the surface to be trimmed from the Surface - Start Trimming menu. (e.g. Hull)

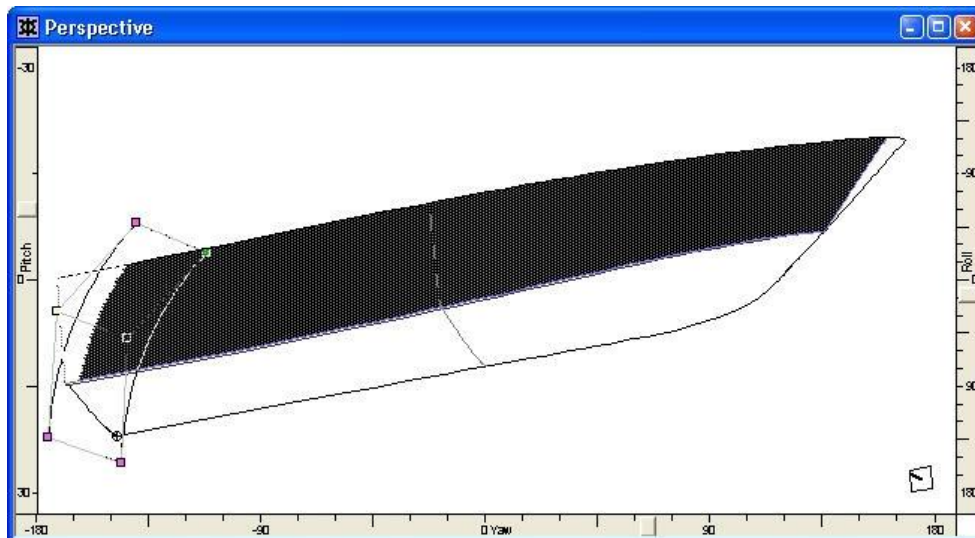
Modeler will show the hull surface with all the currently visible regions selected. In this case since the whole surface is visible, the whole surface will be selected. A selected region is shown with Gray shading.



To change the visibility of a region, click inside that region. The shading will be turned off or on as appropriate to show whether it is selected or not.

In this case, to trim off the area behind the transom (i.e. make it invisible), simply click on the region defined by the edges of the hull, and the hull's intersection with the transom. When this region is clicked on, the shading will disappear.

After clicking in the region to be trimmed off, the screen looks like this:



The window may be zoomed, panned or rotated while regions are being selected.

Now that all the regions to be trimmed off have been selected

- Choose Trim from the Surface menu (or use Ctrl T).

The contours on the design will now be recalculated and trimmed to the visible regions of the hull surface.

Trimming is often easiest in the perspective window with the "Half" display option turned on. This way you will only see the actual surfaces and not their mirror images. Rotate the view direction so that you can select the desired region. You can zoom and pan the design whilst trimming; however, you cannot change window.

Displaying Trimmed Surfaces

Modeler provides the option of displaying trimmed contours in Gray or ignoring trimming altogether.

The Trimming command in the Display menu controls how Modeler will display trimmed surfaces. If Trimming Off is selected, Modeler will not display or recalculate any trimmed surfaces. This provides a method for making changes to the design without the overhead of recalculating the surface trimming after each change.

If Trim Gray is selected, Modeler will trim off any parts of the contours that are not in the visible regions of the surface, and show the trimmed off parts in grey.

If Trim Invisible is selected, Modeler will trim all contours to the visible regions of the surface and not display the trimmed contours.

If the trimming mode is set to off, and then turned on (or Gray), Modeler will find all the intersections, and attempt to reform any trim regions that you have selected. For example, you could work in low precision with trimming turned off, and switch to high precision with trimming turned on to have a more accurate look at the design without having to re-trim the design.

In some cases, if the design has been substantially changed, it may be necessary to re-trim some surfaces. It is possible for a complete surface to be trimmed away and hence invisible when trimming is turned on, if this happens use Trim Gray to make the surface visible or untrim the surface; see [Untrim Surface](#) in the next section.

A good design will trim consistently at all precisions. If you find that the trimming behaviour changes at different precisions, your surfaces may not intersect consistently as you change the precision. The best way to overcome this type of problem is to increase the amount by which surfaces overlap.

Also be aware that you may lose or change trimming information if you add new surfaces or move existing surfaces or control points so that the trimming regions change considerably. It is generally best to do the trimming when you have virtually completed the design.

Untrim Surface

The trimming information for any surface may be removed by selecting Untrim - Surface name from the Surfaces menu. Only surfaces that have been trimmed are listed in the Untrim menu.

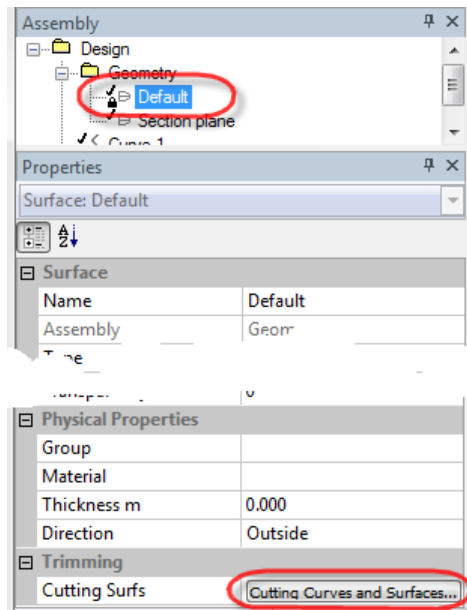
Note

Only visible surfaces are listed in the trim menus.

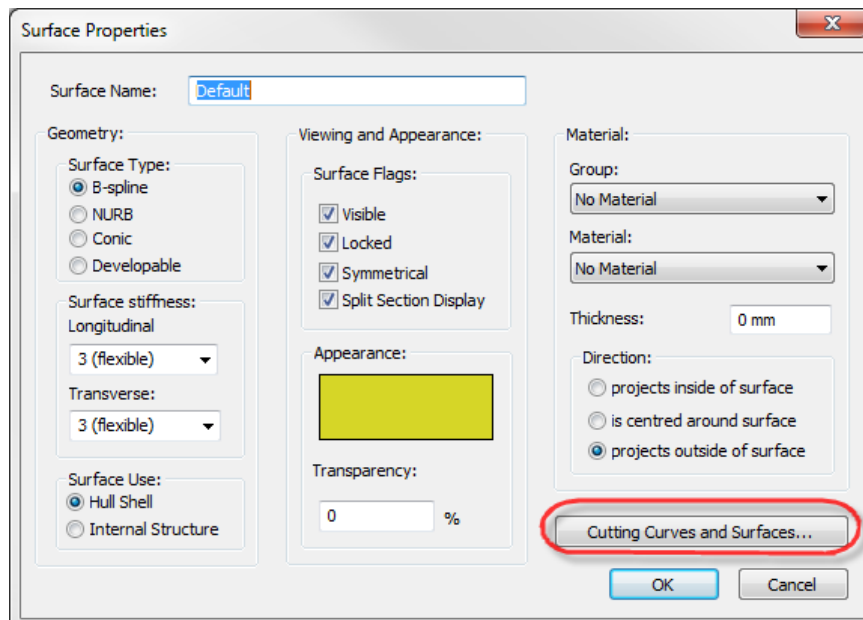
Also see the section on Cutting Surfaces.

Cutting surfaces

To help facilitate designs with more complex surface configuration Modeler has the concept of "cutting surfaces" and "Cutting Curves". For each surface a list of cutting surfaces and curves is assigned; these surfaces and curves are used to divide the surface to be trimmed into regions. By default each surface has all other surfaces in the design assigned as cutting surfaces and none of the curves. The user can manually select the cutting surfaces and curves for each surface that is going to be trimmed. This can greatly reduce the complexity of trimmed designs with large numbers of surfaces. Cutting surfaces are assigned from the Surface properties dialog or from the surface property sheet by clicking on the "Cutting Curves and Surfaces" button.

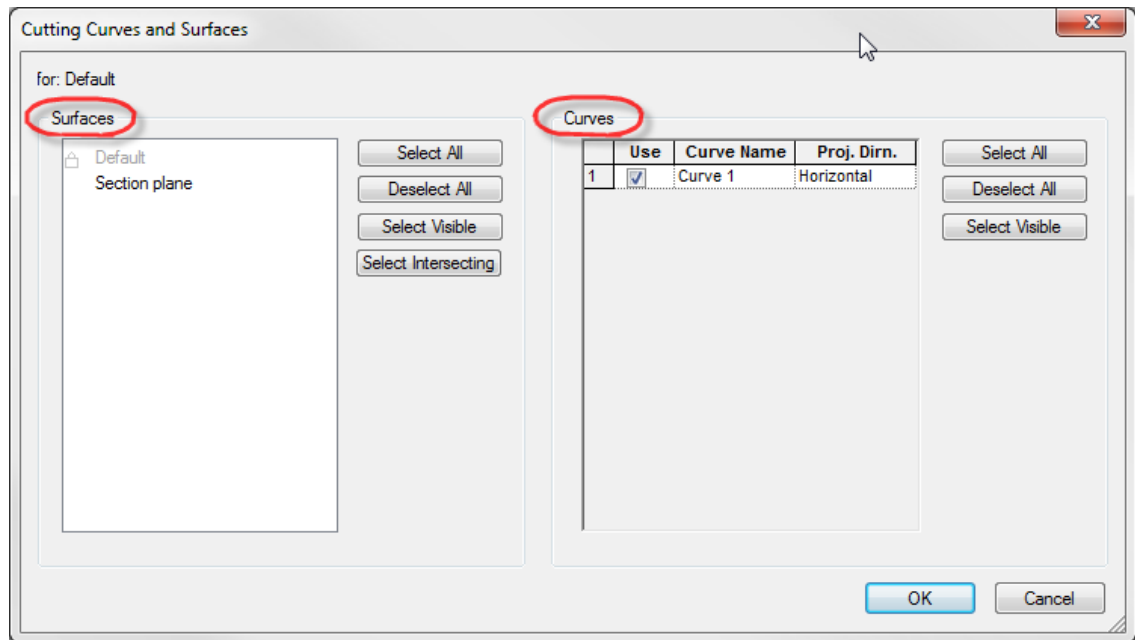


Accessing the Cutting Curves and Surfaces dialog from the Properties pane



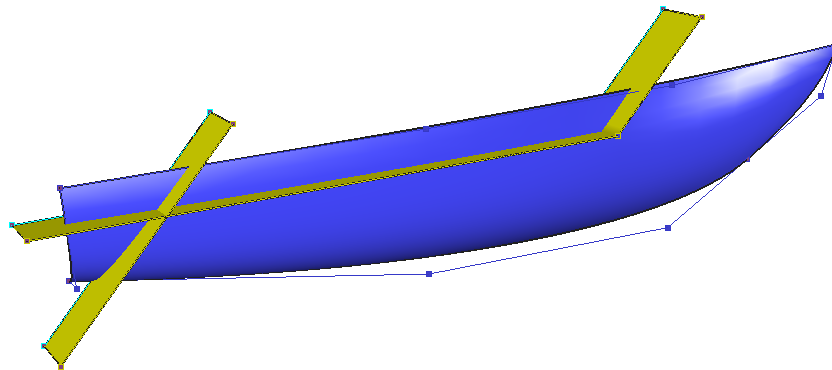
Accessing the Cutting Curves and Surfaces dialog from the Surface Properties dialog

This will load the Cutting Surfaces dialog box from which the user can check which surfaces will be used as cutting surfaces; for the cutting curves you must specify the projection direction that is used to form the intersection with the surfact:

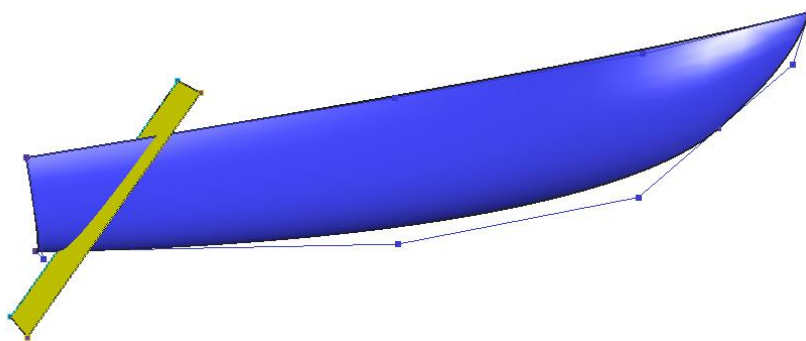


Cutting surface and curve selection dialog

As a simple example the main hull surface (blue) in the following diagram 4 trimming regions if all the surfaces are used as cutting surfaces:

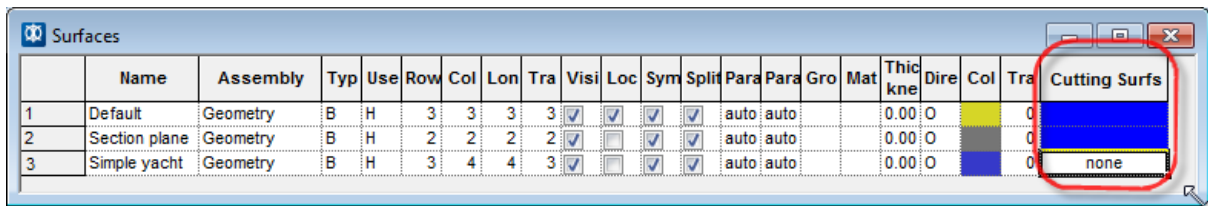


However if the hull surface has only the Transom surface assigned as a cutting surface then the hull will be divided into two trimming regions only.



Copying Cutting Surfaces and Curve Lists

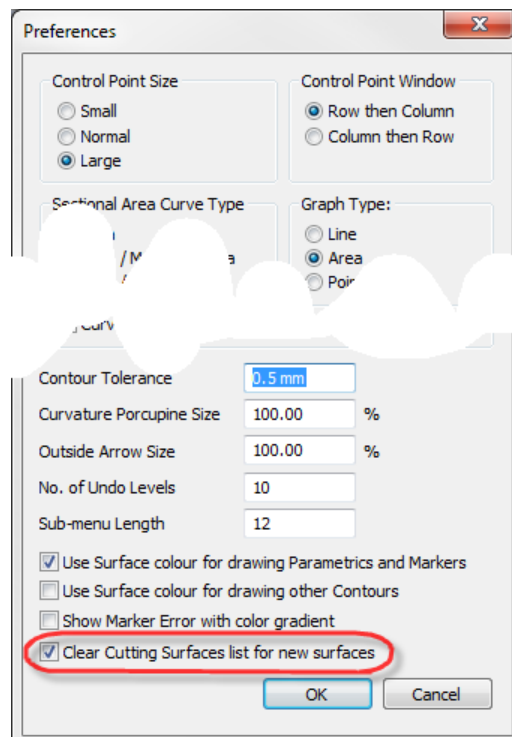
If several surfaces require the same cutting list, you may copy these data using the "Cutting Surfs" column in the Surfaces data table:



Copying Cutting properties

Cutting Preferences

By default all surfaces do not use all other surfaces as potential cutting surfaces. However this can be inconvenient for simple models with few surfaces. If desired you can modify the preferences so that new surfaces are automatically added to the cutting surface list of the other surfaces in the design. This option may be found in the Edit | Preferences dialog:



Ensure that the cutting list for new surfaces is empty and that new surfaces are not added to the cutting list of existing surfaces in the design

Bonding Surfaces

Modeler allows you to join two surfaces together along a common edge so that the two surfaces behave as one larger surface. This procedure is referred to as Bonding. The edges may be in the same surface, or in separate surfaces.

Bonding is an extremely powerful tool to help you create all kinds of features such as areas with specific radiuses, chined transitions etc. Aside from that Modeler has the capability to enforce specific tangential continuity across a bonded edge.

- [Bonding Surfaces - Concepts](#)
- [When to Use Bonding](#)
- [Bonding Surfaces - Procedures](#)
- [Bonding Types](#)
- [Viewing bonding types](#)
- [Changing bonding types](#)

- [Error! Reference source not found.](#)
- [Bonding within a Surface](#)
- [Commands Affected by Bonding](#)

Bonding Surfaces - Concepts

Two rules govern which edges may be bonded together:

1. The two edges that are to be bonded must share the same number of control points along their respective edges.
2. The two edges must share the same stiffness in the direction of the edge.



Two surfaces prior to Bonding

When to Use Bonding

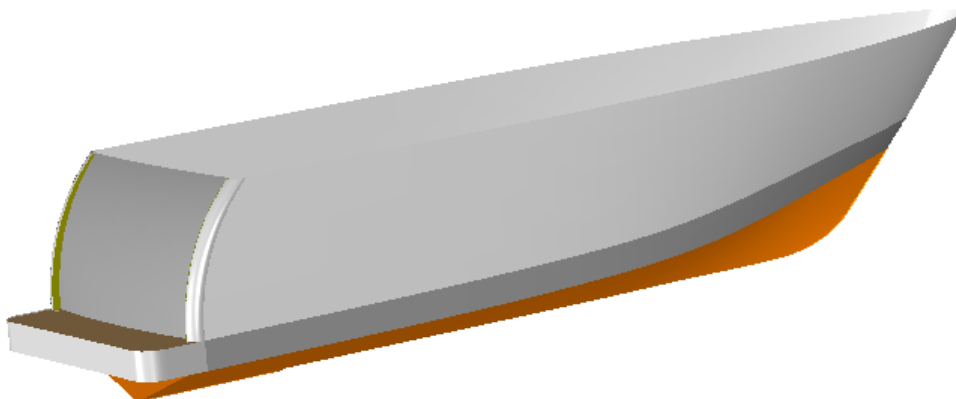
Bonded surfaces are typically used to achieve particular design features where different types of surfaces or surfaces with different properties have to be used.

Typically a bonded surface is used to achieve a combination of regular hull surfaces:

- Areas with a specific radius using conic surfaces: bilge, bowcone, transom-hull transition
- Areas with no curvature using linear surfaces: flat plate
- Areas where you require a different stiffness in the direction other than the direction of the edge. For example a transition from transverse stiffness 5 to transverse stiffness 3 along a longitudinal bonded edge.

Example applications

- Knuckles in the hull, chines
- Bowcones
- Bilge radiuses
- Flat of sides, flat bottom plate
- Box shaped models, deckhouses that require plate expansion
- Bulbous bow transition into hull
- Tangential keel – hull transitions



Example of a model using bonded surfaces; bow cones, swim platform, radiuses and large fillets.

The advantages of using bonding are:

- Tangent transitions between surfaces with different properties.
- Control over the shape of the bonded edge.
- Knuckles can be created without adding a lot of extra control points as opposed to using compacted control points.
- Model transfers into other programs more easily (no trimming problems)
- By dividing the model up into different panels, the plate development can be a lot easier in Structure. You can use the Develop Surface command in Structure without having to subdivide the surface.

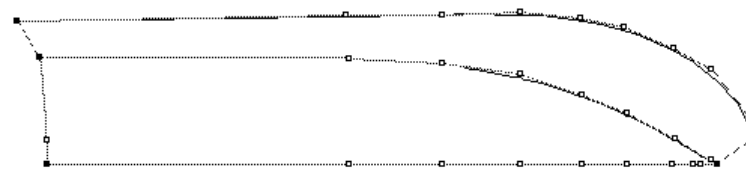
The disadvantages of using bonding are:

- It can sometimes be difficult to change the surface shape of the bonded surfaces due to tangency constraints. Applying the tangent transition at a later design stage and some practice can solve this.
- Certain changes, such as changing the number of control points on the edge or the stiffness of the edge, affect both surfaces. This may result in:
 - A large number of surfaces (use the Assembly Pane to manage large number of surfaces)
 - A large number of control points

Bonding Surfaces - Procedures

Two edges need to be selected for bonding. The first edge selected will remain stationary in space while the second edge is moved to exactly match it. The process of bonding may be performed by using the following steps:

- Select the edge that you wish to remain stationary first,
 - While holding the Shift key down, select the second that you wish to move and bond to the stationary edge,
 - Choose Bond Edges from the Controls Menu and select the type of tangency you wish to apply to bond the edges together.
- Or,
- Select any control point, excluding corner points, from the edge that you wish to remain stationary first,
 - While holding the Shift key down, select any control point, excluding corner points, from the edge that you wish to move and bond to the stationary edge,
 - Choose Bond Edges from the Controls Menu and select the type of tangency you wish to apply to bond the edges together.



Two surfaces Bonded together

Corner points cannot be used to select edges, as they fall on two edges simultaneously, and so do not uniquely define one particular edge. Hence edges that you wish to bond must contain at least one intermediate control point.

Only two edges may be bonded together. However, if you wish more surface edges to be bonded you can manually bond the third edge by compacting and grouping the control points and make the surface stiffness's the same.

Also see:

[Advanced - Transpose](#) on page 221 for information on how to swap rows and columns.

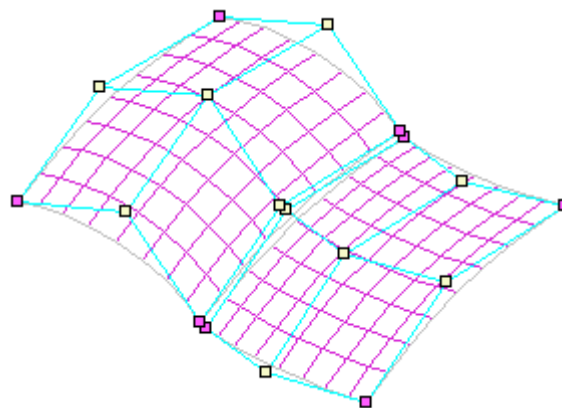
Bonding Types

Modeler has additional options for bonding together two surface edges while maintaining overall fairness across the surface edge. This is particularly useful in situations where it is necessary to create a number of surfaces in an area of the hull but no discontinuities are desired. Modeler has three options to bond together two edges

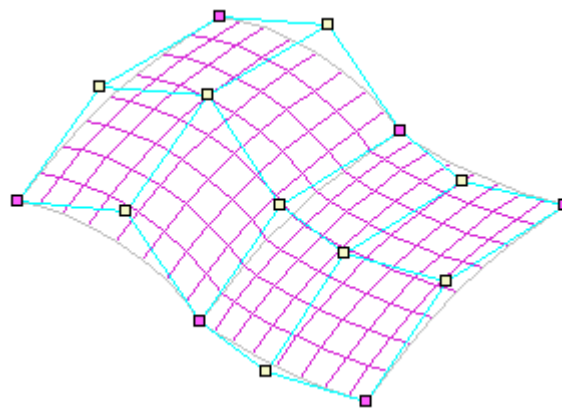
- No Tangency
- Approximate Tangency
- Strict Tangency

No Tangency:

Ensures that each control point along the bonded edge maintains the same location as the corresponding point on the other surface and the surface has the same stiffness in the direction of the bonded edge.



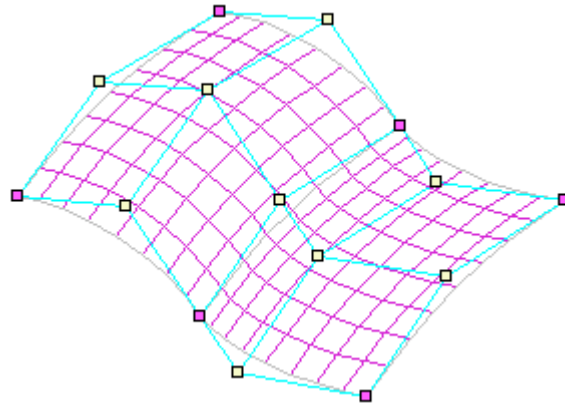
Original surfaces to be bonded



Bonding with No Tangency, points on edge share same location

Approximate Tangency:

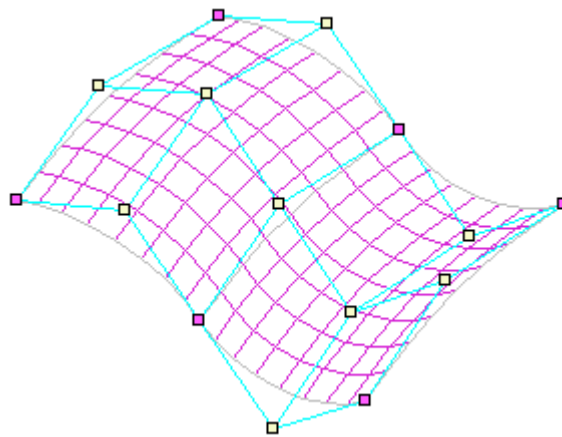
Ensures that the two control points in the same row or column perpendicular to the edge remain in a straight line. This is the same effect you can get by using the Align to Vector command and it helps ensure that the surfaces share a common tangent direction across the edge. In some cases, particularly when there is significant twist in the surface, this is not sufficient to guarantee tangency and the third bonding option must be used.



Approximate Tangency – Rows are colinear across the surface edge

Strict Tangency

Ensures both that each row or column running across the surface edge is kept in alignment with the corresponding row or column on the other surface, and also that the next control point on each side of the edge is kept at an equal distance from the edge. This guarantees tangent continuity in all cases.



Strict Tangency – Rows are co-linear across surface edge and points are equal distance from the edge

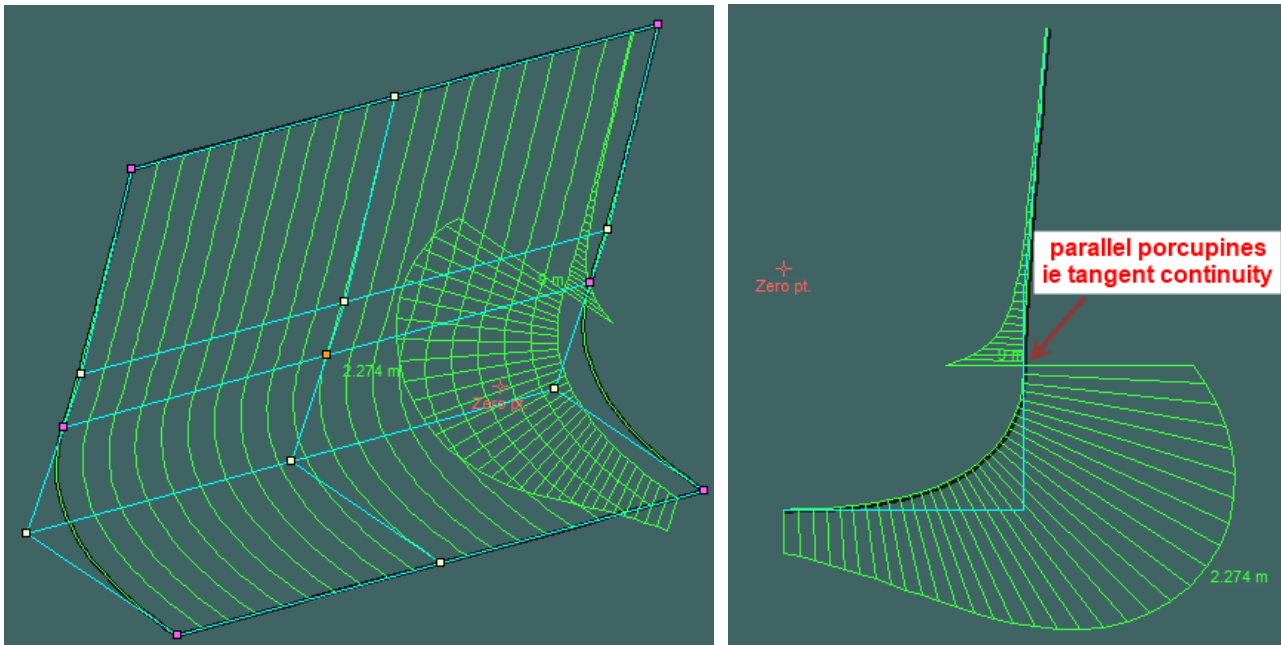
After bonding with a tangency constraint, and points either side of the edge are moved, the points on the opposite side of the bonded edge will move in the opposite direction to maintain continuity. If a point on the edge is moved, then both adjoining points will be moved in the same direction and by the same amount.

When bonding two edges together using tangency constraints, some control points will be moved by Modeler to ensure continuity. In this case, the rule as to which points remain unchanged and which points are modified is the same as that used with simple bonding. The first surface selected is treated as the master surface and is not modified in any way. The second surface selected is regarded as the slave surface, meaning that Modeler will modify points in this surface as required to meet the constraints.

Whether you need the "Strict Tangency" constraint or can use the "Approximate Tangency" constraint depends on whether the surfaces curve in both longitudinal and transverse directions at the same time or are "twisted". For prismatic surfaces (that do not change cross-section along their length, the "Approximate Tangency" constraint is sufficient. If there is significant longitudinal curvature then the "Strict Tangency" constraint may be required to ensure *exact* tangent continuity.

Prismatic surfaces

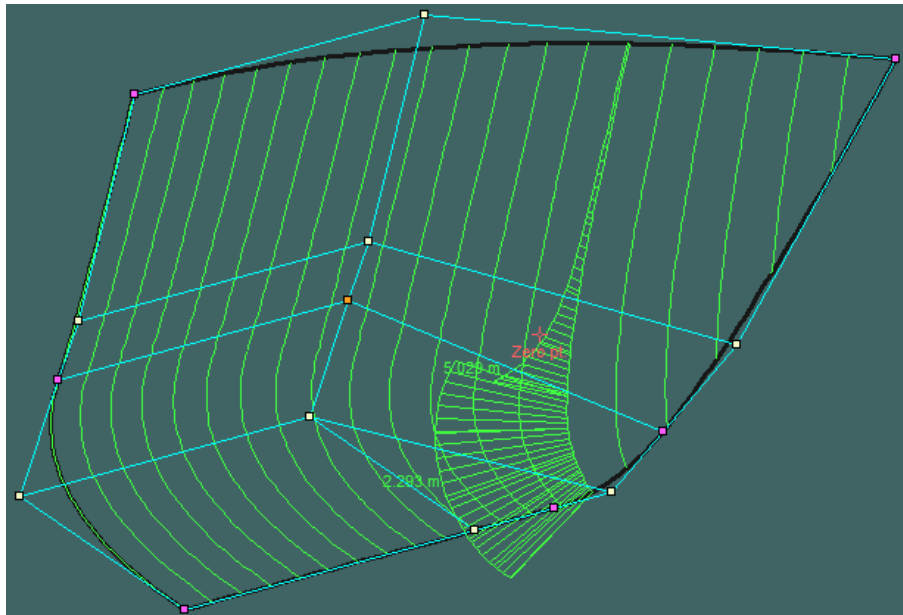
If the surfaces have constant sections along their length (prismatic) then the equidistance "Strict Tangency" constraint is not needed for C1 tangent continuity:

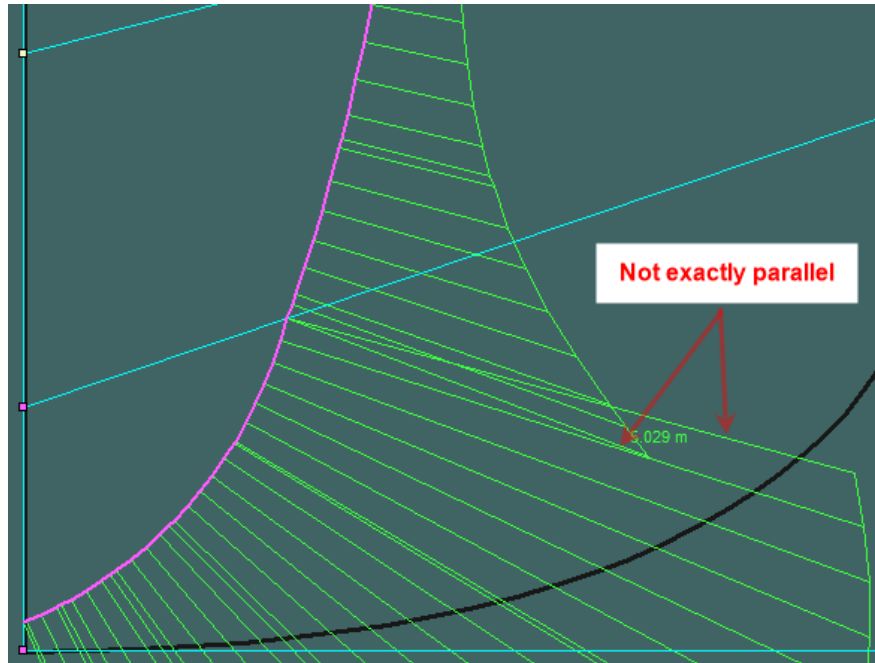


Prismatic surfaces, "Approximate Tangency" constraint is sufficient

Twisted surfaces

If the Surfaces are twisted or curved in the longitudinal direction as well as the transverse, then the tangent continuity will not be *exact* (as shown by the porcupines not being exactly parallel at the bonded edge) unless the "Strict Tangency" (equidistant control points) condition is met:

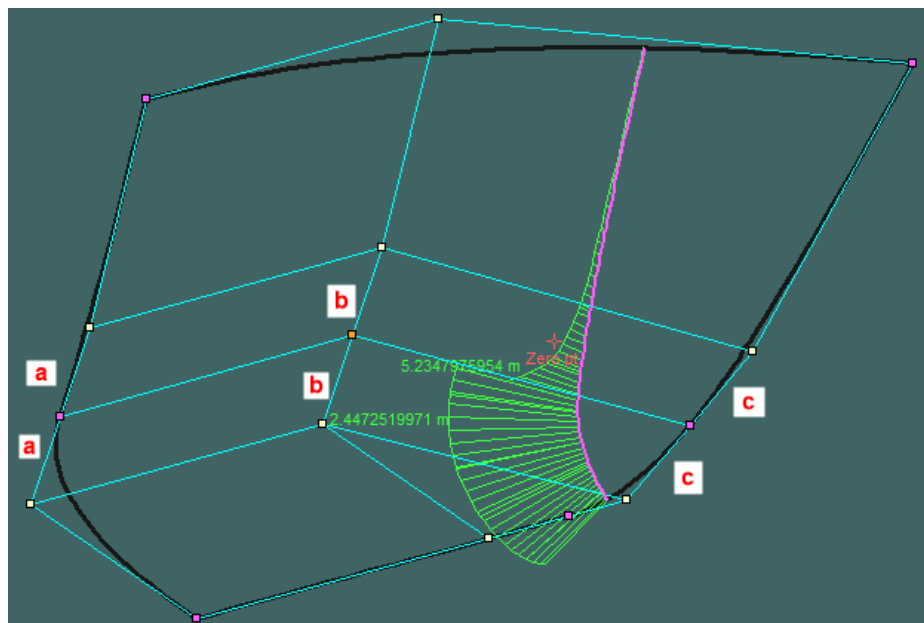


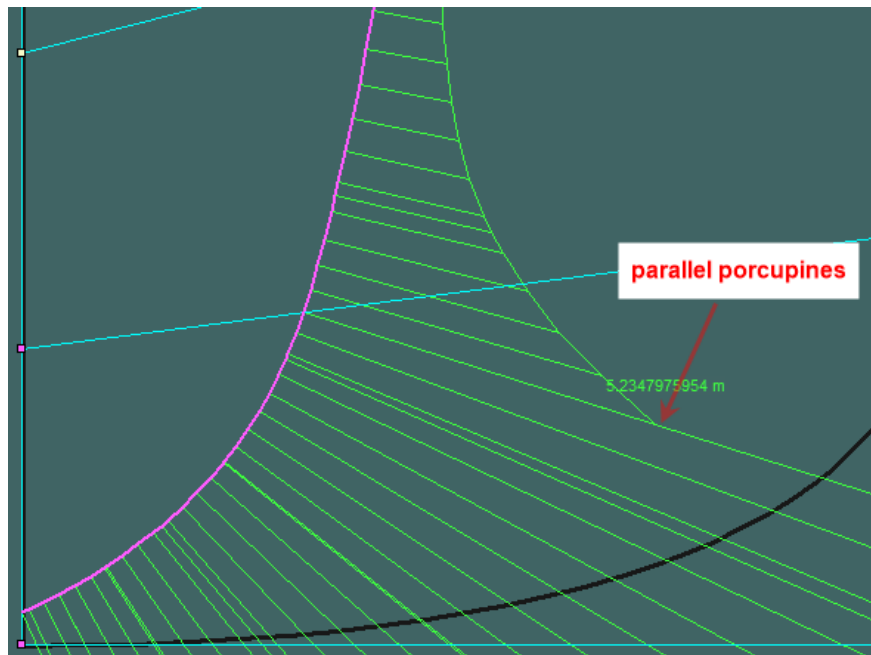


Approximate Tangency:

Control points immediately adjacent to the bonded edge *are not* equidistant for each surface:

With the bonded edge control points moved so that they are equidistant from the next row of control points: ie the two 3D lengths at [a] are the same; and both length [b] are the same and both lengths [c] are the same then the tangency is exact.



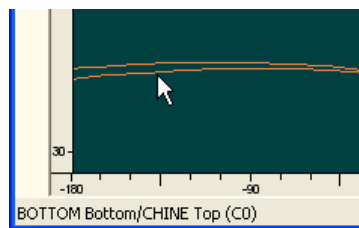


Strcit Tangency:

Control points immediately adjacent to the bonded edge *are* equidistant for each surface:

Viewing bonding types

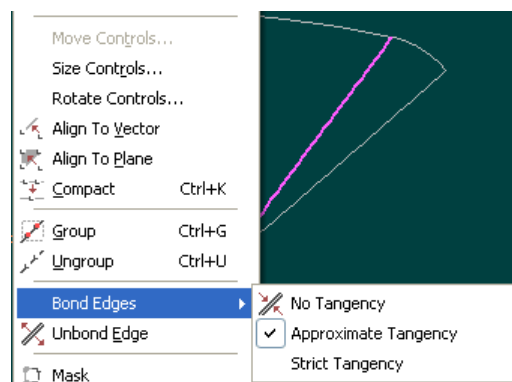
Positioning the mouse cursor over a bonded edge will display both edge names and the continuity constraint in the bottom left of the status bar:



The following types of bonding annotations are possible:

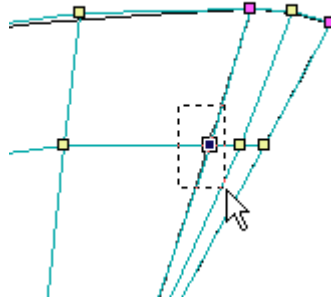
- C0 = No Tangency
- C1 Lenient = Approximate tangency
- C1 Strict = Strict Tangency

If a bonded edge is selected, the bonding constraint will be displayed ticked in the Bonding menu:



Changing bonding types

You can change the type of tangency easily by simply selecting a bonded edge or a control point on that edge, excluding corner points, and re-apply the bonding.



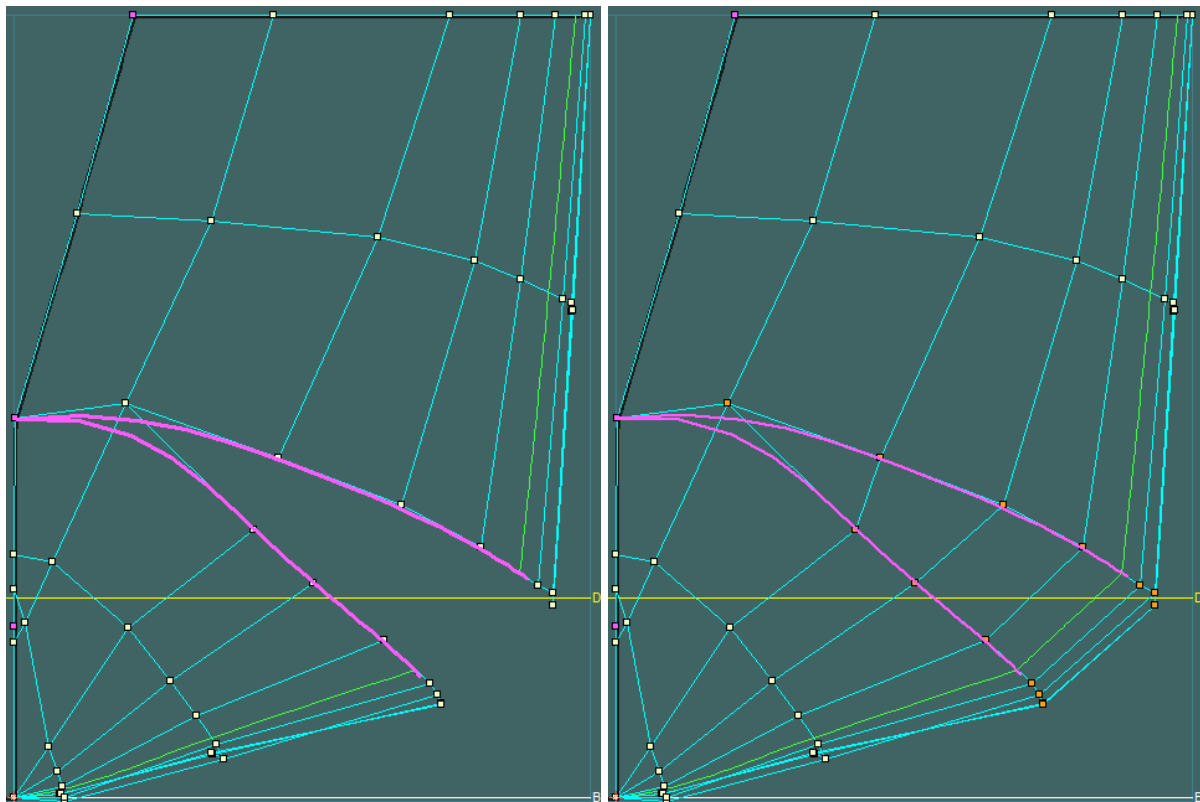
Selecting a shared control point on a bonded edge.

Tip:

In practise you may find it more convenient to use "no tangency" in early stages of design and then apply tangency later when you get to the fairing stage of design. This is mainly because the control points move on both sides of the bonded edge when tangency constraints are used.

Bonding with Inserted surface

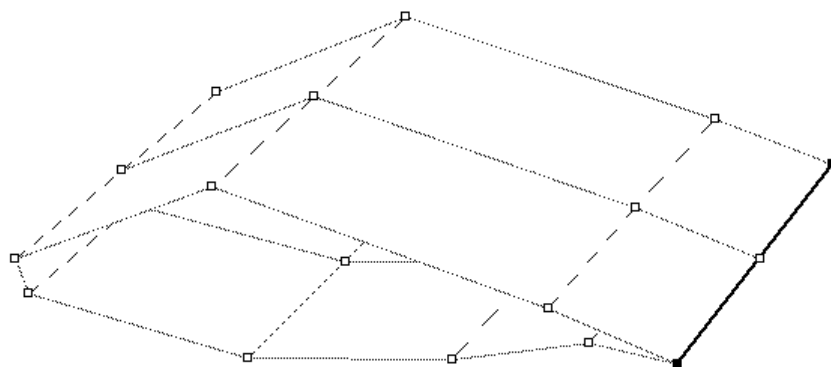
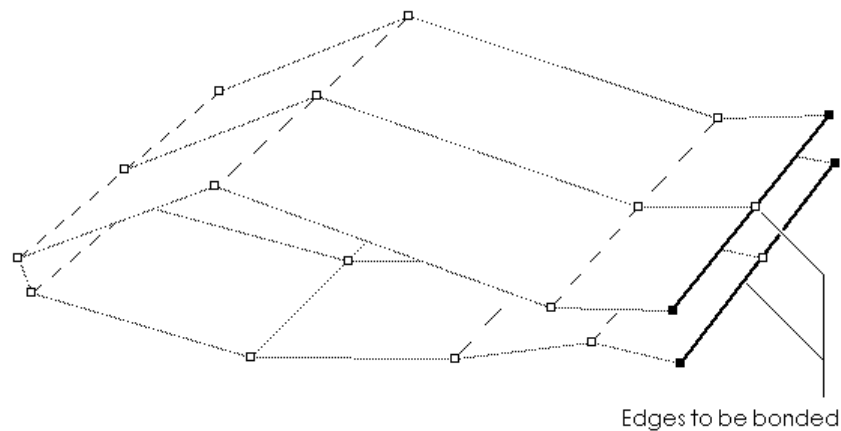
Two surfaces may be bonded by inserting a new surface in the gap between the edges. In this way the two original surfaces are unaffected. For example inserting a bilge chine surface between a topside and bottom surface. To do this, hold the shift key down while selecting the desired bond continuity.



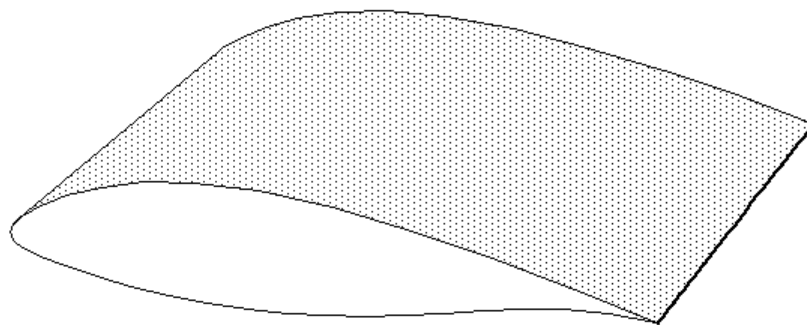
Bonding by inserting a new surface

Bonding within a Surface

Edges within a surface may also be bonded to one another as long as they follow the requirements of number of control points and stiffness. If you wished to form a closed surface, you might start off with the following net arrangement, with the object being to bond the two edges as indicated.



This net would generate a closed surface joined along the bonded edge. To highlight the bonded edge as shown, select the Bonded Edges option from the Contours dialog.



Unbonding

Any two edges may be unbonded by selecting the bonded edge or a control point on that edge, excluding corner points, and choosing Unbond from the Controls menu. Once the edges are unbonded they are free to move independently.

Commands Affected by Bonding

Bonding can effectively combine two surfaces to form a larger surface. As a result, commands that previously worked on single surfaces treat bonded surfaces as one:

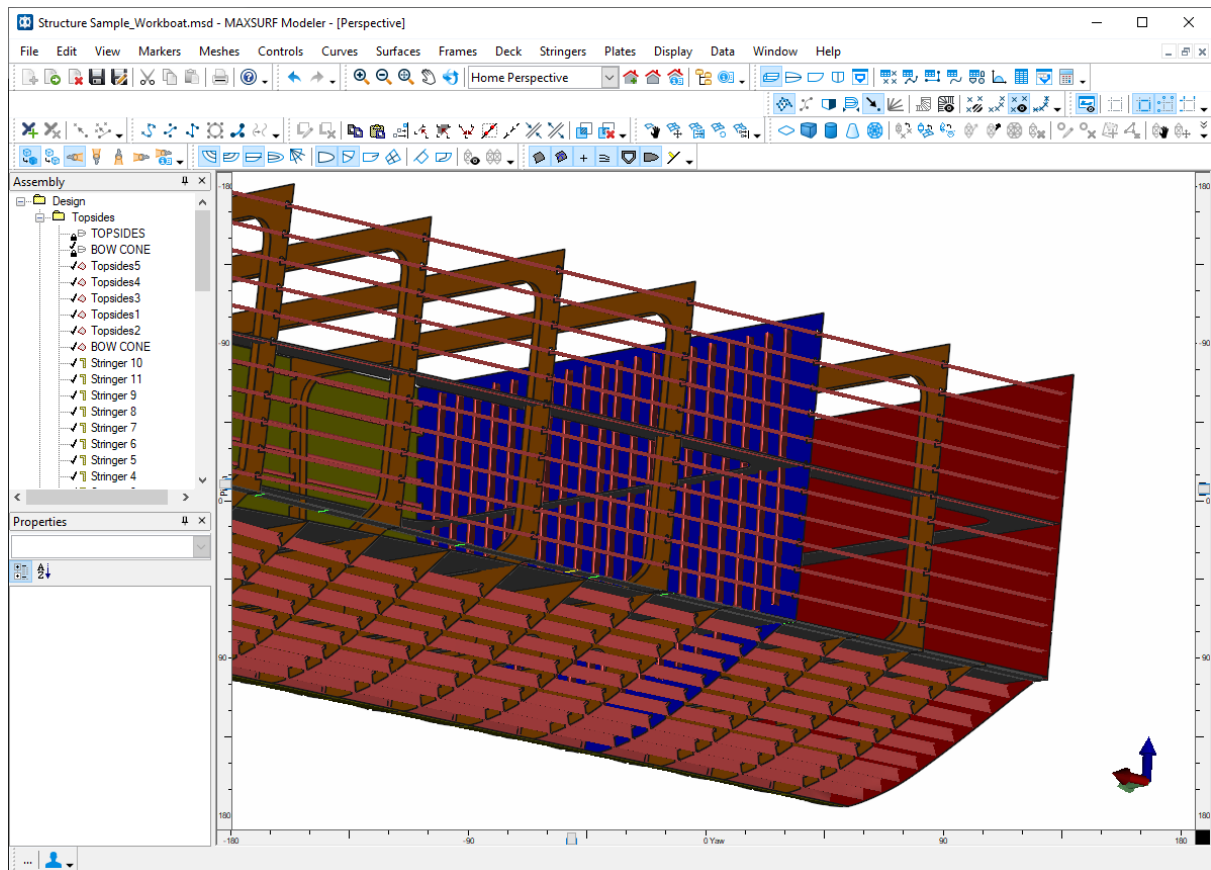
- Adding and Deleting of controls, where the row or column extends across a bonded edge.
- Surface locking, where all surfaces that are bonded together are locked or unlocked in unison.

- The Size Surface function, i.e. bonded surfaces cannot be sized independently.
- Surface movement commands such as Move, Flip, Rotate and Align.

Also see the section on [Trimming Surfaces](#).

Generating NURBS Surfaces from Parts

NURBS surfaces can be generated from any part in the design file (frames, decks, stringers, plates). The functionality is accessed from the Menu item "Surfaces | Add from visible parts...". This item is only enabled when a part or parts are visible in a drawing window. Adding a surface on a Frames/Bulkhead is a useful way to generate stiffened frames/bulkhead by being able to add a stringer on the generate surface.



Working with Mesh Surfaces

A Mesh surface is a linear surface made up of triangular facets. It is not a NURB surface, it does not have control points, and it cannot be trimmed with NURB surfaces.

The TriMesh surface is not designed to be smooth – it follows the input data points exactly and is linear between the data points. The Markers should provide sufficient detail for the purpose to which the model is going to be put such as, for example, hydrostatic and/or seakeeping analysis in Stability, Motions and Resistance.

It is often the case that line or point data for a vessel exist but a full NURB surface model does not exist. Reverse engineering an accurate NURB surface model from point or line data can be laborious and time consuming. Often the level of smoothness obtained from a NURB surface is not required and a surface made up of linear, triangular facets is sufficient. The aim of the TriMesh surface is to provide a quick route from line

or point data to a vessel surface model that may be used for analysis. The accuracy of a TriMesh surface representation of the hullform is dependent on the number of data points used to generate the TriMesh.

Because a TriMesh surface is a linear surface, it is generally not suitable for construction purposes. This means that if for example plate expansions and longitudinal stiffener geometry outputs are expected to be required at a later stage in the design process, the Modeler and Fitting tools for fitting NURB surfaces to markers should be used.

See [Mesh Surface](#) on page 235 for more information.

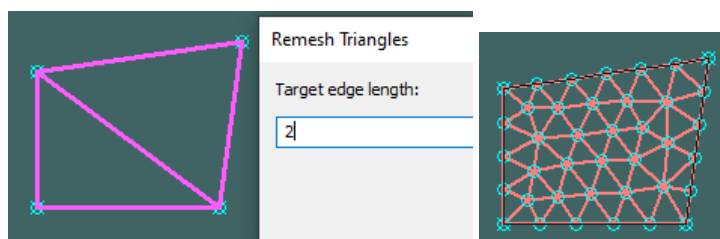
Add Mesh from 3 or 4 markers

A very simple Mesh can be added by selecting 3 or 4 Markers. If 3 Markers are selected a single triangle will be added. The simple one- or two-triangle mesh can be remeshed with smaller triangles



Marker selection and resulting Mesh surface

The simple one- or two-triangle Mesh can be re-meshed with smaller triangles using the "Remesh Triangles command":

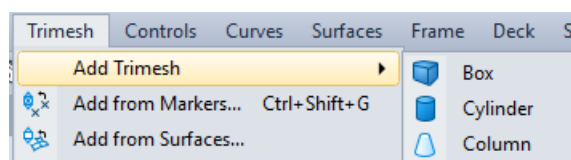


The mesh may be remeshed if required

Much more complex mesh models can be created from structured Marker data defining multiple cuts through the model. For full details, please see [Mesh Surface](#) on page 235.

Mesh from simple geometry

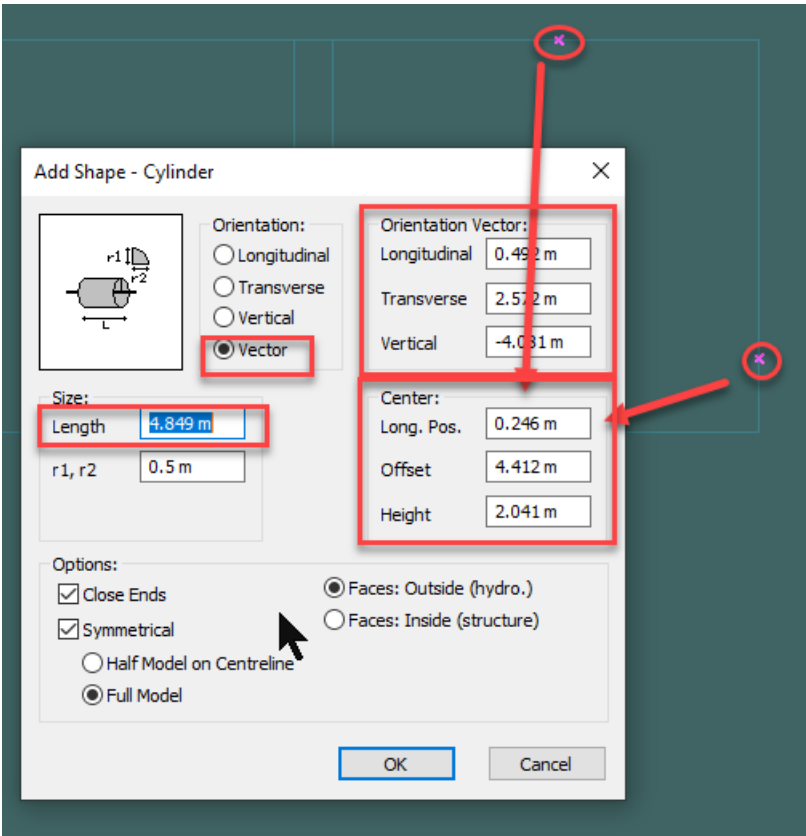
Meshes for planes boxes, circular cylinders and combinations of cylinders (columns) may be created directly. From the Mesh menu, choose Add mesh | Box, Cylinder or Column as required:



Adding Meshes for simple geometries

For boxes and cylinders there are similar dialogs for specifying the parameters of the required mesh. An arbitrary direction vector can be defined by selecting two marker points prior to adding the Mesh. In this

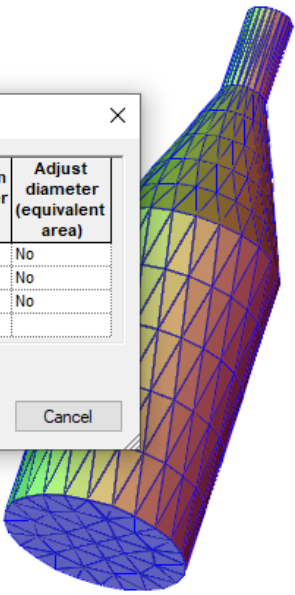
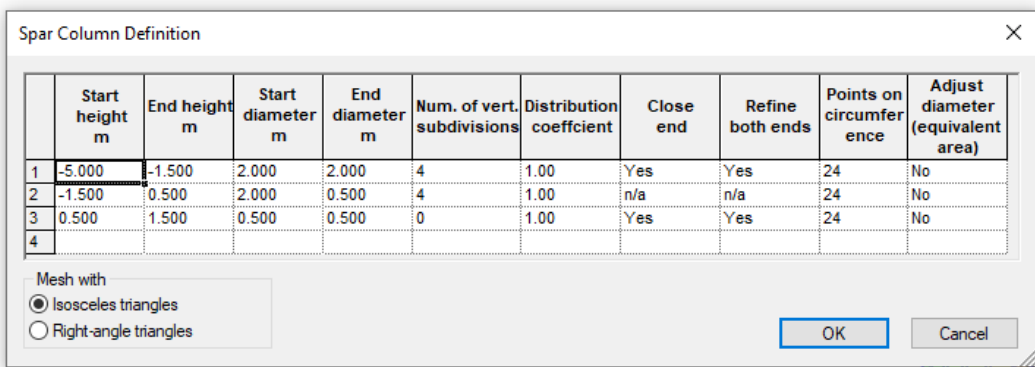
case the direction vector, length and center will be automatically computed and the data entered as defaults in the dialog:

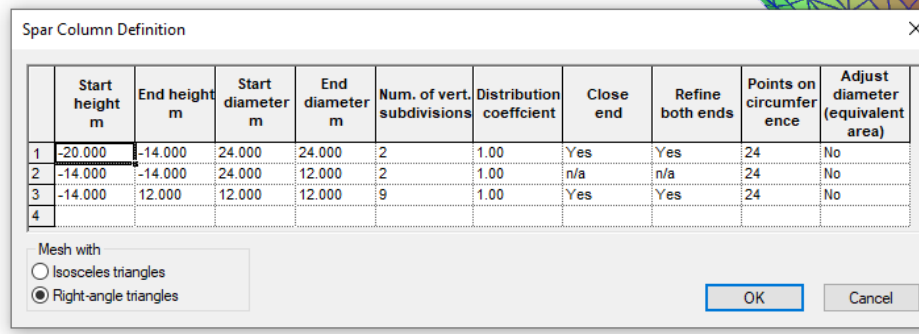


Using markers to define the position, direction and length of the mesh

The outside arrows direction can be specified and whether or not to close the ends of the mesh.

The "Column" mesh tool is used for generating vertical, circular cross-section columns (spar) mesh objects and uses a slightly different procedure as outlined below:





Mesh Spar Column Generating Tool

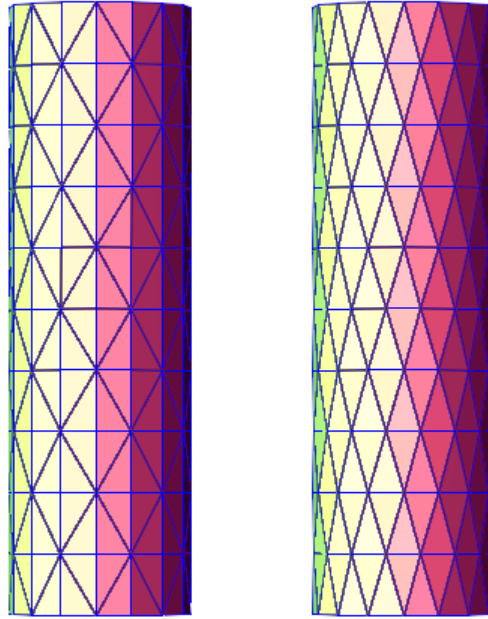
The newly generated mesh column is always vertical and of circular cross-section centered about the model origin longitudinally and transversely; furthermore, a full (asymmetric) mesh is generated. The resulting mesh can then be moved to the desired final location.

The mesh is generated by specifying a number of segments; each row defines one segment. For each segment you must specify:

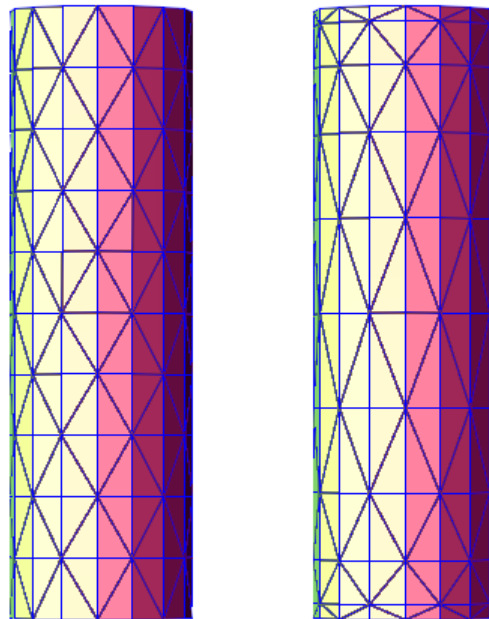
- Start height: vertical position of the start of the segment
- End height: vertical position of the end of the segment
- Start diameter: diameter of the cylinder at the start of the segment
- End diameter: diameter of the cylinder at the end of the segment
- Num. of vert. subdivisions: The number of extra divisions to be inserted along the height of the segment. If '0' is specified triangles will be the full height of the segment; if '2' is used then two extra subdivisions will be inserted resulting in three triangles along the height of the segment.
- Distribution coefficient: This is the ratio between a linear and cosine distribution of the intermediate divisions inserted along the height of the segment. '1.0' will give a linear (equal) spacing between the subdivisions; '0.0' will give a fully cosine distribution with the triangles concentrated near each end of the segment.
- Close end: This applies only to the first and last segment definition and specifies whether the start of the first segment or the end of the last segment should be closed.
- Refine both ends: If the refine option is selected, then the ends will be meshed in using concentric rings of triangles with each successive ring having half the number of triangles as the previous until no further halving of the number of triangles is possible. Otherwise a simple mesh of radial triangles will be used to close the ends.
- Points on circumference: All segments must have the same number of points.
- Adjust diameter (equivalent area): If selected, the diameter of the circle on which the triangle nodes are placed will be increased so that the *areas* of the polygon and the circle match. Otherwise the nodes will use the specified diameter.

The mesh can be generated using either isosceles or right angle triangles.

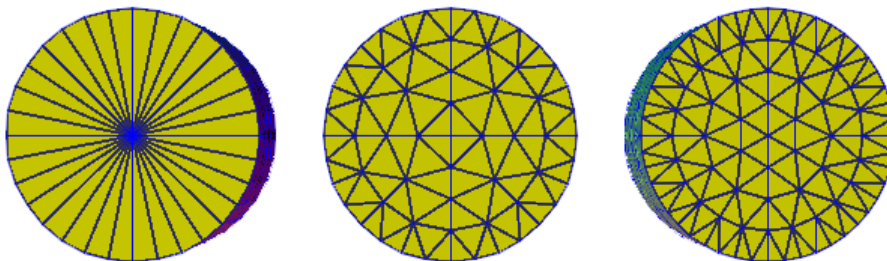
- Isosceles triangles: the nodes on each successive vertical ring are rotated by half a triangle.
- Right-angle triangles: the nodes are not rotated. This means that the triangles for vertical strips of triangle that all lie in the same plane.



Difference between Isosceles and Right-angle triangle mesh



Difference between Linear (1.0) and Cosine (0.0) distributions

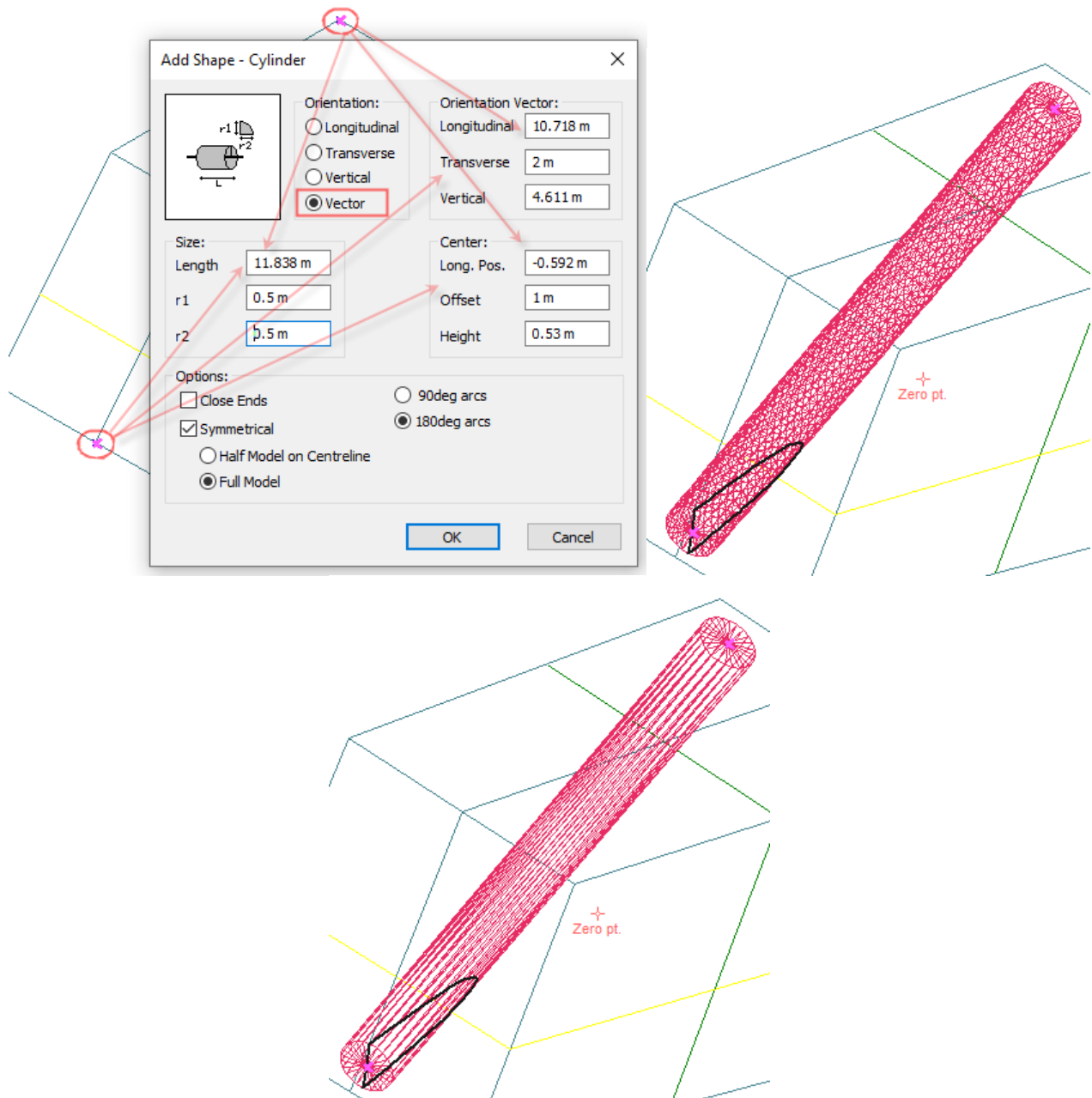


Difference between end mesh: Unrefined and Refined (with 32 and 48 nodes on circumference)

When accessing this tool through the Automation interface (COM) – for example via VBA – it is possible to specify the orientation and location of the column; it is not restricted to being vertical and centered on the model zero point.

Furthermore, it is possible to specify:

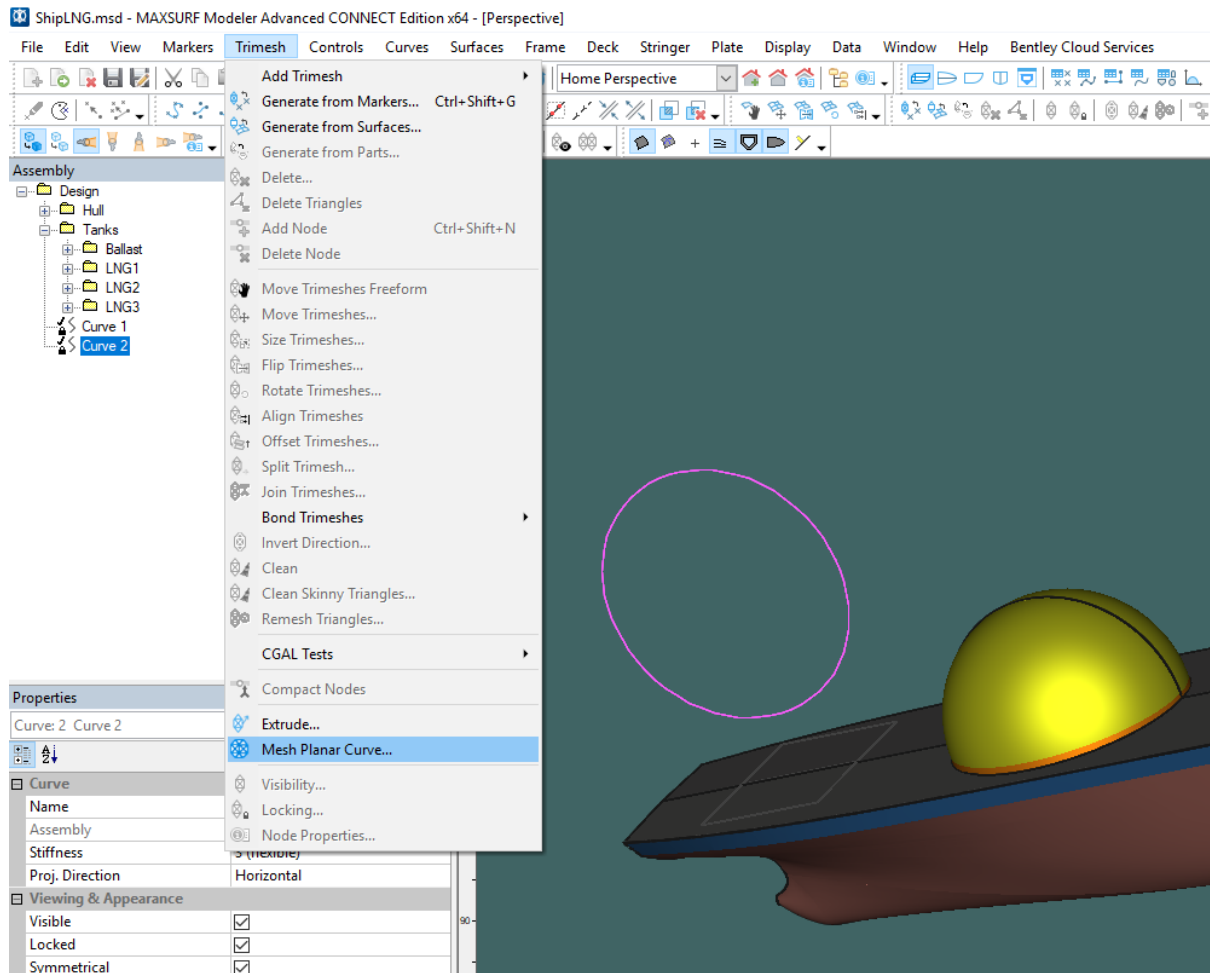
- Shape center (rather than defaulting to model center always)
- Shape orientation other than along one of the principle axis directions.
- Orientation, center and length will be set automatically if you first select two marker points which define the ends of the shape.



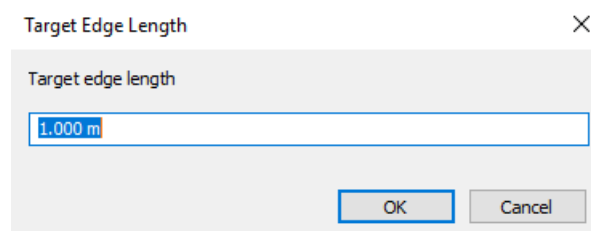
For meshes holding down shift key when clicking the OK button in the dialog will result in a single triangle along the whole length of the cylinder (right)

Add Mesh from planar curve

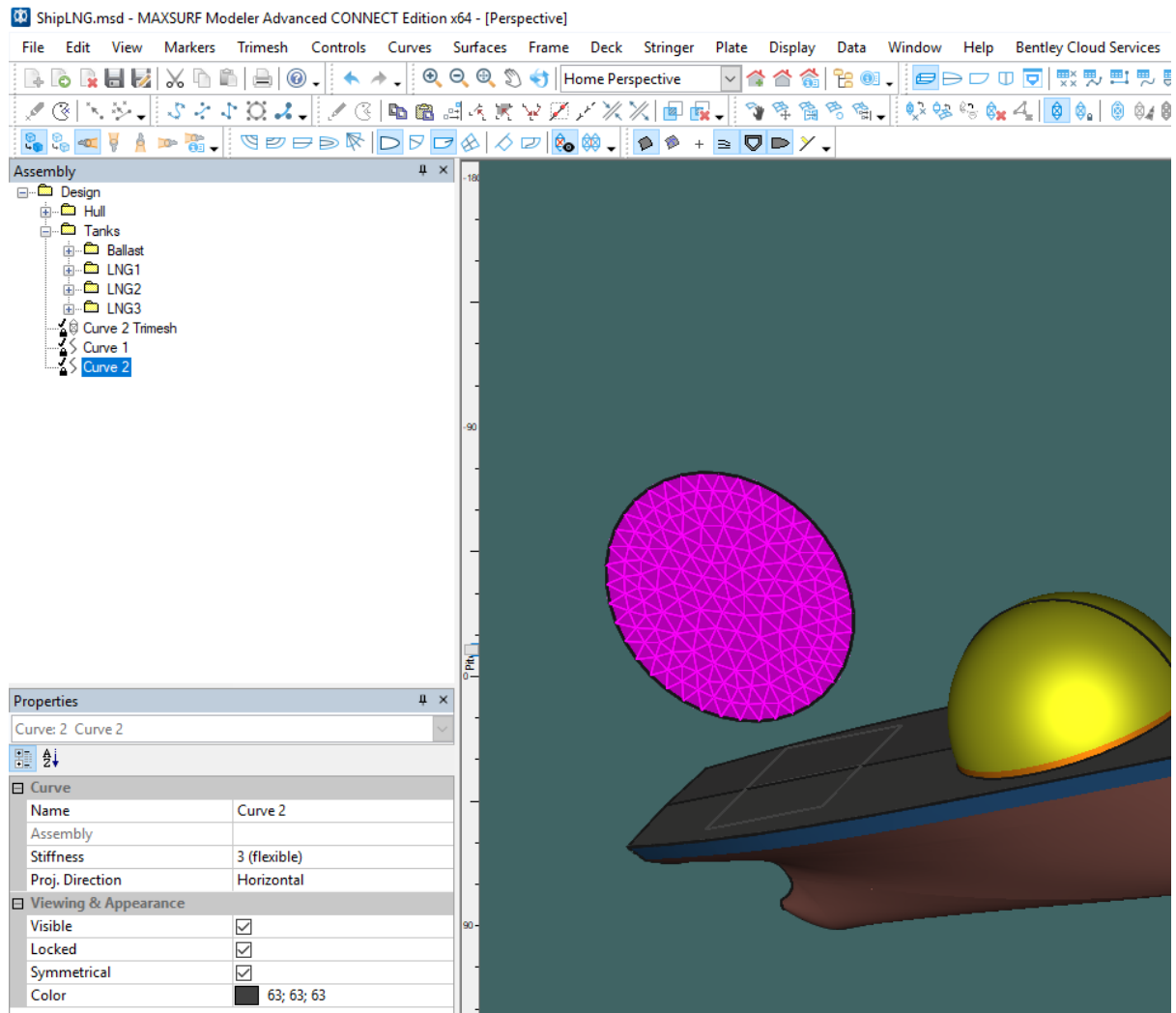
A mesh can be created from a planar curve. The function is available from the mesh surface when a planar curve is chosen (all points on the curve must lie in a plane):



If the curve is a NURBS curve the user will be asked to specify the target edge length along the boundary:



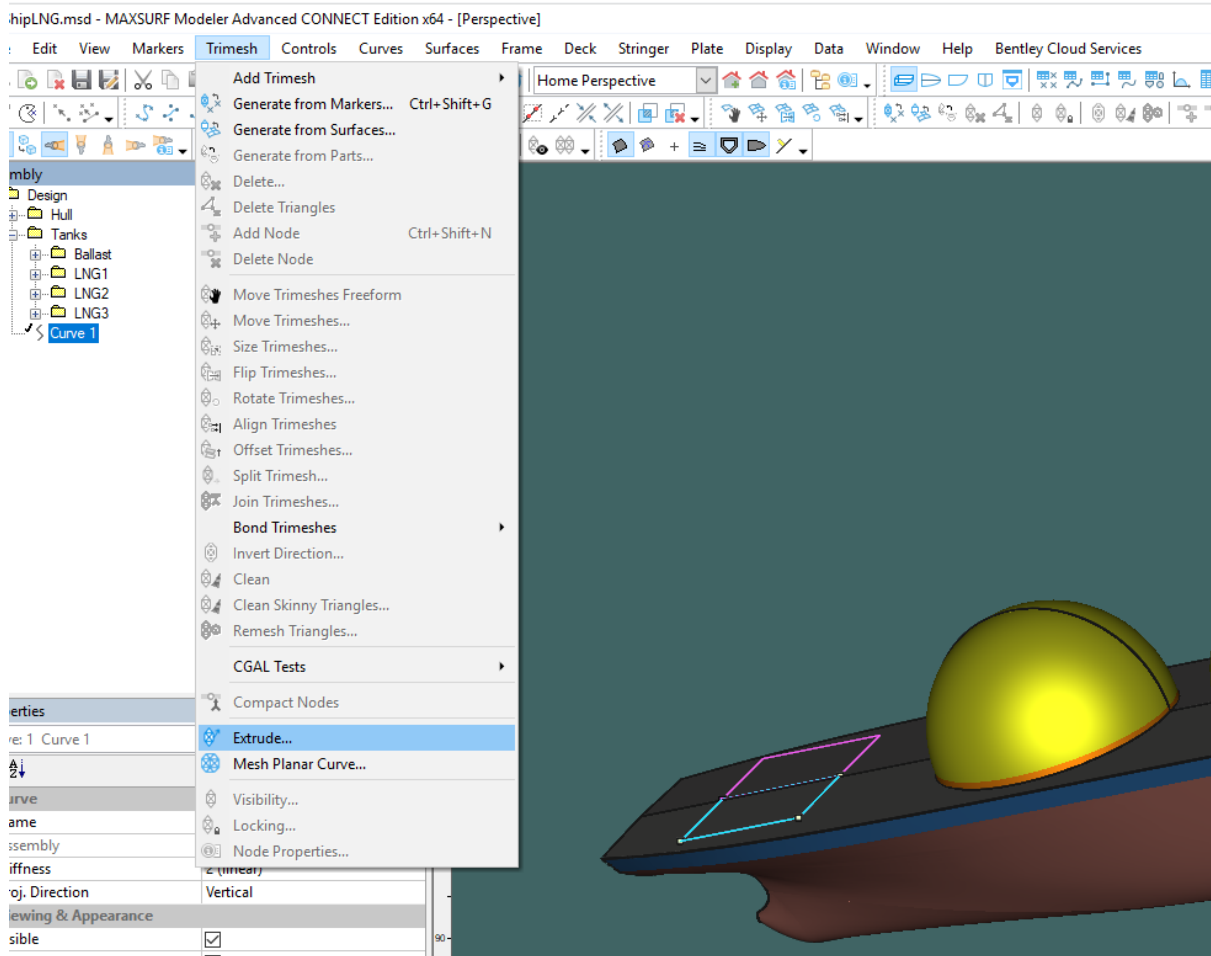
The curve will then be meshed at that density. If the curve is not closed then Modeler will close it with a linear segment between the first and last points.



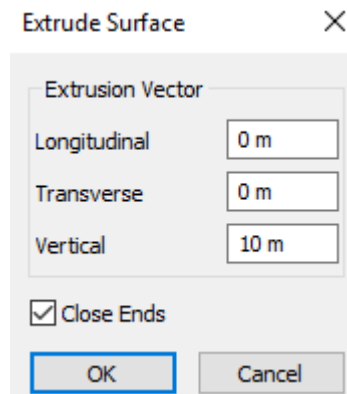
If the chosen curve is linear then the simplest mesh possible will be created. If a denser mesh is required, select the mesh and remesh, the user will then be asked for a target edge length.

Add Mesh from extruded curve

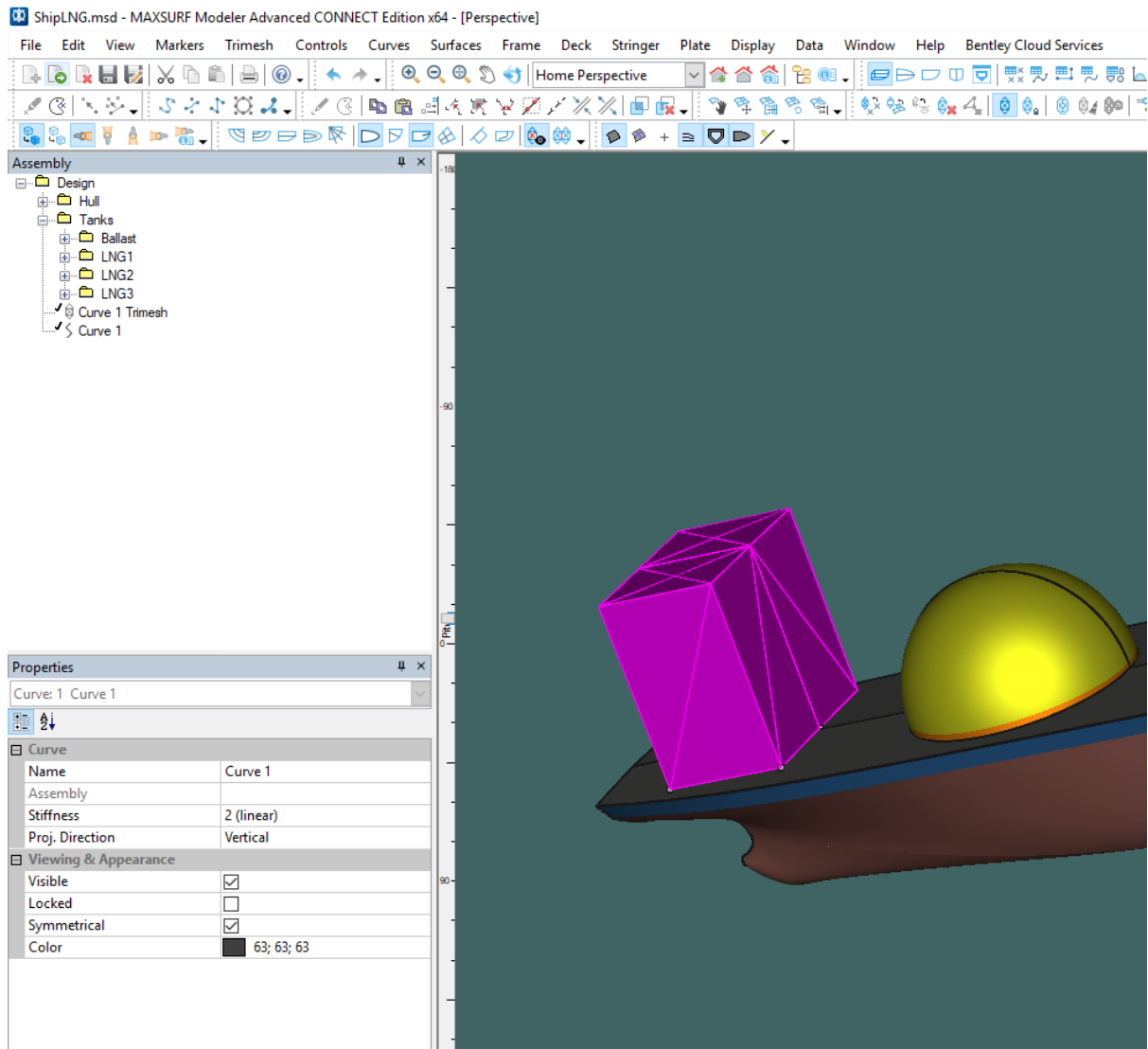
A mesh can be defined by extruding a curve along a vector, similar to the function for extruding a nurbs surface from a curve. The function is in the Mesh menu called "Add by Extruding Curve..." :



The function is enabled when a curve is selected, the following dialog will then appear:



Enter the extrusion vector and whether or not you want the ends to be meshed to create a closed volume ("Close Ends"), click ok:

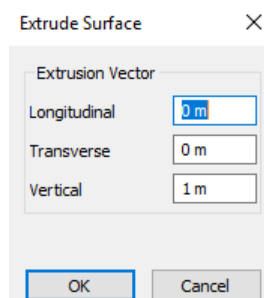


In the case of a linear curve being extruded the simplest mesh possible will be created. To create a denser mesh use the remesh command and you will be asked for a target edge length, the mesh will be remeshed to this specified density.

If a NURBS curve is chosen to be extruded, the process is the same except that the user will be asked to specify a target edge length for the triangle edges along the chosen curve.

Extrude Mesh

A closed mesh can be created by extruding a non-closed mesh along a vector. Once a mesh is chosen from the assembly tree, select "Extrude Mesh" from the Mesh Menu. A dialog will appear into which the extrusion vector is entered:



Click Ok and the mesh will be extruded along that vector to form a closed mesh. This command is useful for creating closed prismatic meshes from irregular planar meshes.

Mesh visibility

As with NURB surface the visibility of a Mesh surface is a property and can be switched on and off. This is done via the assembly tree, properties pane or Surfaces Window. In the assembly pane, right click on the Mesh surface and the visibility option can be toggled on or off.

Locking/Unlocking a Mesh

As with NURB surface a Mesh surface can be locked and unlocked. If a mesh is unlocked then a small square box will appear on each triangle vertex, this box may be dragged or double clicked. See Manipulating a Mesh surface below. Locking/unlocking a surface is done via the assembly tree. Right click on the Mesh surface and toggle the lock option.

Deleting a Mesh

To delete a mesh choose Delete Mesh from the Surfaces Mesh surface menu. A dialog will then appear in which the user can select the mesh they wish to delete.

Importing a Mesh

Meshes may be imported from a number of file formats. To import a mesh choose either .stl,.ply or .dat file formats from the File Import Trimehes menu item.

Exporting a Mesh

To export a mesh choose the file format from File Export Meshes menu. Only visible meshes will be exported. Formats currently supported are .stl and .ply.

Adding a node to a Mesh

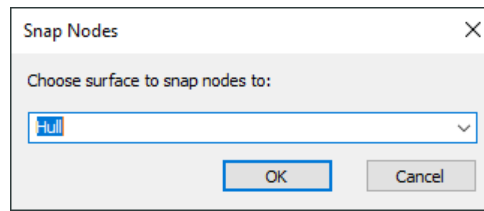
A node may be added to the mesh via the user interface. Click on the add node button from the mesh toolbar. If the node is added to a mesh surface that was created from a NURBS surface that is still in the design then the new node will be pinned to the surface (i.e. lie on the surface). Otherwise the added node will lie in the plane of the triangle in which it is added. When a node is added within the regions of a triangle edge then the node will be added on the edge and the adjacent triangles will be split in two. If the node is added on the interior of a triangle then that triangle will be replaced with 3 new triangles each containing one vertex that is the new node. In the case of a mesh that wraps back on itself it may be hard to add the node in the desired triangle. To specify which triangle the node should be added to select the triangle before adding the node.

Deleting a node from a Mesh

A node may be deleted from the mesh via the user interface. Click on the delete node button from the mesh toolbar. The cursor will then change to a pacman icon, move the cursor over the mesh node to be deleted and click the left mouse button. The node will be deleted from the mesh and new triangles will be created to fill the void left by the deleted node, the resulting mesh will always create less triangles than the initial mesh.

Snap Node(s) to Surface

This function will be enabled when one or more mesh vertices are selected. Once executed the user will be asked which NURBS surface the nodes should be snapped to:



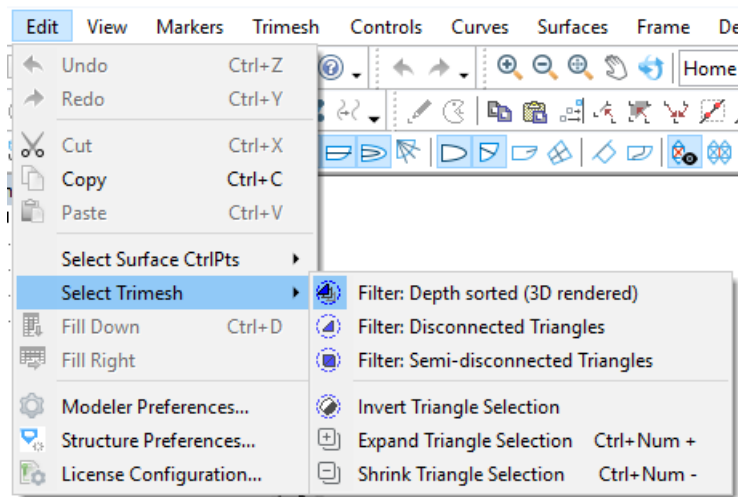
All selected nodes will then be snapped to the nearest point on the chosen nurbs surface.

Selecting a triangle from a mesh

A triangle can be selected in non-rendered mode by clicking on the triangle or dragging a selection rectangle over a group of triangles. Non-contiguous triangles can be selected by holding the shift key down to add to the selection. Triangles can only be selected when the mesh is locked, when unlocked the mesh vertices will take precedence for selection. To increase or decrease a selection by 1 ring of neighbouring triangles press ctrl + or ctrl -. The increase/decrease functionality can be accessed from the Edit | Select pop up menu.

Triangle selection options

Several selection tools have been added to aid with selecting triangles have been added. These tools comprise filters and selection manipulation and are available from the Edit menu:



Triangle selection options

- Filter: Depth sorted (3D rendered) – When enabled only the front visible triangles will be selected; if disabled then all triangles in the selection rectangle will be selected irrespective of whether they are visible or not. (Available in rendered perspective view only.)
- Filter: Disconnected triangles – Selects only triangles with no other connected, adjacent triangles (i.e. geometrically isolated triangles only)
- Filter: Semi-disconnected triangles – Selects only triangles with at most one other connected, adjacent triangle (i.e. only one edge connected to another triangle)
- Invert Triangle Selection – change the selection so that all the non-selected triangles are now selected.
- Expand Triangle Selection – Add all triangles directly connected to the perimeter current selection to the selection.
- Shrink Triangle Selection – Remove all triangles on the perimeter of the current selection from the selection.

Deleting a triangle from a mesh

Select the triangle(s) you wish to delete. Either hit the delete key or select Mesh | Delete Triangles.

Flip adjacent tri pair

When two adjacent triangles are selected the "Mesh | Flip Tri. Pair" function becomes enabled. When run it changes the connectivity from:



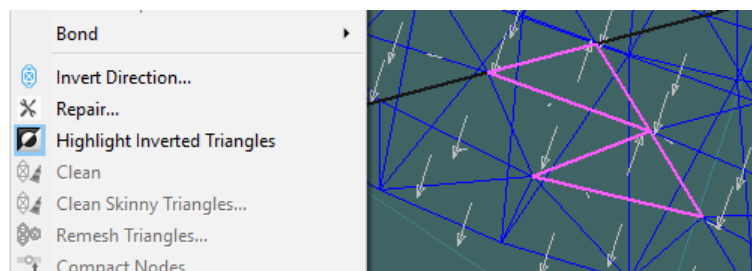
Mesh node properties

The location of a mesh node can be altered numerically by double clicking on a node in one of the view windows. A dialog will then appear in which the node coordinates can be altered.

Outside Arrows

As with NURB surfaces mesh surfaces have an inside and outside direction. Visually the outside arrows can be switched on and off via the same outside arrows button used for the NURB Surfaces (or choose Outside Arrows from the Display menu). An outside arrow will be displayed for each triangle in the mesh and they can be reversed by clicking on the tip of the arrow.

The "highlight Inverted Triangles" command can be used to facilitate the identification of mesh triangles which have opposite orientation to their neighbors:



Inverting a Mesh

Mesh triangles have directions associated with them (this is particularly important to downstream analysis applications such as Stability and Motions). An entire mesh can be inverted (switch directions) by selecting the Invert option from the Surfaces Mesh menu.

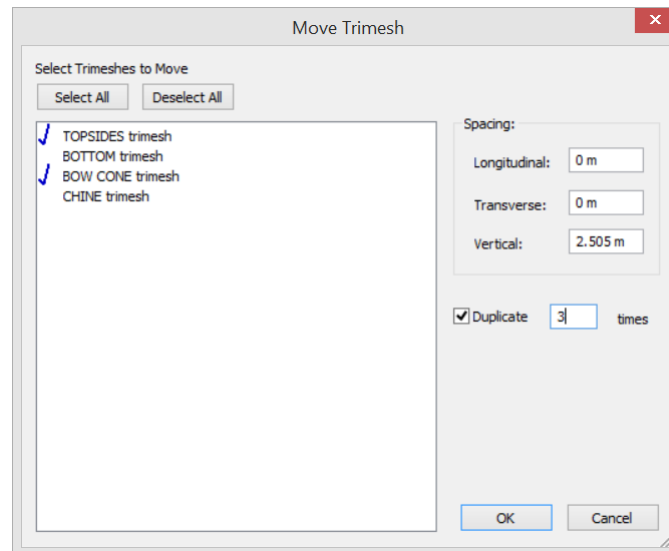
Compact Mesh nodes

If a mesh is unlocked and more than one node is high lighted then the the compact nodes options will be enabled. This will compact all the nodes onto the first selected node. Any degenerate triangles created (zero area) will be deleted, any duplicate nodes created will be deleted.

Moving a Mesh Surface

Meshes can be moved in two ways, either numerically or freeform by dragging the surfaces with the mouse.

The Move mesh function can be used to move a mesh and/or duplicate a mesh. Select Mesh | Move Meshes... from the menu. The following dialog will appear:



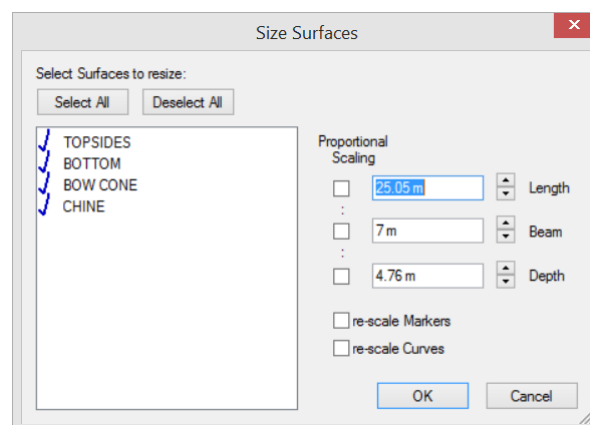
Select the mesh that is to be moved by clicking on its name, a tick will then appear beside it (multiple meshes can be moved at once). Specify the distance the Mesh is to be moved, a distance can be specified for each of the 3 orthogonal directions. To duplicate the surface while moving check the “Duplicate” checkbox, and specify the number of times the surface(s) is to be moved. If duplicate is not chosen then the surface will just be moved, if it is selected then the original surface will remain unchanged and duplicate surfaces will be created at “Spacing” intervals the specified number of times. The “Select All” and “Deselect All” buttons will select all and deselect all meshes in the mesh list.

Move Mesh Freeform

Moves a mesh using the mouse. To execute unlock the mesh and choose move mesh freeform from the mesh menu, next click the mouse near a vertex on the mesh you wish to move and holding the left mouse button down drag the mesh. When the left mouse button is released the mesh will be placed in that location.

Size a Mesh Surface

To size a mesh select Mesh | Size Mesh from the menu. The following dialog will appear:

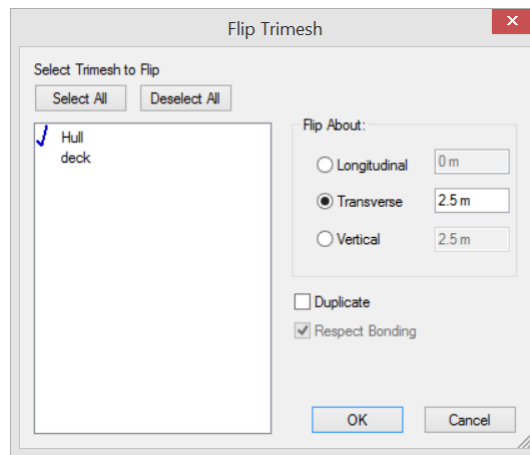


Select the mesh that is to be sized by clicking on its name, a tick will then appear beside it (multiple meshes can be sized at once). Next specify the size you wish the surface(s) to be in the edit boxes for Length, Beam and Depth. The default values are the values for the combined surfaces currently selected. If the proportional scaling box is checked then the values in linked direction will be scale proportionally when one value is changed. For example if the length and beam are 10m and 5m respectively and the proportional scaling check box is selected for both, then when the length is changed to double (20m) then the beam will automatically be set to double also (10m).

The “Select All” and “Deselect All” buttons will select all and deselect all meshes in the mesh list.

Flip a Mesh Surface

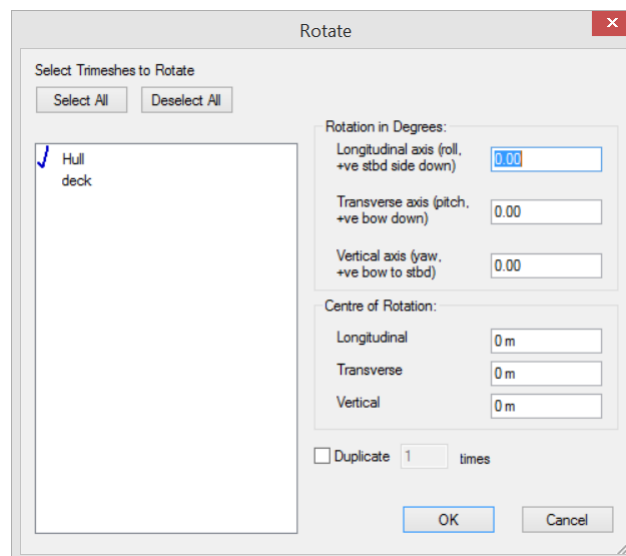
A mesh surface can be flipped about an orthogonal axis. To flip a mesh choose Mesh | Flip... from the menu, the following dialog will appear:



Select the mesh surface and which axis you would like it flipped (mirrored about). If the Duplicate check box is selected, then the original surface will remain and a new surface will be created. If the surface is locked, then it cannot be flipped if not duplicating (it will be greyed in the surface list).

Rotate a Mesh Surface

A mesh surface can be rotated about an arbitrary axis. To rotate a mesh choose Mesh | Rotate... from the menu, the following dialog will appear:



Select the surface you wish to rotate and enter then rotation angle (about each of the 3 orthogonal axes) and centre of rotation.

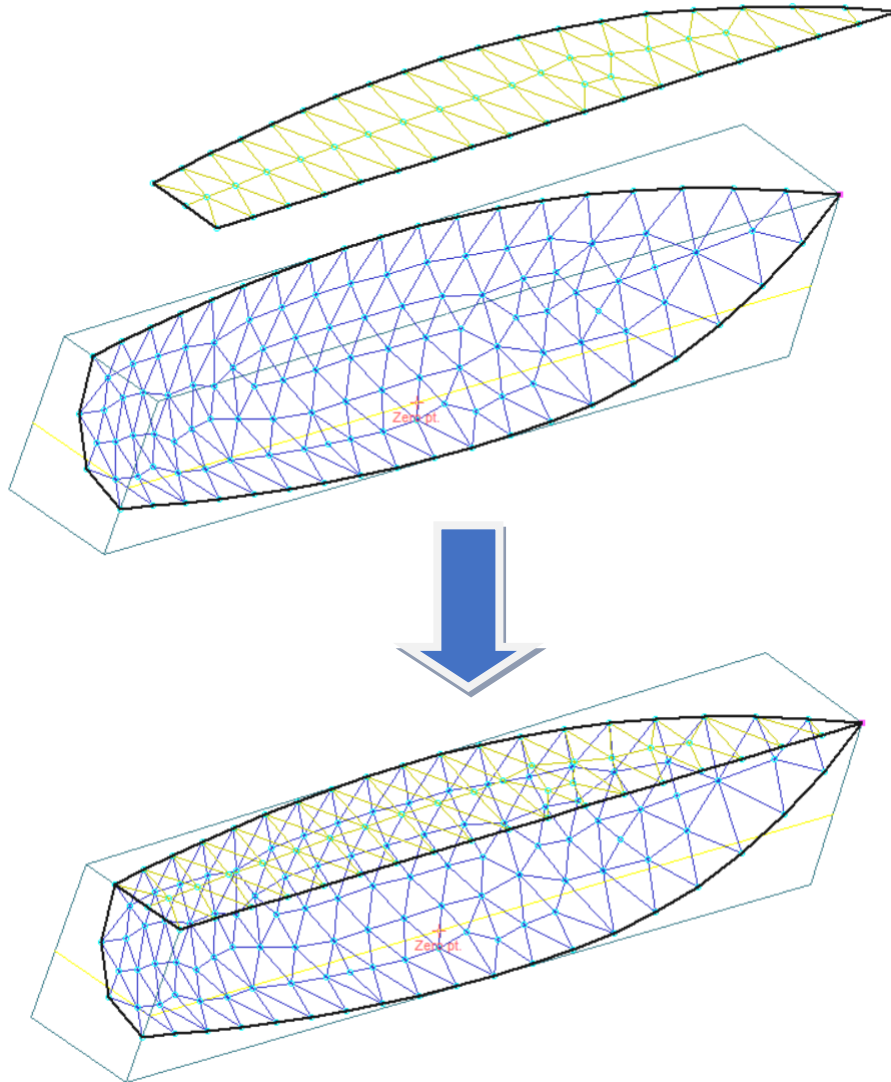
If the Duplicate check box is selected, then the original surface will remain and a new surface will be created. If the surface is locked, then it cannot be rotated if not duplicating (it will be greyed in the surface list).

Align Mesh Surfaces

Two meshes can be aligned by selecting a node (vertex) on each of the mesh surfaces. To have the nodes displayed the mesh must be unlocked and the internal edges must be visible. The first node selected is the

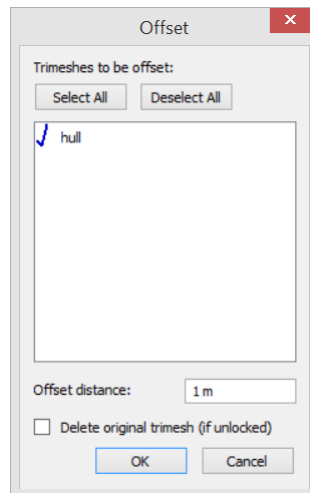
Master and the second is the slave. Choose Mesh Align from the menu and the slave mesh will be moved so that the two selected nodes are coincident.

The following illustrates the align command when the two bow nodes are selected:

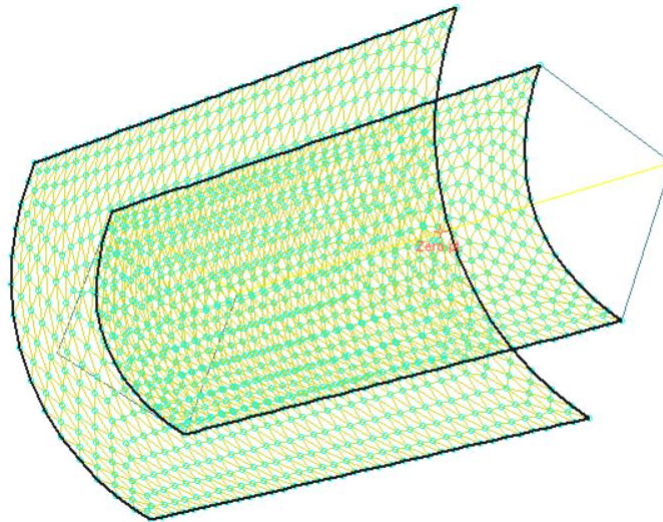


Offset a Mesh Surface

An offset mesh surface can be created using the Mesh | Offset... command from the menu. When chosen the following dialog will appear:

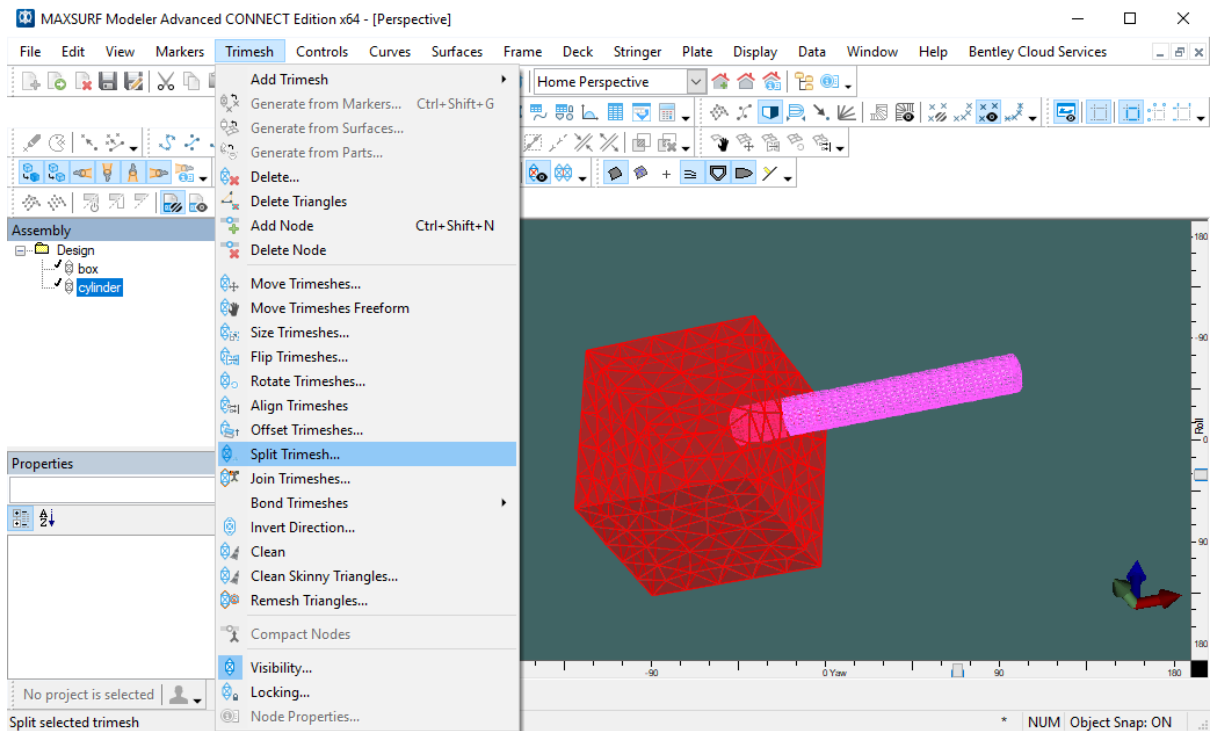


Select the mesh(es) you wish to be offset from the list and set the offset distance. The mesh will be offset in the direction of the outside arrows by the distance specified. If the mesh surface is unlocked then the original mesh can be deleted after the offset operation by checking “Delete original mesh” option.

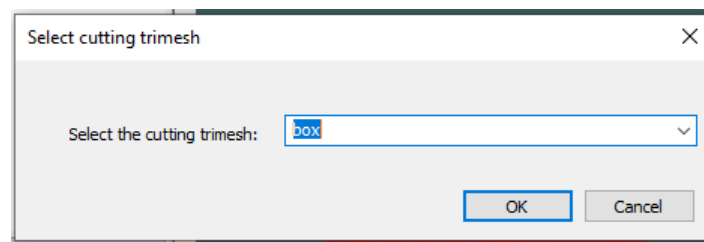


Split a Mesh

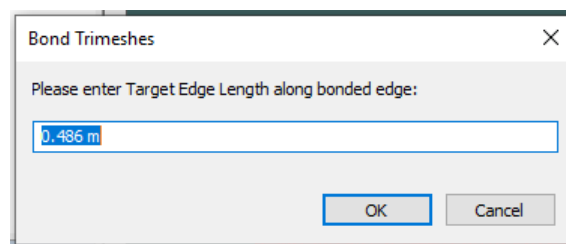
Meshes can be split into regions by a cutting mesh surface. To split a mesh select the mesh in the assembly tree and choose “Split Mesh” command from the Mesh menu:



You will then be asked to select a cutting mesh:



Next you will be asked to set the target edge length for the triangles along the intersection, the default is set to the default of the parent mesh:



The parent mesh will then be split into a number of meshes according to how many regions were formed by the cutting surface. Properties for the child meshes will be inherited from the parent mesh.

Join Mesh Surfaces

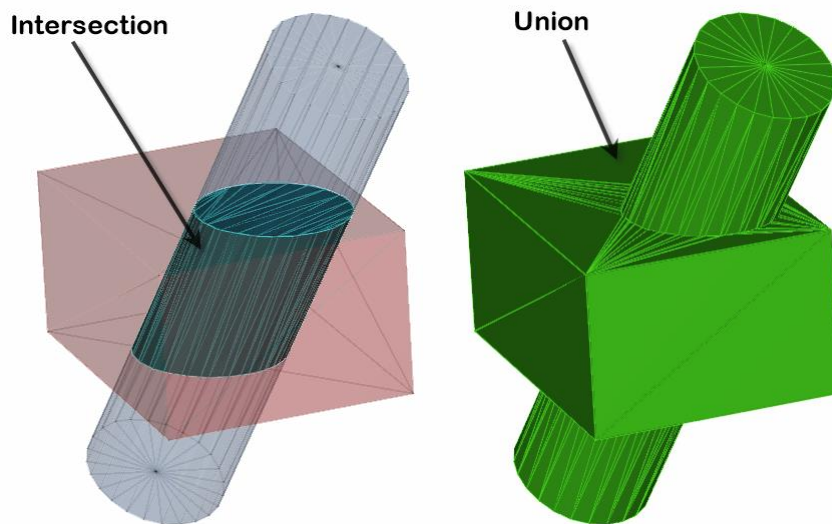
Two or more meshes can be joined into a single mesh by selecting the meshes (for example in the assembly tree by holding the shift key down) and choosing Mesh | Join from the menu. The first selected mesh will be the parent and the subsequent meshes will be joined to it. This means all the properties of the joined mesh will be inherited from the parent.

Boolean operations for closed Meshes

Several Boolean operations which may be applied to closed meshes have been included in this version. Two or more closed meshes may have Boolean operations applied. The two operations which may be performed are:

- Union: the single volume created by joining all selected meshes together
- Intersection: the volume common to all selected meshes

For instance:



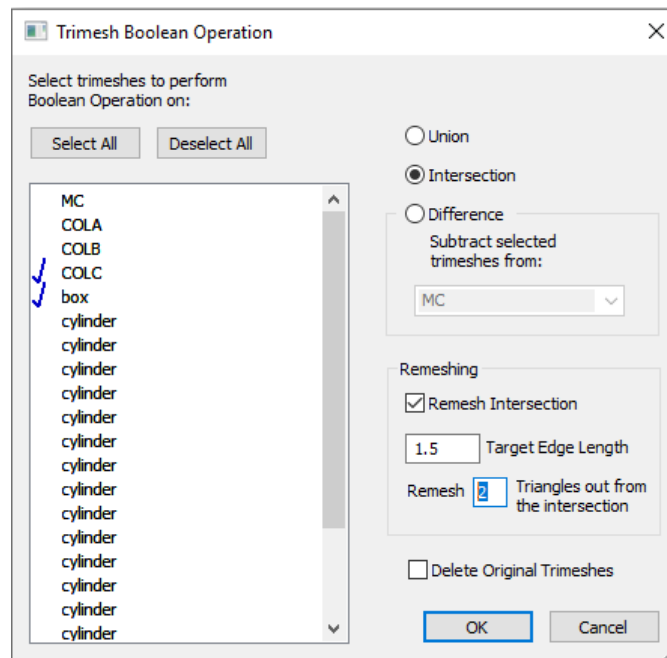
Boolean Intersection and Union of box and a cylinder meshes

In a ship allocation:

Union might be used to create a single mesh of a vessel from multiple individual meshes modeling the hull and wheelhouse.

Intersection might be used to create a tank mesh by finding the volume inside the hull limited by a simple bounding box which may extend beyond the hull.

The Boolean Operations dialog is launched from the Mesh menu:

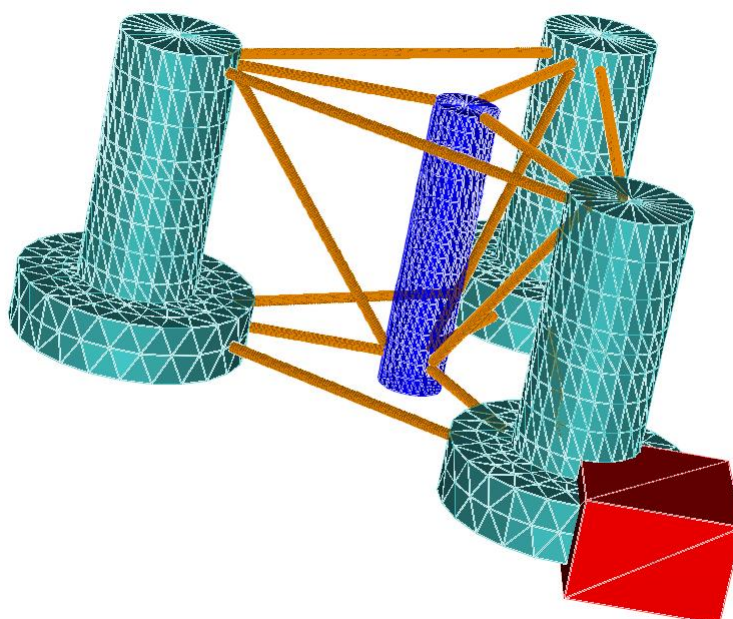


Select 2 or more mesh surfaces from the list you wish to perform the boolean operation on. From the radio controls select what type of Boolean Operation is to be performed. If difference is chosen then the user needs to select the mesh which is to have the other selected meshes subtracted from it (using the drop down list of surfaces).

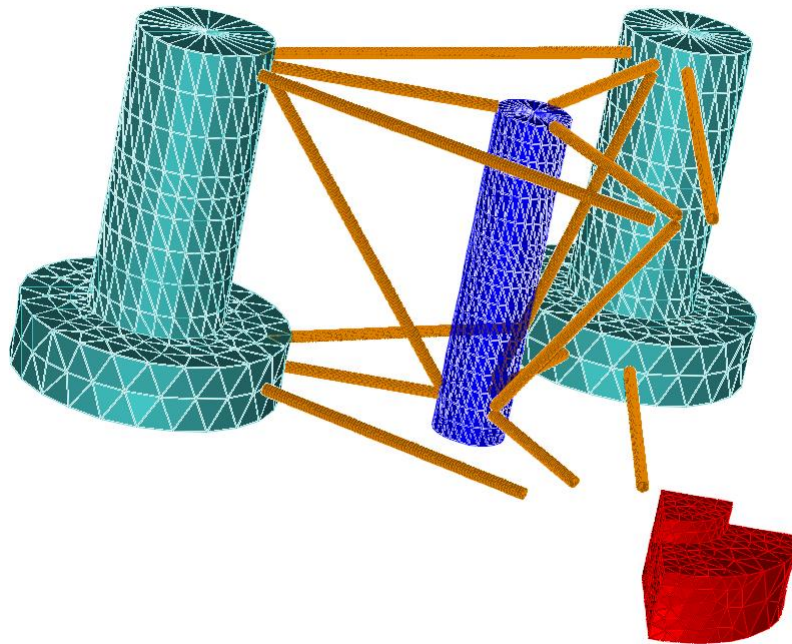
If the user would like the intersection to be remeshed to form 'nice' triangles then select the "Remesh Intersection" checkbox. Two parameters can then be specified to control the output from the remeshing:

- (1) Target edge length is the final edge length of the triangles that are remeshed and
- (2) The zone to be remeshed. It is specified by the number of triangles out from the intersection of the meshes.

In the example below a compartment is formed within the legs of a tripod floated boundaed by the red box mesh:



The settings used are those in the dialog above. The resultant mesh (compartment) is:



When Boolean operations do not work

If the Boolean operation fails, verify that the meshes are closed without degenerate faces. Detailed statistics are calculated for the selected mesh by selecting Mesh Statistics from the Mesh menu (select the mesh in the Assembly, or in rendered mode, by clicking on the actual mesh).

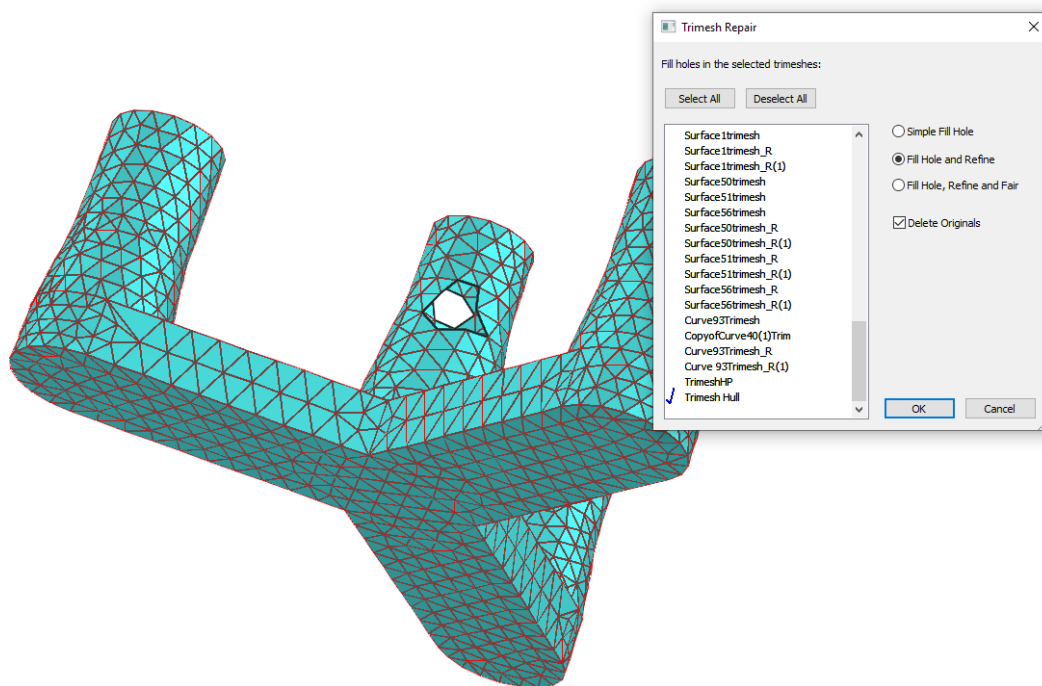
To determine if the mesh is properly closed and is suitable for Boolean operations verify that:

num.of Connected Components	=	1
num.of Border Edges	=	0
num.of degenerate faces	=	0
Volume	>	0.0
Self-intersecting	=	no
num.of Holes	=	0

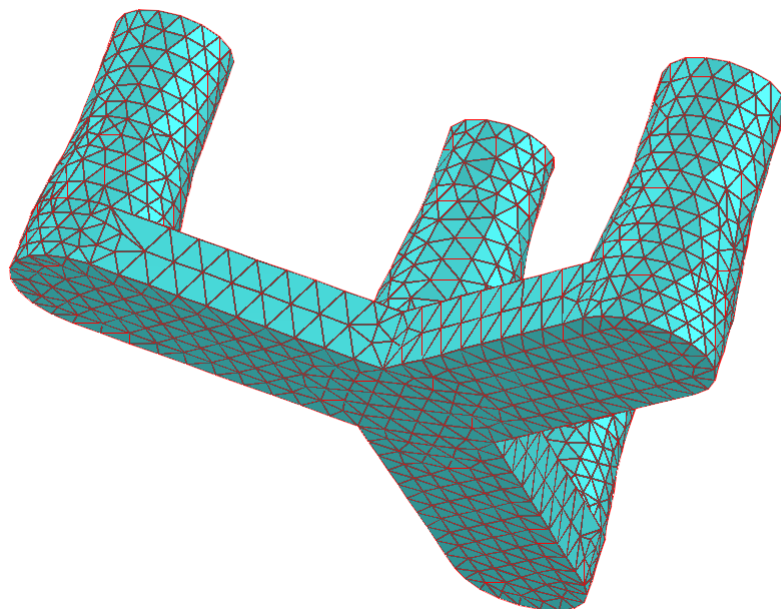
For further details, see the Section "Detailed Mesh Statistics" below.

Mesh Repair

Holes in non-manifold meshes can be filled automatically to form a manifold mesh. Choose "Repair..." from the Mesh menu:



Select the meshes to be repaired, the type of repair and whether or not to delete the original mesh. Type of repair include a simple fill hole that will fill the hole with the minimum number of tris, fill and refine that will produce a “nicer” interior mesh, fill, refine and fair that will also take into account orientation of surrounding tris and try to fair into those. The result on the above mesh is:



Mesh bonding

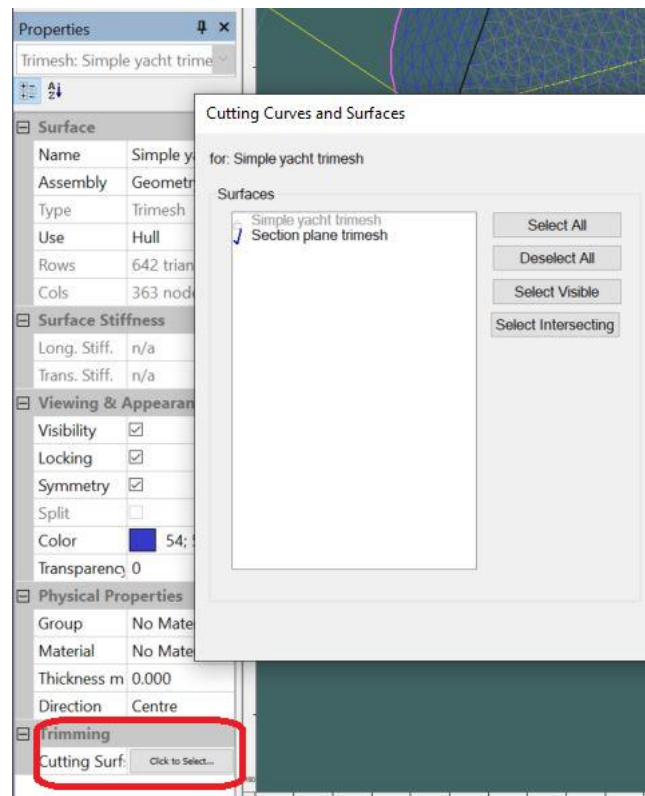
Intersection contours for meshes

Intersection contours between meshes can be displayed in a similar way to those for NURBS surfaces.

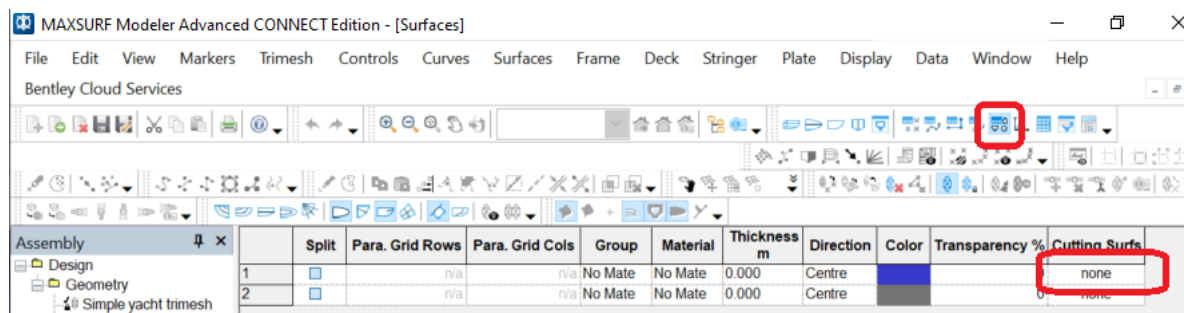
The intersection visibility is turned on via the intersection contour button or the contour visibility dialog (Display | Contours...). Intersections will only occur between meshes that are linked via “cutting surfaces” (see below).

Cutting surfaces for meshes

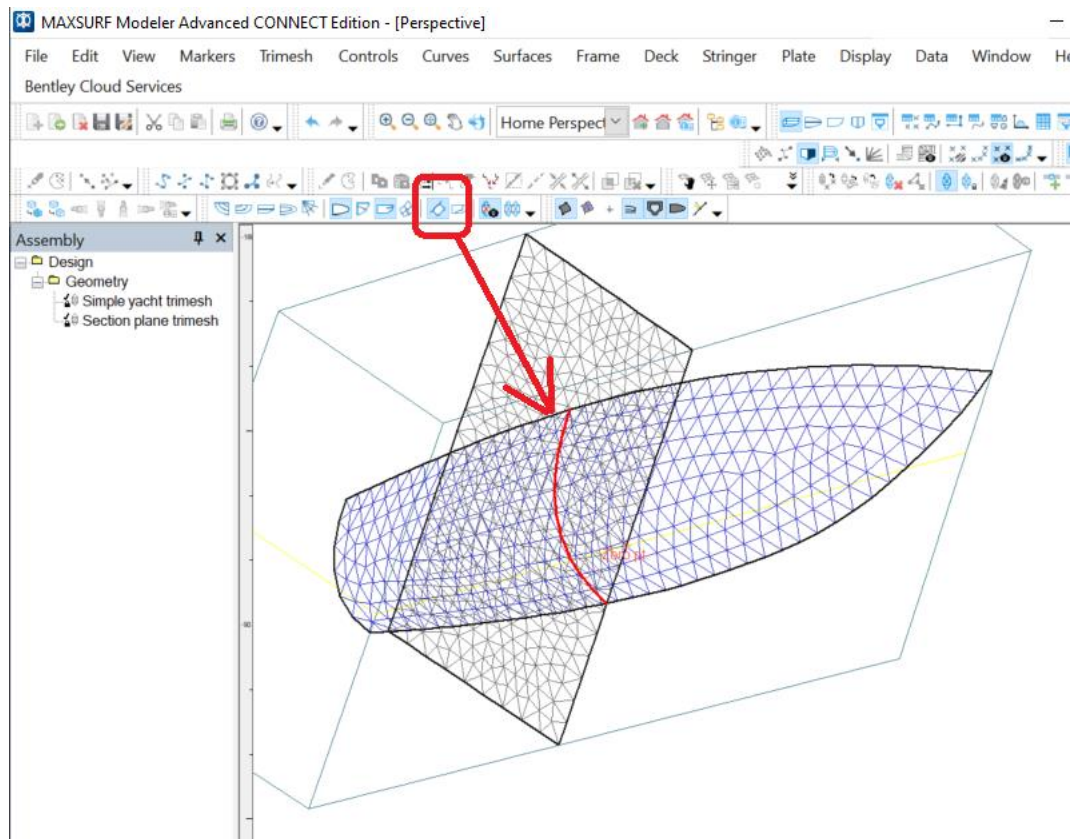
An intersection contour will only be displayed if it represents the intersection with one of the “cutting surfaces” listed in the mesh properties, as with NURBS surfaces. The cutting surfaces are listed in the cutting surface dialog, accessed via the properties pane:



Or by double clicking in the cutting surfaces cell of the Mesh in the Surfaces window:

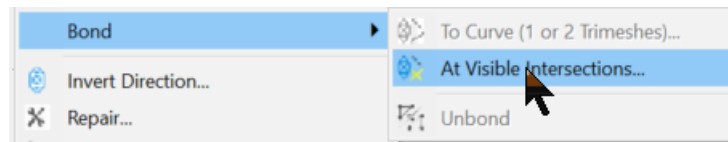


The intersection(s) will then be displayed in the drawing window:

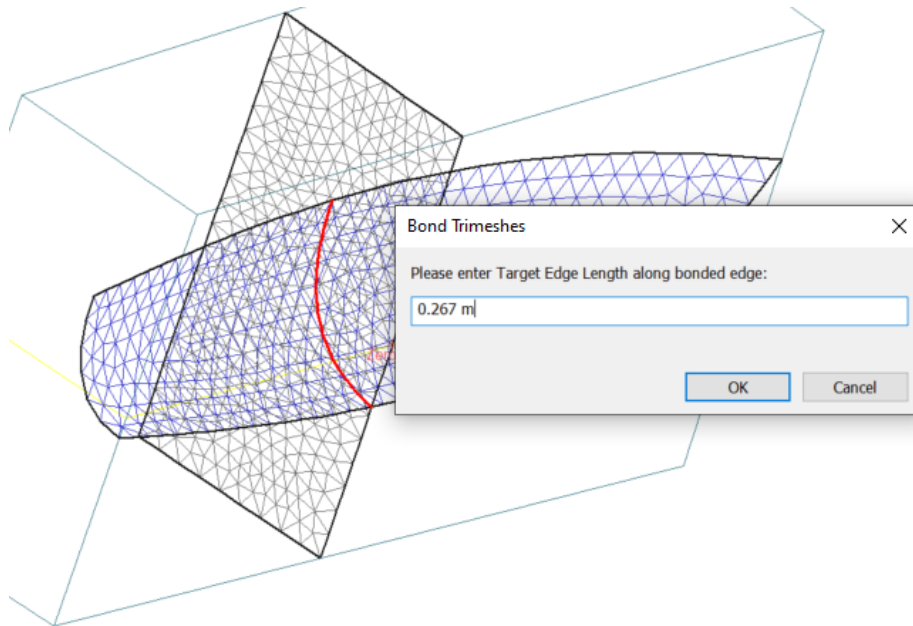


Bond meshes at visible intersections

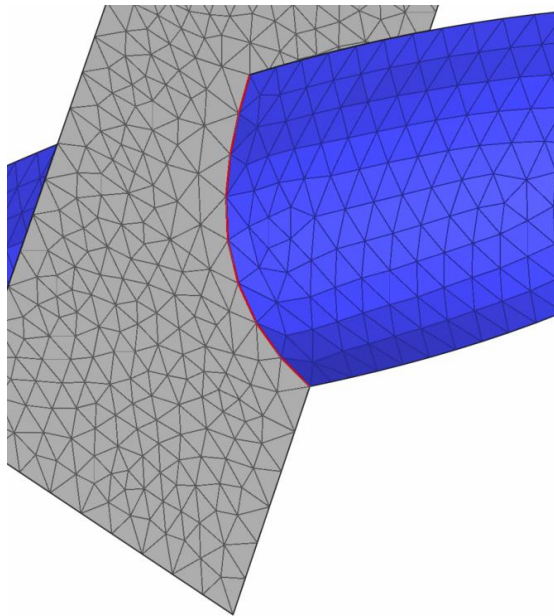
The Mesh bond commands are available from the Bond pop up sub menu in the Mesh menu:



To bond at visible intersections turn on only the intersections that you wish bonding to occur (using the cutting surfaces list as outlined above). Make sure the intersection forms a continuous contour in the regions required to be bonded. Choose Bond | At Visible Intersections from the Mesh menu. You will then be asked for a target edge length along the bond :



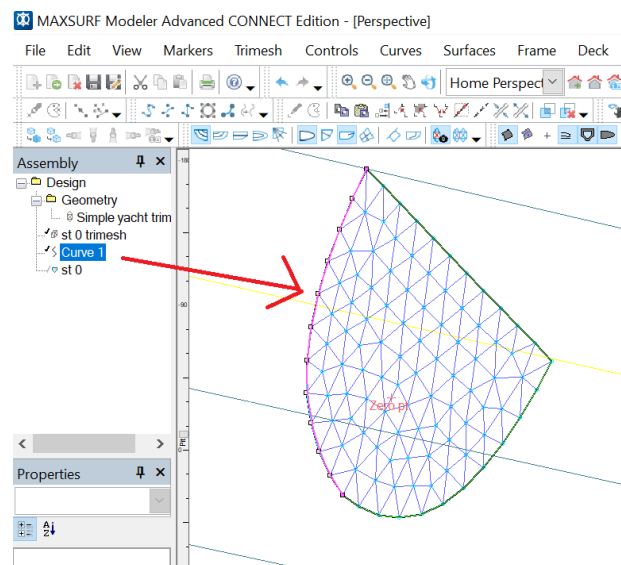
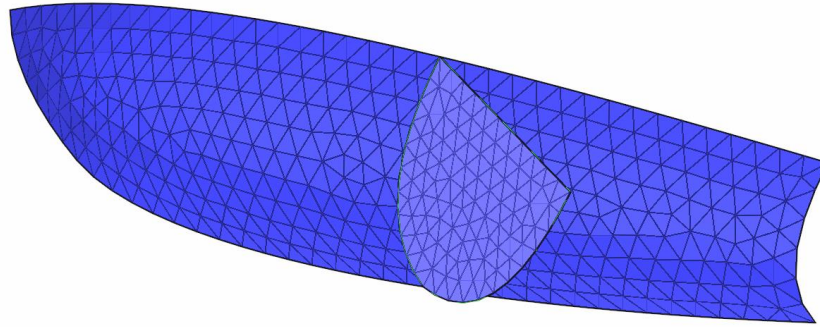
The default value is the average edge length of triangles of the intersecting meshes. Click OK:



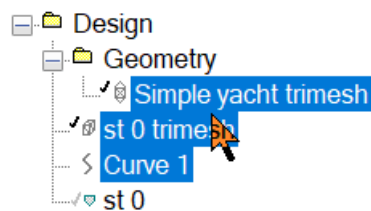
The meshes will now be bonded at the intersection with coincident nodes.

Bond mesh to curve

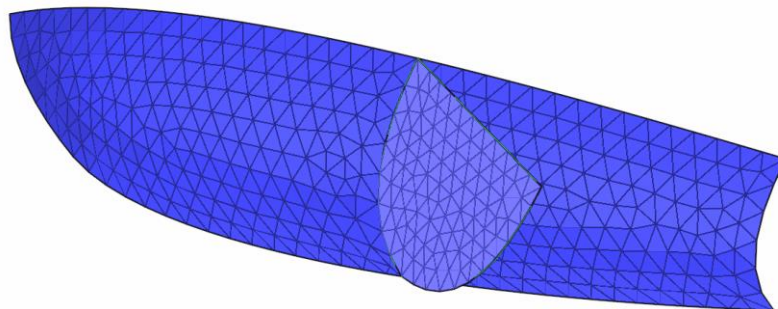
For meshes that 'almost touch' Modeler requires the user to tell it where the bond should occur. This is done using the "Bond | To Curve" function. First add a curve where the bond is to occur, in the case below we will add a curve along the edge of a bulkhead mesh:



Then selecting the two meshes and curve in the assembly pane:



We can then execute the Mesh | Bond | To Curve (1 or 2 meshes) command (the user will be asked for target edge length of tris along bond):



In summary, when bonding meshes, if a clean unambiguous intersection contour exists between two meshes use the “Bond | At visible intersections...” option. If the meshes “just” touch or “almost” touch define where the bond is to occur by adding a Curve and then use the “Bond | To Curve (1 or 2 meshes)...” option.

Mesh meshed from a NURB surface

Meshes may be created from a NURB surface (or a number of NURB surfaces). To generate a mesh surface from an existing NURB surface select “Mesh from Surface” from the Mesh menu. The following dialog will then appear (alternatively you may right-click with a surface selected in one of the drawing views):

Generate Trimesh

Mesh Type

☒ Optimised
☐ Regular Parametric

Generate Trimesh for

☐ Single mesh for all selected Surfaces
☒ Individual mesh for each selected Surface

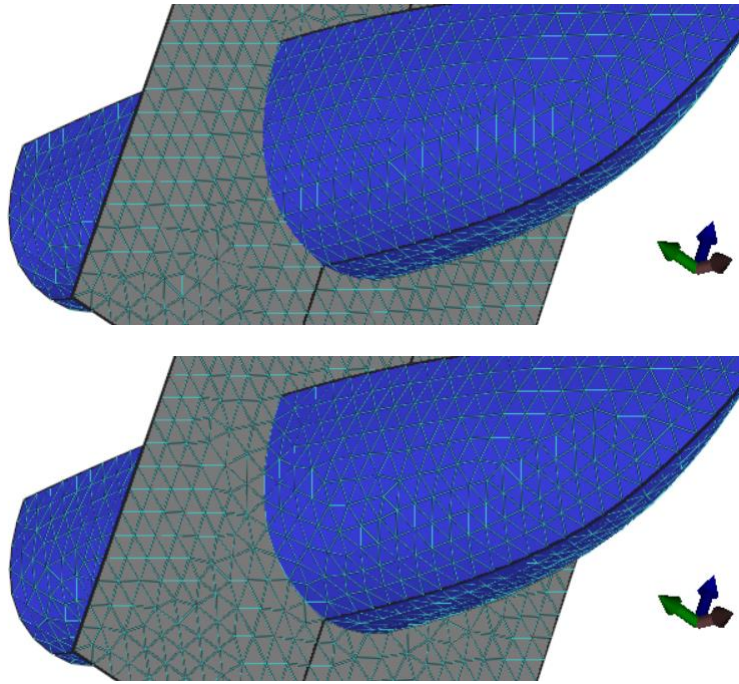
☐ Bond trimeshes at visible intersection curves
☐ Bond trimeshes at NURB bonded edges
☒ Preserve feature lines

Select All Deselect All Select Visible

	Surface Name	Create Trimesh	Delete existing	Min. edge length m	Max. edge length m	Hotspots/ Hotlines	Estimated num. triangles
1	bow	☒	☐	0.450	0.550	none	1499
2	Simple yacht	☐	☐	0.225	0.550	none	305

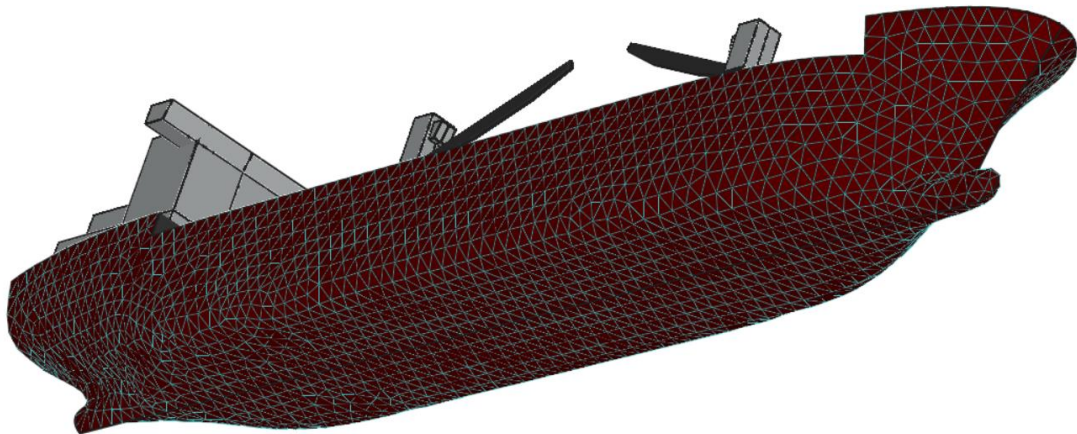
OK Cancel

A single mesh can be created over the entire model or a mesh can be created for each surface with a tick in the Mesh column of the table. “Min. edge length” is the minimum edge length required in each triangle generated in the mesh. “Max. edge length” is the maximum edge length required in each triangle generated in the mesh. The target edge length for the meshing algorithm will be the average of these values. The user can choose to “Bond trimeshes at intersection curves” or “Bond meshes at NURB bonded edges” by ticking the appropriate checkboxes. Bonding surfaces ensures that the triangles share common vertices (and hence triangle edges) along the contour. The two images below show two meshes with (bottom) and without (top) bonding across the intersection contour:



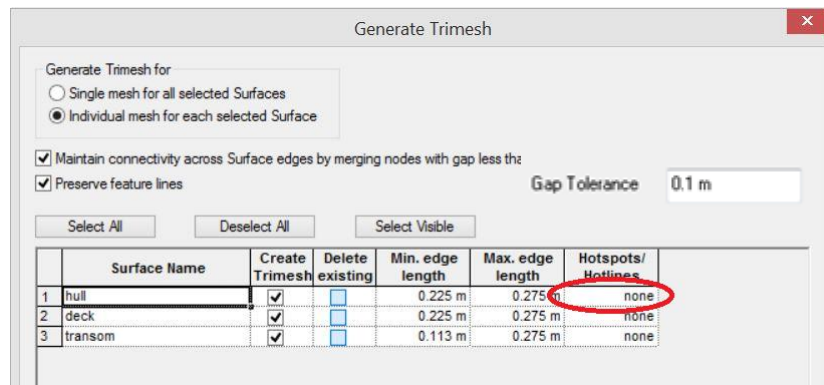
Notice in the bottom image that the triangle vertices and edges are coincident across the surfaces intersections. For meshes to be bonded across intersection contours, the intersections must be turned on (from the Visibility toolbar).

If the "Enforce feature line preservation" checkbox is ticked then any featurelines on the NURBS surface will be maintained in the mesh with triangle edges running along the featureline. This is useful to ensure that feature lines do not get 'lost' in the final mesh (i.e. chamfered out). If higher mesh density is required in a specific area one or more hotspots and/or hotlines may be defined for each mesh.

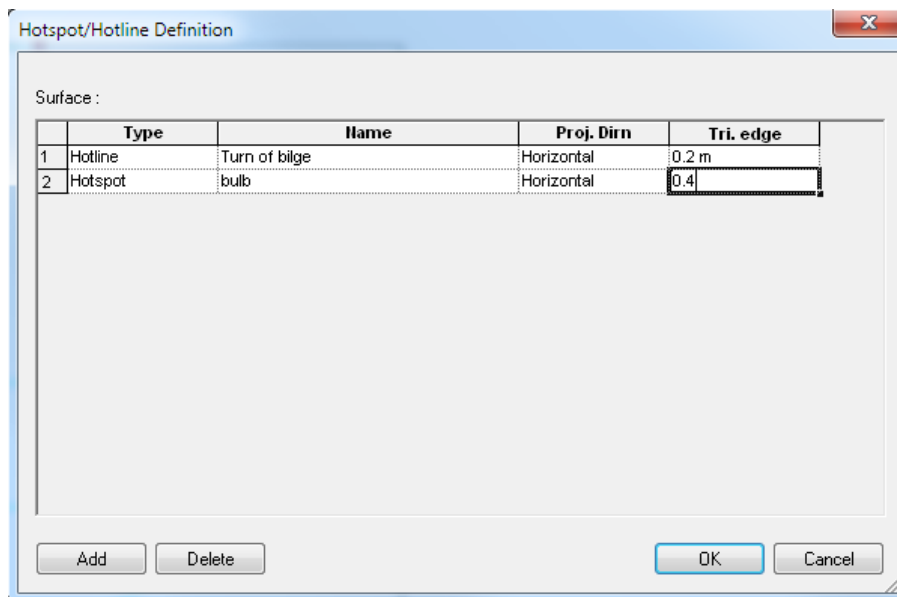


Hotspot/Hotline definition

In the generate mesh dialog the last column of the table is for defining hotlines or hotspots in the mesh. A hotline or hotspot is a point or line in the mesh where the user requires a different mesh density to that for the rest of the mesh. Hotspots can be defined at Marker locations, similarly hotlines can be defined at curve locations on the surface. To activate the Hotspot/Hotline definition dialog double click in the Hotline/Hotspot cell of the appropriate mesh in the generate mesh surface dialog:



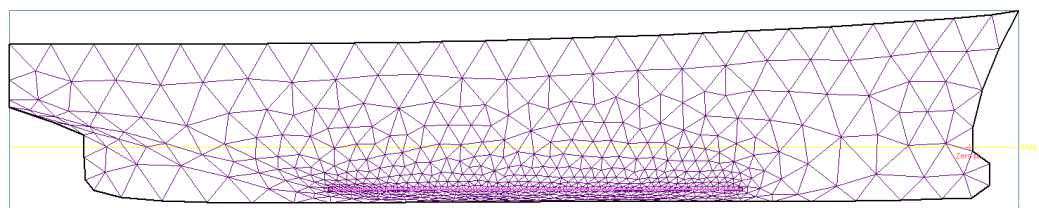
This will activate the definition dialog:

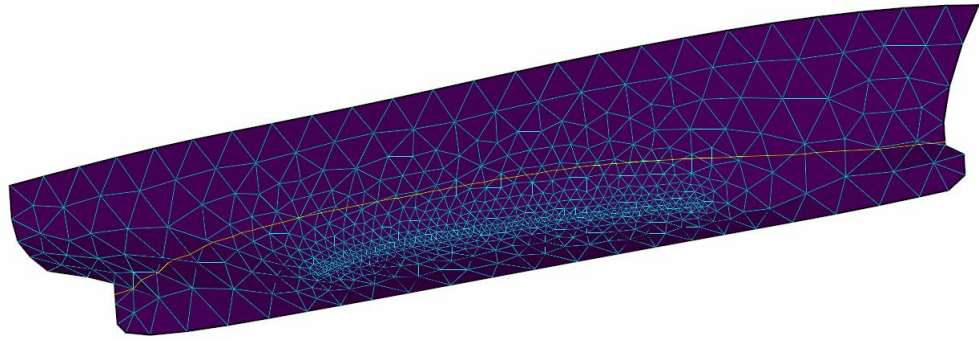


To add a hotspot or hotline click the Add button in the bottom left. This will add a blank row. In the first column specify the type (hotspot or hotline) from the combo box. The Name combo box will then be populated with all named curves or markers in the design (note if the marker is not named it will not appear in the list). Since a curve or marker can be specified anywhere in space they need to have a projection direction associated with them, this will tell Modeler which way you want the curve/marker projected onto the surface. Finally the Tri. edge column is where the user specified the target mesh density for that region of the mesh.

A hotspot/hotline may be deleted by clicking on the row number which will highlight the row and hitting the delete button.

An example may be where a user would like to have a higher mesh density along the bilge radius of the hull:





Mesh from Parametric-based NURB Surface mesh

The “Add Mesh from Surface” dialog has an option to mesh the NURB Surfaces parametrically. The resulting Trimesh is based on a grid of the Surface’s iso-parametric curves which are then subdivided into pairs of triangles. The basis diagonal can be selected to be the 1st or 2nd diagonal or have the diagonal selected so as to give a symmetrical layout. Furthermore, the alternative diagonal can be selected if it is shorter than a specified percentage of the basis diagonal. These options offer a great deal of flexibility in specifying how the surface’s parametric mesh should be subdivided.

Generate Trimesh [X]

Mesh Type

☐ Optimised

☒ Regular Parametric

Generate Trimesh for

☐ Single mesh for all selected Surfaces

☒ Individual mesh for each selected Surface

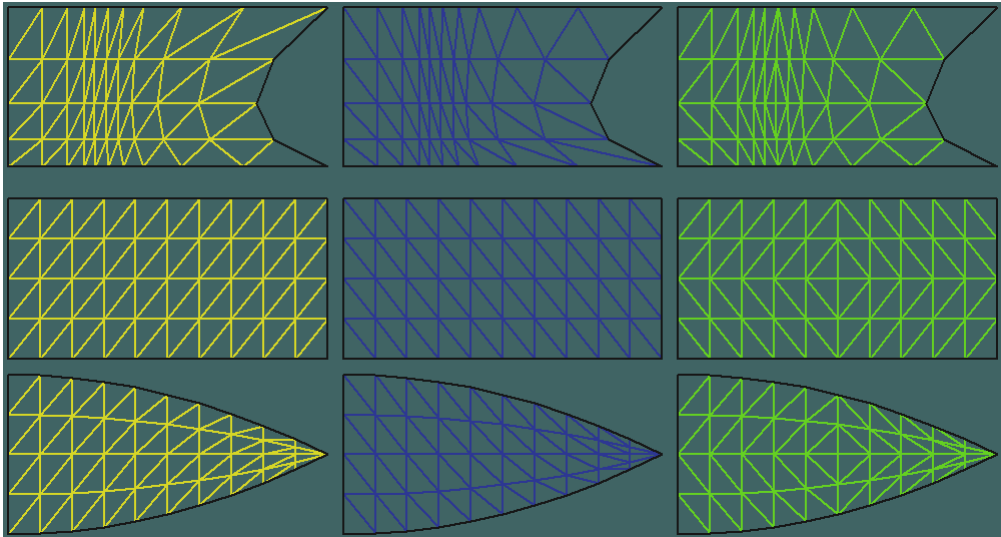
☐ Bond trimeshes at visible intersection curves

☐ Bond trimeshes at NURB bonded edges

☐ Preserve feature lines

	Surface Name	Create Trimesh	Delete existing	Num. longitudinal triangles	Num. transverse triangles	Diagonal layout	Use shortest diagonal if < x% of basis	Estimated num. triangles
1	Cres1stDiag	☒	☐	10	4	1st diagonal	0%	80
2	Cres2ndDiag	☒	☐	10	4	2nd diagonal	0%	80
3	CresSymDiag	☒	☐	10	4	symmetrical	0%	80
4	Rect1stDiag	☒	☐	10	4	1st diagonal	0%	80
5	Rect2ndDiag	☒	☐	10	4	2nd diagonal	0%	80
6	RectSymDiag	☒	☐	10	4	symmetrical	0%	80
7	Point1stDiag	☒	☐	10	4	1st diagonal	0%	80
8	Point2ndDiag	☒	☐	10	4	2nd diagonal	0%	80
9	PointSymDiag	☒	☐	10	4	symmetrical	0%	80

New “Add Mesh from Surface” dialog with parametric meshing option

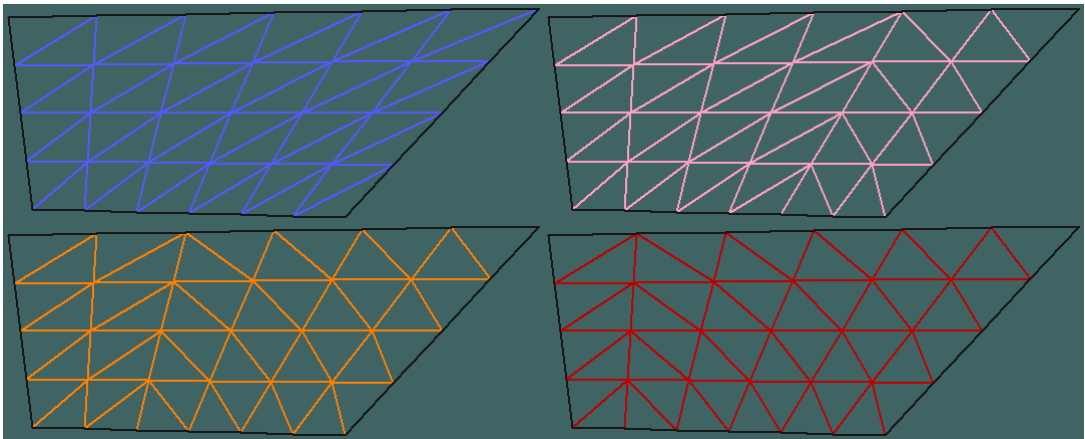


Resulting Meshes with basis layouts:

1st diagonal (yellow); 2nd diagonal (blue); symmetrical diagonal (green)

With the “Use shortest diagonal” option, skinny triangles with small minimum angles can be avoided. Use the percentage value to determine at which point the shortest diagonal will be selected in preference to the basis diagonal selected in the “Layout” column:

- 0% : Always use the basis diagonal
- 100% : Always use the shortest diagonal
- 80% : Use the shortest diagonal if it is less than 80% of the length of the basis.



Meshes with progressively increasing priority of shortest over basis diagonal:

Always (0%) use basis diagonal (blue); Use shortest if less than 60% of basis (pink)
Use shortest if less than 80% of basis (orange); Always use shortest (100%) diagonal (red)

Meshing parameter details

Details of the basis NURB Surface and meshing parameters are shown in the “Type” column of the Surfaces table:

Surfaces			
	Name	Assembly	Type
1	QuadSurface	Geometry	B-Spline
2	QuadSurfTM diag.ovr 90%	Geometry	Trimesh (basis: NURBS QuadSurface Parametric mesh: 6x4; Layout 1st diagonal; Diag.override 90%)
3	QuadSurfTM diag.ovr 80%	Geometry	Trimesh (basis: NURBS QuadSurface Parametric mesh: 6x4; Layout 1st diagonal; Diag.override 80%)
4	QuadSurfTM diag.ovr 60%	Geometry	Trimesh (basis: NURBS QuadSurface Parametric mesh: 6x4; Layout 1st diagonal; Diag.override 60%)
5	QuadSurfTM diag.ovr 40%	Geometry	Trimesh (basis: NURBS QuadSurface Parametric mesh: 6x4; Layout 1st diagonal; Diag.override 40%)

Meshing parameters and basis geometry information for Meshes

Tips for creating a NURB surface that will mesh well

The automated meshing tool is aimed at assisting the user in generating a mesh that will be useful for Analytical applications (BEM, CFD and FEA). The automated mesh will need to be reviewed by the user and manual adjustments made where required.

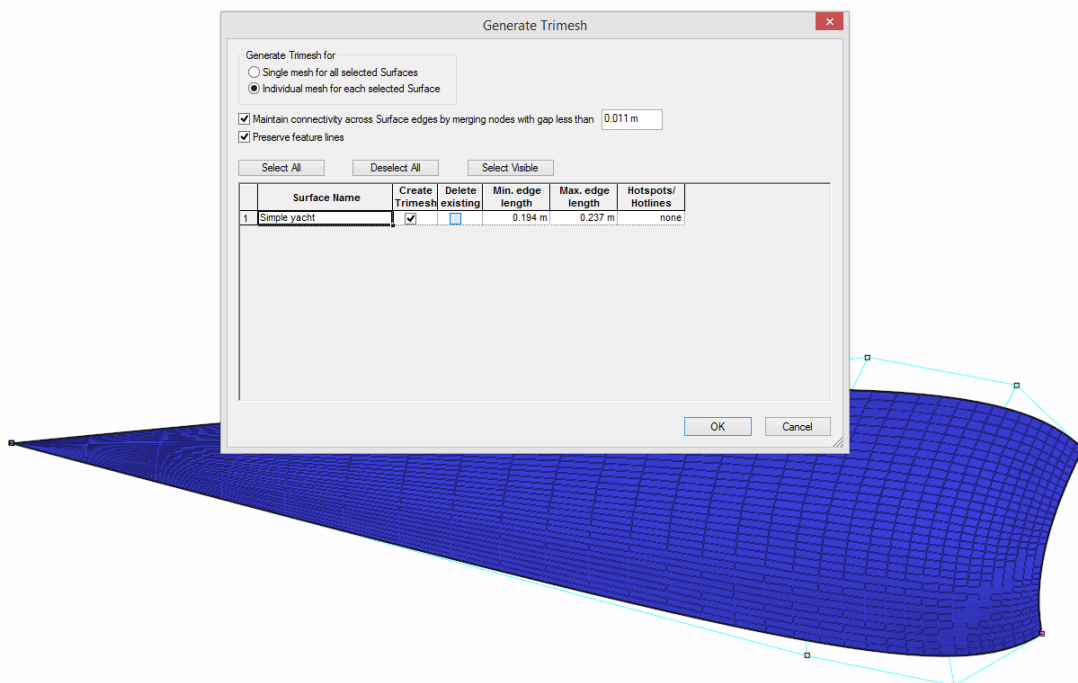
Below are a list of rules to follow when modelling NURBS surfaces for mesh generation.

- Don't try to model surfaces to a level of detail smaller than the minimum edge length set in the generate mesh surfaces dialog

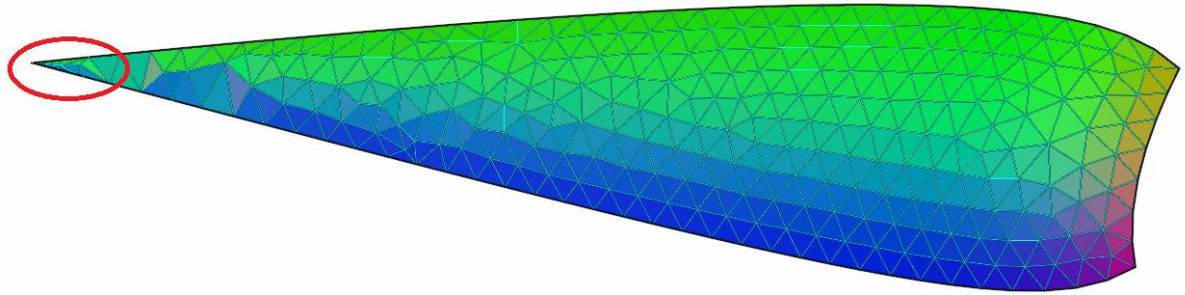
If the minimum edge length set for a mesh is 1m, and the model contains a gunwale radius of 0.5m then there is no point in modelling this.

- Avoid discontinuities in the NURBS surface

The meshing is a two stage process, the initial mesh is done in parametric space and the final mesh is done in x,y,z space. A surface with discontinuities (i.e. a triangular surface with one edge collapsed to a single point), may not generate an even mesh in the vicinity of the discontinuity. For example the surface below is collapsed to a singularity on the left hand edge:



This produces an uneven mesh in the vicinity of the singularity that will need manual editing:

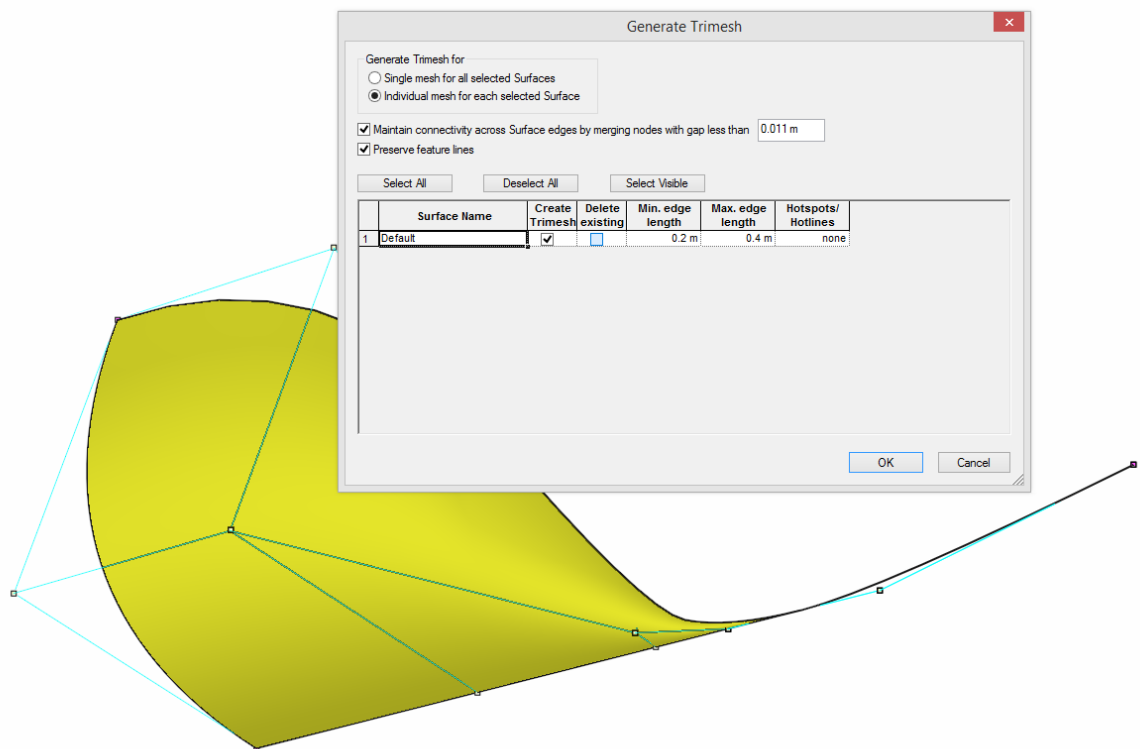


Unless using the “preserve featureline” option, featurelines (rows/coumns of over compated control points) will cause an uneven automatically generated mesh.

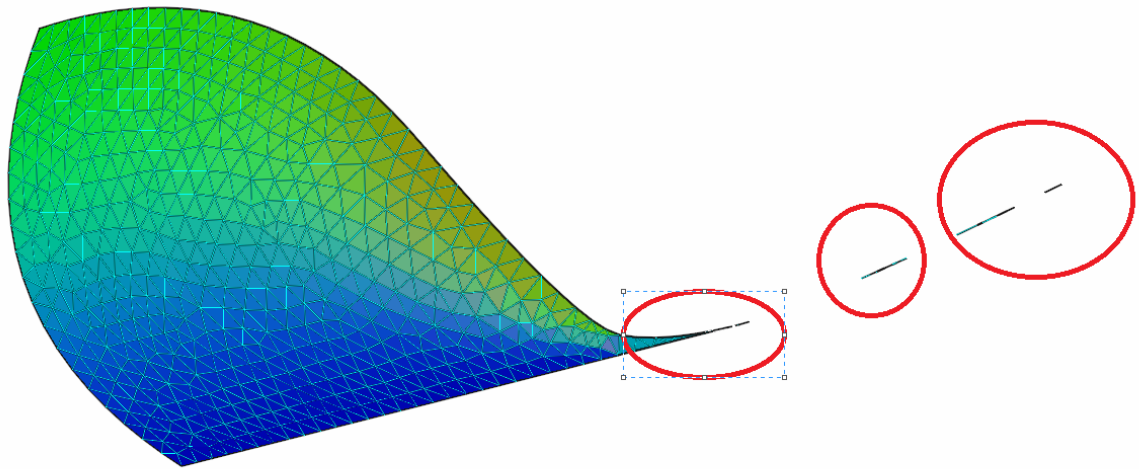
- Avoid high aspect ratio surfaces

Long skinny surfaces do not tend to mesh evenly especially if opposite side edges approach each other at a distance less than the minimum edge length of the mesh.

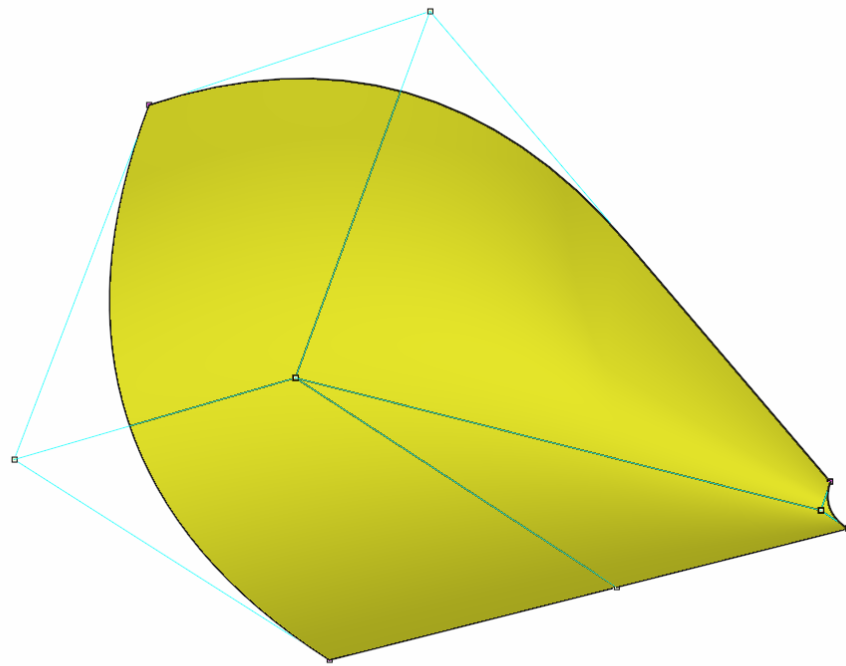
For example the following high aspect ratio surface that has the top and bottom edges converging into a single line:



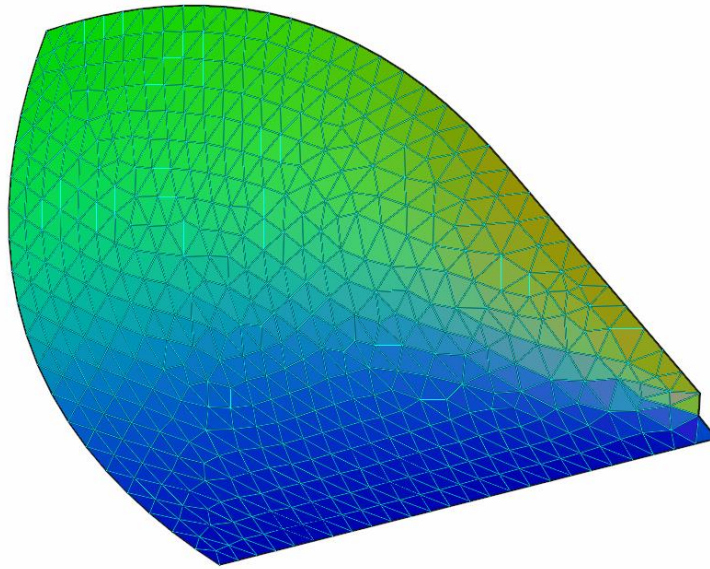
Auto generates this undesirable mesh:



In this case it would be best to model the NURB surface to the level of detail on the min edge length (0.2m), as shown below:



Which auto generates this mesh:



To reduce the aspect ratio of a NURBS surface the surface may be split in two. Use the Surfaces | Split surface command. This command become available when an edge control point (at the position where you want the surface split) is selected.

- Where possible bond the edges of adjacent surfaces.

This is a good NURBS surface modelling practice in general. It also aids when trying to generate multiple meshes across multiple surfaces where the “maintain connectivity” option is chosen in the generate mesh dialog.

- Bad autogenerate meshes are often a result of too many constraints.

Try simplifying the constraints by unchecking “maintain connectivity” option in the generate mesh dialog. A more even mesh can be obtained by setting the minimum and maximum edge lengths to be equal (or close to), doing this however may result in some topological features getting lost due to edge lengths being too long.

- Use manual editing tools to adjust mesh post auto generate.

Use compact nodes function (highlight 2 or more nodes when mesh is unlocked) and press ctrl+K to compact the nodes. The first node selected will be the ‘master node’ to which other nodes will move. Highlight one or more nodes and hit delete to remove that node. Comapcting and deleting are the tools most often used when aligning nodes along edges of adjacent surfaces.

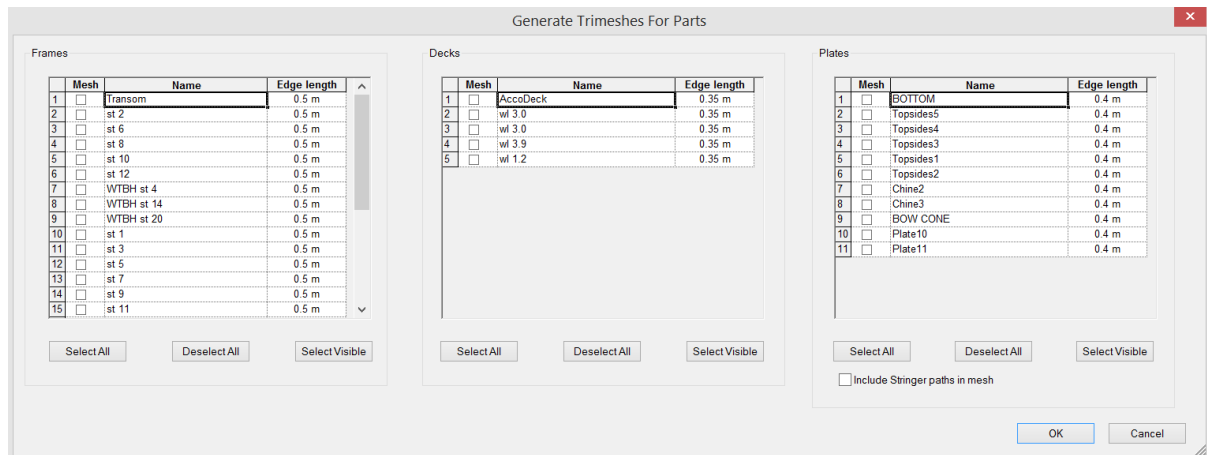
Ctrl+shift+N will add a new node to the mesh, move the add node cursor to location you wish the node to be added and left click. Adding a node in the centre of a triangle will create three inner triangles. Adding a node close to a triangle edge, will split the triangle edge and form new triangles.

- Use hotlines/hotspots to capture geometric anomalies

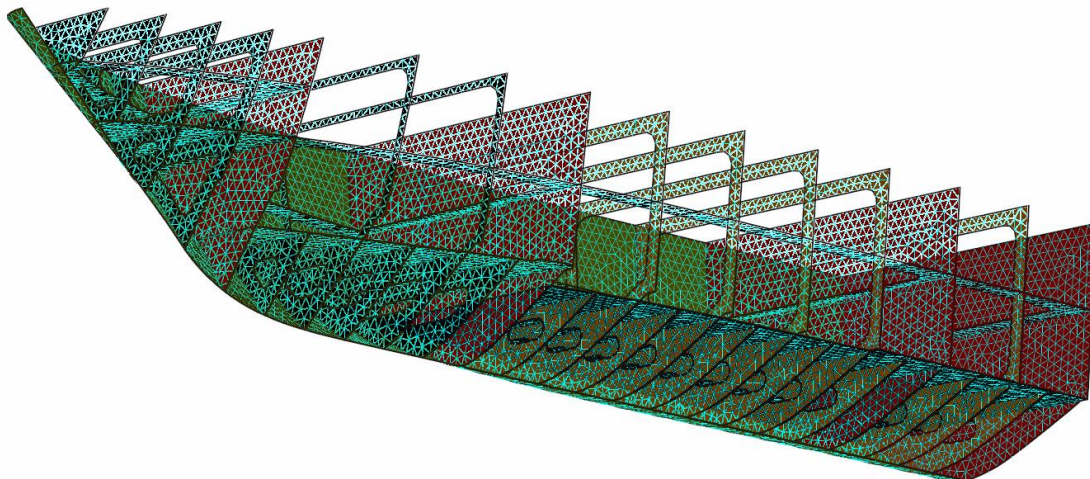
If a mesh is too coarse to capture a feature that you require in the mesh, use either a hot spot or a hotline to adjust the mesh density in the vicinity required.

Mesh from Structural Parts

A mesh surface can be generated from a part (Frame, Deck or Plate). To do this choose Mesh | Generate from Parts... from the mesh menu. The menu item will only be enabled if you have at least one part in the design. The following dialog will then appear:



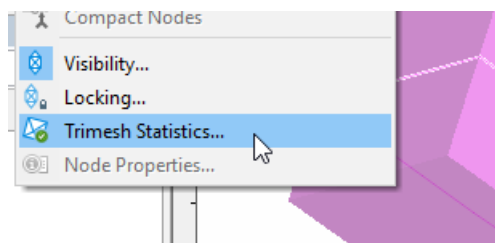
The parts to be meshed should be checked and the target edge length for the generated mesh should be entered. Note that the generated meshes may not be coincident along common boundaries with other meshes in the design. To ensure coincidence the meshes would need to be bonded. If the user would like to have mesh nodes coincident with the stringer path (useful for FEA mesh generation), the "Include stringer paths in mesh" option should be selected.



Meshes generated from Structural parts

Detailed Mesh Statistics

Detailed statistics are calculated for the selected mesh by selecting Mesh Statistics from the Mesh menu (select the mesh in the Assembly, or in rendered mode, by clicking on the actual mesh):



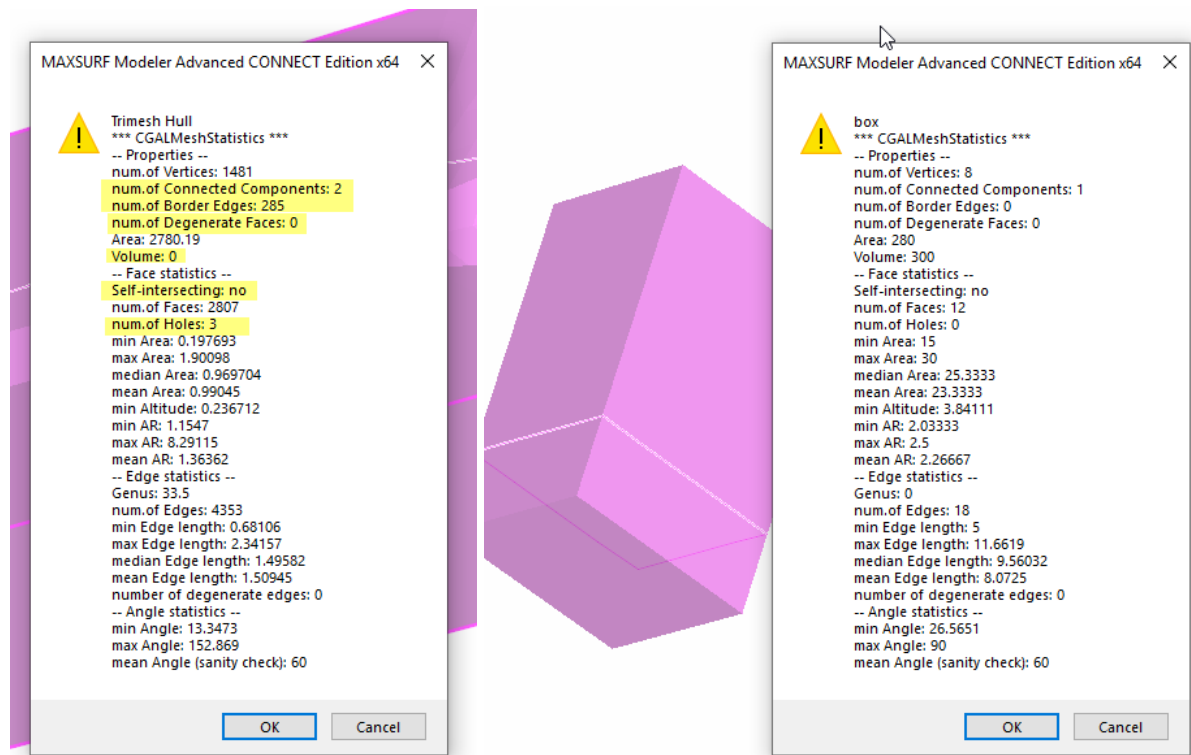
Show detailed statistics for the selected Mesh

Verifying Mesh Properties for Successful Boolean Operations

If the Boolean operation fails, verify that the meshes are closed without degenerate faces. Detailed statistics are calculated for the selected mesh by selecting Mesh Statistics from the Mesh menu (select the mesh in the Assembly, or in rendered mode, by clicking on the actual mesh):

To determine if the mesh is properly closed and is suitable for Boolean operations verify that:

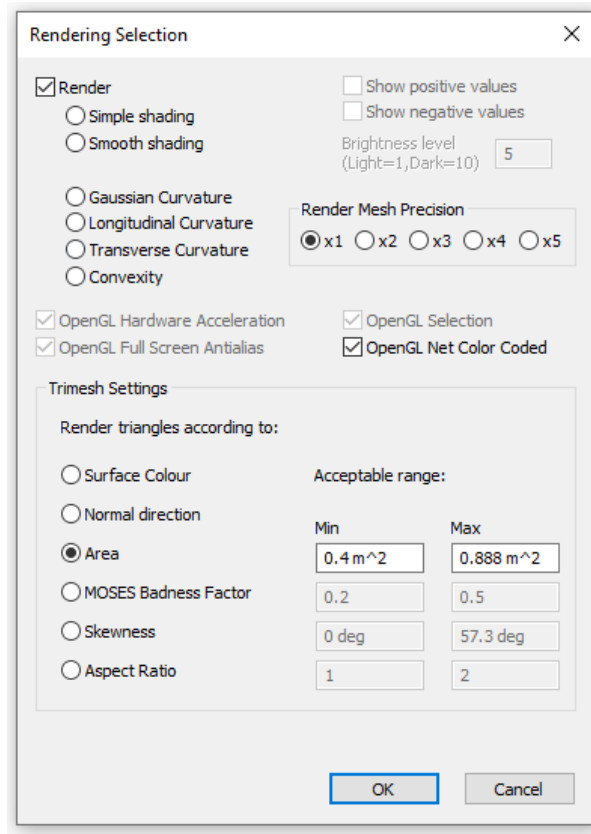
num.of Connected Components	=	1
num.of Border Edges	=	0
num.of degenerate faces	=	0
Volume	>	0.0
Self-intersecting	=	no
num.of Holes	=	0



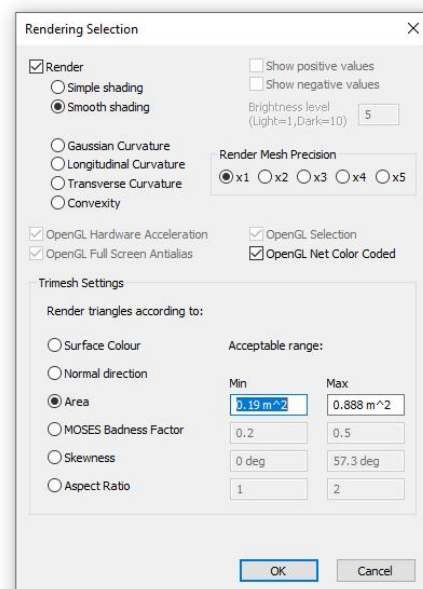
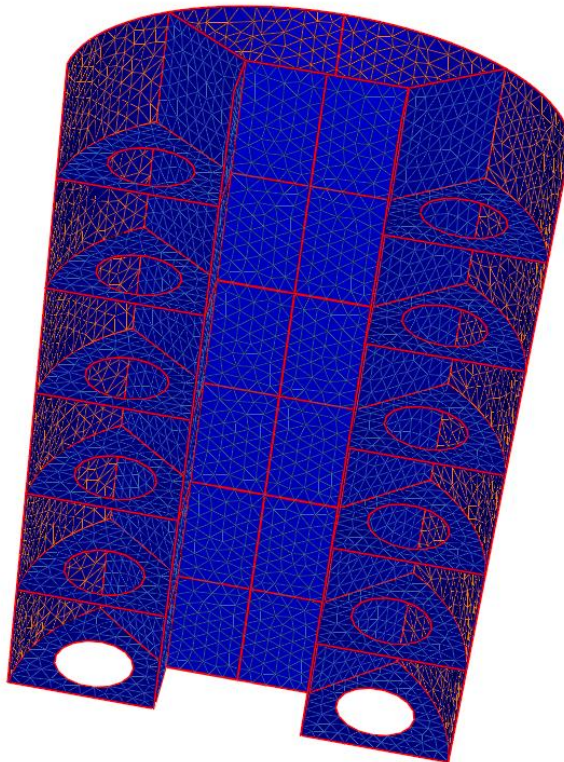
Bad(left) and Good (right) statistics for Mesh which is suitable for Boolean operations

Mesh rendering options

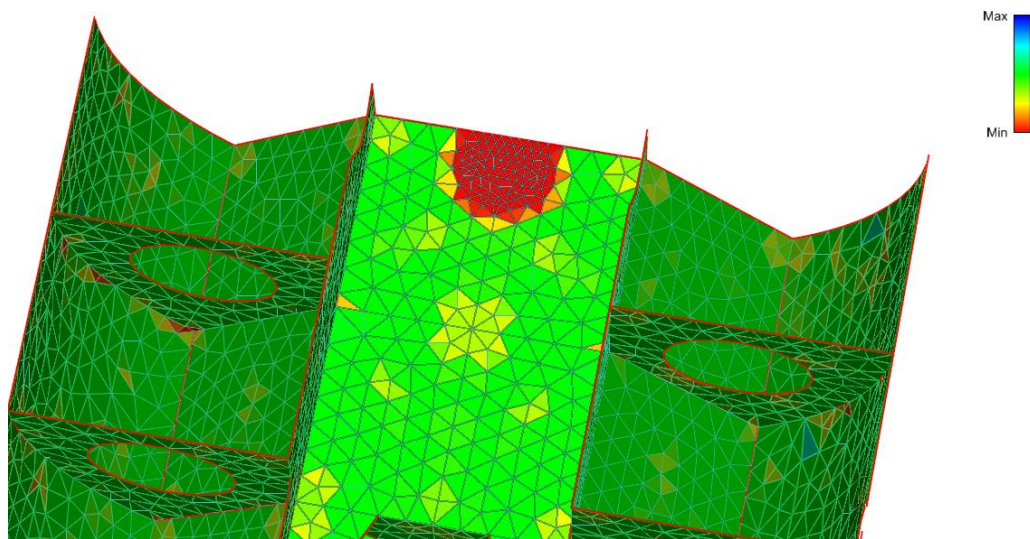
There are a number of rendering options for meshes to help the user to get an even regular mesh. If the rendering dialog is launched from Display Render the following will appear:



Whether you are meshing for FEA or Hydrodynamic analysis, the quality of the mesh used is important. Triangles can be rendered according to 6 parameters: Surface color, Normal direction, Area, MOSES Badness factor, Skewness and Aspect Ratio. The last four the user can set a range on the minimum and maximum acceptable values:

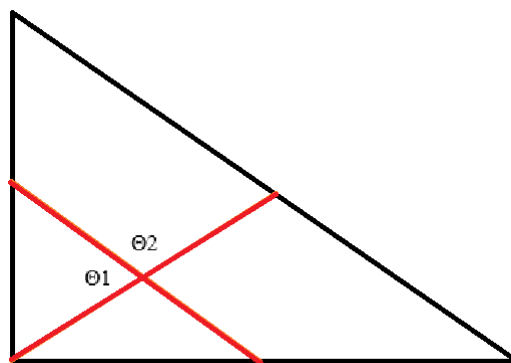


Any triangle that falls within the acceptable range will be colored green with triangles outside this range being shaded according to the legend:



This gives a quick visual reference as to which area(s) need to be worked on and how severely the triangles are outside the acceptable range the user has set. The rendering is updated in real time as the triangles are manipulated giving feedback on their quality.

Skewness gives a measure of how 'skewed' the triangle is. The definition chosen can be seen below:



-Draw a line from each node to mid point of its opposite side, Draw another line joining mid-points of other two sides measure the angles between two lines. Repeat the step for all the three nodes and find all six angles (Θ_1 to Θ_6).

-Skewness (in degrees) is calculated by subtracting minimum angle from 90:

-Skewness = $90 - \min(\Theta_1, \Theta_2, \dots, \Theta_6)$

The triangle aspect ratio definition chosen for Modeler is:

Aspect Ratio = the circumradius to twice its inradius, so $AR = abc / (8(s-a)(s-b)(s-c))$ where a, b, c are the lengths of sides of the triangle and $s = (a+b+c)/2$.

Please refer to the MOSES manual for an explanation of the MOSES badness factor.

Quadmeshes

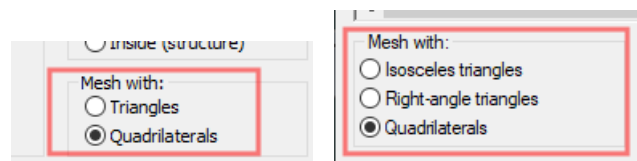
Modeler can produce meshes with quadrilateral elements (facets) to Mesh objects.

Quadrilateral and Triangle mesh elements

The Modeler Mesh objects supports quadrilaterals as well as triangles. This is done by tracking the link between pairs of adjacent triangles which are to be considered as a single quadrilateral mesh element. Since the quadrilaterals are composed of two triangles there is no requirement (or enforcement) that quadrilateral element be planar. Under most circumstances, the common edge between the adjacent triangles is not displayed, however it is apparent in the shaded, rendered view if the two triangles are not in the same plane. It is also for this reason that you will see two mesh normal arrows on a quadrilateral element (one for each triangle). However clicking on either will invert the direction of the whole quadrilateral.

Generating quadrilateral meshes

Primitive geometrical shapes (plane, box, cylinder, column) can be added from the Meshes menu. In the dialog you are given the option to mesh using triangles or quadrilateral elements. Where possible the resulting mesh will use quadrilateral elements (triangle elements will be used to close the ends of cylinders and columns).



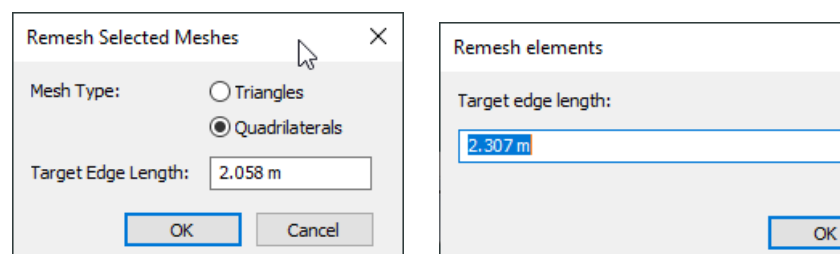
Mesh element options for primitive Mesh objects

Meshing individual NURB surfaces can be done using quadrilateral elements. Use the Meshes | Add Mesh from Surfaces dialog. You can then select whether to create a mesh or a quad-dominant mesh:

	Surface Name	Create Trimesh	Create Quadmesh	Delete all existing meshes linked to Surface	Min. edge length m	M
1	Topsides	☐	☒	☐	0.527	
2	Chine	☒	☐	☐	0.473	
3	Bottom	☐	☐	☐	0.479	
4	Deck	☐	☐	☐	0.530	

When creating a Quadmesh using the Optimised mesh type option, although the mesh will be quad-dominant, triangles may also be added if necessary. The Regular Parametric mesh layout will produce a quad-only mesh provided the NURBS surface does not have a zero-length edge. The Regular Parametric mesh layout will mesh the whole NURBS surface, even if it is trimmed.

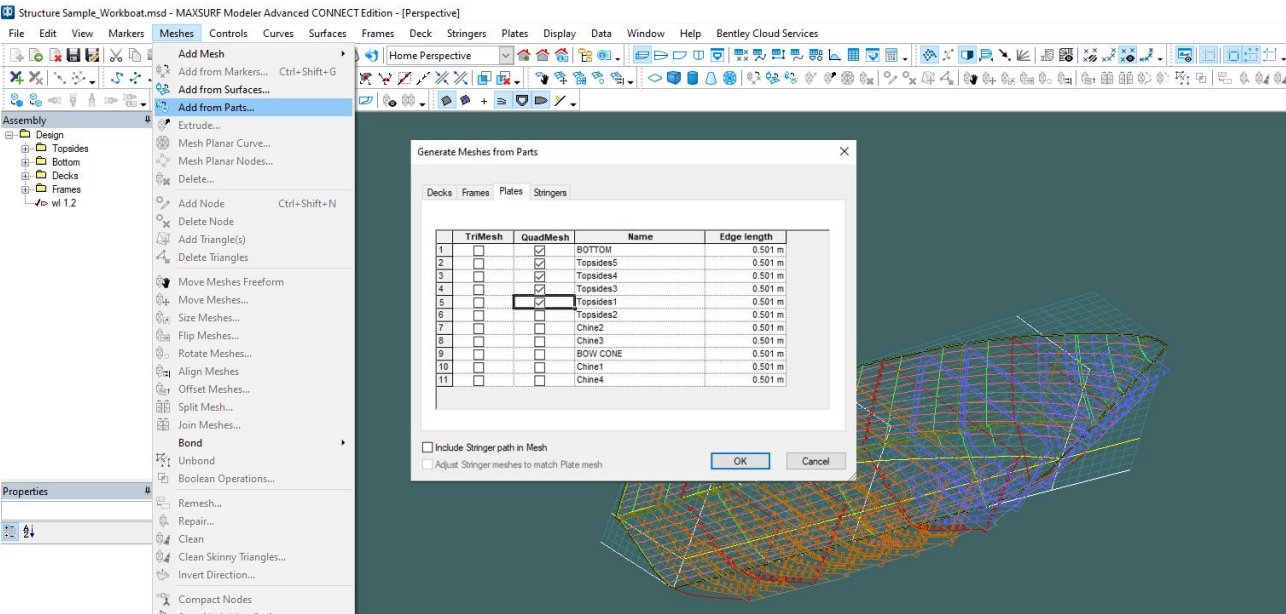
When remeshing an existing mesh, you can select whether the mesh should be triangular or quad-dominant. Note that this only applies when remeshing an entire mesh. When remeshing a part of an existing mesh, this will convert the entire mesh to triangular elements.



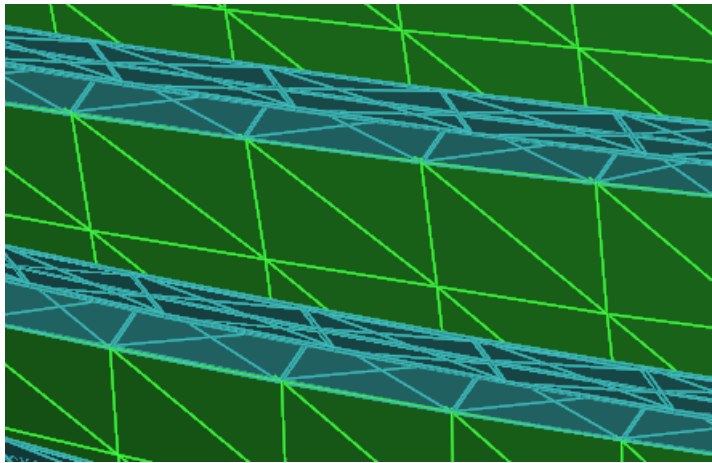
Remeshing dialog for: a) whole mesh and; b) selected mesh elements

Quadmeshes can be added on structural parts

As with adding a mesh on a structural part (Frames, Plates, Decks and Stringers), Quadmeshes can also be added on structural parts:



If include stringer path is checked then the stringer path edges will be coincident with the Plate Mesh quad element edges:



Meshing plates with exact stringer paths included in the mesh

Import and export

The following import and export file formats now support quadrilateral elements: WAMIT, 3DM, PLY and OFF. (STL file format supports only triangular mesh elements and any quadrilateral elements will be subdivided into two triangles about their common internal boundary).

Mesh operations

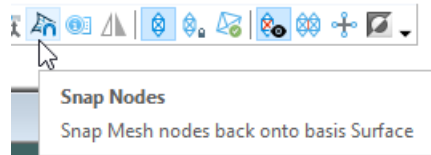
A large proportion of the Mesh operations will convert the mesh to a pure triangular element mesh. However these may be remeshed to a quad-dominant mesh if required.

User interface changes

We have stopped referring to “Mesh” and now use “Mesh” since the can now include both triangular and quadrilateral elements. This is reflected in the Menu, Ribbon and Toolbar. We are updating the manual. However to prevent breaking automation scripts, we have not changed the COM interface objects; even though they retain the same “Mesh” name they *do* support both type of elements.

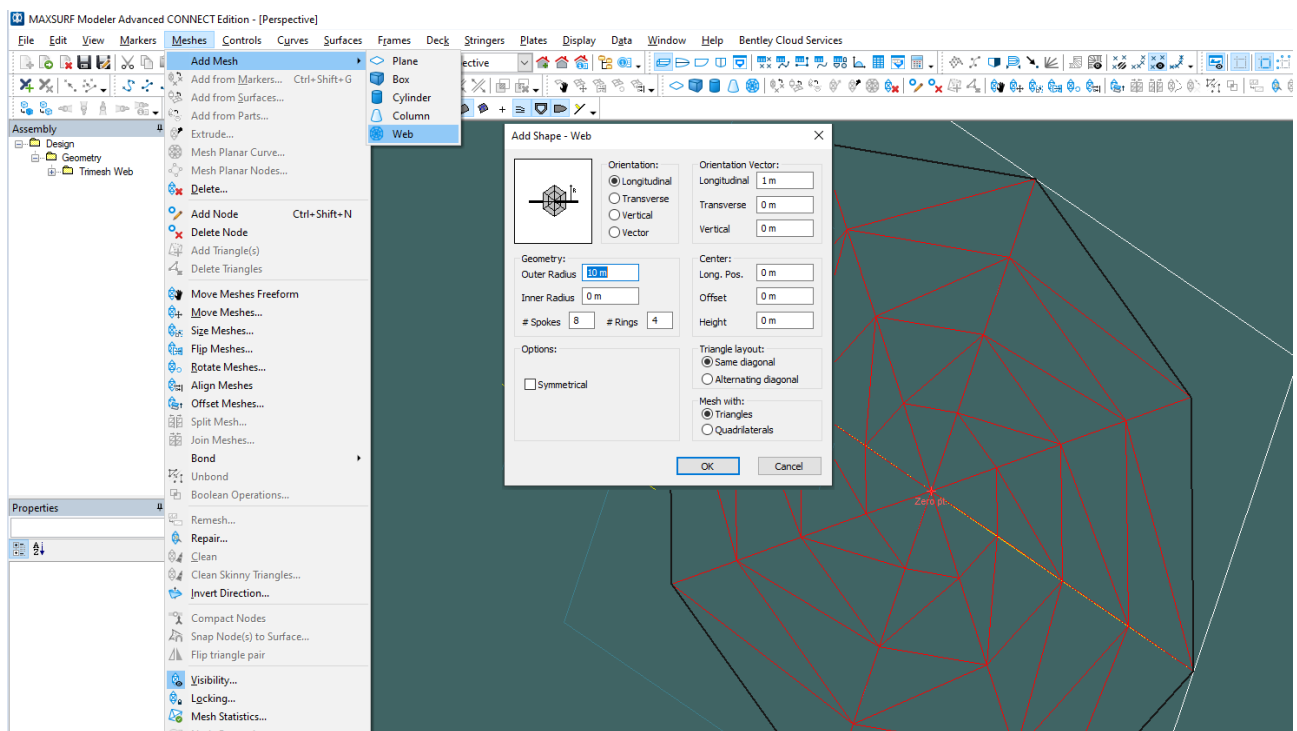
As noted earlier, the Mesh creation dialogs which now support an option for quadrilateral meshes.

The Mesh toolbar has been extended to include all the commands available in the Mesh menu. Furthermore, the icon for the snap node to surface button has been changed:



Web Meshes

Modeler can auto-create "Web meshes" polygonal tri/quad meshes:



Different element layouts can be user specified depending on the use of the mesh (ie hydrodynamics vs. FEA vs hydrostatic calculations).

Working with Control Points

Within Modeler, a design is made up of a number of [NURBS Surface](#). The shape of the surface is defined only by the:

- Control points positions
- Control points weight
- Surface stiffness

In the previous section we have seen how surfaces may be manipulated as whole objects. Here we will look at the tools available for manipulating a control point or group of control points to achieve the desired surface shapes.

In this section:

- [Adding Control Points](#)
- [Deleting Control Points](#)
- [Moving Control Points](#)
- [Aligning Control Points](#)
- [Smooth Controls or Straighten Controls](#)
- [Manipulating Groups of Control Points](#)
- [Compacting Control Points](#)
- [Grouping Control Points](#)
- [Displaying Control Points](#)
- [Control Point Weight](#)
- [Advanced - Transpose](#)

Adding Control Points

To increase the density of a control point net, columns of control points may be added in the Plan and Profile windows, and rows may be added in the Body Plan window.

To add a control point row or column

- Select the Plan, Profile or Body Plan window, as appropriate.

In the Body Plan window you will also need to select a current column in which to perform the insertion. The current surface's current column is indicated by a highlighted column marker in the Control Box. You can select a different column by clicking on a different column marker, or by selecting a control point in the desired column.

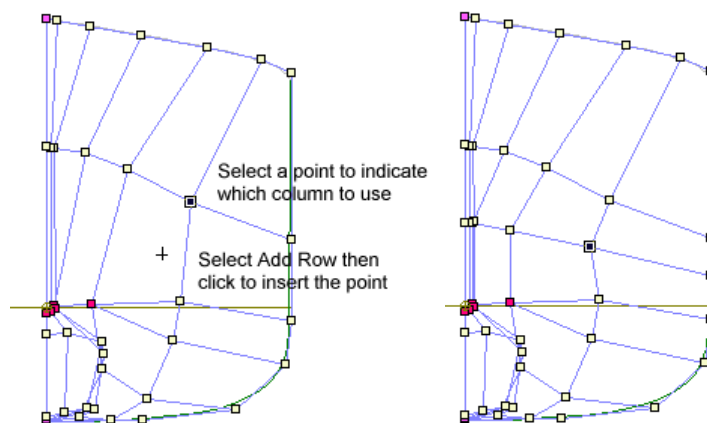
Note:

In cases where there is ambiguity as to which surface is being added to, you can give Modeler a hint by first clicking on an existing point in the required surface. Selecting a point in this way tells Modeler to begin its search for the correct insertion point in this surface.

- If necessary, select the surface and column into which you want to insert the new control point.
- Select Add from the Controls menu.

Position the Add cursor at the point where you wish to insert the new row or column.

- Click the mouse button.



Modeler will insert the new control point and also insert all other points in the column or row based on an average of the positions of existing points in the net. The new shape of the [Current Surface](#) will then be calculated.

If Modeler is unable to determine into which surface you wish to add the points, it is a good idea make all other surfaces invisible or to lock them to remove the source of ambiguity.

Deleting Control Points

The Delete command allows you delete a control point row or column to decrease the density of a control point net. Columns of points may be deleted from the Plan and Profile windows, while rows of points may be deleted from the Body Plan window.

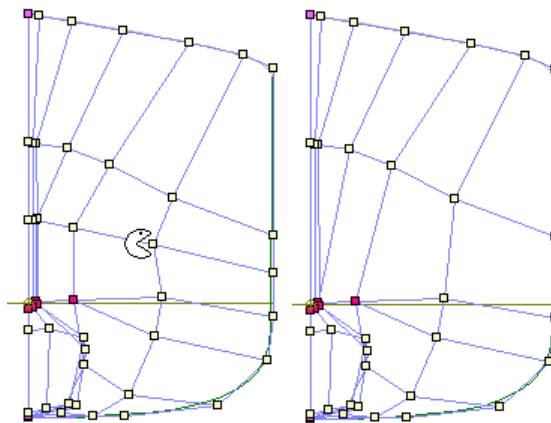
To delete a control point row or column

- Select the Plan, Profile or Body Plan window as appropriate.

Control point rows may be deleted in the Body Plan; control point columns may be deleted in the Plan or Profile windows.

- Select Delete from the Controls menu.

Position the jaws of the Delete cursor around the control point.



- Click the mouse button.

Modeler will delete the control point row or column and calculate the new shape of the [Current Surface](#).

Deleting a row or column that contains one or more control points in a bonded edge will impact the bonded surface. If a row or column, which is part of a bonded edge, is chosen for deletion, you will be alerted to this fact by a dialog, through which you may then choose between cancelling, and deleting the row or column from both surfaces.

Control Point copy and paste

It is now possible to copy the control point position and weight ready for pasting into another control point or several other control points.

Copy Control Point

- Select the control point in one of the drawing views (if several are selected on the first will be used)
- Select Copy Positon from the Controls menu.

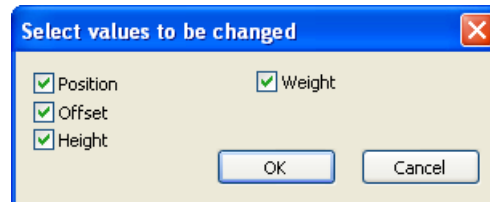
Paste Control Point

- Select the control points you wish to update in one of the drawing views

- Select Paste Position from the Controls menu.

The selected control points will have their position and weight updated to the value in the Control Point Clipboard. In some ways this is quite similar to the Compact command, the advantage being that you can move the destination control point to another location before pasting its value into the other control points.

Further options which allow you to select which data are updated can be accessed by holding down the Shift key when you select the Paste Position command.



Moving Control Points

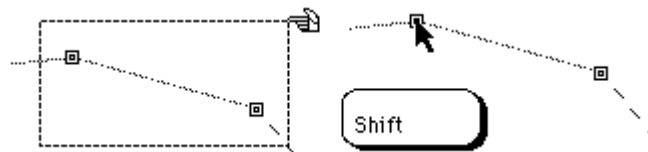
Control points may be moved, either individually, or as a group.

To move an individual control point:

- Click and drag the control point to a new location.

To move multiple control points:

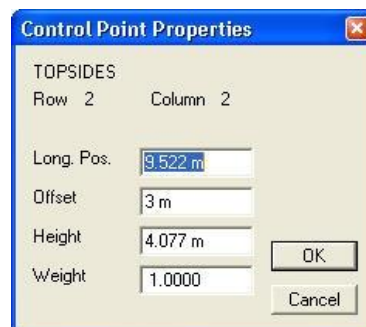
- Select multiple control points using the selection box or the mouse and Shift key



- Click on any control point from the selection and drag the selection to a new location.

Control Point Properties

It is possible to change values for control points numerically by double clicking on the control point.



A small dialog box appears containing data about the control point. You may select and edit any of the control point's coordinates, or its weighting value. You can also change these values in the table in the Control Points window.

See also [Moving a Group of Control Points](#) on page 210.

Constraining Movement

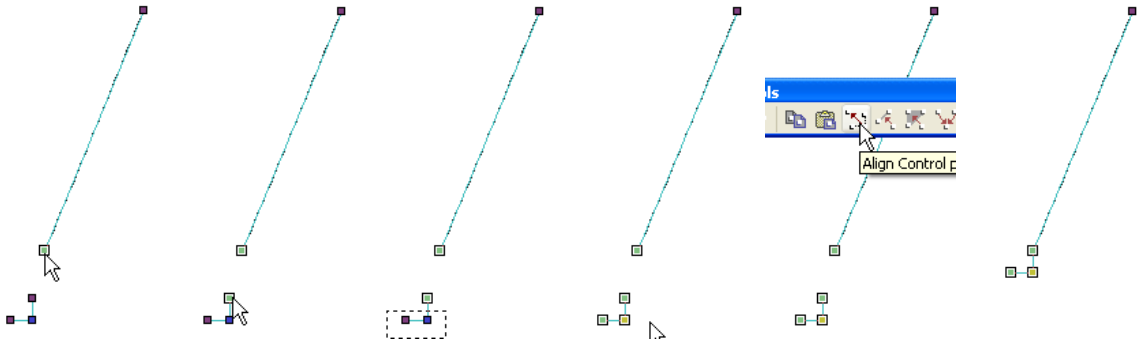
Any group or individual control point may be constrained in its movement by holding down the Shift key while dragging the control points. The constraint restricts movement to the vertical or horizontal directions. If you release the shift key, you are returned to unconstrained movement.

If a surface is symmetrical, its control points are constrained to lie all on one side of the centreline. You can override this constraint by holding down the Ctrl key while dragging the point.

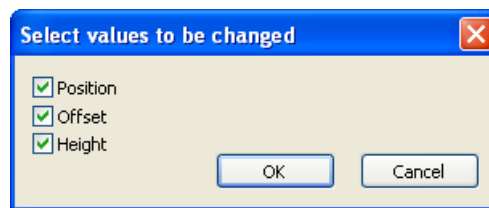
Align Control Points To First Selected

This command is similar to the Align Surfaces command. It allows you to move a group of control points whilst maintaining their position relative to one another:

- Select the destination control point in one of the drawing views
- Select the first control point in the group to be moved. The group of control points will be translated so that this control point is coincident with the first selected control points.
- Select the remaining control points in the group



Further options which allow you to select which data are updated can be accessed by holding down the Shift key when you select the Align To First Selected command; the weights of the control points are never changed.



Aligning Control Points

The align commands are for making groups of control points co-linear or co-planar. The Align to Vector command aligns points along a 3D line in space, while the Align to Plane command aligns points to a specified plane.

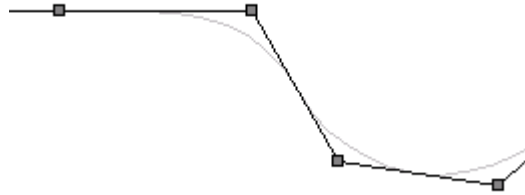
Align to Vector differs from Smooth and Straighten in that the points selected do not have to be in the same row or column, or even in the same surface. You may select any control points to define the vector or plane, and then align any control points with that vector or plane; the control points need not lie in the same surface. You must also select *all* the control points to be aligned, patches may *not* be selected as they are in the Smooth and Straighten commands.

To use Align to Vector:

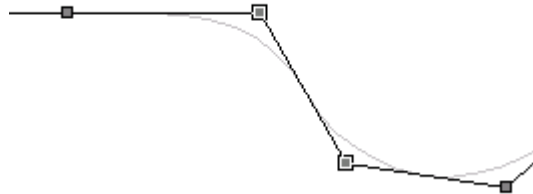
- Select two control points with the shift key held down.
- While still holding down the shift key, select one or more additional points.
- Select the Align to Vector command.

The first two selected control points define the vector; all subsequently selected points are moved to the point on that vector closest to their original position:

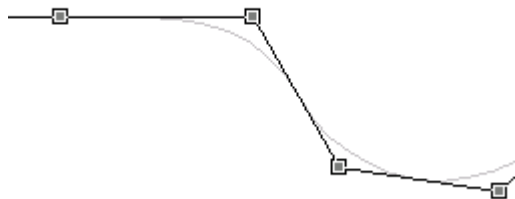
- Before aligning, endpoints with vector defined by two middle points



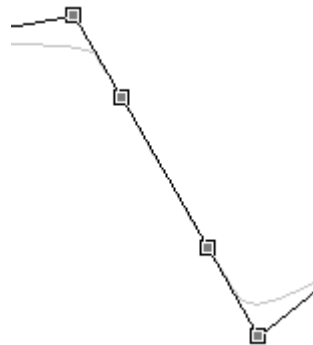
- With shift key held down, select two middle points to define vector



- With shift key held down, select two end points to be aligned

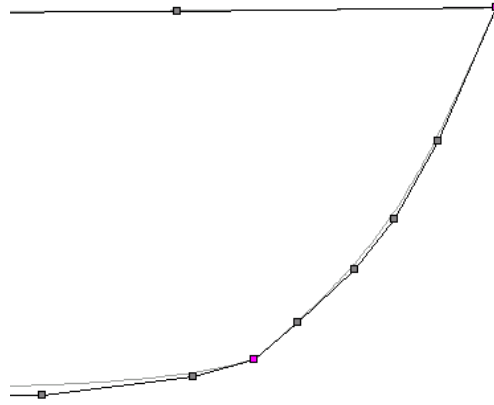


- Select Align to Vector; end points are moved to closest corresponding points on the vector

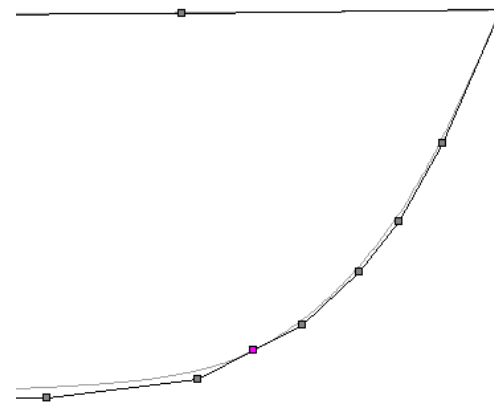


This feature is very useful for ensuring co-linearity of control points either side of a bonded edge to ensure that the surface is slope continuous across that edge, or for creating a straight bow. Align to vector can also be used to ensure a smooth forefoot, going from the surface's bottom edge to the bow; ensure that the corner control point and the two control points on each side, immediately adjacent to it, are co-linear:

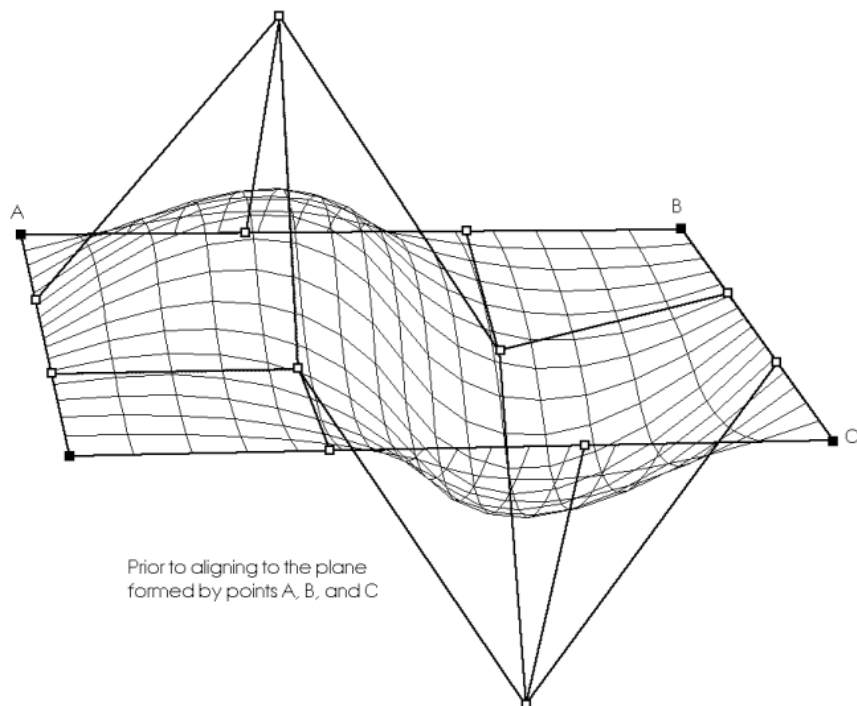
- Before alignment, discontinuity at forefoot

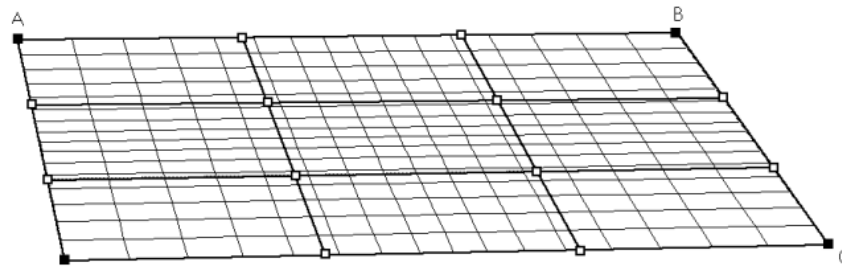


- After alignment, smooth transition at forefoot



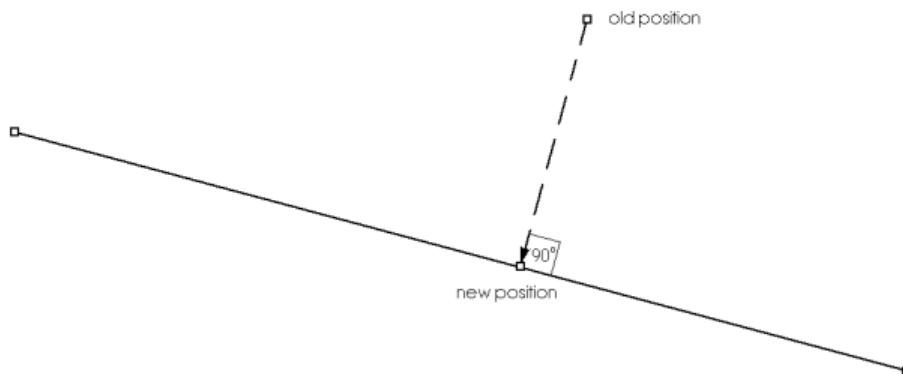
Align to Plane works in a similar fashion to the Align to Vector command, except that the first three points selected define a plane in space that all subsequently selected points are moved on to.





After aligning all points to the plane formed by points A,B and C

Control points, other than those defining the plane, are moved to the closest position in the plane.



This means that a line between the new and old positions of the point would form a line that was perpendicular to the plane.

Smooth Controls or Straighten Controls

Modeler has some commands that help you manipulate a group of control points. The Smooth Controls and Straighten Controls can be used on one row or column, or on a selection of control points from one surface (this is called a "patch").

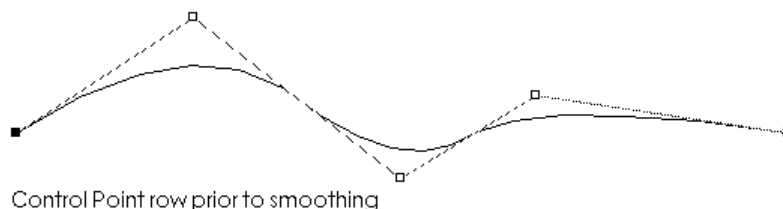
- [Smoothing and Straightening - Rows or Columns](#)
- [Smoothing or Straightening a Patch](#)

Smooth commands can only be executed on control points that belong to surfaces.

Other commands that let you manipulate groups of control points are discussed in the section [Manipulating Groups of Control Points](#) on page 209.

Smoothing and Straightening - Rows or Columns

Any complete or partial row or column of control points can be straightened into a straight line, or faired into a smooth curve.



To smooth or straighten three or more control points

- Click on the surface control point at one end of the group to be straightened.

This control point will be the control point at one end of the selection to be smoothed or straightened. If after selecting one point you wish to change your selection, simply click anywhere in the background of the window.

- Hold down the shift key.
- Click on the surface control point at the other end of the group.

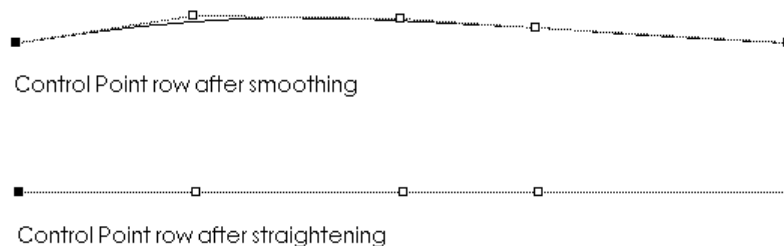
This control point will be the control point at the other end of the selection to be smoothed or straightened. It should lie in the same row or column as the first selected control point. If you wish to change your selection after selecting a second point, release the shift button and start again with the first point.

- To fair the control points, select Smooth Controls from the Controls menu and select the desired stiffness.

The stiffer the smoothing, the straighter the line will become. Smoothing is done in three dimensions.

- or to straighten the control points, select Straighten Controls from the Controls menu and select whether you wish to straighten in 2D or 3D.

If you straighten in 2D, the control points will be forced to a straight line in the current view direction, leaving the third direction unchanged. This can be particularly useful for producing straight bow profiles or transoms; this is done by using the Straighten in 2D command in the profile view. Straighten in 3D will produce a straight line in all three dimensions.



You can choose to undo a smooth or align operation by selecting Undo from the Edit menu.

Smoothing and Straightening may only be performed in a single surface; it may not be performed over two or more surfaces.

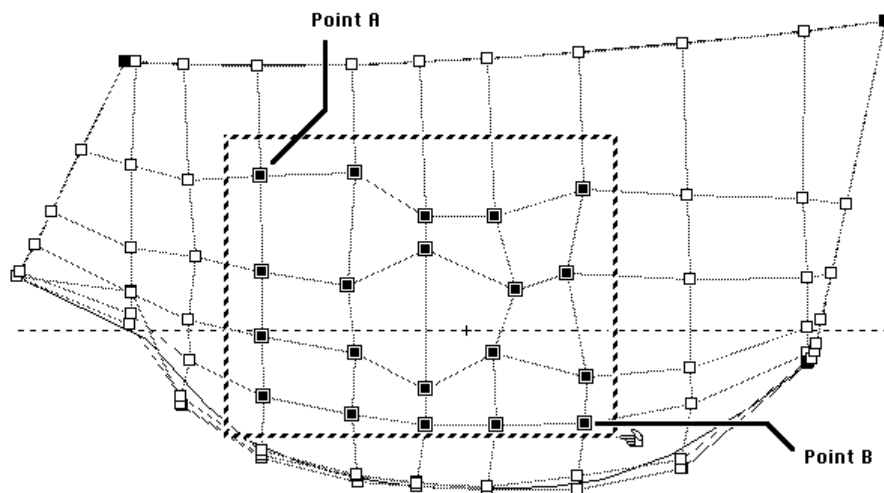
Note

The Straighten Controls and Smooth Controls commands only work for control points that belong to the same surface.

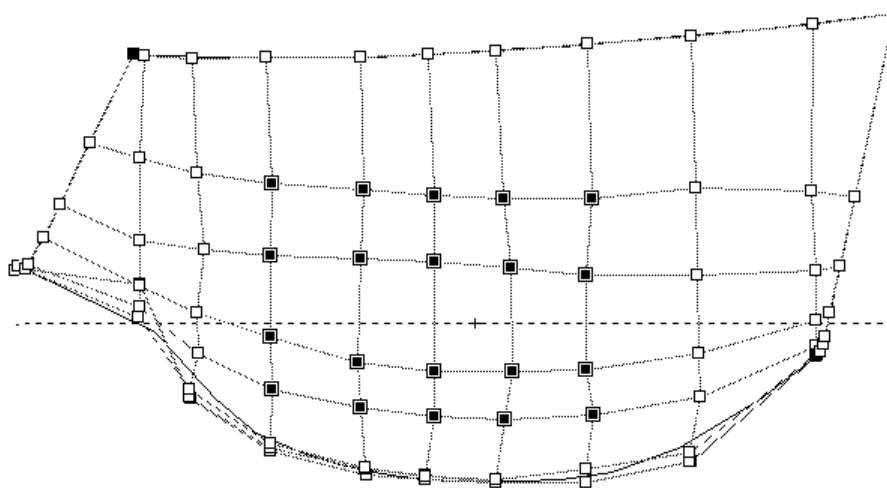
Smoothing or Straightening a Patch

The Smooth and Straighten commands in Modeler can also be applied to a patch of control points that spans more than a single row or column.

A patch is a rectangular grid of control points that can be as small as one row or column, or as large as the entire surface. Not all the points to be smoothed need to be selected as Modeler will determine the size of the patch from the points that are selected.



For example, in the diagram above, if only points A and B had been selected (instead of using the selection box) Modeler would have smoothed exactly the same group of control points, since points A and B define the limits of the patch.



To smooth or straighten a patch of control points, select them and choose Smooth Controls or Straighten Controls from the Controls Menu. An example of a smoothed patch is shown above. If straighten had been applied in the previous example, the control point rows and columns in the selected patch would lie on straight lines.

Manipulating Groups of Control Points

There are several commands that are useful for fairing a design. These operate on groups of control points: either controls within a particular row or column; a patch of control points in a surface or across bonded surfaces, or arbitrary control points.

The smoothing and straightening commands are particularly useful for manipulating the edges or features of a surface, e.g. creating a straight bow profile, producing a smooth, fair chine line, etc.

Smoothing may also be useful when applied to a patch of control points for maintaining a fair net and hence a fair design.

The align functions are useful when you wish to place control points in the same vector or plane. The control points may be from the same or different surfaces. These functions are also useful for fairing across the edges of bonded surfaces. If the Align to Plane or Align to Vector functions are called while holding the shift

key down a constraints dialog box will appear. In this dialog you may specify movement constraints in certain planes.

The size function may be used to enlarge or reduce a portion of a design, by re-scaling a group of control points. In a similar way, certain areas of a design can be rotated by rotating the control points in that area. To re-size or rotate complete surfaces, please refer to the Size and Rotate commands in the Surfaces menu, these are described in the Surfaces section.

In this section:

- [Moving a Group of Control Points](#)
- [Resizing a Group of Control Points](#)
- [Rotating a Group of Control Points](#)

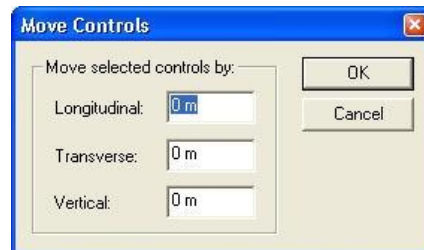
Also of interest may be:

- [Smoothing and Straightening - Rows or Columns](#)
- [Smoothing or Straightening a Patch](#)
- [Aligning Control Points](#)
- [Compacting Control Points](#)
- [Discontinuities and Feature Lines](#)

Moving a Group of Control Points

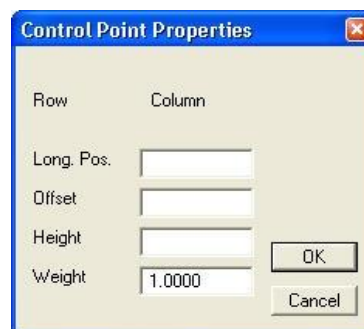
This command may be used to move an area of a surface.

- Select the control points you wish to move.
- Choose Move Controls from the Controls menu.
- In the dialog type in the desired movements.
- Click OK.



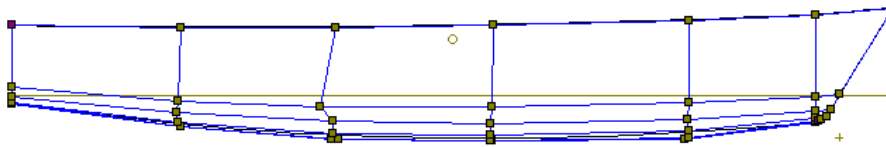
It is also possible to change the properties of a group or selection of control points using the Controls Properties.

- Select a group of control points
- Select Control Properties from the Controls menu.

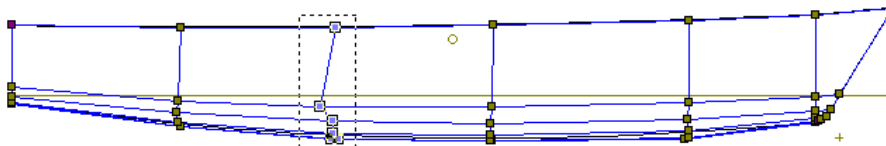


The same dialog will appear, but only those values that are common between all the selected control points will be filled in. You can change any of the values and all the selected control points will have their respective values updated.

This function is useful for aligning columns of control points; see below:



- Select the control points you wish to align:



- Select Control Properties from the Controls menu:

Row	Column
Long. Pos.	
Offset	
Height	
Weight	1.0000

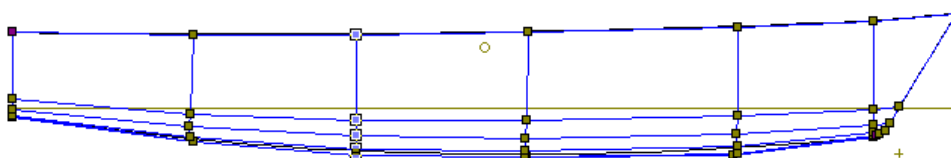
OK Cancel

Any data fields, which are common for all the selected control points will be filled in; any data fields which differ for the selected control points will be left blank. In this case the surface name and column number are the same for all the selected control points.

Row	Column
Long. Pos.	-5.26 m
Offset	
Height	
Weight	1.0000

OK Cancel

- Type in the longitudinal position you wish to set for all the control points and click OK.



The selected control points are now all set to the same longitudinal position of -5.26m.

Resizing a Group of Control Points

The Size Controls command may be used to enlarge or reduce an area of a surface.

- Select the control points you wish to resize.
- Choose Size Controls from the Controls menu.

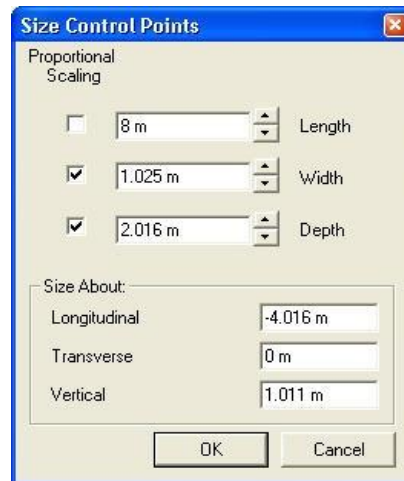
- In the dialog type in the desired length, width and depth dimensions; proportional scaling may be achieved by ticking the Proportional Scaling boxes next to the dimensions.

Note that the values that appear when the dialog is first displayed are the dimensions of the bounding box containing the desired control points. To scale the control points, type in the dimensions of the new bounding box you wish the control points to occupy.

- Choose the origin for the re-scaling in the Size About fields.

This will control the direction in which the control points are moved. The Size About point will remain unchanged when the control points are resized.

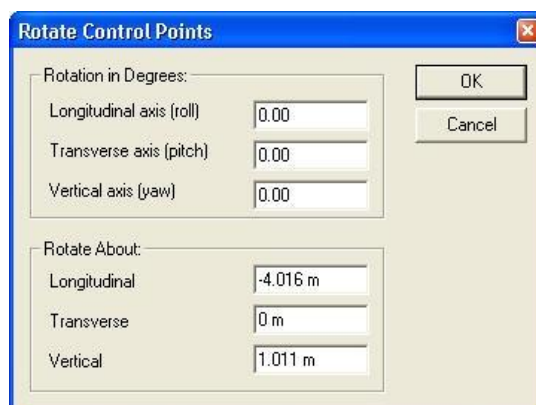
- Click OK.



Rotating a Group of Control Points

To rotate a group of control points:

- Select the control points you wish to rotate.
- Choose Rotate Controls from the Controls menu.
- Type in the rotations about each of the axes.
- Type in the rotation centre.
- Click OK.



Compacting Control Points

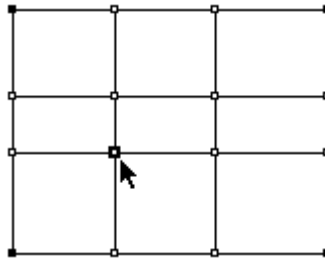
Compacting Control Points – Concepts

The Compact function is used when you wish to position one or more control points precisely over an existing control point. This can be used to create a knuckle or discontinuity in the surface or to collapse a side to a point e.g. at the bottom of a bow cone (see [Discontinuities and Feature Lines](#) on page 214).

Compacting Control Points – Procedures

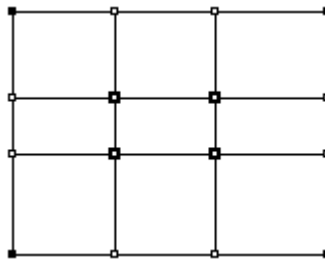
To compact two or more control points

- Select one control point,



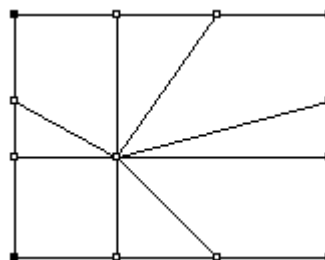
This control point is the one that you wish all other control points to compact onto. It will remain stationary when you select the Compact function.

- Hold down the Shift key and select one or more other control points.



These are the control points that you wish to compact to the first selected control point.

- Select Compact from the Controls menu or type Ctrl+K.

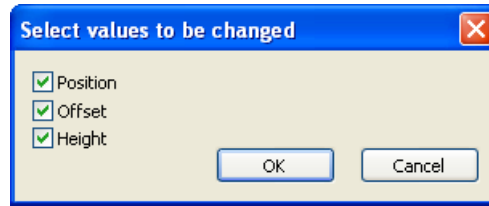


This will compact all selected control points to the first selected point.

Note:

The Compact command does not hold the compacted points together permanently, i.e. if you select one with the mouse and move it, the other points in the same position will not move with it. If you wish all the points to move as one, select all the compacted points (with a selection box) and then select the Group command.

Further options which allow you to select which data are updated can be accessed by holding down the Shift key when you select the Compact command; the weights of the control points are never changed.

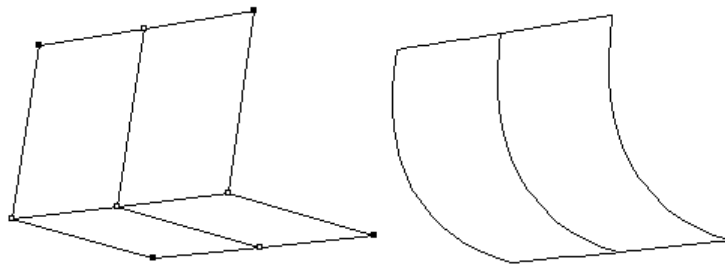


Discontinuities and Feature Lines

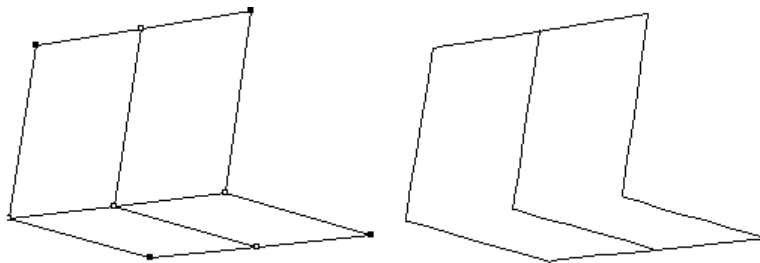
It is possible to use compacted grouped control points as a means of introducing a discontinuity into a surface.

Consider a surface whose stiffness is set to flexible (order 3) in the transverse (column) direction. The following example shows how the use of two rows of control points can form a discontinuity in the surface.

Illustrated below is a Perspective view of the net and shape of a simple surface.

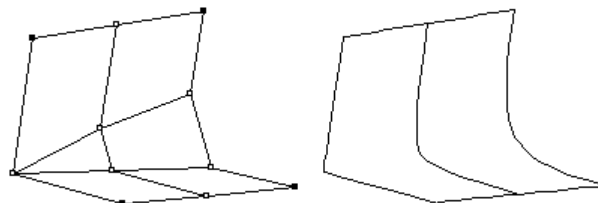


If an additional control point row is added, and its points compacted directly on top of the existing intermediate control point row, a hard corner is created running along the surface, as shown in the following illustration.



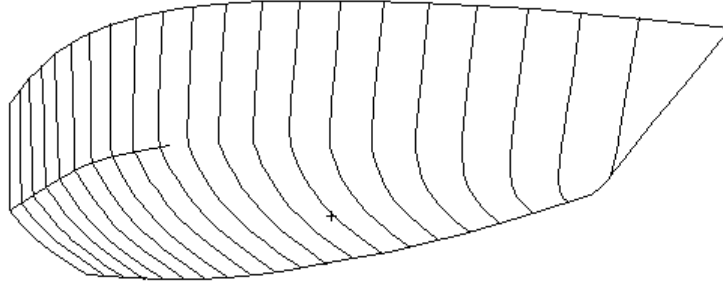
The scope of the discontinuity may be changed by splitting the superimposed row in some of the columns.

By splitting the row at the right hand edge of the surface as shown it is possible to fade the hard edge out. The surface shows a smoothly curved section at its right hand edge and a section with a hard corner at its left hand edge.



To achieve a discontinuity of this type you need to compact several control points together. The number of points required to achieve a hard corner is always one less than the order of the surface in that direction. For example, if you have a surface that is flexible in the transverse direction (Order 3), you will only need to superimpose two control points to achieve a hard corner in the section. If, on the other hand, the transverse stiffness is set to stiff (Order 6), five control points will need to be compacted together to form the corner.

If you wish to display the line along the corner you have created, select Feature Lines from the Contours menu item. The effect will be similar to the following:



Note:

Do not compact more control points than necessary to create a knuckle. The required number of control points is the surface stiffness minus 1. For example: a stiffness 4 surface requires 3 compacted control points to create a knuckle.

See also [Grouping Control Points](#) on page 215.

Grouping Control Points

Control point grouping is provided to help with the movement of multiple control points. Grouping allows the movement of a number of control points that move as a whole whenever a single control point from the group is moved. To group several control points simply select the points that you wish to be in the group and choose Group from the Controls menu or type Ctrl + G.

Note: It is possible for you to group together several groups that have been previously created, to create a single group of control points.

The Ungroup function performs the reverse of the Group function. Choose the group that you wish to ungroup by selecting a control point from within the group. Choose Ungroup from the Controls menu.

Single control points cannot be marked as grouped – you must select at least two control points to group them. If a control point is grouped to another which is subsequently deleted (by, for example, deleting the surface, row or column in which the control point lies) the remaining control point will become ungrouped unless there are other, undeleted, control points in the same group.

Group and Compact

It is important not to confuse the functions of the Group command and the Compact command. Group allows you to move control points simultaneously, even if they occupy different positions in space. Compact forces several control points to the same position in space, but does not force them to move simultaneously. If you wish to have several control points occupy one position and move together, you will need to select the Compact command followed by the Group command.

See also [Compacting Control Points](#) on page 213.

Displaying Control Points

To control the net display, select the items under the Net submenu in the Display menu.

Show Net

The entire network of control points is shown by itself.

Hide Net

In Perspective, Profile and Plan window only the edges of the surface will be shown. If you wish to hide the edges you should hide or lock the surface.

In Body Plan window, the current control points column of the [Current Surface](#) will be displayed. This allows you to adjust the shape of one column at a time. See also [Body Plan Window](#) on page 43.

Rows

Only Rows are displayed.

Columns

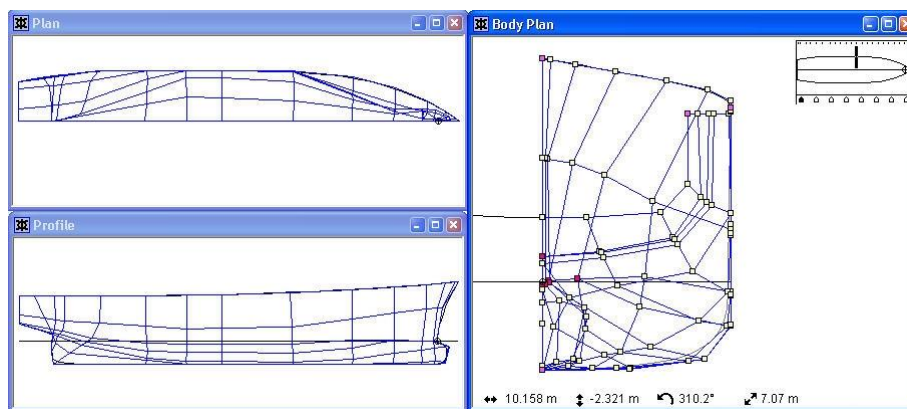
Only Columns are displayed.

Masking the Net

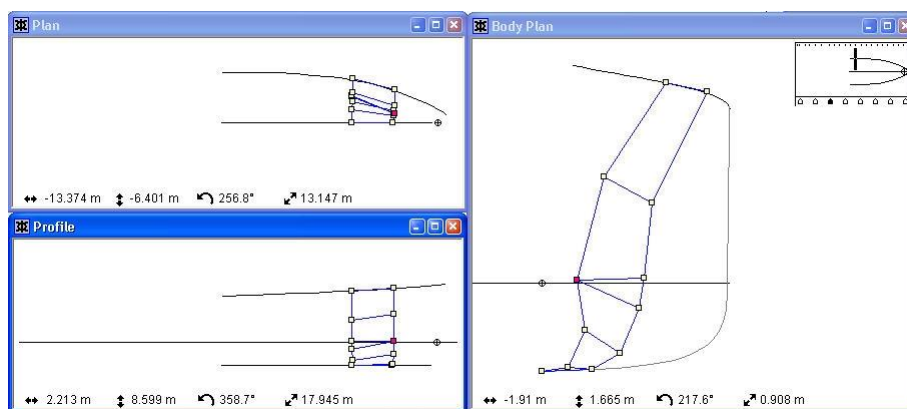
In cases where you are dealing with a complex net of control points, it is sometimes very helpful to be able to select a particular group of control points to remain visible while the remainder of the net is hidden. This is the function of the Mask command.

It can be particularly useful to select the control points you wish to mask in one view and then to manipulate them in another view. For example, when fairing a bulbous bow, it can be helpful to mask the control points in the Perspective or Profile views and then manipulate the surface shape in the Plan view.

When working with a control point net with a large number of columns, it may be very confusing to try to work on particular points in the Body Plan window.



By selecting all the points on one or two columns and then selecting Mask, only the portion of the net selected, and the area of the surface that it exerts an influence on, will be displayed. It will then be much easier to view and modify in the Body Plan or any other window.



When the Unmask command is selected, the entire net will be made visible again.

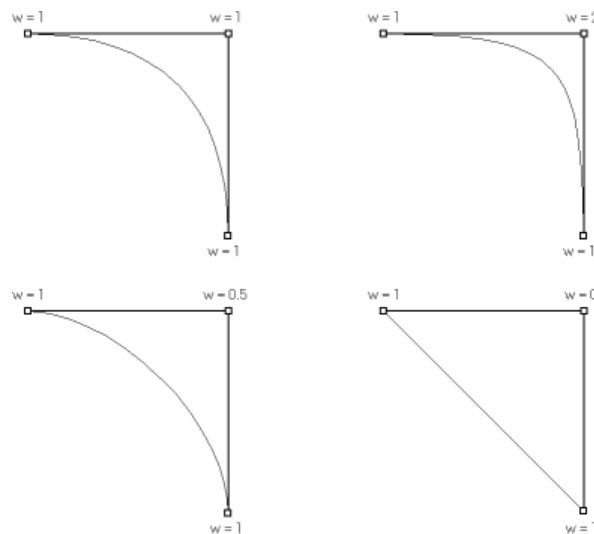
Control Point Weight

Non Uniform Rational B-Spline (NURB) surfaces have an additional shape control parameter, called a weighting value, available at each control point. In Modeler, the weighting value is shown in the last column of the Control Point window.

This section explains the effect of changing the weight of control points and gives some examples where the weight of a control point is used to achieve a specific shape such as [Circular Arcs](#)

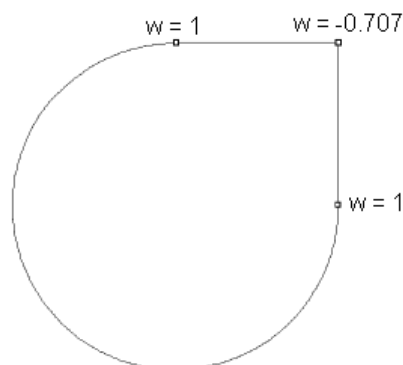
Control Point Weight Concept

When a control point has its weighting value increased, the surface, in the vicinity of that control point, will be attracted towards the control point. Conversely, a decrease in the weighting value will result in the surface being pushed away from that control point:



Effect of increasing / decreasing the weight of the middle control point.

This is even true to the extent that a negative value will give a curve that balloons outward from the polygon of control points:



Negative control point weight

Note:

In general you should only use weighting values to exactly model a conic curve. It is not usually a good idea to use the weighting value to control the shape of free form curves.

Circular Arcs

The primary use for the weighting values in a NURB surface is to allow you to create true arcs of circles and ellipses. There are two standard formulations for these curves, one using three control points and Order 3 (quadratic) stiffness, and one using four control points and Order 4 (cubic) stiffness. In both cases, there is an equation that describes what the weighting value should be, based on the angle of the arc.

Note that when a surface is defined as a conic, Modeler only automatically computes the control point weights correctly for three-point-forms. Weights will be incorrect for four-point-forms and you must change the surface type to NURB so that you can enter the weights manually.

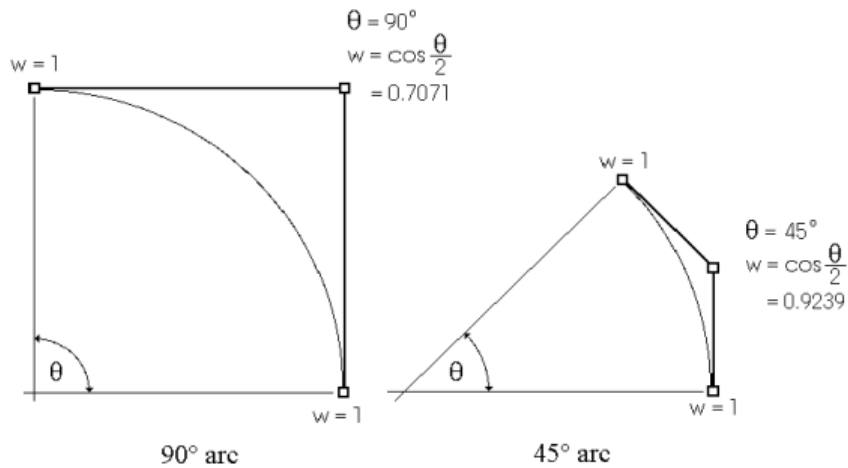
For the three point case, the distances between control points are kept equal and the weighting value for the centre point is set to

$$w = \cos(\theta/2)$$

where θ is the included angle of the arc.

For example for a 90° arc,

$$w = \cos 45^\circ = 0.7071.$$



For a 45° arc,

$$w = \cos 22.5^\circ = 0.9239.$$

The four point form again has the control points forming an equally spaced polygon around the curve. In this case, the equation for the weighting value is

$$w = (1 + 2 \cos(\theta/2)) / 3$$

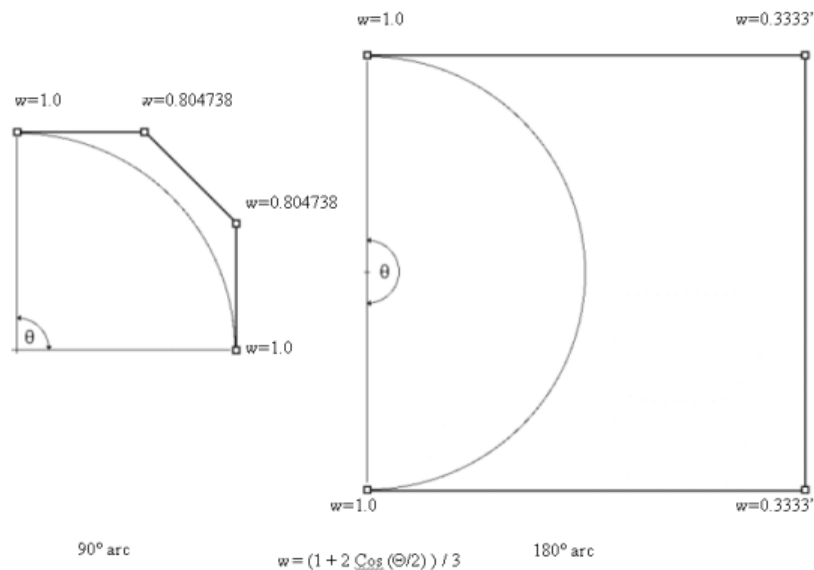
where θ is the included angle of the arc.

For example, for a 90° arc,

$$w = (1 + 2 (\cos 45^\circ)) / 3 = 0.804738$$

For a 180° arc,

$$w = (1 + 2 (\cos 90^\circ)) / 3 = 0.3333'$$

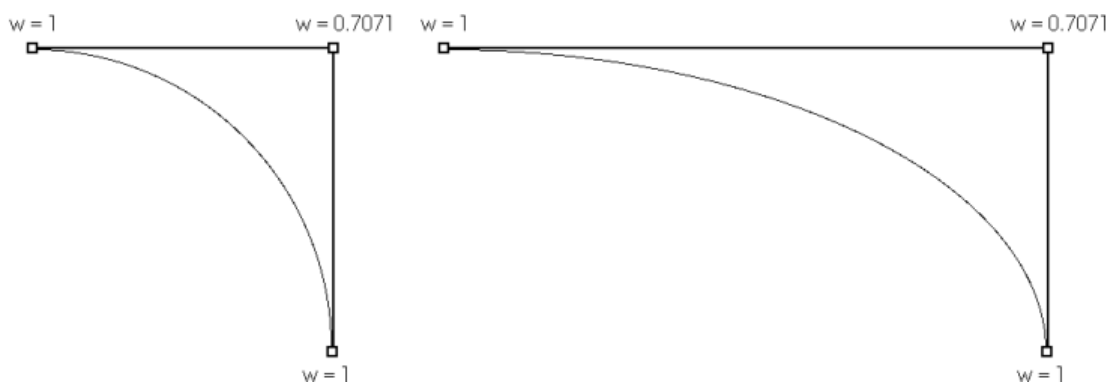


The three point form of circular arc can be used for any arc from 0° to 180°. However, as the angle approaches 180°, the weighting value approaches 0 and the position of the centre control point approaches infinity. It is best applied to arcs between 0° and 90°.

The four point form has a more complex equation for the weighting values, but is an elegant form for 180° arcs. It is best used for arcs between 90° and 180°.

Elliptical Arcs

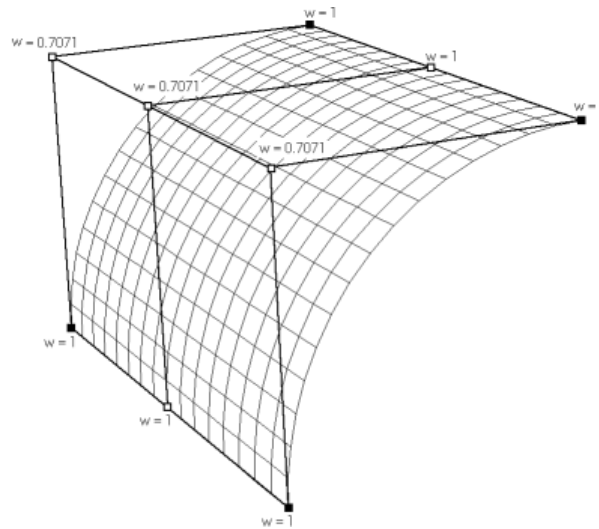
NURBS can represent elliptical as well as circular arcs simply by stretching the control point polygon in a linear fashion.



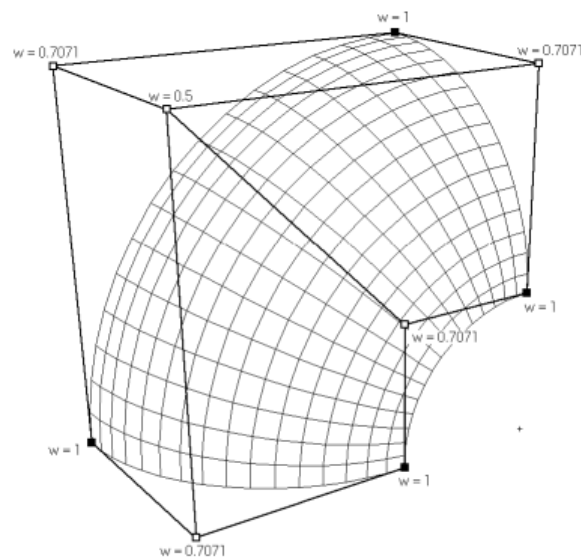
When this is done the weighting values remain the same as those for the circular arc on which the ellipse is based.

Circular and Elliptical Surfaces

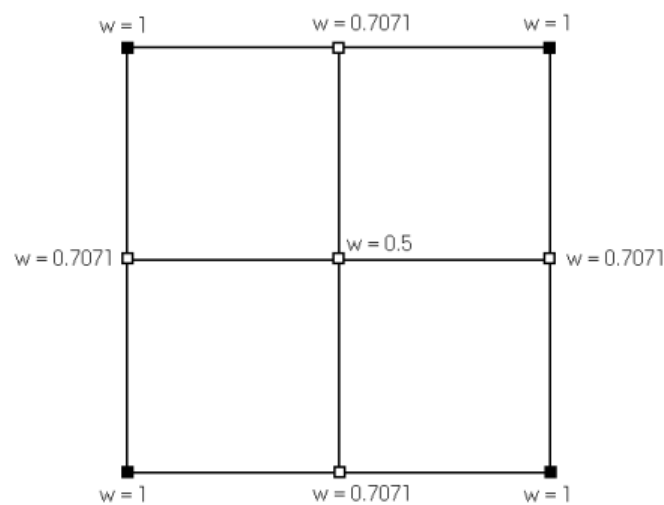
The extension of the NURB curve to the surface allows cylinders, spheres, cones etc. to be modelled exactly. For a cylinder or cone, one row is set to the weighting value required for the arc.



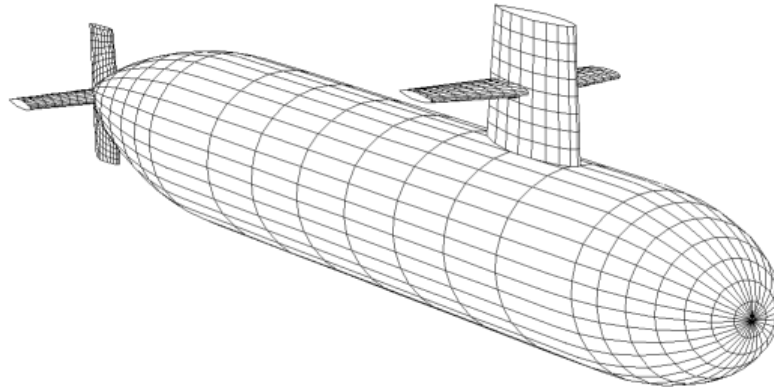
In the case of a toroidal or spherical surface where curvature runs in both directions on the surface, the weighting values are set as follows:



The central control point has its weighting value set to the product of the weighting values of the row and column that intersect it. In this case $0.7071 \times 0.7071 = 0.5$.



By combining these surfaces, it is possible to build up complex designs made up of true circular and elliptical segments, such as a submarine. The submarine example is in the Sample Designs folder, which comes with Modeler. (Modeler\Sample Designs\Naval\Submarine\Submarine.msdl)

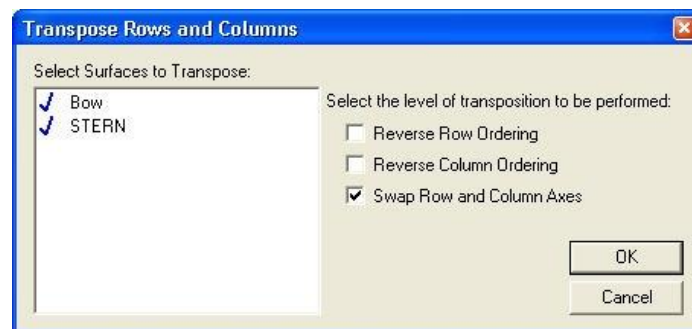


Advanced - Transpose

The transpose function may be used to flip the column or row ordering and also swap the rows and columns of a surface. This operation will not affect the actual shape of the surface, merely the ordering of the control points. This can be particularly useful if you have imported a NURB surface which uses a different control point orientation from Modeler, or if you have rotated a surface and the column selection in the Body Plan view selects columns which are now closer to being longitudinal than transverse.

The Reverse Row Ordering and Reverse Column Ordering functions change the order in which the control points in a row or column are indexed.

The Swap Row and Column Axes function will make rows columns and columns rows. This can be useful if the columns have become longitudinal rather than transverse through surface rotations.



Surface Fitting

This section explains all the concepts and steps to fit a surface to an existing design. There are two types of surfaces in Modeler which you can use to fit to existing design data: NURBS and TriMesh surfaces. This section explains when to use which and what tools should be used in the process.

You may wish to read the section on [Working with Markers](#) on page 80 before starting a surface fitting process.

In this section:

- [Surface Fitting – Concepts](#)
- [NURB Surface Fitting – Procedures](#)
- [Fitting a NURB Surface to Markers](#)
- [Mesh Surface](#)

Surface Fitting – Concepts

Often a Naval Architect has to produce hydrostatic or other data from an existing design. For example: a modification to the hullshape has been made and a stability booklet has to be re-submitted; or a similar vessel is to be built from existing plans. The existing design may be in various formats: paper lines plan, a CAD drawing, or an offsets file. Depending on the type of output required and the design data available there are different tools available and they are used in different orders.

Surface Fitting Concepts – Theory

1. Data Representation of Shapes

You can represent a shape in space in the following ways:

- Points
- Lines
- Surface(s)

A cloud of points can describe a shape in 3D, but it is very difficult to see whether all points are exactly **on** that shape. Or: whether the shape that runs through the cloud of points is fair.

By connecting the points with fair lines, the shape can be better visualised. The problem with this method is that there is no explicit relationship between the individual lines thus making changes to the 3D shape is time consuming because changing one line means that all other lines must be updated so that the lines are all consistent with one another.

The highest order to represent a 3D shape is using a 3D surface. The surface can have different levels of stiffness which can be seen as the stiffness of the traditional fairing battens. Any required line (e.g. section, waterline, buttock) is taken directly from the surface ensuring that all lines are consistent. This is what Modeler NURB surface modelling does. For more information, see the Modeler manual chapter 2.

The greatest advantage of using a surface to represent a shape is that you can **derive** points and lines from the surface instantaneously and this allows you to produce lines plans on the fly and make changes easily. The Modeler suite takes this concept one step further and derives the input required for stability, resistance and seakeeping analysis as well as production information directly from the surface shape. This enables a naval architect to make last minute changes to the hullshape as the design iteration cycle progresses.

2. Fully automatic fitting

Before you continue reading this document, it is important to establish that a fully automatic fitting tool that fits a NURB surface to a set of points or lines does not exist. Human interaction to judge the input, the hullshape and decide on the output information required from the surface model is necessary.

Having said that, there are some cases where fully automated surface fitting is possible: a round bilge simple hullshape where only Hydrostatic output information is required can be fitted automatically.

3. Modeler Tools

Modeler has two options to assist with Naval Architecture tasks that start with existing design data in any format:

- NURBS Surface

- **Mesh Surfaces**

Which one to choose and how to get from a specific type of input data to a specific output is described in the next step-by-step sections.

Surface Fitting Concepts - Choice of Tools

Step 1: Think Ahead

There are many different considerations you need to make before starting on a surface fitting process.

1a. Types of input

Typically you can have the following types of input:

- Pictures of the vessel
- Paper lines drawings
- Table of offsets
- 2D DXF lines plan
- 3D DXF lines plan
- NURB surface description in the form of an IGES (or similar) file

The type of input you have does not really determine whether the TriMesh or the NURB surface method is suitable, but it does determine how much work is required to produce a hull model by each method.

1b. Hull Shape

The hull shape of the existing design is a very important factor when determining how to create a surface model. Some hull shapes like catamarans and trimarans can be difficult to achieve with TriMesh surfaces. These can then best be done using NURB surfaces.

Whenever you use NURB surfaces to reproduce an existing design you will have to determine the required topology and how best to lay out the surfaces at an early stage. To some extent this is also dependent on Step 3: what you want to do with the surface model. For example: in case you want to use the surface geometry to develop plates, you may need to use a different topology than when you are producing a hydrostatic model.

Basically, when you are talking about surface fitting, you can divide hullshapes into the following categories:

- Simple: Round bilge
- Complex: Chined hulls, stern bulbs, bulbous bows, specific features in the design (bilge radius, flat of sides, bow cones), multihulls

Interpreting the hullshape and deciding how to arrange your surface model is the most difficult part of surface modelling and requires some experience. Some general guidelines to surface topologies (surface lay outs or arrangements) are given in [Determining Surface Topology](#) on page 226.

Note: when you fit surfaces to an existing design, always try to break down the hullshape into the simplest shape. Do not include keels or other appendages at this stage and also ignore steps in the deck or specific transom features. These can be modelled later with separate surfaces and possibly using surface trimming (in, for example, the case of producing a vessel with a stepped deck).

1c. Types of output

What do you want to do with the surface model? Do you want to do some hydrostatic calculations, resistance analysis etc.; or will you be delivering production information from the model as well?

Basically there are two types of output data that can be derived from a surface model:

- Hydrostatic calculations, resistance analyses, seakeeping analyses.
- Production information

This determines the required model fairness and accuracy and thus also the tools you will have to use to create it. A Hydrostatic model generally requires less fairness and accuracy than a production model.

1d. Ability to make changes

Another important question that you have to ask yourself before producing a surface model is whether you will need to be able to make changes in the future. Also the type of change that is required later on may be relevant. For example, if the existing design is going to be lengthened by inserting a parallel midbody, you will want to split your design in to a fore and aft part and bond them in the middle.

Step 2: Choice of tools

The process that you are about to start is called "surface fitting"; you will be fitting a 3D surface to a set of lines or points. The Modeler range offers different tools:

Tool Selection criterion	Fitting as plug-in inside Modeler	Generate TriMesh
Surface quality, fairness and accuracy.	Highly accurate fitting while maintaining control point and surface fairness. Modeler does not add extra control points. The model can be faired and adjusted later and can be used for production purposes.	The Mesh is made up of triangular or planar quadrilateral panels and is by definition not fair. However it fits <i>exactly</i> through the data provided.
Hullshape	Any hullshape including model which use multiple surfaces.	A wide range of hullshapes as long as marker data is sorted intelligibly.
Time required to create model	- Surface preparation requires planning, and good modelling skills. - Need to link markers to surfaces - Running the Genetic Algorithm fit can take a long time (typically several hours) to iterate to acceptable error values.	Fast, requires some marker preparation (see below).
Manual fitting work required	Yes. User needs to prepare the surface first: # control points, stiffness & edge shape.	No, except for the data preparation: all data points have to be sorted into transverse contours and correctly ordered on each contour before the surface can be generated.
Surface model usage	Structural and hydrostatic/hydrodynamic analysis. Plate development possible if surface topology set up correctly.	Hydrostatic only; may also be used in Resistance and Motions for resistance and seakeeping predictions.
Modifications possible	Yes.	- Cannot easily modify the Mesh surface model, although a parallel midbody can be created by duplicating the midship markers station and moving all fwd stations fwd.
Other attributes	+ Useful to make a surface developable, Fitting as a plug-in is great for fitting the internal control points to the ruling	Multiple Trimeshs Can be combined with NURBS surfaces in

	lines that describe the developable surface.	the same model "Use" property as per NURBS surfaces
--	--	--

Next: see [NURB Surface Fitting – Procedures](#) on page 225.

NURB Surface Fitting – Procedures

After you read the [Surface Fitting Concepts - Choice of Tools](#), you basically have three options:

1. Use Fitting Standalone; See Fitting Manual. Also see the section on [Mesh Surface](#) since it is often more convenient to organise and sort the marker data in Modeler and import it into Fitting.
2. [Fitting a NURB Surface to Markers](#); manually fit surfaces to markers – possibly assisted by the genetic algorithm surface fit function enabled by Fitting as a Plug-in to Modeler.
3. Use Modeler to [Mesh Surface](#).

Inserting Markers - Procedures

First step is to import the existing design and create marker points for it. Use the links in the table below to navigate to the section in the manual describing this procedure.

Data input format	Instructions / Link
Digital images	See Importing Background Images on page 331. This section explains the procedure from scanning through to digitising a lines plan in Modeler.
Paper lines plan	Scan in the lines plan and follow image instruction above.
Marker files in Excel or other	See Pasting Markers from Excel .
Offset text files	Import as coma separated or tab delimited into Excel and then follow Pasting Marker files in Excel instructions above.
2D CAD drawing	Convert to 3D by moving the sections in the Z-direction one by one and then follow 3D CAD instructions. See Converting a 2D DXF to a 3D DXF on page 337.
3D CAD drawing	Save as DXF and then see Importing DXF Markers .

It is important to have a reasonable number of data points defining the edges of the surface. If there is a discontinuity in an edge it is better to create some extra data points to form a fair edge and model the discontinuity using trimmed surfaces at a later time.

There is a maximum of 30000 markers you can use, however the more points you use the longer it will take to find a solution. It is unlikely that using a very large number of points will improve the overall quality of fit, so it may be worth pruning down the original table of offsets data if there is more than about 3000 points.

Setting up the Grid for the Markers

If you haven't done so already, run the Generate Grid From Markers command [Generating a Grid from Markers](#) on page 34 to automatically generate marker stations. This will allow you to view one markers station at a time.

Next go to either:

[Fitting a NURB Surface to Markers](#)
[Mesh Surface](#)

Fitting a NURB Surface to Markers (Manual fit)

After you have inserted your marker point and set up your grid, you are ready to start manually fitting surfaces to your marker files. For this you may need to manually add surfaces, or you can select a similar vessel from the Modeler Sample Designs directory and resize that as a starting point. When you are doing this, it is important to think about the Surface Topology.

Determining Surface Topology

Surface topology is the way that surfaces are organised in order to form a 3D Shape. A number of hullshapes and the recommended surface topologies are given in the following section.

To a large extent, the surface topology recommendation depends on what you want to do with the surface model. For example: if you only want to do hydrostatic calculations you generally do not need to worry about the surface topology too much, whereas in case you need to calculate plate expansions and deliver production information, the surface topology is critical.

Deciding on a surface topology is one of the most difficult parts in surface modelling and requires a certain level of experience in surface modelling. Unfortunately the surface topology is also something that has to be decided before starting a surface model because it cannot be changed easily later on in the modelling process.

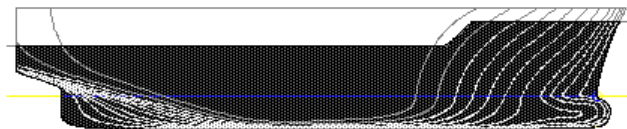
When determining the surface topology it is important to recognise hullshape features:

Knuckles and chines

Knuckles and chines in the hullshape are very easy to recognise. They can either be: partial knuckles or full-length knuckles or chines. Especially when a structural model has to be created you will want to use bonding along the chine edge. Alternatively you can use compacted control points.

Deck step-ups

It is important to recognise deck step-ups. In these situations, extend the hull surface beyond the step-up and then trim the hull back.



Deck step-up at the forecastle.

Transom

To model a transom you can do two things:

- In profile view, simply end the hull surface at the transom and then either:
 - Don't model the transom. Only when the transom is a vertical plane.
 - Bond the transom surface to the aft hull surface edge.
- Extend the hull to beyond the transom and use a trimming surface to trim the hull.

Maindeck

To a large extent Decks are very similar to Transoms and the same options apply. Do not model 'tween decks if you need a hydrostatic model or set the 'tween deck to internal structure.

Bulwarks

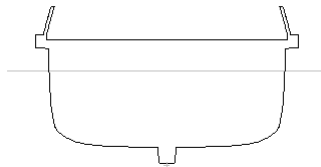
Bulwarks are generally not included in the surface model when the model is intended solely for hydrostatic purposes. The reason for this is that the hydrostatic sectioning

routine cannot identify whether the bulwark surface encloses a volume or not and will sometimes include the volume between the bulwarks as part of the hull volume. This will affect hydrostatic analysis such as large angle stability and damaged stability analyses.

When the model is used for production information only (not for hydrostatic analysis) and the bulwarks are part of the hull plating you can simply model them by extending the hull upwards from the deck to the top of the bulwark. If the bulwark has step-up in it, you can use a trimming surface to trim it back, same as for deck step-ups. (To simplify modelling for hydrostatics, you could include a weather deck which you would trim the topsides back to for Hydrostatic analysis.)

In case the model is intended for both hydrostatic analysis as well as production purposes, you have the following options:

- Bond a separate bulwark surface to the top of the hull and set it to an internal structure surface type. That way it will not be included in the hydrostatic section.
- Offset the bulwark surface inwards by the plate thickness and add a top-plate. This is to make sure that the hydrostatic sections will be closed.



The bulwark of this tug has been offset inwards and closed at the top to form one closed section line.

When determining surface topology, it is highly recommended to read: [When to Use Trimming](#) on page 141 and [When to Use Bonding](#) on page 150.

Corner Control Points

After you have determined the surface topology, it is time to do a rough manual fit of the surfaces to the marker cloud data. The order in which you should fit a surface to markers is strictly from top to bottom:

- Corner control points
- Edge control points
- Internal control points

The corners of the surface will usually be marked with a marker point. A useful command to set the position of the corner control points is the Snap Control Point to Markers. To automatically snap a control point to a marker:

- Select the marker in any design view
- Select the control point whilst holding down the Shift key.
- Markers | Snap Control Point to Marker

The control point will be made coincident with the marker. In addition, the marker's properties will be updated so that it is now linked to the surface to which the control point belongs and, if the control point lies on a corner or edge, the marker will be linked to the corresponding corner or edge, otherwise the control point will be linked to the surface's interior.

Fitting a Surface Edge

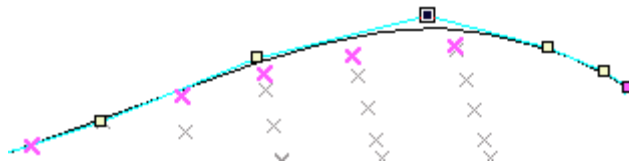
Edge fitting - concepts

Often, for best results, it is recommended to do the edge fitting manually. This gives you the opportunity to get a feel for the required surface stiffness and number of control point columns/row along that edge and preparing the surface appropriately before fitting the surface automatically.

The Fit Edge to Markers command will automatically fit the selected edge to the selected markers. This command requires an appropriate number of control points and surface stiffness along the edge direction to have been defined.

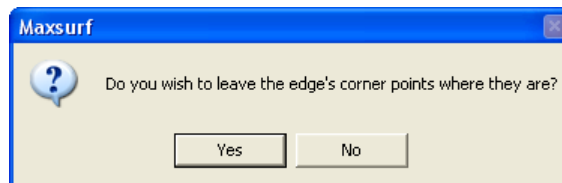
Edge fitting - procedures

- Select the markers that mark the surface edge shape
- Select the surface edge or *one* control point that lies on the surface edge (not a corner point) while holding down the shift key



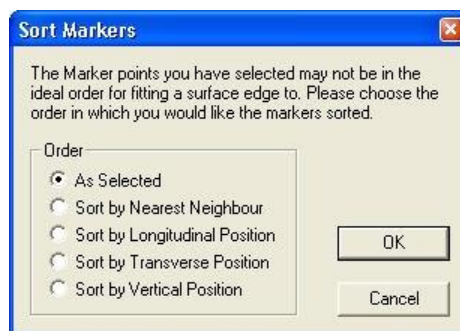
- Select Markers | Fit Edge to Markers

This will bring up the following dialog:



Usually, you will want to leave the corner control points where they are, because you should have manually fitted those in the previous step.

The next dialog allows you to specify the marker order:



The Sort options allow you to override the order in which the markers have been selected.

For a successful fit, the order in which you select the markers is important and it is therefore recommended to select the markers in the correct order manually. If you are fitting a longitudinal edge (control point row), then you must select the markers from left to right. If you are fitting a transverse edge (control point column) then you must select the markers starting at the centreline outwards. In the fitting process the control point in the first row or column of the edge will go to the first selected marker and the control point in the last row or column of the edge will go to the last selected marker.

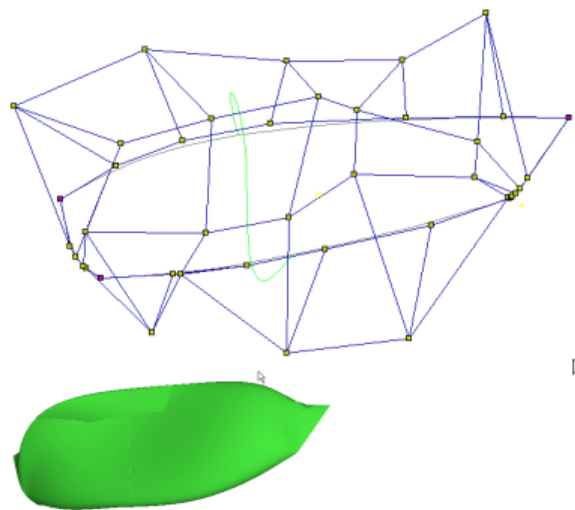
When a surface edge has been fitted to markers, the markers are associated with that particular surface edge. Once markers have been associated with a particular surface edge, you can fit the surface edge to its markers by simply selecting a control point in the edge (other than a corner control point) and selecting Markers | Fit edge to markers. This is very useful as it allows you to easily change an edge's properties (number of control points and stiffness) and refit the edge to obtain the best possible fit to the markers without having to always reselect the markers. It also makes it very easy to add or remove markers to those used for the edge fitting.

Smooth Surface Interior

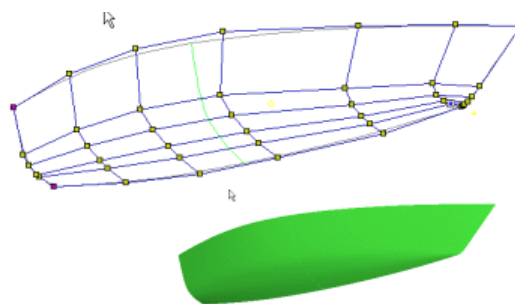
Once you have fitted the surface's edges, the interior control points, displayed by turning on the surface net, may be very irregular. This command uses a three dimensional smoothing procedure to smooth the surface's interior control points to follow the shape defined by the surface's edges. This will give you a much better starting point for faring.

Smooth Interior Controls - procedures

- Select the surface for internal control point smoothing (smoothing is done on the current surface)
- Turn on the control point net –(net must be turned off for this command to be active)
- Markers | Smooth Interior Controls



Before smoothing, having fitted the edges



After smoothing the interior

It is a good idea to “help the Genetic Algorithm” and provide it with a good starting point prior to set it off running. This means that it will pay off if you spend some time manually adjusting the shape of the internal sections to fit to the markers. To do this effectively, you should have used the [Generating a Grid from Markers](#) command and automatically linked the Markers to the sections, or – if you prefer to do this manually, you should read the section on [Linking Markers to a Section](#).

Fitting a NURB Surface to Markers (Automated fit)

The Fit Surface to Markers command in the Markers menu uses a Genetic Algorithm search to minimise the measured error between the Markers and the Surface by modifying the shape of an existing surface by changing the position of position of its control points.

Genetic Algorithm – Concepts

The Genetic Algorithm (GA) is an advanced optimisation method based on a simulation of the process of evolution. Although slow, GAs are excellent for solving problems with large numbers of dimensions and constraints. The surface fitting problem falls into this category, as it is not sufficient to simply create a surface that is a close fit to the data points provided, it is also desirable that the surface be fair and that the control point net be smooth and regular.

Genetic Algorithms start by creating a population of individuals with a small amount of random variation. In the case of surface fitting, a population of usually 25 to 50 surfaces is created based on an initial example surface. Each individual in the population is made unique by randomly moving one or more control points a small distance.

Once the initial population has been created the system simulates evolution by repeating the following steps until a solution is found that meets the fitting criteria sufficiently well.

Measure the fit of each individual surface to the data points and give it a score based on its quality of fit, fairness and net regularity. Rank all the individuals from best to worst.

Repeatedly choose two individuals as parents, with individuals being chosen in proportion to how highly they were ranked in step 1. This means that the most successful individuals produce the most offspring in the next generation.

Create members for the next generation by choosing some surface control points from one parent and the remainder from the other parent. Occasionally include a small amount of random control point movement to provide some mutation from generation to generation.

The result is a population that progressively gets closer to an ideal solution of the problem, with poor solutions being eliminated. The highest ranked individual in the final generation is the best solution.

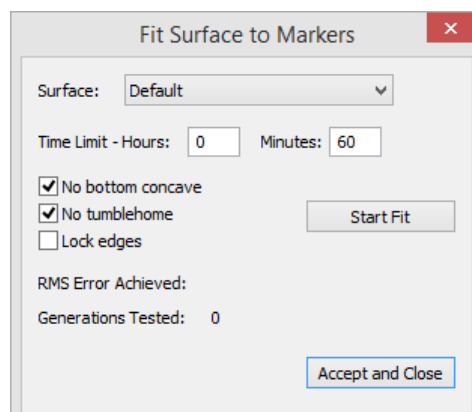
Genetic Algorithm – procedures

Before you can fit a surface to markers it is necessary to link the markers to a surface, see the section on Linking Markers to a Surface.

To start fitting a surface using the Genetic Algorithm:

- Select Fit Surface to Markers command from the Markers menu

You will see the following dialog appear:



- Specify Surface

At the top of the dialog you can specify which surface you want fitted to the available markers. This requires that (a selection of) the markers have to be linked to the surface you wish to fit. The markers do not need to be ordered in any specific way. If your design only has one surface than no options will be shown in the Surface pull down menu.

- Specify Time limit

You can set a time limit for the search. It is recommended to first search for approx 5 – 10 minutes and see whether the initial surface definition is capable of being fit to the markers. If, for example, the markers require a very flexible surface with a lot of control points columns and you have used a very stiff surface with only a few control point columns, you will never get an accurate fit even if you let Modeler search for 24 hrs.

The four corner control points of the selected surface are always assumed to be in their correct locations and Modeler will not move them during the fitting procedure. You can use the [Snap Control Point to Marker](#) function in the Markers menu to help you fit the corners.

- Lock edges

If you do not wish Modeler to attempt to fit the surface edges, select Lock Edges. You would do this if you were happy that the surface is already adequately fitted to the edges of the data points or when you are using the Fit Surface to Markers command to fit a developable surface to the marker points generated from the ruling lines. You can use the [Fit Edge to Markers](#) function in the Markers menu to help you fit the edges.

The two remaining options constrain the surface so that **bottom concave** and **tumblehome** at the hull side are not permitted. If you know that your design contains these features then one or both of these boxes should be unchecked to minimise the search option and get better and faster results.

Once you have specified these options it is simply necessary to click on the **Fit** button. Once Modeler has fitted the surface it will add values to the RMS Error Achieved (see [Measure Surface Error](#)) and Generations Tested fields.

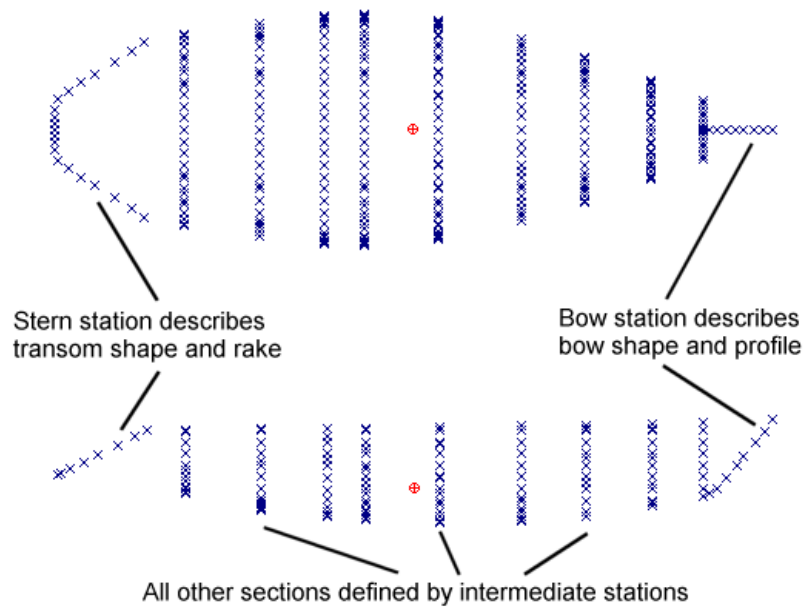
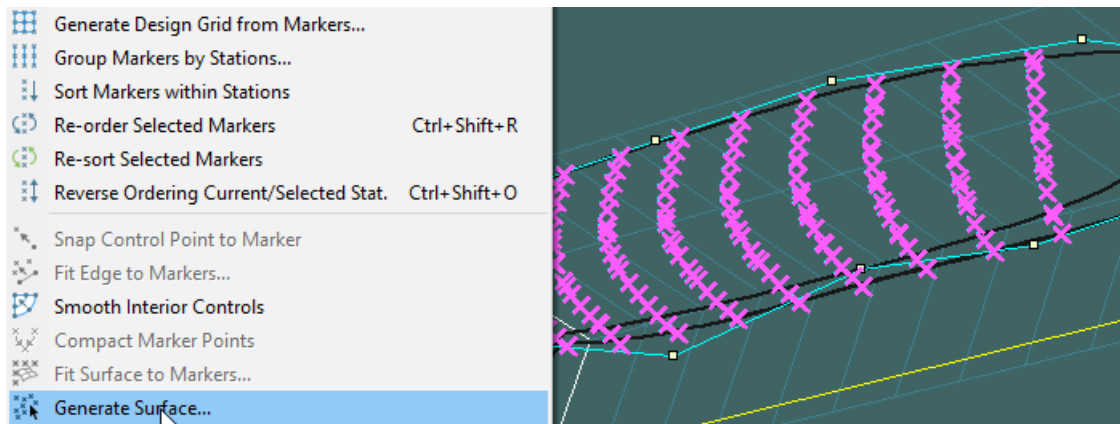
Once you have clicked the Finish button you will be returned to the Modeler viewing windows where you will see the new fitted surface.

This command is only available if Markers in the design have been assigned to a surface. Once a series of markers have been created that define the shape that is required, the markers need to be assigned to a surface (the surface that will be fitted). If there is currently no surface in the design one will need to be added (Surface | Add Surface). To assign a surface to the markers, highlight the markers and select "Markers | Marker Properties..." from the menu. Set the "Linked to:" Surface. The "Markers | Fit Surface to Markers..." functionality should now be enabled.

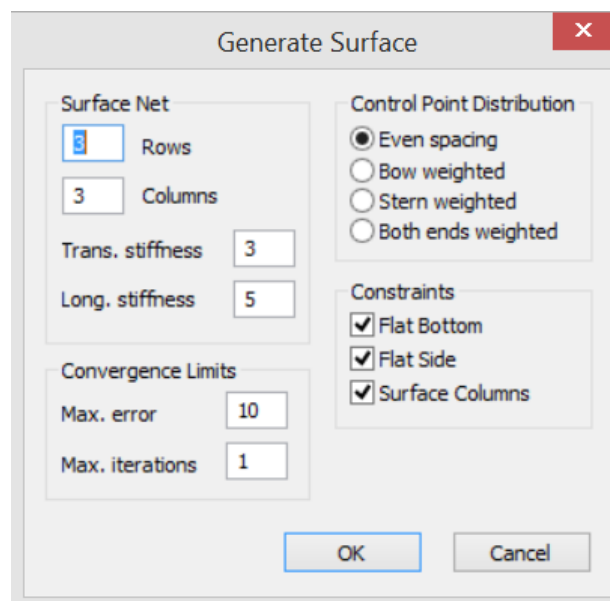
Fitting a NURB Surface to Marker stations (Automated fit)

If the Markers are structured, ideally defining a number of planar sections, and transom and bow profiles, then a faster automated fit may be used. The algorithm works by first fitting NURB curves through each section and then lofting a surface through those sections. This is quicker than the Genetic Algorithm method but requires ordered markers.

Once a series of markers have been created that define the shape that is required the markers will need to be assigned station indices for the Markers | Generate Surface command. The fitting algorithm is similar to a skinning algorithm where by a curve will be fitted to each set of markers sharing the same station index as well as a 'top' (eg gunwale) and 'bottom' (eg keel) curve. The surface will then be fitted to this family of curves. The Generate Surface dialog is accessed from the "Markers | Generate Surface..." menu item.



Once the function is called a Generate Surface Dialog will appear:



The dialog allows you to enter the number of control points transversely across the surface (rows) and longitudinally along the surface (columns).

Convergence Limits

The Control Points dialog also contains several other items, headed Limits and Constraints. These limits control how much time will be spent trying to fit splines or a surface to your markers. You can automatically iterate to improve the quality of fit until the maximum error at any marker point is less than a certain level. At the same time it is necessary to specify the maximum number of iterations to make before it gives up.

For the moment, set maximum error to 10 mm. and iterations to 5.

Iterations

After the initial surface fitting, further iterations of the surface fitting procedure may be carried out to obtain a better fit. To do this, select Regenerate Surface from the Surface menu. The surface net may also be manipulated manually between iterations to achieve a smoother and more useable net.

The net may be displayed by selecting Net from the Display menu. The control points may be changed by directly dragging the points in the Plan, Profile or Body Plan views or by typing their positions in the Control Points table. Dragging may be constrained to either the vertical or horizontal direction by holding the shift button down.

Constraints

Constraints restrict control points to lie within certain areas to improve the regularity of the control point net that is generated.

The constraints available are:

Flat Bottom.

This prevents any control points in a particular column being below the level of centreline at that position. This is a reasonable constraint for the vast majority of hull designs, the only exceptions being those tunnel hulls or concave bottoms.

Flat Side.

This prevents control points from going further from the centreline than the widest point at the sheerline. Only hulls with tumblehome in their topsides would need to have this option turned off.

Surface Columns.

This option constrains columns in the fitted surface net to a particular arrangement. For example, if a design has a lot of longitudinal curvature in the bow it is reasonable to constrain the control point columns so that they are concentrated more in the bow. If a hull shape has fairly even curvature from bow to stern, Even Spacing or no column constraint at all is acceptable.

If the Surface Columns constraints box is ticked, the desired control point distribution may be selected from the Control Point Distribution radio buttons.

For the sample hull, all the Constraints may be left on.

Flexibility

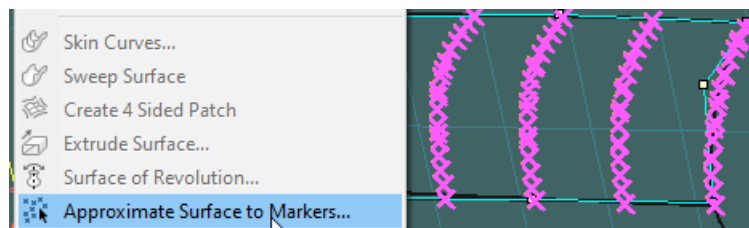
Before generating splines through the markers, it is necessary to set the flexibility of the splines to be used. Order 4 splines are most suited to the generation of surfaces, as you gain experience with the program you may wish to experiment with other spline stiffnesses.

Click the OK button in the Control Points dialog to accept the changes, close the dialog and generate the surface. The calculation of the surface may take some time, as it is a complex mathematical problem. The time taken depends on the size of the required control point net. The larger the net, the greater the time taken to fit the surface. To speed up the process, fewer initial iterations can be specified. At a later stage, subsequent iterations may be calculated to refine the surface if greater accuracy is required.

Fitting a NURB Surface to Markers (Least squares fit)

If 3 or 4 Markers are selected, this command will automatically fit a 3- or 4-sided NURB surface with control points at the Markers. Markers should be selected: bottom-right; bottom-left; top-left; top-right.

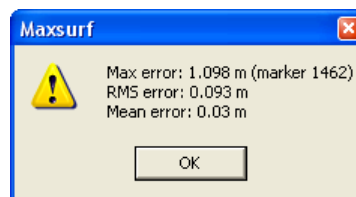
This command is found in the Surfaces menu and does a least squares approximation of a surface to the selected markers. Optionally 4 curves may be selected that define the edges of the surface to be fitted. To enable the command at least 5 markers must be selected and optionally 4 edge curves. The command will display a dialog asking the user to enter the number of control point rows and columns as well as the longitudinal and transverse stiffness of the surface to be created. The number of markers required to fit a surface is equal to the number of control point columns multiplied by the number of control point rows defined by the user when the command is executed.



NURB Surface from Markers using least-square approximation

Measuring Surface Error

This command measures the distance between the marker and the closest point on the surface with which it has been associated. After measurement, the distance for each individual marker is entered in the Markers table, and summary data is displayed in a dialog.



Max error: is the maximum error between the marker that is further away from the surface. This marker is displayed in brackets behind the error value and can be easily located by sorting the marker table in the [Markers Window](#) by error (see also image below).

RMS error: is the Root Mean Square. This is the square root of the sum of the squares of all of the errors between every data point and the fitted surface. It is a good measure of overall fit, and it should be regarded as an average error rather than a worst error value.

Mean error: the average error of all errors.

The error of each marker associated with a surface will be calculated and displayed in the Error column of the Markers Table.



	Station Inde	Long. Pos. m	Offset m	Height m	Surface	Kind	Name	Error m
1	1	0.490	1.121	0.916	Surface 2	Internal		0.004
2	1	0.490	1.121	0.916	Surface 2	Internal		0.004
3	1	0.490	1.121	0.916	Surface 2	Internal		0.004
4	1	0.490	1.121	0.916	Surface 2	Internal		0.004
5	1	0.490	1.105	0.827	Surface 2	Internal		0.004
6	1	0.490	1.105	0.827	Surface 2	Internal		0.004
7	1	0.490	1.078	0.672	Surface 2	Internal		0.002
8	1	0.490	1.078	0.672	Surface 2	Internal		0.002
9	1	0.490	1.072	0.660	Surface 2	Internal		0.001

The Markers window displays the surface error for each marker.

Measure Error can be called up with any of the drawing windows or the markers window topmost. Note that the error is calculated using the current surface precision setting.

Mesh Surface from Markers

Generating a Mesh Surface from marker points – Concepts

A description of a Mesh surface is given in the [Surface Types in](#) section on page 107.

Generating a TriMesh surface – Procedures

Before you can generate a TriMesh it is necessary to order the markers. You can do this marker ordering in many ways, but there is a particular work order which generally seems to work quite well:

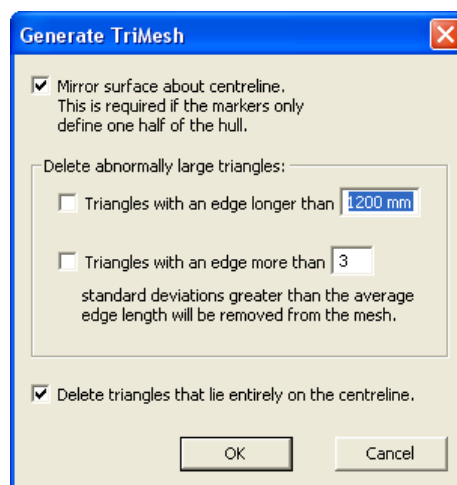
1. [Generating a Grid from Markers](#)
2. [Group Markers by Stations](#)
3. [Delete Duplicate Markers](#)
4. [Sort Markers within](#) Stations (automatic sorting)
5. [Re-order Selected Markers](#) (manual sorting)

You may have to delete spurious markers that for example do not form a complete station from bottom to deck edge. Also see other commands in the [Marker Menu](#) that may be of use.

After ordering your marker data, you may generate a TriMesh:

- Markers | Generate Mesh Surface

This brings up the following dialog:



Mirror surface about centreline

This will mirror the surface generated from the markers across the transverse centreline of the model. Tick this box if you have defined just one half of the vessel. If you have defined markers for both port and starboard halves of the vessel then leave this option un-ticked.

Delete abnormally large triangles

You can specify how large the triangles are allowed to be before they are deleted, or by specifying the number of standard deviations from the mean edge length.

Delete Triangles that lie entirely on the centreline

This option is useful if the vessel has a bulbous bow. If this is the case then there will be large triangles generated on the centreline between the top of the bulb and the bow above the bulb. These triangles will only affect wetted surface area calculations but if desired, they can be detected and deleted

Notes:

- The Mesh Markers command works on the selected markers.
- May be combined with other TriMeshes and/or other NURB surfaces in the same model.
- A TriMesh surface cannot be trimmed with regular NURB surfaces.
- A TriMesh surface can be saved as a .msd file and read directly into Stability, Motions and Resistance for analysis. For complex tanks and compartments, additional TriMeshes and/or NURB internal structure surfaces should be defined.

TriMesh from Markers Troubleshooting

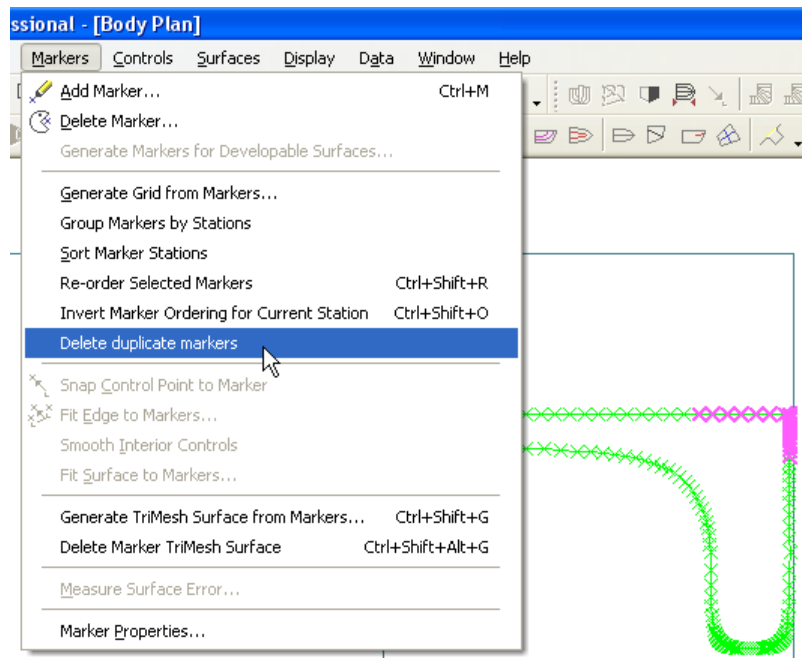
Unwanted duplicate markers

In most cases duplicate markers cause problems. For instance in this catamaran models, duplicate markers on the sharp corner cause the corner detection algorithm to fail. However in some cases duplicate markers are required. To delete duplicates from a range of markers, simply select the range of markers with the selection box and choose Markers | Remove duplicate markers. If no markers are selected duplicates from the entire marker list will be deleted

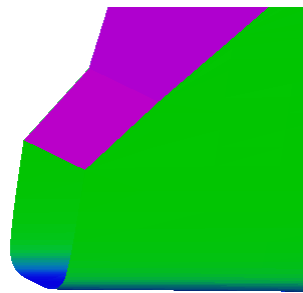


Duplicate markers causing corner detection to fail

Delete duplicate markers in the corner region of these sections:



Select the markers from which you want to delete duplicates

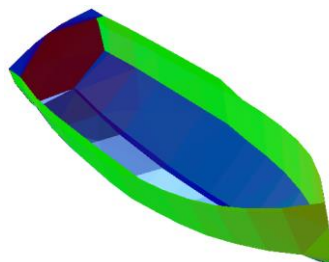


Duplicate markers removed: corner detection now works correctly

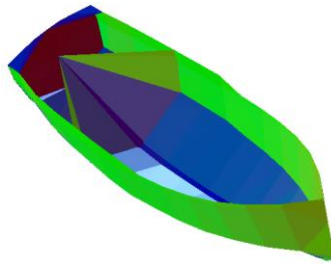
Connections appear between non-consecutive sections

In some cases if you insert markers graphically, although they are inserted into the correct section (ie given the correct station index, they may be in the wrong order in the marker table). This is easily solved with the Markers | Group Markers by Station command which will reorganise the marker table so that the markers will be grouped by station index. The order of the actual markers on a given section will not be changed. For this to work correctly it is imperative that the section grid spacing is monotonic.

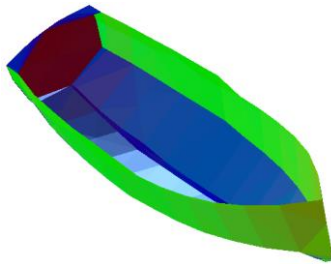
In the example below, an extra station was added – these markers were appended to the end of the Marker table, out of sequence, hence the TriMesh is trying to connect the sections in the wrong order.



Before adding new section



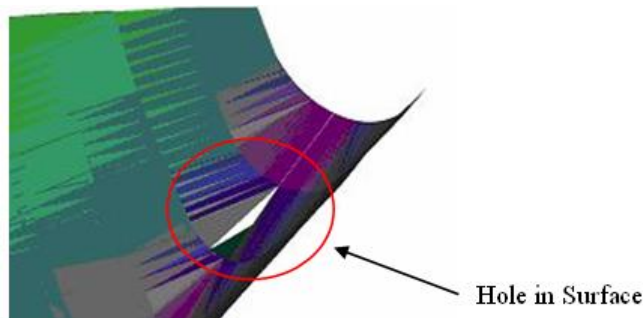
After adding section, markers are out of sequence because added to the end of the Marker table



After Markers | Group Markers by Station -
markers appear in the correct order in the Marker table

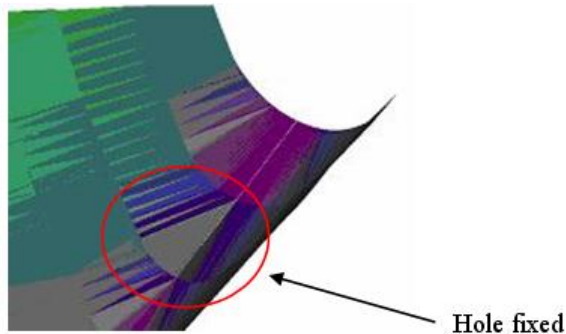
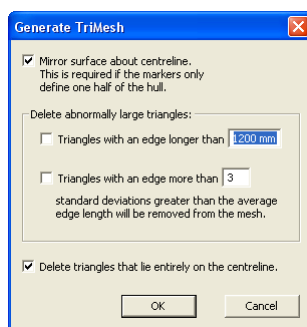
An unwanted hole appears in the TriMesh surface

Generally it is desirable to remove unusually large triangles from the mesh – this enables bulbous bows and yachts with fin-keels and bulbs to be defined (not to mention multihulls). However if the marker spacing is very irregular, valid triangles that should not be removed are removed resulting in unwanted holes in the surface:



There are two solutions to fix this problem:

- a) When generating the mesh surface either increase the size of acceptable triangle sizes or turn off the "Delete abnormally large triangles" option:



- b) Insert an extra section of markers that dissects the hole in the surface (this will create triangles that have edges that lie within the acceptable range).

Displaying a TriMesh surface

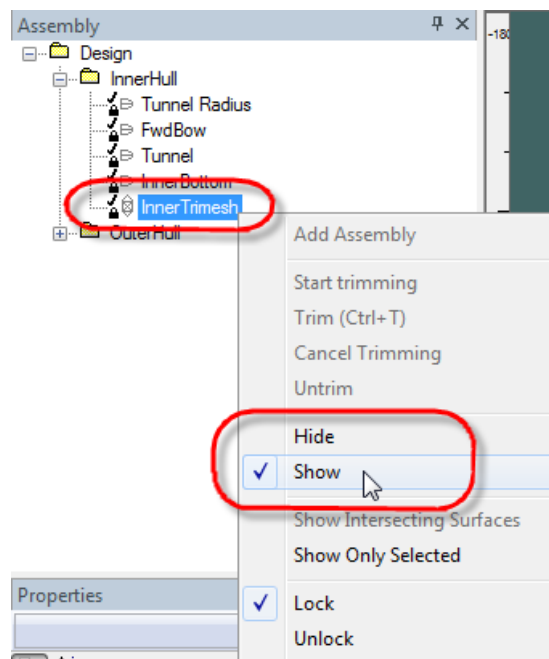
A TriMesh surface can be rendered in Perspective view. The color of the rendered TriMesh panels is dependent on their orientation in the model.

When "Show all Markers" and "Connect Marker Stations" is selected from the Display | Markers menu, the individual TriMesh panels may be displayed in all four view Windows.

Just like for NURB surfaces you can display all regular contours on a TriMesh surface.

Hiding a TriMesh surface

As with NURB surfaces, the visibility of meshes can be controlled in the Assembly tree by right clicking to show the context-menu:

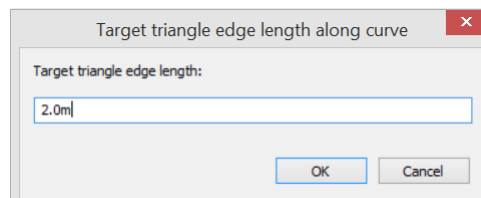


Mesh properties can be accessed through the Assembly right-click context menu.

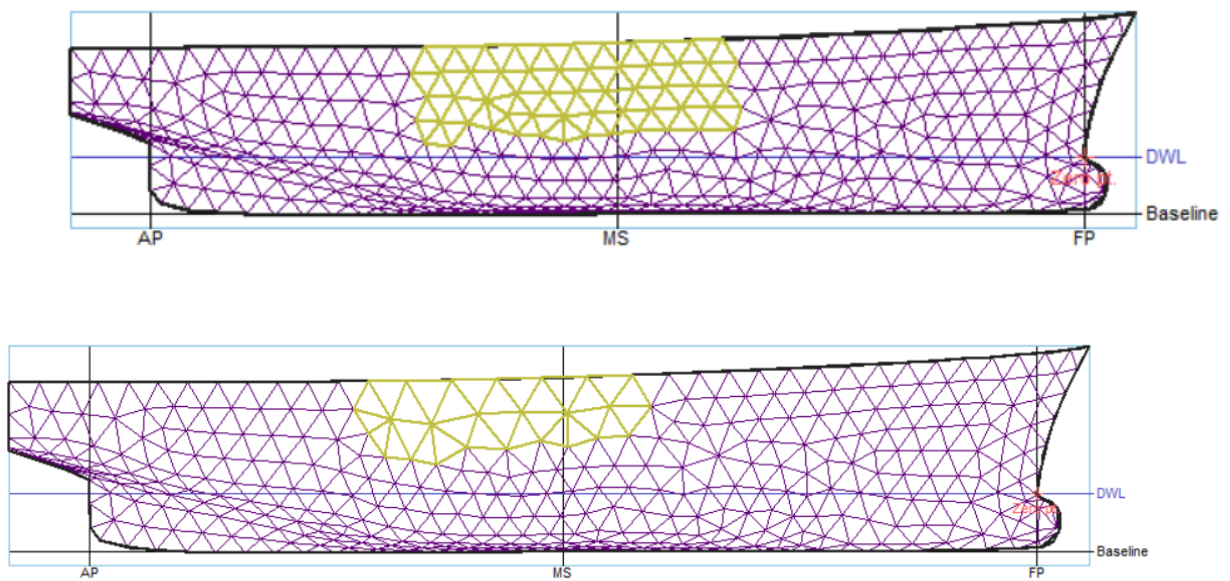
Alternatively you can use the TriMesh visibility dialog from the TriMesh menu.

Remeshing triangles

A selected set of meshes can be remeshed to produce a lower or higher density of triangles. The function is executed by first selecting the triangles (see [Selecting a triangle from a mesh](#) above) and then choosing "Remesh Triangles" from the Mesh menu. A dialog asking for the target edge length for remeshing will then be displayed:



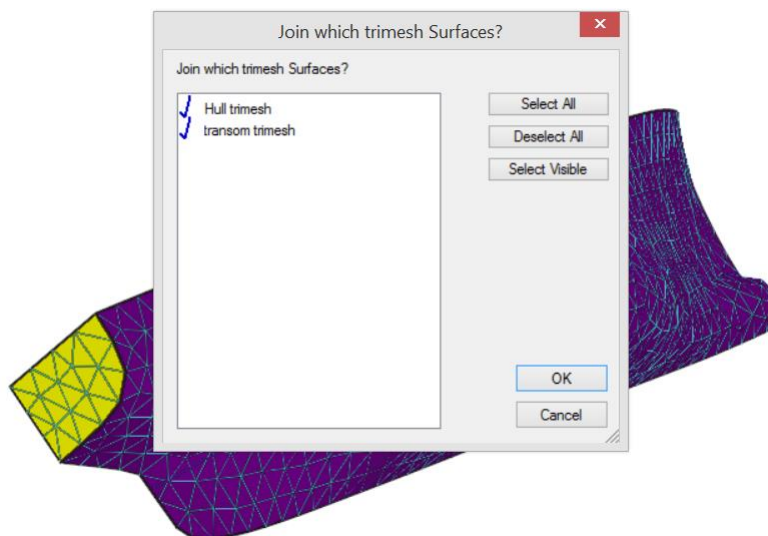
In the images below the selected triangles are remeshed from a target edge length of 1.0m to 2.0m:

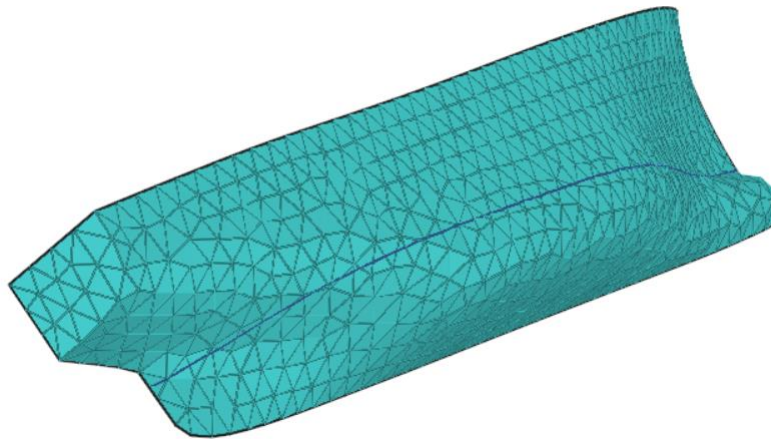


This function is useful to increase the mesh density in areas of high curvature (ie around bulbous bows) and decrease the mesh density in areas of low curvature (i.e. flat of sides and bottom)

Joining Meshes

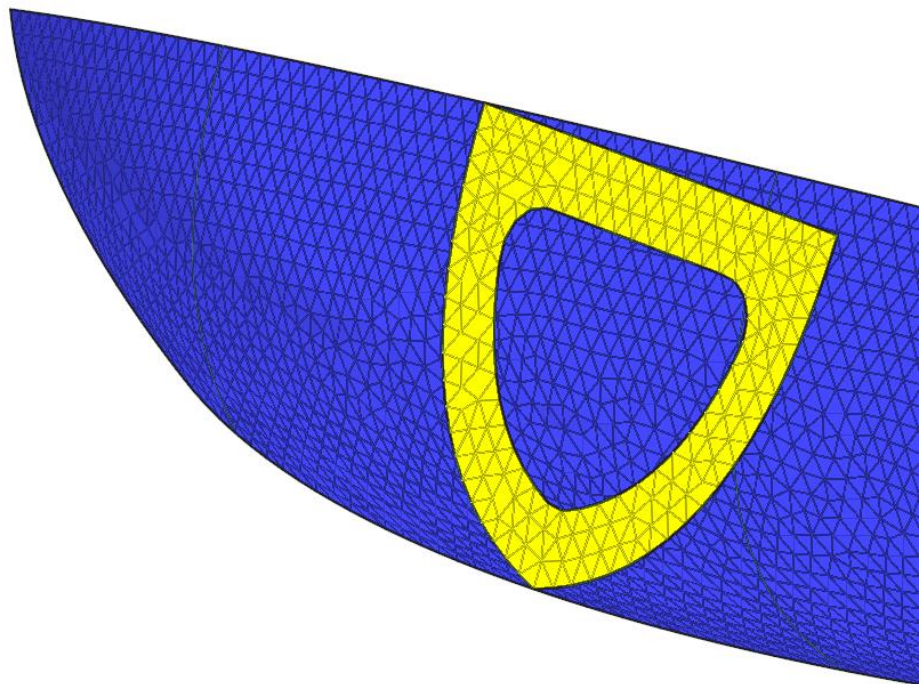
Two or more meshes can be joined into a single mesh using the Join function from the Mesh menu. The function will be enabled when there is two or more trimeshes in the design:



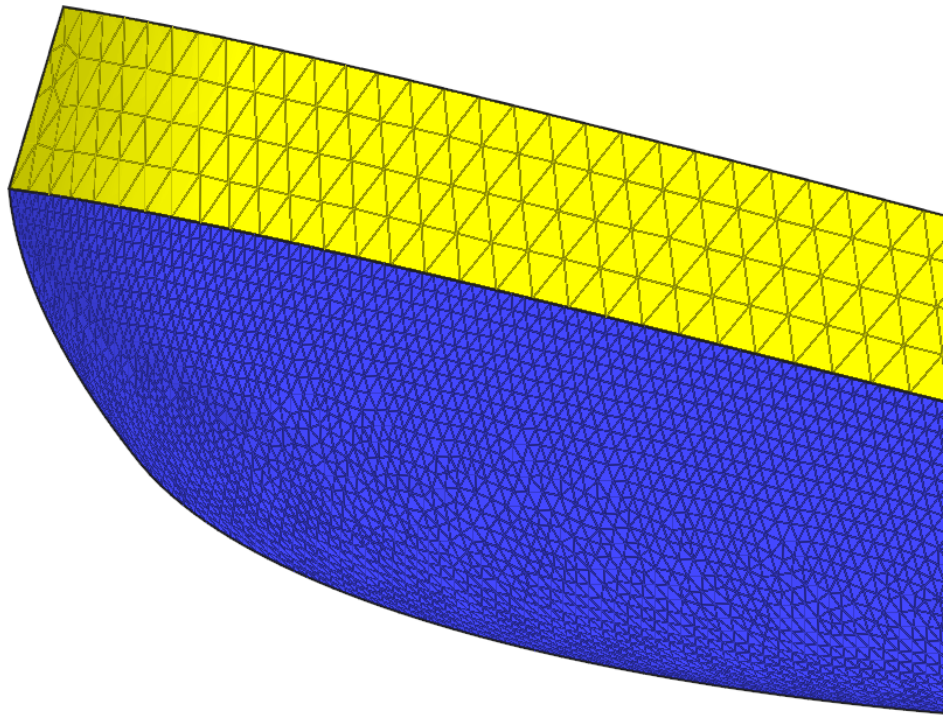


Bonding Meshes

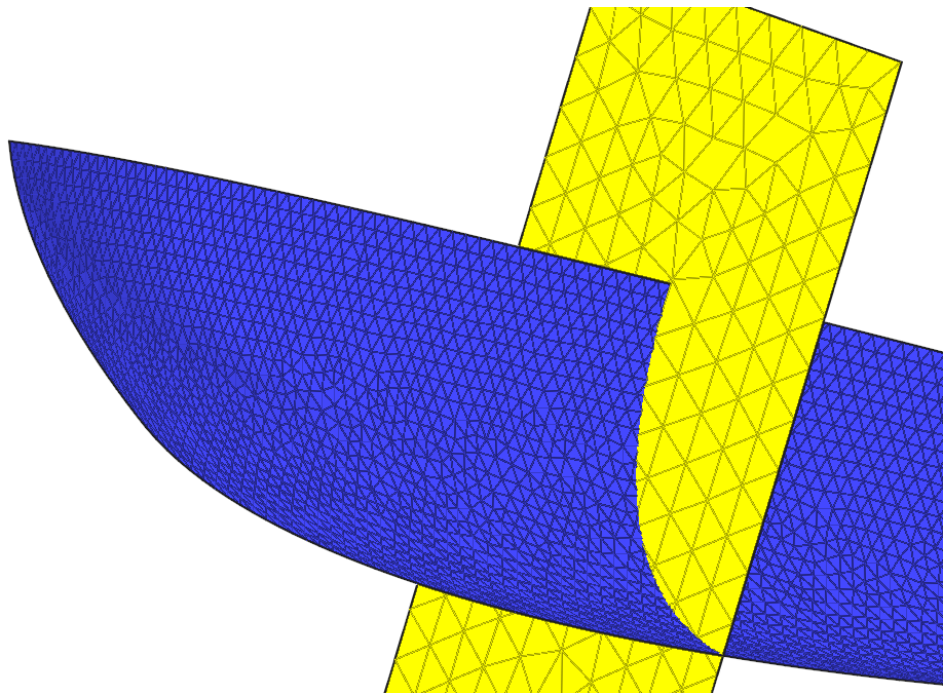
A bond is usually performed on two mesh surfaces or two mesh surfaces and a curve. Mesh bonding functionality is categorized into four options; a Tee bond:



A Butt bond:

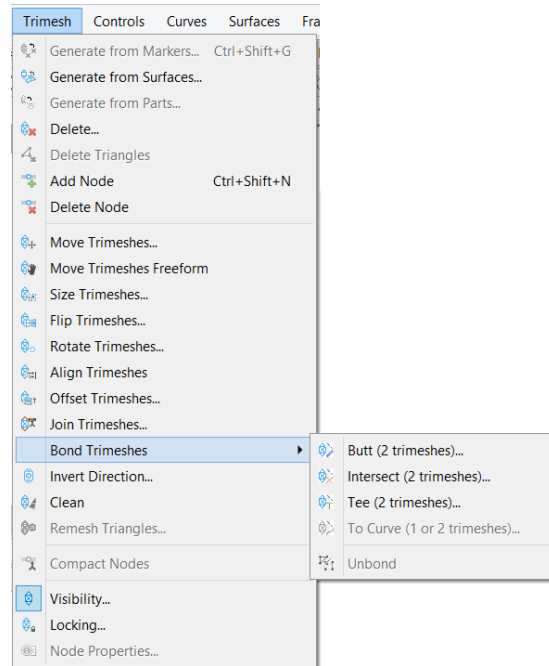


An Intersection bond:

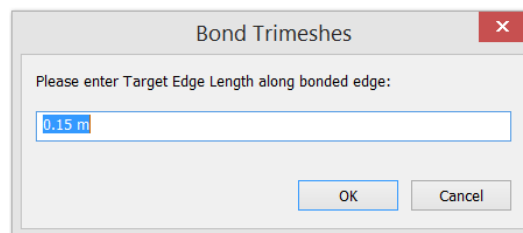


And bonding a mesh to a curve.

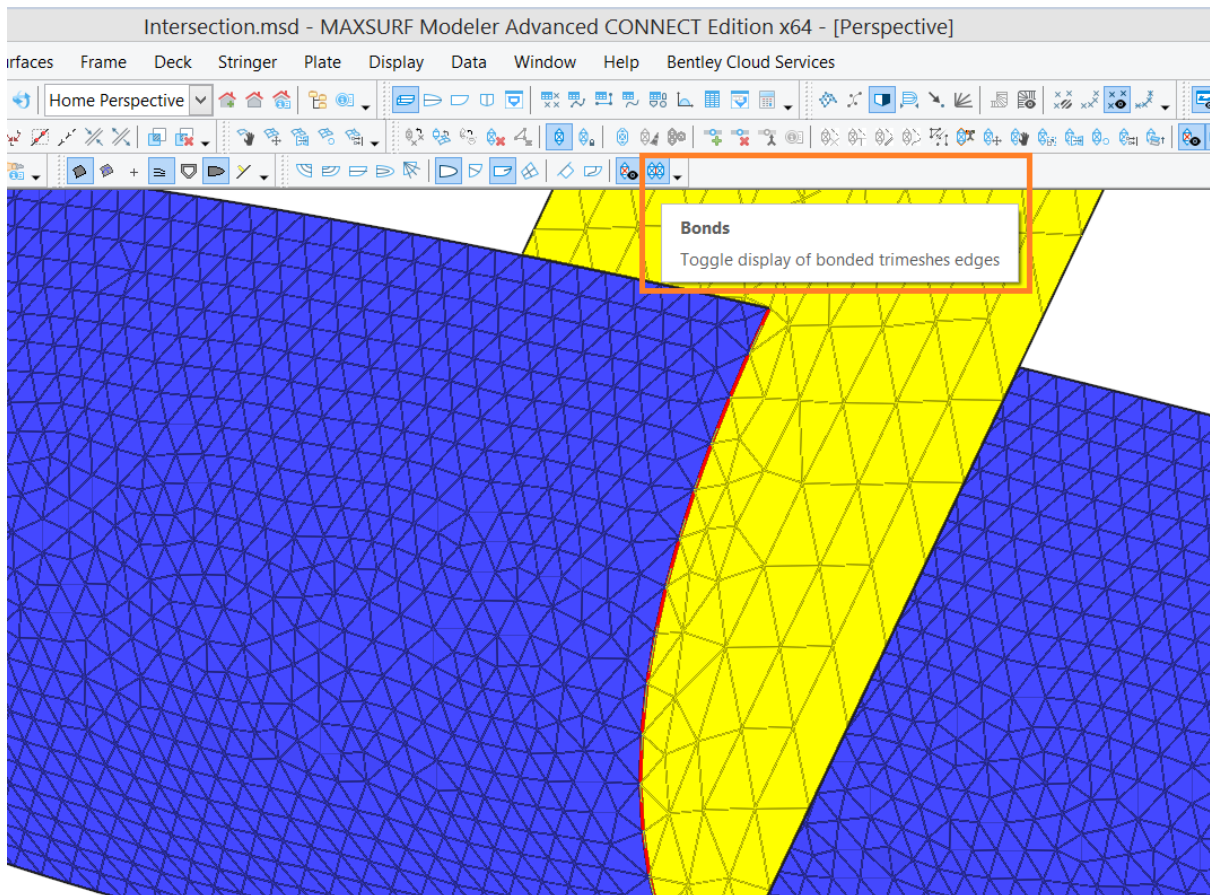
The options can be accessed from the Meshes | Bond Meshes pop up menu:



The options are enabled when 2 meshes are selected, either in a drawing window or from the assembly window. To enable bonding to a curve either a curve and one or two meshes needs to be selected. Choose the option that best represents the configuration of the mesh(es) you are trying to bond. The user will then be asked to specify a target edge length along the common bonded edge. The default value is the average edge length of the mesh(es) that are being bonded:



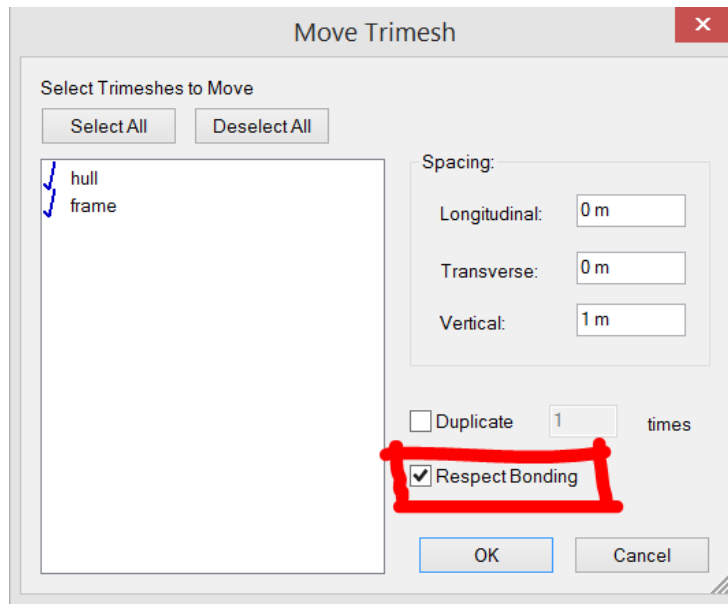
Once the meshes are bonded, the bonds can be viewed with the mesh bonded edge visibility turned on:



The bonded edge contour color and thickness is set in the Colors and line thickness dialog (View | Colors and Lines...).

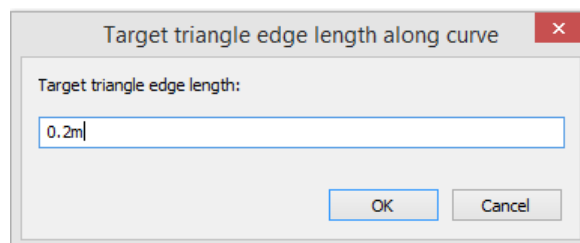
Once two meshes are bonded changes to any vertices on the bonded edge will automatically be reflected in the bonded vertices of the Bonded mesh. To unbond two meshes, in non-rendered mode select the bond contour and choose Meshes | Bond Meshes | Unbond or simply hit the delete key.

As with normal NURBS surfaces if a mesh manipulation function is performed on a bonded mesh the user can select whether the manipulation is to be performed on the associated bonded mesh. For example, when moving a mesh the user can select the "respect bonding" checkbox in the dialog. If this is chosen then any bonded surfaces to the selected surface will also be moved the specified translation:

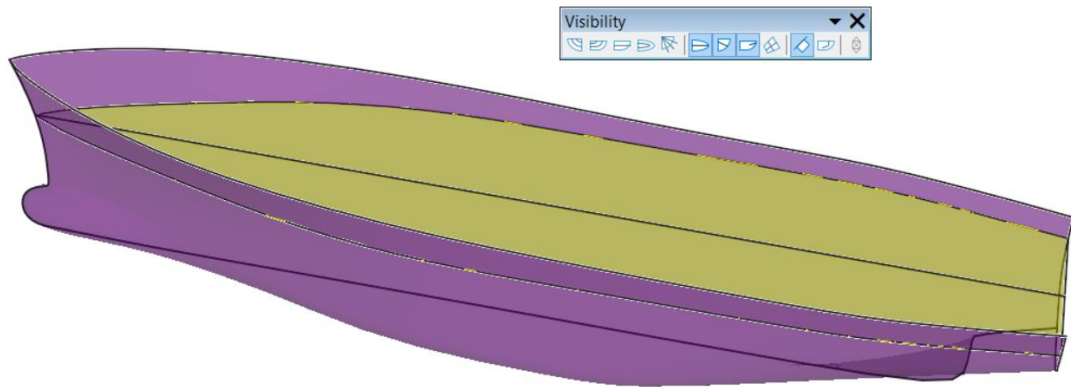


Bonding a mesh along a curve

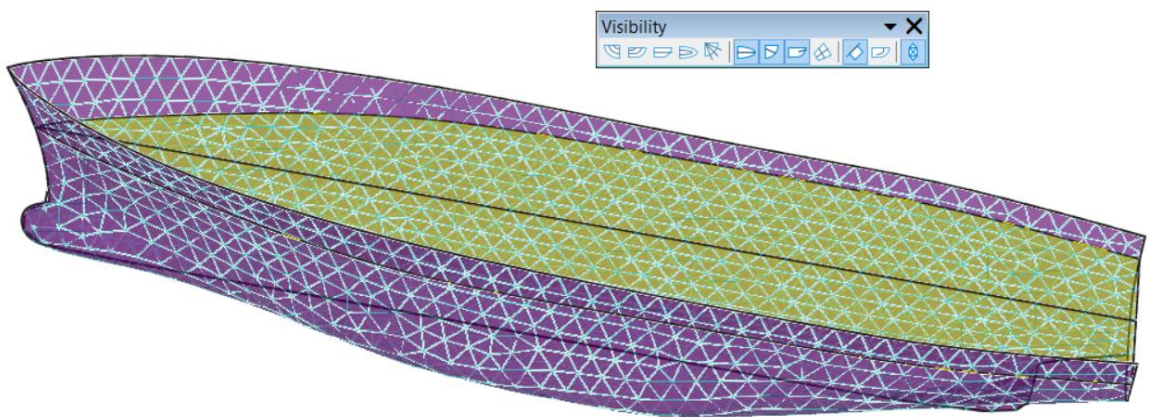
The curve defines the line along which you would like the two meshes to be “bonded”. As when generating meshes a bond will ensure that the triangles from each mesh share common vertices (and hence triangle edges) along the contour. From the assembly tree select the two meshes you wish to bond and the bond contour (holding the shift key down). The selected contour defines the line along which the bond will occur. Next choose “Bond Meshes | To curve...” from the mesh menu. A dialog will appear asking the user to specify the target edge length of the tri edges along the bond.



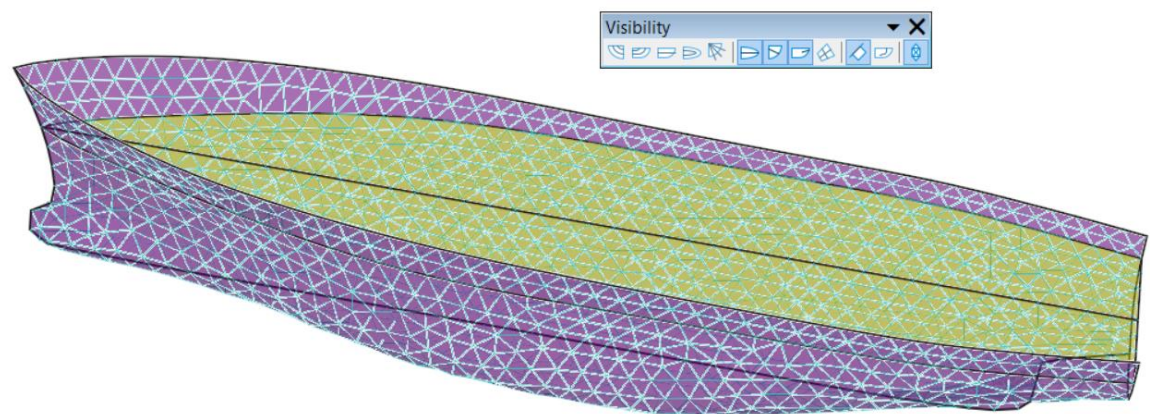
This function is particularly useful in cases where surfaces “just” or “almost” touch each other and bonding along the intersection line(s) doesn’t work as there is not one continuous intersection line. An example might be a deck surface that joins the hull surface below the gunnel, where the deck is not trimmed to the hull surface and doesn’t form a clean intersection line (broken yellow line):



In this case a curve could be added to the deck edge by highlighting the edge and adding a curve using the Curve | Fit Curve to Contour... command. Once the meshes have been created we have the following (Note how the triangle edges do not match along the apparent intersection):



If the bond command is then executed on the two meshes and the contour that was added on the deck edge we will get triangle continuity across the bond:



In general target edge length along the bond is advised to be equal to or less than the target edge length of the meshes being bonded. This will ensure no detail is lost along the bond contour as well as avoiding any other unexpected bond results.

Note:

Meshes need to have the NURBS surface that they were created from still present in the design for a bond to work correctly.

A bond can also be performed on a single mesh and curve. This generates a similar outcome to a hotline with the main advantage being that only the triangles in the vicinity of the bond will be altered.

Calculations

Modeler is able to calculate a variety of data from your design. Before making calculations with your Modeler design, it is essential that sections can be properly formed. This means that sections must be closed, or have only one opening which is then closed with a straight line. If you are using intersecting surfaces and trimming, ensure that all unwanted portions of all the surfaces are removed.

This section explains how to calculate:

- [Hydrostatics](#)
- [Calculate Girth](#)
- [Calculate Areas](#)

For information on how to calculate offsets see:

[Offsets Window](#) section on page 66.

Also see:

[Calculations Window](#) on page 72.

Hydrostatics

The hydrostatics dialog, in the Data menu, will compute the upright hydrostatics at the design waterline. More sections are used for the computations as the precision is increased. As well as the normal hydrostatics, the righting moment at one degree is also given. This is computed as: $GM_t \Delta \sin(1^\circ)$. The precision at which the calculations were done and the number of sections used are given in the last row.

Hydrostatics at DWL

	Measurement	Value	Units
1	Displacement	549.544	tonne
2	Volume	536.14	m ³
3	Draft to Baseline	2.43	m
4	Immersed depth	2.43	m
5	Lwl	39.837	m
6	Beam wl	10.4	m
7	WSA	402.693	m ²
8	Max cross sect area	22.663	m ²
9	Waterplane area	277.644	m ²
10	Cp	0.594	
11	Cb	0.533	
12	Cm	0.897	
13	Cwp	0.67	
14	LCB from zero pt	-20.246	m
15	LCF from zero pt	-21.723	m
16	KB	1.376	m
17	KG	2.43	m
18	Bmt	3.513	m
19	Bml	38.312	m
20	Gmt	2.459	m
21	Gml	37.258	m
22	Kmt	4.889	m
23	Kml	39.688	m
24	Immersion (TPc)	2.846	tonne/cm
25	MTc	5.14	tonne.m
26	RM at 1deg = Gmt.Dis	23.586	tonne.m
27	Precision	Medium	50 stations

Density: Recalculate

VCG: Close

To change the density or VCG, type in the new value and press tab or the recalculate button to refresh the data.

Note:

The wetted surface area displayed in this dialog is calculated using the same method as that used in the Calculate Areas dialog in the Data menu. This method is more accurate than that used in Stability; Modeler divides the surfaces into triangular elements to calculate the surface area, whereas Stability does a lengthwise integration of section girth.

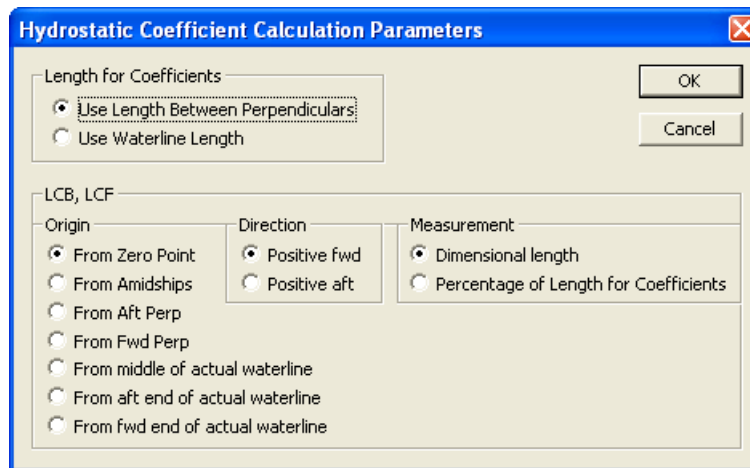
When using the Calculate Hydrostatics dialog from the Data menu, Modeler displays the sections used to calculate the hydrostatic properties in the window in the background. You can move the hydrostatics dialog around on screen to view the sections to ensure that they are correct. Note that this dialog calculates based on your current settings of trimming, surface visibility and precision.

Note: Skin thickness and hydrostatic calculations

In Modeler skin thickness is used for creating the table of offsets. For hydrostatic calculation purposes, skin thickness is ignored. Stability can be used to calculate the effect of skin thickness on the vessel's hydrostatics by selecting "include skin thickness" in the opening dialog.

Hydrostatic Coefficient Calculation Parameters

This dialog is available in Modeler in the Data menu.



Default settings for coefficient length and LCB/LCF measurements

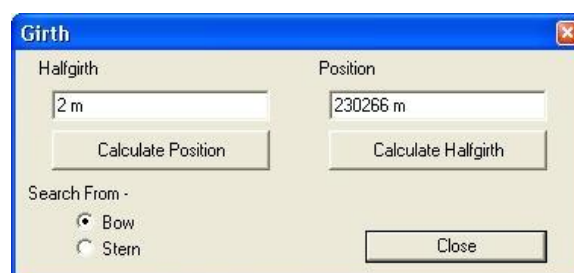
The first part of the dialog is used to define the length to be used for calculating Block, Prismatic and Waterplane Area coefficients; the same length may also be used to non-dimensionalise the LCB and LCF measurements. (In previous versions of Stability, the length had to be set *before* an analysis was carried out. In this version, the changes may be made at any time and the relevant results will update automatically without the need for redoing the analysis – this is the same behaviour as the Units options).

The second part of the dialog is used to define how LCB and LCF (but *not* LCG) measurements are presented. The range of origins for the measurements has been extended to include forward and aft perpendiculars. Also you may define whether positive values should be in the forward or aft direction (relative to the chosen origin); finally you may choose to non-dimensionalise the value by dividing by the coefficient length (chosen in the first part of the dialog) and displaying as a percentage.

These options are used in all Stability results and in the Modeler Hydrostatics dialog. The LCB options are *not* used in the Modeler Parametric Transformation dialog (though the coefficient length is used for the coefficients in this dialog).

Calculate Girth

Girths may be found for all visible surfaces by using the Girth function from the Data Menu.



To locate the longitudinal position where a given half girth occurs:

- Enter the half girth length.
- Click the Calculate Position button.

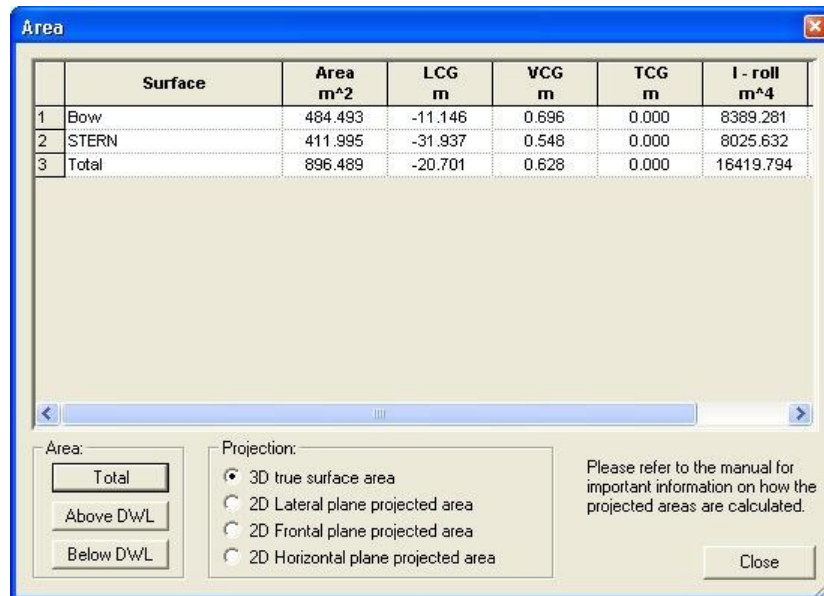
To find a half girth length at a given longitudinal position:

- Enter a position where the half girth is required.
- Click the Calculate Half girth button.

You may search from the bow or stern. When you have finished, click Finished.

Calculate Areas

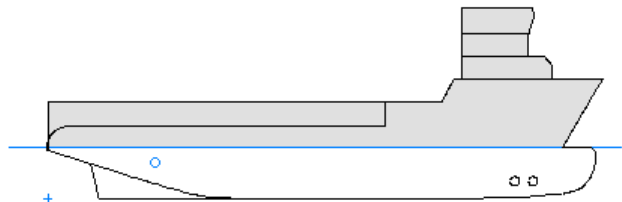
The Calculate Areas function in the Data menu allows the area of surfaces to be accurately calculated, together with their longitudinal, transverse and vertical centre of gravity. The dialog also includes the surfaces' second moments of area about their centres of area. This can be useful when estimating gyradii of the designs.



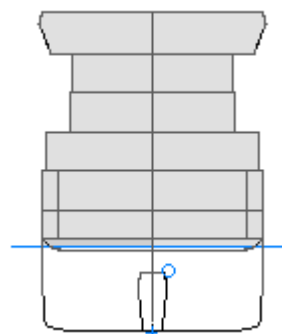
It is possible to compute the total, above water or below water area and inertias. The values account for **trimming** and **surface symmetry** (i.e. all surface area results, except 2D lateral plane projected, are duplicated for symmetrical surfaces). The DWL, as defined by the Frame of Reference, is used.

Some examples:

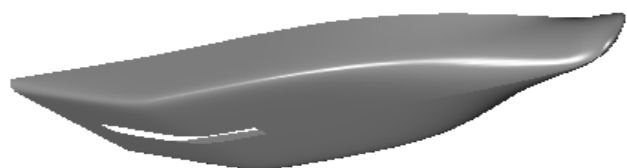
2D Lateral Plane Projected area
above DWL



2 D Frontal plane Projected area
above DWL



3D True surface area of trimmed,
visible surfaces



A fine mesh triangulation of the surface is used, so errors inherent in the use of Simpson's rule are avoided. The higher the precision selected, the denser the triangulation, and the more accurate the answer will be. The CG positions are given relative to the current zero point.

The projected transverse areas and centroids may also be computed, and this is useful for estimating centres of pressures. Note that the total projected area is computed, so if you have a propeller tunnel for example, both sides of the tunnel will be included.

It is important to distinguish between the CG of the surface area and the centre of lateral area provided by the Calculations window. In the first case the centre is of the entire 3D area, whereas in the second case the centre is of the 2D projection of the area.

It is possible to copy the columns of data from the dialog by clicking and dragging over the cells you require and then using the Ctrl C Copy command to copy the data to the clipboard.

Click the Finish button when you wish to close the dialog.

Using Parametric Transformation

Parametric Transformation - Concepts

Modeler Advanced has the ability to perform parametric transformations of hull shapes using the Parametric Transformation command in the Data menu. Essentially, it uses the Lackenby method. This process involves moving the columns fore and aft, while not changing the section shapes (unless scaling them) i.e. all y-coordinates move by ratio of beams, all z by ratio of drafts etc. The transformation moves stations fore and aft until the required parameter(s) specifications are met. A key quality of this function is that it maintains the fairness of the hull to a very high degree during the transformation process.

The Parametric Transformation command displays a dialog that is used to perform the transformation on the design's visible surfaces.

Parametric Transformation

AP MS FP B

☐ Parallel midbody

Aft midbody limit 0 m Forward midbody limit 0 m

Search For:

☐ Block Coefficient 0.494 LCB aft of FP 51.07 % D/WL

☒ Prismatic Coefficient 0.717 ☐ LCF aft of FP 54.01 % D/WL

☒ Midship Area Coefficient 0.718 ☐ Topside Flare 1.22 Degrees

☐ Waterplane Area Coefficient 0.878

Scale To:

☐ Displacement 14455.2 kg ☒ Beam on D/WL 2.696 m

☒ Length on D/WL 14.504 m ☒ Immersed Depth 0.73 m

Search OK Cancel

The parameters that can be specified are divided into two groups:

- [Search Parameters](#)
- [Scaling factors](#)

After doing a parametric transformation you may want to be able to do a [Hull Shape Comparison](#). Also see the section on [Parametric Transformation Restrictions](#) on page 255.

Parametric Transformation – Procedures

See:

- [Search Parameters](#)
- [Scaling factors](#)

Search Parameters

Search Parameters are those that require a non-linear transformation of the hull shape. These are -

- Prismatic or Block coefficient
- LCB – Longitudinal Centre of Buoyancy (also see the section on [Parametric Transformation Restrictions](#) on page 255.
- Parallel Midbody – In a merchant vessel that has a parallel midbody it is important to be able to perform parametric transformations on the fore and aft portions of the vessel without affecting the midbody. To use this feature, enter the positions of the fore and aft limits of the parallel midbody. The Parametric transformation will not change the location of any control points between these two locations.
- Midship Area Coefficient. This parameter determines the fullness of the midship section and the magnitude of any bilge radius. A change to the Midship Area Coefficient will also result in the sections in the bow and stern becoming fuller or slacker.
- Topside Flare. This parameter is unlikely to be used for commercial ships but is useful in the design of leisure craft, both power and sail. The angle of the topsides at the position of greatest beam can be specified, resulting in more or less flare. This flare is also distributed in a fair manner forward and aft, tending to taper out near the extremities.

To vary these values, some form of non-linear transformation of the hull must take place. Because the non-linear transformation is not an explicit function of these parameters, Modeler must perform an iterative search to achieve the specified values.

Note that the only required search parameters are LCB, and either Prismatic or Block coefficients. All other parameters and scaling factors are optional and can be combined arbitrarily.

Also note that the values for Waterplane Area coefficient and LCF are provided for informational purposes only; i.e. it is not possible for them to be specified as search parameters.

Using the search parameters for a parametric transformation

Before performing a parametric transformation of the hull, make sure you do the following steps first;

- Ensure that you have a coherent model for which hydrostatic data can be calculated – see the Stability user manual for details.
- Unlock all surfaces (tip: use the right click function in the [Tool Pane](#) to do this).
- Hide all superstructure and other surfaces that are not used to describe the hullshape.

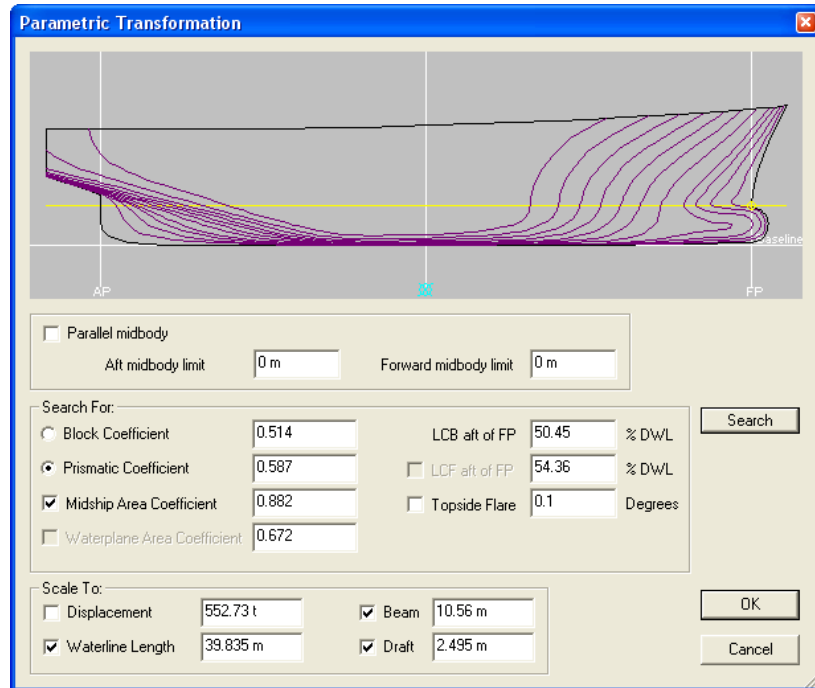
You are now ready to do a parametric transformation.

- Specify the new LCB position and either a new Block or Prismatic Coefficient, followed by constraint values for a maximum of three of Displacement, Waterline Length, Beam and Draft.
- Once these values and constraints have been specified, click on the Search button.

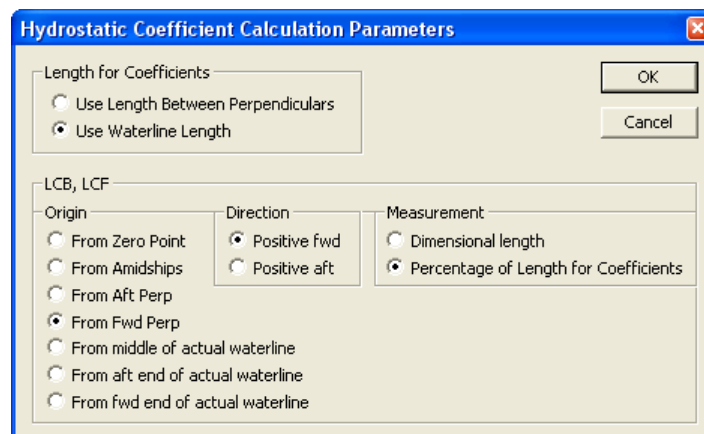
Modeler will iterate to achieve the required parameters and will display the results when it is finished. If the required parameter values could not be achieved, Modeler will beep at the conclusion of the calculation process.

Example: LCB variation on fishing trawler

- Open the Modeler Sample_Trawler.ms3 file from the sample designs folder on your hard drive.
- Save the design to another location with a different filename
 Tip: it is generally good practice to save to another filename before doing a parametric transformation or working with the standard sample designs.
- Go to Data | Parametric transformation



- Type LCB = 49 to search for a LCB of 49% of Lwl aft of FP
- Leave the Length, Beam and Immersed Depth fixed, but allow the displacement to vary by using the "Scale to:" settings as in the image above.
- Press Search and wait for the hourglass to disappear.
- Press Ok to accept the new hullshape
- Do a Data | Calculate hydrostatics to check the new LCB position
 Tip: use the following Data | Coefficients dialog settings:



- Go to profile view and do a Ctrl+Z to undo the parametric transformation and a Ctrl+Y to redo it. This quickly gives you an idea of what has changed.

Also see:

- [Scaling factors](#)
- [Hull Shape Comparison](#)

- [Parametric Transformation Restrictions](#)

Scaling factors

Scaling Factors are those parameters that can be calculated directly using a linear scaling of the hull, namely Displacement, Waterline Length, Waterline Beam and Draft. These parameters can be constrained to particular values, or left to vary as other parameters change by selecting the appropriate check boxes and setting the required values.

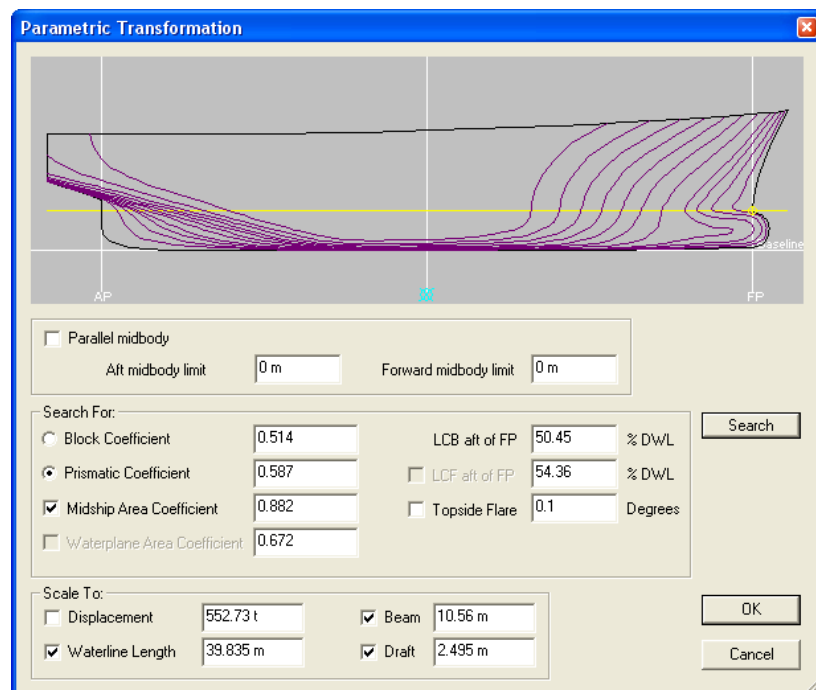
As an example, if no scale factors are specified and the Block Coefficient is increased, the displacement will also increase, while the length, beam and draft will remain constant. If the displacement is the only constraint, the length, beam and draft will decrease proportionally to accommodate the redistributed volume. If displacement, length and beam are constrained to particular values the draft will be changed accordingly.

Note

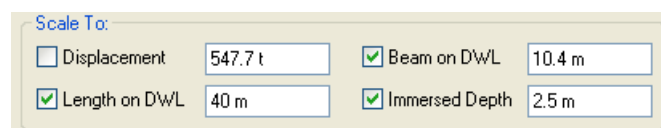
It is not possible to constrain all the values simultaneously – of the four values, displacement, length, beam and draft, at least one must be allowed to vary at all times.

Example: specifying the waterline dimensions

- Open the Modeler Sample_Trawler.msd file from the sample designs folder on your hard drive.
- Save the design to another location with a different filename
 Tip: it is generally good practice to save to another filename before doing a parametric transformation or working with the standard sample designs.
- Go to Data | Parametric transformation



- Specify the following scaling factors and leave the displacement un-constrained:



- Press Search and wait for the hourglass to disappear.
- Press Ok to accept the new hullshape

- Do a Data | Calculate hydrostatics to check the changes have been made correctly.
Note that the displacement has changed, but the LCB % and hull form coefficients are still the same. The draft to baseline is not updated until you specify the new baseline and you may also want to reset your AP and FP in the Frame of Reference dialog to match the new overall dimensions of the hull.

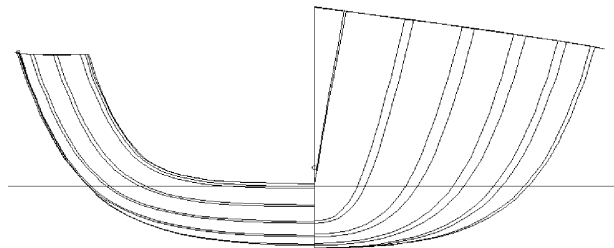
Also see:

- [Search Parameters](#)
- [Hull Shape Comparison](#)
- [Parametric Transformation Restrictions](#)
- [Hydrostatics](#)

Hull Shape Comparison

For comparison purposes it may be a good idea to duplicate the surfaces prior to modification (the duplicate surfaces can be made to lie exactly on top of the originals by specifying zero for all three spacings). The duplicate surfaces should then be made invisible so that they are not affected by the transformation. If, after the transformation, you make the original surfaces visible, you will have a direct comparison of changes to the waterlines, buttocks and sections.

Tip: by holding the mouse cursor over the contour lines you can see the surface name and the contour line name in the bottom left of the Modeler window. Note that this method of comparison will not work if the model is trimmed as the coincident surfaces will corrupt the trimming information.



Hull comparison after 2% change to Prismatic and 1% change to LCB

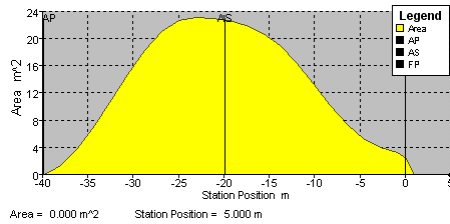
Another quick way to compare the hull is by simply doing Ctrl+Z to undo the Parametric Transformation and Ctrl+Y to redo.

It can also be a good idea to make a copy of the hydrostatic data before the parametric transformation into Excel and compare it with the data afterwards.

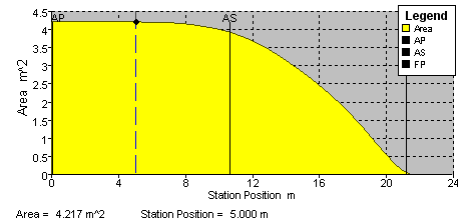
Parametric Transformation Restrictions

Transformations are applied to visible surfaces only. Surfaces selected should form a coherent model capable of being analysed in Stability. This means that care should be taken to trim intersecting surfaces where necessary and make sure that potential ambiguities such as multiple openings in the hull are resolved.

It is important that the parent model be reasonably close in form to the required final design as it is not feasible to make major changes to the search parameters before the hull starts becoming unreasonably distorted. Experience suggests that reasonable variations of Block, Prismatic or LCB are of the order of up to plus or minus 3 to 5%. These limits may be lower for vessels that have moderately high prismatic coefficients – as the prismatic increases the scope for variation seems to improve.



Normal hullshape with relatively low prismatic coefficients allow 3 to 5% LCB variation.



Highly prismatic hullshapes only allow small % LCB variations.

The parametric variation tool is ideal for making small adjustments from a suitable parent design. It is not intended to be used for gross design modifications. The scope for parameter variation is greatest for designs that are not extreme; as the design and its form parameters become more extreme the scope for parametric variation is reduced.

Because of the non linear nature of the transformation and the requirement that fairness be maintained, it is often found that long fore and aft overhangs distort proportionally more than the underwater body. The shorter the overhangs of the vessel are the less noticeable this effect will be.

It is not possible to constrain Displacement, Length, Beam and Draft simultaneously; the maximum permissible is any three parameters out of the four available.

Americas Cup Yachts

Special functionality has also been added specifically for the designers of Americas Cup class yachts. The ACRule can be accessed from Data menu.

Parametric Transformation

Parallel midbody
Aft midbody limit: 0 m Forward midbody limit: 0 m

Search For:
☐ Block Coefficient 0.447
☒ Prismatic Coefficient 0.553
☒ Midship Area Coefficient 0.795
☐ Waterplane Area Coefficient 0.739

LCB aft of FP 54.5 % DWL
☐ LCF aft of FP 55.96 % DWL
☒ Topside Flare 0 Degrees

Scale To:
☒ Displacement 22 t
☒ AC rule LBG 20.2 m
☒ Beam 3.4 m
☐ Draft 0.82 m

Search OK Cancel

In the case of Americas Cup class yachts the constraint applying to LWL has been replaced by one applying to the AC LBG value, a length measured 200mm above the DWL. In addition, the Parametric Transformation automatically constrains the forward and aft girths to the minimum allowable value, as measured by a chain girth at the forward and aft girth stations.

Working with Structure

The following sections describe the tools and workflows for defining the primary structural parts (frames, stringers, decks and plates) of your vessel. We shall first look at some of the requirements for preparing your model to facilitate the generation of structural parts and then examine the tools available for each type of part in greater depth.

Preparing the design for generating structure

Before working on structure a number of checks should be made. These are very similar to the model requirements for calculating hydrostatics and analysis in the other modules, and consists of checking:

- [Frame of Reference and Zero Point](#)
- [Check trimming](#)
- [Check surface Use](#)
- [Check Surface Thickness](#)
- [Check Outside Arrows](#)

Frame of Reference and Zero Point

It is considered best practice to ensure that the model has the correct frame of reference and zero point defined before starting to add structure and to not change the frame of reference after you have started defining parts.

Check trimming

Ensure that all surface intersections are correct and that the model is trimmed completely. In order to be able to form frames correctly you should have closed transverse sections (or sections with at most one opening).

In order to be able to form decks correctly, the model's waterlines should be made up from closed polylines.

Check surface Use

Surfaces are divided into two types:

- Hull shell
- Internal structure

When modeling structure these different surface types are treated differently. For example, an internal surface cannot be used to define a frame boundary.

Use for Structural Parts	Hull Shell	Internal Structure
Frame boundaries	✓	✓
Deck boundaries	✓	✗
Plate development	✓	✓

Check Surface Thickness

The thickness assigned to the surfaces, is used to form frames and stringer paths. For more information, see:

[Check Outside Arrows](#) on page 257

[Deduct Skin Thickness](#) on page 267

[Plate Thickness](#) on page 310

Check Outside Arrows

To define which side of each surface is outside and which is inside, you need to set the outside arrows direction. The direction of the outside arrows is used to determine the direction of stringers (cutouts through frames) and throw direction of skin thickness (plate development, frame perimeter deduction).

You can check the direction of the outside arrows by turning on the outside arrows in the display menu or using the toolbar item:



You can change the direction of the outside arrows by clicking on the circle at the end of the arrow.

Precision for Structural Parts

There is a balance between speed and accuracy depending on the performance of your computer. Especially when developing plates the user has to be aware of the trade off between speed and accuracy. In this section some recommendations on precision are made to make sure you get maximum speed and accurate results.

When you are creating frames, decks and stiffeners, it is recommended to use medium precision.

When defining plates, it is generally best to use high or highest precision because the plate edge countours are created with the precision at the time the plate was defined and are not recomputed. However, when developing plates the user has to be aware of the trade off between speed and accuracy and it is sensible to make a rough check of the plate development at medium precision before re-developing at a higher precision.

Before you export the structural parts to downstream structural detailing programs, you should set the precision to highest precision and recalculate all parts.

Working with Frames

Working with Frames - Concepts

You will usually begin creating your structural model by creating the frames. Frames are added at the location of sections along the hull. A Frames Properties dialog allows you to clip a frame, assign material and add openings.

Working with Frames - Procedures

Working with frames consists of the following:

- Adding a Frame
- Adding a Frame Opening
- Calculating Frames
- Frame Visibility
- Deleting Frames
- Modifying Frames
- Moving Frames
- Copying Frames

See also Exporting Parts on page 322.

Adding a Frame

To add a frame or frames you first need to select one or more sections to specify the position of the frames. Note that if you do not have sections at the location where you require them, you can use the Design Grid command to move or add sections to the required locations before using the Add Frame command.

If the sections are not visible on the hull, use the Contours command from the Display menu to turn on the display of sections.

To select a section

- Click on the section

The section will be drawn with a bold line to indicate that it is selected.

To select a number of sections

- Hold down the shift key while clicking on each section
- or Drag a selection rectangle around the sections to be selected

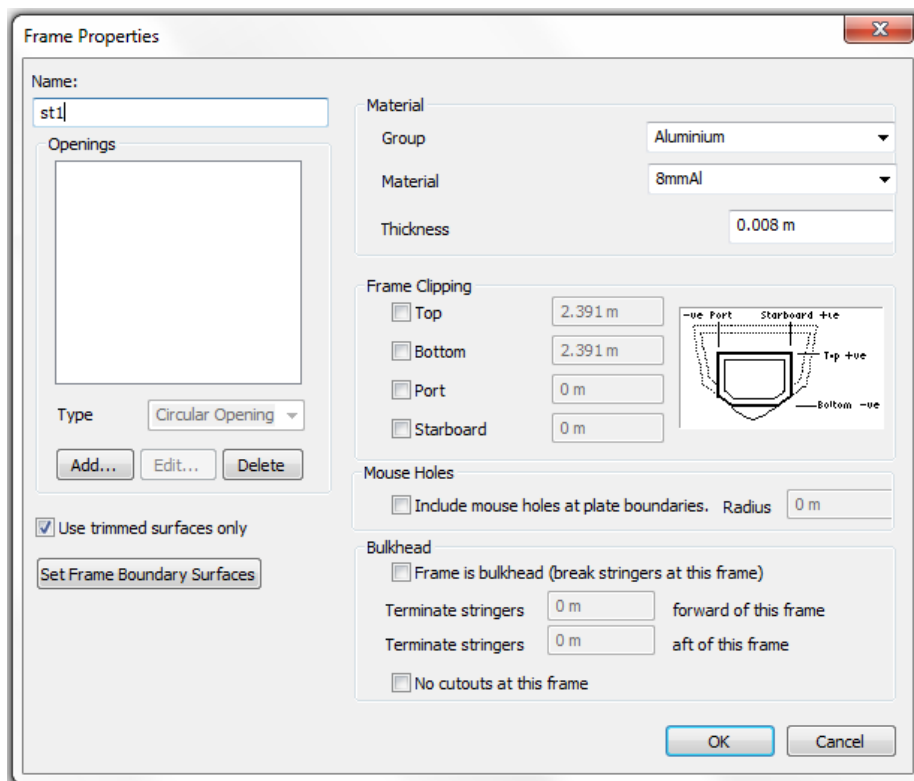
To select all of the sections in the design

- Choose Select All from the Edit menu or drag a selection rectangle around all of the sections

To add frames at the selected sections

- Choose Add Frame from the Frame menu

A dialog will appear allowing you to specify the shape of the frames:



Frame Material

If there is not an appropriate material available in the pop-up menu, you will need to use the Edit Materials | Add Material... item from the Edit menu to add the required material to the library before using the Add Frame command.

- Choose the frame's material group from the Material group pop-up menu
- Choose the Frame's material from the Material pop-up menu

Frame Clipping

The Frame dialog displays information for the whole frame. The Frame Clipping parameters define the limits to which the frame will be drawn.

- Check the check boxes of the clipping limits you wish to set for the frame
- Type in the dimensions of any limits you wish to set

Mouse Holes

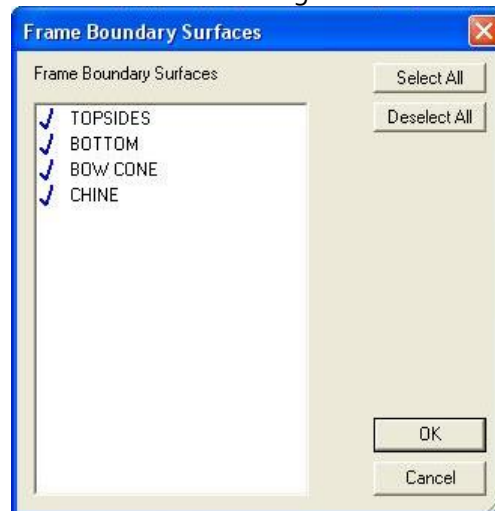
At the intersection between the frame and each plate seam you can add a mouse hole of specified radius by selecting the checkbox under Mouse Holes.

Frame Openings

The current opening is the opening selected in the list on the left. You can add new openings by clicking the Add button. You can delete the current opening by clicking the Delete button. You can edit the current opening by clicking on the Edit button. When the Add or Edit button is clicked, the Frame Opening Properties dialog box will appear (see the "[Adding a Frame Opening](#)" section below).

Frame Boundary Surfaces

You can choose which surfaces will be used to form the frame. To do this you need to click on the Set Frame Boundary Surfaces button in the Frame Properties dialog box. The Frame Boundary Surfaces dialog will then appear; here you select the surfaces to be used for forming the frame.



Toggle surfaces that you wish to use for forming the frame. This information will be stored with the frame and if the properties are copied to another frame the surface association will remain.

- Click OK to confirm the selected boundary surfaces

Note

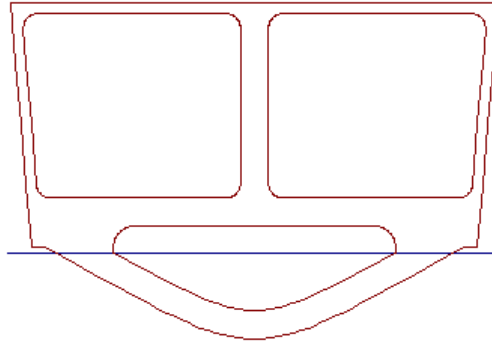
The way Modeler-Structure forms frames along a section is very similar to the way that Hydromax forms hydrostatic sections. Same as in previous versions of Modeler-Structure, it is still important that the selected surfaces form one closed contour with a maximum of *one* opening. See the Hydromax manual on [section forming](#) for more information.

After clicking OK in the Frame Properties dialog, an approximate shape for each frame will be calculated and displayed in the drawing windows. Initially, Modeler-Structure just calculates the exterior shape of the frames, not taking into account any cutouts or fillet radii. The final frame shape will be calculated when you use the Calc All Frames, Calc Selected Frames or Calc All Parts commands. Frames work with trimmed surfaces if trimming is turned on and frames will be calculated using the currently visible surfaces.

If you do not turn on any attributes, the frame will be a simple bulkhead with no internal cutouts and extending to the limits of all of the visible surfaces.

If you do turn on any attributes, you can type in the dimensions of these parts of the frames. The frame and opening limits are all specified as locations relative to the zero point of the design. These limits can have positive or negative values and use the usual coordinate system.

Use of the frame limits allows you to have a number of frames at one location. You can add a number of frames at one longitudinal position by repeatedly selecting a section and using the Add Frame command. This allows you to build up complex frames as a combination of frames.



Adding an Inclined Frame

Frames may also be added where you have defined inclined sections. When the inclined sections have been defined use them in the same way as normal sections to make frames. Only web-openings are available for inclined frames.

Note:

The sections may only be inclined fore and aft.

See [Working with Inclined Sections](#) on page 297 for details on how to define inclined sections.

Other frame procedures:

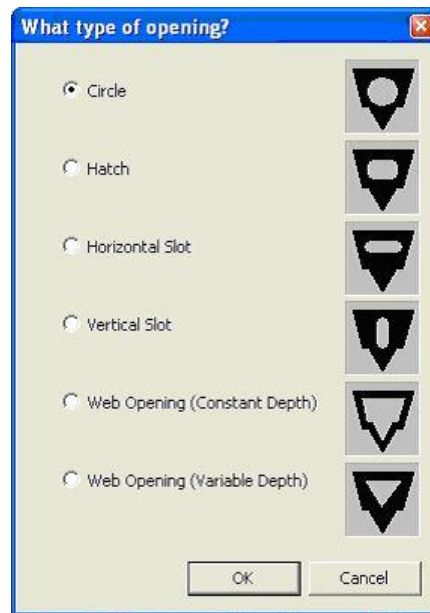
- Adding a Frame
- Adding a Frame Opening
- Calculating Frames
- Frame Visibility
- Deleting Frames
- Modifying Frames
- Moving Frames
- Copying Frames

Adding a Frame Opening

You can add, edit and delete openings in frames while you are in the Frame Properties dialog box. To add a Frame opening

- Click on the Add button.

The following dialog box will appear:



In this dialog you specify the type of opening you wish to be added to the frame. If you choose any of the first 5 opening options the following dialog box will appear.

In this dialog you specify the shape and location of openings in the current frame(s). If the dialog box is opened using the Add button the existing fields will be set. The values of these fields will be the same as those of the opening highlighted in the Openings field of the Frame Properties dialog box when the Add... button was clicked.

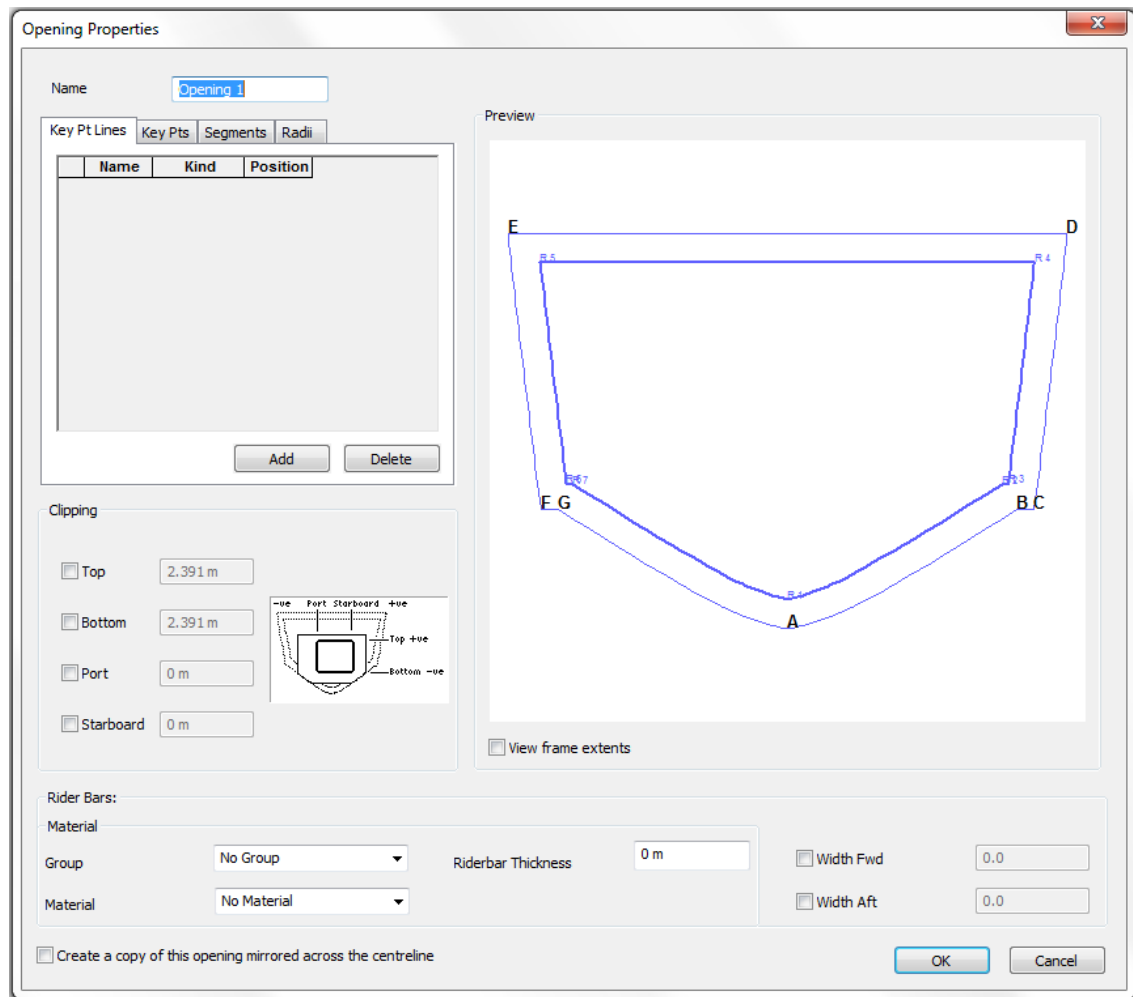
Opening Location defines where the opening will be situated in the frame. The opening insertion point is defined by an Offset and Height and also where the opening insertion point is in relation to the opening. This insertion point is referred to as the anchor point.

The Shape radio buttons define the shape of the opening to be added. Five shapes are available and an unlimited number of shapes may be added to a frame.

The dimensions fields are relative to the opening and the fields required (for the particular shape chosen) are editable, while those not required are greyed.

See [Adding a Rider Bar to a Frame Opening](#) below if you wish to add a rider bar (also called a Face Plate) to your opening.

If you chose to add a Variable Web-Depth opening in the opening type dialog, the following dialog will appear:



The preview in the right hand side of the dialog will display the opening you are currently working on. It can display just the opening or the entire frame. To view the entire frame, check the View frame extents box.

Adding a Variable Web-Depth Frame Opening

The general process when defining a Variable Web Depth (VWD) opening is to define the web depth at a number of key points around the opening. You do this by following the tab sheets from left to right. When the opening is first created, a number of key points will be displayed at the centreline and at any corner points on the opening. Next, you can optionally add additional key points. These points are defined from the intersection of Key Pt Lines and the frame contour. To enter a Key Pt Line

- Click on the Key Pt Lines tab and then click the Add button.
- Define the kind of the line (vertical or horizontal) and the numerical position (height or offset)

Once entered the line will appear in the preview on the right. Key points will also appear at the intersection of the line and the frame.

Next, you can review the key points in the Key Pts tab.

- Click on the Key Pts tab to bring it to the front

If there are any redundant key points you can delete them from the Key Pts tab sheet using the Delete button.

Next, click on the Segments tab where you can review and modify the web depth along each segment of the opening. Each segment runs between two key points.

- Enter the web depths at the start and end of each segment

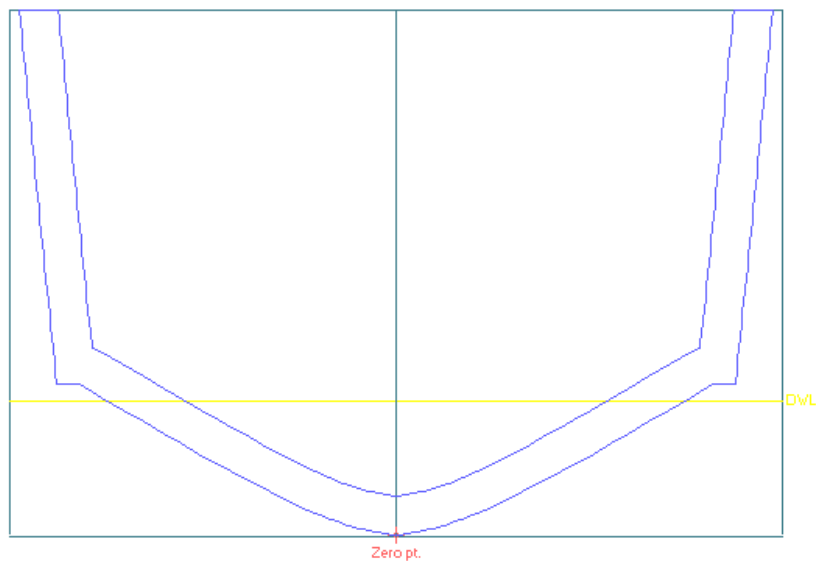
Note: If you set the 'type' to 'smooth', the web depth will vary linearly along the segment from the start depth to the end depth. If 'linear' is chosen, then a straight line will be drawn from the start point to the end point of the segment.

Finally you can set the radius at each of the corner points around the opening. Hard points in the opening will automatically be marked with a R1,R2etc. You can set a fillet radius at these points. Hard points are defined at intersections of line segments that form an angle greater than 15 degrees. To view the radii

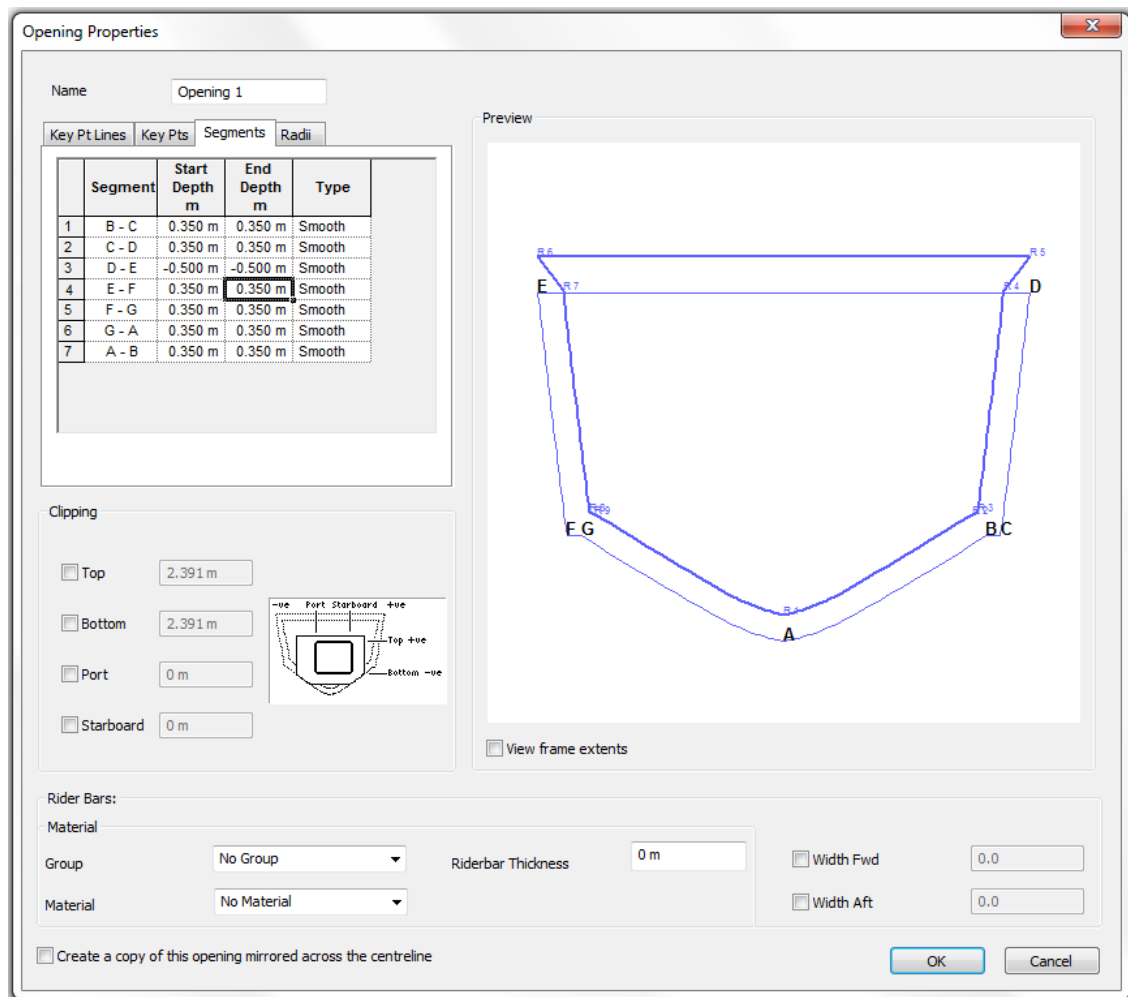
- Click on the Radii tab to bring it to the front
- Fill in the value of the radius required at each of the corners

Adding an 'open' webframe

To create an open frame, similar to this:



Use the variable web depth opening option. In the dialog box set the segment you wish to be 'open' to a negative depth:



Clipping and Symmetry

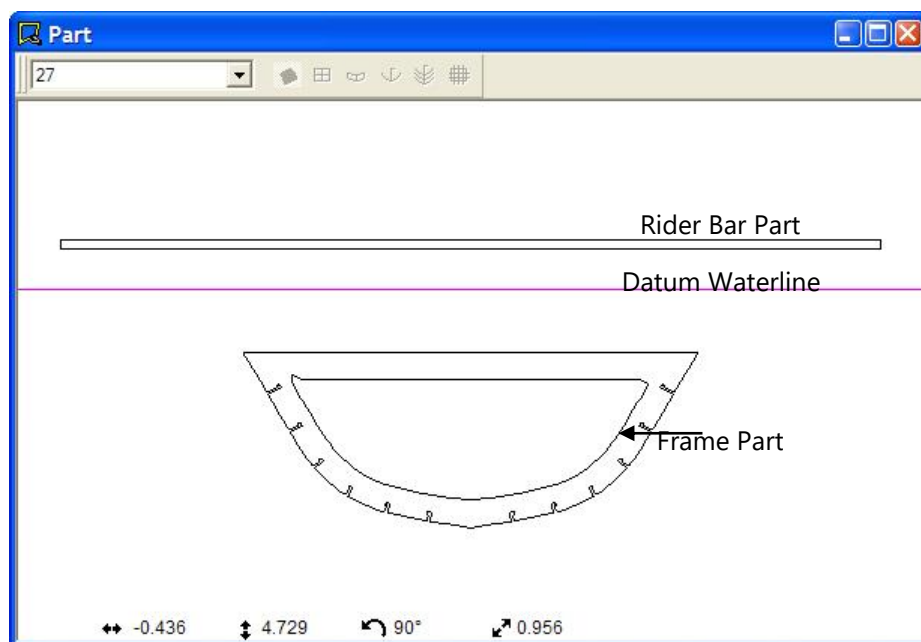
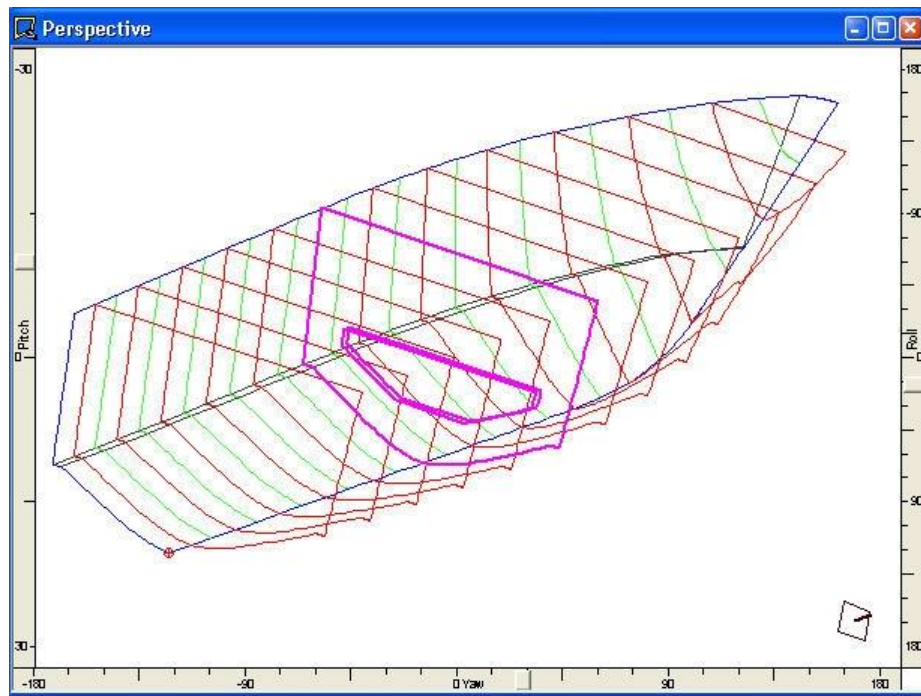
In the same way that you can use clipping to define the extents of a frame, you can use clipping to define the limits of your opening. Turn on the check boxes beside the clipping limits you wish to set and enter a value for the clipping location.

An opening may also be mirrored about the centreline. To create an identical opening on the opposite side of the vessel centreline,

- check the Symmetrical box

Adding a Rider Bar to a Frame Opening

You can optionally generate rider bars on the openings in your frames. Rider bars form a flange on the web of the frame. If you turn on the rider bar check box and enter dimensions for the forward and aft dimensions of the rider bar, it will be generated along with the frame when you use the Calc Frames command. If you select a material for the rider bar, the weight of the rider bar will be included in the weight of the frame listed in the parts table in the Data window.



Calculating Frames

You can use the Calc Frames commands to calculate the frame shapes including any internal openings. If you have finished setting up all of your upstands and stringers, the calculated frame shapes will also include cutouts to allow stringers to pass through the frames.

Note:

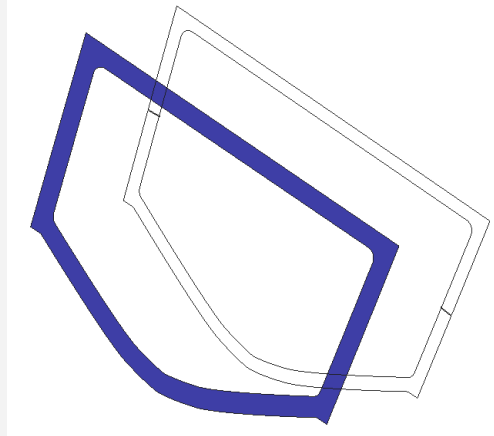
Frame shapes will be calculated using the precision setting at the time you use the Calc command, not the precision setting when you define the frame. This means you can define your frames with precision set to medium, then change to highest precision to calculate the final shape.

Once the frame shapes have been calculated, the 3D shape of the frame will be displayed in the drawing windows and the 2D shape of the frame will be displayed in the Part window. You can use the pop-up menu at the top left corner of the Part window to choose which frame is displayed.

If stringer display is turned on in the Structure command from the Display menu, the stringer shapes will be displayed along with the frames in the Part window.

Note:

If the outside contour of a frame self-intersects or intersects with one of its opening contours then the frame will appear as a wireframe in rendered view (as opposed to a solid).



Deduct Skin Thickness

By selecting "Deduct Skin Thickness" in the Frames menu, the skin thickness is taken into account. If the skin thickness throw direction is set to inside or centred and the "Deduct Skin Thickness" is selected in the Frame menu, the skin thickness will be deducted from the perimeter of the frame. Stringers and stringer cutouts will also be moved to the correct position on the frame.

Note:

The plate thickness that is assigned during plate development is independent to the skin thickness of the surface and is used only to compute the mass of the plate.

The surface thickness as assigned is however deducted from the frame perimeter. The plating thickness as assigned during development is ignored in the "Deduct Frame Thickness" command.

Also see:

[Check Outside Arrows](#) on page 257

[Plate Thickness](#) on page 310

Frame Visibility

You can control the visibility of Frames and Frame Names using the Structure visibility dialog or the Frame icon on the Visibility toolbar.

You can also modify some Frame properties numerically in the Frame table of the Data window.

Frames appear in the list of parts in the Part table in the Data window.

Individual frames may be shown or hidden. This is done by right-clicking on the part in the assembly tree view, and choosing Show or Hide from the pop-up menu, or by changing the check box in the Visibility column in the Data window.

Deleting Frames

You can delete frames by selecting them in the graphics windows and then using the Delete Frames command or hitting the delete button. You select a frame or frames by clicking on it or by dragging a selection rectangle around it. You can also select the row of the frame in the Frames table in the Data window and use the Delete Frames command.

Modifying Frames

You can modify a single frame by double clicking on it in one of the graphics window. You can also select a frame or frames and use the Frame Properties command to modify its properties. You can also edit some frame properties in the Frames table in the Data window. An individual frame's color may be set by clicking on the colored cell in the Color column in the Data window.

Moving Frames

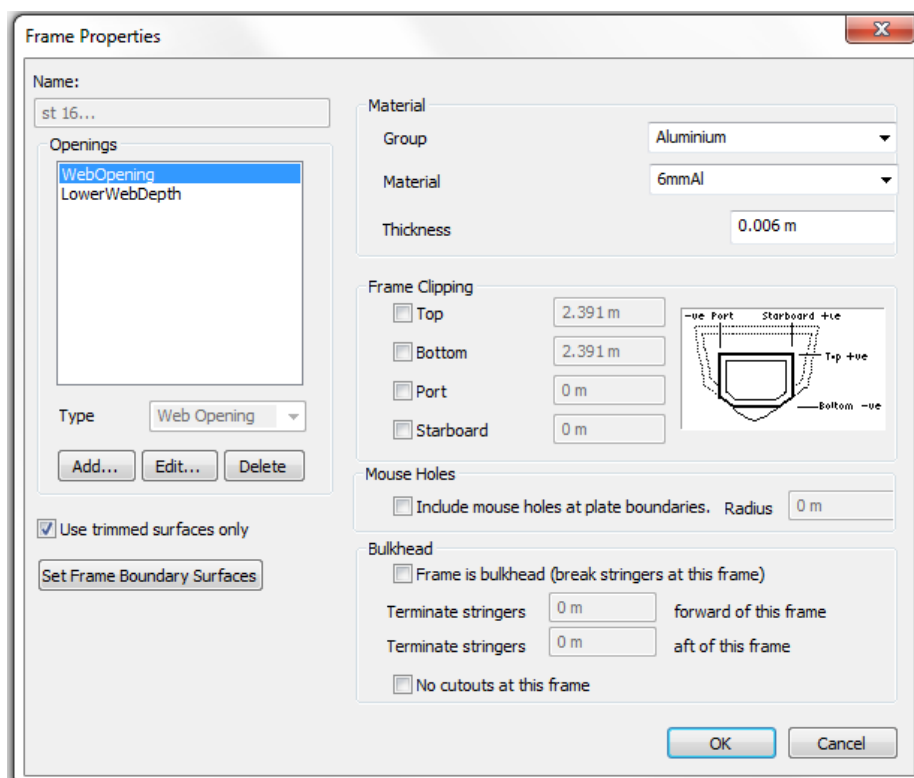
Frames can be moved by changing the longitudinal position in the Frames tab in the Data table from the Windows menu.

Copying Frames

The properties of a frame (opening, rider bars, material etc) can be copied as follows:

- Select the frame you wish to copy the properties from
- Select the frames you wish to copy the properties to
- Select Frame Properties from the Frame menu

The properties you are about to copy are displayed, except for the name of the frames. This field is greyed out with the original frame name displayed:



- Press <ok>
- Select Calculate Selected Frames from the Frame menu

Note:

Make sure trimming display and the surfaces displayed are the same as when you defined the parent frame.
Copying frame properties does not work when the destination frames' section lines are a very different shape compared to the parent frame. For example: when the parent frame crosses a knuckle and the destination frames do not, Modeler-Structure cannot copy the frame properties to the destination frames.

Working With Decks

Modeler-Structure includes a capability for adding deck structures to a Modeler-Structure model. A deck is a horizontal structure at a defined height which is trimmed to the boundaries of the visible surfaces in the design. A deck may also have a number of openings in it.

Working with decks consists of the following:

- Adding a Deck
- Deleting a Deck
- Adding a Deck Opening
- Drawing Decks
- Deck Visibility
- Modifying Decks
- Moving Decks
- Copying Decks
- Deck Limitations

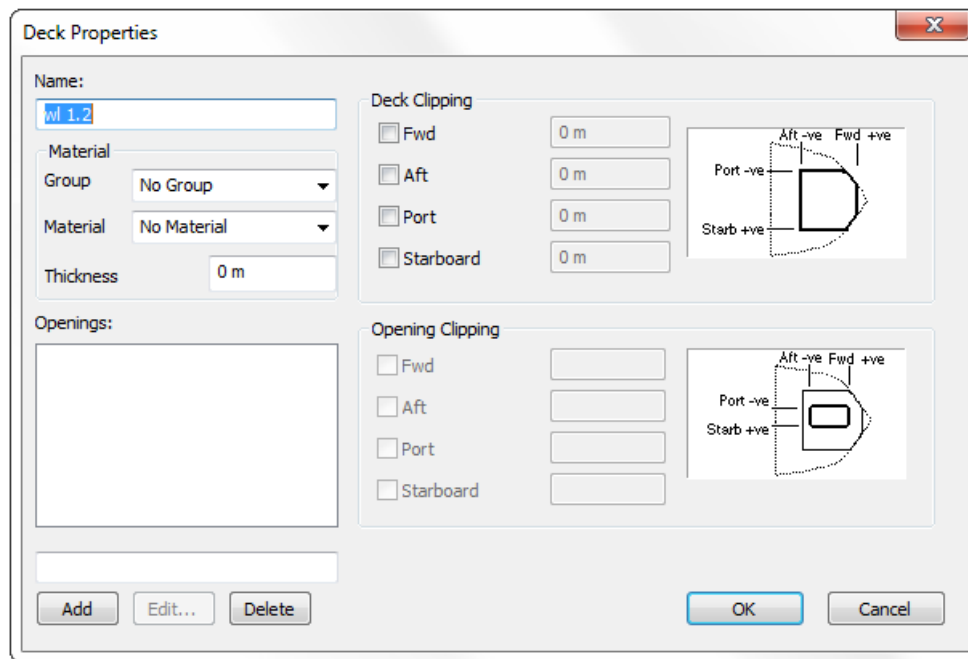
Working with decks is very much the same as working with frames.

See also Exporting Parts on page 322.

Adding a Deck

A deck is added by selecting a waterline at the location of the deck and then using the Add Deck command from the Deck menu.

Properties for each deck can be specified in the Deck Info dialog.



Deck properties are similar to those for frames with the user being able to specify name, material, thickness, and optionally trim the deck to specified forward, aft, port and starboard limits.

Deleting a Deck

If you wish to remove a deck, you first need to select it.

To select a deck

- Click on the deck

To select a number of decks at once

- Drag a box surrounding the decks to be selected

To delete a deck once it is selected in the drawing windows

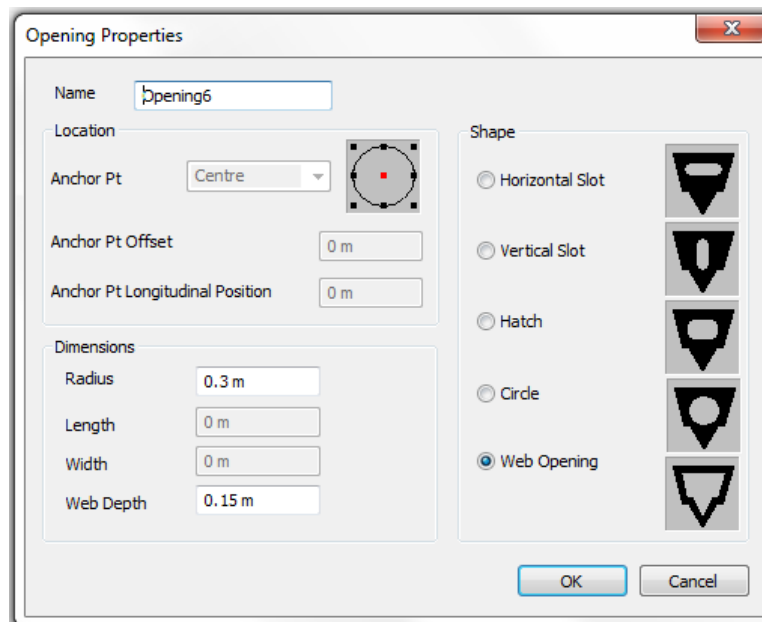
- Choose Delete Deck from the Deck menu or hit the delete button.

In the Deck table in the Data window

- Click and drag to select the rows of the decks to be deleted
- Choose Delete Deck from the Deck menu

Adding a Deck Opening

To add a Deck opening click on the Add... in the Deck Info dialog box. The following dialog box will appear:



In this dialog you specify the shape and location of openings in the current deck(s). If the dialog box is opened using the Add button the appropriate fields will be set. The values of these fields will be the same as those of the opening highlighted in the Openings field of the Deck Info dialog box when the Add... button was clicked.

Opening Location defines where the opening will be situated in the deck. The opening insertion point is defined by an Offset and Longitudinal Position and also where the opening insertion point is in relation to the opening.

The Shape radio buttons define the shape of the opening to be added. Five shapes are available and any number of shapes may be added to a deck.

The dimensions fields are relative to the opening and the fields required (for the particular shape chosen) are editable, while those not required are greyed.

Drawing Decks

Once calculated, decks are displayed in the Plan, Profile, Perspective and Body Plan windows as well as in the Part window. You can select a deck by clicking on it and you can change its properties by double clicking on it.

Deck Visibility

You can control the visibility of Decks and Deck Names using the Structure visibility dialog or the Deck icon on the Visibility toolbar.

You can also modify some Deck properties numerically in the Deck table of the Data window.

Decks appear in the list of parts in the Part table in the Data window.

Individual decks may be shown or hidden. This is done by right-clicking on the part in the assembly tree view, and choosing Show or Hide from the pop-up menu, or by changing the check box in the Visibility column in the Data window.

Modifying Decks

You can modify a single deck by double clicking on it in one of the graphics window. You can also select a deck or decks and use the Deck Info command to modify its properties. You can also edit some deck properties in the Decks table in the Data window. An individual decks color may be set by clicking on the colored cell in the Color column in the Data window.

Moving Decks

Decks can be moved by changing the height value in the Decks tab in the Data table from the Windows menu.

Copying Decks

The properties of a deck (openings, material etc) can be copied as follows:

- Select the deck you wish to copy the properties from
- Select the decks you wish to copy the properties to
- Select Deck Info from the Deck menu

The properties you are about to copy are displayed, except for the name of the decks. This field is greyed out with the original deck name displayed.

- Press <ok>
- Select Calculate Selected Decks from the Deck menu

Deck Limitations

You should be aware of the following limitations when creating decks.

The waterline you select to define a deck must form a single continuous closed contour around the surfaces of the design. For best results you should be sure to define a transom surface when defining decks which extend to the stern of your design.

If you are using trimming, the deck shape will correspond to the trimmed shape of your design if trimming is turned on when calculating decks.

Skin thickness is not deducted from the deck's perimeter.

Working with Stringers

Stringers are defined using contours which lie in a surface of the design. As their name suggests, they are usually associated with stringers which run along the hull between frames however they can also be used to define other curves which lie in a surface such as longitudinals for engine beds, stiffeners on a plate or boundaries for a plate.

Working with stringers consists of the following:

- [Stringer Tips and Tricks](#)
- [Adding Stringers](#)
- [Generating Stringers](#)
- [Modifying Stringers](#)
- [Longitudinal Girders](#)
- [Adding Points To Stringers](#)
- [Moving Stringer Points](#)
- [Deleting Stringer Points](#)
- [Modifying Stringer Points](#)
- [Duplicating Stringers](#)
- [Breaking Stringers](#)
- [Joining Stringers](#)

- Mirroring Stringers
- Girth Centreline
- Calculating Stringers

Also see:

[Working with Stringer Shapes](#) on page 294
[Exporting Parts](#) on page 322.

Stringer Tips and Tricks

Before starting there are a few basic points to remember when working with stringers.

- When working with complicated surfaces, especially catamaran forms, hide all the surfaces apart from the one you are adding the stringers to. This will prevent the stringer point from being projected onto the wrong surface.
- When defining a stringer that goes across multiple surfaces, ensure that there is at least one stringer point on each surface.

Adding Stringers

You add stringers to your structure, either by drawing them in the Plan, Profile or Body Plan windows or by generating them automatically. A stringer is defined by a polyline of points, each of which lies in the surface of the design. When you add a point in one of the two dimensional views, Modeler-Structure will project the point through the surface and adjust the points coordinates so that it lies in the surface. Each stringer may lie wholly within one surface of the design, or it can run across a number of surfaces. You will usually find it easiest to have each stringer contained within one surface.

To add a stringer to a surface

- Choose Add Stringer from the Stringer menu

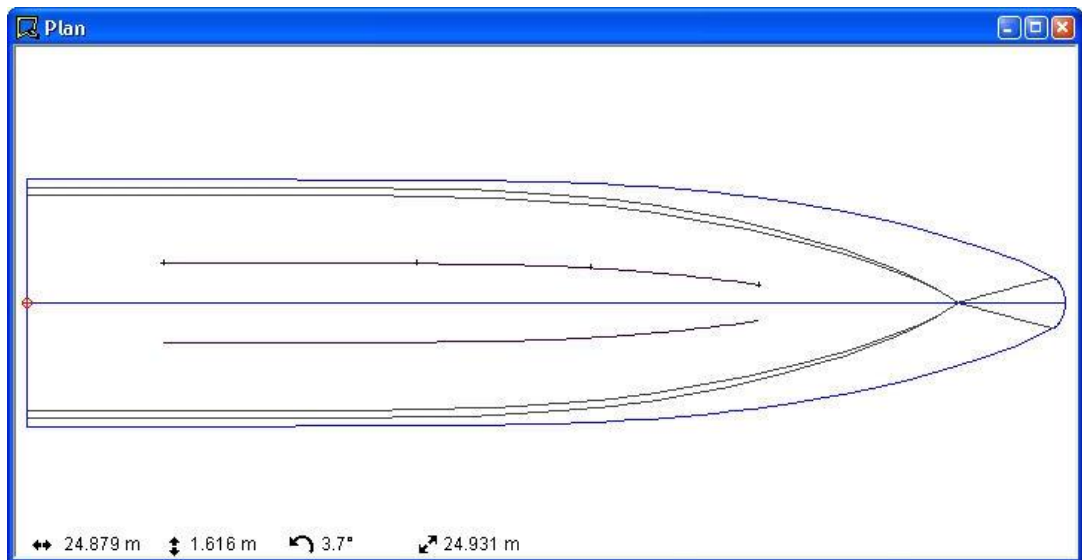
The cursor will change to a cross

- Click at the location of the first point in the stringer
- Click at the locations of each of the following points along the stringer
- Double click at the last point to finish adding the stringer.

If you wish to add a point, or entire stringer, on the port (negative) side of the centreline then hold the ctrl key down while placing the point. If this is not done then the point will automatically snap to the centreline of the vessel.

Alternatively you can press Return or Enter after adding the last point to your stringer or right-click and select OK.

As you add each point, Modeler-Structure will place a marker on the surface underneath the point you add. The stringer will be fitted as close as possible to these markers.



Immediately after you have added your stringer, Modeler-Structure will calculate an approximate fit of the stringer shape. This is just to give you an idea of the shape of the stringer and to give you a curve to select to set other attributes. You will still need to use the Calc Stringer, Calc Selected Stringer or Calc All Parts commands to accurately calculate the final stringer shape.

You can repeat the above process for each stringer you wish to add to the hull.

The display of stringers and points is controlled using the Structure... dialog from the Display menu. You can turn on and off the display of stringer points, the net between stringer points, the stringer curves and the numbers of stringers using the check boxes in this dialog.

Altering the shape of the stringer can be done graphically by dragging stinger points in one of the orthogonal views. Multiple stringer points can also be dragged.

Generating Stringers

Adding stringers by hand can be time consuming. Modeler-Structure provides a feature which allows you to automatically add stringers at a specified girth spacing around the hull.

To automatically generate stringers on a surface

- Choose Generate Stringers from the Stringer menu

A dialog will appear allowing you to enter the longitudinal and girth positions of the points on the stringer.

Generate Stringer Points

Number of Stringers: 8

Surface: TOPSIDES

Kind:
☐ 2 (linear) ☐ 4 (medium)
☒ 3 (flexible) ☐ 5 (stiff)

Longitudinal Spacing:

	Stringer	Start m	End m
1	1	0.999	21.00
2	2	0.999	21.00
3	3	0.999	21.00
4	4	0.999	21.00
5	5	0.999	21.00
6	6	0.999	21.00
7	7	0.999	21.00

Transverse Spacing:

	Frame	Long. Pos. m	No. Of Stringers	Start Girth m	Girth Spacing m	End Girth m	Total Girth m
1	st 1	1.000	8	0.342	0.342	2.735	3.077
2	st 2	2.000	8	0.342	0.342	2.735	3.077
3	st 3	3.000	8	0.342	0.342	2.734	3.076
4	st 2	4.000	8	0.342	0.342	2.733	3.074
5	st 3	5.000	8	0.341	0.341	2.730	3.071
6	st 4	6.000	8	0.344	0.344	2.727	3.068

Graph showing stringer positions along the hull.

- Type in the number of stringers you require
- Choose the stringer surface by choosing one of the names from the pop-up surface menu
- Type in the stringer start and end locations in the Start and End columns of the Longitudinal Spacing table
- If you wish you can specify the start girth, end girth and girth spacing by typing numbers into the Transverse Spacing table. These values will override the values automatically calculated by Modeler-Structure
- Select the type of stringer you require
- Click OK to generate the stringers along the hull.

Each time you make a change to the longitudinal position of a stringer, the Transverse Spacing table will be automatically updated to show you the girth positions which will result from the values you have entered.

These girth positions, which are generated automatically by Modeler-Structure, are displayed in the Transverse spacing table. These girth positions will also be displayed in the girth expansion drawing at the bottom of the window. The automatically generated stringer points will be calculated by evenly spacing around the girth of the surface, all of the stringers which pass through the frame at that position.

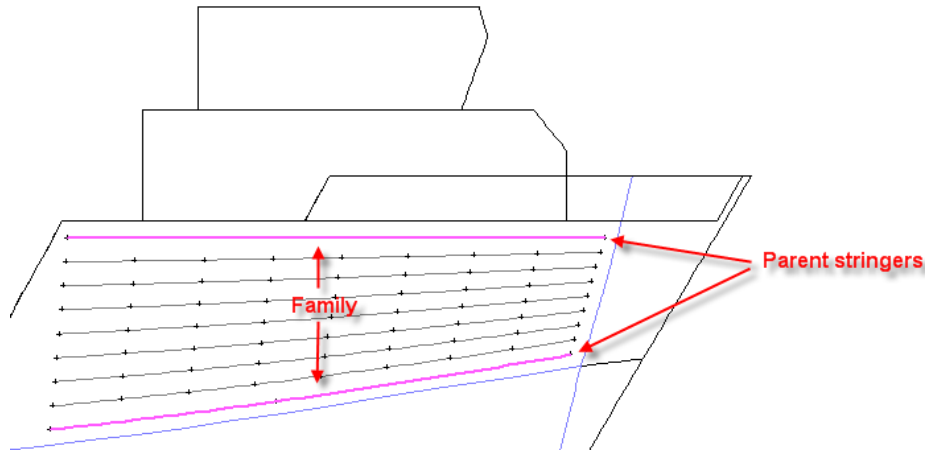
Note: You can use the Help button to get information on how to use the stringer generation dialog.

When entering positions, you can use mathematical expressions such as 12.5-2.5 and you can also use the names of frames as variables in place of their coordinates. For example, if you want a stringer to start 200mm aft of a frame names Fr4 you would enter Fr4-0.2 in the Start column.

Generating a Family of Stringers

Generating a Family of Stringers - Concepts

By selecting two stringers and choosing the "Generate Family..." command from the stringers menu you can generate a series of smoothly transitioning stringers between the two 'parent' stringers:



Generating a Family of Stringers - Procedures

- Select the first parent stringer
- Hold the shift or ctrl key down and select the second parent stringer
- Select Stringers | Generate Family...

When the command is invoked a dialog box will appear asking you how many stringers you would like to generate. This is the number of stringers that will be created between the two 'parent' stringers. 'Child' stringers will inherit the properties of the first of the parent stringers selected including shape, cutout, color, flexibility and orientation.

The command will only be enabled when two stringers are selected. The command does not work with two parent stringers that have stringer points with different projection directions. In this instance a dialog box will appear notifying you of this error.

Generating a family of stringers across multiple surfaces is possible.

Modifying Stringers

After you have added a stringer you can modify its surface or the stringer type. The stringer type may have a range of stiffness's in a similar manner to the way you can specify the stiffness for NURBS surfaces and curves.

To modify a stringer in the drawing windows

- Double click on the stringer

A dialog box will appear with the number of the stringer displayed.

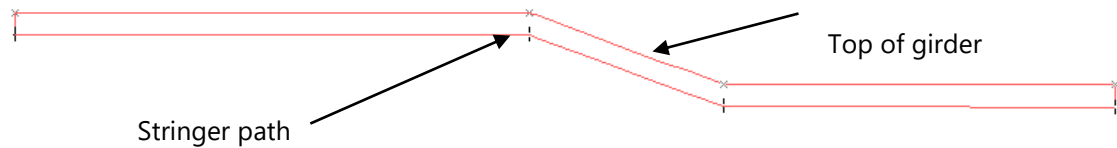
- Change the stringer type (Kind) by clicking one of the radio buttons.
- Change the Group, Shape or Cutout via the appropriate drop down list box. This will only be available if the stringer is not set as a Longitudinal Girder.
- Change whether the stringer is symmetrical by checking the “Is symmetrical” check box.
- Change the stringer stiffness by clicking on the appropriate radio button.
- Change the material of the stringer by selecting the Group and Material from the drop down lists.
- The stringer may be set as a longitudinal girder by checking the “Is longitudinal girder” check box and entering appropriate properties in the Long. Girder group of controls (see [Longitudinal Girders](#) below).
- Click the OK button to confirm your changes.

Longitudinal Girders

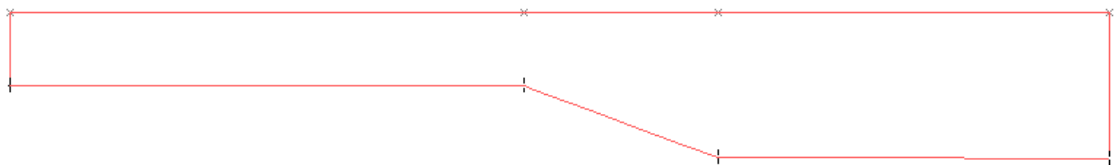
A longitudinal girder is a special type of stringer, mainly used to create engine beds and other longitudinal structure. A longitudinal girder can be created in three ways. Directly by choosing the Stringer | Add Longitudinal Girder (also found in the Stringer toolbar) or by setting the Is longitudinal girder check box to true in the Stringer Properties dialog (see above) or via the data sheet of a regular stringer. If a girder is created from the Stringer | Add Longitudinal Girder or Longitudinal girder button the stringer default Kind and Stiffness values will be set to 2 (Linear).

The projection direction is the direction in which the girder web will project from the base stringer path. By default this is set to most common direction of the stringer point’s projection directions. The Stiffness is the stiffness of the curve defining the top of the stringer path. By default this is set to the same as the Kind value of the underlying stringer.

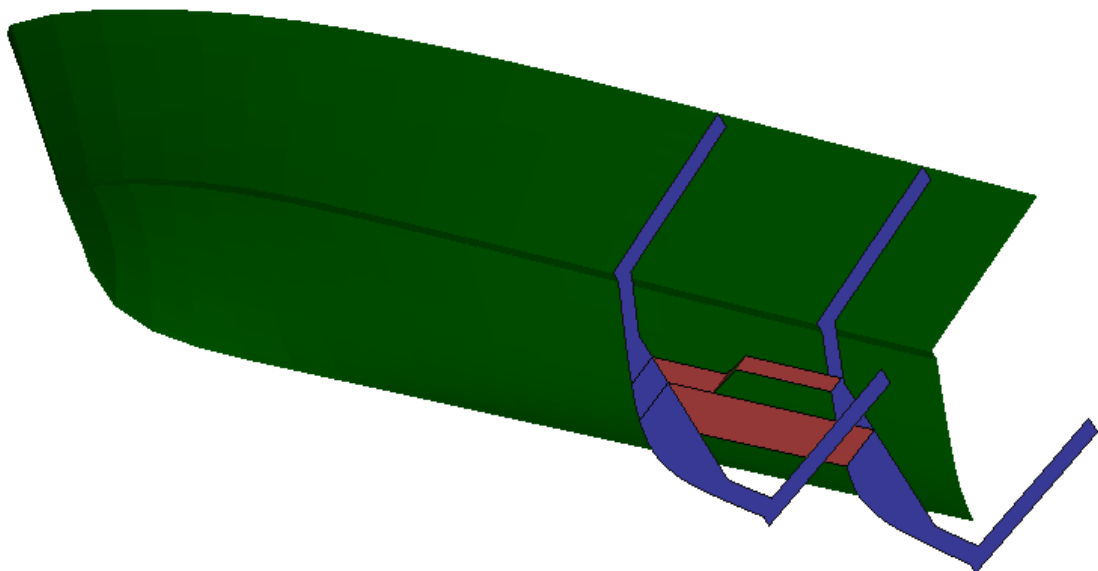
A longitudinal girder is defined by 2 curves, the usual stringer path, and a top curve which defines the top of the girder. When the stringer is set to be longitudinal then Modeler-Structure will produce an outline of the stringer perimeter similar to the following (shown in profile):



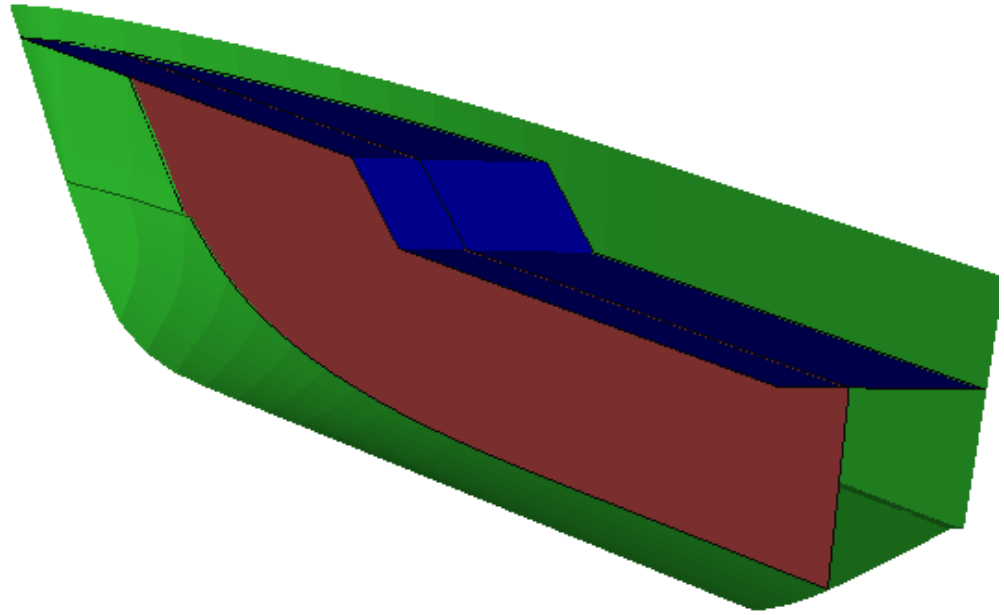
The shape of the top of the stringer may then be customised by the user. The marker points defining the top curve may be dragged by using the mouse in either of the orthogonal views or the position can be set precisely by double clicking on the marker or highlighting the marker(s) and choosing Stringer | Marker Properties from the Stringer menu. Movement of the marker point is restricted to the cranked girder's projection direction. Shown below is the girder created above with all the top points set to a common height:



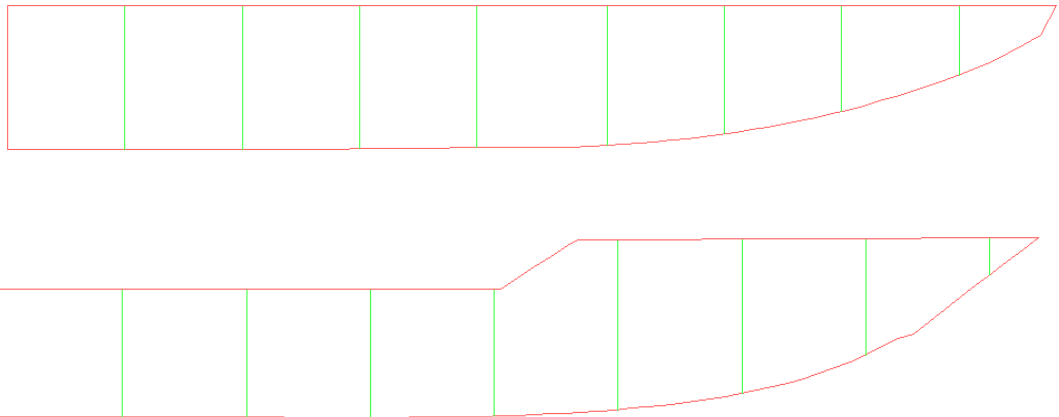
And in 3 dimensions two girders defining the engine bed are shown in red:



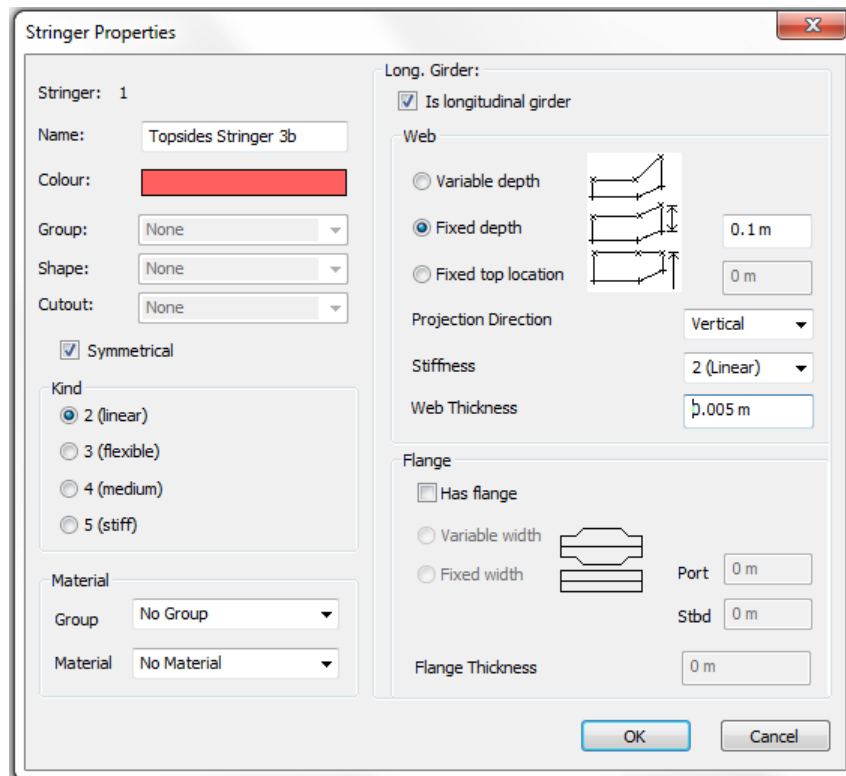
Girders can be projected in any of the three orthogonal directions. Thus generic structure can be created such as longitudinal bulkheads and stepped decks:



These parts may then be developed for production information:



Longitudinal girders are developed in the same way as normal stringers. Part, weight and centre of gravity information can be read from the data sheet as usual. Stringer material is set in Stringer Properties dialog box. The flange and web materials and thicknesses can be set independently of each other.



Stringer Properties

Stringer: 1

Name: Topsides Stringer 3b

Colour:

Group: None

Shape: None

Cutout: None

☒ Symmetrical

Kind

☒ 2 (linear)

☐ 3 (flexible)

☐ 4 (medium)

☐ 5 (stiff)

Material

Group: No Group

Material: No Material

Long. Girder:

☒ Is longitudinal girder

Web

☐ Variable depth

☒ Fixed depth

☐ Fixed top location

Projection Direction: Vertical

Stiffness: 2 (Linear)

Web Thickness: 0.005 m

Flange

☐ Has flange

☐ Variable width

☐ Fixed width

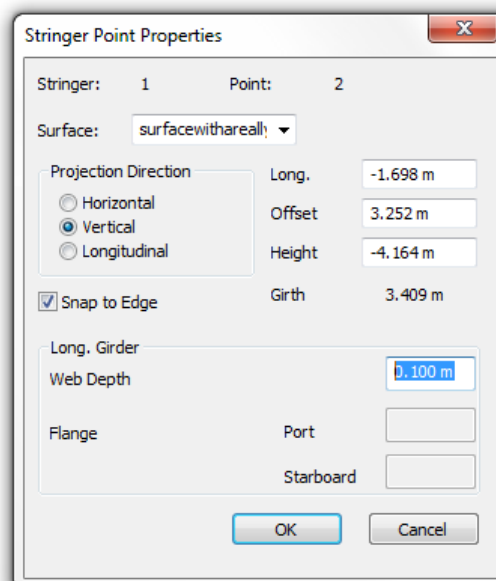
Port: 0 m

Stbd: 0 m

Flange Thickness: 0 m

OK Cancel

There are three options available for the type of Web the girder has. A variable depth girder means that the depth of the girder can vary along the length of the girder. If this option is chosen then the girder will appear in the drawing windows with a marker above each stringer point defining the top of the girder. This marker can be dragged by the user to set the depth at this location. The marker is constrained in all directions except the projection direction of the girder (for example a marker belonging to a girder that has vertical projection direction can only be altered in the z axis). Alternatively the marker position can be altered via its properties dialog box, accessed by double clicking on the marker. The web depth at each stringer point may also be set by double clicking on the stringer point and altering the web depth in the stringer point properties dialog:



Stringer Point Properties

Stringer: 1 Point: 2

Surface: surfacewithareall

Projection Direction

☐ Horizontal

☒ Vertical

☐ Longitudinal

☒ Snap to Edge

Long. Girder

Web Depth: 0.100 m

Flange

Port:

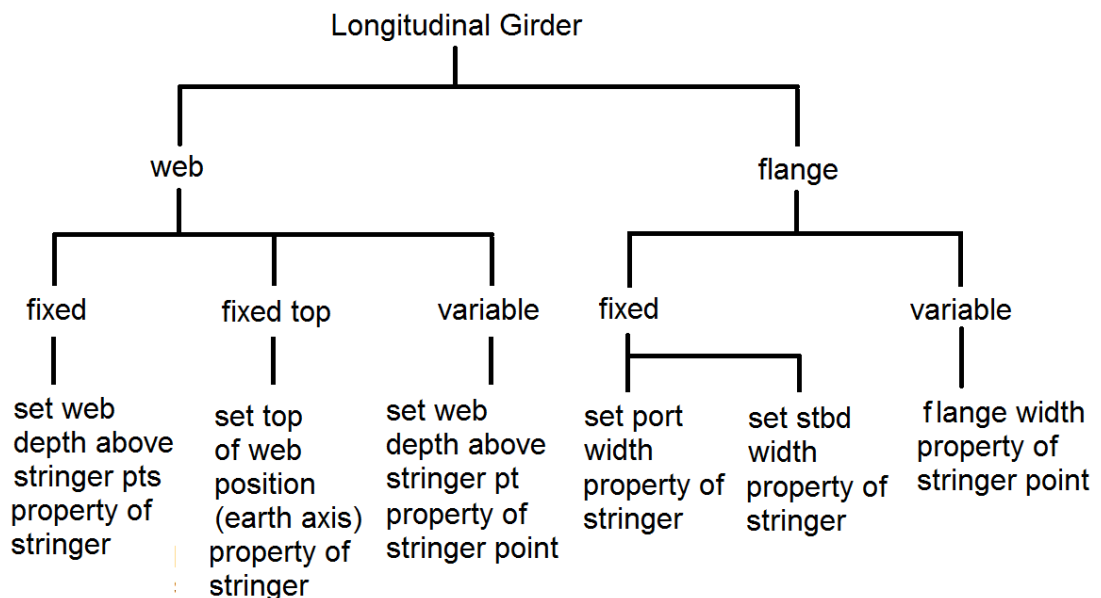
Starboard:

OK Cancel

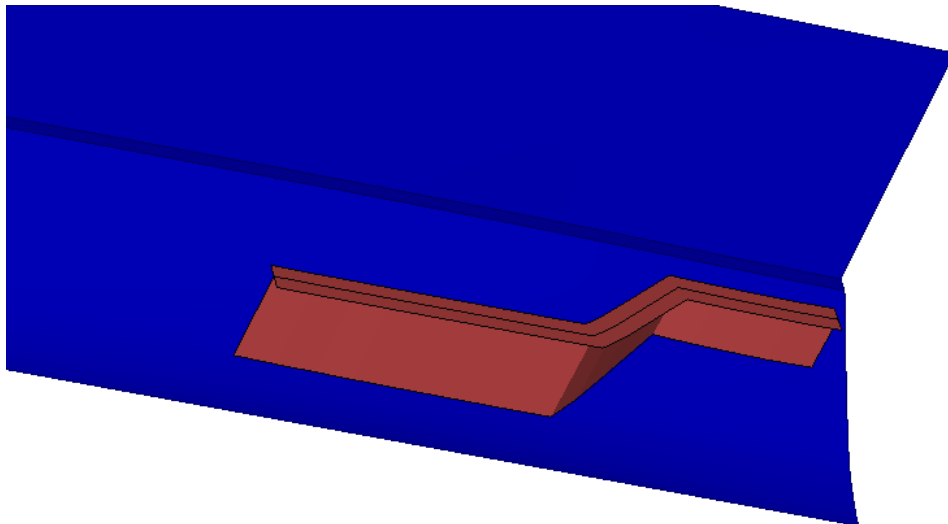
A fixed depth web means that the top of the girder is a fixed distance above the bottom at each stringer point location. This depth is defined in the fixed depth edit box. A web set to "Fixed top location" means

that the top of the girder is at a fixed height (in the case of a vertically projected girder) in space. This is useful for girders being used as engine beds when the top of the girder needs to be flat. The fixed top location is set in the edit box to the right of the "Fixed top location" radio button.

A flange may be added to the top of a longitudinal girder. The flange may be either "Variable width" or "Fixed width". If variable width is chosen then the width of the flange to port and starboard may be set at each individual stringer point (by altering the property of the stringer point in the Properties dialog or the Stringer Point data sheet). If fixed width is chosen then the width of the flange is set along the length of the girder according to the values set for port and starboard in the edit boxes.

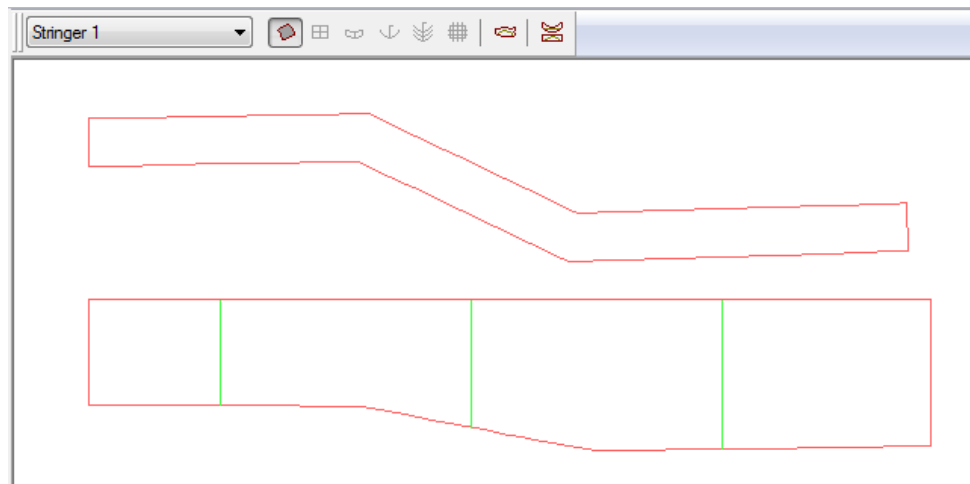


All of the girder data may also be edited/alterd in the Stringer and Stringer Point data sheets in the Data Window.



The above figure shows an engine bed girder with flange.

Flanges will be developed when the stringer is calculated. The developed flange is displayed at the top of the part window and the web in the bottom:



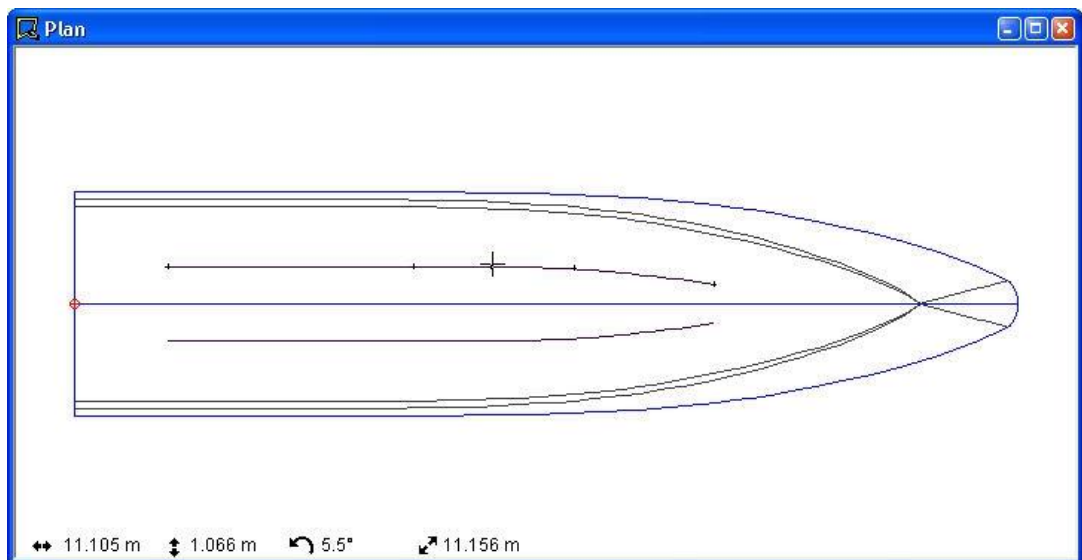
Adding Points To Stringers

If after adding or generating a stringer you find you need additional points in the stringers, you can insert them in the middle or add them at the ends of the stringer using the Add Point command from the Stringer menu.

To add a stringer point

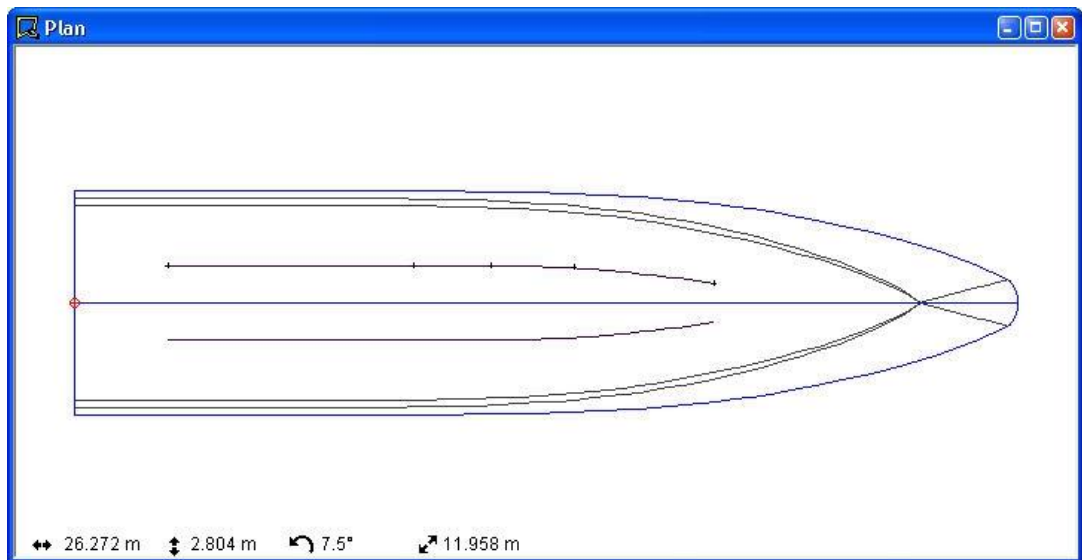
- Choose Add Point from the Stringer menu

The cursor will change to a cross and you can then click on the surface at the location of the stringer point. The point will be added to the closest stringer. You will need to use the Calc Stringer commands to calculate the actual stringer shape resulting from the addition of the point.



If you wish to add a point on the port (negative) side of the centreline then hold the ctrl key down while placing the point. If this is not done then the point will automatically snap to the centreline of the vessel.

You can repeat this procedure for as many stringer points as necessary until you have the number of points you require. As you add each point, a marker will be projected on the surface underneath the point you add. The stringer will be fitted as close as possible to these markers.



When the stringer point is projected onto a surface, it needs to know whether the point should be projected horizontally, vertically or longitudinally onto the surface. If you add a point in the Plan view the projection direction will be set to vertical, in the Profile view it will be set to horizontal and in the Body Plan view it will be set to Longitudinal. You can change the projection direction of a point by editing it in the Stringer Data window or by double clicking on the point and changing the setting in the Stringer Point dialog. It's a good idea to have all the points in a stringer use the same projection direction.

Moving Stringer Points

If you wish to move a stringer point after you have added it, you can either drag it in a graphics window, double click on it to change its properties and position or type in new coordinates in the Stringer Data window. You can also select stringer points and nudge them up and down, backwards and forwards using the corresponding arrow keys. This works in much the same way as the similar function for control points.

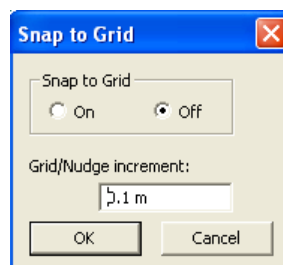
To move a stringer point in the Plan, Profile or Body Plan windows

- Point to the stringer point
- Press the mouse button and hold it down
- Move the mouse to drag the point to its new location

The coordinates of the point will be displayed in the bottom left hand corner of the window as you move it

- Release the mouse button

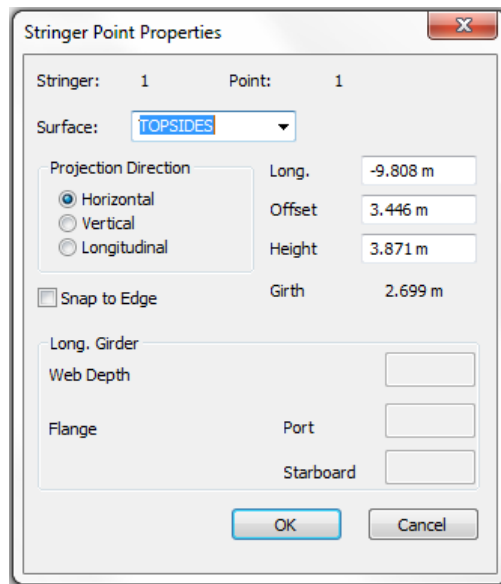
You can also nudge the location of stringer points by selecting the point and then using the up, down, left and right arrows keys to move the point. The distance of the movement is the same as the Design Grid setting set using the Snap to Grid command from the Stringer menu.



Alternatively, to change a stringer point in the Plan, Profile or Body Plan windows

- Double click on the stringer point

The stringer point details dialog will appear.



- Enter new values for the stringer point position or properties
- Click the OK button

Note : Changing properties of multiple stringer points

If multiple stringer points are chosen and the Stringer Point Properties dialog is activated fields that have the same value for all the selected stringers will be filled in and the remaining fields will be left blank. Any edited values will be applied to all selected stringer points. This behaviour is consistent with changing properties of multiple control points or markers in MODELER.

To move a stringer point numerically

- Bring the Stringer Data window to the front

	Stringer Number	Name	Assembly	Visible	Colour	Long. Pos. m	Offset m	Height m	Girth m	Surface Name	Projection Direction
1	1			✓		2.508	0.980	0.246	1.014	BOTTOM	Vert
2	1			✓		8.999	0.980	0.256	1.017	BOTTOM	Vert
3	1			✓		11.057	0.980	0.266	1.025	BOTTOM	Vert
4	1			✓		13.286	0.931	0.333	0.994	BOTTOM	Vert
5	1			✓		17.010	0.514	0.314	0.601	BOTTOM	Vert

- Click on the coordinate you wish to change
- Type in a new value
- Type Enter to confirm the new value

If you now move back to the graphics window you will see the stringer point has moved to its new position. You can cross reference between stringers in the table and stringers in the graphics views by displaying the stringer numbers in the graphics views using the Structure dialog from the Display menu.

You can change all of the stringer point coordinates in this window. You can also use the Copy, Paste, Fill Down and Fill Right commands in the table of stringer points. Remember that you can right click on the heading of any column to sort the contents of the table.

Note : snap to edge

When drawing a stringer the snap to edge property of each stringer point is off by default unless the point is placed within the snap tolerance of a surface edge. This also applies to adding a stringer point to an existing stringer. If a stringer point's snap to edge property is turned on then the stringer point will snap to the nearest visible point on the surface associated

with the stringer point. If a stringer point has snap to edge switched off but is dragged or created off the surface then it will move to the nearest point on the surface.

Deleting Stringer Points

If you wish to remove a stringer point or points, you first need to select the points.

To select a stringer point

- Click on the point

To select more stringer points

- Hold down the shift key while clicking on the points

To select a number of stringer points at once

- Drag a box surrounding the points to be selected

To delete a stringer point or points

In the drawing windows

- Select the point or points to be deleted.
- Choose Delete Point from the Stringer menu or hit the delete button.

In the Stringer table in the Data window

- Click and drag to select the rows of the points to be deleted
- Choose Delete Point from the Stringer menu

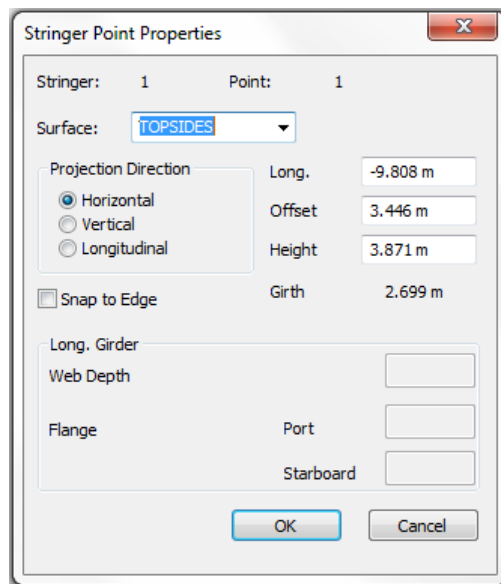
Modifying Stringer Points

When you add a stringer point, it is assumed you want to project the point perpendicular to the view you added it in. For example, if you add points in the Plan window, the points will be projected vertically on to the surface. If for some reason you want to change the projection direction of a point you can do this in either of two ways.

To modify a stringer point in the drawing windows

- Double click on the point

A dialog box will appear



The dialog box is titled "Stringer Point Properties" and contains the following fields and controls:

- Stringer:** 1
- Point:** 1
- Surface:** TOPSIDES (dropdown menu)
- Projection Direction:**
 - ☒ Horizontal
 - ☐ Vertical
 - ☐ Longitudinal
- Long.:** -9.808 m
- Offset:** 3.446 m
- Height:** 3.871 m
- Girth:** 2.699 m
- ☐ Snap to Edge
- Long. Girder:** (text field)
- Web Depth:** (text field)
- Flange:** (text field)
- Port:** (text field)
- Starboard:** (text field)
- Buttons:** OK, Cancel

- Choose the projection direction from the radio buttons

- If you wish you can also change the position of the point by entering new values for the Position, Height and/or Offset of the point
- Click the OK button

To modify a stringer point in the Stringer Data window

- Click in the Projection Direction column of the row of the points you want to modify
- Type in H for Horizontal, V for Vertical or L for Longitudinal
- Type Enter to confirm your change

If the stringer is a longitudinal girder then the web depth and flange widths may be set at the current stringer point. These options are only available if the longitudinal girder has a variable web depth and variable flange width (set in the stringer properties dialog).

A horizontal entry indicates that the point will be projected horizontally, this means you want it to be projected horizontally onto the surface. A Vert indicates that the point will be projected vertically onto the surface and indicates that you want it to be projected down onto the surface. A Long indicates that the point will be projected longitudinally onto the surface and indicates that you want it to be projected aft onto the surface.

You can change which surface the points will be projected onto by typing in the name of the surface in the Surface name column.

Note: If you want to change the values for a number of consecutive points, you can select the values to be changed and use the Fill Down command from the Edit menu or from the right-button mouse menu to set the rest of the points to the same value as the first point.

Duplicating Stringers

If you have a number of stringers which form a regular pattern on the hull, it may be easier to duplicate an existing stringer rather than drawing or generating stringers.

To duplicate a stringer or stringers

- Select the stringer or stringers to be duplicated
- Choose Duplicate Stringer from the Stringer menu

A dialog box will appear with fields for the number of stringers and their spacing

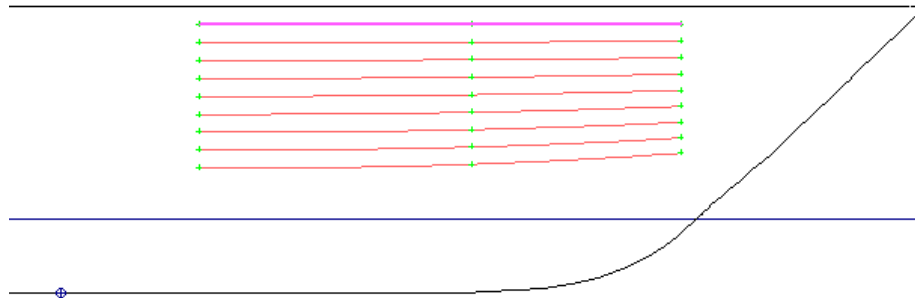


- Enter the number of duplicates and the spacing
- Click a radio button to indicate how the stringers are to be spaced
- Click the OK button

The spacing of the duplicated stringers may be different at the forward from the aft end of the stringer. This allows you to taper stringers or fan them out as the girth changes.

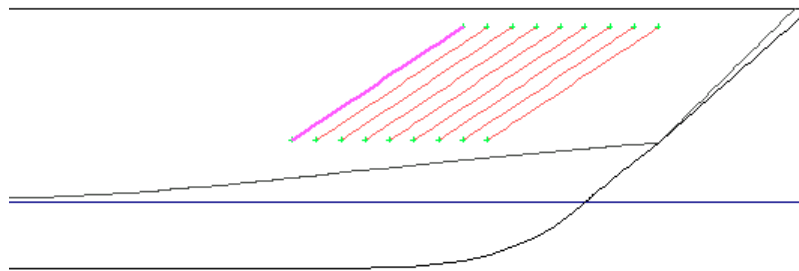
You can duplicate the stringer by position, offset, height (longitudinal, transverse and vertical directions respectively) or girth. You may enter a negative spacing to space in the reverse direction.

Duplicating by girth means the stringers will be spaced around the girth of the currently visible surfaces.



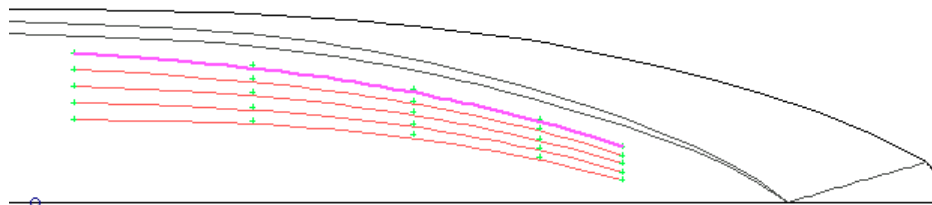
The highlighted stringer has been duplicated 8 times with a girth spacing of -0.3m at each end.

Duplicating by position means the stringers will be spaced in the longitudinal direction.



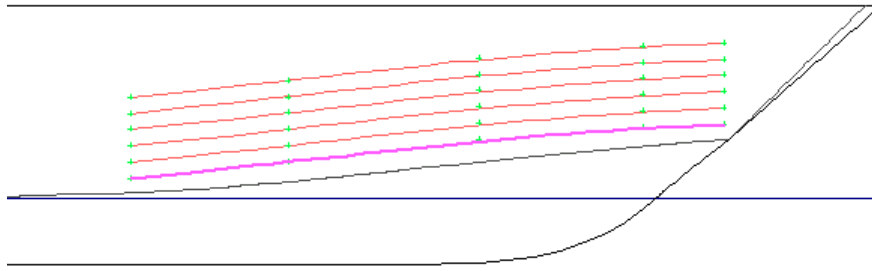
The highlighted stringer has been duplicated 8 times with a position spacing of 0.45m at each end.

Duplicating by offset means the stringers will be spaced in the transverse direction.



The highlighted stringer has been duplicated 4 times with a offset spacing of 0.3m at the aft end and 0.15m at the fwd end.

Duplicating by height means the stringers will be spaced in the vertical direction upwards from the original stringer.



The highlighted stringer has been duplicated 5 times with a height spacing of 0.3m at each end.

Breaking Stringers

If you have drawn a long stringer and then find you wish to break it into a number of smaller stringers, you can do so using the Break Stringer command from the Stringer menu.

To break a stringer or stringers

- Select the stringer points at the locations where you wish to break the stringer
- Choose Break Stringer from the Stringer menu

The stringers with selected points will be split at the selected locations.

In general you will want to break your stringers up into different zones along the hull. It is better to do this rather than trying to run stringers all the way from one end of the design to the other.

Joining Stringers

This command allows you to join two separate stringers to form one continuous stringer. This is the opposite of breaking a stringer. The procedure is to

- First select the end of the stringer you wish to remain stationary and, while holding down the shift key, select the end of the second stringer.
- From the menu choose Stringer | Join Stringers.

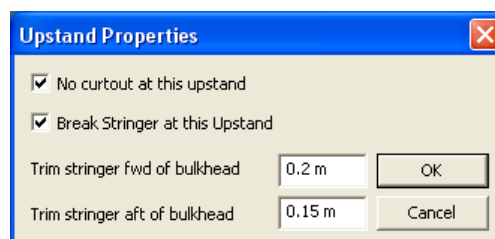
The end of the second stringer will move to the end of the first stringer and there will be strict tangency across the join. Stringer attributes (color, stiffness etc) of the new stringer will be the same as those of the stringer of the first selected point.

Breaking Stringers at Bulkheads

Stringers may be broken at any frame. The stringer can be trimmed a defined distance forward or aft of the frame position. An individual stringer may be broken at a frame or all stringers passing a frame may be broken.

To break an individual stringer at a single frame:

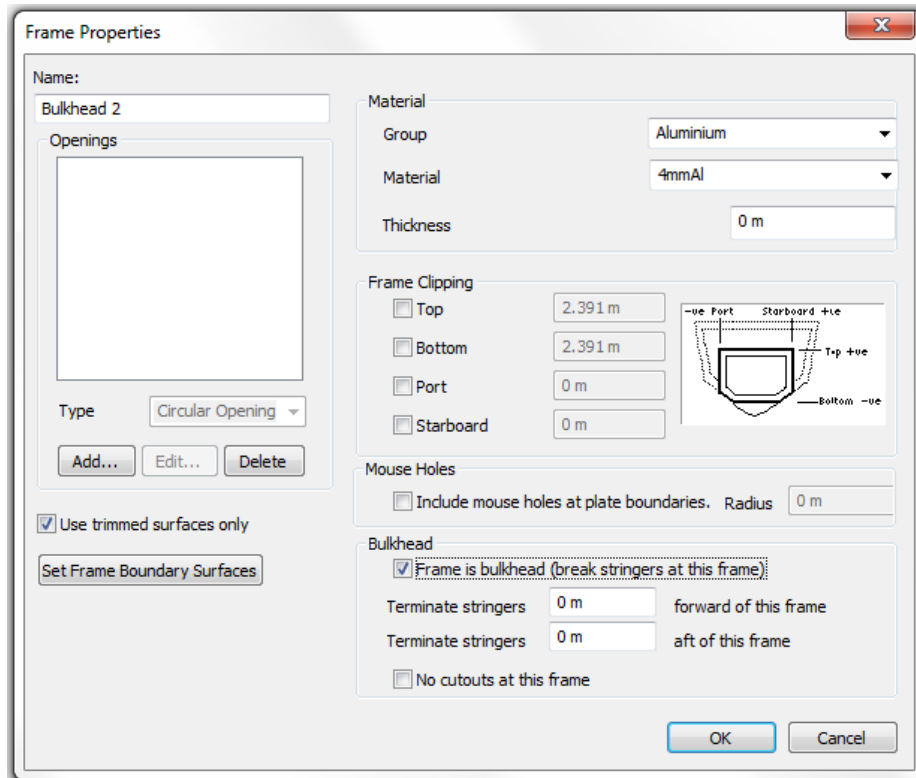
- Turn on upstand visibility.
- Double click on the upstand that defines the intersection of the frame and stringer (or select the upstand and choose Stringer | Upstand Properties...)



- Select “Break Stringer at this upstand” and set the fwd and aft spacing of the break. You can also set the upstand to have no cutout at this frame.
- Calculate stringer(s).

Alternatively you may wish to set an entire frame so that all stringers passing the frame are broken. This is done through the Frame Properties dialog box:

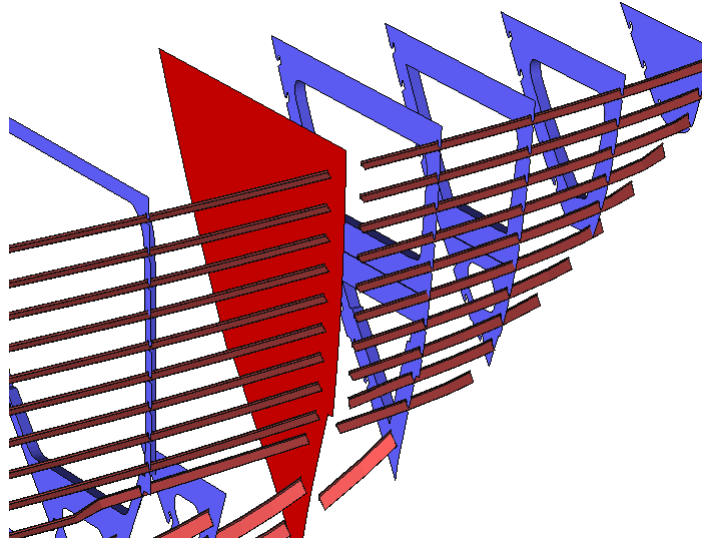
- Turn on frame visibility
- Double click on the frame of interest (or select the frame and choose Frame | Frame Properties...)



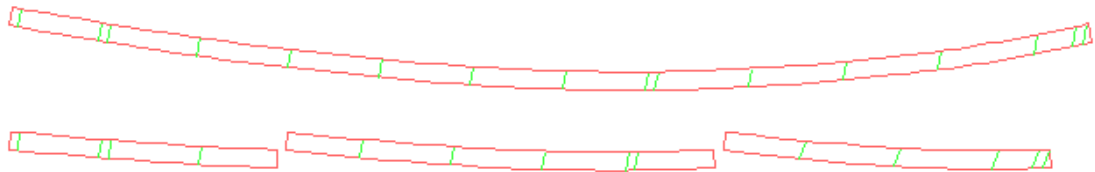
- Select “Frame is bulkhead” and set the fwd and aft spacing of the break. You can also set the frame to have no cutouts.
- Calculate stringer(s).

The break stringers at bulkhead variables may also be set in the Upstand data sheet.

The output is displayed in both the rendered view:



As well as in the expanded stringer view in the Part window. The stringer will be displayed twice. The full expanded stringer (unbroken) is displayed above and the expanded broken stringer is displayed below:

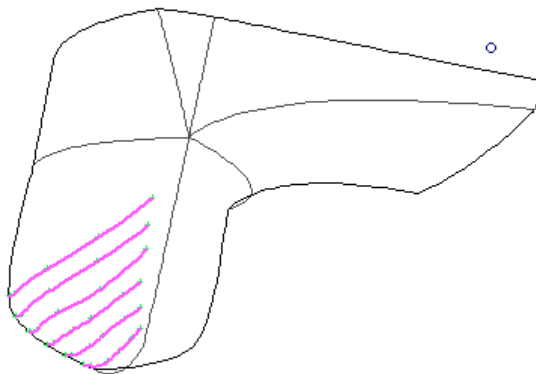


Mirroring Stringers

If you are designing a catamaran you may find it useful to use the Mirror Stringers command to duplicate the stringers on one side of the catamaran hull onto the other side of the hull. Mirroring can also be used to mirror longitudinally about a station or vertically about a particular height.

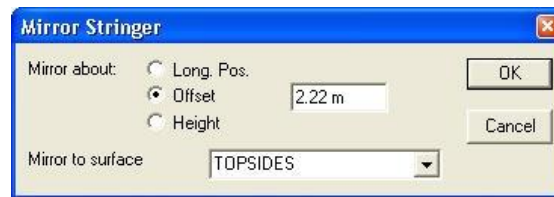
To mirror a stringer or stringers:

- Select the stringer or stringers to be mirrored



- Choose Mirror Stringer from the Stringer menu

A dialog box will appear with fields for the centre of mirroring, the direction of mirroring and the surface the stringer is to be mirrored on to



- Enter the position the stringer is to be mirrored about

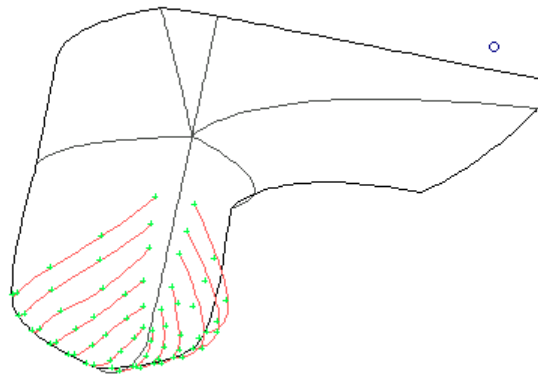
In the case of a catamaran this will be the offset of the demi-hull centreline

- Click a radio button to indicate how the stringers are to be mirrored
- Select the surface from the pop-up menu that the stringer is to be mirrored on to

In the case of a catamaran this will be the inner hull if the original stringers are on the outer hull and the outer hull if the stringers are originally on the inner hull

- Click the OK button

The stringers will be mirrored and placed on the selected surface



Also see:

[Working with Stringer Shapes](#) on page 294

[Stringer Orientation](#) on page 295

Girth Centreline

Catamaran designers must use the Girth Centreline function in the Stringer menu to specify the demi-hull centreline location. Girths are measured from the end of the section nearest this point.



Calculating Stringers

Once you have added the desired number of stringers, you can calculate the final shape of the stringers through the points. Each stringer will be generated as a smooth B-Spline curve which lies in the surface of the hull.

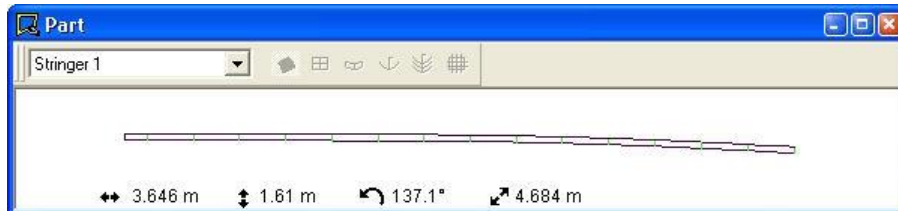
To calculate a stringer:

- Select the stringer
- Choose Calc Selected Stringer from the Stringer menu

To fit all of the stringers:

- Choose Calc All Stringers from the Stringer menu

The stringers you have generated will be displayed in the drawing windows. If you have specified a shape for the stringer, the plate shape of the web of the stringer will also be developed and displayed in the Part window. The height of the stringer is taken from the total height of the stringer shape you have selected. The stringer development is based on the height of the stringer shape you have specified.



Note : Stringer development orientation

The stringer development calculation does not take upstand orientation into account. This means that when a stringer has an upstand orientation that is different to the orientation of the stringer, this will not be taken into account during the development calculations. The Stringer orientation overrides the upstand orientation for development.

You can use the drop down menu at the top left of the Part window to choose which stringer to display.

The stringers generated will pass exactly through all of the points you have specified. The fitted curve is smoothed to ensure its longitudinal fairness and depending on the position of the stringer points you have entered.

The stringer curves can be copied, printed and plotted in the same way as the other contours. You can turn the display of these contours on and off with the Stringers check box using the Structure... command from the Display menu.

Note : warning – open stringer profile detected

When calculating a stringer a dialog may appear warning that an open stringer profile is detected. This means that the profile assigned to the stringer in the shapes library is not a closed contour. This means that it will not be able to calculate an enclosed volume for the weight of the stringer. If you wish to get an accurate weight for the stringer you will need to edit the shape of the stringer in the Shape Editor module.

When developing a stringer, the weight and centre of gravity of the stringer will be calculated provided its material properties are set. To do this, add a material to the shape associated with the stringer in the Shapes Library (Edit | Shapes Library). Modeler will then calculate the weight of the stringer based on the density of the assigned material multiplied by the shape cross sectional area multiplied by the length of the stringer. The density of the material is calculated from the material properties set in the Materials library (Edit | Materials Library). Density is derived from the thickness and t/sq.m values set for the material.

Calculating Stringer Inverse Bending Lines

Inverse Bending Lines – Concepts

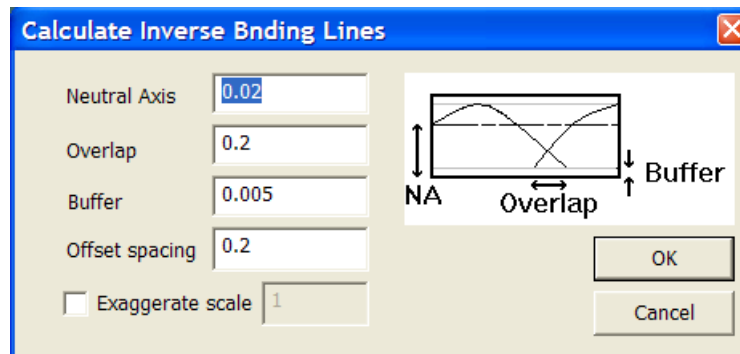
Stringer inverse bending lines are used to assist in the production of the correct shape of a stringer from a straight piece of stringer material (for example flat bar). To calculate the inverse bending lines select a stringer or stringers and choose Calc Inverse Bending Lines from the Stringer menu.

Note : Calculating stringer inverse bending lines

It is recommended to calculate the stringer at highest precision before calculating the inverse bending lines.

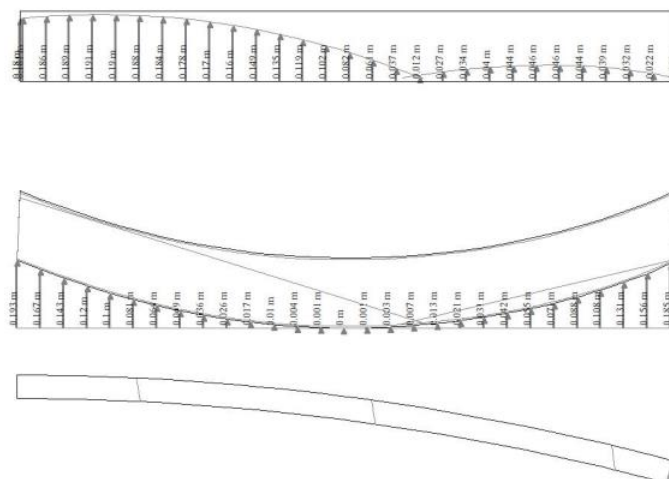
Inverse Bending Lines – Procedures

The following dialog box will appear:



The "Neutral Axis" dimension is the height from the base of the stringer section shape to the neutral axis of the stringer. The "Overlap" dimension is the distance that each of the subsequent inverse bending lines will overlap by. The "Buffer" line signifies the area inside which the inverse bending line contours will be drawn. The "Offset spacing" denotes the distance between offset readings of the inverse bending line dimensions in the display (see output diagram below). To better illustrate the inverse bending lines in the output the vertical scale may be exaggerated, to do this check the exaggerate scale check box and the desired scaling factor

If there are no currently selected stringers and the active view is the part view with a stringer displayed then the inverse bending lines for that stringer will be calculated. To display the Inverse Bending Lines, go to the parts window, select the stringer and click on the Inverse Bending Lines Button from the Parts toolbar: .



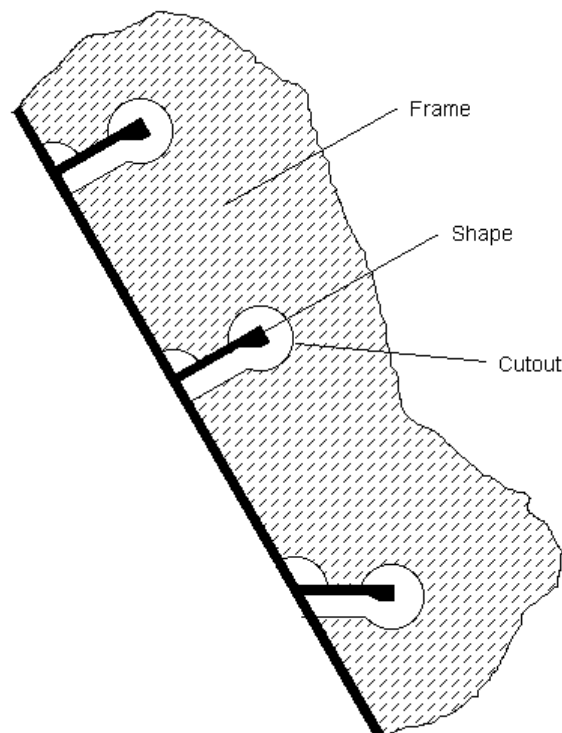
The bottom figure is the developed shape of the stringer, the middle figure is the shape of the stringer when the inverse bending lines are straight (i.e. the shape required after bending) and the top figure is the straight piece of stringer material prior to bending with the inverse bending lines drawn on (note the middle and top figures have a vertical exaggeration of 2.0). The measurements in grey show the dimensions from the baseline to the edge of the curved stringer (in the middle figure) and from the edge of the stringer to inverse bending line for the mark up of the stringer prior to bending. Dimensions are spaced according to the value entered in the dialog box. To turn off the inverse bending line diagram, click on the part button : .

Note: double curvature of stringers

Inverse bending line information is only useful if the stringer bending is planar (i.e. in one plane). In practice, however, a small amount of double curvature in the stringer will not invalidate the inverse bending line data. For this reason you can calculate the inverse bending lines of the stringer even if it has double curvature. This is done by first projecting the stringer shape onto the "closest" arbitrary plane. All the inverse bending line calculations are then done on this projected shape. A good way to check how much double curvature is in the stringer is to compare the stringer shapes on the expanded stringer (bottom figure) and the projected stringer (middle figure) in the inverse bending line expansion drawing. For a stringer with no double curvature the shapes will be the same.

Working with Stringer Shapes

In order to calculate the cutout when a stringer passes through a frame, it is necessary for you to specify what shape the stringer is and what cutout shape is to be used to make the opening for the stringer to pass through each frame.



Working with stringer shapes consists of the following:

- Sections Library
- Setting Shapes & Cutouts
- Stringer Orientation

Also see:

[Working with Stringers](#) on page 272

Sections Library

Modeler-Structure has a library of shapes and cutouts. You can add, delete and edit standard sections in the library using the Shape Editor application that is shipped with Modeler. For more information please refer to the Shape Editor Manual.

Setting Shapes & Cutouts

When a stringer passes through a frame, a cutout to allow the stringer to pass through the frame will be inserted. You specify which stringer shape and cutout to use for a stringer by using the Shape & Cutout command from the Stringer menu.

To set the shape and cutout for stringer or stringers

- Select the stringers
- Choose Shape and Cutout from the Stringer menu

A dialog will appear listing the shape and cutouts you have added to the Sections Library

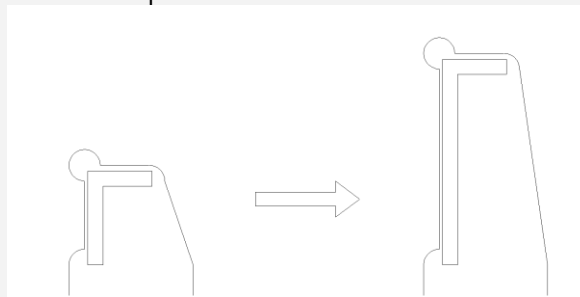
- Click on the name of the group you wish to use
- Click on the name of the shape you wish to use
- Click on the name of the cutout you wish to use
- Click OK

The next time that you calculate frames, the cutouts you specify will be deducted from any frames that the selected stringers pass through.

When choosing cutouts you may specify that the stringer has no cutout by choosing "None" from the Cutout list. In this case, no cutouts will be inserted for the selected stringers.

Note:

When a stringer intersects a frame at an angle to the horizontal the cutout size will need to be increased in order for the stringer to pass through the frame. This 'stretching' is done by extending the midsection of the cutout by a distance proportional to one on cosine of the angle the upstand makes with the frame plane:



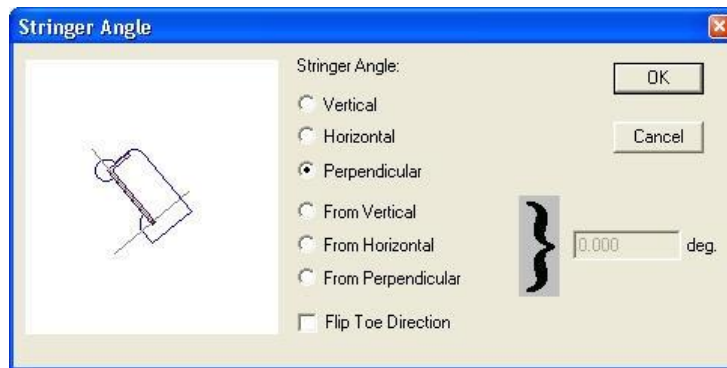
Stringer Orientation

When a stringer is run down the hull, you need to specify the angle between the stringer and the surface at each frame. This allows Modeler to calculate the orientation of any cutouts which may need to be made in the frames to accommodate the stringer. The default orientation is to have the stringer perpendicular to the surface. If you want to change this orientation you can do so using the Orientation command from the Stringer menu.

To set the orientation of a stringer or stringers

- Select the stringers you want to change
- Choose Orientation... from the Stringer menu

A dialog box will appear showing the orientation options.



- Click on one of the radio buttons to select your orientation

Either:

- Type in an angle for the orientation

Or:

- Click and drag on the stringer diagram to rotate it to the desired location
- Click OK to set the orientation of the selected stringers

Changing Stringer Toe Direction

By default the Section Library is set up so that asymmetrical stringers (for example L sections) are inserted with their toe facing towards the centreline of the vessel. If this is not desired you may flip the stringer so that the toe direction is pointing outboard. To do this highlight the stringer(s) you wish to flip and activate the Stringer Angle dialog box (Stringer | Orientation), check the "Flip Toe Direction" box and click on OK.



Working with Upstands

At each point where a stringer crosses a frame, an upstand is generated. Upstands are used to override the stringer orientation and cutout shape. An upstand is a short line that runs from the intersection of a stringer and a frame in the orientation-direction of the stringer.

The orientation of the upstand is used to calculate the shape and location of the cutouts through the frames. The upstand defines the following:

- location of the intersection of stringer and frame (base point of the upstand)
- orientation of the stringer
- stringer and cutout shape

Changing these properties allows you to specify different properties for each frame. Information on how to edit any of the upstands properties is given below.

- [Calculating Upstands](#)
- [Upstand Cutouts](#)

- [Upright Orientation](#)

Calculating Upstands

To calculate the position and orientation of the upstands

- Choose Calc Upstands from the Stringer menu

The upstands will be generated and displayed as a short line in the direction of the stringer orientation at the intersection between each stringer and each frame in the design.

If Upstands are not visible, choose the Structure command from the Display menu and turn on the Upstands box.

Note that to ensure that the direction of the upstands is correct relative to the surface, you should check the direction of the surface [Check Outside Arrows](#) (see page 257 of this manual). The arrow should point in the opposite direction as the upstands.

Upright Cutouts

You can change the stringer cutout at any frame by selecting the upstand at that frame and using the Shape and Cutout command from the Stringer menu. At this frame, this will override the cutout specified for the stringer. You can set the cutout back to the default for the stringer by choosing Default in the Cutout list in the Shape & Orientation dialog.

Upright Orientation

In the same way as cutouts can vary from frame to frame, stringer orientation may also be varied by setting the orientation of the upstand.

You can change the orientation of the cutout at any upstand by selecting the upstand and using the Orientation command from the Stringer menu. This will override the orientation specified for the stringer.

Note:

Changing the shape and orientation properties for a particular frame does not change the stringer's properties. This means that, in theory, it is possible to go from horizontal stringer orientation to vertical and back again between three frames.

You can set the orientation back to the default for the stringer by typing a "-" (minus sign) in the last column of the stringer table in the Data window.

Working with Inclined Sections

Arbitrarily, longitudinally-inclined sections may be defined. These are contours defined by the intersection of an arbitrarily inclined plane with the surface. The arbitrarily inclined plane is defined by three points which lie in the plane. Inclined sections can be used to define inclined frames and can also be used as plate boundaries.

Working with inclined sections consists of the following:

- Adding Inclined Sections
- Displaying Inclined Sections

Also see:

[Working with Frames](#) on page 258

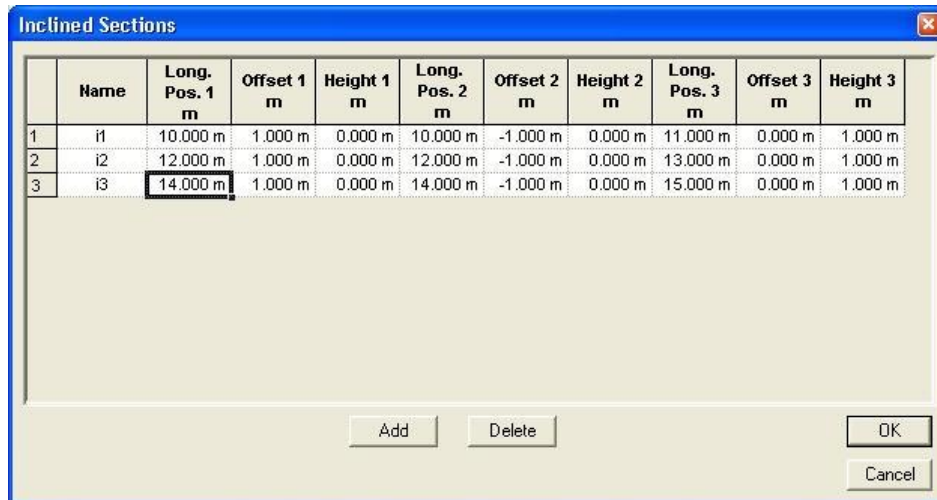
[Working with Plates](#) on page 299

Adding Inclined Sections

To add, edit or delete inclined sections

- Choose Inclined Sections... from the Data menu

A dialog will appear allowing you to add, edit or delete sections



To add an inclined section:

- Click on the Add button

To edit an inclined section:

- Click on the coordinates to be changed and type in new values

To delete an inclined section:

- Click in the row of the section to be deleted and click the Delete button

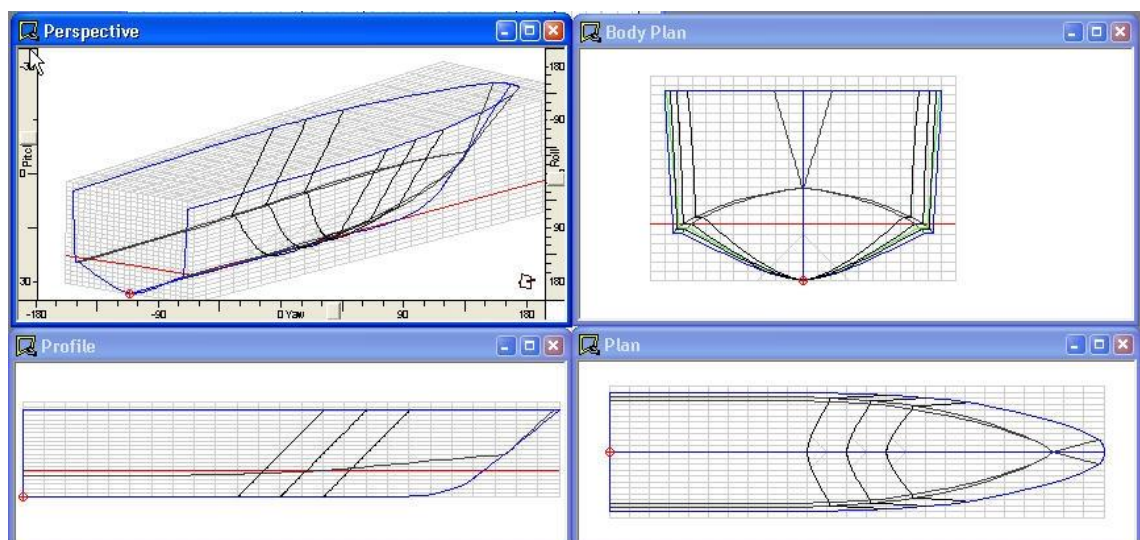
To confirm your changes to inclined sections

- Click the OK button

Displaying Inclined Sections

You can display both the grid of planes defining the inclined sections and the inclined sections themselves.

The three point planes defining each of the inclined sections will be displayed whenever you have the grid turned on.



To display the inclined sections:

- Choose Contours... from the Display menu
- Click the Inclined Sections check box to turn it on
- Click the OK button

The inclined sections will be displayed in all the views and can be copied, printed and plotted in the same way as other contours.

Working with Plates

Working with Plates - Concepts

Modeler-Structure allows you to expand a three dimensional hull surface into its equivalent two dimensional expanded shape. This provides the basis for the template you would need to cut out a plate in order to construct the surface. You can create plates which cover an entire surface or cover just a part of a surface. You can define a plate on part of a surface as the area between four selected contours.

Working with Plates – Procedures

Working with plates consists of the following:

- Plating a Whole Surface
- Adding a Plate
- Deleting a Plate
- Defining Triangular Plates
- Plate Development Method
- Calculating Plates
- Girth Differences
- Plate Strain
- Plate Development and Rolling Jigs
- Plate User Coordinate Systems (UCS)
- Plate Inset Lines
- Plate Templates
- Plate Pin Jigs
- Shell Expansion
- Plate Displays

See also Exporting Parts on page 322.

Plating a Whole Surface

To add a plate that covers an entire surface:

- Choose a surface from the Add Surface Plate sub-menu under the Plate menu
- Type in a name for the Plate
- Select a material for the plate from the pop-up menu
- Select a [Plate development method](#) for the plate from the pop-up menu. If necessary specify any additional information for the plate development
- Click OK

The plate will be displayed in the drawing windows and it will be selected. Plates are highlighted when they are selected by inverting the interior of the plate with a grey pattern. You can control the visibility of plates by using the Plate check box in the Structure... dialog from the Display menu.

Note:

Switching to a lower precision significantly speeds up the plate development calculations, especially if you are using the General development method. However, you should still define

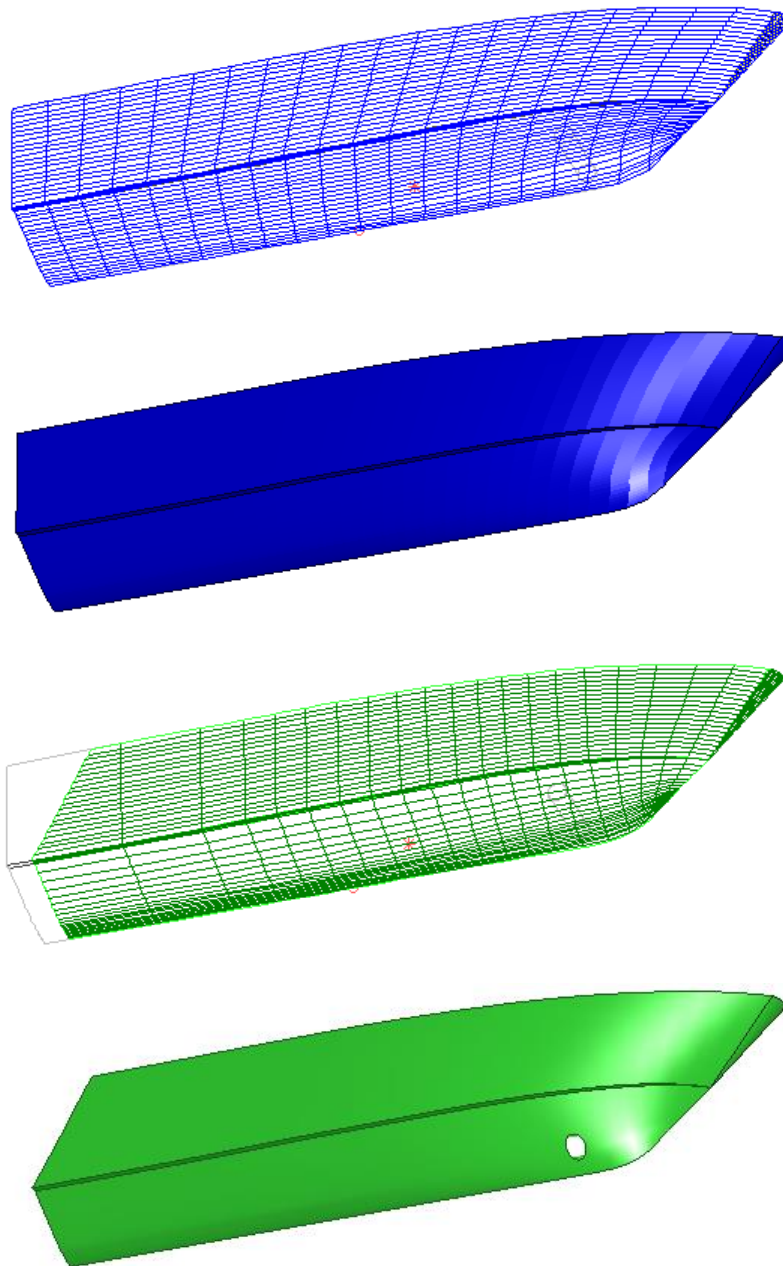
the plate at highest precision and develop the final plate at highest precision before exporting.

See [Plate Precision on page 310](#)

Developing Trimmed Surfaces

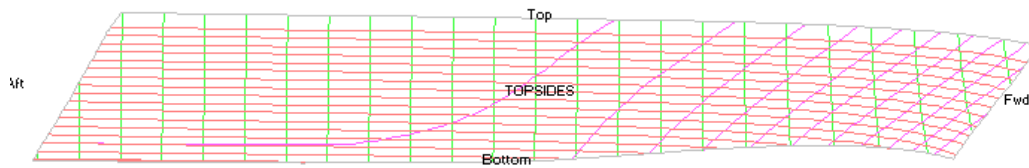
When a plate is added on a surface via the Plate | Add Surface Plate menu item, the plate will obey the trimming of the surface. If trimming is switched on the plate boundary will be the same as the boundary of the untrimmed region(s) of the surface. If trimming is switched off then the plate will be added on the entire surface.

Below is an example of the workboat sample design with surface plates added with trimming turned off (blue) and trimming turned on (green)



The developed plate of a trimmed surface will show only the untrimmed region and associated contours:

Girth Aft Length = 3.59 m Difference = 0 m
 Girth Fwd Length = 3.449 m Difference = 0 m
 Girth Top Length = 25.218 m Difference = 0 m
 Girth Bottom Length = 23.147 m Difference = 0 m
 Max Girth Diff. from Aft to Fwd = -0.013 m
 Max Girth Diff. from Top to Bottom = -0.004 m
 Plate Area = 147.023 m²
 Method = General (Longest plate axis 100%, Preserve plate edge lengths)



Definition of triangular trimmed plates is described in [Defining Triangular Plates](#) on page 304.

Plates Expansion Mirrored Across Ship Centreline

Expanded plates may be mirrored across the centreline for plates that stretch across the centreline such as bow cones, deck plates and transoms etc.

Plates may be defined so that they are mirrored port and starboard across the ship centreline. To do this, the plate must have at least one edge on the ship centreline; in the case of several edges lying on the centreline, the longest edge will be taken as the mirroring axis. To define a mirrored plate, select the plate edges as normal (but only on one side of the vessel) – or choose Add Surface Plate from the Plate menu – and tick the “Mirror plate expansion across centreline” option at the bottom of the dialog.

Plate Properties

Plate Name:

Material

Group:

Material:

Thickness:

Method:

Forming direction

☒ Longest plate axis
☐ Ideal
☐ Shortest plate axis
☐ Ship longitudinal axis
☐ Ship transverse axis

Strain Ratio: % 100% isotropic
0% uni-directional

Forming process

☒ Preserve plate edge lengths
☐ Expand only (e.g. rolling)
☐ Shrink only (e.g. line heating)

Mesh

☒ NURB
☐ Cartesian (use if surf parametrics skewed)

Precision

☒ Automatic Seam Direction:
☐ Manual Butt Direction:

☐ Smooth plate edges
☐ Mirror plate expansion across centreline

OK Cancel

Note that when a plate is first defined, the mirror option will always be enabled, even if there is no suitable edge for mirroring. Once the plate is defined, however, this option will not be available if the plate does not have an edge on the vessel centreline.

Adding a Plate

You can add a plate which covers just part of a surface by selecting four contours (for a quad plate) or three contours (for a triangular plate) defining the boundary of the plate.

To select a contour:

- Click on the contour

To select additional contours:

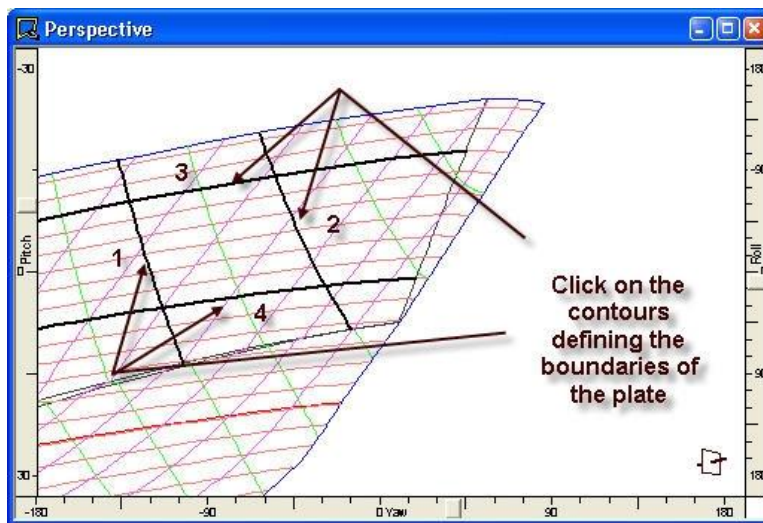
- Hold down the shift key while clicking on each contour

Selected contours are highlighted with a bold line to indicate that they are selected.

All of the four (or three) contours defining the plate must lie in the same surface. You may find that it is necessary to turn off the display of other surfaces to allow you to select edges which butt up against another surface.

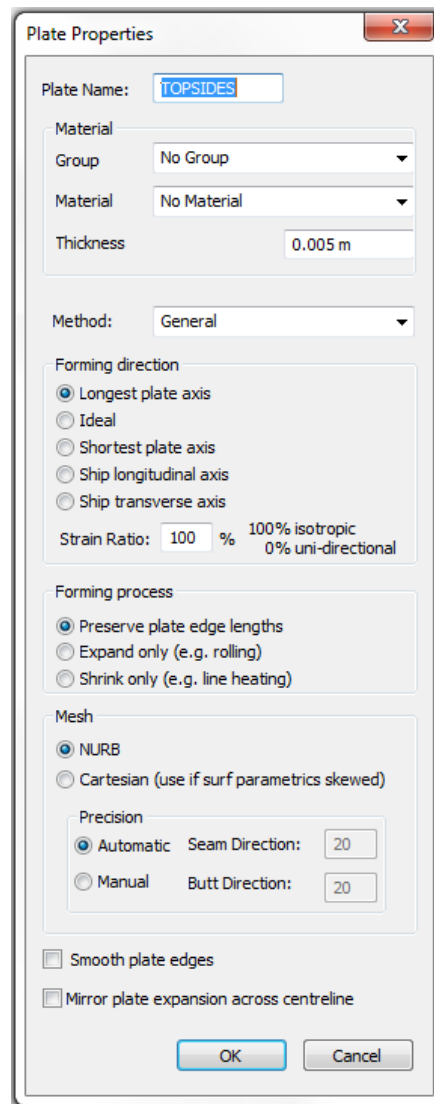
To add a plate:

- Select the four contours which define the boundary of the plate
- These should be selected in the order aft butt, forward butt, top/outer seam, bottom/inner seam



- Choose Add Plate from the Plate menu

The Plate Info dialog will appear allowing you to specify the plate's name, material and the development method to be used.



- Type in a name for the Plate
- Select a material for the plate from the group and material drop-down menus.
If you do not have the material you require available, use the Add Material command from the Edit Material menu to add materials to the library.
- Select the plate expansion method
See [Plate Development Methods and Options](#) on page 305 for details of the different methods; if you have chosen the General method, select the additional options you require.
- Click OK

The plate precision may be manually specified or automatic. If you choose Automatic, the actual mesh precision used will depend on your precision settings at the time that the plate is calculated.

The default plate material and thickness will be set to the material and thickness of the underlying surface.

When adding a plate a dialog may appear stating

"Multiple closed regions found. Structure is unsure which region you wish to create the plate on. Would you like to continue?".

This means that 4 contours selected by the user create more than one closed region on the surface. In this case Modeler is unsure which region the user wishes to create the plate on. So this does not occur ensure that each plate edge contour only intersect each other contour once. This may require using "dummy"

stringers to define the plate edges. Dummy stringers are stringers inserted purely for defining the plate edge and may be deleted once the plate has been formed.

You will also be warned when the calculated plate mesh deviates more than 2 mm from the selected plate contours or edges and then turns the Plate Mesh visibility on for the user to check the plate mesh.

Note : Plate mesh problem detected

After opening a design or adding a plate an error message entitled "Plate mesh problem detected in plate:" may appear. This indicates that one of the plate mesh edges in the named plate deviates from the plate edge by more than the build tolerance of 2mm. When this happens Modeler automatically turns on the plate mesh display for the user to review. The plate mesh is the mesh upon which the development is based, if it is incorrect the plate development will be incorrect. A faulty plate mesh is usually caused by either trying to add a triangular plate (see [developing triangular plates](#)) or adding a plate using 4 contours which define more than one closed region.

Also see: [Plate Precision](#) on page 310

Deleting a Plate

If you wish to remove a plate, you first need to select it.

To select a plate

- Click on the plate

To select a number of plates at once

- Drag a box surrounding the plates to be selected

To delete a plate once it is selected in one of the drawing windows

- Choose Delete Plate from the Plate menu or hit the delete button

In the Plate table in the Data window

- Click and drag to select the rows of the plates to be deleted

Choose Delete Plate from the Plate menu

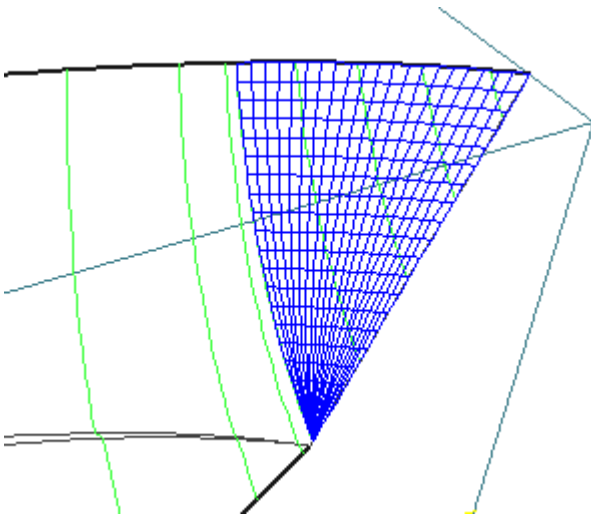
Defining Triangular Plates

There are two scenarios that may cause triangular plates:

- The surface itself is triangular
- The surface is trimmed to a triangular shape

Surface itself is triangular

When two corner control points have been compacted in a triangular surface has been created. Develop this surface by selecting "Add Surface Plate" from the plate menu. Due to the development algorithm being based on quad mesh elements the Triangular plate is approximated by a Quad mesh with one of the Apexes being approximated by a very small edge. You can see this by turning on the plate meshes for the triangular plate and zooming in on the Apex:



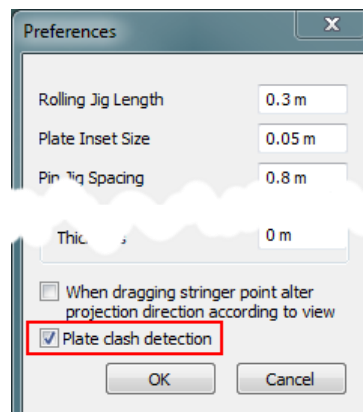
The collapsed edge is formed at the apex between the 1st and second contours selected.

Trimmed surfaces in a triangular shape

A triangular plate may also be created by selecting boundary contours, in a similar fashion to a quad plate, but by selecting 3 boundary contours. The tri plate is approximated by the quad mesh in a similar way as for the surface triangular plate explained above.

Plate clash detection

A warning dialog will now be displayed when two plates created on the same surface overlap. Plate clash detection can be turned on and off from the preference dialog (Edit | Preferences...).



Structure will warn which plates are overlapped by the newly correctly plate:

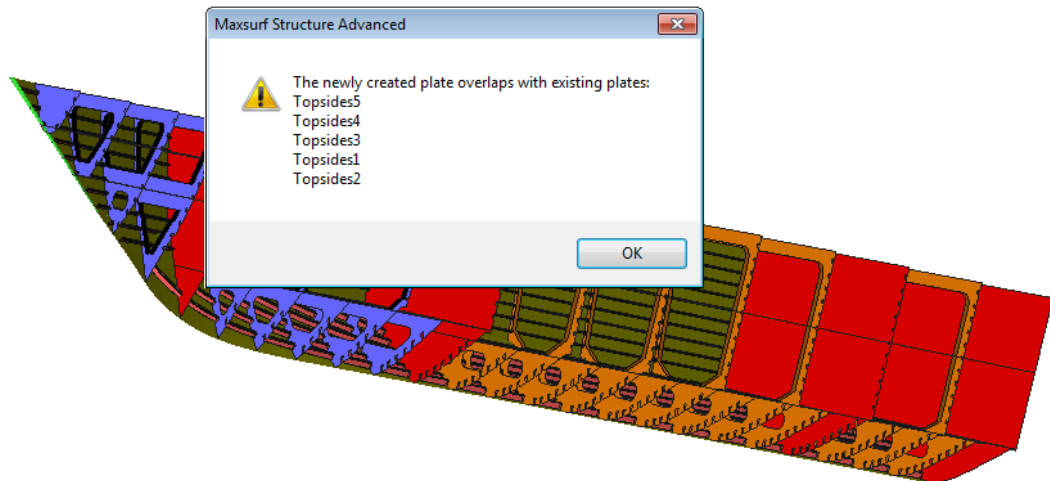


Plate Development Methods and Options

There are three different plate expansion methods which you can choose from. In most cases the General method should be used, however, there may be times when either of the other two methods may be more suitable. Details of the different methods are given below:

General Method

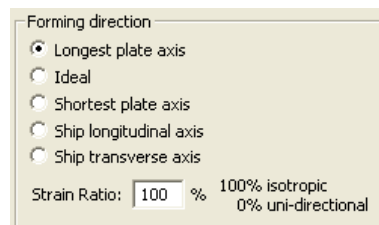
The General method is the most versatile and is designed to work with plates which have either positive, negative or zero Gaussian curvature. A greater degree of control over the development is available with this algorithm allowing you to match the development to your plate forming machinery.

Positive Gaussian curvature occurs in plates with double curvature in the same direction, either inwards or outwards so that the plate is purely convex or concave. Negative Gaussian curvature occurs in plates which are twisted or saddle shaped. Zero Gaussian curvature plates are developable and can be developed with requiring any stretching or shrinking.

The [General Method - Forming Direction](#), [General Method - Strain Ratio](#) and [General Method - Forming Process](#) items in this dialog will only be available if the General method is selected.

General Method - Forming Direction

This option is used to specify the direction in which the plate is to be rolled; this is the direction in which the maximum strain will be put into the plate.



The Ideal option will try to form the plate in the direction of minimum principal curvature. This is calculated as weighted average over the whole plate. Theoretically, forming in this direction will require the minimum amount of plate stretching or shrinking. Due to constraints of plate forming machinery or other production reasons, you may wish to expand in a different direction.

To give you this control, you are able to specify either the plate's longest axis or its shortest axis directions. (These directions are local to the plate and do not necessarily correspond to the global ship axes.) Note that

if you select Ideal, the forming direction will snap to the plate's longest or shortest axis if the direction of minimum principal curvature is within 10 degrees of these directions.

The forming directions "Ship longitudinal axis" and "Ship transverse axis" are useful for mirrored plates when you wish the expansion direction to be parallel or perpendicular to the mirrored edge; otherwise the expansion direction will also be mirrored for the two halves of the plate.

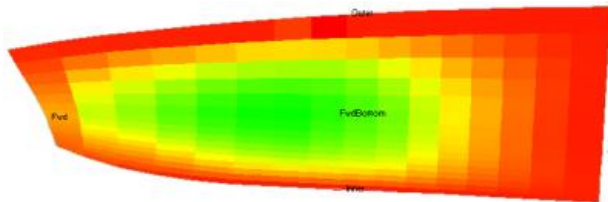
Note that the forming direction will only affect the result if the strain ratio (see below) is less than 100%. Also note that if the plate is being formed by line heating, the maximum shrinkage will be perpendicular to the heating lines. The forming direction, described above, refers to the direction in which the maximum strain magnitude will occur; thus this will be perpendicular to the direction of line heating but parallel to the plate rolling lines.

The forming direction used is indicated by an arrow which is displayed along with the plate in the Part view.

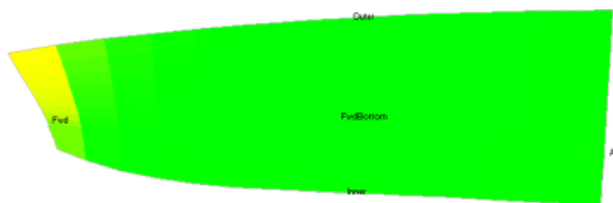
General Method - Strain Ratio

The strain ratio relates the strain in the direction perpendicular to the direction of maximum strain to the maximum strain. If the strain ratio is 100% then the strain does not depend on the orientation, it is isotropic. If the strain ratio is 0%, the expansion is purely uni-directional and there will be no strain in the direction perpendicular to the direction of maximum strain. Forming methods such as peening and planishing produce isotropic strain distributions, whereas line heating and rolling produce a more uni-directional strain distribution.

The figures below show the effect of strain ratio and forming direction on the expansion of a plate with negative Gaussian curvature. For this plate the forming direction was set to longitudinal with 0% strain ratio. As one would expect, the results show positive longitudinal strain on the edges, with the transverse strain remaining virtually zero. In the figures below, red areas show high positive strain and green areas have zero strain.



Longitudinal strain for negative Gaussian curvature plate: Expand only with 0% Strain Ratio.



Transverse strain for negative Gaussian curvature plate: Expand only with 0% Strain Ratio.

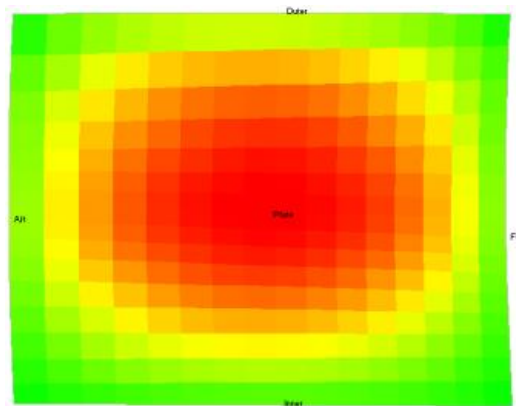
General Method - Forming Process

You may choose whether you wish to preserve the plate edge lengths, stretch the plate (expand only) or shrink the plate. Most processing methods will expand the plate, however line heating will shrink the plate.

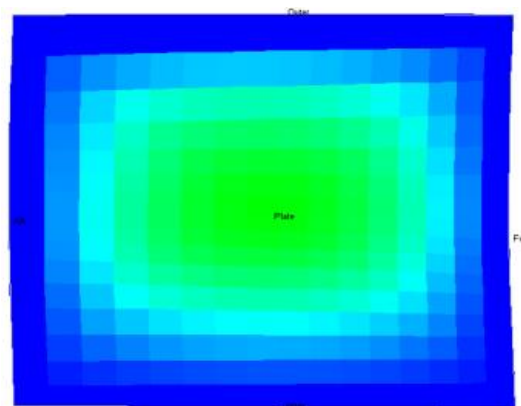
Using the Preserve Lengths method will ensure that the expanded plate edges are always the same length as the 3D surface plate. This may result in positive, negative or a mixture of positive and negative strain in the interior of the plate.

If you choose the Expand or Shrink options, the strain in the plate will be uniformly increased or decreased so that the overall strain is either all positive or all negative, depending on the selection made. (Note that there are small in built tolerances on the strain values.)

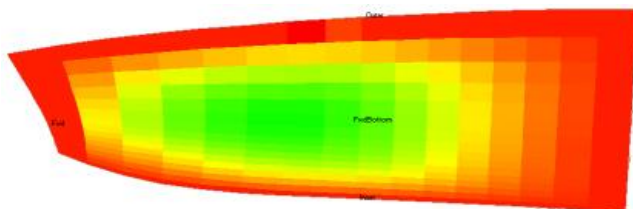
For plates with positive Gaussian curvature, selecting Expand only, will result in a plate where the middle requires stretching whilst the perimeter will remain unchanged. For a similar plate, choosing Shrink only, will result in the middle of the plate being unchanged whilst the edges will need to be shrunk (In the figures below, green areas have no strain; red areas positive strain and blue areas negative strain:



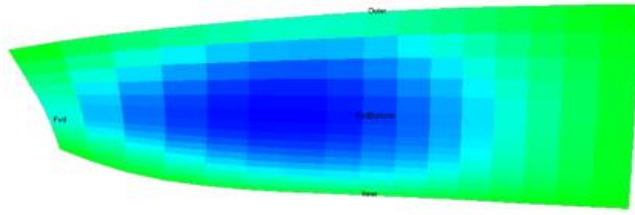
Positive Gaussian Curvature Plate: Expand Only; centre of plate requires stretching, edges remain unchanged.



Positive Gaussian Curvature Plate: Shrink Only; centre of plate remains unchanged, edges require shrinking.



Negative Gaussian Curvature Plate: Expand Only; centre of plate remains unchanged, edges require stretching.



Negative Gaussian Curvature Plate: Shrink Only; centre of plate requires shrinking, edges remain unchanged.

Whilst the General method is robust, it may have problems with surfaces which have a very large variation in local surface normal direction, e.g. a complete hemisphere. If such problems are encountered, the plate should be subdivided further.

The computations required for the General method can be quite intensive. It is advisable to check the results at lowest or low precision before completing the calculations at medium precision. For the General method, high and highest precision use the same plate mesh as medium precision. See: [Plate Precision](#) on page 310

Note:

The new method uses fewer points in the plate mesh and consequently there will be fewer points along each plate edge. You may need to turn on the Smooth Plate Edges option to get smooth plate edges for your final plate shapes.

Old Developable Method

This method may be useful for plates which have zero or slightly negative Gaussian Curvature. It has the advantage of being quicker to compute than the General method.

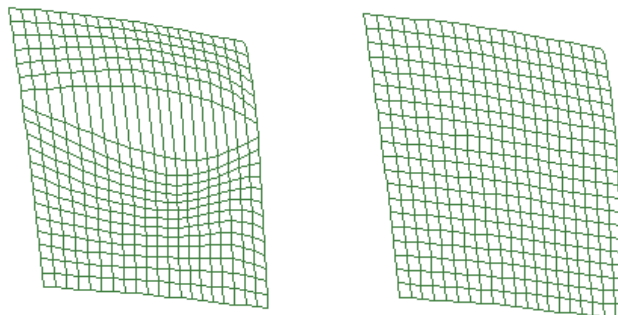
Fabric Method

This method should be used if you are to developing fabric panels. The method models the way in which a fabric stretches most along its bias direction. With this method angles will tend to distort whilst the lengths remain constant; with the other methods, angles will tend to remain constant whilst the lengths change.

Plate Mesh Options

The plate meshing can be done in either parametric (NURB) space or Cartesian space. If NURB is chosen then calculating the plates will be quicker, however this method does not work well if the underlying surface parametrics are not uniform (e.g. over-compacted control points and feature lines). The Cartesian method is more reliable but development times may be longer.

The figure below shows an example of a NURB mesh over a discontinuity (over-compacted control points on the underlying surface) and the same plate with a Cartesian Mesh:

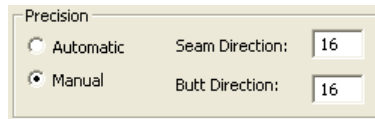


NURB mesh (left); and Cartesian mesh (right)

Setting Plate Precision Manually

You may specify the number of line segments that is used to draw the expanded plate. For very long and skinny plates you may wish to use the Manual plate precision setting on the bottom of the Plate Info dialog. Decreasing the number of line segments on the short edge, can reduce computation time while not compromising the accuracy.

The automatic precision uses the NURBS surface precision as set in the program's Data menu. In general it is recommended to use the Automatic setting for the plate precision.



Calculating Plates

Once you have added the plates you require, you can calculate the developed shape of each plate using the Calc Plate commands from the Plate menu. After the developed shape of a plate has been calculated, it is displayed in the Part window. You can use the pop-up menu in the Part window to choose which plate is displayed. Also you can switch on any contour lines, frames and stringer paths.

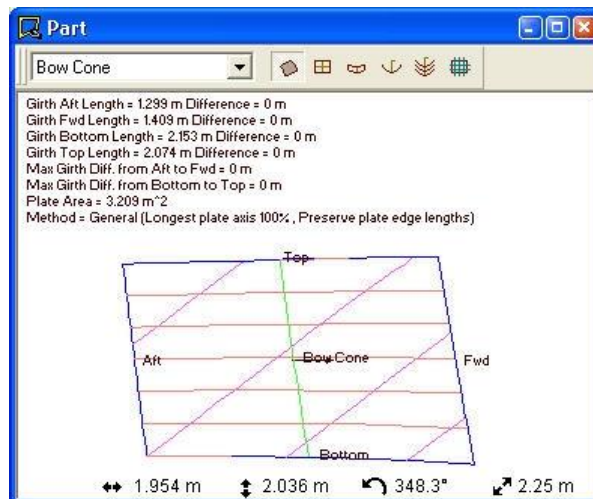


Plate Precision

The accuracy of the plate shape will depend on the precision setting at the time you choose the Calc command. However, you should always add plates using highest precision. If you are developing very long plates, you may need to turn on the Smooth Plate Edges option (found in the plate properties dialog or the plate data sheet). This will fit a smooth spline along the edges of the plate after development to remove any faceting that may occur.

To achieve accurate plates while being able to work fast we recommend the following work order:

- Set precision to highest when adding the plate (use Smooth Plate Edges for long plates)
- Set precision to lowest– medium to quickly see if the plate is developed correctly
- Set precision to highest when calculating the final plate.

Important!

Always use high or preferably highest precision when selecting "Add Plate" or "Add Surface Plate".

Also see: [Old Developable Method](#) on page 309.

Plate Thickness

The plate expansion calculation is based on the NURB surface geometry ignoring thickness. The specified plate thickness will be used to calculate the plate mass only.

You can view the expanded plate in the Part window using the Zoom, Shrink, Pan and Home View commands. You can also copy, print and plot the drawing in this window including copying the expansion into a CAD system via a DXF file.

Also see:

[Check Outside Arrows](#) on page 257

[Deduct Skin Thickness](#) on page 267

Development Failed to Converge

For surfaces that have a lot of curvature, the surface cannot be developed using only one plate. This happens when a surface wraps around an angle greater than 90°. The following error message will appear when you try to calculate (read: develop) the plate:



For information on subdividing surfaces, see: [Adding a Plate](#) on page 302.

Girth Differences

If the plate is developable, the expanded shape will be an exact two dimensional development of the three dimensional plate shape. If the plate is not developable, there will be a girth length difference between the original and the developed plate shape. This girth difference is displayed at the top left corner of the Part window and gives some indication of how developable the plate is.



The first four items displayed are the girth lengths and differences of the developed plate along the aft, fwd, top and bottom edges of the plate. The second two rows, the Max girth differences, are the greatest girth errors found along the mesh used to develop the plate and so represent the worst girth errors within the plate between opposite edges. Then the plate area and forming method information are given. The final two rows give the maximum and minimum strain which will be required to stretch the 2D expanded plate into the 3D hull plate, this strain is given in the forming direction.

The girth difference indicates how much in-plane strain will be required to deform the developed shape to match the true plate shape. If the girth error indicates that too much strain will be required to lay up the plates accurately, you may wish to break the plate up into a number of smaller pieces.

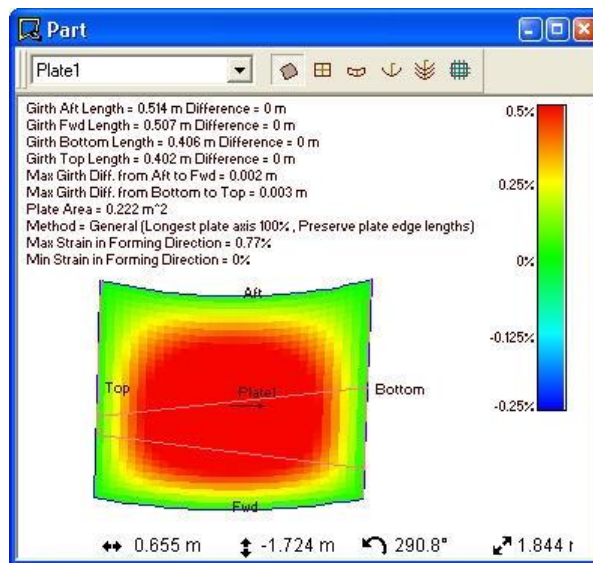
The difference values are calculated as follows (a girth positive difference indicates that the 2D plate must be stretched, whilst a negative girth difference indicates that it must be shrunk to fit onto the hull):

$$\text{Girth difference} = \text{Actual Girth} - \text{Developed Girth}$$

When determining whether the plate can be feasibly formed, it is important to look at all six values. With the General forming method it is always possible to create an expanded plate which has no strain in the plate edges. However, you must also check that the internal girth differences and maximum and minimum strains are within acceptable limits.

Plate Strain

Another way to view how much deformation will be required to accurately position the plate on the hull is to display a map of strain in the plate. You can display this by turning on items in the Strain menu. A color map of strain in the plate will be displayed; areas of positive strain (stretching) are shown in red, areas of negative strain (shrinking) in blue and areas of no strain in green.



The strain is calculated by dividing the plate up into small quadrilateral elements. In each area the principal direct strains and the shear strains are computed. The strains are computed by comparing the elemental lengths of the expanded and unexpanded plate. Negative strains indicate that the expanded plate must be shrunk for it to fit onto the vessel; positive strains indicate that it must be stretched. There are a number of relationships between the strains:

- The strain in the forming direction is always greater than or equal to the strain perpendicular to the forming direction.
- The strain in the forming direction is equivalent to the maximum magnitude principal strain.
- The strain perpendicular to the forming direction is equivalent to the minimum magnitude principal strain.
- If the strain ratio is 100%, the strain is isotropic and the maximum and minimum principal strains are equal. In this case the shear strain is zero.
- If the strain ratio is 0%, all the strain is in the forming direction; the minimum principal strain is zero and the shear strain is a maximum.
- If the plate is developable, all strains are zero.

The palette of colors at the top right corner of the Part window indicate the percentage strain corresponding to the colors in the map.

The amount of strain allowable must be decided by the designer and builder and will depend on the stiffness and thickness of the plate material. The maximum strain at any point in the plate is displayed at the bottom of the list of girth information at the top right corner of the Part window.

Three different of strains may be displayed, these are essentially the principal maximum and minimum direct strains and the shear strain. Each is chosen by selecting the appropriate item from the strain sub-menu under the Display menu.

Forming Direction

(Maximum principal direct strain) This displays the strain in the direction which the plate will be formed. This is the direction in which the maximum direct strain will occur. Note that if the General method is used, the forming direction is displayed by an arrow; if either Old Developable or Fabric methods is used, the forming direction is the direction from the first to second plate edge selected when defining the plate (i.e. from butt to butt).

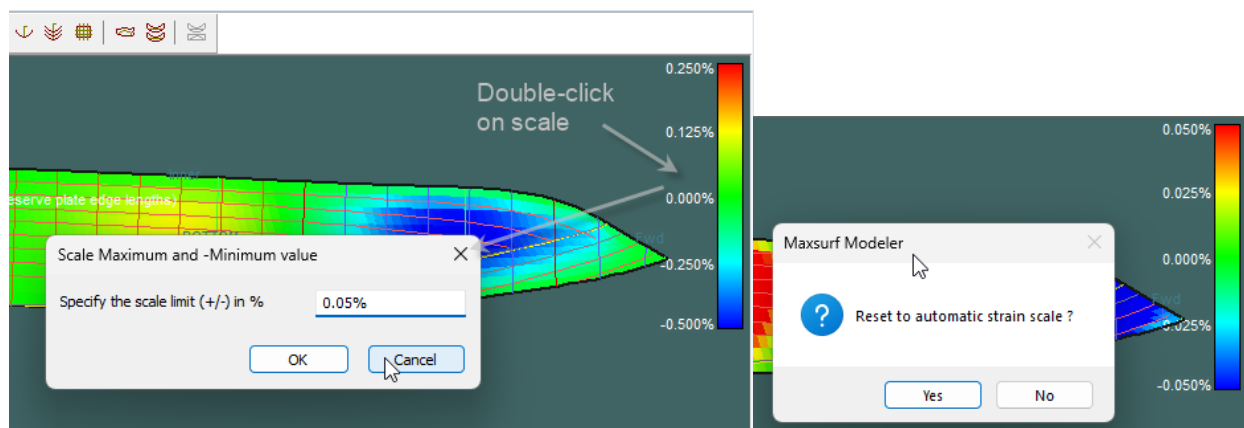
Perpendicular to Forming Direction

(Minimum principal direct strain). At 90° to the maximum principal direct strain will be the minimum principal direct strain. If the strain ratio is 100% then both will be equal, otherwise the minimum principal strain will be reduced, if the strain ratio is 0%, the minimum principal strain will be zero.

Shear strain

If the principal strains are not the same, there will be some shear strain present. This is a measure of how much the angles are distorting in the plate.

To change the scale range used when displaying the strain on an expanded plate, double left-click on the scale and specify the new range (symmetrical about zero) you wish to use. To revert back to automatically scaling, hold down the shift key when double-clicking.



Double click to set the scale limits; Shift-Double click to return to automatic scaling

Plate Development and Rolling Jigs

These features enable better control of the rolling and forming of doubly curved aluminium and steel plates.

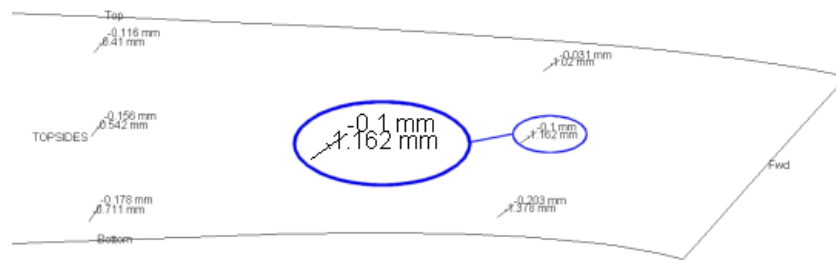
- Rolling Jigs
- Measuring Curvatures

Rolling Jigs

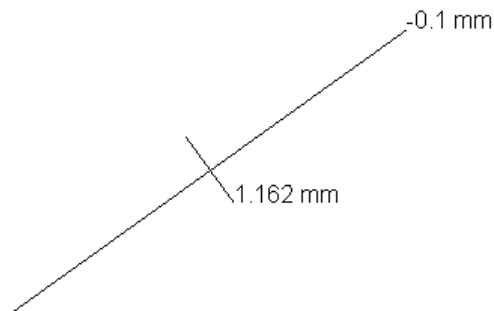
When a plate is developed, you can display rolling lines on the developed plate that provide information about the directions of the principal curvatures (or radii) and the magnitude of these curvatures (or radii).

Note:

If these are not displayed after loading a model file, you should re-calculate the plates.



You turn on these rolling lines using the Structure dialog from the Display menu. Each one of these markers has two numbers associated with it, the major and minor principal curvatures. The magnitude of the number associated with the small arm of the cross will always be greatest - this is the major principal curvature. The magnitude of the curvature associated with the long arm of the cross is the minor principal curvature. This is the direction along which plates are generally rolled or stretched to introduce double curvature.

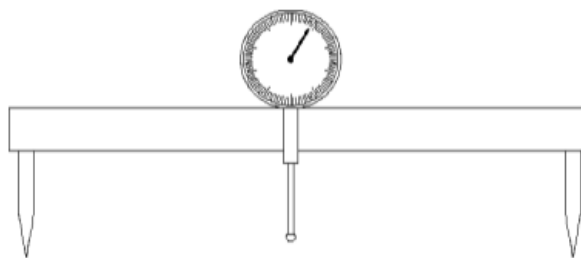


If both curvatures are zero, then the plate is locally flat. If the minor curvature is zero or close to zero, the plate can be considered to be developable, and the long axis of the cross will correspond with the direction of the developable ruling line.

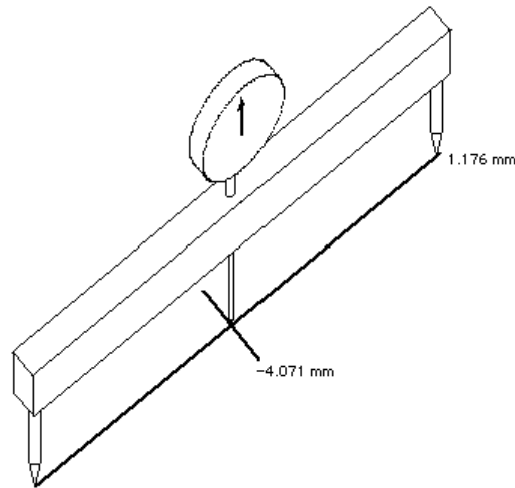
If the sign of the major and minor principal curvatures is the same, then the plate has spherical (or positive Gaussian) surface curvature. If these curvatures are different, as in the example above, this indicates that the plate has saddle shaped, or negative Gaussian curvature.

Measuring Curvatures

Curvatures are measured on the plate using a jig similar to the following.



A dial gauge is placed centrally on a beam that has two pins 300 mm apart and the dial gauge is calibrated to read zero when the jig is placed on a flat surface. Sufficient travel should be allowed for so that the dial gauge can read both positive and negative deflections.

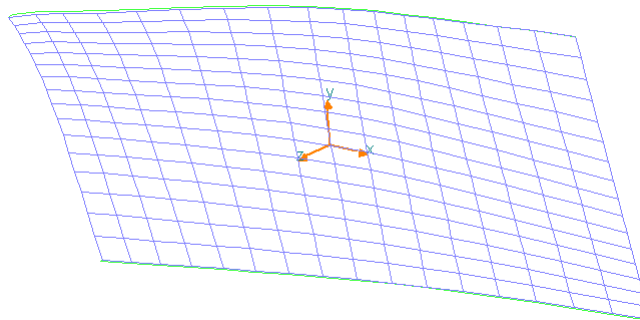


To check the minor curvature on a plate being rolled the jig is placed with the tip of the dial gauge on the centre of the marker and the pin tips at the end of the long axis of the cross. To measure the major curvature the jig is rotated 90° while keeping the dial gauge centred on the cross.

This method has the advantage of being lightweight, portable and accurate. As a result it is a more flexible approach than using templates and pin jigs, but as it only measures curvature locally should be used in conjunction with these methods on large or complex plates.

Plate User Coordinate Systems (UCS)

Users of AutoCAD and other CAD systems will be familiar with the User Coordinate System (UCS) concept. This allows you to define a local coordinate system on each plate which is then used for displaying views of that plate.



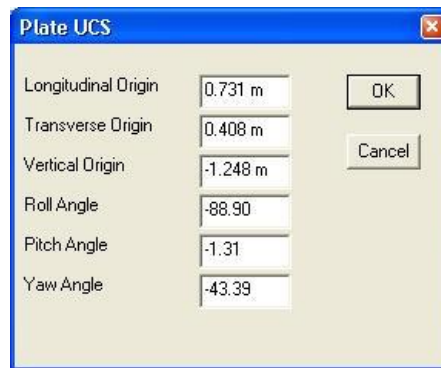
To set the UCS of a plate –

- Turn on display of Plate meshes (Display | Structure...)
- Select 2 points on the plate mesh (shift or Ctrl click to select the 2nd point)

The first point selected will be the origin of your UCS. The second point selected determines the direction of the Longitudinal (x) axis of the UCS. The surface normal direction of the plate will automatically be used for the Vertical (z) axis of the plate and the Transverse (y) axis will be set to the perpendicular of the other two axes.

- Next, choose Plate UCS from the Plate menu

This will display a dialog with the location and rotations of this UCS relative to the global coordinate system.



You can modify these settings if you wish before clicking OK. This makes it easy for you to set the orientation of the UCS to lie in a vertical or horizontal plane for example. The positions and rotations shown in the UCS dialog show transformations required to go from the global to the local coordinate system. You can display the plate UCS by switching on the Plate Axes in the Display | Structure dialog.

All of the plate displays described below use the UCS to display the plate.

Plate Inset Lines

A contour inset a constant distance from the boundary of the developed plate may be displayed. This plate inset line can be turned on using the Plate Inset Lines item in the Structure dialog. The distance of the line from the plate edge can be set using the Preferences dialog in the Edit menu. Japanese users may know this line as a "Sashikoshi" line. It provides a useful line for checking dimensions and alignment of plates independent of any edge distortion which may occur while the plate is being cut or shaped.

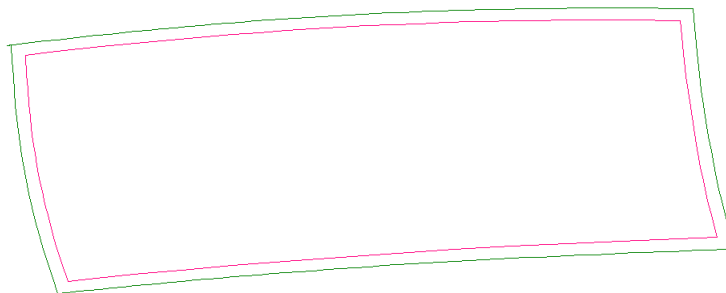


Plate Templates

Plate templates, to assist with accurate forming of the plate, can be caulated automatically. Plate forming templates run across the plate and show the shape of the plate at that location. You can add any number of templates at any location along the plate. The location of each template is defined as a longitudinal (x) position in the UCS of the plate.

To set the templates for a plate

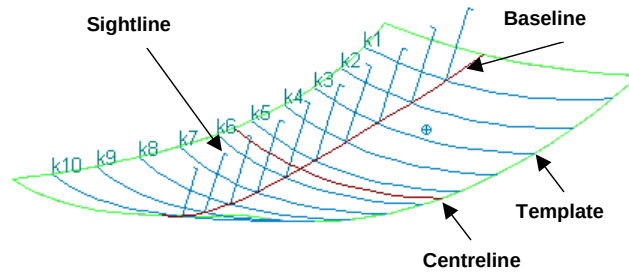
- Select the plate you wish to work on
- Choose Plate Templates from the Plate menu

A dialog box will appear that allows you to set up the location of the plates. This dialog works the same way as the Design Grid dialog.

To quickly view plate butt templates of a non-surface plate simply

- Choose Plate Butt Templates from the Structure Visibility dialog (Display | Structure)

Each plate also has a centreline and a baseline. The centreline is the intersection of the UCS yz plane with the plate, the baseline is the intersection of the UCS xz plane with the plate.



Each plate template also has a sightline marker at a constant height in the UCS and situated vertically above the baseline of the plate. Once the set of templates are set up along the length of the plate, the fabricator can sight along the sightline to ensure all of the templates are at the correct height.

Plate templates can also be rendered when rendering is turned on in the perspective window.

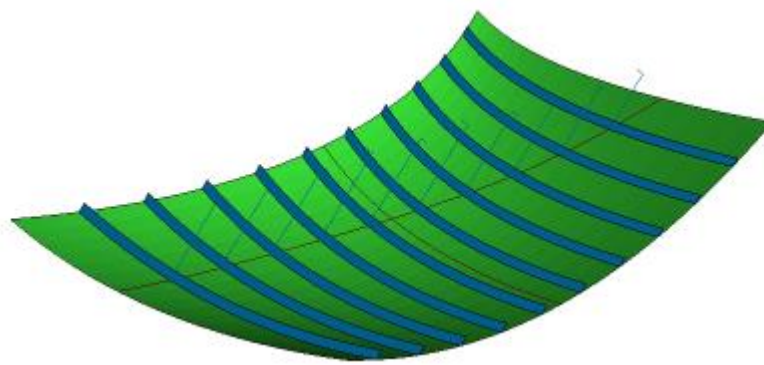


Plate Pin Jigs

Pin jigs, to assist with accurate setup of a plate or assembly of plates for attachment of internal structure, can be caulated automatically. Pin jigs are a series of vertical pins each of which is at a height to just support the shape of the 3D assembly. The pins run in the direction of the vertical z axis of the assembly's UCS.

Pin Jigs are most commonly used for assemblies of plates. You create a plate assembly by dragging the plates into the same folder in the Assembly window.

You set the spacing of the pins in the jig using the Structure Preferences item from the Edit menu; the location and height of each pin will be calculated automatically. A table of heights for each of the pins can be viewed in the Pin Jig table in the Data window. This displays the data for the plate assembly currently being viewed in the Part window.

	Edge Dist. m	Edge Ht m	Edge Ht m	Edge Dist. m	C m	D m	E m	Edge Ht m	Edge Dist. m
1	-	-	-	50	48	46	43	-	-
2	-	-	-	1033	1018	1012	1015	-	-
3	68	1039	-	1029	1019	1009	1018	1011	26
4	61	1028	-	1020	1009	999	1007	1002	31
5	62	1026	-	1017	1007	997	1005	998	30
6	71	1030	-	1019	1009	998	1006	1002	23
7	89	1043	-	1032	1021	1010	1021	1012	8
8	-	-	-	1033	1019	1014	1017	-	-

The pins can be displayed graphically by turning on the Plate Pin Jigs item in the Structure dialog.

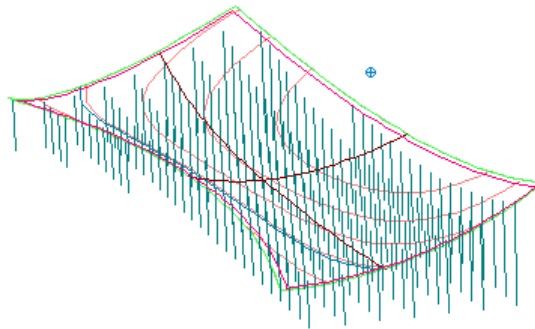


Plate pin jigs can also be rendered when rendering is turned on in the perspective window. The most commonly used view of the pin jig is the Pin Jig View described below. Note that the pin jig table is a tool to check the correct shape of the plate or plate assembly.

Shell Expansion

Shell Expansion – Concepts

A shell expansion drawing may be created for any design. The drawing shows the girth positions of all parts (plate edges, stringers, frames and decks) from a specified datum position (usually, but not always, the baseline).

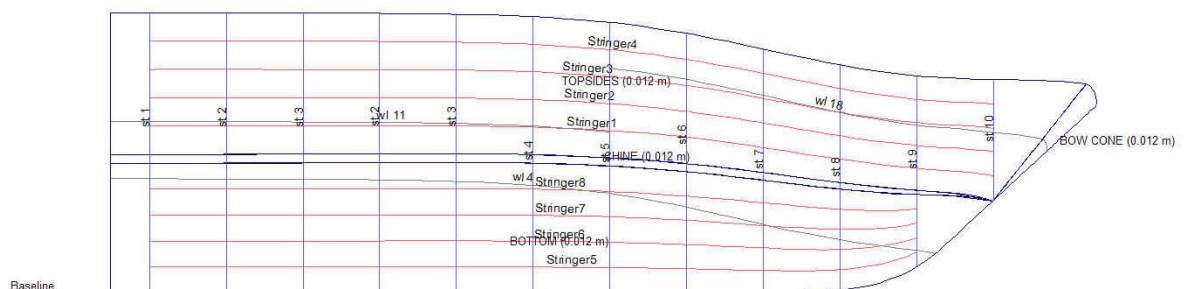
Shell Expansion - Procedures

Before generating the shell expansion drawing, the model must be set up correctly. The shell expansion drawing is generated by calculating the girth distance around the surface of the vessel from the baseline to the part. Hence it is important that all "superfluous" surfaces be either set to "internal use" or made invisible so they are not included in the girth calculations. Surfaces normally switched off during the shell expansion calculations include the superstructure surfaces, tank boundaries, railings etc. Surfaces that are normally on include the hull surfaces, any chine surfaces and the main decks. Normally the trimming is also turned on prior to a shell expansion so that the trimmed sections of the surface are not included in the girth calculations.

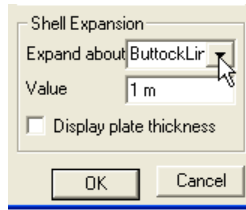
Once the model is set up correctly the shell expansion drawing can be generated by choosing Calc Shell Expansion from the Plate menu.

Note:

Only parts that are visible will be shown in the shell expansion drawing.



To change the datum about which the contours are expanded go to Display | Structure and specify the datum in the bottom right of the Structure Visibility dialog that appears. If the datum is a waterline or a buttock line then you will need to enter a value. For example if you wish to expand the shell about a buttock line 1.0 m from the centreline, you will need to select Buttock Line and enter the 1.0 m value in the Value field.



Displaying a Shell Expansion

Go to [Shell Expansion View](#) on page 321 for information on how to display Shell Expansion drawings.

Plate Displays

Once you have created a plate, defined its UCS, added any required templates and calculated the developed plate shape, there are different views in the Part window which can be used to check the plate's shape. The Plate sub-menu in the Display menu can be used to control which view is displayed in the Part window.

The possible views are –

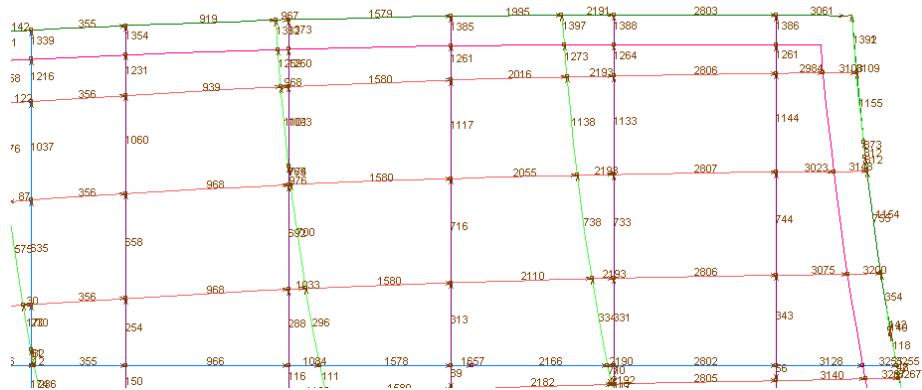
- [Development View](#) –shows the 2D developed shape of the plate.
- [Plan View](#) – plan view in the UCS of the plate. Also allows display of girth length and diagonal information on the plate.
- [Profile View](#) – profile view in the UCS of the plate. Also allows display of the sightline along the tops of the templates.
- [Body Plan View](#) – body plan (cross-section) in the UCS of the plate. Also allows display of templates on the plate.
- [Template View](#) –similar to the Body Plan view but offsets each plate template downwards so that you can clearly view the shape of each individual template.
- [Pin Jig View](#) –displays a plan view of the plate and displays the location and height of each pin supporting the plate.
- [Shell Expansion View](#) – displays the shell expansion.

Development View

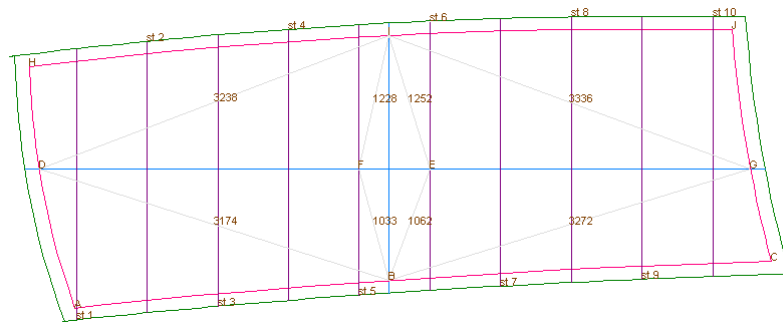
Shows the 2D developed shape of the plate. This is the shape that you would cut out of a flat sheet of metal. The plate will only appear in this window if the plate has been developed by one of the Calc Plate commands from the Plate menu.

Plan View

The Plan view allows you to view the edges, centreline, baseline, girth distances and diagonal distances along the plate. If the option to display Plate Girths is turned on in the Structure dialog, then a series of labels on the diagram which show the cumulative girth measurements away from the baseline and the centreline with the distances marked to each contour will be displayed. The girth display will use whichever contours are currently displayed. This means you can display any of the standard sections, waterlines etc as well as the plate templates. The result is a complete summary of girth information for the plate.

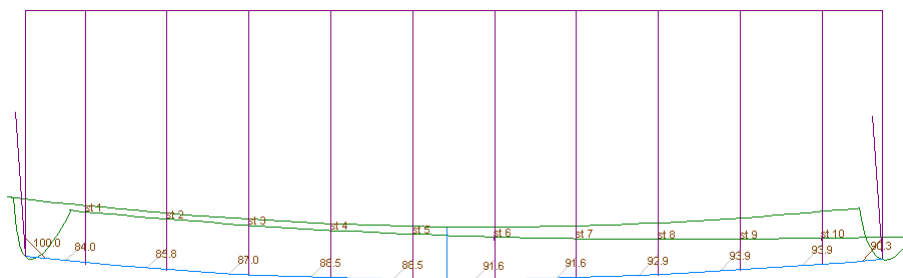


If the option to display Plate Diagonals is turned on in the Structure dialog, a number of 3D straight line diagonal measurements are shown on the plate. This provides a useful set of checks when checking the accuracy of plate forming. The diagonals run from plate corners to centreline and baseline ends and to the first template each side of the centreline.



Profile View

The Profile view allows you to view the plate from side on and displays the longitudinal locations of the plate forming templates, the height of the sight line of the templates above the plate, and the angles of the templates to the plate baseline. If frames are displayed, this view will also display lines showing the orientation of each frame relative to the plate.

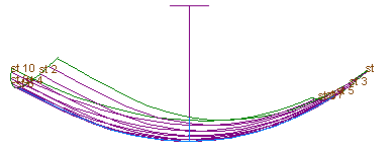


Also see

[Plate Templates](#) on page 316, for the definition of the plate baseline.

Body Plan View

The Body Plan view shows a cross section view of the plate in the plate UCS. This is useful for checking the shape of the templates of the plate.

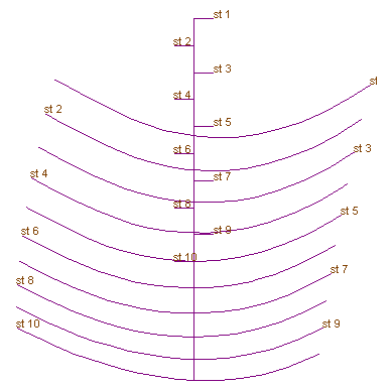


Also see:

[Plate User Coordinate Systems](#) on page 315.

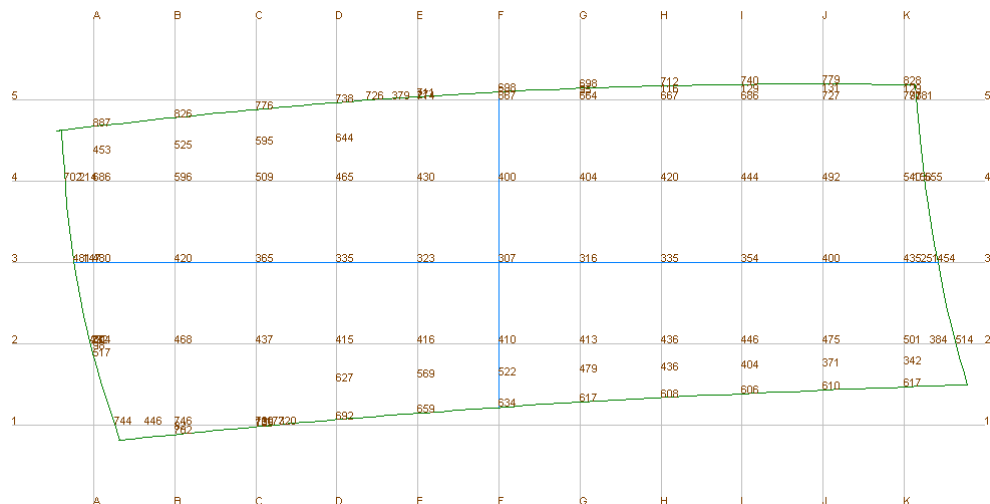
Template View

The template view shows the same orientation as the body plan view but each template is offset downwards so that you can clearly see the shape of each template. When exported to DXF for cutting or plotted at full size, this view provides shapes which can be cut out to create the templates for the plate.



Pin Jig View

The pin jig view is a plan view on the plate or plate assembly. The view displays the height of each of the pins in the grid of pins which support the assembly. You can set the spacing of the pins using the Preferences dialog in the Edit menu.

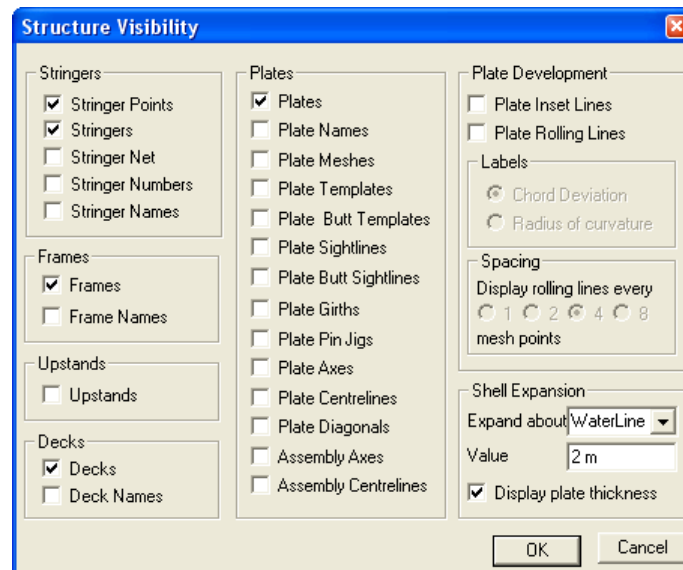


In addition to the height of each pin, this view also displays the height and the horizontal distance of each row and column from the edge of the plate. All of these heights and edge distances are also displayed in the Pin Jig table in the Data window.

Shell Expansion View

To display the shell expansion drawing, activate the Parts window and then select the Display | Plate | Shell Expansion or use the  button on the Part toolbar.

If you require the plate thickness displayed in the shell expansion drawing check the “Display plate thickness” box in the bottom right of the Structure Visibility dialog. Plate names, stringer names, frame names and deck names may also be displayed in the shell expansion drawing, this flag is also set in the Structure visibility dialog.



All colors in the shell expansion are taken from the associated part colors. For example the stringer expanded contours are the same color as stringers. These colors are set from the Colors and Lines dialog (View | Colors and Lines...).

Go to [Shell Expansion](#) on page 318 for information on how to create shell expansion drawings.

Working with Parts

Once you have developed all of the parts you require, you can view the parts in the Part window. Plates, frames, decks and stringers are all displayed in this window. You can choose which part is displayed by choosing from the pop-up menu at the top left corner of the Part window.

Calculating Part Weights

In the Parts Tab of the Data Table the weights of all parts are automatically calculated. Note that when a Part has no material assigned to it, it will not have any weight. To ensure that the weights table is up to date, select Calc All Parts from the Data menu.

Note

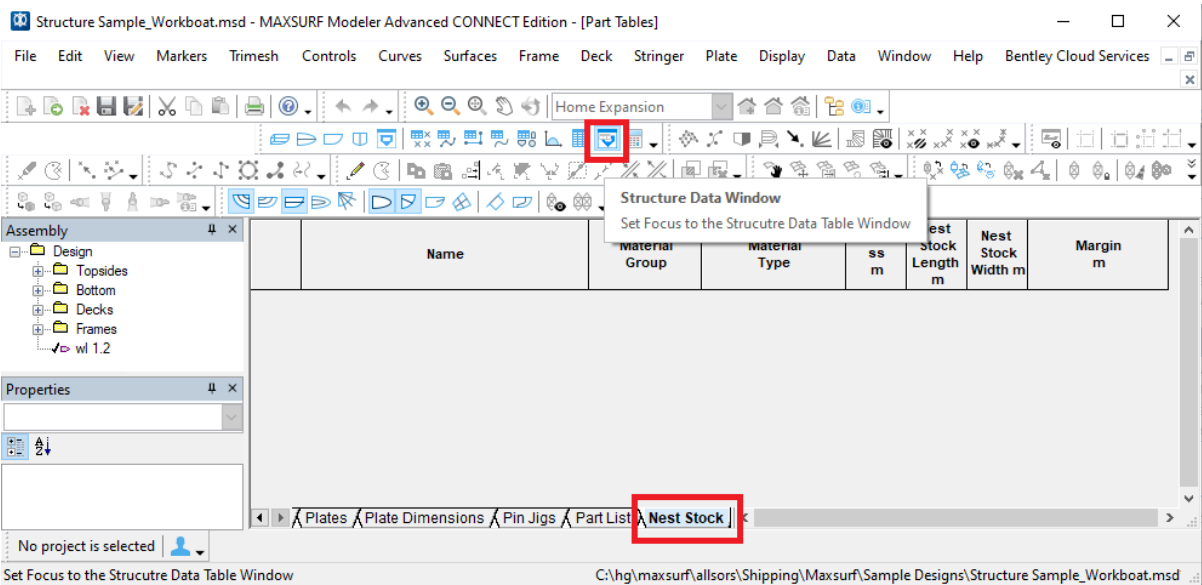
Stringer weights are calculated from the volume of the stringer multiplied by the material density. The volume is calculated from the stringer cross sectional area multiplied by the stringer path length. The density is a property of the material assigned to the stringer.

Nesting Parts

Once you have calculated all your parts you can assign them to stock and have them nested onto that stock for export to dxf file.

Creating Stock

Switch to the Structure Data window and click on the Nest Stock tab:



There are two ways to add stock to this list. If you have already assigned materials from the material library to each part, then you can automatically populate the Nest Stock list by choosing Data | Calculate Parts | Create Stock from the menu. Modeler will then look at all the materials used for each of the parts and add them to the stock list:

	Name	Material Group	Material Type	Thickne ss m	Nest Stock Length m	Nest Stock Width m	Margin m
1	Aluminium 5083-H111 7 mm	Aluminium	5083-H111	0.007	6.000	2.500	0.005
2	Aluminium 5083-H111 5 mm	Aluminium	5083-H111	0.005	6.000	2.500	0.005
3	Aluminium 5083-H111 6 mm	Aluminium	5083-H111	0.006	6.000	2.500	0.005
4	Aluminium 5083-H111 8 mm	Aluminium	5083-H111	0.008	6.000	2.500	0.005
5	Aluminium 6082-T6	Aluminium	6082-T6	0.000	6.000	2.500	0.005

The default stock sheet size is set to a length of 6m and width of 2.5m, the user is left to alter these values to what their physical material sheet sizes are.

Alternatively a new stock entry can be manually created by pressingjg Ctrl +A and editing the fields individually.

	Name	Material Group	Material Type	Thickne ss m	Nest Stock Length m	Nest Stock Width m	Margin m
1	Aluminium 5083-H111 7 mm	Aluminium	5083-H111	0.007	6.000	2.500	0.005
2	Aluminium 5083-H111 5 mm	Aluminium	5083-H111	0.005	6.000	2.500	0.005
3	Aluminium 5083-H111 6 mm	Aluminium	5083-H111	0.006	6.000	2.500	0.005
4	Aluminium 5083-H111 8 mm	Aluminium	5083-H111	0.008	6.000	2.500	0.005
5	Aluminium 6082-T6	Aluminium	6082-T6	0.000	6.000	2.500	0.005
6	Stock6	No Group	No Material	0.010	6.000	2.500	0.005

To choose a material group, click in the cell and choose from the drop down list, repeat for material type, add stock thickness, length and width.

Margin is the minimum allowable distance between parts when nested and between a part and the edge of the sheet.

Assigning parts to stock

Once stock has been created you can assign individual parts to the stock they are to be cut from. Again, this can be done either automatically or manually.

To automatically allocate the stock choose Data | Calculate Parts | Allocate Parts to Stock. Modeler searches through the stock for matching material stock for each part and assigns the best match.

Alternatively it can be done manually, click in the “Part List” tab:

Structure Sample_Workboat.msd - MAXSURF Modeler Advanced CONNECT Edition - [Part Tables]

FileEditViewMarkersTrimeshControlsCurvesSurfacesFrameDeckStringerPlateDisplayDataWindowHelpBentley Cloud Services

Home Expansion

	Name	Assembly	Ty	Qty	Material Group	Material Type	Thickness m	Area m²	Perimeter m	Weight tonne	CG Long. Pos. m	CG Offset m	CG Height m	Nest Stock
1	BOTTOM	Bottom	Pla	2	Aluminium	5083-H111	0.007	65.962	49.510	1.247	-2.391	1.376	0.660	excluded from nesting
2	Topsides5	Topsides	Pla	2	Aluminium	5083-H111	0.005	21.142	18.959	0.285	-9.604	3.390	2.969	excluded from nesting
3	Topsides4	Topsides	Pla	2	Aluminium	5083-H111	0.005	21.146	19.028	0.285	-3.668	3.366	3.000	excluded from nesting
4	Topsides3	Topsides	Pla	2	Aluminium	5083-H111	0.005	19.786	18.633	0.267	2.289	3.037	3.140	excluded from nesting
5	Topsides1	Topsides	Pla	2	Aluminium	5083-H111	0.005	7.305	11.397	0.099	10.082	1.013	3.582	excluded from nesting
6	Topsides2	Topsides	Pla	2	Aluminium	5083-H111	0.005	10.858	13.451	0.147	7.129	2.115	3.446	excluded from nesting
7	Chine2	Bottom	Pla	2	Aluminium	5083-H111	0.006	1.211	12.636	0.020	2.287	2.680	1.531	excluded from nesting
8	Chine3	Bottom	Pla	2	Aluminium	5083-H111	0.006	1.301	12.435	0.021	-3.675	3.129	1.240	excluded from nesting
9	BOW CONE	Topsides	Pla	1	Aluminium	5083-H111	0.005	2.806	8.497	0.038	11.572	0.000	3.931	excluded from nesting
10	Plate10	Bottom	Pla	2	Aluminium	5083-H111	0.006	1.303	12.238	0.021	-9.604	3.161	1.177	excluded from nesting
11	Plate11	Bottom	Pla	2	Aluminium	5083-H111	0.006	0.497	10.297	0.008	7.024	1.459	2.030	excluded from nesting
12	Transom	AttFrames	Fr	1	Aluminium	5083-H111	0.008	28.452	21.201	0.615	-12.552	0.000	2.657	excluded from nesting
13	st 2	AttFrames	Fr	1	Aluminium	5083-H111 0.006 (excludin	7.506	56.788	0.122	-10.554	0.000	2.371	excluded from nesting	
14	st 6	AttFrames	Fr	1	Aluminium	5083-H111 0.006 (excludin	7.510	56.694	0.122	-6.554	0.000	2.368	excluded from nesting	

PlatesPlate DimensionsPin JigsPart ListNest Stock

No project is selected

ReadyC:\hq\maxsurf\allisors\Shipping\Maxsurf\Sample Designs\Structure Sample_Workboat.msd*NUM Object Snap: OFF

Click in the parts “Nest Stock” cell and choose the stock from the drop down. Alternatively you can start typing the name of the stock and autocomplete will fill the rest. If multiple parts have the same stock you can copy and paste or fill down:

Structure Sample_Workboat.msd - MAXSURF Modeler Advanced CONNECT Edition - [Part Tables]

FileEditViewMarkersTrimeshControlsCurvesSurfacesFrameDeckStringerPlateDisplayDataWindowHelpBentley Cloud Services

Home Expansion

	Name	Assembly	Ty	Qty	Material Group	Material Type	Thickness m	Area m²	Perimeter m	Weight tonne	CG Long. Pos m	CG Offset m	CG Height m	Nest Stock
1	BOTTOM	Bottom	Pla	2	Aluminium	5083-H111	0.007	65.962	49.510	1.247	-2.391	1.376	0.660	Aluminium 5083-H111 7 m
2	Topsides5	Topsides	Pla	2	Aluminium	5083-H111	0.005	21.142	18.959	0.285	-9.604	3.390	2.969	Aluminium 5083-H111 7 m
3	Topsides4	Topsides	Pla	2	Aluminium	5083-H111	0.005	21.146	19.028	0.285	-3.668	3.366	3.000	Aluminium 5083-H111 7 m
4	Topsides3	Topsides	Pla	2	Aluminium	5083-H111	0.005	19.786	18.633	0.267	2.289	3.037	3.140	Aluminium 5083-H111 7 m
5	Topsides1	Topsides	Pla	2	Aluminium	5083-H111	0.005	7.305	11.397	0.099	10.082	1.013	3.582	Aluminium 5083-H111 7 m
6	Topsides2	Topsides	Pla	2	Aluminium	5083-H111	0.005	10.858	13.451	0.147	7.129	2.115	3.446	Aluminium 5083-H111 7 m
7	Chine2	Bottom	Pla	2	Aluminium	5083-H111	0.006	1.211	12.636	0.020	2.287	2.680	1.531	Aluminium 6082-T6
8	Chine3	Bottom	Pla	2	Aluminium	5083-H111	0.006	1.301	12.435	0.021	-3.675	3.129	1.240	excluded from nesting
9	BOW CONE	Topsides	Pla	1	Aluminium	5083-H111	0.005	2.806	8.497	0.038	11.572	0.000	3.931	excluded from nesting
10	Plate10	Bottom	Pla	2	Aluminium	5083-H111	0.006	1.303	12.238	0.021	-9.604	3.161	1.177	excluded from nesting
11	Plate11	Bottom	Pla	2	Aluminium	5083-H111	0.006	0.497	10.297	0.008	7.024	1.459	2.030	excluded from nesting
12	Transom	AttFrames	Fr	1	Aluminium	5083-H111	0.008	28.452	21.201	0.615	-12.552	0.000	2.657	excluded from nesting
13	st 2	AttFrames	Fr	1	Aluminium	5083-H111 0.006 (excludin	7.506	56.788	0.122	-10.554	0.000	2.371	exclud	exclud
14	st 6	AttFrames	Fr	1	Aluminium	5083-H111 0.006 (excludin	7.510	56.694	0.122	-6.554	0.000	2.368	exclud	exclud

Plates / Plate Dimensions / Pin Jigs / Part List / Nest Stock

No project is selected

Copy

Paste

Fill Down

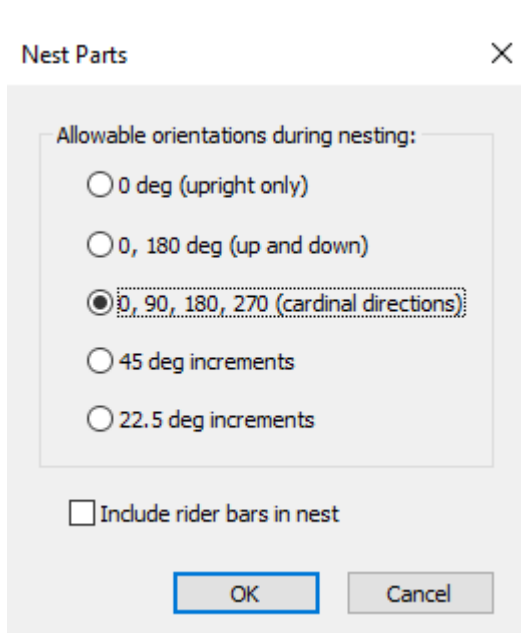
Fill Right

C:\hq\maxsurf\allisors\Shipping\Maxsurf\Sample Designs\Structure Sample_Workboat.msd* NUM Object Snap: OFF

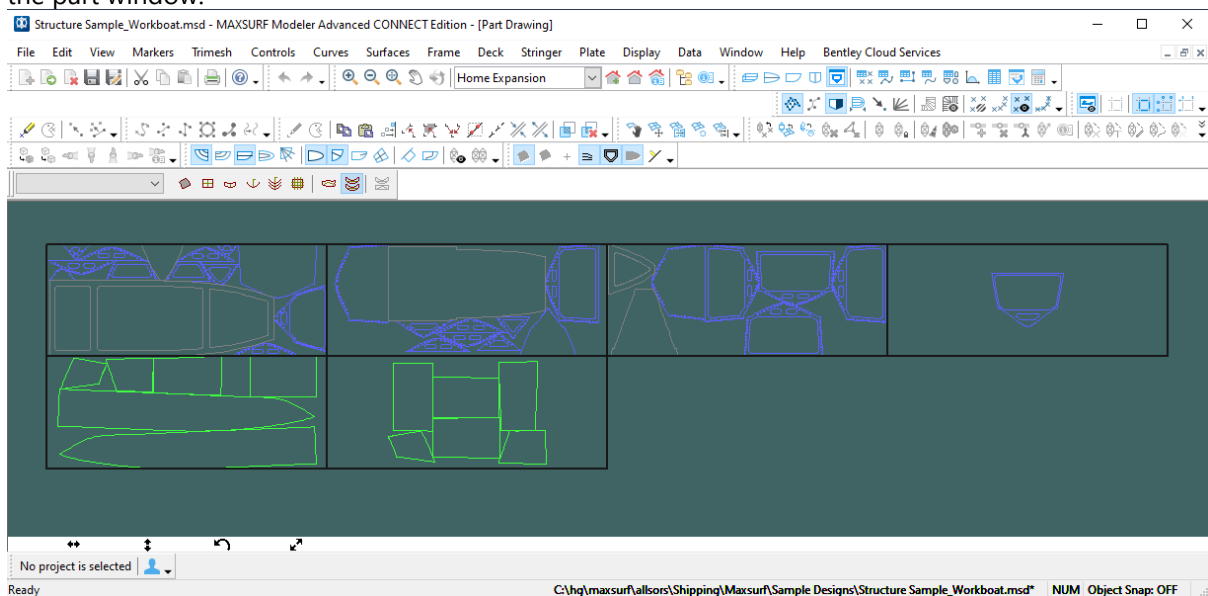
Creating the nest

Once all the parts you wish to nest have been assigned stock you are ready to run the nesting function.

Choose Data | Calculate Parts | Nest Parts... and the following dialog will appear:



Here you define the angles the parts can be rotated during the nesting algorithm iterations as well as whether to include the frame rider bars in the output. After hitting OK the resulting nest will be displayed in the part window:



Each row consists of stock plates packed with parts assigned to the stock. Once a sheet is filled a new column is added.

To export the nest diagram to dxf, choose File | Export | DXF and IGES...

Exporting Parts

You can copy, print or plot the parts in the same way you produce output for other lines in MODELER. The most common option is likely to be export of a DXF file for use by a CAD or CAM system.

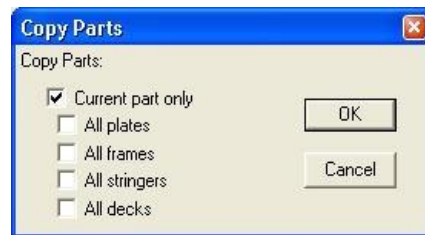
You can export one or more parts from the Part window

To save a DXF file of a part or parts:

- Bring the Part window to the front

- Select the part from the pop-up menu at the bottom left
- Choose Export from the File menu

A dialog will appear allowing you to copy one or more parts

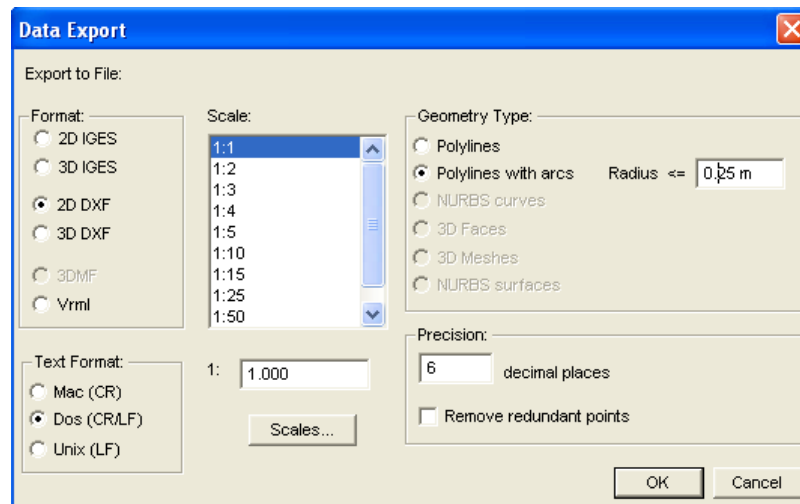


- Click the check boxes to specify which parts to copy
- Click OK

The usual copy dialog will appear allowing you to specify the format and scale of the exported parts

- Choose the DXF and 2D copy options
- Click OK
- Type in a name and select a destination folder for the DXF file

You will then be able to read this file into any DXF compatible CAD or CAM system. You can also export the parts into a 3D DXF file by going to the Perspective window and using Export DXF and choosing the 3D DXF option.



When exporting data as a 2D DXF file, there is an option to export the line as a series of polylines or a series of polylines and arcs. Controllers of different cutting machines work better with different polyline data. If you wish to export arc segment data you need to specify the maximum radius of arc that you wish to be exported as arc data rather than polyline data. For example if a radius of 0.25 m is specified in the radius edit box, all curves with an arc radius of less than 0.25 m will be exported as arc data and all those above will be exported as a series of straight line segment data. Units used are those currently set in the units dialog box.

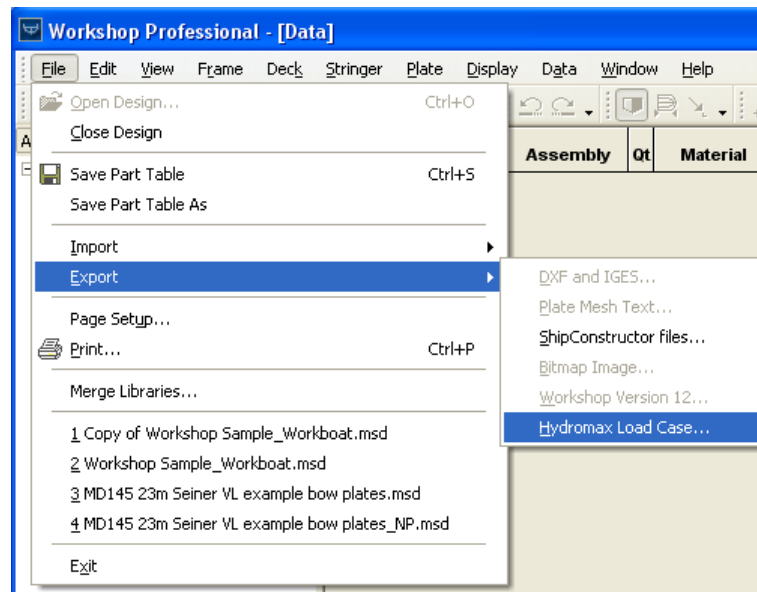
The remove redundant points option will delete all points that lie on a straight line apart from the start and end points. This reduces file size and aids in downstream processing.

Exporting parts to Stability Load case file (.hml)

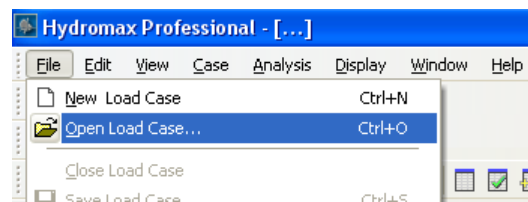
It is possible to export structural part weight and position information as a Stability Load group file (.hml file). This file may then be loaded directly into Stability for stability calculations. Prior to exporting, check all parts

are present in the "Data" window, "Parts" tab. If a part has no weight then this will be because it does not have a material assigned.

To export the information simply choose File | Export | Structure Parts | Stability Loadcase... from the menu.



To import the information into Stability go to the load case window and choose File | Open Load Case... from the menu.



The part information is automatically set as a load group (see the Stability manual for more information). This enables the structural information to be cross referenced into all the load cases (e.g. as a lightship weight).

As well as part weight and centre of gravity information, fwd and aft extents of each part are included in the file which can be used for longitudinal strength calculations in Stability.

Materials and Section Libraries

Materials Library

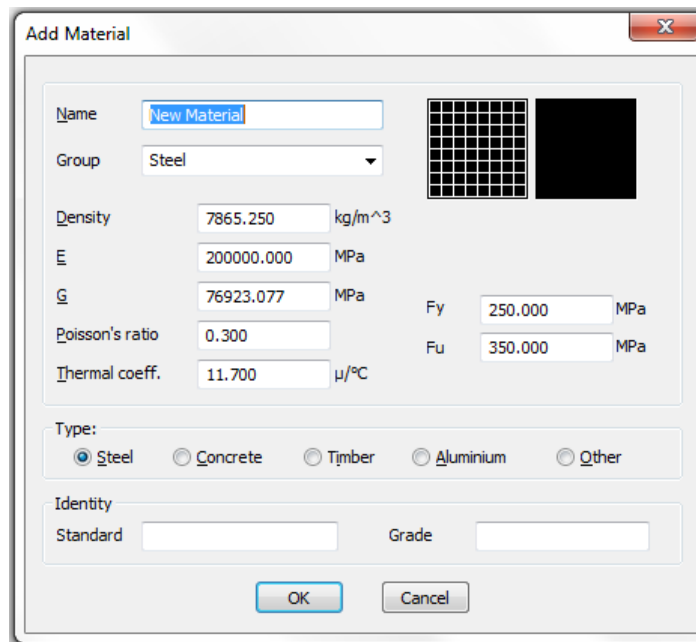
In order to determine the thickness offset used when calculating plate shapes and to determine weights of parts in the design, it is necessary to specify material types for frames, decks, stringers and plates. These materials are stored in the Sections Library and can be edited using the Edit Materials... command from the Edit menu.

To add, edit or delete a material:

- Choose the Edit | Edit Materials command from the menu
- Choose Add Material, Edit Material or Delete Material from the submenu

To add a Material:

- Choose the Edit | Edit Materials | Add Material from the menu and the Add Material dialog will appear



The 'Add Material' dialog box is used to define a new material. It includes fields for Name, Group, Density, E, G, Poisson's ratio, Thermal coeff., Fy, and Fu. The 'Type' section has radio buttons for Steel, Concrete, Timber, Aluminium, and Other. The 'Identity' section has fields for Standard and Grade. There are OK and Cancel buttons at the bottom.

Add Material

Name:

Group:

Density: kg/m³

E: MPa

G: MPa

Poisson's ratio:

Thermal coeff.: μ/°C

Fy: MPa

Fu: MPa

Type:

☒ Steel ☐ Concrete ☐ Timber ☐ Aluminium ☐ Other

Identity

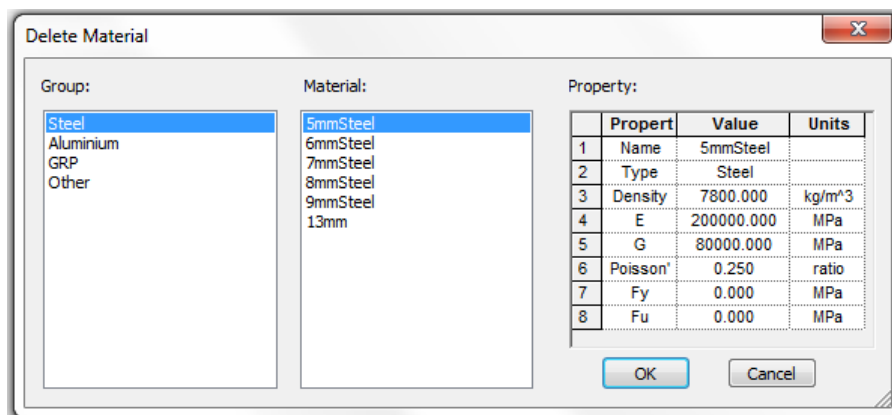
Standard: Grade:

OK Cancel

This is the standard Add Materials dialog used across all Applications in the suite. Only the Density field is used. However if the design is to be processed in a downstream application such as Multiframe it is advisable to fill in the other property fields.

To remove a Material:

- Choose the Edit | Edit Materials | Delete Material from the menu and the Delete Material dialog will appear



The 'Delete Material' dialog box shows a list of materials grouped by category. The 'Group' list includes Steel, Aluminium, GRP, and Other. The 'Material' list includes 5mmSteel, 6mmSteel, 7mmSteel, 8mmSteel, 9mmSteel, and 13mm. The 'Property' table shows the material's properties and units. There are OK and Cancel buttons at the bottom.

Delete Material

Group:

- Steel
- Aluminium
- GRP
- Other

Material:

- 5mmSteel
- 6mmSteel
- 7mmSteel
- 8mmSteel
- 9mmSteel
- 13mm

Property:

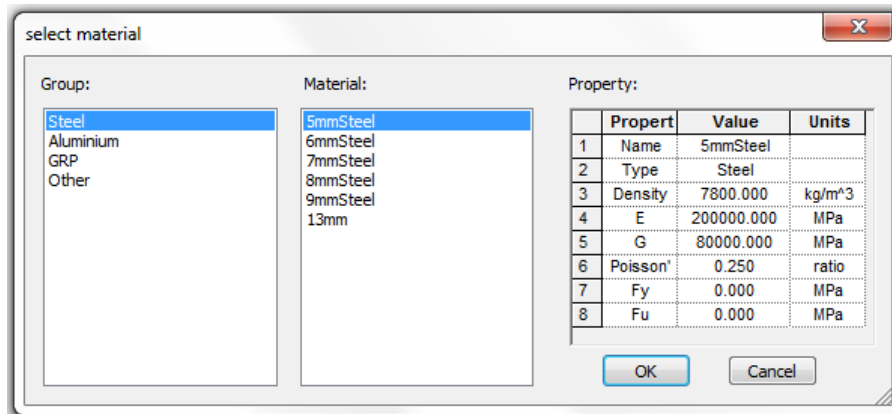
	Propert	Value	Units
1	Name	5mmSteel	
2	Type	Steel	
3	Density	7800.000	kg/m ³
4	E	200000.000	MPa
5	G	80000.000	MPa
6	Poisson'	0.250	ratio
7	Fy	0.000	MPa
8	Fu	0.000	MPa

OK Cancel

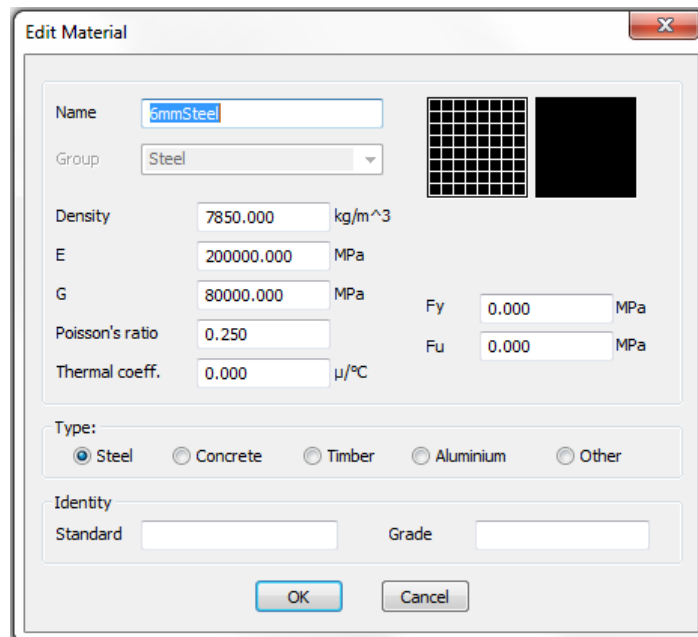
- Choose the material you wish to delete and Click the OK button

To edit a Material:

- Choose the Edit | Edit Materials | Edit Material from the menu and the Select Material dialog will appear



- Select the material you wish to edit and click OK. The Edit Material dialog will then appear



- Alter the fields you wish to edit.

To save your changes to the materials:

- Click the OK button

The materials will be saved in the Sections Library automatically. They are saved along with the stringer shapes and cutouts, and will be available next time you use the program.

Sections Library

Modeler uses a Sections Library .slb file to store the stringer shapes & cutouts and materials. Working with this library is completely automatic. Every time you start Modeler, the application will look for the default file path (see below) and load the library automatically.

If you don't want Modeler to load the library automatically on start-up, you should rename the library file in the default directory. Modeler will then ask you which library you wish to load each time it starts up. The library will be automatically saved to this file when you close Modeler.

Default library location:

- Windows Vista/Windows 7 – C:\Users\Public\Documents\Maxsurf\Maxsurf##
- Other Operating Systems - C:\Program Files\Maxsurf ##

, where ## is the version number. The default library name is "Maxsurf StructureLibrary.slb". A sections library is common across all the applications including Multiframe. This is to allow ease of transfer of design data between the applications.

Once Modeler is running, a new library may be opened via the file menu item File | Load Sections Library.... Modeler will save and close the currently open library and open the new library.

In previous versions of Modeler a separate library was used the *.wsl file. To upgrade a *.wsl file to a *.slb file choose the File | Convert Library command from the menu. Modeler will then ask you to locate the *.wsl file you wish to convert, then it will ask you for the name and location you wish to save the converted *.slb file to.

Sharing library files

It is possible to share library files across a network. However, you should be aware that changes to the Library file can easily be overwritten by someone else if that person closes Modeler after you did. You can create a shortcut Section Library .slb file to the shared library file on the server to share the library. Please note that most of the other applications in the suite use the sections library file and will also automatically save to this shared library. It may be easier to occasionally distribute the physical library file.

Data exchange

Data exchange is a key requirement in the modern design office and Modeler supports a wide range of industry standard formats. You can copy and paste numerical tables to and from Microsoft Excel for custom calculations and formatting. Graphical views can be copied and pasted to other applications to help create presentation materials and reports.

The following is a list of file formats supported for data exchange:

Importing		Exporting	
MicroStation DGN	NURBS surfaces and curves	MicroStation DGN	NURBS and mesh surfaces and curves
IGES	NURBS surfaces and curves (untrimmed)	IGES	NURBS surfaces and curves
Rhino 3DM	NURBS surfaces and curves	Rhino 3DM	NURBS and mesh surfaces and curves
IMSA	NURBS surfaces	IMSA	NURBS surfaces
USNA / Fastship	NURBS surfaces	USNA / Fastship	NURBS surfaces
DXF	Curves and Markers, Background	DXF	Polylines
GHS	Marker sections	BMT Microship	Sections
PIAS ascii	Marker sections	MHCP	Sections
Seaway	Marker sections	HYDROS	Sections
Wolfson LHF	Marker sections	IHI	Sections
nuShallo PAN	Mesh surface	IMSA	Hull parameters
Polygon mesh PLY	Mesh surface	GHS	Sections
Stereo Lithography mesh STL	Mesh surface	nuShallo	Mesh
WAMIT med GDF	Mesh surface	Parametric	Sections
Image PNG, JPEG, GIF	Background image	PD Strip	Sections
MicroStation DGN	NURBS surfaces and curves	PIAS ascii	Sections
		SHCP	Sections
		Shipflow	Sections and waerlines
		Stereo	
		Veres (MASHIMO)	Sections

		Wintech	Sections
		Wolfson LHF	Sections
		Polygon mesh PLY	Mesh surface
		Stereo Lithography mesh STL	Mesh surface
		WAMIT med GDF	Mesh surface
		Direct X	Mesh surface
		Wavefront	Mesh surface
		Image BMP	Background image

Input of Data

Modeler can import a range of data formats to assist you in creating a Modeler design to match existing data. Point and line data can be imported via DXF, surface data can be imported via IGES and Rhino 3dm files and image data can be imported in a range of bitmap formats such as GIF, BMP and PNG.

In this section:

- [Pasting](#) (General)
- [Importing Background Images](#)
- [Importing DXF](#)
- [Importing DXF Markers](#)
- [Importing IGES Curves](#) and Surfaces
- [Importing Rhino .3dm files](#)

Before continuing this section, you may wish to read the [Modeler Coordinate System](#) on page 20.

Note

If your File | Import submenu is greyed out, make sure you create a new design first from File | New.

Pasting (General)

You can paste data from any spreadsheet, word processor or text editor into the tables in Modeler. See [Paste](#) in the Table Windows – General section on page 58.

Importing Background Images

Modeler has the ability to display background images in each of the design views. Those in the Plan, Profile and Body Plan views may be re-positioned and resized to match the design and can then be used to assist in fairing to an existing design. The image in the perspective view is always displayed at full image size and from the top left corner; by using the zoom and pan control you can re-position your vessel in the scene.

Modeler has no built-in image manipulating commands, other than to control the position and scale of the image. Hence you must ensure that the image is in the correct orientation prior to loading into Modeler. It is also helpful if you make the background color of your image similar to that of the Modeler background color, since it will be easier to see the control points and other lines which are drawn over the top of the image by Modeler.

Scanning an Image

When scanning the lines plan it is important to do this at a reasonably high resolution to minimize any digitizing errors. In this document, scanning refers to the process of creating a bitmap image (such as a jpeg or bmp file) from a paper drawing, while the term digitizing refers to the creation of a point in 3D space (i.e. having x,y and z coordinates) from information on a scanned drawing.

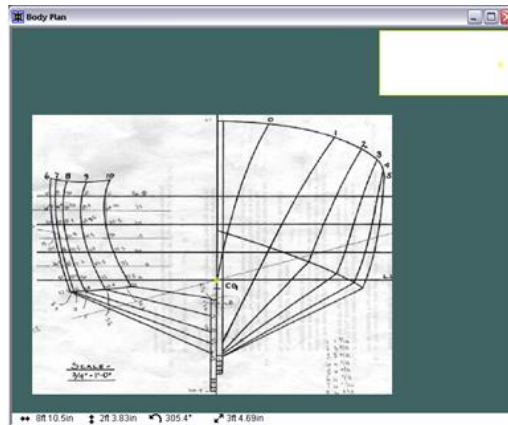
It is also important to make sure that the scanned image is not rotated relative to the screen, this will make it extremely difficult to digitize accurately. If you find that your image does have a slight rotation, it is necessary to correct this in a paint program such as Photoshop or Paintshop Pro prior to importing it into Modeler.

Importing an Image

Modeler can load image files of the type jpg, gif, bmp and png. Once you have a suitable image of for example your body plan

- Open Modeler and select File| New Design... from the file menu.
- To import the image, switch to the body plan view and select File|Import|Image Background...

This will import the image in the Body Plan window.



When first opened, the image will be scaled so that it is displayed at full size at the current window zoom level. Once you have loaded the image, you can re-position it by locating the zero point in the image. The image will then be moved so that the image zero point is in the same position as the design zero point.

Setting the Zero Point

The next step is to centre the image in Modeler's coordinate system. To do this,

- Select Display | Background | Set Image Zero Point.
- Zoom in and click on the point on the image that you wish to be the zero point.

The most likely location for this example image is the intersection of the centreline with the DWL or the baseline. For an image in Profile you may pick the Aft Perpendicular and the same DWL or Baseline.

When you click the mouse, the background image will move so that the point you just clicked lines up with the design's zero point:

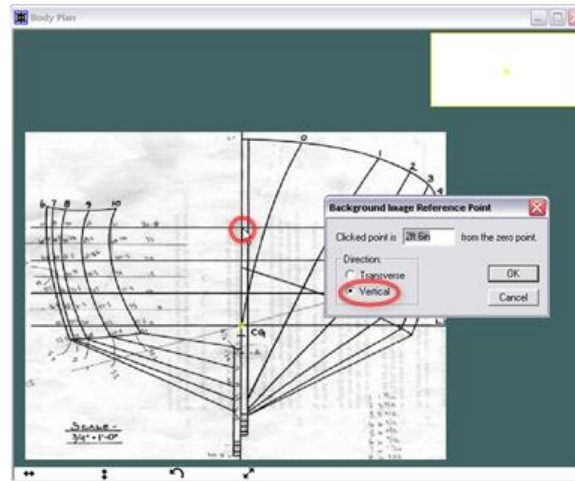
Setting the Scale

You can specify a known position on the image to scale the image. To do this you only need to know the value of a single dimension on your drawing in either the horizontal or vertical direction. For most drawings this is trivial as they will have a known waterline and/or buttock spacing and these can be directly measured.

- Select Display| Background | Set Image Reference Point.
- Zoom in and click one of the waterlines, preferably the one that is the greatest distance from the DWL.

A dialog will appear asking you to specify a value for this distance. For example, if you know that your waterline spacing is 1.5 meters, and you clicked on waterline 3, enter 1.5 m.

- Make sure that the Direction item has Vertical selected and click OK.



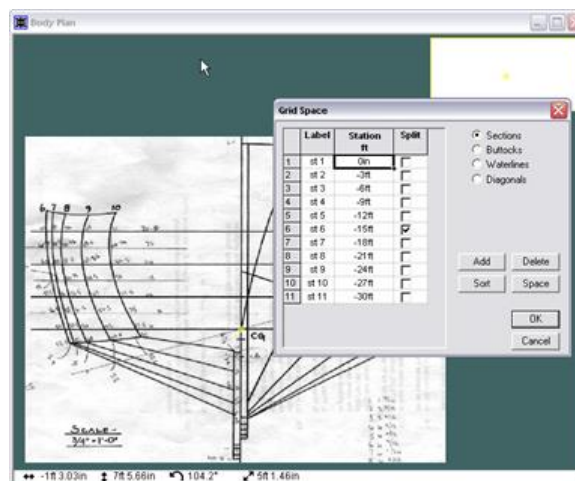
See the [Modeler Coordinate System](#) on page 20 for information about UCS directions and sign conventions in Modeler. Note that the aspect ratio of the image is always maintained.

Images in all the views are independent so you can load a profile and buttock lines sketch into the Profile view, a sketch of the sections into the Body Plan view and a sketch of the waterlines into the Plan view.

Setting up a Grid

The next step is [Setting up the Grid](#), see page 30 of this manual.

Note that Modeler uses a coordinate system that increases in value towards the bow, so if your station zero is at the bow, and you wish your profile view to have the bow on the right hand side, you will need to have the negative values for the station locations. If you place your zero point at the transom, station locations will be positive values.



Once you have specified the station spacing and clicked OK, it is likely that the Inset Box in the top right corner of the Body Plan window will be blank. This is because there are no surfaces specified and Modeler needs to be forced to rescale its internal viewing parameters so that the station locations will appear, scaled to the length of the Inset Box. To force this to happen,

- select View | Home View.

Small marks should now appear at the top of the Inset Box corresponding to the locations of the stations. Click on the right most of these marks – a black vertical line should appear below it –



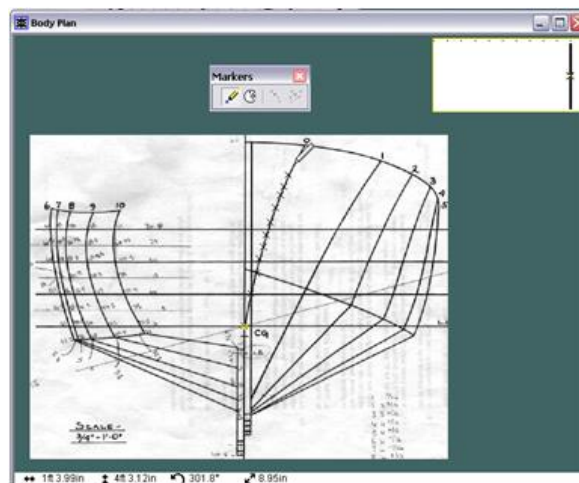
Inset box in top right of Modeler Body plan window

You have successfully selected station zero of your background grid as the current station. This sets the longitudinal position of any point you digitize in the Body Plan view, providing the third value of the x, y, z coordinates for the point.

Digitising a Background Image

The next step is to select the Add Marker tool and click on a point on the section at station zero. The most efficient way to do this is to drag the Markers Toolbar onto the Body Plan window so that you can repeatedly select Add Marker. Alternatively, Add Marker can be selected by using the Ctrl-M key combination.

You can now add marker points for as many points on the section as you like. Remember that you can use the zoom tool to zoom in as far as you like when adding markers to get the most accurate points possible.



Once you have finished digitizing the points for station zero, click on the indicator for station 1 in the inset box. This sets the longitudinal position of any subsequent markers that are inserted to the location of Station 1. In this way we build up a three-dimensional set of marker points, rather than just a two dimensional dataset.

Again, digitize the points for the section at that station location. It helps if the drawing is clearly labelled with the station number, as in this example, as it is very easy to get the longitudinal location confused and accidentally digitize a section at the wrong longitudinal position.

Continue digitizing points for each section, then selecting the next station location until all stations have been digitized. If you now switch to the perspective view, you should find that you have a complete, three-dimensional set of data points entered for all the stations.

If you have a profile view scanned, you can now switch to the Profile View window and perform the same process. In this case you are probably just interested in the bow profile, and this is easy as the marker points you insert will be added on the centreline.

Import your image as before, set the zero point to the intersection of the DWL and Station zero, then set your scale by selecting the position of one of your aft stations and entering its location in the Reference Point Dialog. In this case remember to select Horizontal for the measurement direction.

You can now proceed with the digitizing of the bow profile, and this will be added to your set of marker points.

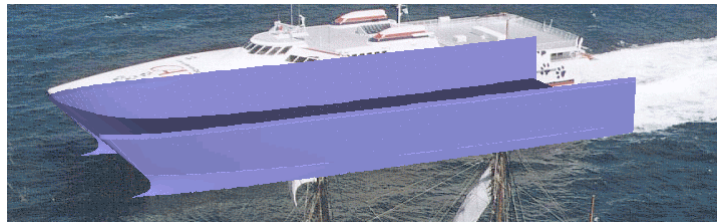
Also see:

[Fitting a NURB Surface to Markers](#) on page 226.

[Working with Markers](#) on page 80

Images in the Perspective View

The background image in the Perspective view can be used for generating presentation pictures. The background image remains in the centre of the screen, use the zoom and pan controls to place the Modeler design in the desired position over the background:



Note that the images are not saved with the design file, only the path name to the image file. If you change the directory of your images or their file names, Modeler will no longer be able to find them and the images will not be automatically loaded when the design is re-opened.

Importing DXF splines and polylines as curves

DXF line, polyline and spline entities can be imported to Modeler as curves. To do this run the File | Import | DXF... command. Lines will be converted to a Modeler curve of stiffness 2 (linear) and have two points (start point and end point). Polylines will be converted to a Modeler curve of stiffness 2 (linear) with $n+1$ points, where n is the number of line segments in the polyline. A dxf spline will open as a Modeler curve with the same stiffness, number of control points and knot vector.

Importing DXF background

Using the Import DXF Background command from the File menu enables you to import a DXF file into Modeler to use as construction lines. The DXF file will be displayed in all design views. DXF backgrounds can be really useful when you have a 3D lines plan in for example AutoCAD that you want to manually fit a surface to.

Note: You can only have one DXF background for your Modeler design. This single DXF background will be displayed in all 4 design windows. In case you want the profile lines displayed in the Profile window, and the section lines displayed in the Body Plan window, you have to convert your 2D lines plan drawing to a 3D lines plan in AutoCAD before saving as a .DXF.

Note

If your File | Import submenu is greyed out, make sure you create a new design first from File | New.

See also:

[Importing DXF Markers](#) on page 336

[Background](#) in the Display Menu section on page 390

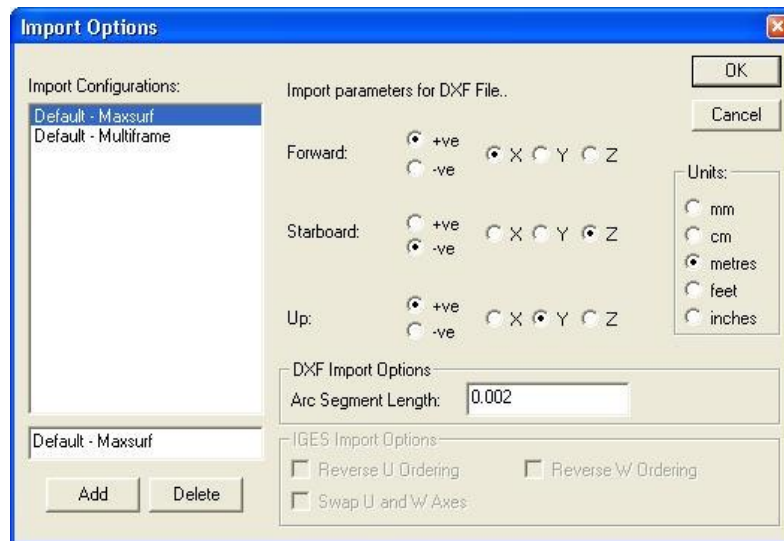
Importing DXF Markers

The importing of marker information is possible from DXF as well as GHS and Seaway section data files.

The Import DXF Markers command from the File menu can be used to import lines, arcs or polylines from a DXF file and display them as markers in the graphics views in Modeler. Each point at the end of a line or along a polyline will be converted into a marker.

To import a DXF file as markers:

- Choose Import->Import DXF Markers from the File menu



- Set the XYZ orientation of the data in the file

Note

If your File | Import submenu is greyed out, make sure you create a new design first from File | New.

You can use the radio buttons in the middle of the dialog to specify the orientation of the points in the file relative to the Modeler coordinate system. For example, choosing +ve (positive) and X in the Forward section means that +ve X coordinates in the DXF file will be aligned with the forward direction of the Modeler model. Also see: [Modeler Coordinate System](#) on page 20

The coordinates in the file are positioned relative to the current Modeler zero point.

- Set the units of the data in the file

Click on the radio button corresponding to the units of the data in the file.

- Optionally set the arc segment length

If you are importing arcs from the file, these will be converted into line segments before importing. This field can be used to specify the length of the line segments along the arc.

Note:

You can only import ARC objects in the DXF file. Arcs within polylines will not be imported, you should explode the polyline first before importing.

- Optionally save the configuration

You can use the Add button at the bottom left of the dialog to add a configuration and save the setting for future usage. For example, if you frequently import from AutoCAD, you might like to set up a configuration, named AutoCAD, to save your import settings.

- Click OK to import the data

Once imported the markers can be manipulated in the usual way, including assigning to a particular station or associating with a surface. See also [Working with Markers](#).

Converting a 2D DXF to a 3D DXF

It is generally recommended to convert a 2D DXF file to a 3D DXF file prior to importing it into Modeler. The simplest way to convert a 2D DXF bodyplan is to shift each station to its correct longitudinal position:

- Choose one section that remains at the Z = zero plane eg the section at the Aft or Fwd perpendicular. This will be your zero point.
- In AutoCAD select the adjacent section line
- Type <M> in the command line and specify <0,0,1500> to move the section for example 1500 mm in the Z-direction.
- Do this for each section.

Importing IGES Curves and Surfaces

To import NURB curves and surfaces into Modeler, use the Import IGES Surfaces and Curves command from the File menu. If you are starting a new design by importing an IGES file, make sure you create a new design first from File | New.

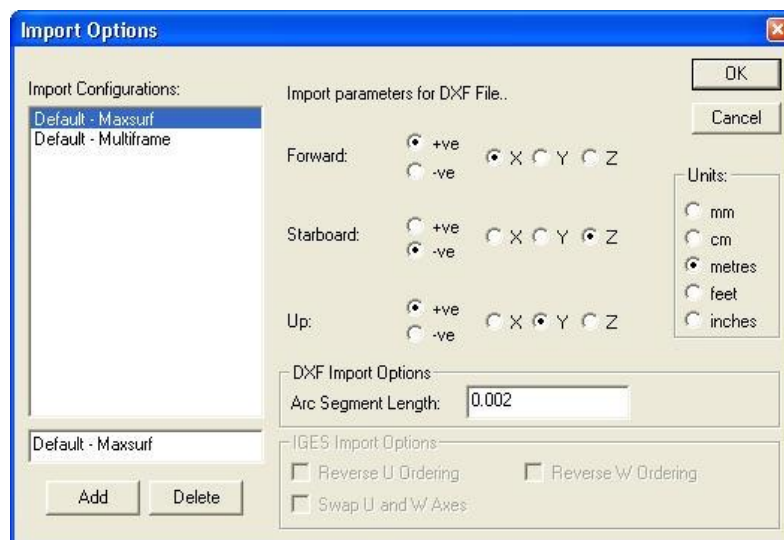
IGES stands for the Initial Graphics Exchange Standard. It is an extremely powerful format and is compatible with the majority of major CAD programs. IGES trimmed NURBS surfaces are represented as an entity 128, which contains the surface geometry, together with an entity 144, which contains only the trimming information.

Modeler can only read the entity 128 data, but does not support importing the entity 144 trimming data.

Note: importing trimming surfaces

Modeler does not support importing trimmed surfaces. All surfaces have to be re-trimmed manually. Also see: [Importing Rhino .3dm files](#) on page 338 for Rhino data exchange options that support trimming.

Each NURB surface found in the file will become a surface in Modeler. Each NURB curves will become a Curve in Modeler.



The orientation of the imported model into the Modeler coordinate system depends on the orientation of the axis in the program where the .IGES file originates from.

Note:

See: [Modeler Coordinate System](#) on page 20

The options at the bottom of the dialog can be used to correctly orientate the control point net with respect to Modeler's usual conventions.

Note:

Not all IGES curve types are supported by Modeler. A dialog will appear warning if an unsupported type is encountered.

Importing Rhino .3dm files

To import NURB surfaces into Modeler, you can use the Import | Rhino .3dm file command from the File menu. When importing a Rhino file which has trimmed surfaces a dialog box will appear asking you whether you wish to try and replicate the trimming.

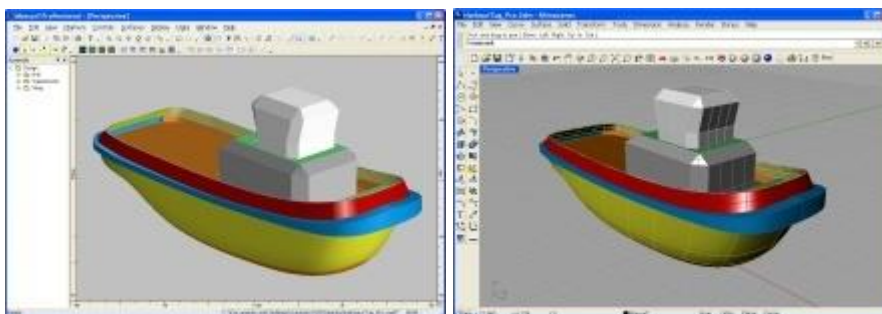
Each NURB/Mesh surface found in the file will become a NURB/Mesh surface in Modeler. Currently NURBS surfaces, NURBS curves, mesh surfaces and Markers are the only entities saved/opened via the .3dm format.

The orientation and dimensions of the imported model into the Modeler coordinate system depends on the orientation of the axis and units set in the 3dm file where the model originates from.

General Guidelines for Importing Rhino models into Modeler

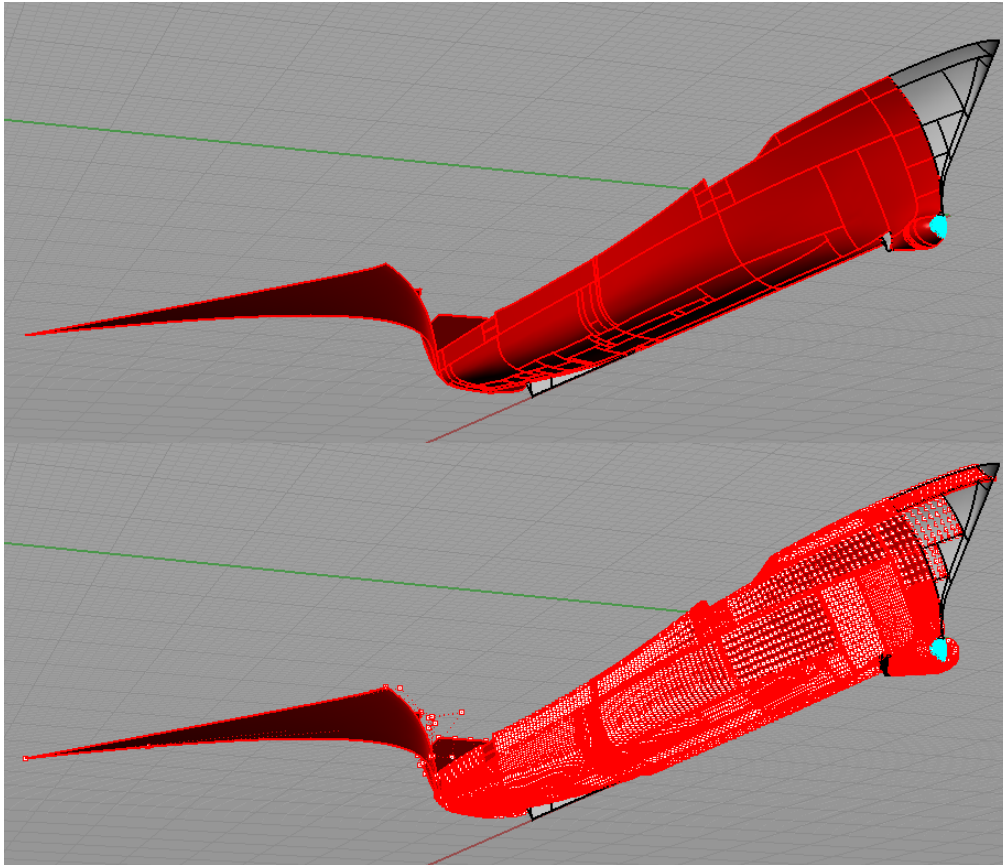
Multipurpose 3D NURBS surface modelling programmes like Rhino are fantastic modellers that allow the user to quickly achieve some impressive shapes. However, these programs are not necessarily ideally suited for the creation of faired hullshapes. Models can easily end up with large numbers of trimmed or hacked up surfaces with 1000's of control points. This is fine for most shapes but for hullshapes that need to be faired it becomes difficult, if not practically impossible to fair the hullshape.

Rhino is however a very useful software tool to complement Modeler. Our recommended workflow approach is to start designs in Modeler to create the major hull surfaces which require highest fairness, and then to use Rhino for superstructure modelling and for modelling of other features like fillets or deck features and other details that are not important from a stability or other performance point of view; in this way you are playing to the strengths of each tool.



Tug model in Modeler.

Tug model in Rhino.



Example of a hull modelled in Rhino large numbers of surfaces with large numbers of control points; this is when it becomes difficult to manipulate and fair the model.

However, if you have existing models in Rhino, the following sections provides some general guidelines on successfully bring Rhino models into Modeler.

Units

Ensure that the model in Rhino is at full scale, because Modeler expects to be dealing with real vessels, models should be of a reasonable size. If you are actually modelling a scale model of a vessel then consider scaling it up so that it is at least a couple of metres long. So check the size and units of the model in Rhino before attempting to read the file in Modeler.

Appropriate level of detail

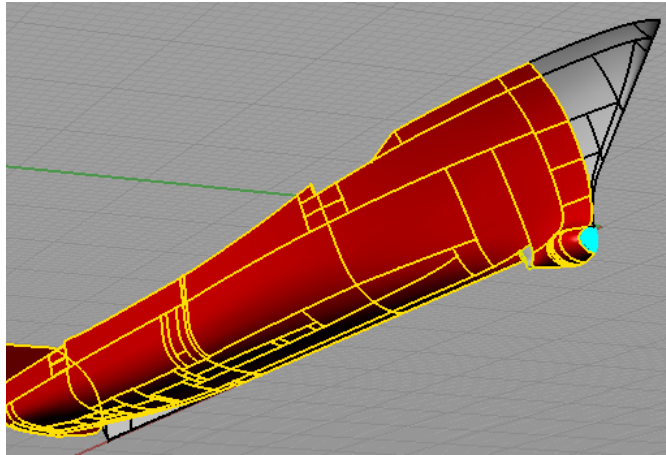
Delete all parts of the Rhino model that do not contribute to the watertight envelope of the hull or those that are required to model complex tank shapes. Details like portholes, deck stanchions, sails, standing rigging, propellers etc should generally be removed from the model: if a surface does not contribute to one of the following then it should generally be removed:

- Contribute significant buoyancy to the watertight envelope of the hull
- Define complex tank geometry (eg spherical or cylindrical tanks, etc)
- Define shapes that are to be expanded for plate-cutting. However if these are superstructure plates, consider saving these into different files.
- Modeler is limited to a maximum of 1200 NURBS surfaces, if the Rhino model exceeds this number they will be ignored by Modeler.

Explode Manifold, Poly and other Compound surfaces

Modeler, can only read standard NURBS surfaces from the Rhino file, so if you have any compound surfaces in the Rhino file, explode them before attempting to read into Modeler. A manifold surface in Rhino is

basically a group of surfaces. Modeler cannot import manifold surfaces and an error message will be displayed.

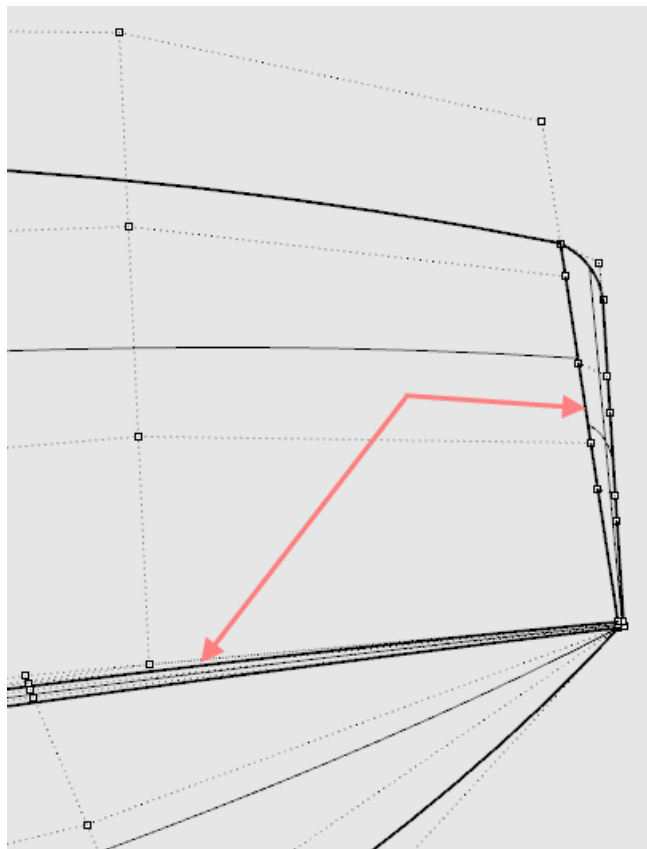


Manifold surface in Rhino

Manifold surfaces need to be exploded into the component surfaces in rhino prior to importing the model into Modeler.

Do surface edges really touch?

In some cases Rhino will approximately fit surface edges; at a distance it seems that the surfaces share and exactly common edge, but actually the surfaces have a different number of control points or the control points do not match. In Modeler, when bonding surface edges, the two surfaces are forced to have identical edges: the number and location of the edge control points must match as must the surface stiffness in that direction:

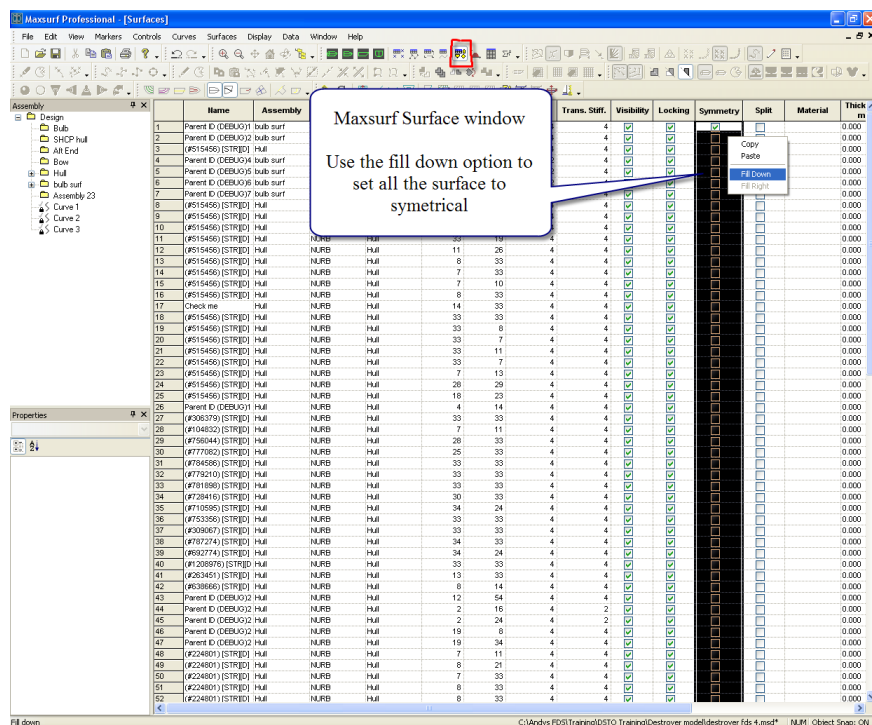


Indicated edges are not necessarily of identical shapes in both surfaces which are supposed to share a common edge

Symmetry

Modeler can automatically replicate port and starboard symmetry. So if the model is symmetrical port and stbd - delete one side of the model in Rhino prior to importing into Modeler, this halves the number of surfaces you have to deal with. You can delete either side of the hull. However, we would recommend deleting the port side simple because the Stbd side is +ve in Modeler and is easier to work with.

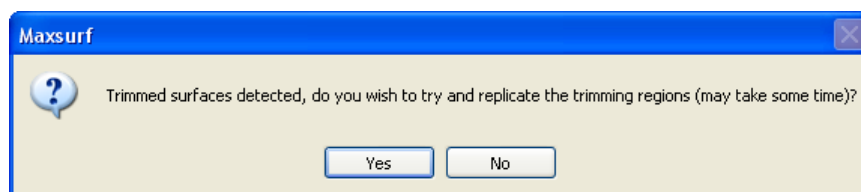
When you bring half the model into Modeler make sure you remember to set the surface properties to 'symmetrical'. If you have a model with 1000's of surfaces then the best way is to go to the surface window and set one surface to symmetrical and then use the right click 'fill down' option.



Trimming Guidelines

There are some fundamental differences with the way the trimming works in Modeler and in other 3d modelling programs like Rhino – Modeler attempts to replicate the rhino or iges trimming but it does not always work because the different programs follow different rules.

When you bring a trimmed rhino model into Modeler a dialog will ask you if you want to replicate the trimmed regions.

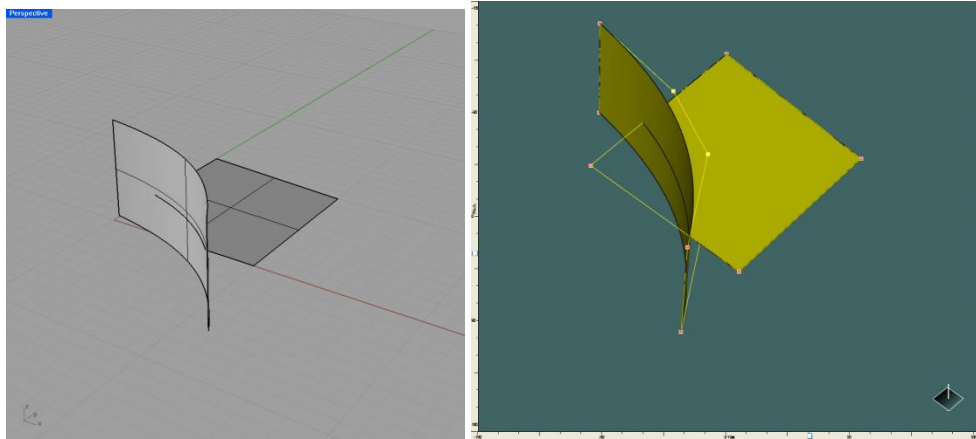


If you click 'Yes' Modeler will attempt to replicate the trimming regions as per the rhino model. If you click 'No' the model will come into Modeler untrimmed and the user can then re-trim the model in Modeler if required.

If at all possible leave the cutting objects in the Rhino file. As you will see below this will make it easier to recreate the trimming in Modeler.

Surface to Surface trimming

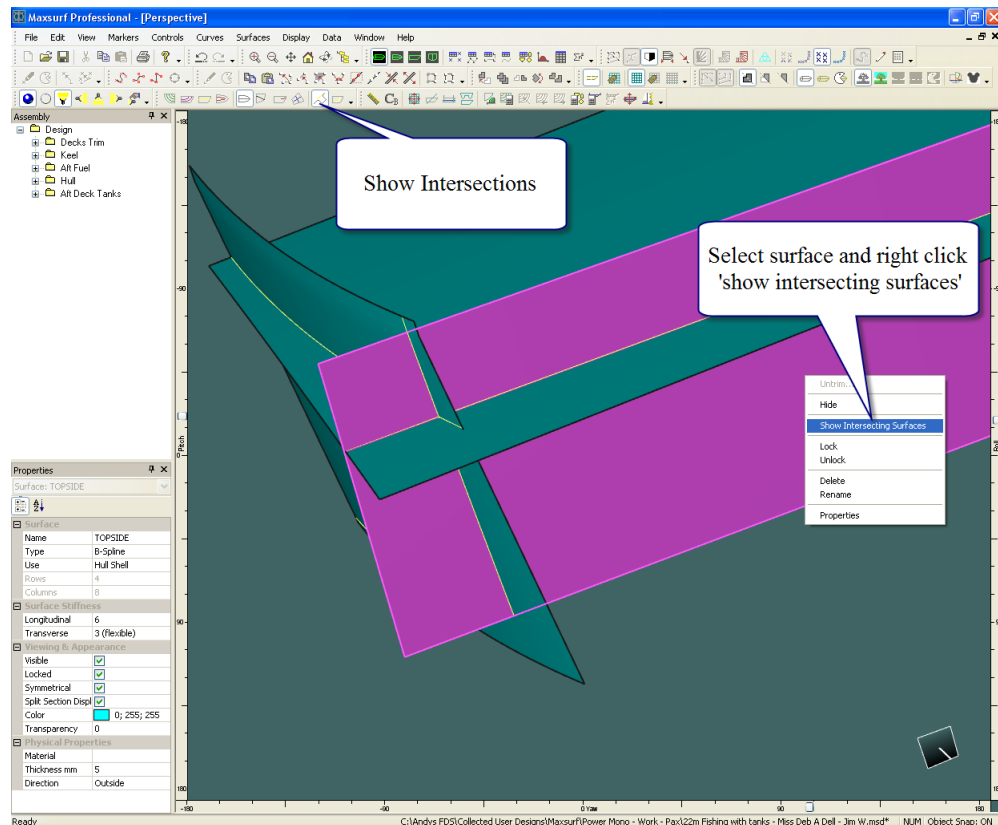
If the rhino model trimming is surface to surface trimming then there shouldn't be a problem when importing the model into Modeler, provided there is an intersection between the 2 surfaces.



Surface to Surface intersection in rhino and Modeler

To check that there is a significant surface intersection you can view the surface intersections in Modeler using the 'show intersections' command (Display – contours – intersections).

Another useful tool is selecting the surface, then right clicking the mouse and selecting 'show intersecting surfaces'. This allows you to concentrate on the surface intersections for one particular area.



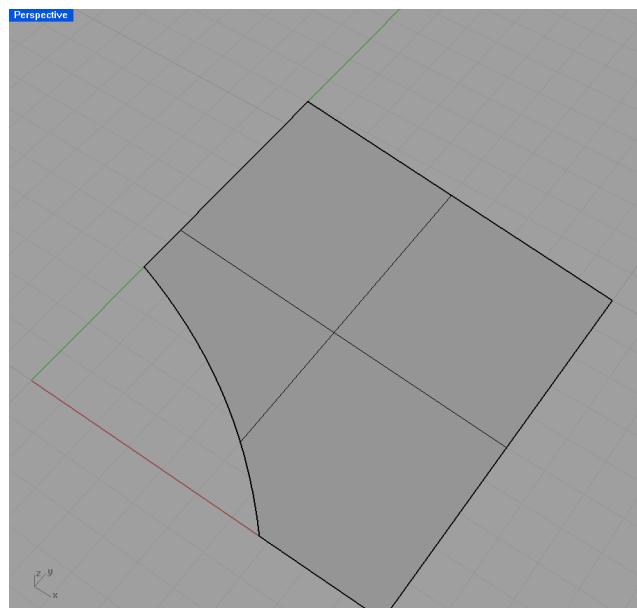
View only those surfaces which intersect the selected surface

Sometimes it is simpler (and you end up with a better model) to ignore the trimming information in the Rhino file and then recreate the trimming in Modeler:

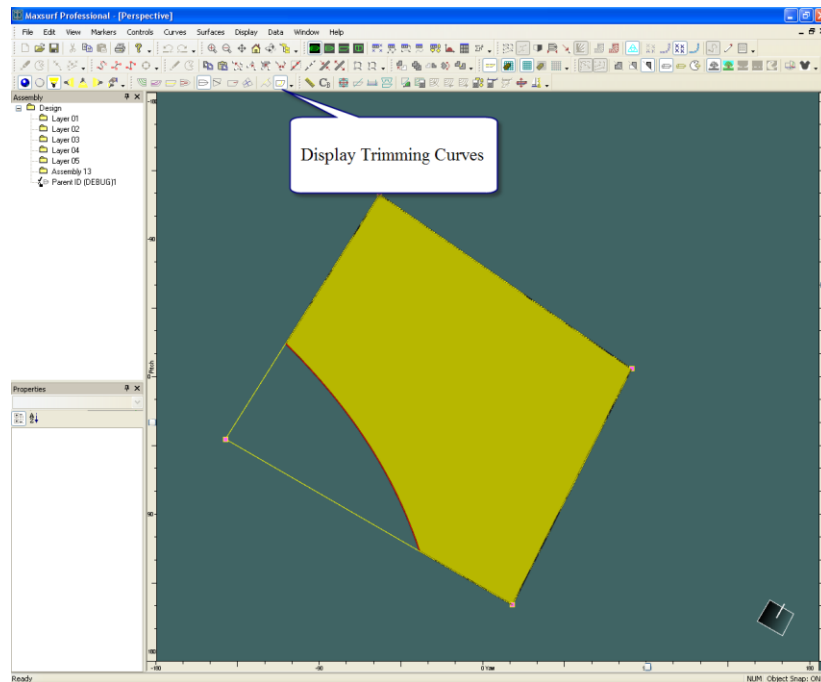
- Read in the Rhino model but choose not to recreate the trimming regions
- Remove all cutting surfaces from all of the surfaces (easily done in the Surfaces table by removing all cutting surfaces from the first surface, then using the fill-down command)
- For each surface that needs to be trimmed, first specify the correct cutting surfaces, then trim the surface.
- Troubleshooting: When surfaces do not trim as expected, check that surfaces intersect properly and that the selected cutting surfaces form closed intersection boundaries which completely subdivide the surface to be trimmed into separate regions.

Rhino trimmed surfaces without surface to surface intersections

For surfaces that are trimmed in Rhino but do not appear to have any trimming curves or trimming surfaces associated with it Modeler will try and replicate the trimming by putting a curve on the surface where the surface is trimmed in Rhino. You can view these curves in Modeler by displaying the trimming curves (Display – contours – trimming curves).



Single trimmed surface in Rhino with no other surfaces



Trimmed surface above in Modeler with the replicate trimming regions option

Modeler by default associates no cutting surfaces with each rhino surface that is imported. This can be manually altered by the user. For more information refer to the section on [cutting surfaces](#).

Note: Importing trimming surfaces

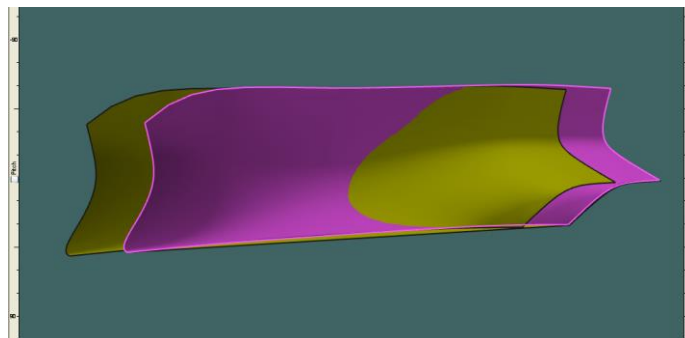
On importation of a trimmed model from Rhino, trimming curves that do not necessarily lie on a surface/surface intersection may be present. These contours can be used to trim the surfaces as any normal intersection line. Visibility of these Rhino trimming contours may be turned on and off via the Display | Contours dialog.

ShrinkTrimmedSrf Command in Rhino

Prior to importing a trimmed-surface rhino model into Modeler it is often very useful to run the ShrinkTrimmedSrf command on all the trimmed surfaces in Rhino. This generally creates a much more robust model and reduces the size of the file.

Duplicate surfaces

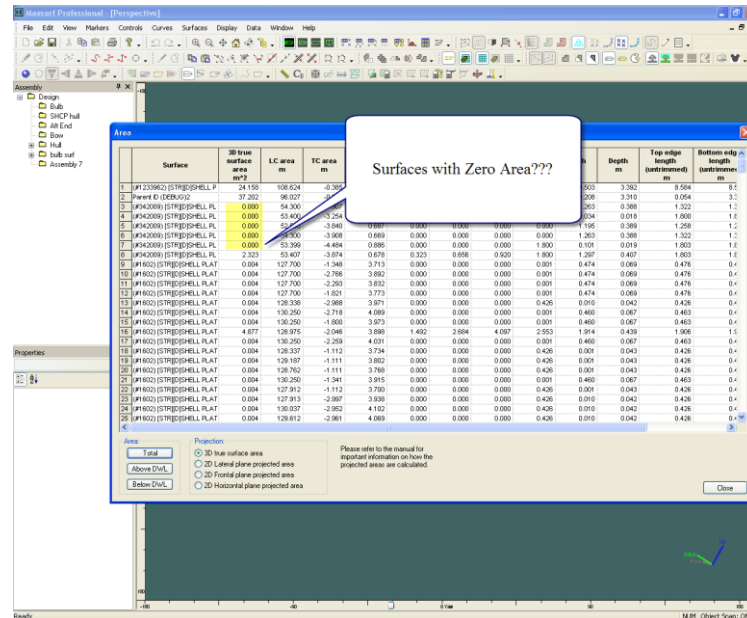
If a model has been brought into Modeler with surfaces lying exactly on top of other surfaces, then when it comes to trimming all bets are off. The Modeler trim engine cannot calculate the trimming regions hence it doesn't know what to trim. You can fix this by carefully selecting the Cutting surfaces for each surface in Modeler



Example of Rhino model imported into Modeler with 2 surfaces lying on top of one another

Zero Area surfaces

When you bring a rhino model with large numbers of surfaces into Modeler it is worth checking the surfaces areas, as it is possible to create surfaces in Rhino with zero area. In Modeler select the 'calculate areas' command from the data menu and check that there are no surfaces with Zero area as this has been know to cause problems with Modeler trimming.



Plan B - Rhino data Points to Markers – Create a TriMesh Surface in Modeler

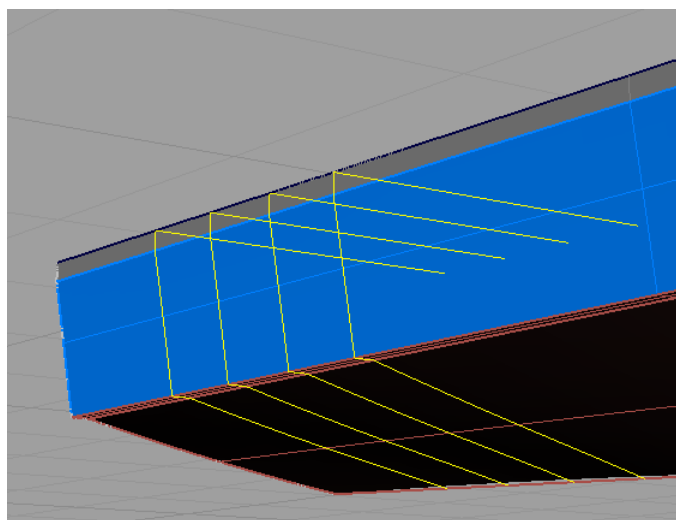
If all else fails and you are not having any success in bringing the Rhino model into Modeler then it is possible to get the offsets out of rhino, then bring these offsets into Modeler as Markers. In Modeler, the Markers can be used either to fit NURBS surfaces (see [Working with Markers](#) on page 80) or to generate a TriMesh (see [Mesh Surface](#) on page 235). The steps below describe one method of creating offsets, exporting them from Rhino and bringing them into Modeler as Markers.

- In Rhino, Step 1

You can create sections on the surface in Rhino by:

Curves – Curve from Object - Sections

You then select the surfaces (that you want a section on) and then draw a vertical line where you want the section. This command can be done on multiple surfaces.



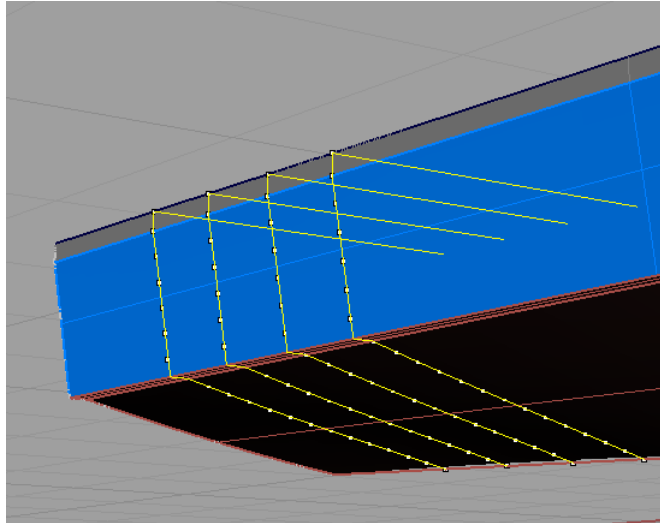
Adding section curves to a Rhino model

- Step 2

Now one you have drawn the sections on the Rhino surfaces you can create data points along this section curve. This can be done by;

Curve – Point object – Divide curve by – (either length of segments or number of segments)

This yields data points along the curve. (Note this can be done for multiple curves in one command)



Data points created along the section curves in Rhino

- Step 3

Now you can export the curves point data from Rhino

You can do this by selecting all the points – (edit – select – points)

Then either – File – export selected - export as a .txt file

This file can then be opened in Excel.

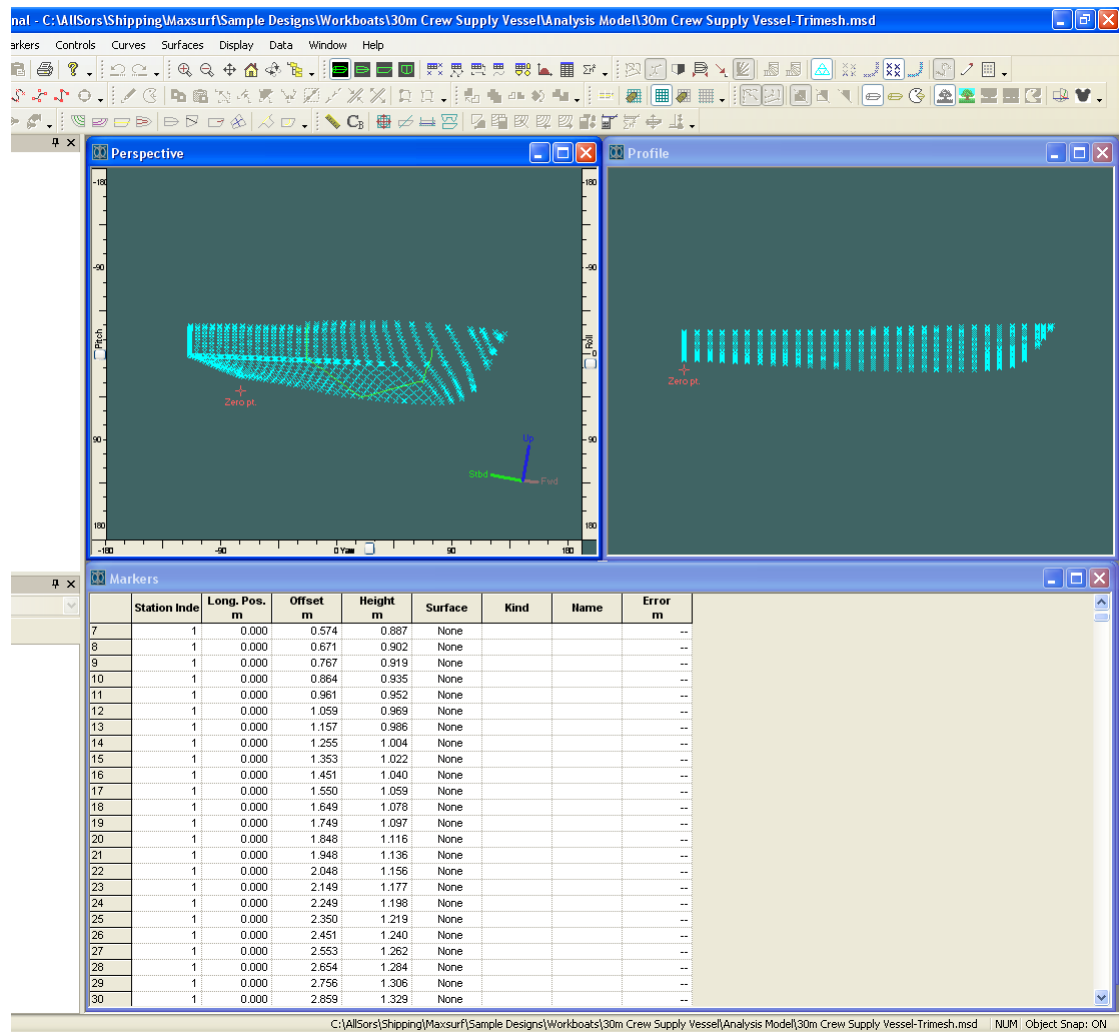
Alternatively find a macro that does this for you ...

<http://wiki.mcneel.com/developer/scriptsamples/exportpointstoexcel>

- In Modeler, now that you have the point data in Excel

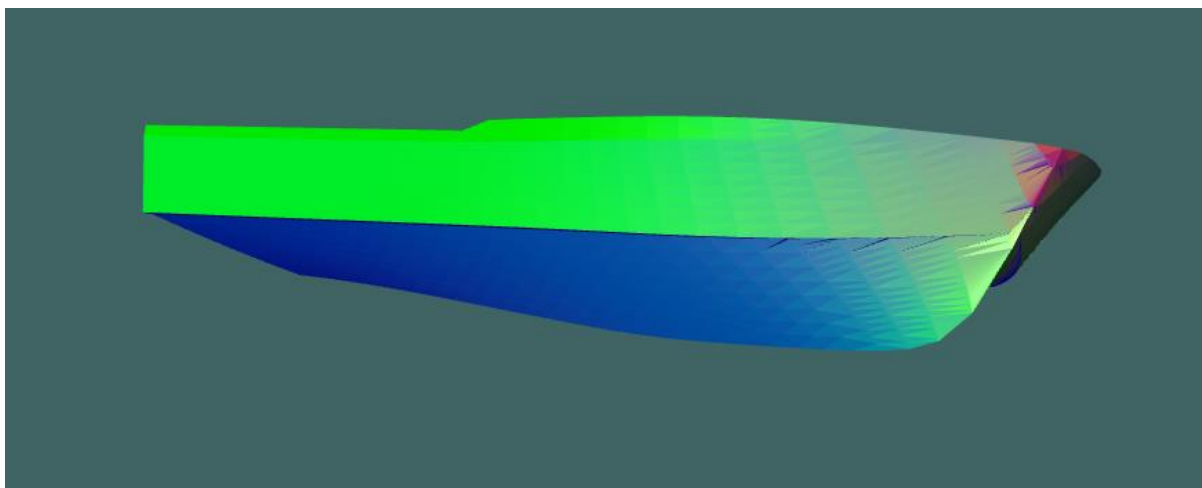
Launch Modeler – Open a new design – Go to the Markers window – add markers (add the number of markers that you have data for in excel)

Then copy and paste the marker data from excel into the Modeler Marker window.



Rhino offset data brought into Modeler as Markers

- Sort the Marker data and generate a TriMesh surface in the usual way
- For more information see, [Mesh Surface](#) on page 235 .



TriMesh Model in Modeler

Output of Data

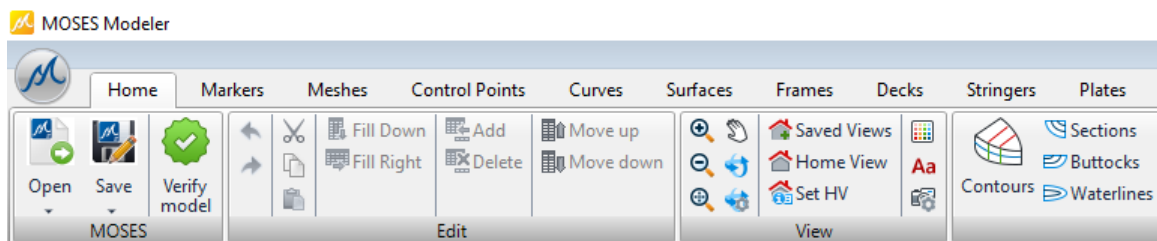
Modeler can output your design lines in a wide variety of file formats and hard copy. File formats available are 2D IGES, 3D IGES, 2D DXF, 3D DXF, 3DMF, 3dm (Rhino) and VRML. Lines from any of the drawing windows may be saved as files or output directly to printers. Data from the Calculations, Markers, Control Points and Offsets files can be printed, saved to text files or copied into the clipboard for pasting into spreadsheets or other applications. Modeler also allows saving of animation files.

This section will describe the following output options:

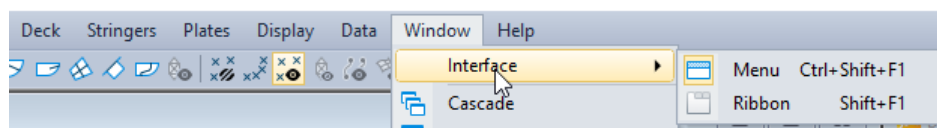
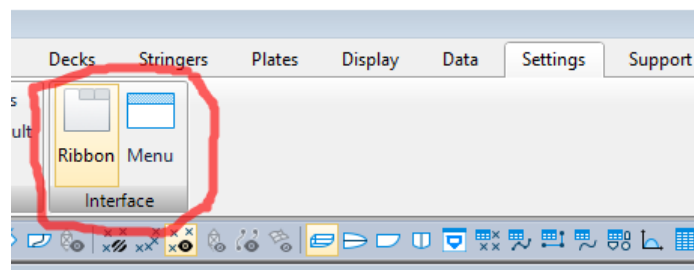
- [Printing](#)
- [Copying](#)
- [Offsets Data Output](#)
- [Animation Files](#)
- [Exporting a Modeler Design](#)

Also see:

- [Ribbon](#)
- A command Ribbon is available in SACS and MOSES versions.



Example of the Ribbon interface



Select the interface you find the most convenient

- Appendix A – Data Export

Printing

Modeler prints the contents of the window that is topmost on the screen. Choosing Print from the File menu will display a print preview, allowing you to view the output, page-by-page, on screen.

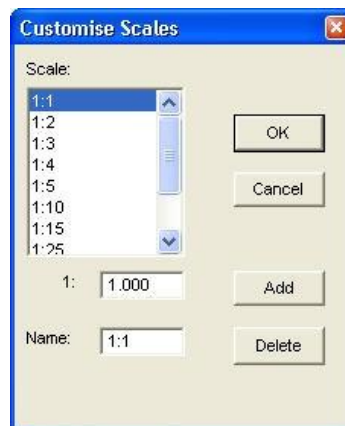
Use Page Setup, also from the File menu, to adjust the paper size, margins, orientation etc.

Standard Views

When printing a standard view of the design, you will be asked select the scale at which you wish to print. There are a number of default scales but you may also enter your own custom scale factors.

- Bring the window you wish to print to the front.

- Choose Print from the File menu.
- Choose the desired Scale.
- If the print preview is as required, click the Print Button.



Printing from Modeler is straightforward and is usually a matter of setting the Page Setup parameters and selecting Print. If you encounter problems, check that the following have been done.

- Make sure your printer is connected to the computer.
- Make sure you have selected a default printer using the Printer control panel

Rendered Perspective Views

Printing a rendered perspective view is the same as printing a standard view except you will not be prompted for a custom scale. Rendered perspective views can only be printed on one page.

The image displayed in the print preview window may have some slight differences from the final printed version, this is only due to scaling in the print preview window.

Also see:

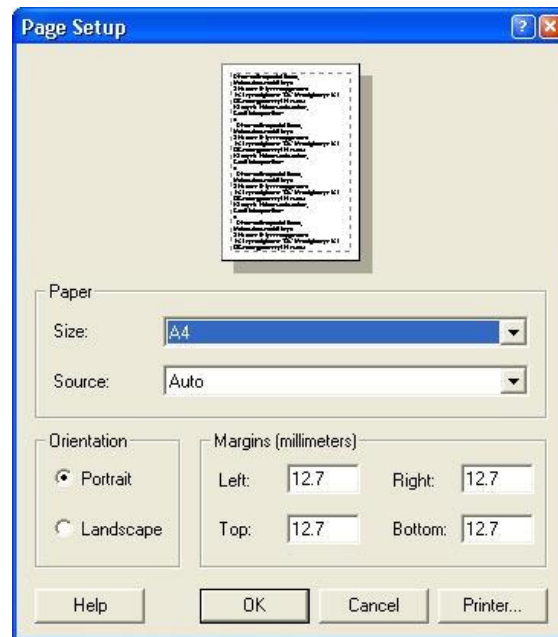
[Exporting a Modeler Design](#), section: [Bitmap Image](#) on page 359.

Color Printing

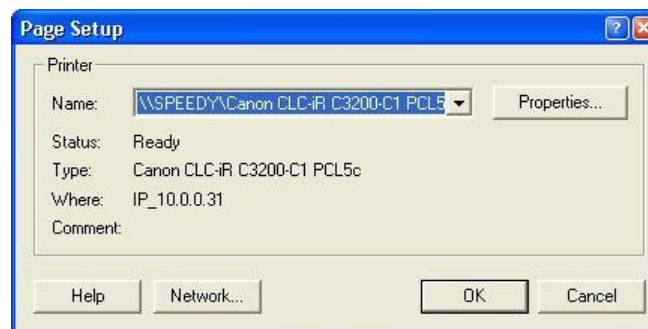
Modeler supports color output to suitable devices. You may choose whether you wish to print in color or black and white in the print preview by clicking the Color button.

Setting Page Size , Orientation and Margins,

The paper size, paper orientation and margins may all be set in the Page Setup dialog accessed from the File menu.



You may choose the printer you wish to print to by clicking the Printer button.



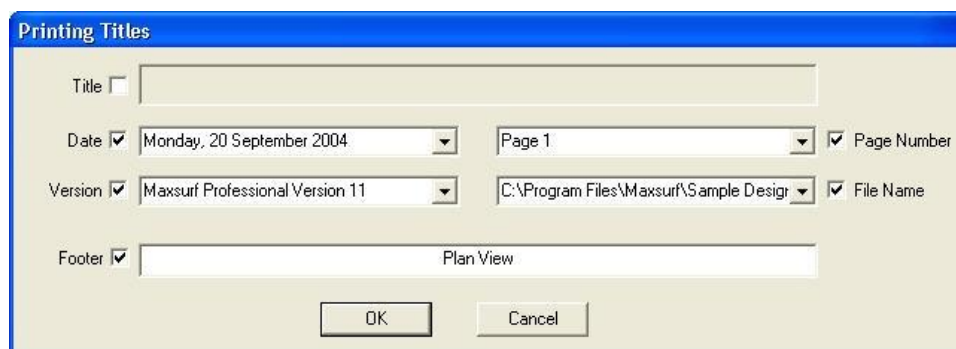
Further printer specific options may be chosen by clicking on the Properties button. The options available will depend on the printer and the printer driver you are using.

Print Titles

You can specify titles to be placed on the head and foot of each page you print. Use the Titles button in the print preview window:



This will bring up the Titles dialog, which allows you to set your print titles.



Other Data Output options include:

- [Copying](#)
- [Offsets Data Output](#)

- [Animation Files](#)
- [Exporting a Modeler Design](#)

Copying

When copying from a drawing window

- Select the window you wish to copy.
- Select Copy from the Edit menu (or use Ctrl C).

To paste this selection into a word processor or spreadsheet

- Select Paste from the Edit menu (or use Ctrl V).

Copying Numeric Data

When copying from a data window:

- Select the appropriate column by clicking in the column title bar.

or

- Select the appropriate row by clicking on the row number.

or

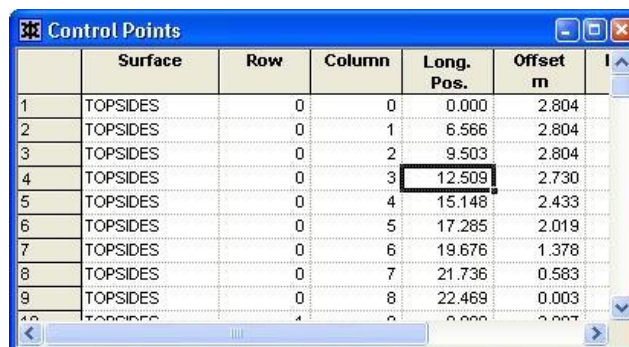
- Select the whole window by clicking in the top left hand corner.



	Station Inde	Long. Pos. m	Offset m	Height m	Surface
1	22	-43.029	4.050	2.900	None
2	22	-43.029	3.950	2.610	None
3	22	-43.029	3.660	2.160	None
4	22	-43.029	2.550	1.900	None
5	22	-43.029	1.820	1.870	None
6	22	-43.029	1.219	1.845	None
7	22	-43.029	0.000	1.805	None
8	21	-41.952	4.320	3.400	None

- Select Copy from the Edit menu (or use Ctrl C).
- To copy the column titles as well as the selected data, hold down the Shift key while selecting Copy (or use Shift Ctrl C)

It is also possible to select arbitrary blocks of text rather than complete rows or columns or individual cells.



	Surface	Row	Column	Long. Pos.	Offset m	I
1	TOPSIDES	0	0	0.000	2.804	
2	TOPSIDES	0	1	6.566	2.804	
3	TOPSIDES	0	2	9.503	2.804	
4	TOPSIDES	0	3	12.509	2.730	
5	TOPSIDES	0	4	15.148	2.433	
6	TOPSIDES	0	5	17.285	2.019	
7	TOPSIDES	0	6	19.676	1.378	
8	TOPSIDES	0	7	21.736	0.583	
9	TOPSIDES	0	8	22.469	0.003	

To achieve a rectangular selection, hold down the mouse button in the cell that you wish to be a corner of the required selection. Drag the mouse in any direction and multiple cells will highlight. If you drag the mouse outside of the table boundaries, the cells will automatically scroll in the correct direction.

Alternatively, click in one cell, then hold down the Shift key and click in a second cell. All cells between the two chosen will be selected.

To paste into the selected cells, choose Paste from the Edit menu (or use Ctrl V).

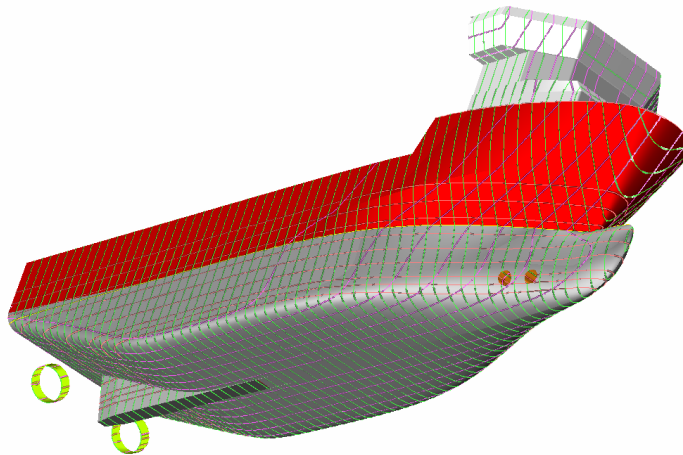
Other Data Output options include:

- [Printing](#)
- [Offsets Data Output](#)
- [Animation Files](#)
- [Exporting a Modeler Design](#)

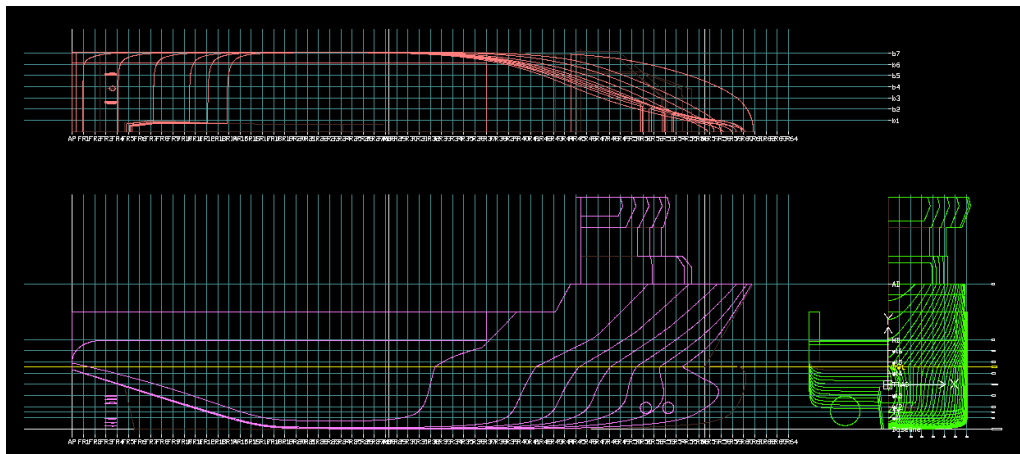
Export Lines Plan

An auto generated dxf lines plan can be generated from the NURBS model. The function is executed by choosing File | Export | Lines Plan... from the menu. All contours that are visible will be displayed in the lines plan. The lines plan will show the three orthogonal views, with each group of contours (sections etc) displayed on individual layers in the dxf file. This function is mainly used to export Water lines, Buttock lines and Section lines (and maybe edges contours) to the lines plan.

A typical example for the NURBS model below:



Produces the following dxf lines plan:



Offsets Data Output

The following output option exist for offset data:

- Copied and pasted into Excel
- Saving the table of offsets as a text file from the Offsets window
- Using Link.

Using Link to Export Offsets Data

Modeler supports saving of offset data in a format suitable for input to other analysis systems. This is done by Link, a separate application in the Modeler suite. Link can read in a Modeler design and save the offset data in a format suitable for input into many other analysis systems.

Other Data Output options include:

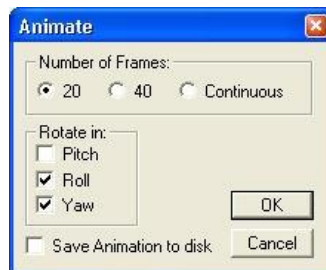
- [Printing](#)
- [Copying](#)
- [Animation Files](#)
- [Exporting a Modeler Design](#)

Animation Files

Animations generated in the perspective view may be saved in avi format for playback at a later date or for use in presentations etc.

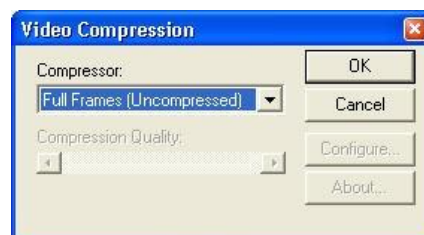
To generate an animation:

- Go to the perspective view and adjust the display, rendering and colors to obtain the desired view.
- Select Animate from the Display menu, set-up the desired animation options and tick the Save Animation to disk box.



- Click OK and then choose a file name for the animation file.
- Click Save and then select the Compressor type.

Please note that some of the compressors will not work with all graphics settings. The Full Frames (Uncompressed option) should work with all graphics settings.



- Click OK

The animation will now be generated and saved to the file. This may take a few seconds, particularly if there are many surfaces to be rendered in the model.

Other Data Output options include:

- [Printing](#)
- [Copying](#)
- [Offsets Data Output](#)
- [Exporting a Modeler Design](#)

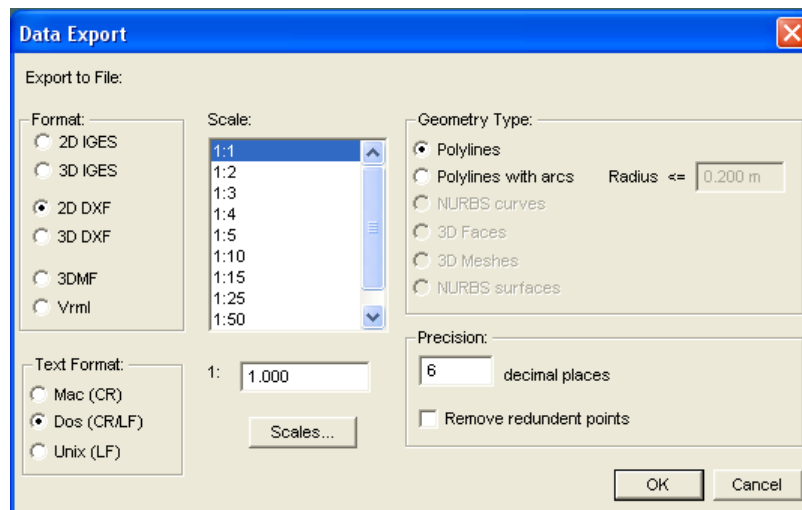
Exporting a Modeler Design

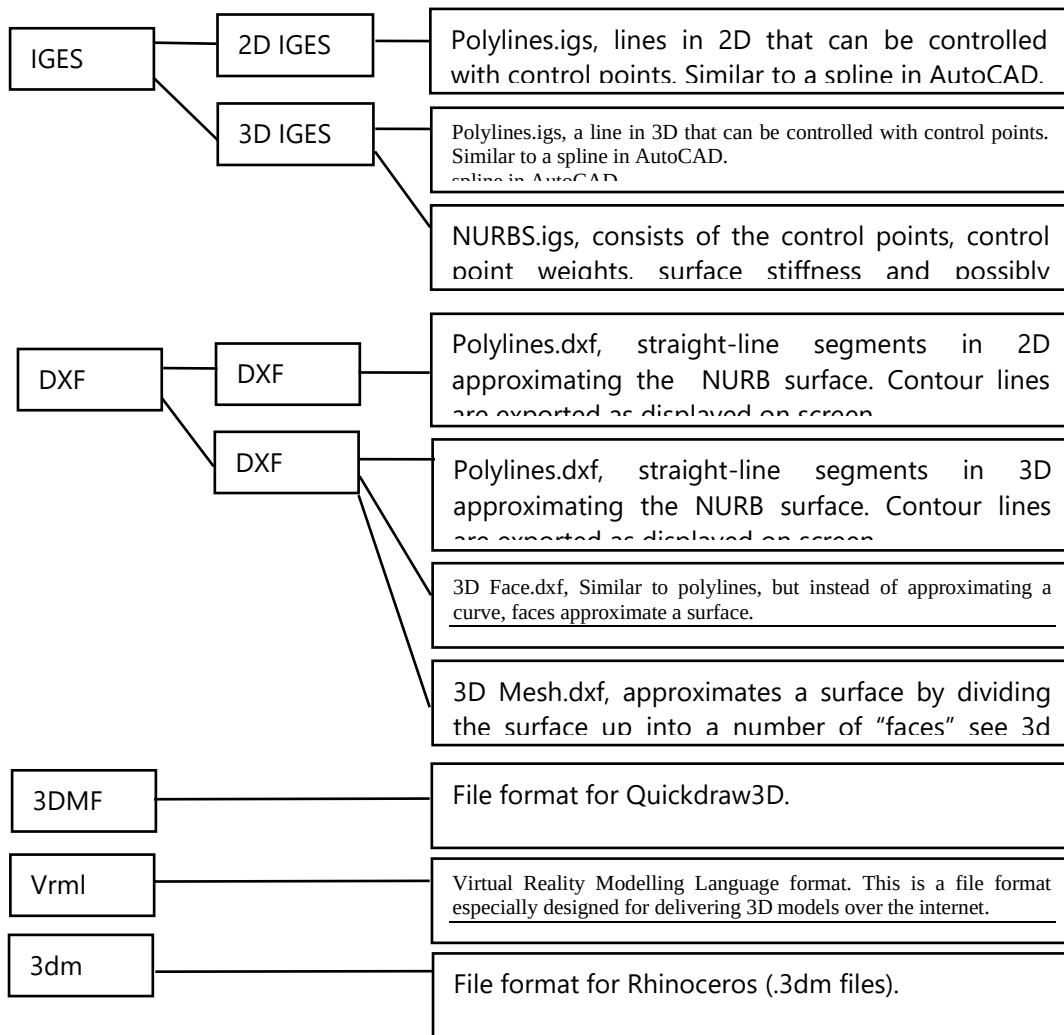
Modeler allows you to export your design file in different formats.

- [DXF File](#)
- [IGES File](#)
- [Rhino .3dm file](#)
- Modeler V8.0 , export to earlier Modeler file formats. Note that this may lead to some loss of data
- Others

You may export the following file formats to other platforms by selecting them from the Data Export dialog box in the File menu:

- IGES
- DXF
- 3DMF
- VrmI
- 3dm

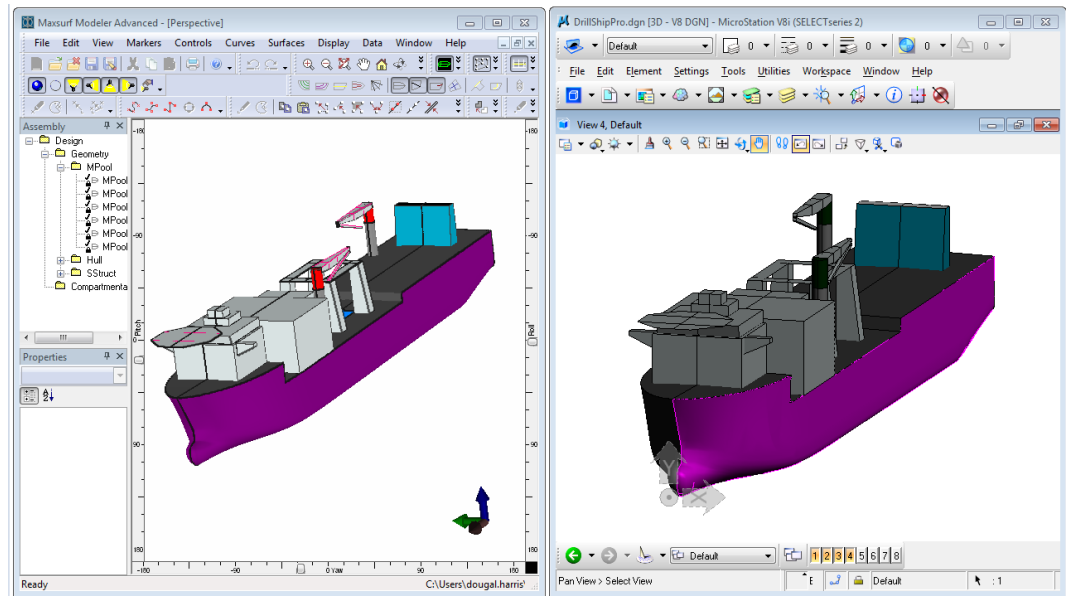




When exporting as a dxf you can choose to "Remove redundant points", this option will remove points that lie on a straight line.

Microstation .dgn File

Round trip interoperability between Modeler and Microstation via the Microstation *.dgn file format is now operational. Entities exchanged include NURBS curves and surfaces as well as mesh surfaces. To import a .dgn model choose File | Import | Microstation.

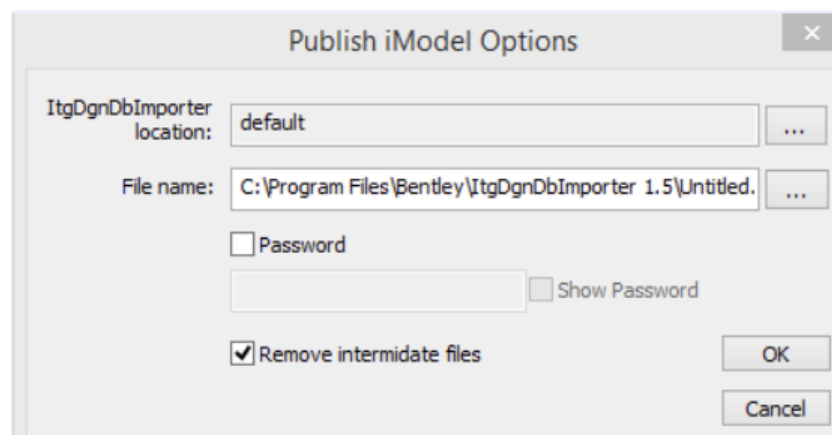


Publish i-model

Bentley Sysytem's i-models can be used to exchange information for projects associated with the lifecycle of infrastructure assets. You can ensure that information flows easily, completely, and accurately between, and within design, construction, and operations environments. The .i-model is compact and optimized for mobile apps. You can review .imodel files with Navigator Mobile or Bentley Navigator Desktop on the Windows platform that contain business data and embedded documents.

To publish an i-model from Modeler:

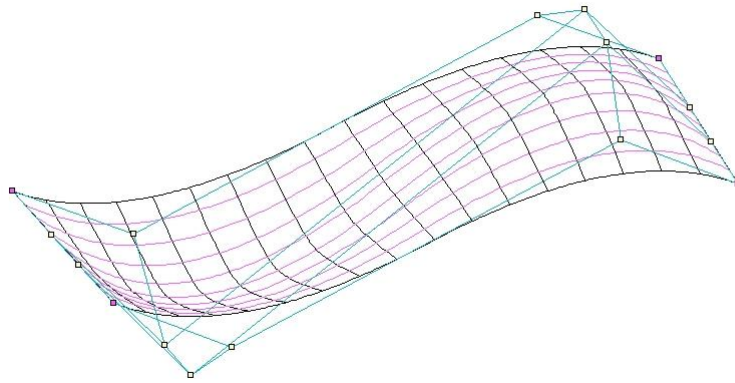
- Choose Publish i-model from the File menu. The following dialog will appear, which allows you to set up i-model options:



Click OK button and an i-model with the .imodel extension will be saved.

IGES File

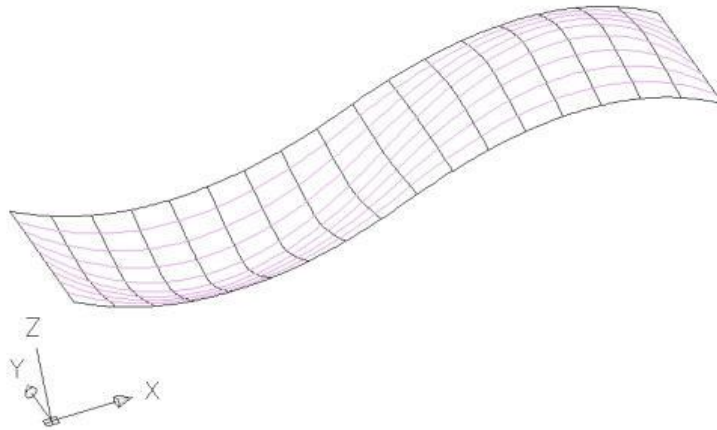
IGES stands for the International Graphics Exchange Standard. It is an extremely powerful format and is compatible with the majority of major CAD programs. Design information may be transferred using a 2D or 3D IGES file. If a 3D IGES file is chosen you have the option of transferring the lines that are visible in the drawing windows (polylines), or using the complete B-spline curve and surface definition (NURBS). For more information on IGES export, please refer to the Appendix on Data Export on page 395.



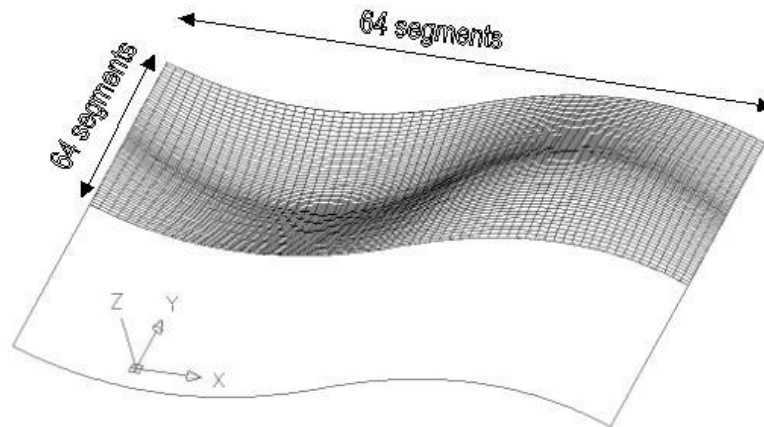
IGES export example. An .IGES file contains all NURBS surface information.

DXF File

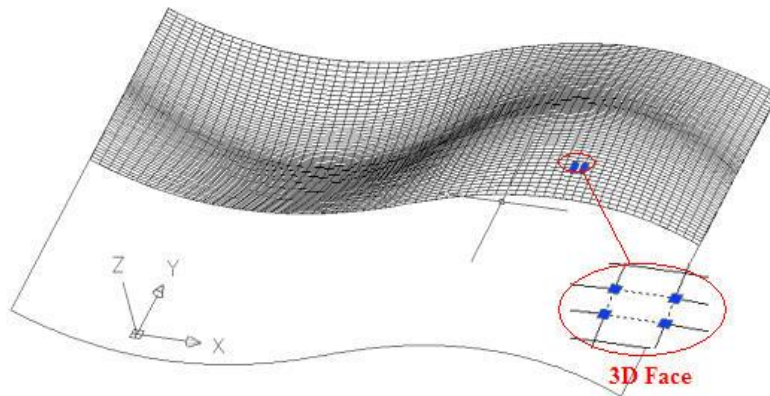
DXF is the Drawing eXchange Format. Design information is transferred using a 2D or 3D DXF file as either a polyline file or a 3D face or mesh. If the "Polylines with arcs" option is chosen a dxf file will be exported that contains lwpolyline, polyline and arc entities. All arcs that have a radius less than that specified in the "Radius" edit box will be exported as arc entities otherwise they will be exported as polyline entities. Only arcs that lie in an orthogonal plane will be exported as arc entities (i.e. XY,YZ or XZ planes).



3D Polyline export example.



3D Mesh export example. The number of segments used to display the surface is dependent on the surface precision.



3D Face export. The mesh is divided up into a number of faces.

3DMF

3DMF is the file format for Quickdraw3D.

VRML

Modeler is able to export models in the ISO standard VRML (Virtual Reality Modelling Language) format. This is a file format especially designed for delivering 3D models over the internet for their display in standard browsers such as Netscape Navigator or MS Internet Explorer. This feature means that Modeler models can be readily included in web pages; this facilitates sharing information on a global (internet) or internal (intranet) level. Further information on VRML is available from the Web3D consortium, whose web site may be found at: <http://www.web3d.org/>

Rhino .3dm file

3dm is the file format for Rhinoceros NURBS modelling software. Both trimmed and untrimmed Modeler NURBS surfaces can be saved in Rhino file format, along with NURB curves, Mesh surfaces and Markers. Because there is no concept of symmetrical surfaces in Rhino, any symmetrical surfaces in Modeler will be duplicated and mirrored about the centreline when exported to Rhino using the Export | Rhino .3dm file command from the File menu in Modeler. 3DM file units will always be in m irrespective of unit settings in Modeler (this doesn't effect the size of the model).

Note

The Modeler assembly tree can be accessed from within Rhino via a Rhino Plug-In. The Plug in is included in the Modeler installer. For more information refer to the Rhino Plug-In Manual (found in the folder \Modeler XX\RhinoPlugInManual.pdf where XX is the version number) which is installed along side the Modeler Manual.

Bitmap Image

Bitmap export gives you the ability to export Rendered Perspective Views to a bmp file. This is similar to Edit | Copy (Copy to Clipboard), however File | Export | Bitmap Image allows you to specify the size of the image; the larger the size, the better the quality. This is a useful feature for creating high resolution promotional material.

This function is not available for the Simple Shading rendered mode.

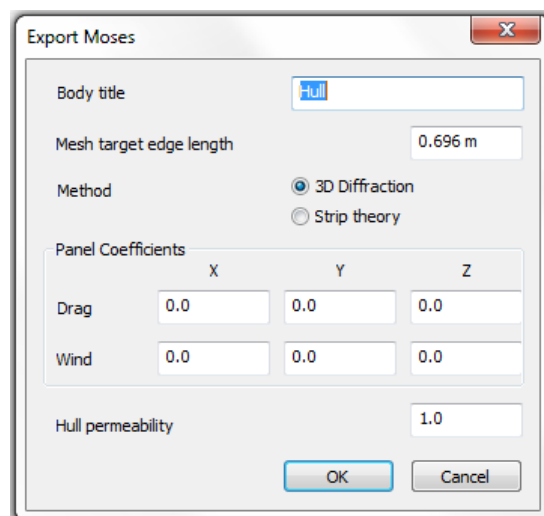
Note

For additional information on data export from Modeler, see: Appendix on Data Export on page 395.

Other Data Output options include: [Printing](#) | [Copying](#) | [Offsets Data Output](#) | [Animation Files](#)

Moses File

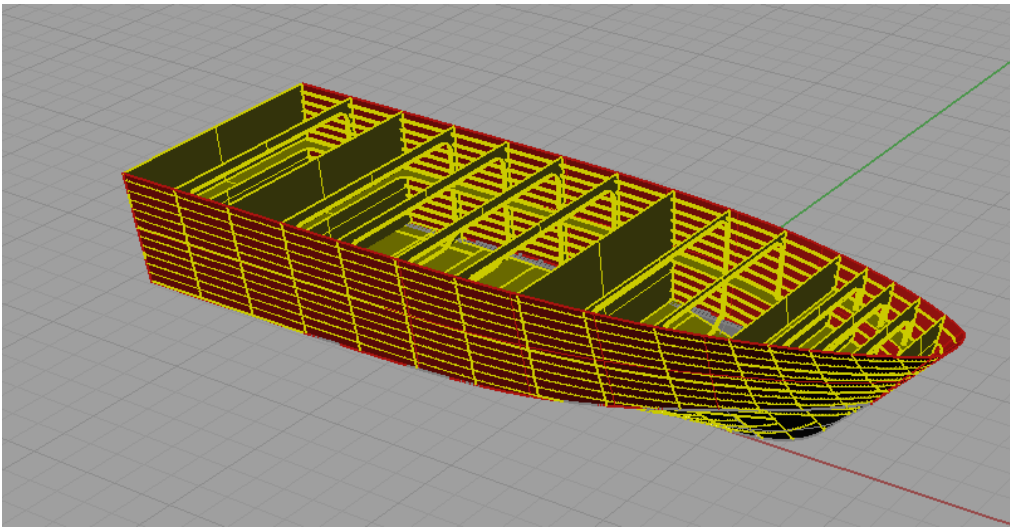
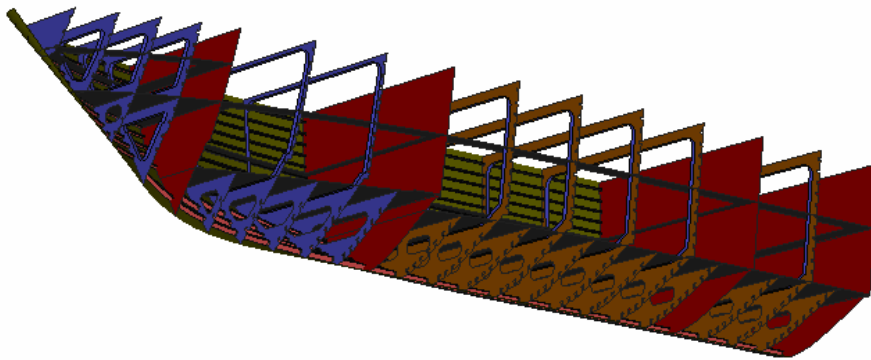
A Modeler file can be exported to the Moses text file format. Choose Moses File from the File menu. The following dialog will appear:



Mesh target edge length refers the average triangle edge length in the mesh the user is targeting. For addition information on the other variables please refer to the Moses manual.

Exporting Parts to Rhino

The parts in a Modeler model can be exported directly to native Rhino .3dm file format. Once a design has been developed past the initial parametric design stage (i.e. hull shape is fixed and all major structural components have been developed) the parts may be exported to a rhino 3dm file for further detailing:



Since Modeler is a parametric modeller (all parts are linked dynamically to an underlying NURBS surface) the data exchange is unidirectional (i.e. from Modeler to Rhino) only. Plates, frames and decks may be exported as trimmed NURBS surfaces or as a mesh to the *.3md file.

Exporting Parts to Multiframe

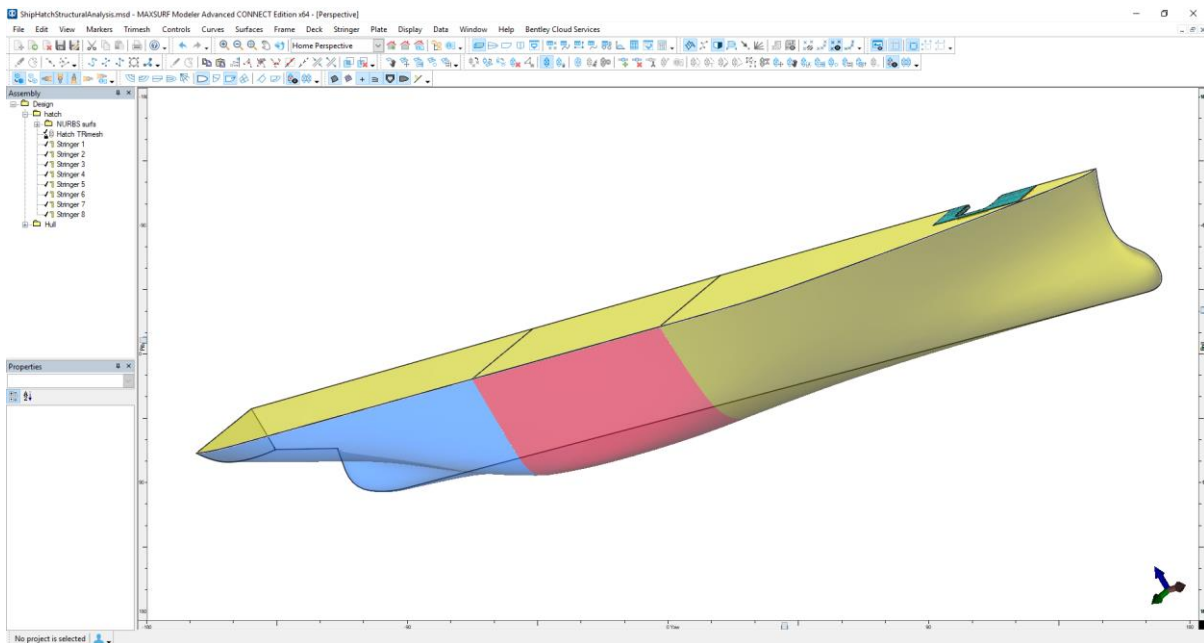
Modeler is able to export the parts (meshes and stringers) defined in the model to the Multiframe text file format. To export to Multiframe

- Show (make visible) only what you want to be exported.
- Choose File | Export | Meshes | Multiframe text file...
- Choose the location of the files to be saved and click OK

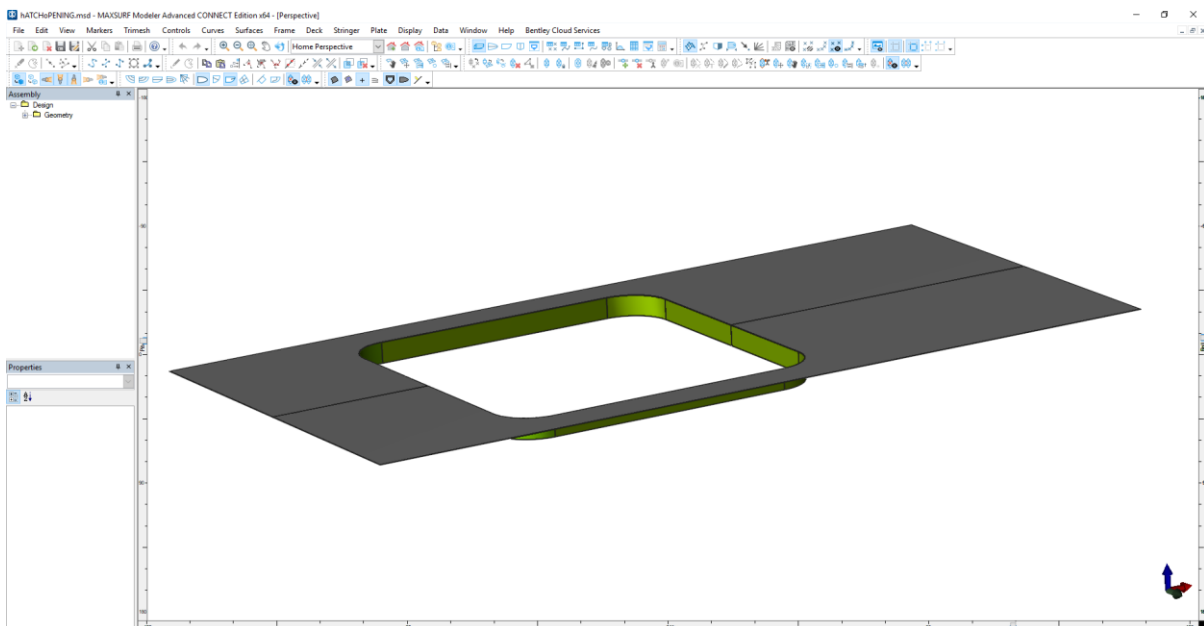
The process will be illustrated in an example below. The basic process is:

1. Model the structure in Modeler Using NURBS
2. Generate meshes on structural parts
3. Assign section shape to stringers
4. Bond stringers to meshes
5. Export to Multiframe text file format
6. Import into Multiframe
7. Assign loads
8. Run analysis

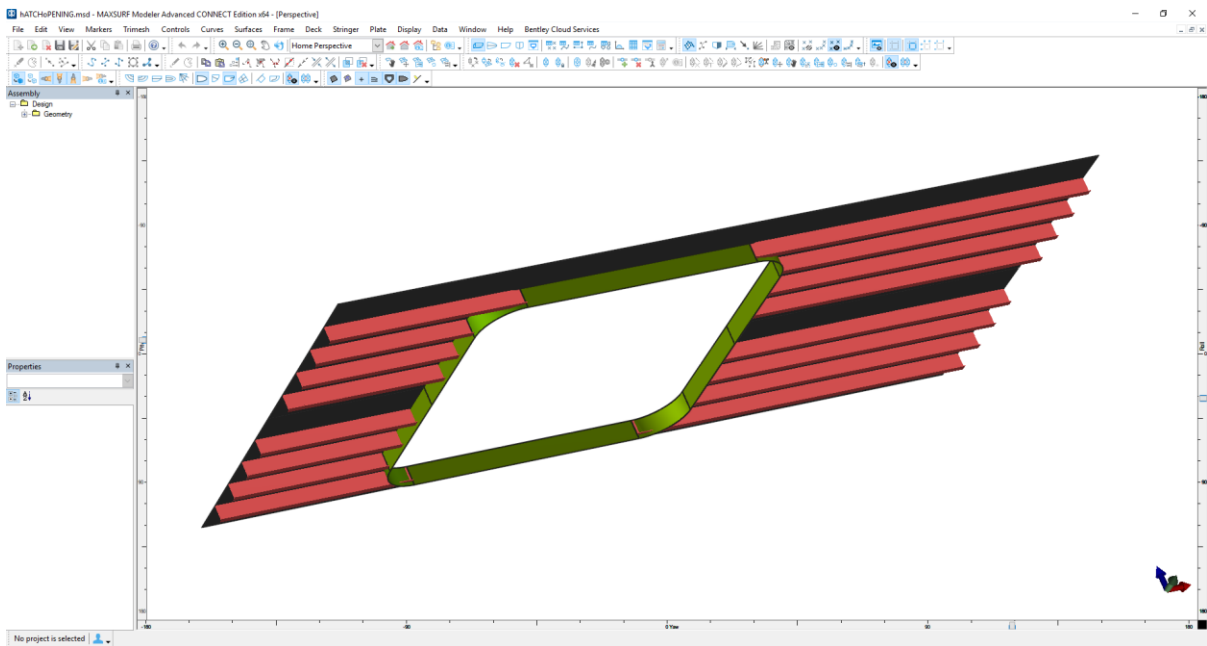
In this example, the deck structure from around a hatch is going to be exported for structural analysis in Multiframe. Here is an image of the complete vessel:



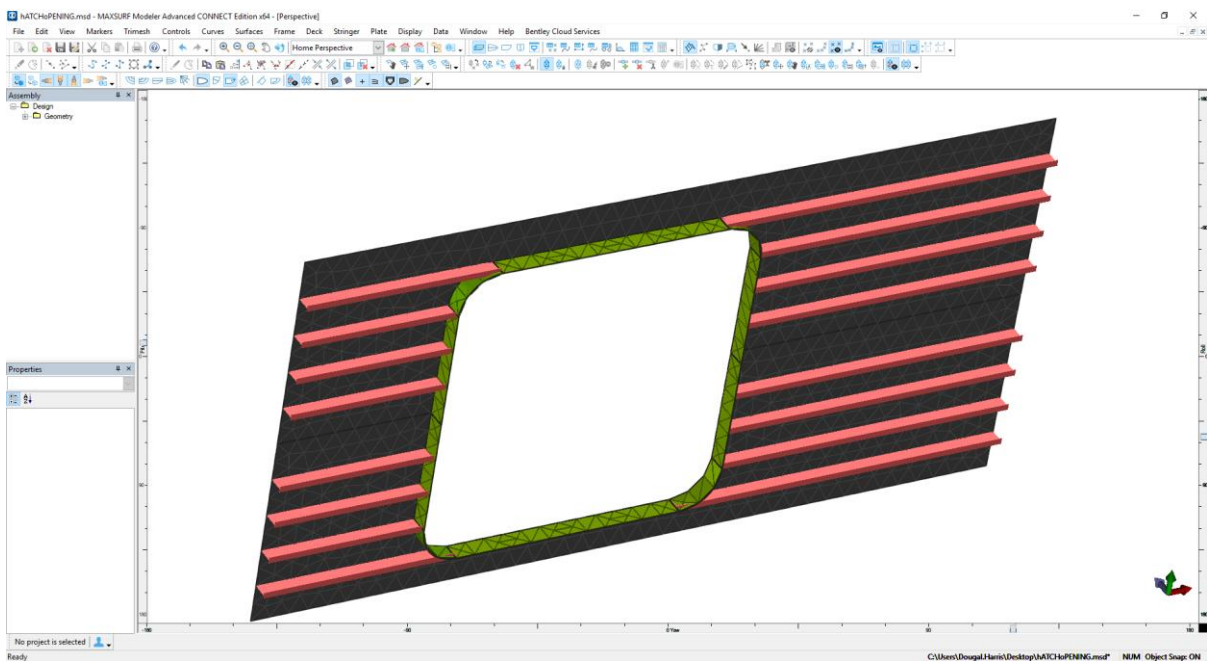
The NURBS surface model of the geometry in the area of interest (hatch at bow):



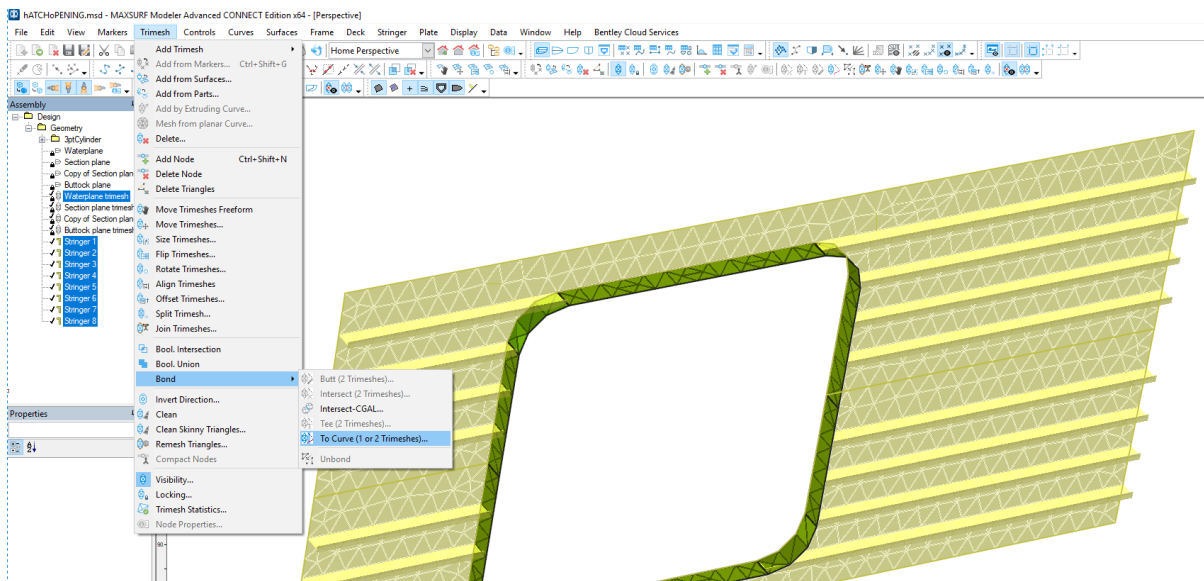
A series of longitudinal stringers with T cross section is added to the underside of the deck:



A mesh can then be added on each of the NURBS surfaces:



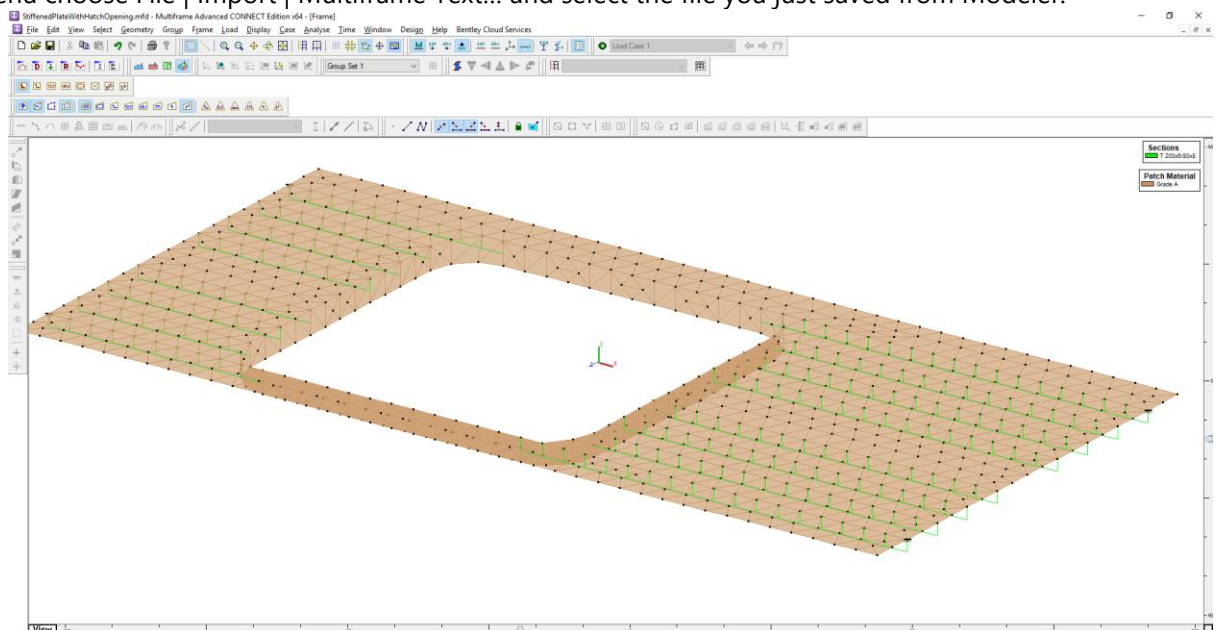
The triangle nodes of the mesh do not necessarily coincide with the stringer nodes, to achieve this the Meshes are "Bonded" to the Stringers using the "Bond to Curve" command:



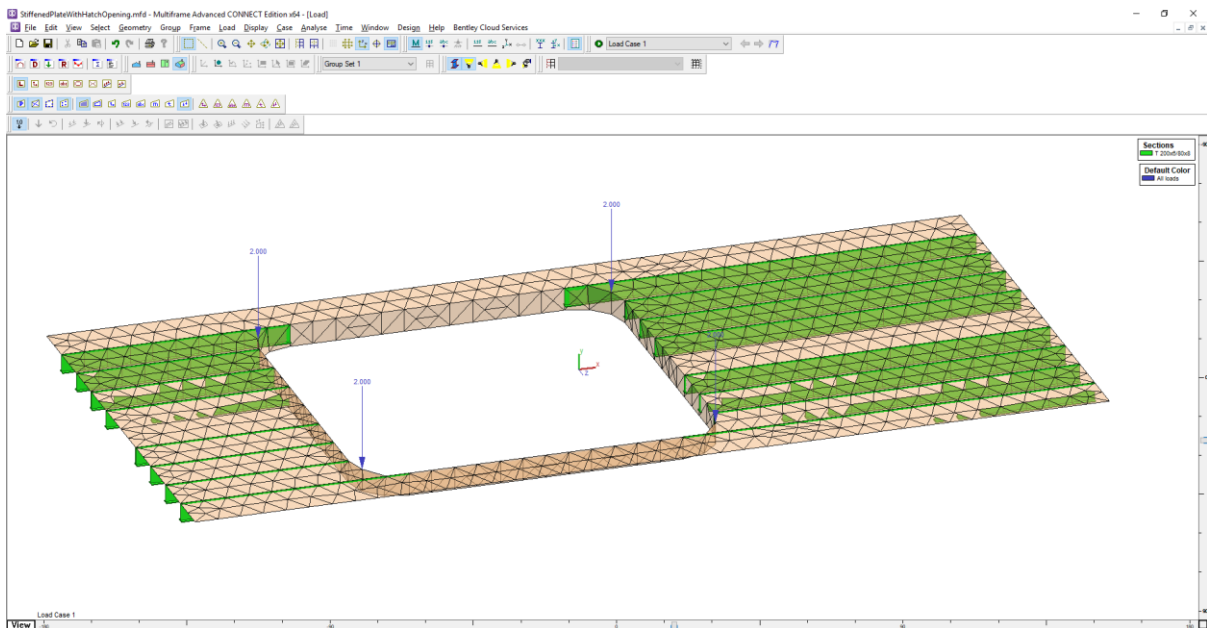
Once the structure geometry has been created then each part to be exported needs to have a material assigned (meshes and stringers). To export the model to Multiframe simply choose "File | Export | Meshes | Multiframe text file..." from the menu.

Before importing the file into Multiframe, ensure that you have the same library (*.slb) loaded into Multiframe as that which was used in Modeler to create the model initially.

From the menu choose File | Import | Multiframe Text... and select the file you just saved from Modeler:

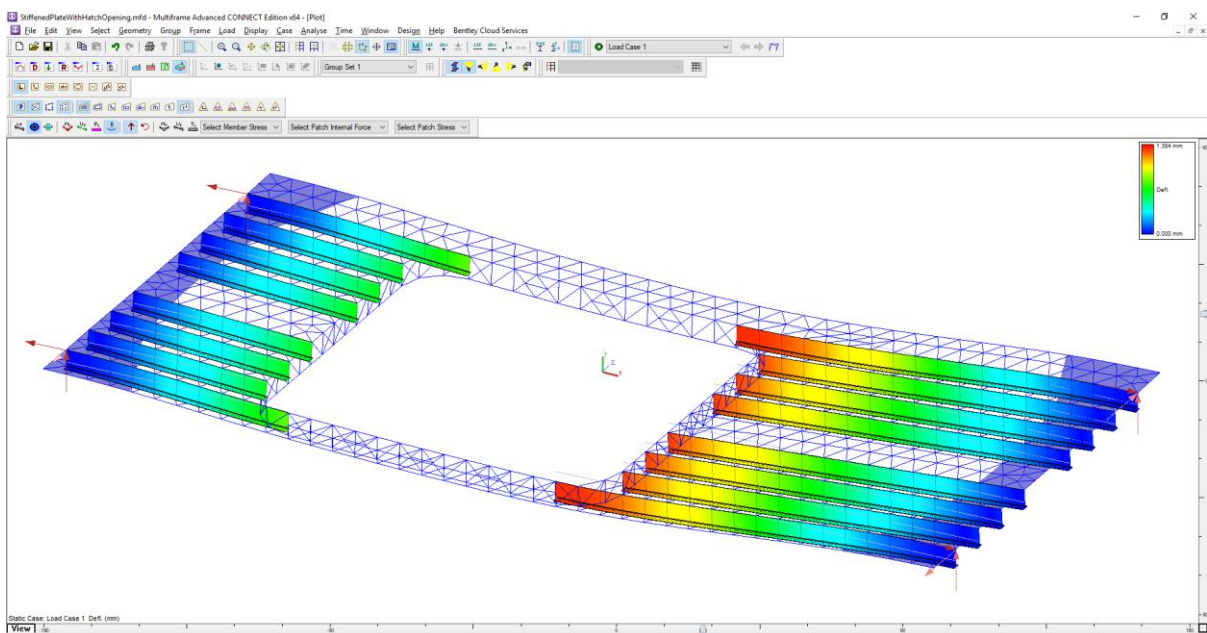


Before an analysis can be performed the restraints and loads need to be applied. In this case we will apply a simple load of 5kN at each of the four corners of the hatch:



A Linear analysis of the structure can now be performed by choosing Analyse | Linear from the menu.

Results can then be viewed either in graphical or tabulated format. The following image shows the deflection of the stiffeners and reaction forces at the restrained nodes:



The results can also be sent to a report or animated for presentation.

This process can be followed for a global analysis of an entire structure or for other local regions of structure for detailed analysis.

Chapter 6 Modeler Reference

This chapter describes the toolbars and menu commands available in Modeler.

- [Toolbars](#)
- [Menus](#)
- [Right Click Menus](#)
- [Ribbon](#)

The functions available in the different Modeler windows are explained in [Chapter 4 Modeler Windows](#) on page 20.

Toolbars

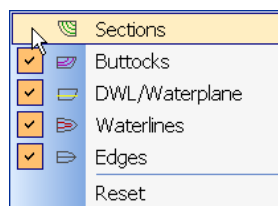
Modeler has a number of icons arranged in toolbars to speed up access to some commonly used functions. You can hold your mouse over an icon to reveal a pop-up tip of what the icon does.

Customising Toolbars

All applications in the Modeler suite contain fully customisable toolbars. This enables you to make your most frequently used commands available with just one click and delete any toolbar buttons that you don't need. The toolbar buttons themselves are also fully customisable, including text and button images.

Adding/Deleting Standard Toolbar Buttons

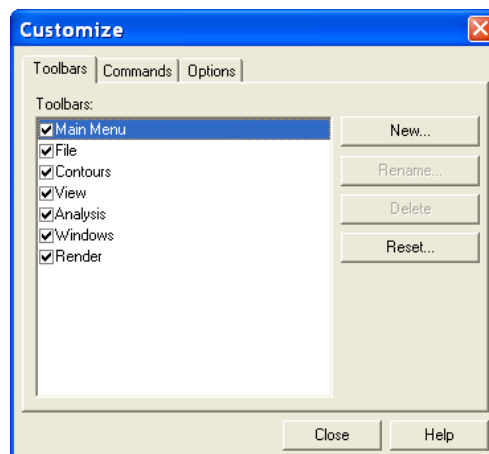
The standard toolbars available in each of the Modeler suite's applications can easily be edited by clicking on the little triangle on the right side of any of the toolbars and select "Add or Remove buttons". This allows you to select which button you wish to add or delete buttons by simply ticking or un-ticking the command.



By selecting "Reset" the Standard toolbar will be restored to its default state. Note: this will delete any extra buttons you may have added to the standard toolbar.

Creating Your Own Toolbars

You can customise all toolbars in all Modeler applications via the customise toolbars function. This is available via the View | Toolbars | Customise Toolbars menu item or by clicking on the little triangle on the right side of each toolbar and selecting "Add or Remove buttons". Alternatively simply right click on a blank space on the toolbar area of the application window and select "Customize". This brings up the following tabbed dialog:

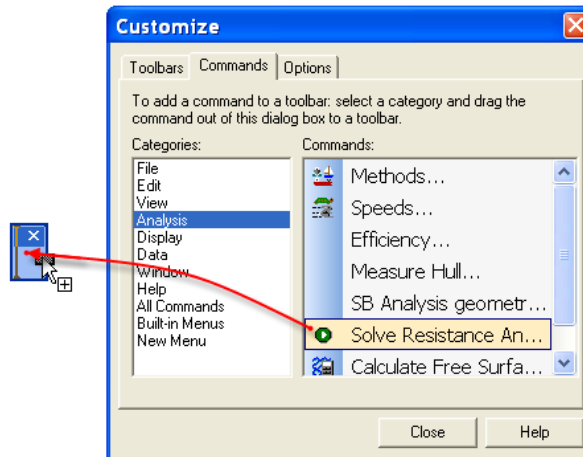


Toolbars tab

Use this tab to add or delete toolbars. Note: the standard toolbars that were installed with the applications cannot be deleted or renamed.

Commands tab

On this tab you will find all commands available in the application sorted per menu. To add a command to a toolbar, simply select the command and drag and drop it onto the toolbar. Note: As long as the Customize dialog is open, you can drag and drop toolbar buttons between different toolbars and the toolbar buttons are inactive; i.e. they will not work.

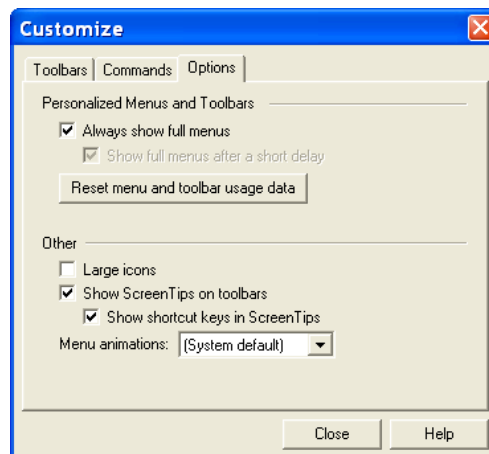


Dragging and dropping a command onto a new toolbar.

Options tab

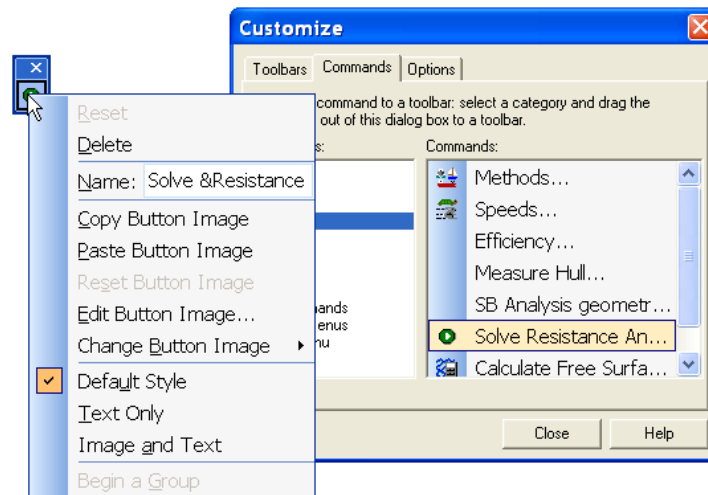
This tab allows you to customise the menu and toolbar display.

- Always show full menus will only show the most commonly used commands in the menu based on the command usage history. This command usage tracking function is not enabled in the Modeler applications; this means you can ignore the top half of this dialog.
- Large icons displays the toolbar buttons larger.
- Show ScreenTips on toolbars will enable the little text boxes that display which command is invoked by that toolbar button.
- Show shortcut key in ScreenTips will also show you the shortcut key (if available). This can be quite handy for you to learn the shortcut keys which allow you to operate the program even faster.
- Menu animations allows you to set how you want the menus to appear.



Editing Toolbars Buttons

When the Customize dialog is open (from View | Toolbar |Customize Toolbars menu), you can right click on a toolbar button to edit it via a right click menu.



A couple of interesting things that this allows you to do are:

- Edit button image will pop up a dialog that allows you to draw your own toolbar button image.
- Change the toolbar button image to one from the standard toolbar button images library.
- Select how you want to display the toolbar button image. For example: if you have trouble distinguishing between different toolbar buttons, you can select the Image and Text option to display both the image and the toolbar button name (the image button name is the command name by default).

Saving a Customised User Interface

Each time you close any of the applications in the suite, the application will automatically save your toolbar settings and windows layout to an ##Settings.xml file in the Modeler root directory. The Modeler root directory is by default C:\Program Files\Bentley\Offshore\Maxsurf *.

For example: After closing Stability, your current user interface will be saved in C:\Program Files\Bentley\Offshore\Maxsurf ## \MSSettings##.xml. The next time you start up Stability, it will look for this file for the application user interface settings.

If you want to save a copy of a user interface settings file, simply use windows explorer to locate the .xml file mentioned above, duplicate it and rename the duplicate to for example: HMSetting_Custom1.xml.

* In Windows Vista the .xml files are stored in the C:\Users\<User>\AppData\Roaming\Maxsurf\.. directory.

Restoring a Customised User Interface

Restoring a toolbar

If you want to restore only one toolbar, use the View | Toolbars | Customise Toolbars command, select the Toolbar you wish to reset and then press Reset.

Restoring the user Interface

Use the Restore default layout from the Windows menu to restore to the default windows and toolbar settings. Alternatively you can hold down the shift key at start-up. Note that the latter clears your [Modeler Settings](#). See the manual for each of the applications in the Modeler suite to find out which settings are reset when you hold down the shift key!

Restoring to a saved settings file

If you want to restore to a saved setting, use the following steps:

1. Rename or delete your current ##Settings.xml in windows explorer. Your current user interface settings file is located in the Modeler root directory and is called MSSettings.xml for Modeler, HMSettings.xml for Stability etc.
2. Rename the custom.xml file to the default name. For example to restore custom1 settings: rename HMSettings_Custom1.xml to HMSettings.xml.

Modeler Toolbars

File Toolbar



The File toolbar contains icons that execute the following commands:

New – Open – Save | Cut - Copy - Paste | Print | Help

Edit Toolbar



The Edit toolbar contains icons that execute the following commands:

Undo – Redo

View Toolbar



The View toolbar contains icons that execute the following commands:

Saved Views combo – Zoom – Shrink – Zoom Extents – Pan – Rotate – Saved Views dialog – Home View – Set Home View – Assembly window – Properties Pane.

The Rotate command is only available in the Perspective window.

Markers Toolbar



The Marker toolbar contains icons that execute the following commands:

Add Marker, Delete Marker | Snap Control Point to Marker, Fit Edge to Markers

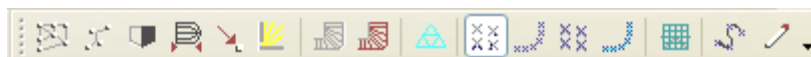
Controls Toolbar



The Controls toolbar contains icons that execute the following commands:

Add Row/Column – Delete Row/Column | Align To Vector – Align To Plane – Compact – Group – Ungroup – Bond - Unbond | Mask – Unmask

Display Toolbar



The Display toolbar contains icons that change the state of the following parts of the display:

Shape – Net – Half – Compress – Outside Arrows – Ruling Lines | Hide Curvature – Show Curvature | Hide Markers – Show Markers for Current Station – Show All Markers – Connect Markers

Window Toolbar



The Window toolbar contains icons that make the corresponding window come to the front:

Perspective – Plan – Profile – Body Plan | Calculations – Control Points – Markers – Surfaces – Graph – Offsets

Render Toolbar



The Render toolbar lets you switch rendering on/off and change light settings. These toolbar commands are only available when you are in the Perspective window.

Render on/off – Light directions (4x) - Lighting Options

Visibility Toolbar



The Visibility toolbar contains icons that can be used to quickly toggle the display of certain contours:

Stations – Buttocks – Design Water Line – Waterlines – Diagonals | Edges – Bonded Edges – Feature Lines – Parametrics | Intersections – Trimming Curves.

Surface Toolbar



The Surface toolbar contains icons that are used to access the surface manipulation commands:

Move – Duplicate – Flip – Rotate – Align

Stringer Toolbar



The Stringer toolbar contains icons which execute the following commands;

Add Stringer Point - Add Stringer

Structure Toolbar



The Structure toolbar contains icons which change the visibility of the following parts in the structure;

Plates - Plate Meshes - Stringer Points - Stringers - Frames - Decks - Upstands

Part Toolbar



The Part toolbar in the Part window contains a number of icons which allow you to quickly switch between different views of the current plate. These views are -

Development - Plan - Profile - Body Plan - Template - Pin Jig | Shell Expansion | Inverse Bending Lines

Menus

Modeler uses the standard set of Windows menu commands for File, Edit and Window operations. It also has a range of menus for control point and surface manipulation and control of data and view displays.

- [File Menu](#)

- [Edit Menu](#)
- [View Menu](#)
- [Marker Menu](#)
- [Mesh Menu](#)
- [Curves Menu](#)
- [Display Menu](#)
- [Data Menu](#)
- [Window Menu](#)
- [Help Menu](#)

File Menu

The File menu contains commands for opening and saving files, importing and exporting data, and printing.

New

Select New when you wish to commence a new design. Alternatively, if the Calculations window is uppermost a new calculations sheet will be created.

Open

Modeler has the ability to save designs to disk and recall them at a later date. By selecting Open, a dialog box appears with a list of available designs. Select the design you wish to recall, click the Open prompt, and the requested design will be read in.

Open will remain active until you are using the maximum number of surfaces available in your version of Modeler. This allows you to read in multiple designs and overlay them, or to read in the same design twice so that any changes can be compared with the original.

When the Calculations or Markers windows are active, Open will allow you to open a calculations sheet or a set of markers from a text file.

Quick Start

A user can open a reference design as a starting point by using the quickstart function. The user can choose one of six standard designs and enter the length beam and draft, Modeler will then create a design to those specifications. The user can then customise their design from there. This function is only enabled when a blank design is open.

Close

Select Close when you wish to finish with the current design. Before closing, a dialog box will appear asking whether you wish to save the current design. If you select 'Yes' the current design will be saved onto the disk.

When the Calculations window is uppermost Close will close the calculations sheet.

Save

Selecting Save will save the current design onto disk.

When the Calculations, Markers, Offsets, Control Points or Surfaces windows are active Save will allow you to save the calculations sheet or Markers, Offsets, Control Points or Surfaces data to a text file.

Save As

Selecting Save As enables you to save the current design under a new name. This is useful if you modify a design, but wish to keep a copy of the old version as well as the new.

When the Calculations window is active Save As will allow you to save a new calculations sheet.

Import

Selecting Import enables you to import a DXF, GHS or Seaway file directly into Modeler as a set of markers, or an IGES file with surface entities directly into Modeler as a surface.

Modeler is able to import a variety of other data formats that do not contain NURB surface information. These are imported as markers and where supported in the incoming data format, connectivity data is maintained. This results in either a full Mesh representation of the vessel (e.g. nuShallo, WAMIT, and STL formats); markers organised into sections (e.g. Seaway) or simply a scattering of markers with no connectivity information (e.g. DXF).

Import DXF Background enables you to import a DXF file into Modeler to use as construction lines. The DXF file will be displayed in the design views.

Import DXF enables you to import a dxf file converting all lines, polylines and splines to Modeler curves. The DXF file will be displayed in the design views.

Note that if you intend to load DXF markers and a DXF background, the DXF background should be loaded *after* loading the DXF Markers. If a zero-point shift of the DXF background occurs due to loading DXF markers, the DXF background should be reloaded.

Import Image Background enables you to import an image file (jpg, gif, bmp or png) file into the background of any of the Modeler design views.

Export

Selecting Export enables you to export a Modeler file as a variety of different file formats such as DXF or IGES.

You may also export to earlier Modeler file formats. Note that this may lead to some loss of data.

Export Render Mesh

Selecting Export | Render Mesh submenu lets the user select from three mesh file formats: wavefront (*.obj), sterolithography (*.stl) or direct X (*.x).

Export Plate Mesh Text

Used to export a special text file containing geometric information about the mesh used to develop the plate.

Export ShipConstructor Files

Used to export all frames, decks, plates and stringers to ShipConstructor.

Export Bitmap Image...

This function exports a bitmap image of the current graphics window. For example the contents of the current profile view.

Export Workshop Version 12...

This function exports the current design to a Workshop Version 12 file. Due to changes in file format some of the parts may need recalculation when opened in Workshop Version 12.

Export Workshop Version 13...

nodeThis function exports the current design to a Work Version 13 file. Due to changes in file format some of the parts may need recalculation when opened in Workshop Version 13.

Export Hydromax Loadgroup File...

Used to export loading information of parts in the design to a Stability Loadgroup file (.hml) this may then be loaded into Stability for analysis.

Export Rhinoceros 3dm File...

Exports all parts to native Rhino 3dm file format. All parts currently visible will be exported. Data exchange is uni-directional (i.e. from Modeler to Rhino only).

Export Multiframe Text File...

Used to export parts to multiframe text file format. Only visible parts will be exported. All visible parts except plates, frame rider bars and non-planar longitudinal girders get exported.

Export Lines Plan...

A dxf lines plan will be automatically generated from the NURBS model. All contours that are visible will be displayed in the lines plan.

Page Setup

The Page Setup dialog allows you to change page size and orientation for printing.

Print

Choosing the Print function prints out the contents of the active Modeler window.

Load Sections Library

Used to load a new library file. Library files have the extension .slb and contain information about the materials that can be assigned to surfaces. The default library is named WorkshopLibrary.slb and is located (by default) in C:\Program Files\Bentley\Offshore\Maxsurf XX \ on pre Vista operating systems and C:\Users\Public\Documents\Maxsurf\ for Vista and post vista operating systems.

Exit

When you have finished using Modeler choose Exit to close the program. If Modeler has a design open that has not been saved to disk, you will be asked whether you wish it to be saved.

Bentley Cloud Services Submenu

Provides access to Bentley Cloud Services resources. You must be logged in with the CONNECTION Centre to access this functionality.

Associate Project

Register the current file you are working on with a CONNECTED Project.

Disassociate Project

Unregister the current file you are working on from a CONNECTED Project.

ProjectWise Projects

Display the CONNECTED Project in your web browser.

Edit Menu

The Edit menu contains commands for copying and pasting data, and working with tables.

Undo

The Undo function will reset the control point net back to its previous state after an accidental or experimental movement of a control point or group of control points.

Redo

The Redo function will reinstate the changes that were discarded with the Undo command.

Cut

Cut cannot be used in Modeler.

Copy

This function performs the standard Copy function.

Paste

Choose the Paste command to Paste data into a table. Paste may also be used with other applications, but cannot be used on Modeler drawing windows.

Select

Facilitates the selection of table cells, columns and rows. The triangle selection commands allow the user to increase or decrease a selection by 1 ring of neighboring triangles.

Fill Down

Choose the Fill Down command when you want to change several cells in a column to the value of the first cell. This function may be used in most of the tables, and may also be available by clicking the right mouse button, which brings up a context menu.

Preferences

The Preferences function allows you to customise some of Modeler's features.

Structure Preferences

Displays a dialog allowing you to set a number of optional settings relating plate forming information and number of undo/redo levels. The default material and thickness to be used for parts can also be set.

View Menu

The View menu contains commands for controlling the appearance of the display in the graphical windows.

Zoom

The Zoom function allows you to work on any part of your design by enlarging the selected area to fill the screen.

Shrink

Choosing Shrink will reduce the size of the displayed image in an active drawing window by a factor of two.

Pan

Choosing Pan allows you to move the image around within a drawing window.

Rotate

Activates the Rotate command, which is a virtual trackball which lets you freely rotate a design in the Perspective view window.

After selecting the Rotate tool, move the mouse to a location in the Perspective window and press the left mouse button. With the left button depressed, you can rotate the image by dragging the mouse around and the rotation is performed by projecting the movements onto a virtual sphere on the screen – essentially like a virtual trackball. Generally speaking, moving the mouse left and right rotates about a vertical axis while moving the mouse up and down rotates about a horizontal axis. The Rotate mode is exited when the left button is released.

Auto rotate cent.

Toggles the automatic update of the view rotation center when the model is panned or zoomed. When enabled, this helps maintain the part of the model near the center of the screen in view whilst rotating the view direction.

Saved Views

Open the Saved Views dialog where you can add, edit and delete the home and other saved views.

Home View

Choosing Home View will set the image back to its Home View size. Modeler starts up with default Home View settings for all its drawing windows. However, the Home View may be set at any time by choosing the Set Home View function.

Set Home View

Choosing Set Home View allows you to set the Home View in each drawing window.

To set the Home View, use Zoom, Shrink, and Pan to arrange the view as you require, then select Set Home View from the View menu.

Perspective

Set the perspective camera angle (field of view).

Colors and Lines

The Colors and Lines function allows you to set the color and thickness of lines, controls, and graphs.

Font

The Font command allows you to set the size and style of the text in the current window.

Toolbars

Allows you to select which toolbars are visible.

Assembly Pane

Displays the Assembly Pane. Shortcut key = F2. See [Assembly Pane](#) on page 53.

Properties Pane

Displays the Properties Pane. Shortcut key = F3. See [Properties Pane](#) on page 55.

Status Bar

Allows you to show or hide the Status Bar at the bottom of the main window.

Marker Menu

The Marker menu contains commands for managing markers and fitting surfaces to existing sets of markers. Also see the section [Working with Markers](#) on page 80 for more detailed explanation of commands.

Add Marker

The Add Marker command is used when you wish to place a reference marker in the Body Plan, Profile, Plan or Markers windows.

Delete Marker

This is used when you wish to delete a marker from the Markers window or from the Body Plan, Profile or Plan windows.

Generate Markers for Developable Surfaces

This command generates markers from the ruling lines on the developable surfaces and is only available if the design contains surfaces with surface type "Developable".

Also see:

[NURBS Surface](#) on page 107 for more information about developable surfaces

[Fit Surface to Markers](#) on page 376 for information on how to use the Fit Surface to Markers command to lock the surface edges and fit the internal surface to the markers generated from the ruling lines.

Move Selected Markers

Translates the selected markers by the specified distance.

Rotate Selected Markers

Rotate the selected markers by the specified angles.

Delete Duplicate Markers

This will delete any duplicate markers with the same position (to within a tolerance of 0.5mm) from the selected group of markers (or all the markers, if none are selected).

Generate Grid from Markers

This command will generate the grid (sections, buttocks and waterlines) based on the marker data and also associate markers with the correct marker station index. Sections are generated at the markers' longitudinal position; buttocks at their transverse offset; and waterlines at their vertical location. See [Setting up the Grid](#) on page 30 for more information.

Assign Markers to Stations

Automatically assigns all markers within a user specified tolerance of a station to that station. Particularly useful when importing large scanned marker datasets.

Group Markers by Stations

This will change the order of the markers in the markers table and collate all markers with the same station index reference number together in the markers table (without changing the order of the markers for a given station index number). This command is useful if a new station of marker points has been added and the new station has been added out of sequence, at the end of the Marker table.

Sort Markers within Stations

After transverse sections have been defined and a Station Index for each marker has been assigned, the markers with the same station index may be sorted into order (using a nearest neighbour sort). This process is automatic on the entire marker set and is independent of current marker selection. See [Sorting Markers](#) on page 87 for more information.

Re-order Selected Markers

This command sorts all markers in a selection in the order of selection. Running this command twice will sort (or: order) the markers in the opposite order. This command can only be run after 2 or more markers are selected. See [Sorting Markers](#) on page 87 for more information.

Re-sort Selected Markers

This command automatically sorts all markers in a selection in a nearest neighbour approach starting from the centreline lowest position. It is possible to run this command for a selection of markers over a number of stations. See [Sorting Markers](#) on page 87 for more information.

Reverser Ordering for Current Station

This will invert or reverse the ordering of the markers on the current section (as selected in the Body Plan view Inset Box). The marker in bold (when Connect Marker Stations is turned on) is the first marker on the station.

Snap Control Point to Marker

See [Corner Control Points](#) on page 227.

Fit Edge to Markers

In a similar way that the Snap Control Point to Marker moved a single control point onto the marker, the Fit Edge to Markers will fit the selected edge to the selected markers. See [Fitting a Surface Edge](#) on page 228 for more information.

Smooth Interior Controls

Smooths the interior control points of the [Current Surface](#) based on the positions of the edge control points. The edges remain unchanged. Useful for getting an initial fitted surface once the edges have been fitted (see [Fitting a NURB Surface to Markers](#))

Fit Surface to Markers

See [Fitting a NURB Surface to Markers](#) on page 230.

Compact Marker points

Compacts the selected markers to the location of the first-selected marker

Measure Surface Error

After Markers have been linked to a specific surface, the distance between the markers and the surface can be measured using this command. See [Measuring Surface Error](#) on page 234 for more information.

Marker Properties

Allows you to edit the properties of the selected marker(s).

Mesh Menu

Add Mesh

Adds atrimesh using one of the standard shapes such as a plane, box or cylinder.

Generate From Markers...

See [Generating a Mesh surface from a NURB surface](#) on page 235. This command will only be enabled when there is at least one marker in the design.

Add From Surfaces...

See [Mesh Surface](#) on page 235. This command will only be enabled when there is at least one NURBS surface in the design.

Add from Parts...

See [Mesh Surface](#) on page 235. This command will only be enabled when there is at least one part (Frame, Deck or Plate) in the design.

Add by extruding curve...

Add a mesh by extruding the selected curve along a user specified vector. The user can also choose to close the ends of the extrusion, this will create a closed volume (if the curve is closed).

Extrude Mesh

This function is enabled when a mesh surface is chosen in the assembly tree. The selected (non-closed) mesh will be extruded along the user specified vector to form a closed volume mesh.

Mesh from planar Cuve...

This function is enabled when a curve is selected in the Assembly tree. The user is asked for a target edge length of the mesh and the curve will have a planar mesh fitted to it. The plane on which the mesh is created is a least squares fit to the selected curve points.

Delete...

This command will display the "Please select the meshes to delete" surface dialog. This command will only be enabled when there is at least one mesh surface in the design.

Snap Node(s) to Surface

This function will be enabled when one or more mesh vertices are selected. The function snaps selected nodes to the closest point on a NURBS surface chosen by the user.

Delete Triangles

This command will be enabled when there is at least one mesh in the design and one or more triangles are selected. To select a triangle, in non-rendered mode lock the mesh and use the mouse to select, either by left clicking individual triangles or dragging a selection rectangle over a group of triangles.

Flip Triangle Pair

This command swaps the common edge of a pair of adjacent triangles is selected (two triangles that share a common edge).

Visibility...

This command will display the "Visible Meshes" dialog. This command will only be enabled when there is at least one mesh surface in the design.

Locking...

This command will display the "Locked Meshes" dialog. This command will only be enabled when there is at least one mesh surface in the design.

Invert Direction...

This command will display the "Please select the meshes to invert" dialog. This command will only be enabled when there is at least one mesh surface in the design. Inverting the direction of a mesh will invert the direction of the outside arrow of every element (triangle) of the mesh. Individual elements (triangles) can be inverted by manually clicking on the tip of the outside arrow for that element.

Mesh Repair...

Repair holes in mesh to form a closed mesh. Make sure all triangle normals are in a consistent direction before performing a mesh repair operation.

Clean

This command will be enabled when there is at least one mesh in the design. The command will remove any duplicate nodes or degenerate triangles as well as recalculating mesh connectivity.

Remesh Triangles...

Remeshes a selected group of triangles at a new mesh density. The command will be enabled when a group of triangles is selected. Once executed a dialog will appear asking for the desired target edge length of the triangles to be remeshed. To select a triangle, in non-rendered mode lock the mesh and use the mouse to select, either by left clicking individual triangles or dragging a selection rectangle over a group of triangles.

Add Mesh Node

Converts the mouse pointer to add mesh node pointer. The next left mouse button click will add a node into the mesh at the pointer location. To cancel the command prior to adding a node hit the escape key. A mesh node can only be added to an unlocked mesh. The command is only enabled if there is a mesh in the design.

Delete Mesh Node

Converts the mouse pointer to delete mesh node pacman. The next left mouse button click will delete the node over which the pacman is currently places. To cancel the command prior to adding a node hit the escape key. A mesh node can only be deleted from an unlocked mesh. The command is only enabled if there is a mesh in the design.

Compact Mesh Node

If two or more mesh nodes are selected (highlighted) then the Compact Mesh Nodes command becomes enabled. All selected nodes will be moved to the location of the first selected nodes. Any resulting degenerate triangles (zero area) will be removed from the mesh.

Node Properties...

This command is enabled when one or more triangle nodes are selected. Nodes will only appear if the mesh is unlocked. Executing the function will launch a dialog displaying the node properties (position etc.) that can be edited.

Move mesh...

Moves selected meshes a defined distance, an array can be created by duplicating a number of specified times. The mesh must be unlocked to be moved. For more information see [Moving a Mesh Surface](#)

Move mesh freeform

Moves a trimesh using the mouse. To execute unlock the mesh and choose move mesh freeform from the mesh menu, next click the mouse near a vertex on the mesh you wish to move and holding the left mouse button down drag the mesh. When the left mouse button is released the mesh will be placed in that location.

Size Meshes...

Function used to scale selected mesh surfaces. The surfaces can be scaled in each of the orthogonal directions and can be scaled proportionally. For more information see [Size a Mesh Surface](#)

Flip Meshes...

Mirrors a mesh surface about an axis. For more information see [Flip a Mesh Surface](#)

Rotate TRimeshes...

Rotates a mesh surface about an arbitrary axis. For more information see [Rotate a Mesh Surface](#)

Align Meshes

Aligns two mesh surfaces. For more information see [Align Mesh Surfaces](#)

Offset Meshes...

Offsets a mesh surface in the direction of the normal. For more information see [Offset a Mesh Surface](#)

Split mesh...

Split a mesh into separate meshes by a cutting surface. For more information see [Split Mesh](#).

Join...

Join command is enabled when there are at least two meshes in the design. Once executed a dialog will appear asking the user which meshes they want joined into a single mesh.

Bond | To Curve(1 or 2 meshes)...

This command will be enabled when at least one curve and at least one mesh is present in the design. Once executed the user will be asked to specify a target edge length for the triangle edges along the bond.

Bond | At Visible Intersections...

This command will bond meshes that currently have visible intersections displayed. The bond will occur at the intersections. The user will be asked for a target edge length. See the Bond ↓ At Visible Intersections...section for more information.

Controls Menu

This menu provides commands to manipulate control points.

Add

To Add a control point, select the add command and click in the current view at the location desired. Control point columns may be added in the Plan and Profile windows; control point rows may be added in the Body Plan window.

Delete

The delete command allows you to delete a row or column of control points from a surface. Select the delete command and click on the desired control points using the delete cursor. Control point columns may be added in the Plan and Profile windows; control point rows may be added in the Body Plan window.

Copy Position

Copies all three parameters of the control point's coordinates and its weight.

Paste Position

Pastes the copied position and weight to the selected control points. Hold down the shift key to select which data are updated.

Smooth Controls

Smooths a complete or partial row or column, or a patch of control points. Smooth is always applied in three dimensions, not simply in the current view.

Straighten Controls

Straightens a complete or partial row or column, or a patch of control points. Straighten can be applied in the plane of the current view, (e.g. straightening a sheer line only in the profile view, while leaving it curved in the plan view), or in all three views simultaneously.

Move Controls

This function allows you to move a group of control points by a specified amount.

Size Controls

This function allows you to scale and re-proportion a group of control points by their principle dimensions.

Rotate Controls

This function allows you to rotate a group of control points about a specified centre of rotation.

Align to Vector

Aligns a number of selected control points to the 3D vector defined by the first two control points selected. To constrain the movement in certain directions hold the shift

Align to First Selected

Similar to Compact, but the geometric relationship of the control points remains unchanged. i.e. The selected control points are translated to so that the first and second selected control points are coincident. Hold the shift key to restrict which ordinates are updated.

Align to Plane

Moves a number of selected control points into the plane defined by the first three control points selected. To constrain the movement in certain directions hold the shift key down whilst selecting this command. A dialog box will appear within which you define the constraints.

Compact

There may be occasions when a number of control points need to be placed at precisely the same point in space. To do this, select the points and choose Compact. The points are all compacted onto the first point in the selection. Hold the shift key to restrict which ordinates are updated.

Group

Any number of control points may be selected from a surface or multiple surfaces and grouped to move as one by selecting the control points and grouping them with the Group command.

Ungroup

This function performs the opposite of the Group function. All control Points that were previously in the selected group will return to being individual control points.

Bond Edges

Two Surfaces may be bonded together by joining along a common edge. The two edges may be selected for bonding by clicking on the edges or by selecting a control point (other than corner points) from each edge.

When Bond Edges is selected the two edges are bonded together and will show a single row or column of control points along their common edge. Bonded edges may be displayed by selecting Bonded Edges from the Contours dialog in the Display menu.

Unbond Edge

Any edge that is bonded may be unbonded. Choose any control point along the desired edge and select unbond. The selected edges will now be unbonded and may be moved independently.

Mask

The Mask command allows you to select a particular group of control points, which will remain visible while the remainder of the net is hidden.

Unmask

The Unmask command restores a masked surface display to one where all control points in a surface net are visible.

Control Properties

This command lets you change the control point properties for a group of selected control points. This command can be quite useful if you want to set a group of control points to the same position or plane; e.g. move a group to the centreline or a specific deck height. See [Control Point Properties](#) on page 203 for more information.

Advanced

This gives you advanced functions that may be used to adjust the control point net. Row and column indexing may be changed and rows and columns transposed. The actual surface shape remains unchanged.

Curves Menu

The Curve menu contains commands for manipulating curves.

Add Curve

Add Curve gives the option of adding 5 different types of curves to the design. The Add curve command is only available when a design is open and one of the four drawing windows is the currently active window. Arc and circle curves can only be added in the three orthogonal views (ie not perspective).

Delete Curve

The delete curve command deletes the currently selected Curve(s) from the design. If a curve is accidentally deleted it can be retrieved via the undo command. Delete command is only enabled when a design is opened and a curve is selected in one of the drawing windows.

Add Data Point

The add data point command will enable the user to add a new data point to the currently selected curve. The curve will pass directly through the data point. The add data point command is enabled when a design is opened and a data point curve is selected.

Add Control Point

The add control point command will enable the user to add a new control point to the currently selected curve. The add control point command is enabled when a design is open and a control point curve is selected.

Move Curve

The move curve command will activate the move curve dialog box. Any number of curves selected in the dialog may be moved numerically longitudinally, vertically or horizontally by this command. The move curve command is enabled when a design with one or more curves is open.

Size Curve

The Size Curve function allows you to scale and re-proportion a curve or group of curves by their principle dimensions. The size curve command is enabled when a when a design with one or more curves is open.

Flip Curve

This command allows you to flip, or mirror, a curve or group of curves about a specified longitudinal, transverse or horizontal plane. The flip curve command is enabled when a when a design with one or more curves is open.

Rotate Curve

The Rotate Curve command rotates one or several curves about a specific centre of rotation. The rotate curve command is enabled when a when a design with one or more curves is open.

Reverse Curve

The reverse curve command will reverse the order of the control points in a curve, so that the first becomes the last and visa versa. This command will only become enabled when a curve is selected.

Project Curve onto Surface

This function is enabled when there is at least one surface and at least one curve in the design. When the function is executed a dialog will appear asking the user to specify which curve is to be projected onto which surface and in which projection direction.

Offset Curve

Offsets a curve in 3 dimensions. If a curve lies in a plane the offset curve will remain in that plane. If the curve is no planar the offset curve is offset in the plane of the local curvature along the curve.

Break Curve

This command will break the curve into two curves at the currently selected control point. The command will only be enabled in the menu if a curve control point is selected.

Join Curves

The join curves command will join the two curves that have end control points selected into one curve. Selection order is important as the second selected control point will be moved to the first selected control point. This command will only be enabled in the menu if two end control points from two separate curves are selected.

Fit Curve To Markers

The fit curve to markers command will add a new curve and fit it to a selected set of markers if only markers are selected. If a curve as well as markers are selected the selected curve will be fitted to the selected markers. The curve will fit to the markers in the order they are selected. The fit curve to markers command is enabled when a design is open and 2 or more markers are selected and one of the four drawing windows is the currently active window.

Fit Edge To Curve Marker Stations

Fit Curve to Marker Stations command will become available if 2 or more markers that are assigned to station indexes are selected. When the command is executed Modeler sorts the markers for each station from centreline lowest marker and then fits a curve through each set of Markers for each station.

Fit Edge To Curve

The fit edge to Curve will fit the selected surface edge to the currently selected markers. The edge will be fitted to the markers in the order that they are selected. The command will be enabled when 2 or more markers are selected and one surface edge is selected.

Fit Curve To Contour

A curve may be fitted on a surface contour. Select any curve (or multiple curves) to enable the command in the Curve menu.

Visibility

This function allows you to select which curves are visible. The visibility of any curve may also be changed in the curve properties dialog and curves window. This function is enabled when a design with one or more curves is open and a drawing view is the current window.

Locking

This function allows you to lock and unlock curves. This function is enabled when a design with one or more curves is open and a drawing view is the current window.

Curve Properties

Each curve that is created in a Modeler design has a unique set of properties associated with it. The Properties command allows you to view and modify these properties. The properties of all the curves may also be edited in the curves window. The properties command is enabled when a design is open and one or more curves is selected.

Surfaces Menu

The Surface menu contains commands for manipulating surfaces.

Add Shape

A number of basic shapes may be added, these include cylinders, boxes and spheres.

Add Surface

A number of basic surface shapes may be added

Add from Visible Parts

Enabled when there is a visible part in a drawing window (frame, deck, plate or stringer). Adds a NURBS surface on the part.

Delete Surface

Choosing Delete Surface displays a dialog that allows you to select the surfaces to be deleted.

Move Surfaces

This command move the selected surfaces. You may select the number and relative positioning of duplicates.

Size Surfaces

The Size function allows you to scale and re-proportion a surface or group of surfaces by their principle dimensions. This function is particularly useful when modelling geometrically similar variations from a parent design, and also allows precise specification of principle design parameters.

Flip Surfaces

This command allows you to flip, or mirror, a surface or group of surfaces about a specified longitudinal, transverse or horizontal plane.

Rotate Surfaces

Choosing Rotate Surface rotates one or several surfaces about a specific centre of rotation.

Align Surfaces

Choosing Align Surfaces allows you to select two surfaces for aligning by way of selecting a control point from each surface. Aligning takes place by movement of the complete surface such that the two control points are coincident. The surface whose control point was selected first is fixed.

Offset Surfaces

The offset surface command from the surface menu is enabled when a design contains one or more surfaces. The offset surface command can only be performed on unlocked surfaces. The original surface may be kept or destroyed. To offset a surface choose "Offset Surfaces..." from the surfaces menu, in the dialog select the surface and enter the offset distance. A positive offset distance will offset the surface in the direction of the outside arrow of the surface, if negative the opposite direction.

Split Surfaces

The Split Surface command from the surface menu is enabled when a parametric curve is selected on an unlocked surface or a control point is selected. The split surface command splits a surface into two separate surfaces along the chosen parametric (or at the parametric position of the chosen control point). To execute choose a parametric curve or edge control point on an unlocked surface, choose Split Surface from the surface menu.

Join Surfaces

The join surfaces command is enabled when 2 edges are selected on separate unlocked surfaces. The join surfaces command joins two surfaces into one surface. To join two surfaces select an edge of the first surface, holding the shift key down select an edge of the second surface. To execute, choose Join Surfaces from the Surfaces menu.

Skin Curves

The skin curves command creates a surface by skinning around a set of selected curves. The order in which the curves are skinned is dependent on the selection order. The skin surfaces command is enabled when a design is open and 2 or more curves are selected.

Sweep Surface

The sweep surface command creates a surface from two curves. The first curve selected is termed the "section curve" and the second curve the "trajectory curve". The new surface is created by sweeping the section curve along the path of the trajectory curve. The swept surface command is enabled when a design is open and two curves are selected.

Create 4 Sided Patch

Creates a four sided surface patch defined by 4 selected Curves. Four curves must be selected to enable the command from the surfaces menu. The last point of the first curve must be coincident with the first point of the second curve and so on until the 4 curves create a closed loop.

Extrude Surface

Creates a surface by extruding a section curve or surface contour. To enable the command at least one curve or surface contour must be selected. When the command is executed the user will be prompted to enter a direction vector for the extrusion.

Surface of Revolution

Surface of revolution command becomes enabled when a curve or a contour on a surface is selected. The curve is then rotated about the axis defined by the user in the dialog box to create a surface. The curve is rotated a user defined number of degrees about the axis.

Approximate Surface to Markers

This command does a least squares approximation of a surface to the selected markers. Optionally 4 curves may be selected that define the edges of the surface to be fitted. To enable the command at least 4 markers must be selected and optionally 4 edge curves. The command will display a dialog asking the user to enter the number of control point rows and columns as well as the longitudinal and transverse stiffness of the surface to be created. The number of markers required to fit a surface is equal to the number of control point columns multiplied by the number of control point rows defined by the user when the command is executed.

Visibility

This function allows you to select which surfaces are visible. The visibility of any surface may also be changed in the Properties dialog and Surfaces window.

Locking

This function allows you to select which surfaces are locked. Locking hides all control points in the surface, which prevents modification of the shape. The locking of any surface may also be changed in the Properties dialog and Surfaces window.

If you hold down the "Shift + Ctrl" keys when clicking OK in the Locking dialog, the surfaces are locked and made read-only forever!

Note that you can *never* unlock them, so be careful! You can use this to send a design to someone, where you do not want them to be able to modify the design or create a new design based on your model. Use Save As to create a new file before using this feature.

Appearance

This dialog allows you to select the surface colors that are used for rendering in the Perspective window, and displaying the surface's parametric curves. This may also be specified in the Surfaces window.

Surface Properties

Each surface that is created in a Modeler design has a unique set of properties associated with it. The Properties command allows you to view and modify these properties. The properties of all the surfaces may also be edited in the Surfaces window.

Start Trimming

Select the name of the surface you wish to trim from this sub-menu and then click on the regions of the surface to turn trimming on and off. When you are happy with the trimmed state of the surface, or if you just want to stop trimming, choose the Trim command.

Trim Surface <name>

Select this command when you have finished clicking on the regions of the surface you are currently trimming, and you want to confirm your trimming changes.

Untrim Surface <name>

Removes all trimming information from the selected surface.

Frame Menu

The Frame menu allows you to add, delete and modify frames.

Add Frames

Add frames at the locations of the sections selected in the front window.

Delete Frames

Delete the frames which are selected in the front window.

Frame properties

Brings up a dialog allowing you to specify the frame characteristics of the selected frames in the front window.

Calc All Frames

Calculates the shape of all the frames you have defined. The shapes includes the properties you have defined using the Frame Shape dialog and the cutouts for any stringers which pass through frames.

Calc Selected Frames

Calculates the shape of the selected frames in the front window.

Deduct Skin Thickness

If this item is ticked, the skin thickness will be deducted from the perimeter of the frame.

Also see:

[Check](#) Surface Thickness on page 257

[Check Outside](#) Arrows on page 257

Deck Menu

The Deck menu allows you to add, delete and modify decks.

Add Decks

Add Decks at the locations of the waterlines selected in the front window.

Delete Decks

Delete the decks which are selected in the front window.

Deck Properties

Brings up a dialog allowing you to specify the deck characteristics of the selected decks in the front window.

Calc All Decks

Calculates the shape of all the decks you have defined. The shapes includes the properties you have defined using the Deck Shape dialog.

Calc Selected Decks

Calculates the shape of the selected decks in the front window.

Stringer Menu

The Stringer menu contains commands for working with stringers and upstands

Add Point

Add a stringer point in the front window. If a graphics window is in front, the point will be projected onto the hull surface underneath it in the current view. The point will be added to the closest stringer. If the Stringer Data window is in front, a new point will be added at the bottom of the table.

Delete Points

Deletes the stringer points that are selected in the front window.

Add Stringer

Add stringer in the front window. You draw the stringer by clicking at the location of stringer point and double clicking at the last point (or hitting return to indicate the last point of the stringer). Each point will be projected onto the hull in the direction perpendicular to the current drawing plane.

Add Longitudinal Girder

Add a longitudinal girder in the front window. You draw the girder by clicking at the location of stringer point and double clicking at the last point (or hitting return to indicate the last point of the girder). Each point will be projected onto the hull in the direction perpendicular to the current drawing plane. A girder's stiffness by default is 2 (Linear).

Delete Stringers

Deletes the stringers that are selected in the front window.

Duplicate Stringers

Duplicates the stringers that are selected in the front window. Stringers can be duplicated in a vertical, transverse, longitudinal or girth direction.

Mirror Stringers

Duplicates and reflects the stringers that are selected in the front window.

Break Stringers

Breaks any stringers with selected points in the front window at the selected locations.

Join Stringers

Joins two stringers into one. Only enabled when two stringer points are selected. Stringer points must lie on different stringers and be at the end of their stringer.

Add Stringer On Contour

Adds a stringer on each of the currently selected contours. Allows you to specify a longitudinal start and end locations for the generated stringers.

Generate Stringers

Shows a dialog allowing you to automatically generate a number of stringers at prescribed girth spacing around a surface. [Generating Stringers](#) on page 274 for more information.

Generate Family

Displays a dialog allowing you to automatically generate a family of smoothly transitioning stringers between two selected 'parent' stringers. See [Generating a Family of Stringers](#) on page 275 for more information.

Point Properties

Displays a dialog allowing you to specify the projection direction for the selected stringers.

Stringer Properties

Displays a dialog allowing you to specify the type and surface for the selected stringers.

Upstand Properties

Displays a dialog allowing you to alter the behaviour of the stringer/frame associated with this cutout.

Shape & Cutout

Displays a dialog allowing you to choose the stringer shape and cutout for the selected stringer or for any selected upstands.

Orientation

Shows a dialog allowing you to specify the orientation of the selected stringers or selected upstands.

Calc All Stringers

Fit all of the stringers through all of the stringer points.

Calc Selected Stringers

Fit the selected stringers through all of their stringer points.

Calc Inverse Bending Lines

Calculates the Inverse Bending Lines for the currently selected stringer(s). If there are no currently selected stringers and the active view is the part view with a stringer displayed then the inverse bending lines for that stringer will be calculated. To display the Inverse Bending Lines, go to the parts window, select the stringer and click on the Inverse Bending Lines Button from the Parts toolbar.

See [Calculating Stringer Inverse Bending Lines](#) on page 292 for more information.

Calc Upstands

Calculate the upstands for all of the stringers. An upstand is a short line that runs from the intersection of a stringer and a frame in the orientation-direction of the stringer. It allows you to override the orientation and cutout shape of a stringer at a particular frame.

Snap to Grid

If Snap to Grid is turned on, stringer points will automatically snap to the specified grid spacing. The Snap to Grid command is used to turn snapping on or off and also to specify the grid spacing. The grid origin is at the zero point.

The Snap to Grid command can also be used when editing shapes in the library window.

Girth Centreline

This may be used to specify a transverse offset from which girth measurements are take. This can be particularly useful for catamaran designers.

Plate Menu

The Plate menu contains commands for working with plates.

Add Plate

Add a plate defined by the four contours selected in the front window. You must first select two contours on opposite sides of the plate and then the other two sides.

Add Surface Plate

Add a surface as a plate. This will automatically use the four edges of the surface as the boundaries of the plate, unless the surface is trimmed and trimming is on. In this case the plate will take into account the trimmed surface boundary.

Delete Plates

Delete the plates currently selected in the front window. You select a plate by clicking inside the plate.

Plate Properties

Displays a dialog for the currently selected plate allowing you to set plate properties.

Plate UCS

Displays a dialog for the currently selected plate allowing you to set the UCS (User Coordinate System) for the plate.

Plate Templates

Displays a dialog for the currently selected plate allowing you to set the location of forming templates along the plate.

Assembly UCS

Displays a dialog for the currently selected plate allowing you to set the UCS (User Coordinate System) for the assembly the plate belongs to.

Calc All Plates

Calculate the developed shape of all of the plates.

Calc Selected Plates

Calculates the currently selected or displayed plate.

Calc Shell Expansion

Creates the shell expansion drawing. This drawing is displayed in the Part window, shell expansion mode (click on the Shell Expansion icon in the Part toolbar).

For more information on creating a shell expansion drawing see [Shell Expansion](#) on page 318.

Flip Vertical

Flip the expanded shape in the Part window vertically.

Flip Horizontal

Flip the expanded shape in the Part window horizontally.

Display Menu

The Display menu contains commands for turning on and off items displayed in the graphics windows.

Net

The Net display shows the complete control point net for all visible and unlocked surfaces.

The net can be manipulated by clicking on a control point vertex and dragging it about the window. Once you are familiar with the results of modifying the net directly, it is an extremely efficient way of modifying a shape to effect global changes.

Control points may be moved in any drawing window, but in the Perspective window the movement is limited to the plane that is most perpendicular to the angle of view. An indicator showing the most perpendicular plane is displayed in the lower right hand corner of the window.

Half

For symmetrical surfaces, the control point net exists only for one half of the symmetrical surface. The Half function controls whether the surface's mirror image is displayed across the longitudinal centreline.

By choosing the 'Split Surface Display' option in the surface Properties dialog, the Body Plan window will display the forward half of a surface on the right hand side and the aft half on the left hand side, when the Half function is selected.

Compress

The vertical and transverse axes are increased by a factor of four relative to the longitudinal axis. This is useful for fairing longitudinal curvature.

Curvature

Displays curvature porcupines for any edge, section, waterline, buttock, diagonal or feature line by clicking on the curve and selecting Show Curvature. Porcupines are displayed perpendicular to the curve and their lengths are inversely proportional to the square root of the radius of curvature at that point on the curve. Note that the smallest radius on a given curve is displayed at the end of the porcupine corresponding to its position on the curve. The porcupines can be scaled via the modeless dialog that appears when the function is active.

Trimming

Turns on or off the trimming option in Modeler. This command also lets you trim surfaces Invisible or Gray.

Precision

Modeler has five levels of calculation and display precision. The appropriate precision may be set at any time depending on the performance of your computer, the complexity of the design and the level of detail required. Also see [Surface Precision](#) on page 115.

Drawing Settings

Sets various drawing parameters, such as Snap, Drawing Grid and Drawing Depth.

See [Drawing Settings](#) on page 35 for more information.

Outside Arrows

Displays the arrows which show the surface's current outside direction. All surfaces should point outwards. The surface direction may be inverted by clicking in the circle at the end of the arrowhead. This is important for rendering, skin thickness direction and correct interpretation of the design when it is loaded into Stability or Structure.

Ruling Lines

Ruling lines can be turned on and off from the Ruling Lines command in the display menu. This function is only enabled if the design contains a surface of type Developable. A surfaces type is set in the surfaces property dialog box.

Markers

You can choose to display all markers or no markers, or you may choose to only show those markers whose station number is the same as the current station. The Connect Marker Stations command displays a connecting line between markers on the same station. See [Sorting Markers](#) on page 87 for more information.

Background

Controls whether the background DXF construction lines and the background images are displayed or not. The background may be loaded from an existing DXF file using the Import function in the File menu. Tools for positioning and scaling the background image are also here.

The commands in the submenu are only available when a background image or DXF has been imported. See:

- [Importing DXF background](#) on page 335
- [Importing Background Images](#) on page 331

Hide DXF

Hides the DXF background.

Show DXF

Shows the DXF background.

Delete DXF background

Deletes the DXF background.

Hide Image

Hides the background image in the current view window.

Show Image

Shows the image in the current view window.

Set Image Zero Point

Sets the image zero point. This command is not available for images in the perspective window.

Set Image Reference Point

Sets the image reference point. This scales the image, see [Importing Background Images](#) on page 331.

Delete Image

Deletes the background image in the current view window.

Grid

When Show Grid Only is selected, the positions of the Sections, Waterlines, Buttocks and Diagonals as specified by the Design Grid function from the Data menu, will be drawn.

When Show Grid and Labels is selected, the names assigned to each grid line in the Design Grid option will also be shown on the screen and hardcopy.

Contours

The Contours option allows you to select which contours are drawn on the screen at any given time. Any combination of contours may be chosen from the contours dialog.

Strain

Used to select which, if any, strain is displayed on plates in the Part window. The strain map shows the strain which will need to be applied to the developed plate in order to make it fit the hull. The color map indicates

the magnitude of the strain; blue denotes compression (shrinking) required and red expansion. Green areas require bending only and no in-plane strain. The color map is limited to +/-0.5%.

None: no strain map displayed

Forming Direction: maximum principal direct strain. For plates developed using the General method, the forming direction will be indicated by an arrow. For the other methods, the forming direction will be from the first to the second selected edge, i.e. butt to butt.

Perpendicular to Forming Direction: minimum principal direct strain.

Shear: maximum shear strain.

Part

Displays a sub-menu allowing you to choose which view of the developed plate is displayed in the Part window.

Render

Render may only be selected when the Perspective window is active. Selecting Render performs hidden line removal on the design with a choice of surface shading and false coloring.

Animate

Animate may only be selected when the Perspective window is active. Modeler will attempt to draw as many views as possible with the available memory, up to the maximum number specified in the animate dialog.

When these views have all been drawn they can be played back as a moving picture by moving the mouse from side to side. Clicking the mouse button will terminate the animation.

Animations may be saved to AVI files for playback at a later time.

Data Menu

The Data menu contains commands for calculating and changing the numerical data that defines the design.

Units

A variety of metric and imperial units may be specified.

Irrespective of the default units specified, the Size, Design Grid, Girth, and Markers options, and data input windows will accept dimensional data in any units. For example, if the default units were metres, all of the following input would be accepted and converted accordingly:

3 (interprets as 3.00 meters)

2.5c

33cm

328mm

650.44 mil

6ft 3.1in

5f 5i

11.25 feet

5'4"

4 inches

3.25"

Coefficients

Allows you to specify how you want to display the LCB and LCF position and how the hydrostatic calculations should be carried out.

See [Hydrostatic Coefficient Calculation Parameters](#) on page 248 for more information.

Design Grid

This menu function allows you to specify the positions of Sections, Waterlines, Buttocks, and Diagonals.

See [Setting up the Grid](#) on page 30 for more information.

Frame of Reference

Choosing this function allows you to set the positions of the Fore and Aft Perpendiculars, the DWL and the Baseline of the design; the Midship position is altered indirectly since it is assumed to be mid way between the Fore and Aft Perpendiculars. All values are entered relative to the Zero Point. However, it may be that the Zero Point is set to the position of one of the elements in the Frame of Reference. In this case the Zero Point is not updated until you have clicked OK in the Frame of Reference dialog.

See [Setting the Frame of Reference](#) on page 21 for more information.

Zero Point

This function sets the longitudinal and vertical reference point for all measurements. See [Setting the Zero Point](#) on page 22 for more information.

Vessel Type

Lets you specify the type of vessel you are modelling. See [Setting the Vessel Type](#) on page 23 for more information.

Windage Surfaces

Lets you specify the surfaces that define the windage and underwater profiles of the vessel. See [Setting the Windage Surfaces](#) for more information.

Girth

Measures halfgirths or locates the position at which a particular halfgirth occurs.

See [Calculate Girth](#) on page 249 for more information.

Calculate Offsets

When Calculate Offsets is selected, the Offsets window will be filled with offset data for the grid that has been specified via the Design Grid command.

See [Offsets Window](#) on page 66 for more information.

Go to Offset

This command allows you to select the station for which the offsets are displayed in the Offsets window. This item is available only when the Offsets window is active.

Calculate Areas

Calculates the areas and centres of the surfaces in the design.

See [Calculate Areas](#) on page 250.

Calculate Hydrostatics

Calculates the upright hydrostatics of the design at the DWL.

See [Hydrostatics](#) on page 247 for more information.

Solve Calculations

Choose this command to solve the calculations currently displayed in the Calculations window. The dialog box that appears can be used to set the longitudinal extent of the hull to be used in the calculations.

See [Calculations Window](#) on page 72 for more information.

Parametric Transformation

Activates the Parametric Transformation dialog, which allows you to manipulate an existing design.

See Using Parametric Transformation on page 241 for more information.

AC Rule

If you have a license for the America's Cup Rule software, this menu item will be available. Modeler will automatically measure your hull and calculate its rating under the America's Cup Rule.

Create Stock

Automatically creates stock from materials used to define parts. See [Nesting Parts](#) for more information.

Allocate Parts to Stock

Automatically allocated stock to parts. Work by matching part material to stock material. See [Nesting Parts](#) for more information.

Nest Parts

Nests all parts that have stock associated with them. See [Nesting Parts](#) for more information.

Calc All Parts

Calculate all of the parts including all stringers, frames and plates. This command is useful for calculating a large number of parts at once.

Note:

All parts will be calculated at the current precision.

Calc All Visible Parts

Calculate all of the visible parts including stringers, frames and plates. This command is useful for calculating a large number of parts at once.

Window Menu

The Window menu allows you to make any window selected from the menu the active window.

Cascade

Displays all the Windows behind the active Windows.

Tile Horizontal

Lays out all visible windows across the screen.

Tile Vertical

Lays out all visible windows down the screen.

Structure Datasheets

Brings the selected table to the front.

Arrange Icons

Rearranges the icons of any minimised window so that they are collected together at the bottom of the Modeler program window.

Help Menu

Provides access to Modeler help and other resources.

Modeler Help

Display the Modeler help manual.

Modeler Automation Reference

Display the Modeler Automation reference help manual.

Online Support

Provides access to a wide range of support resources available on the internet. You must have internet access to access this functionality.

Service Request Manager

Submit and manage your technical support requests with Bentley Systems. You must have internet access to access this functionality.

CONNECTION Centre

Display your personal CONNECTION Centre web portal in your web browser.

License wiki

Web portal for help on Bentley licensing.

Online Support

Provides access to a wide range of support resources available on the internet. You must have internet access to access this functionality.

Service Request Manager

Submit and manage your technical support requests with Bentley Systems. You must have internet access to access this functionality.

Idea Feedback

Access the web portal for submitting feedback and suggestions relating to Modeler.

Licensing Feedback

Access the web portal for submitting feedback and suggestions relating to Bentley licensing.

Check for Updates

Provides access to our website with the most recent version listed. You must have internet access to access this functionality.

News

Display the Bentley Systems Offshore news feed. You must have internet access to access this functionality.

Legal Notice

Display the Bentley Systems Legal Notice.

About Modeler

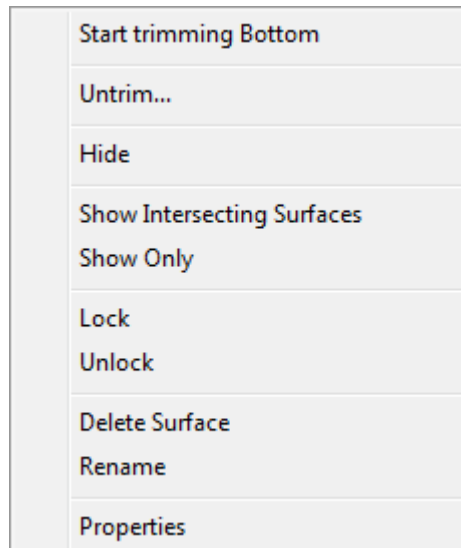
Displays information about the current version of Modeler you are using and other diagnostic information. Use this to obtain version and diagnostic information when reporting a problem to the Support Staff at Bentley Systems, Incorporated.

Right Click Menu

Some of the most commonly used commands can be accessed from a right click menu that is dependent on the object currently selected. These so called right click commands are exactly the same as the commands in the regular menus of the application. Please consult the [Menus](#) section on page 369 for an explanation of the individual commands.

Surface Selection Right Click Menu

A floating menu will appear if the right mouse button is clicked when a surface contour is selected in any of the view windows or a rendered surface is selected in the perspective window. The following menu will appear next to the cursor:

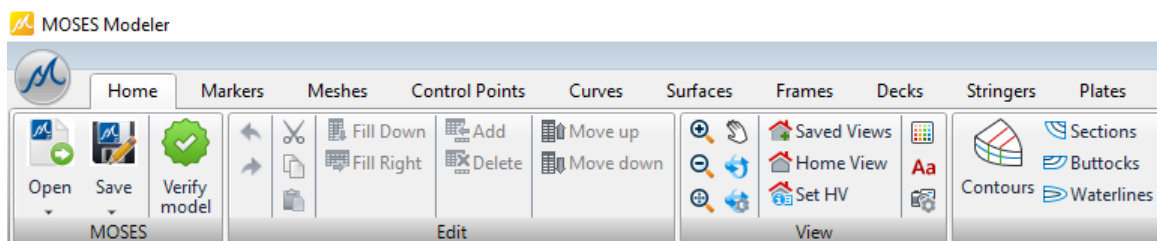


Right click menu after a surface has been selected.

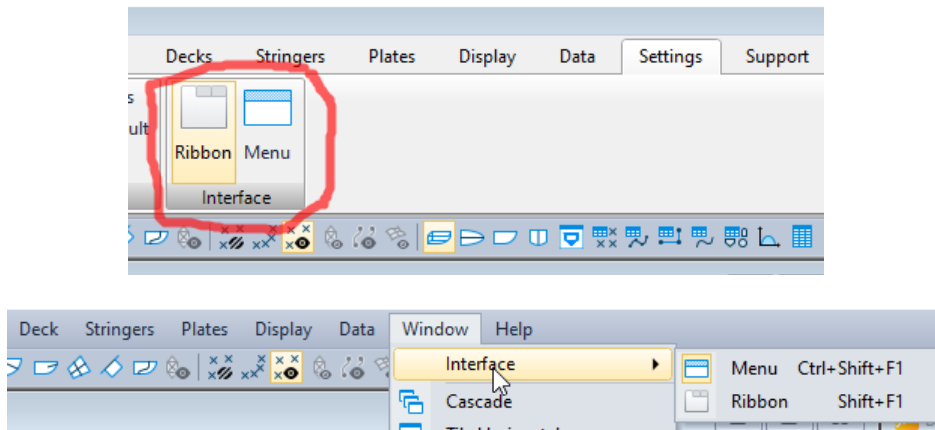
Menu items will be enabled, disabled or hidden depending on the context of the right click and application settings. For example, when trimming has been set to Trimming Off in the Display menu and a surface is selected by right clicking, the trimming commands in the right click menu will be hidden. You will have to switch trimming to Trimming Invisible or Gray first to be able to trim the surface.

Ribbon

A command Ribbon is available in SACS and MOSES versions.



Example of the Ribbon interface



Select the interface you find the most convenient

Appendix A – Data Export

Modeler provides a number of alternative methods for transferring data between Modeler and other software applications on other computers. Each of the different methods of data transfer has various advantages and disadvantages. This appendix will help you determine which method is most suitable for your work.

Broadly speaking the data interchange can be divided into five types.

- [Pictures and Text](#)
- [2D Drafting Data](#)
- [3D Drafting Data](#)
- [3D Surface Definition](#)

Also see: [Surface Precision](#) on page 115.

Supported File Formats

Modeler supports the following file formats:

Importing

MicroStation DGN	NURBS surfaces and curves
IGES	NURBS surfaces (untrimmed)
Rhino 3DM	NURBS surfaces and curves
IMSA	NURBS surfaces
USNA / Fastship	NURBS surfaces
DXF	Curves and Markers, Background
GHS	Marker sections
PIAS ascii	Marker sections
Seaway	Marker sections
Wolfson LHF	Marker sections
nuShallo PAN	Mesh surface
Polygon mesh PLY	Mesh surface
Stereo Lithography mesh STL	Mesh surface
WAMIT med GDF	Mesh surface
Image PNG, JPEG, GIF	Background image

Exporting

MicroStation DGN	NURBS surfaces and curves
IGES	NURBS surfaces
Rhino 3DM	NURBS surfaces and curves
IMSA	NURBS surfaces
USNA / Fastship	NURBS surfaces
DXF	Polylines
BMT Microship	Sections
MHCP	Sections
HYDROS	Sections
IHI	Sections
IMSA	Hull parameters
GHS	Sections
nuShallo	Mesh
Parametric	Sections
PD Strip	Sections
PIAS ascii	Sections
SHCP	Sections
Shipflow	Sections and waerlines
Stereo	
Veres (MASHIMO)	Sections
Wintech	Sections
Wolfson LHF	Sections
Polygon mesh PLY	Mesh surface
Stereo Lithography mesh STL	Mesh surface
WAMIT med GDF	Mesh surface
Direct X	Mesh surface
Wavefront	Mesh surface
Image BMP	Background image

Pictures and Text

Copying the drawing

The drawing in any of the views may be copied in two ways:

The first method will copy a scaled drawing to the clipboard which may then be pasted into another application such as MS Word. To do this select Copy from the Edit menu.

Modeler supports the transfer of pictures and tables of text on the clipboard for use by other applications. (If you are not familiar with copying information to the clipboard, consult your computer's owner's guide.) In Modeler you can copy pictures by using the Copy command from the Edit menu or by using Ctrl C.

When copying a design, the Copy dialog allows you to choose what scale you would like to copy the picture at. A picture on the clipboard can be pasted into word processors, page layout applications or drawing applications.

The main disadvantage of pictures is their limited resolution. This effect may be minimized by copying at a large scale and then reducing the size of the picture in the destination application. Some CAD programs have

a 'Re-scale when Pasting' option that allows you to reduce a set of lines copied into the Clipboard from Modeler at a 1:1 scale, down to the scale you are using at the time.

The second method makes a copy of the actual screen image (similar to a copy made by screen capturing software or the print screen button). To activate this method, press Ctrl I whilst in the view. Alternatively you can directly save the window image to a bitmap file by holding down the Shift key whilst pressing Ctrl I. This will then prompt you for the file name (by default the file name contains the design name and the view name).

Copying Text

As well as copying pictures to the clipboard, Modeler allows you to copy text from the Control Points, Markers and Offsets windows or any of the tables in the dialogs to the clipboard. It is also possible to Paste text into these windows.

Before copying or pasting you must select the range of data you wish to copy from or paste into. You can select a single cell, row, column, arbitrary rectangular range of cells or the whole table of data. You can click at the top of a column to select the whole column, click at the left of a row to select a whole row, and click at the top left of a table to select the whole table.

Text copied from Modeler will be placed on the clipboard in "Tab Delimited Format". This means each number is separated from the next number in the row by a Tab character, and each row of numbers is terminated by a Return character. If you paste this text into a word processor or text editor such as Microsoft Word, the data will be displayed with each row on a separate line, and tabs between the numbers. If you paste the text into a spreadsheet such as Excel, the numbers will be displayed in individual cells exactly as they are shown in the tables in Modeler.

If you hold down the Shift key while using the Copy command in a table, the headings of the columns will also be included in the clipboard.

It is also possible to save the contents of the Control Points, Markers and Offsets windows as text files. With the appropriate window frontmost on the screen, choose Save As... from the File menu and Modeler will save the contents of the window into a text file. This data will also be in Tab Delimited Format, i.e. the contents of the text file will be exactly the same as the text placed on the clipboard if you copy the whole table. Text files created in this way can be read into a word processor, text editor or spreadsheet.

If you wish to write a program that reads data from Modeler, these text files are an easy way to access the data. If you want to reconstruct a Modeler surface exactly, you can read in the control point positions and use them to generate the surface coordinates from the standard B-spline algorithm (see [Appendix B – Surface Algorithms](#) on page 401).

Note:

Surface stiffness and other properties may be saved from the Surfaces window.

2D Drafting Data

Modeler includes support for IGES and DXF files, the two most commonly used interchange formats for computer-generated drawings. These files are the most effective way of transferring lines drawings to a CAD system for further drafting work.

Two-dimensional output is always in the $z=0$ plane. The positive y -axis corresponds to the upward vertical screen axis and the positive x -axis to the right-hand-side horizontal screen axis.

DXF (Drawing Exchange File) was originated by AutoCAD and is supported by almost all CAD systems on most platforms.

IGES (Initial Graphics Exchange Standard) is widely supported by CAD systems on all types of computers and is published by the US Dept of Commerce, National Bureau of Standards. IGES is widely supported by workstation and mainframe based CAD systems.

These files are the most accurate way to transfer lines drawings from Modeler to a CAD system with data accurate to 5 decimal digits.

Note:

Modeler designs exported in this way are at the current Modeler precision. For best quality, set the Modeler precision to High or Highest before writing the file.

When you choose Export from the File menu, with a drawing window frontmost, Modeler will present you with the export dialog, allowing you to choose whether you wish to save DXF or IGES format data. If you choose either of these options and click OK, Modeler will ask you for the name of the file and allow you to save it to disk. This file will contain the DXF or IGES format data, which can then be read into your CAD system (e.g. using the DXFIN command in AutoCAD). Use the 2D option in the Export dialog to copy drawings in this way.

Note:

Many programs (such as AutoCAD) require a file extension for both DXF and IGES files. File names should have either the .DXF or .IGS extension appended to them.

Modeler supports the DXF format and is compatible backwards to release 10. It supports the Line and Polyline entities. Each contour from Modeler will be saved as a polyline in the DXF file.

If the export “polylines with arcs” option is chosen for 2D drawings the user needs to set a maximum radius value. All arcs in the polyline with radius smaller than this value will be exported as polyline entities with dxf bulge values set for the arc segments. All arcs with radius greater than this value will be exported as polyline vertices only.

Modeler supports IGES, version 4.0. It supports IGES entity number 106, the copious data entity. Each contour from Modeler is saved as a linear path of copious data, with x, y and z coordinates.

Also see: [Surface Precision](#) on page 115.

3D Drafting Data

If you use your CAD system in 3D, you can use the DXF or IGES methods to save the lines from the Modeler design as three-dimensional lines rather than just 2D. This allows you to do things like interior space planning in three dimensions. Choose the 3D IGES or 3D DXF file format as required.

Modeler exports 3D poly-line and surface data so that it will be correctly interpreted by a CAD system which has the following right-handed co-ordinate system: x-axis, positive along the forward longitudinal axis of the vessel; y-axis, positive along the port transverse axis of the vessel; and z-axis, positive along the upward vertical axis. This means that the y-ordinates will be the negative of the Modeler “Offset” values since, in Modeler, “Offsets” are measured transversely, but positive to starboard.

Modeler supports the 3DLine, Arc and 3DPolyline entities. Each contour from Modeler will be saved as a 3D polyline in the DXF file.

If the export “polylines with arcs” option is chosen for 3D drawings the user needs to set a maximum radius value. All arcs with radius smaller than this value will be exported as dxf arc entities. All arcs with radius greater than this value will be exported as polylines.

Modeler supports IGES entity number 106. Each contour from Modeler is saved as a linear path of copious data, with x, y and z coordinates.

Also see: [Surface Precision](#) on page 115.

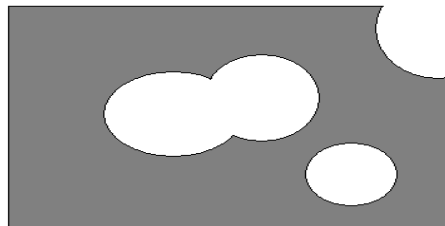
3D Surface Definition

Modeler supports IGES NURB surfaces, which allow you to transfer an exact surface description from Modeler, if you use a CAD system, such as ShiCpConstructor, SolidWorks or Catia which supports B-spline surfaces. This is the most advanced and accurate way to transfer the model and ensures that accuracy is preserved. Use the Export function to export an IGES NURBS surface model in 3D. When your CAD system reads in the data it will accept the surfaces and will then allow you to manipulate them in any way that the system allows.

As well as the usual contours, which are saved as IGES entity number 106, Modeler will save an IGES entity number 128 for each Modeler surface and any associated trimming data as entity 144. This entity is a NURB (Non Uniform Rational B-spline) surface, which has the positions of the control points and the stiffness of the surface stored within it.

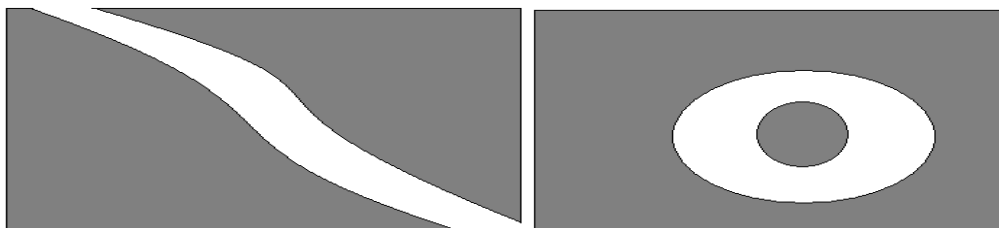
If you wish to write custom software that needs to read accurate Modeler surface descriptions, we recommend that you support input of the IGES file and this entity in particular as the most effective way of accessing the Modeler surface description.

Please note that there are some surface topologies that may cause difficulties when they are exported as trimmed IGES surfaces. In these cases the trimming may have to be re-done in the importing application. In most cases the trimmed IGES export should work correctly: For example the topology shown below with multiple edge and interior trims, including intersecting trim regions, will work correctly.



There are two cases where IGES NURBS surface trimming is known to break down. The topology shown below (right), where a trim region contains an untrimmed region, will display correctly in Modeler but may not transfer correctly through IGES – this may be dependent on the receiving program.

The major limitation of IGES NURBS transfer is in the interpretation of a surface that has been divided into two or more independent regions, as in the following examples:



Exporting to ShipConstructor Hull

If you wish to export surfaces to ShipConstructor's Hull module, you should follow the following sequence -

Set Units in Modeler to mm using the Units command from the Data menu.

If you wish to export Trimmed surfaces, then set Precision to Highest and turn trimming On. This is to ensure any trim curves are accurate. Precision does not affect the surface data itself.

Use Export IGES from the Export sub-menu in the File menu

Choose Format: 3D IGES, Scale: 1:1, Geometry Type: NURBS Surfaces, Precision: 2 decimal places.

This last precision setting is important to avoid roundoff error in larger numbers of decimal places when using mm for dimensions.

In ShipConstructor Hull module, Use Import Iges to Import the file. In the Import dialog box, set the tolerance to 0.5mm. This will usually give you a good quality surface mesh. You could go down to 0.25mm if you want a really fine mesh but its best to avoid tolerances finer than this as the resulting meshes in Hull will likely be too large.

Appendix B – Surface Algorithms

This appendix describes the mathematical basis of the Modeler models, and will be of interest to users who wish to write software to recreate Modeler surfaces.

Modeler constructs its shapes using the Rational B-spline formulation as described by the IGES standard. Modeler uses a uniform knot vector. However, non-uniform knot vectors are permissible.

B-spline curves and surfaces are dependent on a set of basis functions, which define the influence of each control point at any given point on the curve. The basis functions are determined by a vector of knots T where $T = \{t_0, \dots, t_i, t_{i+1}, \dots, t_m\}$ and $m = \text{number of control points} + \text{order of the curve}$. The parameter t varies from zero to $n - k + 2$. In general, the B-spline basis function on a curve of degree p (order $k = p + 1$) is given by

$$N_{i,0}(t) = \begin{cases} 1 & \text{if } t_i \leq t < t_{i+1} \text{ and } t_i < t_{i+1} \\ 0 & \text{otherwise} \end{cases}$$

$$N_{i,p}(t) = \frac{t - t_i}{t_{i+p} - t_i} N_{i,p-1}(t) + \frac{t_{i+p+1} - t}{t_{i+p+1} - t_{i+1}} N_{i+1,p-1}(t) \quad \text{assuming } 0/0=0.$$

The elements of the open uniform knot vector, t_i , are given by

$$\begin{aligned} t_i &= 0 & 1 \leq i \leq k \\ t_i &= i - k & k + 1 \leq i \leq n + 1 \\ t_i &= n - k + 2 & n + 2 \leq i \leq n + k + 1 \end{aligned}$$

where there are $n+1$ control points on the curve, and k is the order of the curve ($k = p + 1$).

A rational B-spline surface is given by

$$\sum_{i=0}^m \sum_{j=0}^n N_{i,p}(u) N_{j,q}(v) W_{ij} P_{ij}$$

$$S(u,v) = \sum_{i=0}^m \sum_{j=0}^n N_{i,p}(u) N_{j,q}(v) W_{ij}$$

where $S(u,v)$ is a point on the B-spline surface at parametric coordinates u,v , $N_{i,p}(u)$ and $N_{j,q}(v)$ are the B-spline basis functions, W_{ij} are the rational weighting values, and P_{ij} are the control point positions.

Appendix C – Command Keys

There are several functions in Modeler that are not obvious to the user as they rely on key combinations to activate them. This is a summary of the various key combinations and their effects:

- [Modifier Keys](#)
- [Special Keys](#)
- [Control Point Weights](#)
- [Menu Shortcuts](#)

Modifier Keys

The following keys modify a function while they are held down.

Shift or Ctrl - select of points.

Adds or removes control points from the current selection.

Shift - drag of points.

Constrains movement to horizontal or vertical planes.

Ctrl - drag of points.

Lets you drag control points on a symmetrical surface over the centreline.

Ctrl - while developable surface is drawing

Calculates markers which can later be used as guides for designing a developable surface.

Shift + Ctrl + C

Copies the column headings along with the data in a data window such as control points or markers.

Shift - While starting Modeler

Will prompt you to reset the Modeler preferences. This will reset the windows registry. For more information see [Modeler Settings](#).

Shift + Save

Adds the current surface to the bottom of the list in the Add Surface menu.

Special Keys

Key	Function
Space bar	Force a redraw of the front window
F1	Help/Contents
F2	Displays Assembly Pane
F3	Toggles Snap On / Off
F4	Displays Properties Pane

F5	Displays Drawing Setting Dialog Box
Ctrl+F6	Next Window
Ctrl+Shift+F6	Previous Window

Control Point Weights

The following keys change the weights of the currently selected control points.

Key	Keyboard symbol	Function
MultKey	*	Increase current weight by factor of 2.0
DivKey	/	Divide current weight by 2.0
PlusKey	+	Add 0.1 to current weight
MinusKey	-	Subtract 0.1 from current weight
ZeroKey	0	Set weight to 0.7071 (value required for mid point on a 3-point circular arc)
1..9	1..9	Set weight to 1.0 to 9.0 respectively

Menu Shortcuts

The following keyboard shortcuts access menu commands

Key	Menu	Function
Ctrl+A	Controls	Add Column/Row
Ctrl+C	Edit	Copy
Ctrl+Shft+C	Edit	Copy, including the column headers
Ctrl+D	Controls	Delete Column/Row
Ctrl+E	View	Zoom
Ctrl+F	Edit	Fill Down
Ctrl+G	Controls	Group
Ctrl+H	View	Home View
Ctrl+I	-	Copy the design screen to a clipboard image
Shift+Ctrl+I	-	Copy the design screen to a bitmap file
Ctrl+K	Controls	Compact
Ctrl+L	Edit	Delete Marker
Ctrl+M	Edit	Add Marker
Ctrl+N	File	New
Ctrl+O	File	Open
Ctrl+P	File	Print Preview
Ctrl+R	View	Shrink
Ctrl+S	File	Save
Ctrl+T	Surfaces	Trim
Ctrl+U	Controls	Ungroup
Ctrl+V	Edit	Paste
Ctrl+W	View	Pan
Ctrl+X	Edit	Cut
Ctrl+Y	Edit	Redo

Ctrl+Z	Edit	Undo
Ctrl+Insert	Edit	Copy
Shift+Insert	Edit	Paste
Alt+Backspace	Edit	Undo
Shift+Delete	Edit	Cut

Appendix D – Moving Files Across Platforms

Modeler design and data files may be moved easily between PC and Macintosh computers. The data file structure used by the two formats is identical on both platforms. The changes required when moving files from Macintosh to PC are only to allow the Open File dialogs to determine the contents of the file.

- [Moving Files From Macintosh to Windows](#)
- [Moving Files from Windows to Macintosh](#)

Moving Files From Macintosh to Windows

Copy your Macintosh file to your Windows computer, either via a network or by inserting a DOS formatted floppy disk into your Macintosh computer and copying the required files onto the floppy disk. Put the disk into your PC and copy the files to the desired location on your PC's hard disk.

It is necessary to add a file extension to the file name so that your PC can determine the contents of the file. If you already have Modeler installed on your PC, once you have added the correct `.msd` file extension to your Modeler design files, these files will have the Modeler icon associated with them.

The following file extensions are given to files saved on the PC. Macintosh users copying files from Macintosh to PC should add the appropriate file extension so that the files are recognised correctly.

	File Extension
Modeler	
Modeler Design Files	<code>.msd</code>
Modeler Calculation Files	<code>.msc</code>
Marker, Offsets, Control Points Files	<code>.txt</code>
Data eXchange Files	<code>.dxf</code>
Iges Files	<code>.igs</code>
Stability	
Stability Loadcase Files	<code>.hml</code>
Damage Case Files	<code>.dcs</code>
Room and Sounding Pipe Definition Files	<code>.htk</code>
Down Flooding Points, Margin Line Points, Modulus Points, Output Data Files	<code>.txt</code>
Report(Rich Text Format)	<code>.rtf</code>
Link	
USNA, IMSA NURBs, Output Data Files	<code>.txt</code>
Nakashima Stereo Files	<code>.nst</code>
Structure	

Structure Design Files	.wsd
Materials Library Files	.slb
Resistance	
Resistance Measurement Files	.hsd
Fitting	
Fitting Offsets Files	.pfd
VPP	
VPP Data Files	.spd
Motions	
Motions Data Files	.sld
Image Formats	
Quickdraw 3D Metafiles	.3dmf
PICT Files	.pic
Renderman Files	.rib
Video for Windows	.avi
Other	
Rich Text Format	.rtf
Text Files	.txt
Temporary Files	.tmp
EPS (expanded PostScript) Files	.eps

Moving Files from Windows to Macintosh

Windows Modeler designs with the file extension .msd will automatically open on the Macintosh. Other files must have their type and creator attributes changed once they have been copied to the Macintosh. This may be done with software such as ResEdit. The following type and creator attributes are required:

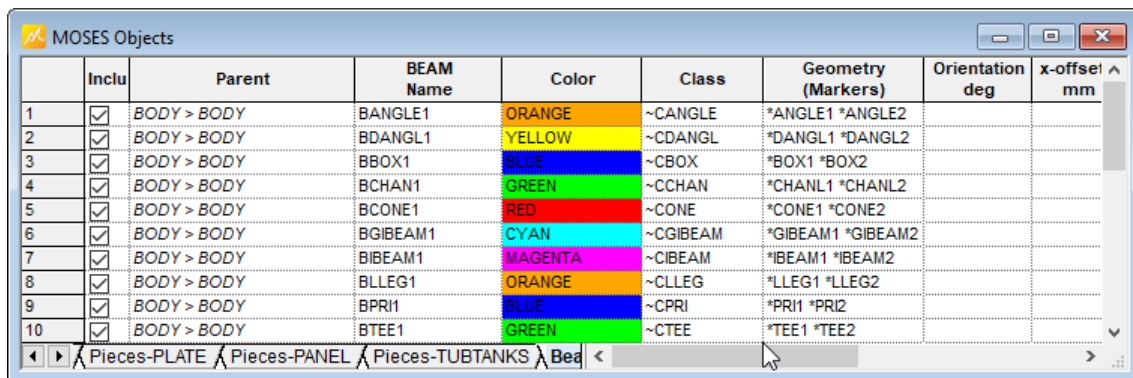
<i>Application and File type</i>	<i>Type</i>	<i>Creator</i>
Modeler		
Modeler Design Files	SHA3	SURF
Modeler Calculation Files	TEXT	SURF
Marker, Offsets, Control Points Files	TEXT	MWRT
Data eXchange Files	TEXT	????
Iges Files	TEXT	????
Stability		
Stability Loadcase Files	CGDA	SHED
Tank Definition Files	TKDF	STAB
Down Flooding Points, Margin Line Points, Modulus Points, Output Data Files	TEXT	MWRT
Report	RPRT	STAB

Structure		
Structure Design Files	SHOP	SHOP
Materials Library Files	WSLB	SHOP
Resistance		
Resistance Measurement Files	ZIS1	ZIST
Fitting		
Fitting Offsets Files	PFIN	PFIT
VPP		
VPP Data Files	QKS1	QKSL

Appendix E – MOSES Objects

User Interface

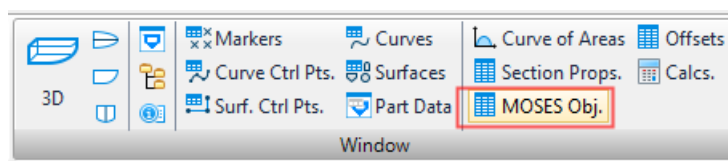
The MOSES Objects are defined in the corresponding tables:



	Inclu	Parent	BEAM Name	Color	Class	Geometry (Markers)	Orientation deg	x-offset mm
1	☒	BODY > BODY	BANGLE1	ORANGE	~CANGLE	*ANGLE1 *ANGLE2		
2	☒	BODY > BODY	BDANGL1	YELLOW	~CDANGL	*DANGL1 *DANGL2		
3	☒	BODY > BODY	BBOX1	BLUE	~CBOX	*BOX1 *BOX2		
4	☒	BODY > BODY	BCHAN1	GREEN	~CCHAN	*CHANL1 *CHANL2		
5	☒	BODY > BODY	BCONE1	RED	~CONE	*CONE1 *CONE2		
6	☒	BODY > BODY	BGIBeam1	CYAN	~CGIBeam	*GIBeam1 *GIBeam2		
7	☒	BODY > BODY	BIBeam1	MAGENTA	~CIBeam	*IBeam1 *IBeam2		
8	☒	BODY > BODY	BLLEG1	ORANGE	~CLLEG	*LLEG1 *LLEG2		
9	☒	BODY > BODY	BPRI1	BLUE	~CPRI	*PRI1 *PRI2		
10	☒	BODY > BODY	BTEE1	GREEN	~CTEE	*TEE1 *TEE2		

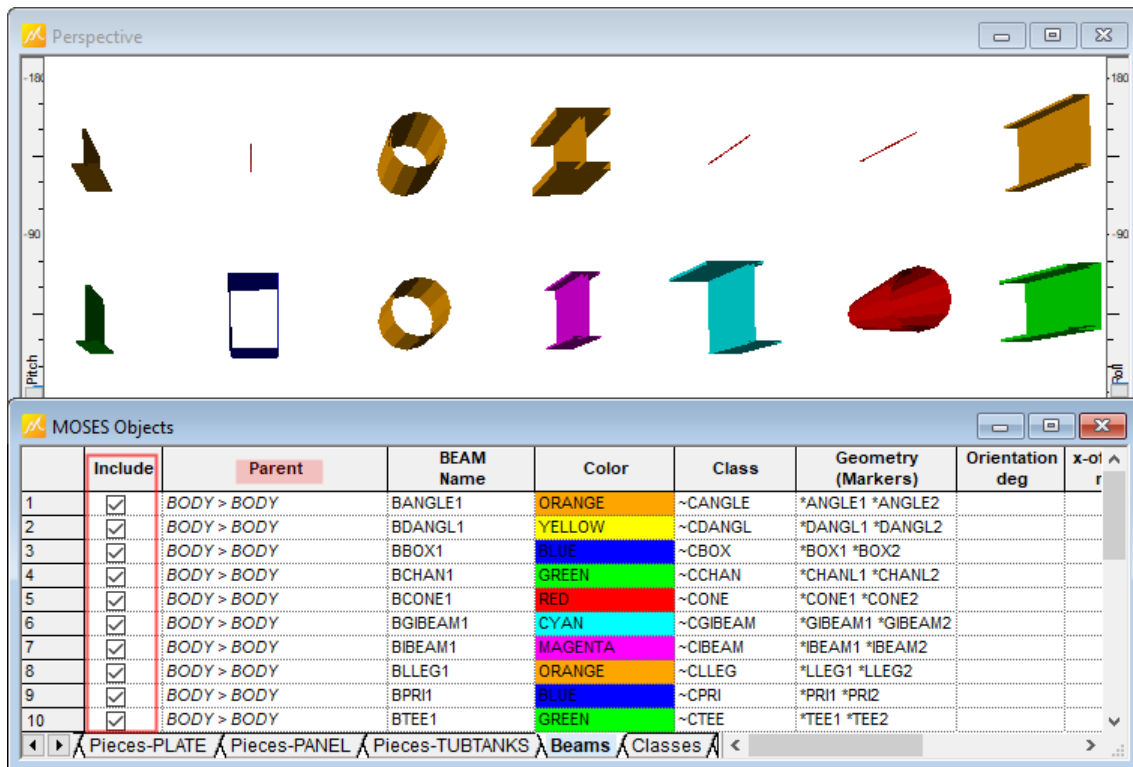
MOSES Object table

This window may be accessed from the following button on the ribbon in the Home tab



Button to access the MOSES Objects window

MOSES Objects are drawn when they and their parent hierarchy are 'Included' in the model.



MOSES Objects are drawn when they and their parent hierarchy are 'Included' in the model.

Introduction

The MOSES Objects table allows you to setup the MOSES model for analysis and link it to geometry elements generated in the Modeler. The MOSES model can then be saved directly as a MOSES.dat file along with an accompanying MOSES.cif file which provides a starting point for setting up your analysis and facilitates viewing the model in MOSES Executive.

When saving the MOSES.dat file all geometry is updated automatically; it is not necessary to manually update any geometry. Furthermore, only selected parts of the full model can be included in the MOSES.dat file adding further flexibility.

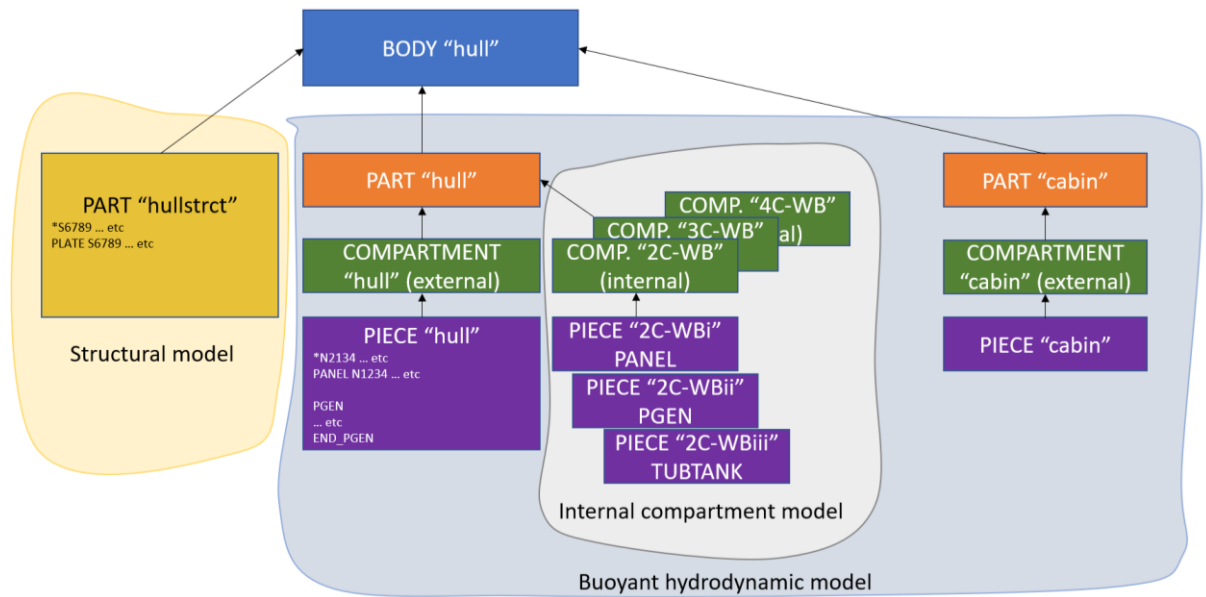
The complete MOSES Object definition along with any customised options are all stored in the normal model.msd file making it very easy to restore a previous work session.

This documentation only explains how MOSES Objects are defined within the MOSES Modeler. Please refer to the MOSES manual and examples for full details of the MOSES Objects and the various options and parameters pertinent to each object type and how they are used.

MOSES Model

The MOSES analysis model objects form a hierarchy as shown below. There are broadly two main parts of the model. The top level of the model is the BODY; several, independent bodies may be defined in a single model file where both the hydrodynamic and structural models may coexist.

- Hydrodynamic model
- Structural model



Schematic of MOSES Object hierarchy

Hydrodynamic model

The MOSES hydrodynamic model is made up of several PARTs. Each PART may contain several COMPARTMENTS and a COMPARTMENT is made up of PIECEs. The actual geometry is defined at the PIECE level and there are three types of hydrodynamic PIECE:

- PANEL meshes
- PGEN sections (definition of PGEN Objects in Hull Modeler is not supported)
- TUBTANK prismatic, circular tubes

A COMPARTMENT is used to define the buoyant hull as well as the internal subdivision of spaces within the buoyant hull.

Structural model

Similarly, the PARTs can contain structural model elements. These are defined two object types:

- PLATE meshes
- BEAM elements

Nodes are defined as the joints between the PLATE mesh elements and beams; these are generated automatically from the specified geometry elements.

Definition of MOSES Objects

The MOSES Objects are all defined in the MOSES Objects window with an individual table for each type of object. The object hierarchy follows from left to right, with the highest level objects, the bodies, on the left. Each row of a table corresponds to a single MOSES Object.

	Include	Parent	PART Name	Type	COMPARTMENT List	PLATE PIECE List	BEAM List	POINT List	Option
1	☐	DEEPCW	COLPANT		DCWL1PT DCWL2PT DCWL3				
2	☐	DEEPCW	COLTUBT		DCWL1TT DCWL2TT DCWL3				
3	☒	DEEPCW	DEEPCWC		DEEPCW			*MRNG1 *MRNG2 *MR	
4	☐	DEEPCW	DEEPCWP			DCWL1PL DCWL2PL DCWL3			
5	☒	DEEPCW	DU	JACKET			DU1 DU2 DU3		
6	☒	DEEPCW	DL	JACKET			DL1 DL2 DL3		
7	☒	DEEPCW	YU	JACKET			YU1 YU2 YU3		
8	☒	DEEPCW	YL	JACKET			YL1 YL2 YL3		
9	☒	DEEPCW	CB	JACKET			CB1 CB2 CB3		
10	☒	DEEPCW	ROTOR	PROP				*NACL *BLDTP1 *BLD	
11	☒	DEEPCW	HELIDK	JACKET		HELIDK			

Navigation: Bodies | **Parts** | Compartments | Pieces-PLATE | Pieces-PANEL | Pieces-TUBTANKS | Beams | Classes | Points

Definition of MOSES Objects

When hovering over a cell in the tables you will notice tooltips which provide a helpful prompt as to the meaning of the data in each column.

PANEL PIECE List	PGEN PIECE List	TUBTANK PIECE List	Options
List of PANEL PIECES in this COMPARTMENT. Double-click for dialog. F2 to edit; space-delimited; use '*' as wildcard			

Hover over cell for help when entering data

Automatic PART, COMPARTMENT and PANEL objects

When you add a new BODY MOSES object a corresponding PART and COMPARTMENT are added with the same name – this reflects what was automatically done by MOSES, and is normally the outer watertight envelope of the BODY. In addition a PANEL object is also created, this should contain the exterior watertight geometry of the BODY. The name of the PART will have 'P' appended to the name of the BODY, but this can be changed if required; furthermore, and more importantly It will have the following options:

`-CS_CURRENT 1 1 1 -TANAKA 0 -DIFTYPE 3DDIF'`

automatically added and will also be given a permeability of 1.0 corresponding to a PIECE providing positive buoyancy. (These options and permeability, as well as the name of the PANEL PIECE may be modified if desired.)

Geometry symmetry of PLATE and PANEL objects

In common with NURBS surface and Curve objects, Mesh surfaces can be defined to have port/starboard symmetry; in this case the original geometry has also a mirrored geometry which is reflected across the centerline plane of $y=0$. It is generally recommended, for these symmetrical geometries, that the original geometry be modeled on the starboard side (with positive y -offsets) with the mirrored geometry appearing on the port side (with negative y -offsets). The geometry's symmetry flag is easily set in the appropriate table:

	Name	Assem	Type	Use	Rows	Cols	Long.	Trans.	Visibility	Locking	Symmetry	Split	Para.	Para.	Thickness m	Direction	Color	Trans	Cutting
8	BoxAft trimesh	Struct	Trimes	Hu	18 tria	15 no	n/a	n/a	☐	☐	☒	☐	n/a	n/a	0.000	Inside	Blue	50	n/a
9	BoxFwd trimesh	Struct	Trimes	Hu	18 tria	15 no	n/a	n/a	☐	☐	☒	☐	n/a	n/a	0.000	Inside	Blue	50	n/a
10	BoxTop trimesh	Struct	Trimes	Hu	90 tria	63 no	n/a	n/a	☐	☐	☒	☐	n/a	n/a	0.000	Inside	Blue	50	n/a
11	BoxBottom trimesh	Struct	Trimes	Hu	90 tria	63 no	n/a	n/a	☐	☐	☒	☐	n/a	n/a	0.000	Inside	Blue	50	n/a
12	BoxStbd trimesh	Struct	Trimes	Hu	124 tria	81 no	n/a	n/a	☐	☐	☒	☐	n/a	n/a	0.000	Inside	Blue	50	n/a
13	Roof trimesh	Struct	Trimes	Hu	54 tria	42 no	n/a	n/a	☐	☐	☒	☐	n/a	n/a	0.000	Inside	Blue	50	n/a
14	Bhead trimesh	Struct	Trimes	Hu	23 tria	19 no	n/a	n/a	☐	☐	☒	☐	n/a	n/a	0.000	Inside	Blue	50	n/a

Mesh symmetry can be set in the Surfaces table

The geometry of MOSES PLATE and PANEL objects created in Modeler is defined by a Mesh surface. If the PLATE or PANEL object's Mesh is symmetrical, then you can select which part of the Mesh should be used. You can select from:

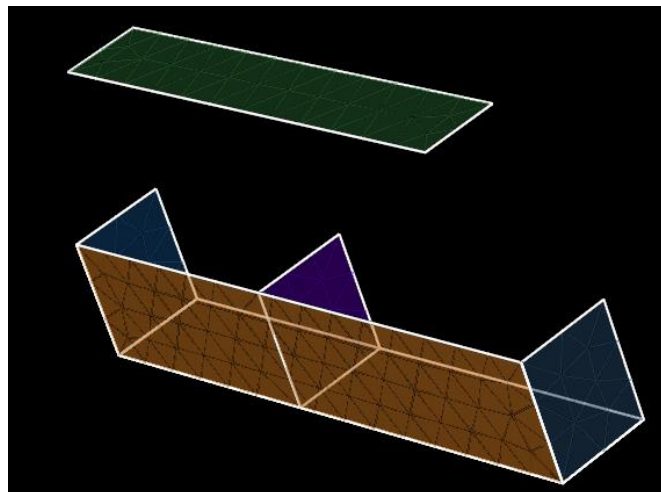
- Original – the side of the Mesh that is actually modeled; it is recommended that this be the starboard side
- Reflection – the mirrored reflection of the 'Original' across the centerline plane of $y=0$; this is typically the port side
- Both – both the 'Original' and the 'Reflection' will be used. Nodes which lie on the centerline plane will not be duplicated. If there are no nodes on the centerline plane then the two halves will be disconnected (the two halves will not be connected across the centerline plane).

MOSES Objects												
	Incl	Parent	Structure Name	Color	Class	Geometry (TMesh)	Symmetry	Node Prefix	DNV stiffened plate	Options	PLATE joint Merge tol. mm	Transparency %
1	☒	BODY001 > BODY001	AFT	STEEL	~PL10MM	BoxAft trimesh	Both	JA	☐		0.100	50
2	☒	BODY001 > BODY001	FWD	SKYB	~PL10MM	BoxFwd trimesh	Original	JE	☐		0.100	50
3	☐		TOP	GRAY	~PL5MM	BoxTop trimesh	Reflection					
4	☒	BODY001 > BODY001	BOTTO	WHEA	~PL10MM	BoxBottom trimesh	Both	JB	☐		0.100	50
5	☒	BODY001 > BODY001	BHDS	GREEN	~PL8MM	Bhead trimesh	Original	JHS	☐		0.100	50
6	☒	BODY001 > BODY001	BHDP	RED	~PL8MM	Bhead trimesh	Reflection	JHP	☐		0.100	50
7	☒	BODY001 > BODY001	SIDES	M_SE	~PL10MM	BoxStbd trimesh	Original	JSS	☐		0.100	50
8	☒	BODY001 > BODY001	SIDEP	BROW	~PL10MM	BoxStbd trimesh	Reflection	JSP	☐		0.100	50
9	☒	BODY001 > BODY001	ROOF	SEA_	~PL5MM	Roof trimesh	Both	JG	☐		0.100	50

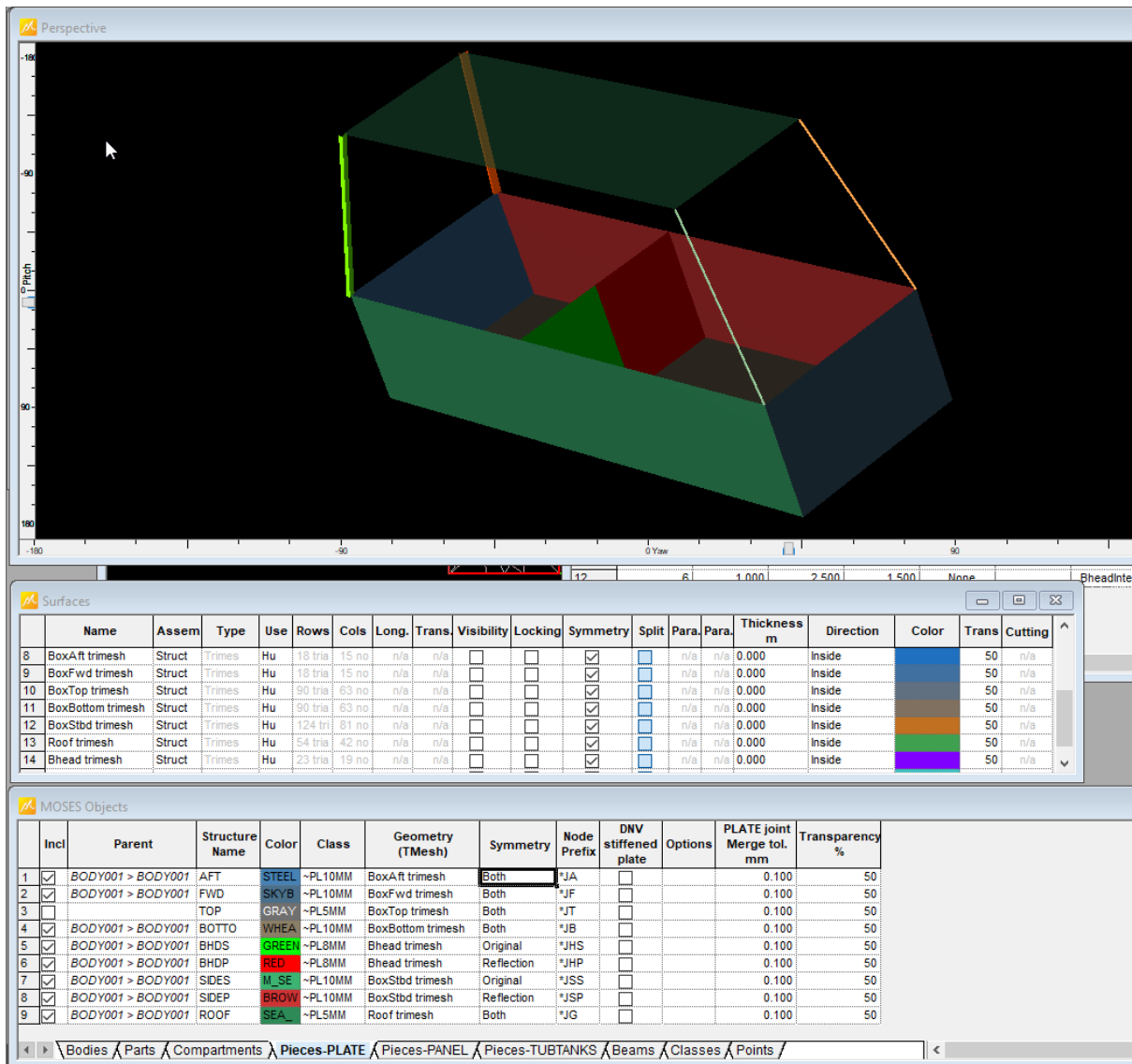
Use original (starboard) or reflected (port) instance of symr

☐ Bodies ☐ Parts ☐ Compartments ☒ Pieces-PLATE ☐ Pieces-PANEL ☐ Pieces-TUBTANKS ☐ Beams ☐ Classes ☐ Points

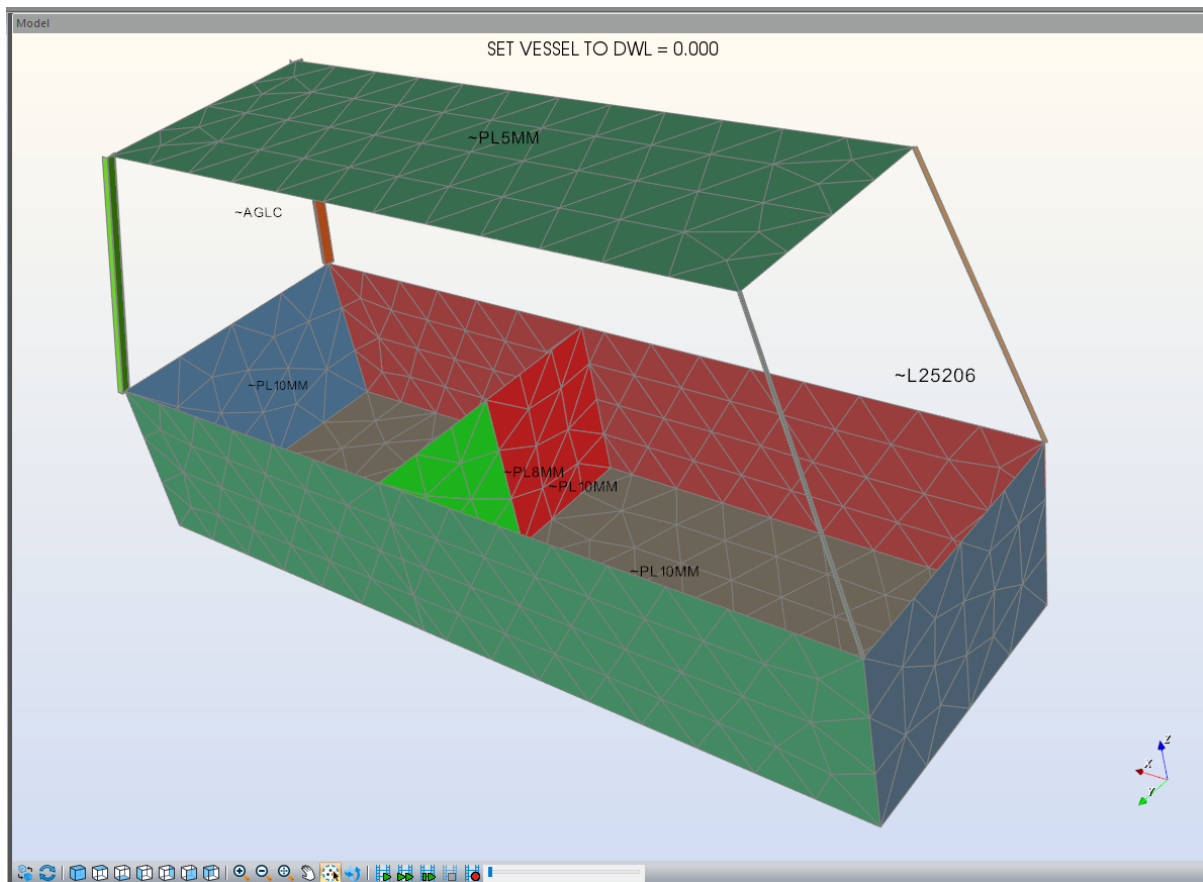
Use the Symmetry column in the PLATE and PLANEL tables to specify which part of the Mesh should define the geometry of the MOSES object.



Symmetrical Meshes of only the starboard side create the full, symmetrical, geometry.



MOSES objects using 'Original', 'Reflected' and 'Both' instances of the Mesh geometry.



Model as seen in MOSES Executive

Having to only model half the geometry of symmetrical parts of the vessel can provide significant time saving and also ensure exact symmetry between port and starboard sides.

Node Reuse

Nodes may be reused for the geometry of PLATE, BEAM and PANEL objects. BEAM objects can be made to use existing nodes from the PLATE meshes if they are within the specified tolerance; similarly PLATE objects can reuse the nodes of existing PLATE objects.

PANEL objects can also be made to reuse existing nodes if they are within the specified tolerance. PANEL objects will first search through the nodes of other PANEL objects and reuse those nodes in preference. If a suitable PANEL node is *not* found, it will then try to find a suitable PLATE node. PLATES and BEAMS will *not* use reuse PANEL nodes.

MOSES Objects											
	Inclu	Parent	PANEL PIECE Name	Color	Permeabil	Geometry (TMesh)	Symmetry	Node Prefix	Options	PANEL, PLATE node Merge tol. mm	Transparency %
4	☒	BARGE > BARGE > HOLDFWD	HOLDFWD	STEEL_B	-0.900	HoldFwd	Both	*HF		0.000	50
5	☒	BARGE > BARGE > WTPAFT	WTPAFT	TOMATO	-0.950	WingAft	Reflection	*TPA		0.000	50
6	☒	BARGE > BARGE > WTPMID	WTPMID	RED	-0.950	WingMid	Reflection	*TPM		0.000	50
7	☒	BARGE > BARGE > WTPFWD	WTPFWD	CORAL4	-0.950	WingFwd	Reflection	*TPF		0.000	50
8	☒	BARGE > BARGE > WTSAFT	WTSAFT	LIME_GR	-0.950	WingAft	Original	*TSA		0.000	50
9	☒	BARGE > BARGE > WTSMD	WTSMD	FOREST	-0.950	WingMid	Original	*TSM		0.000	50
10	☒	BARGE > BARGE > WTSFWD	WTSFWD	O_OLIVE	-0.950	WingFwd	Original	*TSF		0.000	50

◀ ▶

Bodies

Parts

Compartments

Pieces-PLATE

Pieces-PANEL

Pieces-TUBTANKS

Beams

Classes

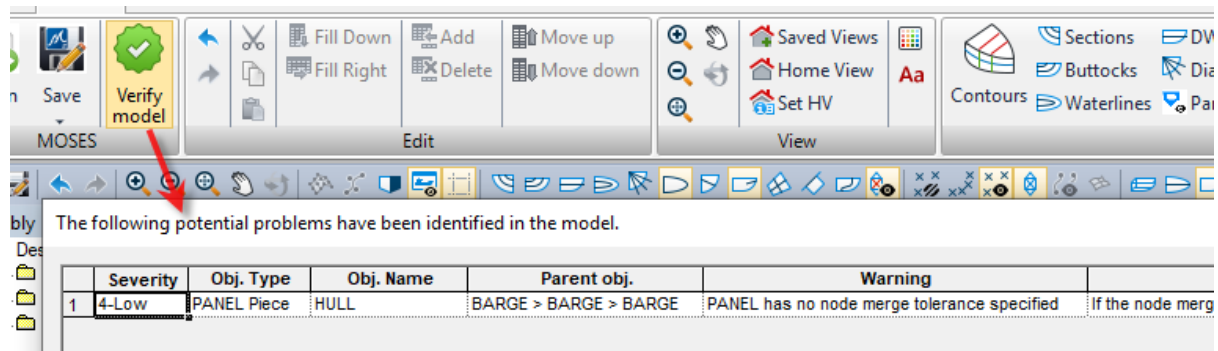
Points

Node Reuse Tolerance

Nodes are only reused within the same PART. The same tolerance specification is used for BEAMS, PLATES and PANELS. A small positive tolerance may be specified, in this case it is possible for there to be some slight change in geometry, because the reused node may not be at *exactly* the same location, only within the specified tolerance. For an *exact* match (to the number of decimal places the nodes are written in the file) specify a tolerance of zero – if an exact match is found, the node will be reused. If the tolerance is left blank (not specified) then the object will create all the nodes it needs and not seek to reuse any existing nodes already defined in the PART.

Verifying the Model and Object data

Some general verification of the MOSES object definition to help ensure a properly defined MOSES model is provided. This is done automatically when the MOSES model is saved or can be done at any time with the Verify Model command:



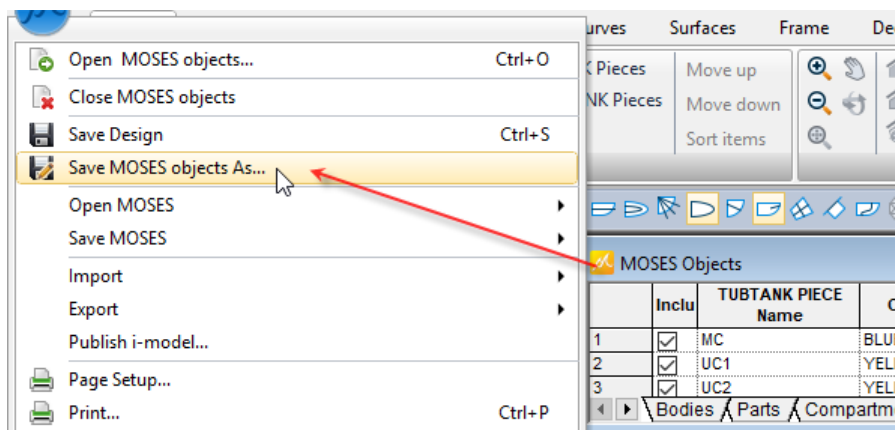
Basic MOSES object model verification

Potential problems are identified and given a severity, as well as a suggested action to be taken to resolve the issue. This is done automatically when saving the MOSES model, but in this case the dialog will *only* be shown if there are warnings.

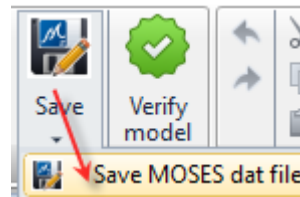
Saving Model and Object data

There are three subtly different ways in which the MOSES Objects are saved.

- As with all model information, all the MOSES Objects defined in these tables are automatically included in the model.msd file when save. This happens when you select "Save" from any window or "Save As" from one of the main drawing windows.
- The MOSES Object tables may be saved as a standalone file by using the "Save As" command when the MOSES Objects window has focus:



- The MOSES Objects and geometry may be saved to a MOSES.dat (and accompanying MOSES.cif) file for analysis in MOSES Executive:



DEFAULT values in the model file

When you save the MOSES model file, the model.dat file generated will include some default settings and set the default material properties to those of generic steel. Full details of these settings can be found in the MOSES documentation.

```

3  $ Exported from MOSES Modeler CONNECT Edition to MOSES geometry (*.dat) file format
4  $ Version: 12.02.00.19
5  $
6  $#####
7  $@
8  $@
9  $@
10 $#####
11 $
12 $
13 $** Save existing and set dimensions
14 &DIMEN -SAVE -DIMEN METRES M-TONS
15 $
16 $** Save existing default values
17 &DEFAULT -SAVE
18 $
19 $ Set new default values for general steel, code check factors, tubular beam flooding and use
20 $ Property      | MO.Imperial | MO.Metric   | Metric      | MO.Option
21 $ Density       | 490 lb/ft^3 | 77000 N/m^3 | 7850 kg/m^3 | -DENSITY
22 $ Young's mod.  | 29000 ksi   | 200000 MPa  | 200 GPa     | -EMODULUS
23 $ Yield stress  | 36 ksi      | 250 MPa     | 250 MPa     | -FYIELD
24 $ Coef.therm.exp. | 6.5e-6 1/F | 1.2e-5 1/C  | 1.2e-5 1/K  | -ALPHA
25 $ Poisson's rat. | 0.3         | 0.3         | 0.3         | -POI_RAT
26 $
27 $ -CMFAC 0.85 0.85 $ CM factors for bending, used in the AISC and API code checks, about Y-axis and Z-axis
28 $ -KFAC 1.00 1.00 $ Effective length multipliers for beam buckling about Y-axis and Z-axis
29 $ -FLOOD NO      $ Do not flood tubular beams (they have buoyancy)
30 $ -USE @         $ Use everything
31 &DEFAULT -DENSITY 77000 -EMODULUS 200000 -FYIELD 250 -ALPHA 1.2e-5 -POI_RAT 0.3 \
32         -FLOOD NO \
33         -USE @ \
34         -CMFAC 0.85 0.85 -KFAC 1.00 1.00
35 $
36 $
37 $** Body BARGE start
38 &DESCRIBE BODY BARGE
39 $

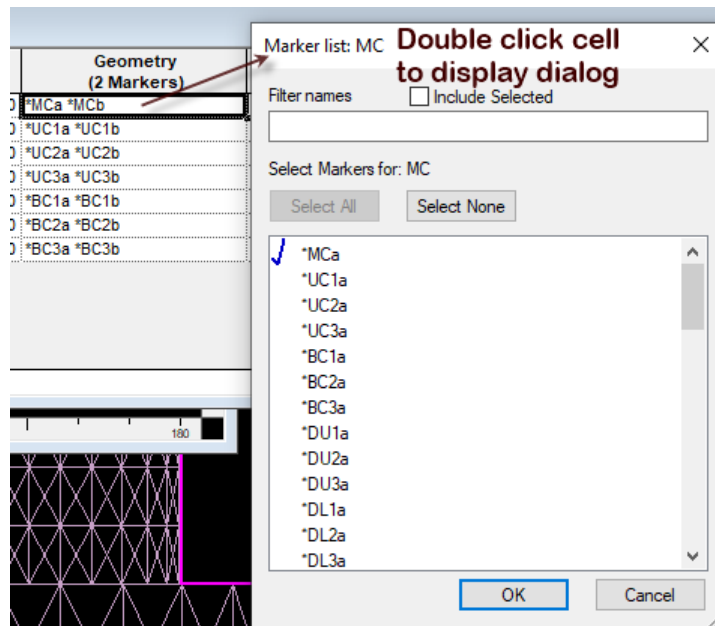
```

DEFAULT settings and set the default material properties added to moses.dat file

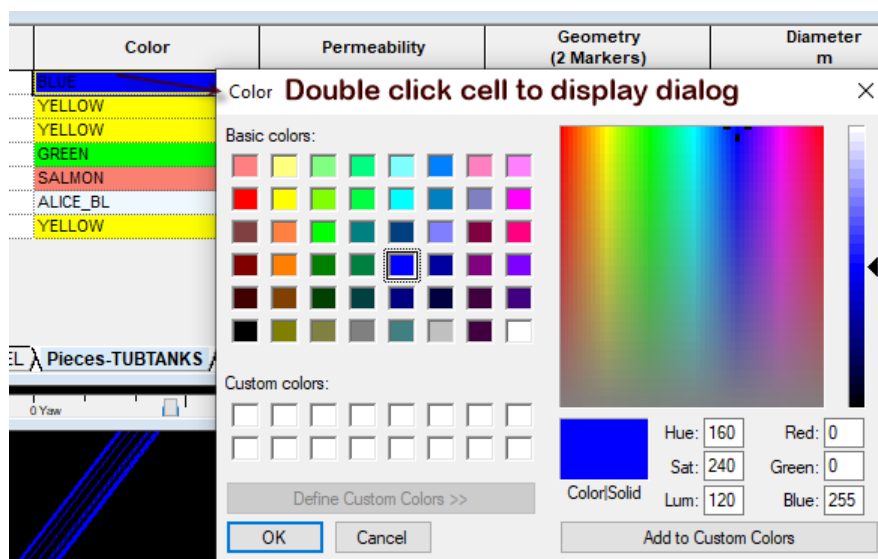
Table UI and Column editing

Most of the columns are simple text fields and support copy and paste. There are two exceptions:

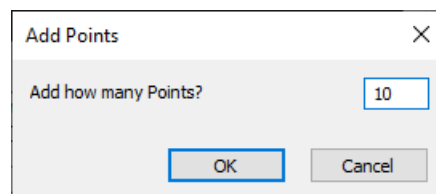
- Include column: This is a checkbox. In all but the CLASS table, this is used to select whether an object is visible and included in the saved MOSES.dat file. In the CLASS table, this column is read-only and indicates whether the particular CLASS is used by visible objects (and hence included in the saved MOSES.dat file). As discussed further below, for an object to be included, its parent objects above it in the hierarchy must also be included; a convenient way to do this is to hold down the Ctrl key when turning ON include for an object. Furthermore, holding down the Ctrl key when selecting a parent object will also select all child objects below it in the hierarchy.
- List selection columns: These columns are used to select MOSES Objects or geometry entities (Surfaces, Meshes or Markers) which already exist in the model. Double clicking in these cells will display the appropriate selection dialog rather than initiating editing of the cell. You can however also edit the text directly, to do so you must press F2 to edit the existing text; typing directly will overwrite the existing data (use the Escape key to abort manual editing). If you edit the text directly you should specify a space-delimited list of the objects. Matching is case-insensitive and you use a '*' wildcard character after the first few characters of the object's name.



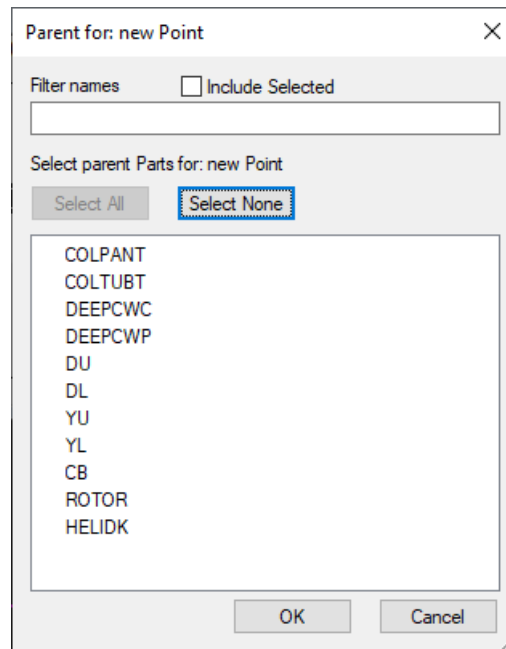
- Color: Double click on the cell to show the Color Picker dialog. The MOSES name of the closest color to that chosen will be used and written in text in the cell. Alternatively simply type the MOSES color name (auto complete is available). Use F2 to edit the existing text.



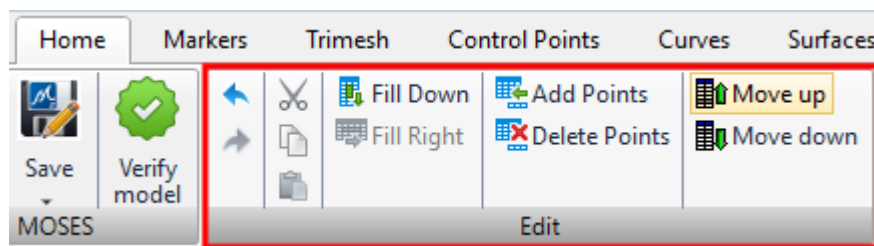
- Adding objects (Ctrl+A) – if a whole Row(s) is selected the number will be used to determine the number of new objects to add. They are inserted below the selection. If the selection is not a whole row you will be prompted for the number of objects to add, and the new objects will be added to the bottom of the table.



- When a new object(s) (unless a BODY or CLASS) is added, you will be prompted for its parent (the new object will then be automatically added to the parent).



- The commands in the Edit menu can be used to Add, Delete objects; move objects within the list and copy, paste, fill data between cells.



Common columns

There are two columns which are common to all the tables:

- Include column:** This checkbox (in the first column of all tables) specifies whether the object should be included in the MOSES.dat file. This feature allows variant objects to be defined and included or excluded from the MOSES.dat file as required (whilst still being saved in the main model.msdl file). This column also controls which objects are displayed in the Modeler drawing windows. Because of the object hierarchy, objects are only visible and included in the MOSES.dat file *if* their parent objects are themselves included. Holding down the Ctrl key when you *turn on* include for an object will also turn on all related parent *and* child objects. This facilitates ensuring that all desired objects are included. For instance, to ensure that all child objects of a specific BODY are included, turn *on* include for the desired BODY with the Ctrl key pressed: this will also turn on all child objects all the way down the hierarchy ensuring that the full BODY definition is included. Conversely, to ensure that a specific child object, for example a BEAM is included, turning *on* include for the desired BEAM whilst holding down the Ctrl key will ensure that all parents are also included, thus ensuring that the BEAM itself is included. This special Ctrl functionality only works when turning objects *on* (not off). The logic behind this is that for an object to be included, all parent objects must also be turned *ON*; however to exclude, it is sufficient just to turn off the specific object.
- Options column:** This is a free-text field where any further options (that do not have their own specific columns) can be included for each object. An understanding of the MOSES language is required to make meaningful use of this feature.

BODY table

When you add a BODY, a PART and (exterior) COMPARTMENT of the same name are automatically added to the corresponding tables, this is done for convenience since a BODY must have at least one PART and will

normally have an exterior COMPARTMENT for the buoyant part of the model. If you modify the name of the BODY, the names of the associated PART and COMPARTMENT will also update. The associated PART and COMPARTMENT may be deleted if desired.

- BODY name: Upper case, maximum eight character name for the MOSES Object.
- PART List: PARTs belonging to the body. Double click to select from the dialog or press F2 to edit manually. The PARTs are referenced by their name and the list is space-delimited.

PART table

- Parent: The BODY to which this PART belongs.
- PART name: Upper case, maximum eight character name for the MOSES Object.
- Type: user-defined or one of the inbuilt types (JACKET, PCONNECT)
- COMPARTMENT List: COMPARTMENTs in the PART
- PLATE Piece List: PLATE mesh PIECEs in the PART
- BEAM List: BEAMs in the PART
- POINT List: POINTs in the PART

COMPARTMENT table

- Parent: The BODY > PART to which this COMPARTMENT belongs.
- COMPARTMENT name: Upper case, maximum eight character name for the MOSES Object.
- Type: Whether the COMPARTMENT is internal or external (not included in MOSES.dat file). This field is non-editable and automatically updated; if the COMPARTMENT name is the same as that of its parent PART and the PART's parent BODY, then it will be an *exterior* COMPARTMENT, otherwise it will be an *interior* COMPARTMENT
- PANEL Piece List: PANEL mesh PIECEs defining the COMPARTMENT
- TUBTANK Piece List: TUBTANK PIECEs defining the COMPARTMENT

Piece: PLATE table

- Parent: The BODY > PART to which this PLATE belongs.
- Structure name: Upper case, maximum eight character name for the MOSES Object.
- Color: Color of the PLATE PIECE
- Class: CLASS describing the plate's structural properties.
- Geometry (mesh): The Mesh surface defining the geometrical shape of the PLATE
- Symmetry: If the Mesh defining the PLATE geometry is symmetrical, the Original, Reflection (about y=0 plane) or Both sides may be used for the PLATE
- Node Prefix: Two-letter prefix to be used for all the NODEs defining the mesh geometry; must include the asterisk character '*' prefix
- DNV stiffened plate: Select this option if you wish the plate to be setup so that it can be included in the DNV stiffened plate code checking (this can only be applied for a limited range of specific plate geometries)
- PLATE node Merge tol. : If specified, the nodes of PLATEs included earlier in the file will be used if they are within the specified tolerance instead of creating new nodes.
- Transparency: Transparency fraction (when rendered in Hull Modeler only).

Piece: PANEL table

- Parent: The BODY > PART > COMPARTMENT to which this PIECE belongs.
- PANEL Piece name: Upper case, maximum eight character name for the MOSES Object.
- Color: Color of the PANEL PIECE
- Permeability: Value for -PERMEABILITY option. (Internal compartments are expected to have negative permeability, so that they may be flooded if required; External compartments contributing to buoyancy are expected to have positive permeability.)
- Geometry (mesh): The Mesh surface defining the geometrical shape of the PANEL. This should normally be a closed mesh with normals facing outwards.

- Symmetry: If the Mesh defining the PANEL geometry is symmetrical, the Original, Reflection (about y=0 plane) or Both sides may be used for the PANEL
- Node Prefix: Two-letter prefix to be used for all the NODEs defining the mesh geometry; must include the asterisk character '*' prefix
- PANEL, PLATE node Merge tol. : If specified, the nodes of PANELs (and then PLATEs) included earlier in the file will be used if they are within the specified tolerance instead of creating new nodes.
- Transparency: Transparency fraction (when rendered in Hull Modeler only).

Piece: TUBTANK table

- Parent: The BODY > PART > COMPARTMENT to which this TUBTANK belongs.
- TUBTANK Piece name: Upper case, maximum eight character name for the MOSES Object.
- Color: Color of the TUBTANK PIECE
- Permeability: Value for -PERMEABILITY option. (Internal compartments are expected to have negative permeability.)
- Geometry (2 Markers): The two ends of the tube. To be used for MOSES Object definition, Marker names must be prefixed with '*'.
- Diameter: The diameter of the tube in mm (if metric units are selected) or inches (if English units are selected)
- Transparency: Transparency fraction (when rendered in Hull Modeler only).

BEAM table

- Parent: The BODY > PART to which this BEAM belongs.
- BEAM Piece name: Upper case, maximum eight character name for the MOSES Object.
- Color: Color of the BEAM
- Class: CLASS describing the BEAM's structural properties
- Geometry (Markers, Curves or Stringers): List of geometry defining the segments of the BEAM. To be used for MOSES Object definition, Marker names must be prefixed with '*'.
- Orientation: Section orientation value in degrees for the -CA option
- Offset 1 (x, y, z): Offset at end 1 of the BEAM
- Offset 1 Axes: Offset axis system is Global (-GO1) or Local (-LO1)
- Offset 2 (x, y, z): Offset at end 2 of the BEAM
- Offset 2 Axes: Offset axis system is Global (-GO1) or Local (-LO1)
- Flood: If selected the -FLOOD YES option is added for the BEAM
- PLATE node Merge tol. : If specified, the nodes of PLATEs included earlier in the file will be used if they are within the specified tolerance instead of creating new nodes.

CLASS table

- CLASS Piece name: Upper case, maximum eight character name for the MOSES Object; must start with a tilde character '~'.
 - Section: Section information may be defined manually by typing (use F2 key to begin editing), for example "TUBE 1600.0 30.0". Double-clicking will display a section dialog from which the section shape may be selected from the standard section shape library (moses.aux) file.
 - Added Plate: Dimensions of added plate as specified by the PT and PW parameters for certain section types
 - Added Mass diameter: Value for the -AMASDIAMETER option
 - Drag diameter: Value for the -DRAGDIAMETER option
 - Tran. Stiffener Number: NUMBER parameter for the -NT_STIFF option
 - Tran. Stiffener Spacing: SPACE parameter for the -T_STIFF option
 - Tran. Stiffener Location: WHERE parameter for the transverse stiffeners
 - Tran. Stiffener CLASS: CLASS name of the transverse stiffeners
 - Long. Stiffener Number: NUMBER parameter for the -NL_STIFF option
 - Long. Stiffener Spacing: SPACE parameter for the -L_STIFF option
-

- Long. Stiffener Location: WHERE parameter for the longitudinal stiffeners
- Long. Stiffener CLASS: CLASS name of the longitudinal stiffeners

POINT table

- Parent: The BODY > PART to which this POINT belongs.
- POINT Piece name: Upper case, maximum eight character name for the MOSES Object; must start with an asterisk character '*'.
- Color: Color of the POINT.
- Geometry (Marker): Marker which defines the location of the POINT; when selected this automatically updates the location data. To be used for MOSES Object definition, Marker names must be prefixed with '*'.
- Location (x, y, z): You can specify the location of the POINT directly without first adding a Marker.

Index

A

About.....	395
AC Rule	393
Add Marker	374
Add Shape	382
Add Stringer On Contour	386
Add Surface.....	126
Adding a Frame Opening	261
Adding an Inclined Frame.....	261
Adding Inclined Sections	297
Adding Points To Stringers.....	282
Aft Perpendicular	21
Align.....	131
Align Surfaces	383
Align to Plane.....	206
Align to Vector.....	204
Americas Cup	256
Amidships	21
Animate	391
Animation Files	353
Appearance	384
Approximate Tangency	152
Arrange Icons	394
Assembly	
Icons.....	55
Assembly Pane.....	53, 374
Context Sensitive Menus	54
Docking	53
Drag and Drop.....	54
Editing.....	54
Hiding.....	54
Associate Project.....	372
Auto rotate center	374
Automation Reference.....	394
Auto-save.....	79

B

Background.....	390
Background Image, Importing	332
Background Image, Perspective View	335
Background Image, Scaling	332
Background Image, Zero Point	332
Background Images	331
Baseline.....	21
Bentley Cloud Services	372
Bibliography.....	18
Body Plan Window	43
Bond Edges	380
Bonded Edges	38

Bonding	151
Types.....	152
when to use.....	150
Bonding with Inserted surface.....	157
Bonding within a Surface.....	158
Bonding, Commands Affected	158
Breaking Stringers	288
Brightness Level.....	121
B-Spline.....	217
B-splines	18
Built-in Functions	74
Buttocks	30, 37

C

Calc All Decks	386
Calc All Frames.....	385
Calc All Parts	393
Calc All Stringers.....	387
Calc Selected Decks	386
Calc Selected Stringers.....	387
Calc Upstands.....	387
Calculate Areas	250, 392
Calculate Hydrostatics	393
Calculating Frames	266
Calculating Part Weights	322
Calculating Plates.....	310
Calculating Stringers	291, 292
Calculating Upstands	297
Calculations Window.....	72
Cascade.....	393
Check for Updates.....	394
Circular and Elliptical Surfaces.....	219
Circular Arcs	218
Close	370
Coefficients.....	392
Color	78, 374
Column Hiding.....	61
Command Keys.....	402
comparison.....	255
Compress	389
CONNECTION Centre.....	394
Constraining Movement	203
Constraints	233
Contour Tolerance.....	77, 116
Contours.....	390
display	36
select.....	39
Control Point	
Display.....	215
Control Point Column Ordering.....	77

Control Point Display Format	65	Copy	373
Control Point Editing	65	Copying	351
Control Point Group		Copying Drawing	397
Moving	210	Copying Numeric Data	351
Resizing	211	Copying Text	398
Rotating	212	Current Surface	124
Control Point Properties	203	Curvature	118, 389
Control Point Weight	217	Curvature Contrast Level	121
Control Point Weights	403	Curvature Display Using Rendering	118
Control Point Window	64	Curvature on Contours	117
Control Point, Adding	201	Curvature Porcupine Size	77
Control Point, Align To First Selected	204	Curve of Areas Window	71
Control Point, Aligning	204	Cut	373
Control Point, Compacting	213		
Control Point, Copy and Paste	202	D	
Control Point, Deleting	202	Data Export	396
Control Point, Grouping	215	Data Interpolation	70
Control Point, Moving	203	Data Menu	391
Control Point, Smoothing	207	Datum Waterline	21, 37
Control Point, Smoothing a Patch	208	Deck Info	386
Control Point, Straightening	207	Deck Limitations	272
Control Point, Straightening a Patch	208	Deck Menu	385
Control Points	200	Deck Opening	270
Add	379	Decks	269
Advanced	380	Add	269
Align to Plane	380	Add	385
Align to Vector	379	Copy	272
Compact	380	Delete	270, 385
Control Points		Modifying	272
Move Controls	379	Move	272
Control Points		Visibility	271
Copy Position	379	Decks table	69
Paste Position	379	Deduct Skin Thickness	385
Control Properties	380	Deducted Sections	39
Delete	379	Delete DXF background	390
First Selected	380	Delete Marker	374
Group	380	Delete Surface	127
Mask	380	Deleting Stringer Points	285
Rotate Controls	379	Developable surfaces	92
Size Controls	379	DGN	355
Smooth Controls	379	Diagonals	30, 37
Straighten Controls	379	Digitising a Background Image	334
Ungroup	380	Disassociate Project	372
Unmask	380	Discontinuities	214
Control Points, Synchronization	65	TriMesh	239
Controls Menu	379	Display	
Controls Toolbar	368	Animate	391
Convergence Limits	233	Background	390
Converting a 2D DXF to a 3D DXF	337	Compress	389
convex hull	18	Contours	390
Convexity	119	Curvature	389
Coordinate system	260	Grid	390

Half.....	389
Markers.....	390
Net.....	388
Outside Arrows.....	389
Render.....	391
Trimming.....	389
Display Area.....	58
Display Menu	388
Display Toolbar.....	368
Display, Part	391
Displaying a TriMesh surface.....	239
Displaying Inclined Sections.....	298
Drafting Data, 2D.....	398
Drafting Data, 3D.....	399
Drawing Decks.....	271
Drawing Depth	389
Drawing Grid	389
Drawing Settings.....	389
Drawing, Copying	397
duplicate markers.....	236
Duplicating Stringers.....	286
DWL.....	21
DXF.....	325

E

Edges	37
Edit Menu	372
Edit Toolbar.....	368
Editing	62
Editing, Copy	62
Editing, Fill Down.....	63
Editing, Fill Right.....	63
Editing, Paste.....	63
Editing, Typing.....	62
Elliptical Arcs.....	219
Exit	372
Export	371
Export Bitmap.....	371
Export Loadgroup.....	372
Export Multiframe text.....	372
Export Plate Mesh Text.....	371
Export Rhino	372
Export ShipConstructor Files.....	371
Export Workshop.....	371
Exporting parts to Multiframe.....	360
Exporting parts to Rhino.....	359
Expressions.....	73

F

Fabric.....	309
Feature Lines	38, 68, 214
Feedback, Ideas.....	394

Feedback, Licensing.....	394
Field of view	51
File Format, 3DMF.....	358
File Format, DXF	357
File Format, IGES	356
File Format, VRML.....	358
File Menu	370
File Toolbar.....	368
Files, Backup.....	79
Files, Recovery.....	79
Fill Down.....	373
Fit Edge to Markers.....	376
Fit Surface to Marker stations, automated	231
Fit Surface to Markers.....	376
Fit Surface to Markers, automated	230
Fit Surface to Markers, least squares	234
Fit Surface to Markers, manual.....	226
Flat Bottom.....	233
Flat Side	233
Flexibility.....	112, 234
Flip Horizontal.....	388
Flip Surfaces.....	383
Flip Toe Direction.....	296
Flip Vertical.....	388
Font	62, 78, 374
Forming Direction.....	306
Forming Process.....	307
Forward Perpendicular.....	21
Frame Boundary Surfaces.....	260
Frame Clipping.....	259
Frame Material.....	259
Frame Menu.....	385
Frame of Reference.....	21, 22, 257, 392
Frame Opening.....	260
Frames.....	258
Add.....	258, 385
Calc All	385
Calculate.....	266
Copy.....	268
Delete	268, 385
Info	385
Modifying	268
Move.....	268
Visibility	267
Frames table	69

G

Gaussian Curvature	118
General.....	306
Generate Stringers.....	387
Generating Stringers	274, 275, 276
Girth	249, 392

Girth Centreline	291, 387
Go to Offset	392
Graph Type	77
Grid	30
Grid Labels	32
Grid Lines, Adding	31
Grid Lines, Copying and Pasting	33
Grid Lines, Deleting	33
Grid Lines, Editing	32
Grid Lines, Sorting	32
Grid Lines, Spacing	31
Grid, Generate from Markers	34
Group and Compact	215

H

Half	45
Help Menu	394
Hidden Surface Elimination	118
Hide DXF	390
Hiding a TriMesh surface	239
Home View	39, 41, 374
Hydrostatic Coefficient Calculation Parameters	248
Hydrostatics	247

I

Image rotation	332
i-model	356
Import	371
Import DXF Background	371
Import DXF Markers	371
Import IGES Surfaces	371
Import Image Background	371
Importing DXF background	335
Importing DXF markers	335, 336
Importing IGES Surfaces	337
Inclined Sections	39
Input of Data	331
Inserting Marker - Procedures	225
Inset Lines	316
Installing	19
Intersections	38
Iterations	233

J

Joining Stringers	288
-------------------------	-----

L

Learning	19
Legal Notice	394
Library	327
Library, Materials	327
Library, Sections	329
License wiki	394

Lighting Options	122
Limits	233
Loadcase, Exporting	326
Loadgroup, Exporting	326
Locking	112, 384
Locking Surfaces	133
Longitudinal Curvature	119

M

Manual Plate Precision	310
Margins	349
Marker distance from surface	93
Marker Menu	374
Marker Station	90
Markers	390
Markers from Excel	86
Markers Toolbar	368
Markers, Adding	85
Markers, Compact	376
Markers, Delete Duplicates	375
Markers, Deleting	86
Markers, Developable Surfaces	92, 375
Markers, Displaying	81
Markers, Fit Surface to	376
Markers, Generate Grid	375
Markers, Generate Grid from Markers	34
Markers, Group Markers by Stations	375
Markers, Importing	92
Markers, Measure Surface Error	376
Markers, Modifying	86
Markers, Move Selected	375
Markers, Properties	376, 377
Markers, Reading Files	91
Markers, Re-order Selected	375
Markers, Re-sort Selected	375
Markers, Reverse Ordering for current station ..	376
Markers, Rotate Selected	375
Markers, Selection	84
Markers, Snap control point to	376
Markers, Snap Control Point to Marker	376
Markers, Sort within Stations	375
Masking	216
Material	112
Material properties	114
Materials	327
Measure Surface Error	234, 376
Menu Shortcuts	403
Menus	369
Mesh Add node	169
Mesh Align	173
Mesh Bonding	180
Mesh Boolean Intersection	177

Mesh Boolean Union	177
Mesh Compact nodes	171
Mesh Delete node	169
Mesh Delete triangle	171
Mesh Direction	171
Mesh Duplicate	171
Mesh Flip	173
Mesh Flip triangle pair	171
Mesh from 3 or 4 markers	160
Mesh from Extruded curve	166
Mesh from Markers	235
Mesh from Planar curve	165
Mesh Hotline	186
Mesh Join	176
Mesh Menu	376
Mesh Move	172
Mesh Offset	174
Mesh Rendering	195
Mesh Rotate	173
Mesh Size	172
Mesh Snap node	169
Mesh Split	175
Mesh Statistics	194
Mesh Triangle selection	170
Microstation	355
Model checking	257
Modifier Keys	402
Modifying Stringer Points	285
MOSES Objects	406
MOSES, skin thickness	115
Mouse Holes	259
Mouse Wheel Support	41
Move Controls	210
Moving Files Across Platforms	404
Moving Files From Macintosh to Windows	404
Moving Files from Windows to Macintosh	405
Moving Stringer Points	283

N

New	370
News	394
No Tangency	152
Nudge	389
NURB	217

O

Offsets	392
Offsets Data	352
Offsets File	68
Offsets Window	66
Offsets, Calculating	67
Offsets, Copying	68

Offsets, Customising	68
Offsets, Displaying	68
Old Developable	309
Online Support	394
Open	370
Orientation	349
Orientation	387
Output of Data	348
Outside Arrow Size	77
Outside Arrows	113, 257, 389

P

Page Setup	372
Page Size	349
Pan	39, 41, 373
Panning	41
Parametric Transformation	251, 393
Parametrics	38
Part Toolbar	369
Part Window	52, 69
Parts table	69
Parts, Exporting	325
Paste	373
Pasting	331
Perspective	374
Perspective camera angle	51
Pictures and Text	397
Pin Jig table	69
Pin Jigs	317
Plate	
baseline	316
centreline	316
Template sightline	317
User Coordinate Systems	315
Plate clash detection	305
Plate development	306
Plate Development	
Plating a Whole Surface	299
Trimmed Surfaces	300
Plate Dimensions table	69
Plate Information	388
Plate Menu	388
Plate Mesh Options	309
Plate Strain	312
Plate Templates	388
Plate UCS	388
Plate, Add Surface	388
Plate, Assembly UCS	388
Plate, Calc All	388
Plate, Calc Selected	388
Plate, Centreline Mirrored	301
Plate, Delete	388

Plate, Strain 390
Plate, triangular 304, 305
Plate, triangular surface 304
Plates
 Add..... 302, 388
 Delete 304
Plates table.....69
Precision 115, 389
Precision, high..... 116
Precision, highest..... 116
Precision, low..... 116
Precision, lowest..... 115
Precision, medium..... 116
Precision, structure..... 258
Preferences.....76, 373
Preferences, Structure..... 373
Print.....372
Print Titles..... 350
Printing 348
Printing, Color 349
ProjectWise Projects..... 372

R

Redo..... 373
Refresh Options.....77
Render..... 391
Rendered Net Color Coded..... 122
Rendered Selection..... 122
Rhino file, Exporting 358
Rhino file, Importing..... 338
Ribbon..... 348, 395
Rider Bar..... 265
Rotate 50, 373
Rotate Controls 212
Rotate Surfaces..... 383
Row Sorting 62
Ruling Lines..... 389

S

Save 370
Save As..... 370
Saved Views41, 374
Saving Calculations76
Saving Markers91
Scaling factors..... 254
scanning 331
Search Parameters..... 252
Section Splitting.....46
Sectional Area Curve Stations 77
Sectional Area Curve Type 77
Sections 30, 37
Select 373

Selection.....59
Selection, Block of Cells60
Selection, Complete Table60
Selection, Contiguous Rows and Columns.....59
Selection, Single Cell59
Selection, Single Column.....59
Selection, Single Row59
Service Request Manager 394
Set Home View 374
Setting Shapes & Cutouts 295
Shape & Cutout... 387
Shapes, Adding New 124
Shapes, Box 124
Shapes, Cylinder 124
Shapes, Hemisphere 124
Shapes, Sphere 124
Sheel expansion 318
Shell Expansion..... 388
 create..... 318
 display 321
Shell thickness..... 114
Shift Key..... 203
Show DXF..... 390
Show Intersecting Surfaces.....54
Show Negative Values 119
Show Only.....55
Show Positive Values..... 119
Shrink..... 39, 40, 58, 373
Simple Shading..... 118
Size Controls..... 212
Size Surfaces 129
Sizing Columns 60
Sizing Columns, Dragging.....60
Sizing Columns, Size to Text61
Skin thickness..... 114
Skin Thickness 267
Skin Thickness Deduction.....67
Skin thickness, modelling for MOSES 115
Smooth Controls.....208
Smooth Interior Controls.....376
Smooth Plate Edges..... 309, 310
Smooth Shading..... 118
Smooth Surface Interior..... 229
Snap to Grid283, 387, 389
Solve Calculations.....75, 393
Split Section Display46, 112
Start Trimming 385
Statement Syntax.....73
station index90
Station Indicators.....44
Status Bar 374
Straighten Controls..... 208

Strain.....	390	Surface Use, Type.....	257
Strain Ratio.....	307	Surface, Adding New.....	126
Strain, Forming Direction	313, 391	Surface, Aligning.....	131, 132, 133, 383
Strain, None.....	391	Surface, Conic.....	107
Strain, Perpendicular to Forming Direction.....	313, 391	Surface, Deleting.....	127
Strain, Principal.....	313	Surface, Flipping.....	130
Strain, Shear.....	313, 391	Surface, Intersections	139
Strict Tangency.....	153	Surface, Mirror	130
Stringer Info.....	387	Surface, Moving.....	127
Stringer Library	294	Surface, Moving, Freeform.....	128
Stringer Menu	386	Surface, Moving, Numerical	128
Stringer Orientation.....	295	Surface, NURB.....	107
Stringer Points		Surface, Rotating.....	131
Add.....	386	Surface, Trimming.....	139
Delete	386	Surfaces	
Info.....	387	Add.....	382
Stringer points table.....	69	Align.....	383
Stringer Toolbar	369	Appearance	384
Stringers	272	Approximate to markers	384
Add.....	386	Delete	383
Break.....	386	Duplicate	383
Delete	386	Flip.....	383
Duplicate.....	386	Locking.....	384
Join	386	Rotate.....	383
Mirror	386	Size	383
Stringers table.....	69	Visibility	384
Stringers, Adding	273	Surfaces list	126
Stringers, Modifying.....	276	Surfaces Menu	381, 382
Stringers, Tips.....	273	Surfaces Window	66
Structure.....	257	Surfaces, Displaying Trimmed	145
Structure Datasheets.....	394	Symmetrical	112
Structure Toolbar.....	369	T	
Sub-menu Length.....	77	Tab	64
Supported File Formats.....	396	Taking Measurements.....	30
Surface		Templates	316
Bonding.....	149	Thickness, skin.....	114
Cutting, trimming.....	146	Tile Horizontal.....	393
Untrim	146	Tile Vertical.....	393
Surface Algorithms.....	401	Toolbars.....	365, 374
Surface Appearance.....	110	Transpose.....	221
Surface Columns	233	Transverse Curvature.....	119
Surface Definition, 3D.....	400	Trim	385
Surface Fitting (NURB) – Procedures	225	Trim Gray.....	144, 146
Surface Name.....	111	Trim Invisible.....	144, 146
Surface Operations	123, 134	TriMesh Surface, Generate.....	376
Surface Properties	111, 385	TriMesh Troubleshooting	236
Surface Thickness	257	TriMesh, Joining	240
Surface Toolbar	369	TriMesh, Remeshing	239
Surface Topology.....	226	TriMesh, Visibility	239
Surface Trimming	257	TriMesh, Welding.....	241
Surface Type	112	Trimmed surfaces	141
Surface Use	112		

trimming
 untrim..... 146
Trimming..... 389
 Concept..... 139
 Rules..... 142
 when to use..... 141
Trimming curves39, 344
Trimming Off..... 146
Trimming, Cutting Curves 146
Trimming, Cutting Surfaces..... 146

U

UCS..... 315
Unbond Edge 380
Unbonding 158
Undo..... 372
Undo Levels77
Units.....29, 75, 391
Untrim 385
Upstand Cutouts..... 297
Upstand Orientation..... 297
Upstands table.....69
Use Surface Color for Drawing Parametrics.....78

V

Variables.....73

Versions, coordinate system11
Vessel Type.....23, 392
Vessel Type, catamaran24
Vessel Type, monohull.....24
Vessel Type, trimaran25
View Menu.....373
View rotation center50
View Toolbar.....368
Visibility 112, 384
Visibility Toolbar.....369

W

Waterlines..... 30, 37
Windage Surfaces.....25, 392
Window Menu.....393
Window Toolbar.....369
Windows.....20
Working With Parts.....322

Z

Zebra..... 120
Zero Point22, 392
Zoom39, 58, 373
Zooming.....39